



U.S. Army Corps of Engineers
Japan District

PN77833 VEHICLE MAINTENANCE SHOP
KADENA AIR BASE, OKINAWA, JAPAN

SPECIFICATIONS

Intermediate Submittal

CONTRACT NO. W912HV-19-D-0010
TASK ORDER NO. W912HV23F0041

Submitted By:
SSFM International
501 Sumner Street, Suite 620
Honolulu, HI 96817

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143	CU133	UTILITY PLAN - 2 (CLIN 0004)
144	CU134	UTILITY PLAN - 3 (CLIN 0004)
145	CU135	UTILITY PLAN - 4 (CLIN 0004)
146	CU141	OVERALL UTILITY PLAN COF SS (CLIN 0004)
147	CU142	UTILITY PLAN - 1 COF SS (CLIN 0004)
148	CU143	UTILITY PLAN - 2 COF SS (CLIN 0004)
149	CU144	UTILITY PLAN - 3 COF SS (CLIN 0004)
150	CU201	DRAINLINE 1 PLAN AND PROFILE - 1
151	CU202	DRAINLINE 1 PLAN AND PROFILE - 2
152	CU203	DRAINLINE 2 AND 3 PLAN AND PROFILE - 3
153	CU204	DRAINLINE 3 AND 4 PLAN AND PROFILE - 4
154	CU205	DRAINLINE 5 6 AND 7 PLAN AND PROFILE - 5
155	CU206	DRAINLINE A1 AND A2 PLAN AND PROFILE - 6
156	CU207	DRAINLINE A3 AND A4 PLAN AND PROFILE - 7
157	CU208	DRAINLINE A4 AN A5 PLAN AND PROFILE - 8
158	CU209	DRAINLINE A6 PLAN AND PROFILE - 9

159	CU210	DRAINLINE B1 AND B2 PLAN AND PROFILE - 10
160	CU211	DRAINLINE C1 AND C2 PLAN AND PROFILE - 11
161	CU212	DRAINLINE C3 PLAN AND PROFILE - 12
162	CU213	DRAINLINE C3 C4 AND C5 PLAN AND PROFILE - 13
163	CU214	DRAINLINE 20 AND 21 PLAN AND PROFILE - 14
164	CU215	DRAINLINE 21 PLAN AND PROFILE - 15
165	CU216	DRAINLINE 22 23 AND 24 PLAN AND PROFILE - 16
166	CU217	DRAINLINE 24 AND 25 PLAN AND PROFILE - 17
167	CU218	DRAINLINE 25A 25B AND 26 PLAN AND PROFILE - 18
168	CU219	DRAINLINE 27 PLAN AND PROFILE - 19
169	CU221	SEWERLINE 1 PLAN AND PROFILE - 1
170	CU222	SEWERLINE 1 AND 2 PLAN AND PROFILE - 2
171	CU223	SEWER LATERAL 1A, 1B AND 2A PLAN AND PROFILE - 3
172	CU224	SEWERLINE 3 AND SLAT 3B PLAN AND PROFILE - 4
173	CU225	SEWERLINE 3 AND SLAT 3B PLAN AND PROFILE - 5
174	CU226	SEWERLINE 20 PLAN AND PROFILE - 6
175	CU227	SEWERLINE 21 AND SLAT 20 PLAN AND PROFILE - 7
176	CU228	SEWERLINE 30 PLAN AND PROFILE - 8
177	CU229	SEWERLINE 30 PLAN AND PROFILE - 9
178	CU230	SEWERLINE 30 PLAN AND PROFILE - 10
179	CU231	WATERLINE 1 PLAN AND PROFILE - 1
180	CU232	WATERLINE 1 PLAN AND PROFILE - 2
181	CU233	WATERLINE 1 PLAN AND PROFILE - 3
182	CU234	WATERLINE 1 PLAN AND PROFILE - 4
183	CU235	WATERLINE 1 PLAN AND PROFILE - 5
184	CU236	WATERLINE 1 PLAN AND PROFILE - 6
185	CU237	WATERLINE 20 PLAN AND PROFILE - 7
186	CU238	WATERLINE 20 PLAN AND PROFILE - 8
187	CU239	WATERLINE 30 PLAN AND PROFILE - 9
188	CU240	WATERLINE 31 PLAN AND PROFILE - 10
189	CU251	FIRELINE 1 PLAN AND PROFILE - 1
190	CU252	FIRELINE 1 PLAN AND PROFILE - 2
191	CU253	FIRELINE 1 PLAN AND PROFILE - 3
192	CU254	FIRELINE 1 PLAN AND PROFILE - 4
193	CU255	FIRELINE 1 PLAN AND PROFILE - 5
194	CU256	FIRELINE 1 PLAN AND PROFILE - 6

195	CU257	FIRELINE 2 PLAN AND PROFILE - 7
196	CU258	FIRELINE 2 PLAN AND PROFILE - 8
197	CU259	FIRELINE 2 PLAN AND PROFILE - 9
198	CU260	FIRELINE 20 PLAN AND PROFILE - 10
199	CU261	FIRELINE 20 AND 21 PLAN AND PROFILE - 11
200	CU262	FIRELINE 21 PLAN AND PROFILE - 12
201	CU271	FIRE HYDRANT LATERAL PLAN AND PROFILE - 1
202	CU272	FIRE HYDRANT LATERAL PLAN AND PROFILE - 2
203	CU273	FIRE HYDRANT LATERAL PLAN AND PROFILE - 3
204	CU274	FIRE HYDRANT LATERAL PLAN AND PROFILE - 4
205	CU275	FIRE HYDRANT LATERAL PLAN AND PROFILE - 5
206	CU276	FIRE HYDRANT LATERAL PLAN AND PROFILE - 6
207	CU277	FIRE HYDRANT LATERAL PLAN AND PROFILE - 7
208	CU501	DRAINAGE DETAILS - 1
209	CU502	DRAINAGE DETAILS - 2
210	CU503	SEWER DETAILS
211	CU504	WATER DETAILS - 1
212	CU505	WATER DETAILS - 2
213	CU506	WATER DETAILS - 3
214	S-001	GENERAL NOTES
215	S-002	DESIGN DATA, DESIGN CODES AND STANDARDS, ABBREVIATION
216	S-003	VMS -COMPONENTS AND CLADDING WIND PRESSURE DIAGRAM
217	S-004	OS AND POL & HZMT -COMPONENTS AND CLADDING WIND PRESSURE DIAGRAM
218	S-005	SSAW -COMPONENTS AND CLADDING WIND PRESSURE DIAGRAM
219	S-006	SS -COMPONENTS AND CLADDING WIND PRESSURE DIAGRAM
220	S-007	FP, CH, AND WT -COMPONENTS AND CLADDING WIND PRESSURE DIAGRAM
221	S-008	COF SS -COMPONENTS AND CLADDING WIND PRESSURE DIAGRAM
222	S-A101	VMS -OVERALL FOUNDATION PLAN (CLIN 0001)
223	S-A102	VMS -OVERALL BRIDGE CRANE, SECOND FLOOR AND LOWER ROOF FRAMING PLAN (CLIN 0001)
224	S-A103	VMS -OVERALL TRUSS BOTTOM CHORD AND ROOF FRAMING PLAN (CLIN 0001)
225	S-A104	VMS -OVERALL UPPER ROOF FRAMING PLAN (CLIN 0001)
226	S-A111	VMS BUILDING -PARTIAL FOUNDATION PLAN (ZONE A) (CLIN 0001)
227	S-A112	VMS BUILDING -PARTIAL FOUNDATION PLAN (ZONE B) (CLIN 0001)
228	S-A121	VMS -PARTIAL SECOND FLOOR FRAMING PLAN (ZONE A) (CLIN 0001)

229	S-A122	VMS -PARTIAL SECOND FLOOR AND LOWER ROOF FRAMING PLAN (ZONE B) (CLIN 0001)
230	S-A123	VMS -TRUSS BOTTOM CHORD FRAMING PLAN (ZONE A) (CLIN 0001)
231	S-A124	VMS -ROOF FRAMING PLAN (ZONE B) (CLIN 0001)
232	S-A125	VMS -UPPER ROOF FRAMING PLAN (ZONE A) (CLIN 0001)
233	S-A201	VMS -EXTERIOR ELEVATIONS (CLIN 0001)
234	S-A202	VMS -EXTERIOR ELEVATIONS (CLIN 0001)
235	S-A203	VMS -EXTERIOR ELEVATIONS (CLIN 0001)
236	S-A301	VMS -BUILDING SECTIONS (CLIN 0001)
237	S-A302	VMS -BUILDING SECTIONS (CLIN 0001)
238	S-A303	VMS -BUILDING SECTIONS (CLIN 0001)
239	S-A304	VMS -BUILDING SECTIONS (CLIN 0001)
240	S-A501	VMS -SECTIONS AND DETAILS (CLIN 0001)
241	S-A502	VMS -SECTIONS AND DETAILS (CLIN 0001)
242	S-A503	VMS -SECTIONS AND DETAILS (CLIN 0001)
243	S-A504	VMS -SECTIONS AND DETAILS (CLIN 0001)
244	S-A505	VMS -SECTIONS AND DETAILS (CLIN 0001)
245	S-A506	VMS -SECTIONS AND DETAILS (CLIN 0001)
246	S-A601	VMS -TRUSS DETAILS (CLIN 0001)
247	S-A701	VMS -MAINTENANCE PIT PLAN AND SECTIONS (CLIN 0001)
248	S-A702	VMS -SECTIONS AND DETAILS (CLIN 0001)
249	S-A703	VMS -ELEVATOR FOUNDATION, FRAMING PLAN AND SECTION (CLIN 0001)
250	S-A704	VMS -ELEVATOR SECTION AND DETAILS (CLIN 0001)
251	S-A705	VMS-STAIR 1 FOUNDATION, FRAMING PLAN AND SECTIONS (CLIN 0001)
252	S-A706	VMS-STAIR 2 FOUNDATION, FRAMING PLAN AND SECTIONS (CLIN 0001)
253	S-A707	VMS -MECHANICAL YARD FOUNDATION PLAN AND ELEVATIONS (CLIN 0001)
254	S-A708	VMS -MECHANICAL YARD BUILDING SECTIONS (CLIN 0001)
255	S-B101	ORGANIZATIONAL STORAGE -FOUNDATION AND ROOF FRAMING PLAN (CLIN 0002)
256	S-B201	ORGANIZATIONAL STORAGE -EXTERIOR ELEVATIONS (CLIN 0002)
257	S-B301	ORGANIZATIONAL STORAGE -BUILDING SECTIONS (CLIN 0002)
258	S-B501	ORGANIZATIONAL STORAGE -SECTIONS AND DETAILS (CLIN 0002)
259	S-C101	POL & HAZMAT STORAGE FOUNDATION AND ROOF FRAMING PLAN (CLIN 0002)
260	S-C201	POL & HAZMAT STORAGE EXTERIOR ELEVATIONS (CLIN 0002)
261	S-C301	POL & HAZMAT STORAGE BUILDING SECTIONS (CLIN 0002)
262	S-D101	SSAW -OVERALL FLOOR FOUNDATION PLAN (CLIN 0004)

263	S-D102	SSAW -OVERALL ROOF FRAMING PLAN (CLIN 0004)
264	S-D103	SSAW -PARTIAL FOUNDATION PLAN (ZONE A) (CLIN 0004)
265	S-D104	SSAW -PARTIAL FOUNDATION PLAN -(ZONE B) (CLIN 0004)
266	S-D105	SSAW -PARTIAL ROOF FRAMING PLAN -(ZONE A) (CLIN 0004)
267	S-D106	SSAW -PARTIAL ROOF FRAMING PLAN -(ZONE B) (CLIN 0004)
268	S-D201	SSAW -EXTERIOR ELEVATIONS (CLIN 0004)
269	S-D202	SSAW -EXTERIOR ELEVATIONS (CLIN 0004)
270	S-D301	SSAW -BUILDING SECTIONS (CLIN 0004) HONO 501 SUMN SSFM I
271	S-D302	SSAW -BUILDING SECTIONS (CLIN 0004)
272	S-D501	SSAW -SECTIONS AND DETAILS (CLIN 0004)
273	S-D502	SSAW -SECTIONS AND DETAILS (CLIN 0004)
274	S-D503	SSAW -SECTIONS AND DETAILS (CLIN 0004)
275	S-D701	SSAW -MECHANICAL YARD FOUNDATION PLAN ANF ELEVATIONS (CLIN 0004)
276	S-D702	SSAW -MECHANICAL YARD BUILDING SECTIONS (CLIN 0004)
277	S-E101	STORAGE SHED -FOUNDATION PLAN (CLIN 0004)
278	S-E102	STORAGE SHED -ROOF FRAMING PLAN (CLIN 0004)
279	S-E201	STORAGE SHED -EXTERIOR ELEVATION (CLIN 0004)
280	S-E301	STORAGE SHED -BUILDING SECTIONS (CLIN 0004)
281	S-E501	STORAGE SHED -BEAM DETAILS (CLIN 0004)
282	S-E502	STORAGE SHED -CHAIN LINK FENCE FOUNDATION SCHEDULE AND DETAILS (CLIN 0004)
283	S-E503	BEAM PROFILE, SECTION AND DETAILS (CLIN 0004)
284	S-F101	FIRE PUMP BUILDING -FOUNDATION AND ROOF FRAMING PLAN (CLIN 0001)
285	S-F201	FIRE PUMP BUILDING -EXTERIOR ELEVATIONS (CLIN 0001)
286	S-F301	FIRE PUMP BUILDING -BUILDING SECTIONS (CLIN 0001)
287	S-G101	CABLE HUT FOUNDATION AND ROOF FRAMING PLAN (CLIN 0001)
288	S-G102	CABLE HUT EXTERIOR ELEVATIONS AND BUILDING SECTIONS (CLIN 0001)
289	S-H101	COF SS -OVERALL FOUNDATION PLAN (CLIN 0004)
290	S-H102	COF SS -OVERALL SECOND FLOOR FRAMING PLAN (CLIN 0004)
291	S-H103	COF SS -OVERALL ROOF FLOOR FRAMING PLAN (CLIN 0004)
292	S-H104	COF SS -PARTIAL FOUNDATION PLAN (CLIN 0004)
293	S-H105	COF SS -PARTIAL FOUNDATION PLAN (CLIN 0004)
294	S-H106	COF SS -PARTAIL SECOND FLOOR FRAMING PLAN (CLIN 0004)
295	S-H107	COF SS -PARTAIL SECOND FLOOR FRAMING PLAN (CLIN 0004)

296	S-H108	COF SS -PARTIAL ROOF FLOOR FRAMING PLAN (CLIN 0004)
297	S-H109	COF SS -PARTIAL ROOF FLOOR FRAMING PLAN (CLIN 0004)
298	S-H201	COF SS -EXTERIOR ELEVATION (CLIN 0004)
299	S-H301	COF SS -BUILDING SECTIONS (CLIN 0004)
300	S-H701	COF SS -MECHANICAL YARD FOUNDATION PLAN AND ELEVATIONS (CLIN 0004)
301	S-H702	COF SS -MECHANICAL YARD BUILDING SECTIONS (CLIN 0004)
302	S-I101	WATER TANK- FOUNDATION AND FRAMING PLAN (CLIN 0001)
303	S-I201	WATER TANK -EXTERIOR ELEVATION (CLIN 0001)
304	S-I202	WATER TANK -BUILDING SECTIONS (CLIN 0001)
305	S-501	TYPICAL REINFORCING STEEL DETAILS
306	S-502	TYPICAL CONCRETE DETAILS
307	S-503	TYPICAL STEEL DECK DETAILS
308	S-504	TYPICAL STEEL NOTES AND DETAILS
309	S-601	CONCRETE PILE CAP SCHEDULE
310	S-602	CONCRETE PILE DETAIL AND BOLLARD DETAIL
311	S-603	CONCRETE GRADE BEAM AND TIE BEAM SCHEDULE
312	S-604	CONCRETE FOOTING SCHEDULE
313	S-605	CONCRETE WALL SCHEDULE
314	S-606	CONCRETE COLUMN ELEVATION
315	S-607	CONCRETE COLUMN SCHEDULE
316	S-608	CONCRETE BEAM DETAIL
317	S-609	CONCRETE BEAM SCHEDULE
318	S-610	CONCRETE COLUMN FOOTING SCHEDULE AND CONCRETE WALL FOOTING SCHEDULE
319	S-611	SITE WALL DETAILS AND SCHEDULE
320	A-001	LEGEND AND ABBREVIATIONS
321	A-002	GENERAL NOTES, SYMBOLS & CLIN SCHEDULE
322	A-200	EXTERIOR FINISH SCHEDULE
323	A-501	DOOR DETAILS
324	A-502	DOOR DETAILS
325	A-503	GATE DETAIL
326	A-504	WINDOW DETAILS
327	A-505	TRANSLUCENT PANEL AND LOUVER DETAILS
328	A-511	CASEWORK DETAILS
329	A-512	CEILING DETAILS
330	A-513	INTERIOR DETAILS

331	A-514	INTERIOR DETAILS
332	A-515	INTERIOR DETAILS
333	A-531	EXTERIOR DETAILS
334	A-532	EXTERIOR DETAILS
335	A-533	EXTERIOR DETAILS
336	A-534	MISCELLANEOUS DETAILS
337	A-541	AIR BARRIER PLANS DETAILS
338	A-551	SIGN TYPES, MOUNTING TYPES AND NOTES
339	A-601	DOOR & GATE SCHEDULE (CLIN 0001)
340	A-602	DOOR & GATE SCHEDULE (CLIN 0002 & CLIN 0004), TYPES AND NOTES
341	A-603	DOOR & GATE SCHEDULES (CLIN 0004)
342	A-604	WINDOW SCHEDULE, TYPES AND NOTES
343	A-612	LOUVER SCHEDULE, TYPES AND NOTES
344	A-621	MATERIAL FINISH SCHEDULE
345	A-622	ROOM FINISH SCHEDULES
346	A-623	EXTERIOR FINISH SCHEDULES
347	A-631	PARTITION TYPES
348	A-632	PARTITION TYPES
349	A-651	SIGNAGE SCHEDULES (CLIN 0001 -0002))
350	A-652	SIGNAGE SCHEDULES (CLIN 0004)
351	A-660	MOUNTING HEIGHTS
352	A-751	STAIR RAILING DETAILS
353	A-752	STAIR RAILING DETAILS
354	A-753	LADDER DETAILS
355	A-A101	VMS -OVERALL FIRST FLOOR PLAN (CLIN 0001)
356	A-A102	VMS -OVERALL SECOND FLOOR PLAN (CLIN 0001)
357	A-A103	VMS -OVERALL ROOF PLAN (CLIN 0001)
358	A-A111	VMS -PARTIAL FIRST FLOOR PLAN -ZONE A (CLIN 0001)
359	A-A112	VMS -PARTIAL FIRST FLOOR PLAN -ZONE B (CLIN 0001)
360	A-A113	VMS -PARTIAL SECOND FLOOR PLAN -ZONE A (CLIN 0001)
361	A-A114	VMS -PARTIAL SECOND FLOOR PLAN -ZONE B (CLIN 0001)
362	A-A121	VMS -OVERALL FIRST FLOOR REFLECTED CEILING PLAN (CLIN 0001)
363	A-A122	VMS -OVERALL SECOND FLOOR REFLECTED CEILING PLAN (CLIN 0001)
364	A-A131	VMS -PARTIAL FIRST FLOOR REFLECTED CEILING PLAN -ZONE A (CLIN 0001)

365	A-A132	VMS -PARTIAL FIRST FLOOR REFLECTED CEILING PLAN -ZONE B (CLIN 0001)
366	A-A133	VMS -PARTIAL SECOND FLOOR REFLECTED CEILING PLAN -ZONE A (CLIN 0001)
367	A-A134	VMS -PARTIAL SECOND FLOOR REFLECTED CEILING PLAN -ZONE B (CLIN 0001)
368	A-A141	VMS -AIR BARRIER PLANS (CLIN 0001)
369	A-A151	VMS -SIGNAGE FIRST FLOOR PLAN (CLIN 0001)
370	A-A152	VMS -SIGNAGE SECOND FLOOR PLAN (CLIN 0001)
371	A-A201	VMS -EXTERIOR ELEVATIONS (CLIN 0001)
372	A-A202	VMS -EXTERIOR ELEVATIONS (CLIN 0001)
373	A-A203	VMS -EXTERIOR ELEVATIONS (CLIN 0001)
374	A-A211	VMS -EXTERIOR ELEVATIONS AND SECTIONS -MECHANICAL YARD (CLIN 0001)
375	A-A301	VMS -BUILDING SECTIONS (CLIN 0001)
376	A-A302	VMS -BUILDING SECTIONS (CLIN 0001)
377	A-A311	VMS -WALL SECTION (CLIN 0001)
378	A-A312	VMS -WALL SECTION (CLIN 0001)
379	A-A401	VMS -ENLARGED FLOOR PLAN AND INTERIOR ELEVATIONS -MEN'S (A108) AND WOMEN'S TOILET AND LOCKER ROOMS (A109) (CLIN 0001)
380	A-A402	VMS -ENLARGED FLOOR PLAN AND INTERIOR ELEVATIONS -MEN'S (RM A211) AND WOMEN'S TOILET (RM A210) (CLIN 0001)
381	A-A403	VMS -ENLARGED FLOOR PLAN AND INTERIOR ELEVATIONS -NURSING ROOM (A140), BTC (A209) AND JANITOR ROOMS (A110, A212) (CLIN 0001)
382	A-A410	VMS -ENLARGED PLAN AND SECTIONS -MAINTENANCE PIT (CLIN 0001)
383	A-A411	VMS -ENLARGED PLAN AND ELEVATIONS -MECHANICAL YARD (CLIN 0001)
384	A-A701	VMS -ENLARGED STAIR FLOOR PLAN AND SECTION -AS01 (CLIN 0001)
385	A-A702	ENLARGED STAIR FLOOR PLAN AND SECTION -VMS STAIR ROOM 2
386	A-A711	ENLARGED LADDER FLOOR PLAN AND SECTION -VMS ACCESS LADDER 1
387	A-A720	ENLARGED FLOOR PLANS AND SECTION -VMS ELEVATOR
388	A-A721	ELEVATOR SPECIFICATIONS
389	A-A722	HOIST WAY PLANS
390	A-A723	HOIST WAY SECTIONS
391	A-A724	HALL POSITION INDICATOR DRAWING
392	A-A725	CAR OPERATING PANEL DESIGN DRAWING
393	A-A726	ACCESS SWITCH DRAWING
394	A-A727	DEDICATED PHONE CAUTION NAMEPLATE DESIGN DRAWING

395	A-A728	ELEVATOR COMMUNICATIONS FAILURE DESIGN DRAWING
396	A-A729	SYSTEM DIAGRAM
397	A-A730	EMERGENCY OPERATION FLOW CHART 1
398	A-A731	EMERGENCY OPERATION FLOW CHART 2
399	A-A732	PROTECTIVE CURTAIN INSIDE ELEVATOR CAR
400	A-A733	CAR DESIGN DETAIL
401	A-A734	LANTERN IN THE BOARDING HALL DETAIL
402	A-A735	FLOOR DISPLAY BOARDS
403	A-A736	FRONT ENTRANCE WALL OPENING DIAGRAM
404	A-B101	OS -FLOOR PLAN (CLIN 0002)
405	A-B102	OS -ROOF PLAN AND REFLECTED CEILING PLAN (CLIN 0002)
406	A-B141	OS -AIR BARRIER PLAN (CLIN 0002)
407	A-B151	OS -SIGNAGE PLAN (CLIN 0001-0002)
408	A-B201	OS -EXTERIOR ELEVATIONS (CLIN 0002)
409	A-B301	OS -BUILDING SECTIONS (CLIN 0002)
410	A-B311	OS -WALL SECTIONS (CLIN 0002)
411	A-B312	OS -WALL SECTIONS (CLIN 0002)
412	A-C101	POL & HAZMAT -FLOOR PLAN, REFLECTED CEILING PLAN AND ROOF PLAN (CLIN 0002)
413	A-C151	POL & HAZMAT -SIGNAGE PLAN (CLIN 0002)
414	A-C201	POL & HAZMAT -EXTERIOR ELEVATIONS (CLIN 0002)
415	A-C301	POL & HAZMAT -BUILDING SECTIONS (CLIN 0002)
416	A-C311	POL & HAZMAT -WALL SECTIONS (CLIN 0002)
417	A-C312	POL & HAZMAT -WALL SECTIONS (CLIN 0002)
418	A-D101	SSAW -OVERALL FLOOR PLAN (CLIN 0004)
419	A-D102	SSAW -OVERALL ROOF PLAN (CLIN 0004)
420	A-D111	SSAW -PARTIAL FLOOR PLAN -ZONE A (CLIN 0004)
421	A-D112	SSAW -PARTIAL FLOOR PLAN -ZONE B (CLIN 0004)
422	A-D121	SSAW -OVERALL REFLECTED CEILING PLAN (CLIN 0004)
423	A-D122	SSAW -PARTIAL REFLECTED CEILING PLAN -ZONE A (CLIN 0004)
424	A-D123	SSAW -PARTIAL REFLECTED CEILING PLAN -ZONE B (CLIN 0004)
425	A-D141	SSAW -AIR BARRIER PLAN (CLIN 0004)
426	A-D151	SSAW -SIGNAGE FLOOR PLAN (CLIN 0004)
427	A-D201	SSAW -EXTERIOR ELEVATIONS (CLIN 0004)
428	A-D202	SSAW -EXTERIOR ELEVATIONS (CLIN 0004)
429	A-D211	SSAW -EXTERIOR ELEVATIONS AND SECTIONS -MECH YARD (CLIN 0004)

430	A-D301	SSAW -BUILDING SECTIONS (CLIN 0004)
431	A-D302	SSAW -BUILDING SECTIONS (CLIN 0004)
432	A-D401	SSAW -ENLARGED TOILET PLANS AND ELEVATIONS (CLIN 0004)
433	A-D411	SSAW -ENLARGED FLOOR PLAN AND ELEVATIONS -MECH YARD (CLIN 0004)
434	A-E101	SS -FLOOR PLAN AND REFLECTED CEILING PLAN (CLIN 0004)
435	A-E102	SS -ROOF PLAN (CLIN 0004)
436	A-E151	SS -SIGNAGE FLOOR PLAN (CLIN 0004)
437	A-E201	SS -EXTERIOR ELEVATIONS AND BUILDING SECTION (CLIN 0004)
438	A-E311	SS -WALL SECTIONS (CLIN 0004)
439	A-F101	FP -FLOOR PLAN, REFLECTED CEILING PLAN AND ROOF PLAN (CLIN 0001)
440	A-F151	FP -SIGNAGE FLOOR PLAN (CLIN 0001)
441	A-F201	FP -ELEVATIONS AND SECTIONS (CLIN 0001)
442	A-F311	FP -WALL SECTIONS (CLIN 0001)
443	A-G101	CH -FLOOR PLAN, REFLECTED CEILING PLAN AND ROOF PLAN (CLIN 0001)
444	A-G141	CH -AIR BARRIER PLANS (CLIN 0001)
445	A-G151	CH -SIGNAGE FLOOR PLAN (CLIN 0001)
446	A-G201	CH -ELEVATIONS AND SECTIONS (CLIN 0001)
447	A-G311	CH -WALL SECTIONS (CLIN 0001)
448	A-H101	COF SS -FLOOR PLANS (CLIN 0004)
449	A-H102	COF SS -REFLECTED CEILING PLANS (CLIN 0004)
450	A-H103	COF SS -ROOF PLAN (CLIN 0004)
451	A-H141	AIR BARRIER PLANS (CLIN 0004)
452	A-H151	SIGNAGE PLAN (CLIN 0004)
453	A-H201	COF SS -EXTERIOR ELEVATIONS (CLIN 0004)
454	A-H211	COF SS -EXTERIOR ELEVATIONS AND SECTIONS -MECH YARD (CLIN 0004)
455	A-H301	COF SS -BUILDING SECTIONS (CLIN 0004)
456	A-H311	COF SS -WALL SECTIONS (CLIN 0004)
457	A-H312	COF SS -WALL SECTIONS (CLIN 0004)
458	A-H401	COF SS -ENLARGED FLOOR PLAN AND INTERIOR ELEVATIONS -TOILETS AND WATER HEATER/JANITOR ROOM (CLIN 0004)
459	A-H402	COF SS -ENLARGED FLOOR PLAN AND INTERIOR ELEVATIONS -KITCHENETTE AND WATER HEATER/JANITOR/STORAGE ROOM (CLIN 0004)
460	A-H411	COF SS -ENLARGED FLOOR PLAN AND ELEVATIONS -MECH YARD (CLIN 0004)
461	A-H721	ENLARGED STAIR FLOOR PLAN AND SECTION -COF SS STAIR 1

462	A-H722	ENLARGED STAIR FLOOR PLAN AND SECTION -COF SS STAIR 2
463	A-I101	WATER TANK -FLOOR PLAN AND ROOF PLAN (CLIN 0001)
464	A-I201	WATER TANK -EXTERIOR ELEVATIONS (CLIN 0001)
465	A-I301	WATER TANK -BUILDING SECTIONS (CLIN 0001)
466	A-I701	LADDER DETAILS
467	A-J101	TRANSFORMER ENCLOSURES -FLOOR PLANS AND ELEVATIONS
468	A-J151	TRANSFORMER ENCLOSURES -SIGNAGE PLAN (CLIN 0001)
469	A-J201	TRANSFORMER ENCLOSURES -EXTERIOR ELEVATION
470	I-A111	VMS - PARTIAL FIRST FLOOR FURNITURE PLAN - ZONE A (CLIN 0001)
471	I-A112	VMS - PARTIAL FIRST FLOOR FURNITURE PLAN - ZONE B (CLIN 0001)
472	I-A113	VMS - PARTIAL SECOND FLOOR FURNITURE PLAN -ZONE B (CLIN 0001)
473	I-D111	SSAW - PARTIAL FURNITURE FLOOR PLAN - ZONE A (CLIN 0004)
474	I-D112	SSAW - PARTIAL FURNITURE FLOOR PLAN - ZONE B (CLIN 0004)
475	I-H101	COF SS - FURNITURE FLOOR PLANS (CLIN 0004)
476	M-001	GENERAL NOTES AND LEGEND
477	M-101	OVERALL MECHANICAL AND COF SS SITE PLAN (CLIN 0001-0004)
478	M-A101	VMS -OVERALL 1ST FLOOR MECHANICAL PLAN (CLIN 0001)
479	M-A102	VMS -OVERALL 2ND FLOOR MECHANICAL PLAN (CLIN 0001)
480	M-A111	VMS -PARTIAL 1ST FLOOR MECHANICAL HVAC PLAN (ZONE A) (CLIN 0001)
481	M-A112	VMS -PARTIAL 1ST FLOOR MECHANICAL HVAC PLAN (ZONE B) (CLIN 0001)
482	M-A113	VMS -PARTIAL 2ND FLOOR MECHANICAL HVAC PLAN (ZONE A) (CLIN 0001)
483	M-A114	VMS -PARTIAL 2ND FLOOR MECHANICAL HVAC PLAN (ZONE B) (CLIN 0001)
484	M-A115	VMS -PARTIAL 1ST FLOOR MECHANICAL HVAC PIPING PLAN (ZONE B) (CLIN 0001)
485	M-A116	VMS -PARTIAL 2ND FLOOR MECHANICAL HVAC PIPING PLAN (ZONE B) (CLIN 0001)
486	M-A301	VMS -MECHANICAL SECTIONS (CLIN 0001)
487	M-A302	VMS -MECHANICAL SECTIONS (CLIN 0001)
488	M-A303	VMS -MECHANICAL SECTIONS (CLIN 0001)
489	M-A401	VMS -ENLARGED MECHANICAL ROOM AND YARD HVAC PLAN (CLIN 0001)
490	M-A901	VMS -MECHANICAL HVAC PIPING ISOMETRICS (CLIN 0001)
491	M-A101	VMS -OVERALL 1ST FLOOR MECHANICAL PLAN (CLIN 0001)
492	M-A102	VMS -OVERALL 2ND FLOOR MECHANICAL PLAN (CLIN 0001)

493	M-A111	VMS -PARTIAL 1ST FLOOR MECHANICAL HVAC PLAN (ZONE A) (CLIN 0001)
494	M-A112	VMS -PARTIAL 1ST FLOOR MECHANICAL HVAC PLAN (ZONE B) (CLIN 0001)
495	M-A113	VMS -PARTIAL 2ND FLOOR MECHANICAL HVAC PLAN (ZONE A) (CLIN 0001)
496	M-A114	VMS -PARTIAL 2ND FLOOR MECHANICAL HVAC PLAN (ZONE B) (CLIN 0001)
497	M-A115	VMS -PARTIAL 1ST FLOOR MECHANICAL HVAC PIPING PLAN (ZONE B) (CLIN 0001)
498	M-A116	VMS -PARTIAL 2ND FLOOR MECHANICAL HVAC PIPING PLAN (ZONE B) (CLIN 0001)
499	M-A301	VMS -MECHANICAL SECTIONS (CLIN 0001)
500	M-A302	VMS -MECHANICAL SECTIONS (CLIN 0001)
501	M-A303	VMS -MECHANICAL SECTIONS (CLIN 0001)
502	M-A401	VMS -ENLARGED MECHANICAL ROOM AND YARD HVAC PLAN (CLIN 0001)
503	M-A901	VMS -MECHANICAL HVAC PIPING ISOMETRICS (CLIN 0001)
504	M-B101	OS -MECHANICAL HVAC PLAN (CLIN 0002)
505	M-B102	OS -MECHANICAL HVAC PIPING PLAN (CLIN 0002)
506	M-B901	OS -MECHANICAL HVAC PIPING ISOMETRICS (CLIN 0002)
507	M-C101	POL & HAZMAT E -MECHANICAL HVAC PLAN (CLIN 0002)
508	M-D101	SSAW -OVERALL MECHANICAL PLAN (CLIN 0004)
509	M-D111	SSAW -PARTIAL MECHANICAL HVAC PLAN -ZONE A (CLIN 0004)
510	M-D112	SSAW -PARTIAL MECHANICAL HVAC PLAN -ZONE B (CLIN 0004)
511	M-D113	SSAW -PARTIAL MECHANICAL HVAC PIPING PLAN -ZONE B (CLIN 0004)
512	M-D301	SSAW -MECHANICAL SECTIONS (CLIN 0004)
513	M-D302	SSAW -MECHANICAL SECTIONS (CLIN 0004)
514	M-D401	SSAW -ENLARGED MECHANICAL HVAC PLAN (CLIN 0004)
515	M-D402	SSAW -ENLARGED MECHANICAL HVAC PLAN (CLIN 0004)
516	M-D901	SSAW -MECHANICAL HVAC PIPING ISOMETRICS (CLIN 0004)
517	M-F101	FP -MECHANICAL FLOOR PLAN (CLIN 0001)
518	M-G101	CH -MECHANICAL FLOOR PLAN (CLIN 0001)
519	M-G901	CH -MECHANICAL HVAC PIPING ISOMETRICS (CLIN 0001)
520	M-H101	COF SS -OVERALL 1ST FLOOR MECHANICAL PLAN (CLIN 0004)
521	M-H102	COF SS -OVERALL 2ND FLOOR MECHANICAL PLAN (CLIN 0004)
522	M-H111	COF SS -PARTIAL 1ST FLOOR MECHANICAL PLAN (ZONE A) (CLIN 0004)
523	M-H112	COF SS -PARTIAL 1ST FLOOR MECHANICAL PLAN (ZONE B) (CLIN 0004)
524	M-H113	COF SS -PARTIAL 2ND FLOOR MECHANICAL PLAN (ZONE A) (CLIN 0004)

525	M-H114	COF SS -PARTIAL 2ND FLOOR MECHANICAL PLAN (ZONE B) (CLIN 0004)
526	M-H115	COF SS -PARTIAL 1ST FLOOR MECHANICAL PIPING PLAN (ZONE A) (CLIN 0004)
527	M-H116	COF SS -PARTIAL 1ST FLOOR MECHANICAL PIPING PLAN (ZONE B) (CLIN 0004)
528	M-H117	COF SS -PARTIAL 2ND FLOOR MECHANICAL PIPING PLAN (ZONE A) (CLIN 0004)
529	M-H118	COF SS -PARTIAL 2ND FLOOR MECHANICAL PIPING PLAN (ZONE B) (CLIN 0004)
530	M-H301	COF SS -MECHANICAL SECTIONS (CLIN 0004)
531	M-H901	COF SS -MECHANICAL HVAC PIPING ISOMETRICS (CLIN 0004)
532	M-501	MECHANICAL DETAILS
533	M-502	MECHANICAL DETAILS
534	M-601	MECHANICAL SCHEDULES
535	M-602	MECHANICAL SCHEDULES
536	M-603	MECHANICAL SCHEDULES
537	M-604	MECHANICAL SCHEDULES
538	M-605	MECHANICAL SCHEDULES
539	M-606	MECHANICAL SCHEDULES
540	M-607	MECHANICAL SCHEDULES
541	M-608	MECHANICAL SCHEDULES
542	M-609	MECHANICAL SCHEDULES
543	M-610	MECHANICAL SCHEDULES
544	MI-701	MECHANICAL SCHEMATICS VAV AHU SCHEMATIC AND CHILLER PLANT DIAGRAM
545	MI-702	MECHANICAL SCHEMATICS AND CONTROL DIAGRAMS
546	MI-703	MECHANICAL SEQUENCE OF OPERATIONS AND CONTROL DIAGRAMS
547	MI-704	MECHANICAL SEQUENCE OF OPERATIONS AND CONTROL DIAGRAMS
548	MI-705	MECHANICAL SEQUENCE OF OPERATIONS
549	MI-706	MECHANICAL SEQUENCE OF OPERATIONS
550	MI-707	MECHANICAL SEQUENCE OF OPERATIONS
551	MI-708	DDC SYSTEM ARCHITECTURE
552	MI-709	DDC POINT LIST
553	MI-710	DDC POINTS LIST
554	MI-711	DDC POINT LIST
555	MI-712	DDC POINT LIST
556	MI-713	DDC POINT LIST

557	E-001	GENERAL ELECTRICAL NOTES
558	E-002	ELECTRICAL SYMBOL LIST AND ABBREVIATION
559	ES101	OVERALL SITE ELECTRICAL DEMOLITION PLAN (CLIN 0001 - CLIN 0003)
560	ES102	PARTIAL SITE ELECTRICAL DEMOLITION PLAN 1 (CLIN 0001 - CLIN 0003)
561	ES103	PARTIAL SITE ELECTRICAL DEMOLITION PLAN 2 (CLIN 0001 - CLIN 0003)
562	ES104	PARTIAL SITE ELECTRICAL DEMOLITION PLAN 3 (CLIN 0001 - CLIN 0003)
563	ES131	OVERALL SITE ELECTRICAL DEMOLITION PLAN (CLIN 0004)
564	ES132	PARTIAL SITE ELECTRICAL DEMOLITION PLAN (CLIN 0004)
565	ES141	OVERALL SITE ELECTRICAL DEMOLITION PLAN - COF SS (CLIN 0004)
566	ES142	PARTIAL SITE ELECTRICAL DEMOLITION PLAN 1 - COF SS (CLIN 0004)
567	ES143	PARTIAL SITE ELECTRICAL DEMOLITION PLAN 2 - COF SS (CLIN 0004)
568	ES201	OVERALL SITE ELECTRICAL PLAN (CLIN 0001 - CLIN 0003)
569	ES202	PARTIAL SITE ELECTRICAL PLAN 1 (CLIN 0001 - CLIN 0003)
570	ES203	PARTIAL SITE ELECTRICAL PLAN 2 (CLIN 0001 - CLIN 0003)
571	ES204	PARTIAL SITE ELECTRICAL PLAN 3 (CLIN 0001 - CLIN 0003)
572	ES231	OVERALL SITE ELECTRICAL PLAN (CLIN 0004)
573	ES232	PARTIAL SITE ELECTRICAL PLAN (CLIN 0004)
574	ES241	OVERALL SITE ELECTRICAL PLAN - COF SS (CLIN 0004)
575	ES242	PARTIAL SITE ELECTRICAL PLAN 1 - COF SS (CLIN 0004)
576	ES243	PARTIAL SITE ELECTRICAL PLAN 2 - COF SS (CLIN 0004)
577	ES501	DUCT SECTION AND DETAILS
578	ES502	ELECTRICAL MANHOLE DETAILS
579	ES503	PAD MOUNTED TRANSFORMER DETAILS
580	ES504	CABLE JUNCTION PEDESTAL (CJP) DETAILS
581	ES505	SITE LIGHTING DETAILS
582	ES601	PRIMARY DISTRIBUTION DIAGRAM - DEMOLITION (CLIN 0001 - CLIN 0004)
583	ES611	PRIMARY DISTRIBUTION DIAGRAM (CLIN 0001 - CLIN 0003)
584	ES621	PRIMARY DISTRIBUTION DIAGRAM (CLIN 0004)
585	ES631	PRIMARY DISTRIBUTION DIAGRAM - COF SS (CLIN 0004)
586	EA101	VMS - OVERALL FIRST FLOOR PLAN (CLIN 0001)
587	EA102	VMS - OVERALL SECOND FLOOR PLAN (CLIN 0001)
588	EA103	VMS - LIGHTNING PROTECTION PLAN (CLIN 0001)

589	EA111	VMS - PARTIAL FIRST FLOOR POWER PLAN - ZONE A (CLIN 0001)
590	EA112	VMS - PARTIAL FIRST FLOOR POWER PLAN - ZONE B (CLIN 0001)
591	EA113	VMS - PARTIAL SECOND FLOOR POWER PLAN - ZONE A (CLIN 0001)
592	EA114	VMS - PARTIAL SECOND FLOOR POWER PLAN - ZONE B (CLIN 0001)
593	EA121	VMS - PARTIAL FIRST FLOOR LIGHTING PLAN - ZONE A (CLIN 0001)
594	EA122	VMS - PARTIAL FIRST FLOOR LIGHTING PLAN - ZONE B (CLIN 0001)
595	EA123	VMS - PARTIAL SECOND FLOOR LIGHTING PLAN - ZONE A (CLIN 0001)
596	EA124	VMS - PARTIAL SECOND FLOOR LIGHTING PLAN - ZONE B (CLIN 0001)
597	EA131	POWER PLAN - ZONE A (CLIN 0001) VMS - PARTIAL FIRST FLOOR MECHANICAL EQUIPMENT
598	EA132	POWER PLAN - ZONE B (CLIN 0001) VMS - PARTIAL FIRST FLOOR MECHANICAL EQUIPMENT
599	EA133	POWER PLAN - ZONE A (CLIN 0001) VMS - PARTIAL SECOND FLOOR MECHANICAL EQUIPMENT
600	EA134	POWER PLAN - ZONE B (CLIN 0001) VMS - PARTIAL SECOND FLOOR MECHANICAL EQUIPMENT
601	EA401	VMS - ENLARGED PLANS (CLIN 0001)
602	EA601	VMS - ONE LINE DIAGRAM (CLIN 0001)
603	EB101	OS - POWER PLAN (CLIN 0002)
604	EB102	OS - LIGHTING PLAN (CLIN 0002)
605	EB601	OS - ONE LINE DIAGRAM (CLIN 0002)
606	EC101	POL & HAZMAT - POWER AND LIGHTING PLANS (CLIN 0002)
607	EC601	POL & HAZMAT - ONE LINE DIAGRAM (CLIN 0002)
608	ED101	SSAW - OVERALL FLOOR PLAN (CLIN 0004)
609	ED102	SSAW - LIGHTNING PROTECTION PLAN (CLIN 0004)
610	ED111	SSAW - PARTIAL POWER PLAN - ZONE A (CLIN 0004)
611	ED112	SSAW - PARTIAL POWER PLAN - ZONE B (CLIN 0004)
612	ED121	SSAW - PARTIAL LIGHTING PLAN - ZONE A (CLIN 0004)
613	ED122	SSAW - PARTIAL LIGHTING PLAN - ZONE B (CLIN 0004)
614	ED131	POWER PLAN - ZONE A (CLIN 0004) SSAW - PARTIAL MECHANICAL EQUIPMENT
615	ED132	POWER PLAN - ZONE B (CLIN 0004) SSAW - PARTIAL MECHANICAL EQUIPMENT
616	ED401	SSAW - ENLARGED PLANS (CLIN 0004)
617	ED601	SSAW - ONE LINE DIAGRAM (CLIN 0004)
618	EE101	SS - POWER AND LIGHTING PLANS (CLIN 0004)
619	EE102	SS - LIGHTNING PROTECTION PLAN (CLIN 0004)
620	EE601	SS - ONE LINE DIAGRAM (CLIN 0004)

621	EF101	FP - POWER AND LIGHTING PLANS (CLIN 0001)
622	EF601	FP - ONE LINE DIAGRAM (CLIN 0001)
623	EG101	CH - POWER AND LIGHTING PLANS (CLIN 0001)
624	EG601	CH - ONE LINE DIAGRAM (CLIN 0001)
625	EH101	COF SS - POWER PLANS (CLIN 0004)
626	EH102	COF SS - LIGHTING PLANS (CLIN 0004)
627	EH103	COF SS - LIGHTNING PROTECTION PLAN (CLIN 0004)
628	EH104	COF SS - MECHANICAL EQUIPMENT POWER PLAN (CLIN 0004)
629	EH401	COF SS - ENLARGED PLANS (CLIN 0004)
630	EH601	COF SS - ONE LINE DIAGRAM (CLIN 0004)
631	E-701	EQUIPMENT SCHEDULES
632	E-702	EQUIPMENT SCHEDULES
633	E-703	EQUIPMENT SCHEDULES
634	E-711	PANEL SCHEDULES
635	E-712	PANEL SCHEDULES
636	E-713	PANEL SCHEDULES
637	E-714	PANEL SCHEDULES
638	E-715	PANEL SCHEDULES
639	E-716	PANEL SCHEDULES
640	E-717	PANEL SCHEDULES
641	E-718	PANEL SCHEDULES
642	E-721	LUMINAIRE SCHEDULES
643	E-722	LUMINAIRE SCHEDULES
644	E-731	LIGHTING CONTROL SCHEDULES
645	E-732	LIGHTING CONTROL SCHEDULES
646	E-733	LIGHTING CONTROL SCHEDULES
647	E-734	LIGHTING CONTROL SCHEDULES
648	E-735	LIGHTING CONTROL SCHEDULES
649	T-001	SITE AND BUILDING TELECOMMUNICATIONS GENERAL NOTES
650	T-002	TELECOMMUNICATIONS SYMBOL LIST AND ABBREVIATION
651	TS101	OVERALL SITE TELECOM DEMOLITION PLAN (CLIN 0001 - CLIN 0003)
652	TS102	PARTIAL SITE TELECOM DEMOLITION PLAN - 1 (CLIN 0001 - CLIN 0003)
653	TS103	PARTIAL SITE TELECOM DEMOLITION PLAN - (CLIN 0001 - CLIN 0003)
654	TS131	OVERALL SITE TELECOM DEMOLITION PLAN (CLIN 0004)
655	TS132	PARTIAL SITE TELECOM DEMOLITION PLAN (CLIN 0004)
656	TS141	OVERALL SITE TELECOM DEMOLITION PLAN - COF SS (CLIN 0004)

657	TS201	OVERALL SITE TELECOM PLAN (CLIN 0001 - CLIN 0003)
658	TS202	PARTIAL SITE TELECOM PLAN - 1 (CLIN 0001 - CLIN 0003)
659	TS203	PARTIAL SITE TELECOM PLAN - 2 (CLIN 0001 - CLIN 0003)
660	TS231	OVERALL SITE TELECOM PLAN (CLIN 0004)
661	TS232	PARTIAL SITE TELECOM PLAN (CLIN 0004)
662	TS241	OVERALL SITE TELECOM PLAN - COF SS (CLIN 0004)
663	TS242	PARTIAL SITE TELECOM PLAN - COF SS (CLIN 0004)
664	TS501	DUCT SECTION AND DETAILS
665	TS502	MAINTENANCE HOLE DETAILS
666	TS503	MAINTENANCE HOLE DETAILS
667	TS511	MAINTENANCE HOLE BUTTERFLY DIAGRAMS
668	TS601	TELECOM DISTRIBUTION DIAGRAM - DEMOLITION (CLIN 0001 - CLIN 0004)
669	TS611	TELECOM DISTRIBUTION DIAGRAM (CLIN 0001 - CLIN 0003)
670	TS621	TELECOM DISTRIBUTION DIAGRAM (CLIN 0004)
671	TS631	TELECOM DISTRIBUTION DIAGRAM - COF SS (CLIN 0004)
672	TA101	VMS - OVERALL FIRST FLOOR PLAN (CLIN 0001)
673	TA102	VMS - OVERALL SECOND FLOOR PLAN (CLIN 0001)
674	TA111	VMS - PARTIAL FIRST FLOOR TELECOM PLAN - ZONE A (CLIN 0001)
675	TA112	VMS - PARTIAL FIRST FLOOR TELECOM PLAN - ZONE B (CLIN 0001)
676	TA114	VMS - PARTIAL SECOND FLOOR TELECOM PLAN - ZONE B (CLIN 0001)
677	TA401	VMS - ENLARGED PLANS (CLIN 0001)
678	TA601	VMS - TELECOM RISER DIAGRAM (CLIN 0001)
679	TB101	OS - TELECOM PLAN (CLIN 0002)
680	TB601	OS - TELECOM RISER DIAGRAM (CLIN 0002)
681	TD101	SSAW - OVERALL FLOOR PLAN (CLIN 0004)
682	TD111	SSAW - PARTIAL TELECOM PLAN - ZONE A (CLIN 0004)
683	TD112	SSAW - PARTIAL TELECOM PLAN - ZONE B (CLIN 0004)
684	TD401	SSAW - ENLARGED PLANS (CLIN 0004)
685	TD601	SSAW - TELECOM RISER DIAGRAM (CLIN 0004)
686	TF101	FP - TELECOM PLANS (CLIN 0001)
687	TF601	FP - TELECOM RISER DIAGRAM (CLIN 0001)
688	TG101	CH - TELECOM PLANS (CLIN 0001)
689	TG601	CH - TELECOM RISER DIAGRAM (CLIN 0001)
690	TH101	COF SS - TELECOM PLANS (CLIN 0004)
691	TH401	COF SS - ENLARGED PLANS (CLIN 0004)

692	TH601	COF SS - TELECOM RISER DIAGRAM (CLIN 0004)
693	T-501	DETAILS
694	T-502	DETAILS
695	T-503	OUTLET AND CONDUIT DETAILSS
696	T-504	GROUNDING AND BONDING DETAILS.
697	T-505	DETAILS
698	TYA112	VMS - PARTIAL FIRST FLOOR SECURITY PLAN - ZONE B (CLIN 0001)
699	TYA114	VMS - PARTIAL SECOND FLOOR SECURITY PLAN - ZONE B (CLIN 0001)
700	TYA401	VMS - ENLARGED SECURITY PLANS (CLIN 0001)
701	TYA601	VMS - SECURITY RISER DIAGRAM (CLIN 0001)
702	TYD112	SSAW - PARTIAL SECURITY PLAN - ZONE B (CLIN 0004)
703	TYD401	SSAW - ENLARGED SECURITY PLANS (CLIN 0004)
704	TYD601	SSAW - SECURITY RISER DIAGRAM (CLIN 0004)
705	TYG101	CH - SECURITY PLANS (CLIN 0001)
706	TYG601	CH - SECURITY RISER DIAGRAM (CLIN 0001)
707	TYH101	COF SS - SECURITY PLANS (CLIN 0004)
708	TYH401	COF SS - ENLARGED PLANS (CLIN 0004)
709	TYH601	COF SS - SECURITY RISER DIAGRAM (CLIN 0004)
710	TY501	SECURITY SYSTEM DETAILS
711	P001	GENERAL NOTES AND LEGEND
712	P101	OVERALL PLUMBING AND COF SS SITE PLAN (CLIN 0001-0004)
713	P-A101	VMS -OVERALL 1ST FLOOR PLUMBING PLAN (CLIN 0001)
714	P-A102	VMS -OVERALL 2ND FLOOR PLUMBING PLAN (CLIN 0001)
715	P-A103	VMS -OVERALL 1ST FLOOR RADON PIPING PLAN (CLIN 0001)
716	P-A104	VMS -OVERALL ROOF PLUMBING PLAN (CLIN 0001)
717	P-A111	VMS -PARTIAL 1ST FLOOR SANITARY PIPING PLAN (ZONE A) (CLIN 0001)
718	P-A112	VMS -PARTIAL 1ST FLOOR SANITARY PIPING PLAN (ZONE B) (CLIN 0001)
719	P-A113	VMS -PARTIAL 1ST FLOOR DOMESTIC WATER PIPING PLAN (ZONE A) (CLIN 0001)
720	P-A114	VMS -PARTIAL 1ST FLOOR DOMESTIC WATER PIPING PLAN (ZONE B) (CLIN 0001)
721	P-A115	VMS -PARTIAL 1ST FLOOR COMPRESSED AIR AND POL PIPING PLAN (ZONE A) (CLIN 0001)
722	P-A116	VMS -PARTIAL 1ST FLOOR COMPRESSED AIR AND POL PIPING PLAN (ZONE B) (CLIN 0001)
723	P-A301	VMS -PLUMBING SECTIONS (CLIN 0001)

724	P-A302	VMS -PLUMBING SECTIONS (CLIN 0001)
725	P-A401	VMS -ENLARGED SANITARY PIPING PLAN (CLIN 0001)
726	P-A402	VMS -ENLARGED DOMESTIC WATER PIPING PLAN (CLIN 0001)
727	P-A403	VMS -ENLARGED 1ST FLOOR COMPRESSED AIR AND POL PIPING PLAN (CLIN 0001)
728	P-A901	VMS -PARTIAL SANITARY PIPING ISOMETRICS (ZONE A) (CLIN 0001)
729	P-A902	VMS -PARTIAL SANITARY PIPING ISOMETRICS (ZONE B) (CLIN 0001)
730	P-A903	VMS -ENLARGED SANITARY PIPING ISOMETRICS -TOILET AND LOCKER ROOMS (RM A108 & A109) (RM A210 & A211) (CLIN 0001)
731	P-A904	VMS -PARTIAL DOMESTIC WATER PIPING ISOMETRICS (ZONE A) (CLIN 0001)
732	P-A905	VMS -PARTIAL DOMESTIC WATER PIPING ISOMETRICS (ZONE B) (CLIN 0001)
733	P-A906	VMS -ENLARGED DOMESTIC WATER PIPING ISOMETRICS -TOILET AND LOCKER ROOMS (RM A108 & A109) (RM A210 & A211) (CLIN 0001)
734	P-A907	VMS -PARTIAL COMPRESSED AIR AND POL PIPING ISOMETRICS (ZONE A) (CLIN 0001)
735	P-A908	VMS -PARTIAL COMPRESSED AIR AND POL PIPING ISOMETRICS (ZONE B) (CLIN 0001)
736	P-A909	VMS -OVERALL RADON PIPING ISOMETRICS (CLIN 0001)
737	P-B101	OS -PLUMBING FLOOR PLAN (CLIN 0002)
738	P-B901	OS -PLUMBING ISOMETRICS -DOMESTIC WATER (CLIN 0002)
739	P-C101	POL & HAZMAT -PLUMBING FLOOR PLAN (CLIN 0002)
740	P-C901	POL & HAZMAT -PLUMBING ISOMETRICS -DOMESTIC WATER (CLIN 0002)
741	P-D101	SSAW -OVERALL PLUMBING FLOOR PLAN (CLIN 0004)
742	P-D102	SSAW -OVERALL RADON PLUMBING FLOOR PLAN (CLIN 0004)
743	P-D103	SSAW -OVERALL ROOF PLUMBING PLAN (CLIN 0004)
744	P-D111	SSAW -PARTIAL SANITARY PLUMBING PLAN -ZONE A (CLIN 0004)
745	P-D112	SSAW -PARTIAL SANITARY PLUMBING PLAN -ZONE B (CLIN 0004)
746	P-D113	SSAW -PARTIAL DOMESTIC WATER PLUMBING PLAN -ZONE A (CLIN 0004)
747	P-D114	SSAW -PARTIAL DOMESTIC WATER PLUMBING PLAN -ZONE B (CLIN 0004)
748	P-D401	SSAW -ENLARGED TOILET PLUMBING PLANS (CLIN 0004)
749	P-D901	SSAW -PLUMBING ISOMETRICS -SANITARY (CLIN 0004)
750	P-D902	SSAW -PLUMBING ISOMETRICS -DOMESTIC WATER (CLIN 0004)
751	P-D903	SSAW -PLUMBING ISOMETRICS -RADON (CLIN 0004)
752	P-E101	SS -PLUMBING FLOOR PLAN (CLIN 0004)
753	P-E901	SS -PLUMBING ISOMETRICS -DOMESTIC WATER (CLIN 0004)
754	P-F101	FP -PLUMBING FLOOR PLAN (CLIN 0001)

755	P-F901	FP -PLUMBING ISOMETRICS (CLIN 0001)
756	P-H101	COF SS -1ST FLOOR PLUMBING PLAN (CLIN 0004)
757	P-H102	COF SS -2ND FLOOR PLUMBING PLAN (CLIN 0004)
758	P-H103	COF SS -ROOF PLUMBING PLAN (CLIN 0004)
759	P-H104	COF SS -RADON PLUMBING PLAN (CLIN 0004)
760	P-H401	COF SS -PARTIAL 1ST FLOOR PLUMBING PLAN (CLIN 0004)
761	P-H402	COF SS -PARTIAL 1ST FLOOR PLUMBING PLAN (CLIN 0004)
762	P-H403	COF SS -PARTIAL 2ND FLOOR PLUMBING PLAN (C0004)
763	P-H404	COF SS -PARTIAL 2ND FLOOR PLUMBING PLAN (CLIN 0004)
764	P-H901	COF SS -SANITARY ISOMETRIC DIAGRAM (CLIN 0004)
765	P-H902	COF SS -SANITARY ISOMETRIC DIAGRAM (CLIN 0004)
766	P-H903	COF SS -DOMESTIC WATER ISOMETRIC DIAGRAM (CLIN 0004)
767	P-H904	COF SS -DOMESTIC WATER ISOMETRIC DIAGRAM (CLIN 0004)
768	P-H905	COF SS -RADON ISOMETRIC DIAGRAM (CLIN 0004)
769	P-1101	WASH RACK - PLUMBING FLOOR PLAN (CLIN 0003)
770	P-I901	WASH RACK - PLUMBING ISOMETRICS (CLIN 0003)
771	P501	PLUMBING DETAILS
772	P502	PLUMBING DETAILS
773	P503	PLUMBING DETAILS
774	P504	PLUMBING DETAILS
775	P601	PLUMBING SCHEDULES
776	P602	PLUMBING FIXTURE SCHEDULE
777	P603	PLUMBING FIXTURE SCHEDULE
778	P604	PLUMBING FIXTURE SCHEDULE
779	FA-001	VMS -FIRE ALARM NOTES (CLIN 0001)
780	FA-A101	VMS -OVERALL 1ST FLOOR FIRE ALARM PLAN (CLIN 0001)
781	FA-A102	VMS -OVERALL 2ND FLOOR FIRE ALARM PLAN (CLIN 0001)
782	FA-A111	VMS -PARTIAL FIRST FLOOR FIRE ALARM PLAN (ZONE A) (CLIN 0001)
783	FA-A112	VMS -PARTIAL FIRST FLOOR FIRE ALARM PLAN (ZONE B) (CLIN 0001)
784	FA-A113	VMS -PARTIAL SECOND FLOOR FIRE ALARM PLAN (ZONE A) (CLIN 0001)
785	FA-A114	VMS -PARTIAL SECOND FLOOR FIRE ALARM PLAN (ZONE B) (CLIN 0001)
786	FAB101	OS -FIRE ALARM PLAN (CLIN 0002)
787	FAC101	POL & HAZMAT STORAGE -FIRE ALARM PLAN (CLIN 0002)
788	FAD101	SSAW -OVERALL FIRE ALARM PLAN (CLIN 0004)
789	FAD111	SSAW -PARTIAL FIRE ALARM PLAN (ZONE A) (CLIN 0004)
790	FAD112	SSAW -PARTIAL FIRE ALARM PLAN (ZONE B) (CLIN 0004)

791	FA-E101	SS -FIRE ALARM PLAN (CLIN 0004)
792	FA-F101	FP -FIRE ALARM PLAN (CLIN 0001)
793	FA-G101	CH -FIRE ALARM PLAN (CLIN 0001)
794	FA-H101	COF SS -1ST FLOOR FIRE ALARM PLAN (CLIN 0004)
795	FA-H102	COF SS -2ND FLOOR FIRE ALARM PLAN (CLIN 0004)
796	FA-501	VMS -FIRE ALARM DETAILS (CLIN 0001)
797	FA-601	VMS -FIRE ALARM SEQUENCE OF OPERATIONS MATRIX (CLIN 0001)
798	FX-001	VMS -FIRE SPRINKLER NOTES (CLIN 0001)
799	FX101	FIRE PROTECTION SITE PLAN (CLIN 0001-0003)
800	FX102	FIRE PROTECTION SITE PLAN (CLIN 0004)
801	FX103	FIRE PROTECTION SITE PLAN SWING SPACE (CLIN 0004)
802	FX-A101	VMS -OVERALL FIRST FLOOR FIRE SPRINKLER PLAN (CLIN 0001)
803	FX-A102	VMS -OVERALL SECOND FLOOR FIRE SPRINKLER PLAN (CLIN 0001)
804	FX-A111	VMS -PARTIAL FIRST FLOOR FIRE SPRINKLER PLAN (ZONE A) (CLIN 0001)
805	FX-A112	VMS -PARTIAL FIRST FLOOR FIRE SPRINKLER PLAN (ZONE B) (CLIN 0001)
806	FX-A113	VMS -PARTIAL SECOND FLOOR FIRE SPRINKLER PLAN (ZONE A) (CLIN 0001)
807	FX-A114	VMS -PARTIAL SECOND FLOOR FIRE SPRINKLER PLAN (ZONE B) (CLIN 0001)
808	FXB101	OS -FIRE SPRINKLER PLAN (CLIN 0002)
809	FXC101	POL & HAZMAT STORAGE -FIRE SPRINKLER PLAN (CLIN 0002)
810	FXD101	SSAW -OVERALL FIRE SPRINKLER PLAN (CLIN 0004)
811	FXD111	SSAW -PARTIAL FIRE SPRINKLER PLAN (ZONE A) (CLIN 0004)
812	FXD112	SSAW -PARTIAL FIRE SPRINKLER PLAN (ZONE B) (CLIN 0004)
813	FX-E101	SS -FIRE SPRINKLER PLAN (CLIN 0004)
814	FX-F101	FP -FIRE SPRINKLER PLAN (CLIN 0001)
815	FX-G101	CH -FIRE SPRINKLER PLAN (CLIN 0001)
816	FX-H101	COF SS -1ST FLOOR FIRE SPRINKLER PLAN (CLIN 0004)
817	FX-H102	COF SS -2ND FLOOR FIRE SPRINKLER PLAN (CLIN 0004)
818	FX-401	VMS -FIRE PUMP SCHEDULE AND EQUIPMENT LAYOUT (CLIN 0001)
819	FX-501	VMS -FIRE SPRINKLER DETAILS (CLIN 0001)

[1.3 SUPPLEMENTARY DRAWINGS

These supplementary drawings may not be a part of the contract but are included with the drawings for information.

[1.3.1 Reference Drawing[s]

The following reference drawing[s] [is][are] intended only to show the original construction. Drawing[s] [is][are] the property of the Government and must not be used for any purpose other than that intended by the contract.[The drawing[s] [is][are] [half-][full-]size. Full-size

drawings are available at the bidder's or Contractor's expense.[
Information on procuring these full-size drawings may be obtained from the
Contracting Officer.][Full size drawings may be inspected during regular
working hours at[the office of the Contracting Officer.][the station.]]

NAVFAC DWG NO.	TITLE
[_____]	[_____]

]][1.3.2 Boring Logs

The Government does not guarantee that borings indicate actual conditions,
except for the exact locations and the time that they were made.[
Subsurface data, not specified or indicated, have been obtained by the
Government at this station. The data are available for examination by
prospective bidders[in the office of the Contracting Officer][at the
station].]

]][1.3.3 Subsurface Data

Subsurface data, not specified or indicated, have been obtained by the
Government at the station. The data are available for examination by
prospective bidders[in the office of the Contracting Officer][at the
station].[The soils report is included as part of the solicitation.]

]] -- End of Document --

SECTION 01 11 00.00 10

GENERAL CONTRACT REQUIREMENTS

PART 1 GENERAL

1.1 DEFINITIONS

The term "Government" refers to "United States Government" whenever the term "Government" appears in this Contract, unless otherwise indicated within the Contract.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE 189.1 (2017) Standard for the Design of
High-Performance Green Buildings Except
Low-Rise Residential Buildings

ASHRAE 52.2 (2017) Method of Testing General
Ventilation Air-Cleaning Devices for
Removal Efficiency by Particle Size

ASTM INTERNATIONAL (ASTM)

ASTM D6245 (2012) Using Indoor Carbon Dioxide
Concentrations to Evaluate Indoor Air
Quality and Ventilation

ASTM D6345 (2010) Standard Guide for Selection of
Methods for Active, Integrative Sampling
of Volatile Organic Compounds in Air

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10 (2022; ERTA 1 2021) Standard for Portable
Fire Extinguishers

NFPA 241 (2022) Standard for Safeguarding
Construction, Alteration, and Demolition
Operations

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION
(SMACNA)

ANSI/SMACNA 008 (2007) IAQ Guidelines for Occupied
Buildings Under Construction, 2nd Edition

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 Safety -- Safety and Health Requirements

Manual

U.S. ARMY GARRISON JAPAN (USAG-J)

Regulation 190-13

(2013) Installation Access and Control
Procedures

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submittals not having a "G" designation are for Contractor Quality Control or Designer of Record approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

[Initial Site Conditions Photos
Key Personnel Qualifications; G
Conformed Drawings Submittal]
[INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN; G]

SD-07 Certificates

Monthly Progress Photos;

SD-11 Closeout Submittals

[Construction Completion Photos;]

1.4 ACCEPTABLE ALTERNATIVE JAPANESE STANDARDS

Where indicated in the technical specifications REFERENCES paragraphs, acceptable Japanese standards are listed which are considered acceptable alternatives to the US Standards within the Contract Specifications. Contract specifications authorize the use of specified acceptable alternative Japanese standards. A limited set of modified UFGS incorporating previously identified and accepted Japanese standards and methods (also known as Japan Edited Specifications, or JES) can be found at <https://www.poj.usace.army.mil/Business-With-Us/References/>. Other portions of the specifications also authorize the use of specified acceptable alternative Japanese standards. The use of products not meeting the applicable US standards of the Contract or alternative Japanese standards specifically authorized by the Contract is prohibited unless authorized by the Contracting Officer. Japanese standards not identified in these specifications as being acceptable alternatives to US standards may be submitted to the Contracting Officer for consideration as acceptable alternatives after award using the variation request process set forth in Section 01 33 00 SUBMITTAL PROCEDURES. Prior to the use of such proposed additional alternative Japanese standards, written approval by the Contracting Officer is required. Incorporation of methods, materials, and equipment that will promote cost-effective and timely maintenance, and that are otherwise authorized by the contract, is highly encouraged.

[1.5 GOVERNMENT FURNISHED CONTRACTOR INSTALLED ITEMS

Attachment 01 11 00.00 10-A "Government Furnished Property" provides the items that shall be Government Furnished, Contractor Installed (GFCI).

]1.6 MANDATORY U.S. TESTED PRODUCTS

The following items shall meet U.S. testing methods (e.g. UL listed, FM approved, ASTM, etc.) and shall be labeled as required. These items shall not be substituted with Japanese testing methods (e.g. JIS, JASS, etc.) for certification:

- a. Fire suppression systems, including valves, alarm valves, sprinklers. Fire suppression systems includes wet chemical, dry chemical, mist, deluge, pre-action, foam, and clean agent, in addition to standard sprinkler systems.
- b. Fire pumps, including motors, controllers, drives, and valves.
- c. Fire alarm and mass notification systems, including panels, initiating devices, notification appliances, smoke alarms.
- d. Engineering technician requirements for development of fire suppression systems and alarm system shop drawings, calculations, and material submittals.
- e. Fire doors, smoke doors, and frames.
- f. Fire dampers and smoke dampers.
- g. Interior finishes with flame spread and smoke development ratings required for installation of rated assemblies.
- h. Insulation with flame spread and smoke development ratings.
- i. Fire proofing and fire stopping materials.
- j. Plenum rated cables.
- k. Domestic water supply piping in the building and plumbing fixtures that directly dispense potable drinking water (NSF stamped and/or labeled). Plumbing fixtures conforming to Japanese standards that do not directly dispense potable drinking water are allowed, such as urinals and water closets, bath and shower faucets, utility/janitor faucets, lavatories and sinks.

1.7 KEY PERSONNEL QUALIFICATIONS

The Contractor's Project Key personnel shall not be assigned duties to any other Contracts (excluding the project manager) without approval from the Contracting Officer. [The Project Superintendent, CQC System Manager, and Site Safety Health Officer shall all be separate persons, and shall not have other duties assigned.] [One person may be assigned the duties of Project Superintendent and CQC System Manager provided they meet all qualification requirements and maintain presence on site as required for the various duties. The Site Safety Health Officer shall not be assigned other duties.] Within five (5) working days after receipt of the Notice to Proceed, the Contractor shall submit in writing to the Contracting Officer an organizational chart, the qualifications and background history of the proposed Key Personnel for approval. The Contracting Officer shall have the explicit right to determine acceptability (or rejection) of the proposed individuals. In addition, the Contractor shall be responsible to

replace said individuals upon notification by the Government should performance become inadequate during the Contract period. Key Personnel shall attend the Preconstruction conference.

[1.7.1 Program Manager

The Program Manager is responsible for the overall management of the Contract. The Program Manager shall have a minimum of [five (5)] years experience in the administration of construction Contracts on construction projects similar in size and scope to this Contract, and shall have a thorough knowledge of the duties of key management personnel assigned to this Contract.

]1.7.2 Project Manager

The Project Manager shall have a minimum of [five (5)] years experience as a Project Manager on construction projects [containing [____] components] similar in size and scope to this Contract. The Project Manager shall maintain oversight of Contract proposals prepared by the Contractor staff and be authorized to negotiate Contract terms and sign Contract documents on behalf of the Contractor. The project manager does not need to be present on the site daily, but shall attend weekly progress meetings and be available on site within 24 hours upon request. Project Manager [shall not have other duties assigned][Project Manager may have other duties assigned].

1.7.3 Project Superintendent (Supervisor)

The Project Superintendent shall be on the work site when on-site work is being performed and shall be available to the Contracting Officer or his representatives upon request. The Project Superintendent shall have overall responsibility for all operations at the job site and be authorized to make decisions, negotiate Contract terms and sign Contract documents on behalf of the Contractor. The Project Superintendent shall have a minimum of [five (5)] years experience as a superintendent on construction projects [containing [____] components] similar in size and scope to this Contract[.][, and have at least one the following qualifications:]

- [a. U.S. Registered Structural Engineer or U.S. Registered Architect or 1 Kyu Kenchikushi (1st Class Qualified Architect).
-] [b. U.S. Registered Civil Engineer or 1 Kyu Doboku Sekou Kanrigishi (1st Class Civil Engineering Works Management Engineer).
-] [c. U.S. Registered Mechanical or Electrical Engineer or 1 Kyu Kankouji Sekou Kanrigishi (1st Class Building Mechanical and Electrical Engineer).
-] [d. U.S. Registered Electrical Engineer or 1 Kyu Denikouji Sekou Kanrigishi (1st Class Electric Construction Management Engineer).
-] [e. U.S. Bachelors Degree in Construction Management or 1 Kyu Kenchiku Sekou Kanrigishi (1st Class Building Construction Management Engineer)
-] [f. U.S. Bachelors Degree in Construction Management or AI*DD Sougoushu (AI*DD Construction Engineer).

]1.7.4 English Speaking Representative (Interpreter)

At all times during the Contract period the Contractor shall have an employee capable of fluent bilingual speech in the Japanese and English languages at the job site. The bilingual interpreter shall have the capability to receive and issue concise and technical explanation and instructions between the Government representative(s) and Contractor supervisory personnel concerning all aspects of Contract administration and construction. Within fifteen (15) days after receipt of the Notice to Proceed, the Contractor shall submit in writing to the Contracting Officer the qualifications and background history of the proposed interpreter for approval. The Contracting Officer shall have the explicit right to determine acceptability or rejection of the proposed individual. In addition, the Contractor shall be responsible to replace said individual upon notification by the Government should performance become inadequate during the Contract period. The interpreter shall attend the preconstruction conference. Interpreter shall not have other duties assigned.

1.7.5 Contractor Quality Control

1.7.5.1 Contractor Quality Control System Manager (CQCSM)

The CQC System Manager is required to be a graduate of an accredited college with an engineering, architecture, or construction management degree, with a minimum of [5] years construction experience on construction of similar size and scope to this Contract. Refer to Section 01 45 00.00 10 QUALITY CONTROL, Paragraph "CQC System Manager" for additional requirements.

1.7.5.2 Contractor Quality Control Personnel

The following Contractor Quality Control (CQC) Personnel are required:[Civil,][Mechanical,][Electrical,][Structural,][Architectural,][Fire Protection,][Communications,][Environmental,][Submittals,][Occupied Family Housing,][Concrete, Pavements, and Soils,][Testing, Adjusting, and Balancing,][Design Quality Control Manager]. All other specialties in section 01 45 00.00 10 not identified in this paragraph does not apply to this project.

See Section 01 45 00.00 10 QUALITY CONTROL for qualification requirements for CQC personnel. These individuals or specialized technical companies [must be directly employed by the prime Contractor and can not be employed by a supplier or subcontractor on this project][may be employed by the prime contractor or a subcontractor on this project]. These individuals [shall have no other duties other than quality control] [can perform other duties but need to be allowed sufficient time to perform the specialized personnel's assigned quality control duties as described in the Quality Control Plan]. [A single person can cover more than one area provided that the single person is qualified to perform quality control activities in each designated and that workload allows.][A single person cannot cover more than one area].

1.7.6 Site Safety and Health Officer (SSHO)

See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

[1.7.7 [SUSTAINABILITY REPRESENTATIVE][TPC SUSTAINABILITY PROFESSIONAL]

See Section 01 33 29 SUSTAINABILITY REPORTING.

] [1.7.8 [Protective Coating Specialist (PCS)]

See Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES.

] 1.7.9 Pass Coordinator

A person on the contractor's staff who is familiar with the requirements of USAG-J Regulation 190-13, Individual and Vehicle Access Pass Procedures, and USAG-J vehicle security inspection procedures. The Pass Coordinator must be able to obtain a DBIDS access pass with escort privileges and may be required to escort other contract personnel. Although a TOEIC score is not required, the Pass Coordinator must be able to understand pass requirements and complete pass application forms in the English language.

1.8 TIME EXTENSIONS FOR UNUSUALLY SEVERE WEATHER

- a. This paragraph specifies the procedure for determination of time extensions for unusually severe weather in accordance with the Contract clause entitled "DEFAULT (FIXED-PRICE CONSTRUCTION)". In order for the Contracting Officer to award a time extension under this clause, the following conditions must be satisfied:

- (1) The weather experienced at the project site during the Contract period must be found to be unusually severe, that is, more severe than the adverse weather anticipated for the project location during any given month.
- (2) The unusually severe weather must actually cause a delay to the completion of the project. The delay must be beyond the control and without the fault or negligence of the Contractor.

- b. The listing below defines the monthly anticipated adverse weather days for the Contract period and is based upon 20th Weather Squadron (MAC), U.S. Air Force; Iwakuni U.S. Marine Corps Air Station, Weather Service; Sasebo Naval Pacific Meteorology and Oceanography Detachment; JASDF (Japan Air Self Defense Force), or similar data for the geographic location of the project. (More specific information may be obtained as specified under Special Contract Requirements Clause entitled "Physical Data.")

MONTHLY ANTICIPATED ADVERSE WEATHER DELAYS
WORK DAYS BASED ON (5) DAY WORK WEEK

	Misawa-area	Kanto Plain	Iwakuni	Okinawa	Sasebo	Osaka
JAN	8	1	2	5	7	4
FEB	9	3	4	7	6	4
MAR	6	6	7	8	7	5
APR	3	6	6	6	7	4
MAY	4	4	6	6	6	4
JUN	5	9	8	6	12	5
JUL	6	6	6	6	9	4
AUG	6	6	5	9	9	3
SEP	6	7	6	7	9	4

OCT	4	5	4	4	8	4
NOV	6	4	4	4	5	3
DEC	7	1	3	5	5	3

The above schedule of anticipated adverse weather days shall constitute the base line for monthly (or portion thereof) weather time evaluations.

- c. Upon acknowledgment of the Notice-to-Proceed (NTP) and continuing throughout the Contract, the Contractor shall record on the daily CQC report, the occurrence of adverse weather and resultant impact to normally scheduled work. Actual adverse weather delay days must prevent work on critical activities for 50 percent or more of the Contractor's scheduled work day.
- d. The number of actual adverse weather delay days shall include days impacted by actual adverse weather (even if adverse weather occurred in previous month), be calculated chronologically from the first to the last day of each month, and be recorded as full days. If the number of actual adverse weather delay days exceeds the number of days anticipated in paragraph (b), above, the Contracting Officer shall convert any qualifying delays to calendar days, giving full consideration for equivalent fair weather work days, and issue a modification in accordance with the Contract clause entitled "DEFAULT (FIXED PRICE CONSTRUCTION)".
- e. For all work under this Contract, adverse weather is defined as:
 - (1) Rainfall - Number of occurrences of precipitation greater than or equal to 0.10 inches (2.54mm).
 - (2) Snowfall - Number of occurrences of precipitation greater than or equal to 1.00 inches (25.4mm). (Not applicable to Okinawa Area)
 - (3) Cold Temperature - Number of occurrences when daily maximum temperature does not exceed the monthly mean low temperature or 32 degrees Fahrenheit (0 degrees Celsius), whichever is lower. (Not applicable to Okinawa Area)
 - (4) Concurrence between snowfall and cold temperature is 80 percent, i.e. 80 percent of the time snow falls, the temperature is "cold".
 - (5) Wind - Number of occurrences when the wind is gusting 30 knots (56 kilometers/hour) or greater.
- f. The Contractor's schedule must reflect the above anticipated adverse weather delays on all weather-dependent activities.

1.9 PERMITS AND RESPONSIBILITIES

The Contractor shall, without additional expense to the Government, be responsible for obtaining any necessary licenses and permits, and for complying with any laws, codes, and regulations (including the requirements of material, prefectural, and local Government of Japan, and associated military installation) applicable to the performance of work. The Contractor shall also be responsible for all damages to persons or property that occur as a result of the Contractor's fault or negligence. These damages shall be repaired or replaced by the Contractor at no cost to the Government. The Contractor shall take proper safety and health precautions to protect all work and workers.

Submit a schedule of planned road closures to the Contracting Officer with the initial project schedule. Notification of specific road closures shall be in writing to the Contracting Officer not less than thirty (30) calendar days in advance of the intended closure. The road closure request shall include the planned traffic control measures as well as the general information about the closure. All road closures shall be coordinated with base officials and are subject to base requirements. No road closure shall be permitted until the Contractor receives written approval from the Contracting Officer. Full road closures is generally not permitted (at least one way traffic shall always be provided).

1.9.1 Naha Port Authority (NPA) Work Permit

The Contractor shall obtain a Harbor Facility Usage Permission Application (Form Number 5), also known as the "Kowan Shisetsu Shiyō Kyōka Shinseisho" permit form addressed to the Naha Port Authority Administrator.

The submission shall include basic information regarding the Contractor's point of contact information, project title, description of work, and usage area information, as well as a work plan (to include scheduling), emergency contact plan, and protection plans. Application shall be hand delivered to:

Naha Port Authority
General Affairs Division
2-1 Tondochō (2nd Floor)
Naha City, Okinawa Prefecture
900-0035

The NPA will require scheduling coordination and due consideration to all Naha Commercial Port end users and coordination with other stakeholders, such as the Local Fisheries Association, to ensure that stakeholder operations are not negatively impacted by the proposed activity and ensure that appropriate coordination is performed by the construction activity proponent. The final Harbor Facility Usage Permit Application will be authorized by the Okinawa Prefectural Governor's office.

1.9.2 Work Application Package

In accordance with Article 31 of the GOJ Act on Port Regulations "Kō Sokuho", the Contractor shall submit a Work Application Package known as a "Kōji Sagyō Kyōka Shinseisho" for review and approval by the Maritime Traffic Division Office of the Japan Coast Guard (JCG) at Naha. A physical printed out copy shall be hand delivered to the following address:

Japan Coast Guard
Naha Coast Guard Office
Maritime Traffic Division
2-11-1 Minatomachi
Naha City, Okinawa Prefecture
900-8548

This application shall contain project safety information, work planning and scheduling, emergency management information as well as copies of all pertinent certifications and licenses (e.g. boat licenses, commercial diver's certifications, etc.) The minimum application review period is 30 days, although in the case of complicated work sites, review of work permit application packages by the JCG may take longer.

The Naha Coast Guard Office normally does not accept a work permit package from a Contractor until approval of work is formalized first with the NPA. Therefore, ample time must be secured for the dredging Contractor to clear both NPA and JCG Naha Coast Guard Office work permitting processes during the pre-construction phase of the dredging project.

1.9.3 Safety Panel and Committee (SPC)

As part of the work permitting process, the Contractor shall also participate in a Safety Panel and Committee (SPC). This SPC is required by the Japan Coast Guard, and shall be facilitated by a consultant hired by the Okinawa Defense Bureau. Intent of the SPC is to provide a forum under which all stakeholder and port operator concerns can be identified and addressed. The Contractor shall attend and present proposed safety, environmental, and operational plans at the forum.

1.10 SPECIAL CONTRACT REQUIREMENTS

1.10.1 Meetings

- a. **Pre-design Conference.** A pre-design conference per FAR 42.503 shall be held. The chairperson shall be the Contracting Officer or his/her designee. The chairperson shall be responsible for providing minutes of the meeting.
- a. **Preconstruction Conference.** A Preconstruction Conference per FAR 42.503 shall be held. The chairperson shall be the Contracting Officer or his/her designee. The chairperson shall be responsible for providing minutes of the meeting.
- a. **Post Award Orientation.** A post award orientation (often referred to as a Preconstruction Conference) per FAR 42.503 shall be held. The chairperson shall be the Contracting Officer or his/her designee. The chairperson shall be responsible for providing minutes of the meeting.
- b. **Weekly Progress Meetings.** A weekly progress meeting shall be conducted, with the day, time, and location to be determined at the Preconstruction Conference. The Contractor's Project Manager, Superintendent, Interpreter, and Quality Control Manager shall attend. Representatives from the Government may include, but are not limited to, Director of Public Works/Facilities Engineering personnel, Project's Resident Office personnel, and Project Manager. The Contractor shall be prepared to discuss work completed during the previous week, work currently in progress, and work forecasted for the following week, as well as the status of any construction issues and open action items. Provide the meeting agenda for review 24 hours prior to the meeting, take notes during the meeting, and provide electronic copies of the meeting minutes in English within 24 hours of the progress meeting for review.
- c. **Red Zone Meeting.** A Red Zone meeting shall be conducted for Contracts with a value in excess of 55,000,000 JPY or those deemed sufficiently complex by the Contracting Officer to warrant one. The purpose of the Red Zone meeting is to discuss closeout requirements for the contract and to establish a timeline to get those items completed (see Section 01 78 00 CLOSEOUT SUBMITTALS for typical closeout requirements). The meeting is a good time for the Contractor to gain approval on format for items such as Operations and Maintenance manuals, the

equipment-in-place list, the warranty plan, and any other items. The Initial Red Zone is typically scheduled when project reached 75 percent completion milestone. Provide the meeting agenda for review 24 hours prior to the meeting, take notes during the meeting, and provide electronic copies of the meeting minutes in English within 24 hours of the Red Zone meeting.

- d. Safety and Quality Control Meetings. Safety and quality control meetings shall be held as needed and determined by the Contractor. See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and Section 01 45 00.00 10 USACE QUALITY CONTROL for specific requirements.
- e. Construction Waste Management Meetings. See Section 01 74 19 CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT for requirements.
- f. Monthly Design Progress Meetings. A monthly design progress meeting shall be conducted, with the day, time, and location to be determined at the project kickoff meeting. The Contractor's Design team, Interpreter, and Design Quality Control manager shall attend. Representatives from the Government may include, but are not limited to, Client, Project Manager, and POJ Engineering personnel. The Contractor shall be prepared to discuss work completed during the previous week, work currently in progress, and work forecasted for the following week, as well as the status of any design issues and open action items. Provide the meeting agenda for review 24 hours prior to the meeting, take notes during the meeting, and provide electronic copies of the meeting minutes in English within 24 hours of the progress meeting for review.

[1.10.2 Conformed Drawings Submittal

Contractor shall submit one (1) hard copy Japanese-style folded and bound set of conformed A3 size drawings. Submittal shall be provided to the Resident Office at the Pre-Construction meeting. Conformed drawings are defined as the original advertised drawings with amended drawing sheets replaced in-kind with the corresponding original sheet number. If applicable, add new sheets and remove deleted sheets in accordance with amendment instructions. If multiple amendments are issued, supersede any previously amended drawing so that only the highest numbered amendment drawing is included.

]1.10.3 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN

Submit an IAQ Management Plan within 15 days after Notice to Proceed and not less than 10 days before the preconstruction meeting. The Plan shall include IAQ management practices implemented during construction and preoccupancy phases and describe how each requirement in ANSI/SMACNA 008 [and TPC requirements] will be met, addressed, and managed on the job site. The plan shall cover the following additional items:

- a. Identify the key players and person(s) responsible for implementing the plan.
- b. Specify procedures for protecting stored and installed absorptive materials from moisture damage.
- c. Prohibit the use of tobacco products inside the building and within 7.5 meters (50 feet) of the building entrance at all times during construction. Consider prohibiting smoking on the entire job site. If

smoking will be allowed, locate on a site plan the designated locations and signage. Comply with the tobacco use policy of the local jurisdiction.

- d. Indicate whether air handlers will be operated during construction, and specify compliant filtration procedures for permanent equipment that will be used.
- e. Take photographs of each IAQ measure for documentation. Annotate photographs to indicate each IAQ measure depicted and its general location. Provide photographs of the methods employed to protect stored and installed absorptive materials from moisture damage during construction and preoccupancy.

Revise and resubmit the Plan as required by the Contracting Officer. Make copies of the final plan available to all workers on site. Include provisions in the Plan to meet the requirements specified below and to ensure safe, healthy air for construction workers and building occupants.

1.10.3.1 Requirements During Construction

Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with ASTM D6245. Provide for evaluation of volatile organic compounds (VOCs) in indoor air in accordance with ASTM D6345. Use filters with a Minimum Efficiency Reporting Value (MERV) of 8 in permanently installed air handlers during construction.

1.10.3.1.1 Control Measures

Meet or exceed the requirements of ANSI/SMACNA 008, Chapter 3, to help minimize contamination of the building from construction activities. The five requirements of this manual which must be adhered to are described below:

- a. HVAC protection: Isolate return side of HVAC system from surrounding environment to prevent construction dust and debris from entering the duct work and spaces.
- b. Source control: Use low emitting paints and other finishes, sealants, adhesives, and other materials as specified. When available, cleaning products must have a low VOC content and be non-toxic to minimize building contamination. Utilize cleaning techniques that minimize dust generation. Cycle equipment off when not needed. Prohibit idling motor vehicles where emissions could be drawn into building. Designate receiving/storage areas for incoming material that minimize IAQ impacts.
- c. Pathway interruption: When pollutants are generated use strategies such as 100 percent outside air ventilation or erection of physical barriers between work and non-work areas to prevent contamination.
- d. Housekeeping: Clean frequently to remove construction dust and debris. Promptly clean up spills. Remove accumulated water and keep work areas dry to discourage the growth of mold and bacteria. Take extra measures when hazardous materials are involved.
- e. Scheduling: Control the sequence of construction to minimize the absorption of VOCs by other building materials.

1.10.3.1.2 Moisture Contamination

- a. Remove accumulated water and keep work dry.
- b. Use dehumidification to remove moist, humid air from a work area.
- c. Do not use combustion heaters or generators inside the building.
- d. Protect porous materials from exposure to moisture.
- e. Remove and replace items which remain damp for more than a few hours.

1.10.3.2 Requirements after Construction

After construction ends and prior to occupancy, conduct IAQ testing in accordance with ASHRAE 189.1 Section 10.3.1.4.

In the event that IAQ testing fails, conduct a building flush-out in accordance with ASHRAE 189.1 Section 10.3.1.3[and TPC requirements]. The space shall be ventilated at a minimum rate of 1.5 Liters per second per square meter of outdoor air while maintaining an internal temperature of at least 15 degrees C (60 degrees F) and no higher than 27 degrees C (80 degrees F) and relative humidity no higher than 60%. Total air volume of outdoor air will be determined by ASHRAE 189.1 Section 10.3.1.4 Equation 10-1. Total volume of outdoor air shall be approved by the Contracting Office prior to building flush-out

Air contamination testing must be consistent with EPA's current Compendium of Methods for the Determination of Air Pollutants in Ambient Air. After building flush-out or testing and prior to occupancy, replace filtration media. Filtration media must have a MERV of 13 as determined by ASHRAE 52.2.

1.11 PROGRESS AND COMPLETION PHOTOGRAPHS

Photographically document site conditions prior to start of construction operations. Provide monthly, and within one month of the completion of work, digital photographs, 1600 by 1200 by 24 bit true color 300 DPI minimum resolution in JPEG file format showing the sequence and progress of work. Take a minimum of [20] digital photographs each week throughout the entire project[and [20] digital photographs at completion of project]. Submit the Initial Site Conditions Photos, Monthly Progress Photos[, and Construction Completion Photos]. Indicate photographs demonstrating environmental procedures. Provide photographs for each month in a separate monthly directory and name each file to indicate its location on the view location sketch. Include a date designator in file names. Cross reference submittals in the appropriate daily report. Photographs provided are for unrestricted use by the Government.

1.12 BASE REGULATIONS

The Contractor and Subcontractor(s) shall become familiar with and obey all base regulations, including fire, traffic, safety, and security regulations. All Contractors shall keep within the limits of the work (and avenues of ingress and egress), and shall not enter any restricted areas unless required to do so and are cleared for such entry. The Contractor's equipment shall be conspicuously marked for identification.

1.12.1 Request for Contractor's Employee Passes and Vehicle Passes

No employee or representative of the Contractor shall be admitted to the work site without a Contracting Officer's furnished authorized admittance. Refer to USAG-J Regulation 190-13. Prior to the start of on-site work, submit applications for base passes to the Contracting Officer for key employees (project manager, site supervisor, interpreter, etc.) for long term DBIDS passes. The Contractor (through the employment of a Pass Coordinator), with assistance from the Resident Office, is responsible for securing sufficient passes for other workers, including subcontractor's employees and vehicles, required to access the base for the project duration. Additional personnel data shall also be furnished.

The Contractor shall use the USAG-J Form 1529 "EZ Pass" as the primary means of obtaining passes for all personnel who do not qualify for DBIDS passes. The USAG-J Form 1529 "EZ Pass" One Time/Multiple Access Roster must be complete and submitted to the Provost Marshall Pass and Vehicle Registration Office a minimum of three (3) U.S. workdays prior to access. The use of the USAG-J "one day" pass (AJ Form 47) should be avoided, and used only for urgent situations. Applicants for a "one day" pass may be required to wait for several hours for their pass to be issued. Contractor personnel applying for more than one (1) "one-day" pass within any 15-day period will not be issued multiple one-day passes, and will instead be required to apply for access via the USAG-J Form 1529, and wait three or more days for approval.

Provide all information required for background checks to meet base access requirements to be accomplished by Base Provost Marshal Office, Director of Emergency Services or Security Office. Contractor workforce must comply with all personal identity verification requirements as directed by DOD, HQDA and/or local policy. The Government reserves the right to make changes to the Contractor security requirements or processes due to a change in the Force Protection Condition (FPCON).

Upon completion of this Contract, return all employee and vehicle passes to the base, and obtain a certification of receipt. Final payment shall be withheld until all passes have been returned.

1.12.2 No Smoking Policy

Smoking is prohibited on installations except in designated smoking areas. This applies to existing buildings, buildings under construction, and buildings under renovation. Discarding tobacco materials other than into designated tobacco receptacles is considered littering and is subject to fines. The Contracting Officer shall identify designated smoking areas.

1.12.3 Munitions and Explosives of Concern (MEC)

Munitions and Explosives of Concern (MEC): Unexploded Ordnance (UXO), Material Presenting a Potential Explosive Hazard (MPPEH), Chemical Agents (CA), or Discarded Military Munitions (DMM) on jobsites shall be treated as extremely dangerous and must be reported immediately. Follow the 3Rs: RECOGNIZE, RETREAT, and REPORT. In the event MEC are discovered or uncovered, immediately stop work in that area and immediately inform the Contracting Officer's Representative. Contractor shall provide dispatch and Contracting Officer's Representative with specific location of the item. Contractor personnel shall stop work in the immediate vicinity of the discovery and maintain a minimum distance of 300 meters from the item.

Contractor shall maintain flexibility in redirecting personnel and work effort in the event that items possessing an explosive hazard are discovered and construction personnel are excluded from areas during the destruction/removal process.

1.12.3.1 Additional Requirements for Camp Foster

All workers on site involved in the earthwork, in any manner, must attend a UXO brief on Marine Camp Foster. Upon completion of the brief, the worker will receive a laminated card verifying that they attended the brief. The card does not expire and is valid for all USACE construction contracts on Okinawa, but is non-transferrable between workers. The USACE team will help coordinate the training and assist with processing base access requests for Marine Camp Foster as needed. The brief is free of charge, but the contractor is responsible for all other costs associated with transport, labor, and all other costs related to the training. The expected duration of this training is two hours.

1.12.4 Unexploded Ordnance (UXO) Procedures on Kadena Air Base

Kadena Air Base 718 CES/CEN assesses the area as UXO unlikely with a low probability of uncovering UXO's on the site. However, since there is always a possibility of encountering UXOs anywhere on Okinawa, albeit unlikely at this location, excavations and other construction activities related earth work on the project site, require a measure of caution. The Ground Penetrating Radar (GPR) survey specified in 01 35 26 paragraph "Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces" for the identification of underground utilities will identify anomalies. Best practices of conducting GPR surveys are to repeat them in areas greater than the used method's level of detection (LOD), down to a depth of 2 meters, to ensure no utility lines or other suspicious objects are present. If suspected UXOs are identified, the construction team needs to follow the standard UXO notification procedures per the Kadena policies identified in attachment 01 11 00.00 10-[B] "UXO Emergency Mitigation Steps and Survey Policies." The contact number for the Emergency Dispatch Center for Kadena Air Base is 098-934-5911. See the following website for additional information:
www.kadena.af.mil/About-Us/Emergency-Actions/.

1.13 ORDER OF WORK

1.13.1 Schedule

Schedule all work to cause the least amount of interference with activity operations. Permission to interrupt any activity or roads shall be requested in writing a minimum of 30 calendar days prior to the desired date of interruption. Interruptions of activities, roads, and utility services shall be allowed only when they will not cause interference with the operations of the activity. The Contractor shall remove and dispose off Government property all Contractor generated debris at the end of each shift, or more frequently if required, to keep the space usable. Upon award of this Contract, the Contractor shall begin and complete all required work; ready for use and including cleanup, within the time period specified on this Contract. All work scheduled in occupied areas shall be accomplished in such a manner as to cause the least possible inconvenience to the occupants.

1.13.2 Working Hours

Normal Working Hours and Days are Monday through Friday, 0800 hours through 1700 hours. Saturday work is allowed on this contract and does not require additional approvals. Notify the Contracting Officer at least three days in advance when scheduling Saturday work.

1.13.3 Noise and Vibration

Noise greater than 70 decibels and vibration producing work in occupied buildings shall be conducted outside core school/business hours (0700 - 1430) unless written permission is given by the Contracting Officer.

1.13.4 Work Outside of Regular Hours

If the Contractor desires to carry on work outside the regular hours, on Sundays, or holidays, a written application shall be submitted to the Contracting Officer or his representative for approval. The Contractor shall allow three working days notice to enable satisfactory arrangements to be made by the Government for inspecting the work in progress. If work is to be accomplished after daylight hours, the Contractor shall illuminate the area in a manner approved by the Contracting Officer or their representative and in accordance with EM 385-1-1. Unless directed by a Contracting Officer, work accomplished outside regular working hours shall be at no additional cost to the Government.

1.14 FIRE PREVENTION DURING CONSTRUCTION

1.14.1 General

Comply with all pertinent fire prevention provisions of the US Army Corps of Engineers Manual EM 385-1-1, NFPA 241 and shall follow the Installation fire regulations. Prior to commencement of welding or other hot work operations, obtain approval from the Installation Fire Chief.

1.14.2 Supply

No more than one day's supply of paint, paint materials, or compounds shall be allowed within the area of the building, and shall be removed from the job site after each working day. No gasoline or similar low flash point flammable liquid shall be allowed within the building area. After proper coordination with the Facility's Emergency Services (to confirm the maximum allowable quantities and location of storage), storage of additional product may be authorized.

1.14.3 Fire Extinguishers

Provide, as a minimum, the number, size, and type of fire extinguishers in accordance with the latest NFPA 10. The Contractor shall comply with the Installation Fire Chief's policies if they are more stringent than NFPA 10. Fire extinguishers shall remain the property of the Contractor and shall be removed upon completion of the project.

1.14.4 Housekeeping

Accumulations of combustible material shall be removed from the building area on a daily basis.

1.14.5 Handling of Gasoline

Gasoline shall be stored in industry standard approved safety containers. Adequate ventilation shall be provided to safely dispose of flammable vapors where flammable liquids are utilized. Gasoline powered equipment shall be refueled a minimum 6 meters away from the building area.

1.14.6 Notification of Fire

Be familiar with methods for notifying the Installation Fire Department. The Installation's fire poster shall be posted in conspicuous locations and at telephones in construction shacks.

1.15 CONTRACTOR FURNISHED MATERIAL AND WORKMANSHIP

The Contractor shall furnish all materials necessary for performance of the work of this contract unless otherwise specified. Materials procured shall be new and shall meet any specifications and standards listed in these specifications. If no specification for a material needed to perform this contract are stated, the material shall be new, of acceptable industrial grade and quality, equal to or better than the manufacturer's original equipment for equipment being repaired or replaced, and will be compatible with existing materials and systems. All materials provided under this contract shall be free of asbestos, lead in paint, and PCB. The Contracting Officer reserves the right to request submittal of any material being provided under this contract. The Contracting Officer shall make the final determination of the acceptability of any material used on this contract.

1.16 CORRESPONDENCE

All correspondence addressed to the Government shall be made through serialized letters furnished with one original and two copies. Serialized letters shall begin with the number S-0001 and shall be continuous without a break in numbering. Serialized letters shall include the contract title and number, date, subject, and shall be signed by an authorized representative of the Contractor.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

Reset Form

Validate

Government Furnished Property

Attachment Number

Attachment File

 of

Attachment Date

Government Contracting Officer

Name

Email

Phone

☐ Contract Award

☐ Contract Modification

☐ Solicitation

☐ Amendment

Modification Number

Amendment Number

Reference Procurement Instrument Number

Custodial Procurement Instrument Identification

Solicitation, Contract or Uniform PIID Number

Issuing Office DoDAAC

Issue Year (YY)

Contract Type

Sequence Number

OR

Order Number (PIID Format FY16+)

Issuing Office DoDAAC

Issue Year (YY)

Contract Type

Sequence Number

ATTACHMENT 01 11 00.00 10-A Page 1

Consolidated GFP Attachment V.M.1.1

Government Furnished Property

SERIALLY MANAGED ITEMS LIST													
ADD													
	Line Number	Item Name	Item Description	NSN	CAGE	Part Number	Model Number	Serial Number	Ull	QTY	Unit of Measure	Unit Cost	Use As Is
Remove		Delivery Date (On or Before)	Duration	Time Unit	Delivery Event								
NON-SERIALLY MANAGED ITEMS LIST													
ADD													
	Line Number	Item Name	Item Description	NSN	CAGE	Part Number	Model Number			QTY	Unit of Measure	Unit Cost	Use As Is
Remove		Delivery Date (On or Before)	Duration	Time Unit	Delivery Event								
Items Authorized to be Requisitioned by the Contractor and Paid for by DoD													
ADD	Line Number	Item Name	Item Description	NSN	CAGE	Part Number				QTY	Unit of Measure	Unit Cost	Use As Is
Remove													



DEPARTMENT OF THE AIR FORCE
PACIFIC AIR FORCES

11 May 17

MEMORANDUM FOR RECORD

FROM: 718 CES/CEN

SUBJECT: UXO Emergency Mitigation Steps on Kadena AB, Okinawa, Japan

1. The purpose of this memorandum is to provide steps for handling unexploded ordnance (UXO) discoveries on Kadena AB, Japan. As an island with a history involving beach invasions during WWII, it is important to properly adhere to the following procedures in order to achieve the highest level of safety during construction.
2. When working on Kadena AB and a UXO is discovered, the following steps shall be followed:
 - a. Cease all construction immediately.
 - b. If Fire Protection (FP)/Explosive Ordnance Disposal (EOD) are not already present, call the Emergency Dispatch Center at (098) 911-1911.
 - c. If possible, safely place a marker near the UXO. Do not mark on the UXO itself or place a marker that will pierce into the ground near the UXO. Instead, place a distinguishing item nearby such as a colored ribbon or flag.
 - d. Remain at least 300 feet clear of the UXO, and direct others to remain clear, until directed further by EOD/FP staff.
3. For questions or more information, please contact 2d Lt Evan Fortney at evan.fortney.1@us.af.mil or DSN 634-3789.

A handwritten signature in blue ink, reading "JCH", is positioned above the typed name of the signatory.

JOEL C. HEARN, Maj, USAF
Engineering Flight Commander
718th Civil Engineer Squadron

SECTION 01 20 00.00 10

PRICE AND PAYMENT PROCEDURES

11/20, CHG 5: 02/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A615/A615M

(2024) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Weight Certificates

1.3 CONTRACT COST BREAKDOWN

The Contractor must furnish within 30 days after the date of Notice to Proceed, and prior to the submission of its first partial payment estimate, a breakdown of its single job pay item or items which will be reviewed by the Contracting Officer as to propriety of distribution of the total cost to the various accounts. Any unbalanced items as between early and late payment items or other discrepancies will be revised by the Contracting Officer to agree with a reasonable cost of the work included in the various items. This Contract cost breakdown will then be utilized as the basis for progress payments to the Contractor.

1.4 CONTRACTOR'S INVOICE AND CONTRACT PERFORMANCE STATEMENT

1.4.1 Submission of Invoices

If DFARS Clause 252.232-7006 Wide Area WorkFlow Payment Instructions is included in the Contract, provide the documents listed in above paragraph CONTENT OF INVOICE in their entirety as attachments in Wide Area Work Flow (WAWF) for each invoice submitted. The maximum size of each WAWF attachment is two megabytes, but there are no limits on the number of attachments. If a document cannot be attached in WAWF due to system or size restriction, provide it as instructed by the Contracting Officer.

Monthly invoices and supporting forms for work performed through the

anniversary award date of the Contract must be submitted to the Contracting Officer within 5 calendar days of the date of invoice. For example, if Contract award date is the 7th of the month, the date of each monthly invoice must be the 7th and the invoice must be submitted by the 12th of the month.

1.4.2 Final Invoice

- a. A final invoice must be accompanied by the certification required by DFARS 252.247.7023 Transportation of Supplies by Sea, and the Contractor's Final Release. If the Contractor is incorporated, the Final Release must contain the corporate seal. An officer of the corporation must sign and the corporate secretary must certify the Final Release.
- b. For final invoices being submitted via WAWF, the original Contractor's Final Release Form and required certification of Transportation of Supplies by Sea must be provided directly to the respective Contracting Officer prior to submission of the final invoice. Once receipt of the original Final Release Form and required certification of Transportation of Supplies by Sea has been confirmed by the Contracting Officer, the Contractor must then submit final invoice and attach a copy of the Final Release Form and required certification of Transportation of Supplies by Sea in WAWF.
- c. Final invoices not accompanied by the Contractor's Final Release and required certification of Transportation of Supplies by Sea will be considered incomplete and will be returned to the Contractor.

[1.5 SINGLE JOB PAYMENT ITEMS

Payment items for the work of this Contract for which Contract job payments will be made are listed in the [BIDDING][PRICING] PROPOSAL SCHEDULE and described below. All costs for items of work, which are not specifically mentioned to be included in a particular job or unit price payment item, are included in the listed job item most closely associated with the work involved. The job price and payment made for each item listed constitutes full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for which separate payment is not otherwise provided.

1.5.1 Mobilization and Demobilization

1.5.1.1 Payment

Payment will be made for costs associated with mobilization and demobilization, as defined in Special Contract [Clause][Requirement] PAYMENT FOR MOBILIZATION AND DEMOBILIZATION.

1.5.1.2 Unit of Measure

Unit of measure: job.

1.5.2 Structure No. 1

1.5.2.1 Payment

Payment will be made for costs associated with operations necessary for construction of the structure at Station XX+XX.

1.5.2.2 Unit of Measure

Unit of measure: job.

1.5.3 Structure No. 2

1.5.3.1 Payment

Payment will be made for costs associated with operations necessary for construction of the structure at Station XX+XX.

1.5.3.2 Unit of Measure

Unit of measure: job.

][1.6 UNIT PRICE PAYMENT ITEMS

Payment items for the work of this Contract on which the Contract unit price payments will be made are listed in the [BIDDING][PRICING] PROPOSAL SCHEDULE and described below. The unit price and payment made for each item listed constitutes full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for each of the unit price items.

1.6.1 Excavation

1.6.1.1 Payment

Payment will be made for costs associated with excavation [for the channel][and][for the structure], which includes performing required excavation and other operations incidental thereto, Contractor-furnished disposal area(s) and disposition of excess excavated material and unsuitable and frozen materials.

1.6.1.2 Measurement

The total quantity of excavated material for which payment will be made will be the theoretical quantity between the ground surface as determined by a survey and the grade and slope of the theoretical cross sections indicated. No allowance will be made for overdepth excavation or for the removal of any material outside the required slope lines unless authorized.

1.6.1.3 Unit of Measure

Unit of measure: cubic meter.

1.6.2 M 90 Riprap

1.6.2.1 Payment

Payment will be made for costs associated with furnishing, transporting, stockpiling (if applicable), placing, and constructing the stone protection as specified.

1.6.2.2 Measurement

Measure riprap for payment by the ton metric by weighing each truckload to the nearest 0.1 ton, and the final quantity of [each truckload][the whole sum] is rounded to the nearest whole ton. Weigh the riprap for payment on approved scales before being placed in the work. Quarry weights will not be accepted. Use scales of sufficient length to permit simultaneous weighing all axle loads. Scales must be inspected, tested and sealed as directed to assure accuracy with 0.5 percent throughout the range of the scales. Certify scales located at the site of the work as to accuracy by an acceptable scales company representative prior to weighing any riprap. Scales will be checked and certified before riprap hauling and rechecked and recertified whenever a variance is suspected. Furnish the scales. If commercial scales are readily available in close proximity, 15 kilometers of site of work, the Contracting Officer may approve the use of the scales. Weigh riprap in the presence of the Government representative. The Contracting Officer may elect to accept certified weight certificates furnished by a public weighmaster in lieu of scale weights at the jobsite.

1.6.2.3 Unit of Measure

Unit of measure: ton metric.

1.6.3 Deformed Steel Bars for Concrete Reinforcement

1.6.3.1 Payment

Payment will be made for costs associated with furnishing, transporting, delivering, and placing deformed steel bars for concrete reinforcement, which includes steel in laps as indicated or as required. No payment will be made for the additional steel in laps which are authorized for the convenience of the Contractor. No separate payment will be made for accessories; include payment in the Contract unit price for the items of work to which the accessories are incidental.

1.6.3.2 Measurement

The measured lengths of deformed steel bars for concrete reinforcement will be converted to weights for the size of bars listed by the use of the nominal weights per lineal meter specified in ASTM A615/A615M.

1.6.3.3 Unit of Measure

Unit of measure: per kilogram in place.

][1.7 CONTAMINATED SOIL REMOVAL

Payment items for the work of this Contract on which the Contract unit price payments will be made are listed in the [PRICING][BIDDING] PROPOSAL SCHEDULE and described below. The unit price and payment made for each item listed must constitute full compensation for furnishing all plant,

labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, [tests and reports,] and for performing all work required for each of the unit price items.

1.7.1 Unit of Measure

Unit of measure: cubic meter.

][1.8 DREDGING

Payment will be made at the Contract job price and must constitute full compensation for performing all dredging, disposal, [and ice and snow removal,] stockpiling, and replacement. Payment will be in accordance with above paragraph, SINGLE JOB PAYMENT ITEMS. In no case will payment be made for material removed from below the maximum pay-line or outside the dredging prism as indicated.

]PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 32 01.00 10

PROJECT SCHEDULE
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AACE INTERNATIONAL (AACE)

- AACE 29R-03 (2011) Forensic Schedule Analysis
- AACE 52R-06 (2006) Time Impact Analysis - As Applied in Construction
- AACE 84R-13 (2015) Planning and Accounting for Adverse Weather

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

- ASCE 67-17 (2017) Schedule Delay Analysis

U.S. ARMY CORPS OF ENGINEERS (USACE)

- ER 1-1-11 (2017) Administration -- Project Schedules

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are [for Contractor Quality Control approval.][for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.] Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

- Project Scheduler Qualifications; G[, [____]]
- Preliminary Project Schedule; G[, [____]]
- Initial Project Schedule; G[, [____]]
- Periodic Schedule Update; G[, [____]]

1.3 PROJECT SCHEDULER QUALIFICATIONS

Designate an authorized representative to be responsible for the preparation of the schedule and all required updating and production of reports. The Designated Project Scheduler must have prepared and maintained at least three previous construction schedules for projects of similar size and complexity to this contract, using Primavera P6. Representative must have a comprehensive knowledge of CPM scheduling

principles and application.

PART 2 PRODUCTS

2.1 SOFTWARE

The scheduling software utilized to produce and update the schedules required herein must be capable of meeting all requirements of this specification.

2.1.1 Government Default Software

The Government uses Primavera P6. Ensure exported schedule files are compatible with the version of P6 used by the Government.

2.1.2 Contractor Software

Scheduling software used by the contractor must be commercially available from the software vendor for purchase with vendor software support agreements available. The software routine used to create the required sdef file must be created and supported by the software manufacturer.

2.1.2.1 Primavera

If Primavera P6 is selected for use, provide the "xer" export file in a version of P6 importable by the Government system. Verify at the SEKO meeting which version of P6 is in use by the Government. Export the schedule in a version of P6 no newer than that used by the Government.

2.1.2.2 Other Than Primavera

If the Contractor chooses software other than Primavera P6, that is compliant with this specification, provide for the Government's use two licenses, two computers, and training for two Government employees in the use of the software. These computers will be stand-alone and not connected to Government network. Computers and licenses will be returned at project completion.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

Prepare for approval a Project Schedule, as specified herein, pursuant to FAR Clause 52.236-15 Schedules for Construction Contracts. Show in the schedule the proposed sequence to perform the work and dates contemplated for starting and completing all schedule activities. The scheduling of the entire project is required. The scheduling of design and construction is the responsibility of the Contractor. Contractor management personnel must actively participate in its development. Designers, Subcontractors, and suppliers working on the project must also contribute in developing and maintaining an accurate Project Schedule. Provide a schedule that is a forward planning as well as a project monitoring tool. Use the Critical Path Method (CPM) of network calculation to generate all Project Schedules. Prepare each Project Schedule using the Precedence Diagram Method (PDM).

3.1.1 Scheduling Expectations Kickoff Meeting(SEKO)

In conjunction with the post award conference or pre construction

conference the Government and the Contractor will hold a separate and distinct Scheduling Expectations Kickoff Meeting(SEKO). The intent of this meeting is for the Government to review with the contractor the requirements of the scheduling specification, communicate the Government's expectations regarding schedule development and management, and answer any questions the Contractor may have regarding the schedule requirements.

3.2 BASIS FOR PAYMENT AND COST LOADING

The schedule is the basis for determining contract earnings during each update period and therefore the amount of each progress payment. The aggregate value of all activities coded to a contract CLIN must equal the value of the CLIN. Match the progress payment "Pay Period Thru" date in RMS to the schedule data date.

3.2.1 Activity Cost Loading

Activity cost loading must be reasonable and without front-end loading. Activities with a negative cost loading is not allowed. Provide additional documentation to demonstrate reasonableness if requested by the Contracting Officer.

3.2.2 Withholdings / Payment Rejection

Failure to meet the requirements of this specification may result in the disapproval of the preliminary, initial, or periodic schedule updates and subsequent rejection of payment requests until compliance is met.

In the event that the Contracting Officer directs schedule revisions and those revisions have not been included in subsequent Project Schedule revisions or updates, the Contracting Officer may withhold 10 percent of pay request amount from each payment period until such revisions to the project schedule have been made.

3.3 PROJECT SCHEDULE DETAILED REQUIREMENTS

3.3.1 Level of Detail Required

Develop the Project Schedule to the appropriate level of detail to address major milestones and to allow for satisfactory project planning and execution. Failure to develop the Project Schedule to an appropriate level of detail will result in its disapproval. The Contracting Officer will consider, but is not limited to, the following characteristics and requirements to determine appropriate level of detail:

3.3.2 Activity Durations

Reasonable activity durations are those that allow the progress of ongoing activities to be accurately determined between update periods. Non-procurement activities Original Durations (OD) are not to exceed 20 workdays or 30 calendar days.

3.3.3 Design and Permit Activities

Include design and permit activities with the necessary conferences and follow-up actions and design package submission dates. Include the design schedule in the project schedule, showing the sequence of events involved in carrying out the project design tasks within the specific contract period. Provide a detailed level of scheduling sufficient to identify all

major design tasks, including those that control the flow of work. Also include review and correction periods associated with each item.

3.3.4 Procurement Activities

Include in the schedule activities associated with the critical submittals and their approvals, procurement, fabrication, and delivery of long lead materials, equipment, fabricated assemblies, and supplies. Long lead procurement activities are those with an anticipated procurement sequence of over 90 calendar days.

3.3.5 Mandatory Activities

Include the following activities in the initial project schedule and all updates.

- a. Submission, review and acceptance of SD-01 Preconstruction Submittals (individual activity for each).
- b. Submission, review and acceptance of features requiring design completion Submission, review and acceptance of design packages.
- c. Submission of mechanical/electrical/information systems layout drawings.
- d. Submission and approval of O & M manuals.
- e. Submission and approval of as-built drawings.
- f. Submission and approval of DD1354 data and installed equipment lists.
- g. Submission and approval of testing and air balance (TAB).
- h. Submission of TAB specialist design review report.
- i. Submission and approval of fire protection specialist.
- j. Submission and approval of Building Commissioning Plan, Controls testing plan, test data, and reports: Develop the schedule logic associated with testing and commissioning of mechanical systems to a level of detail consistent with the contract commissioning requirements. All tasks associated with building testing and commissioning will be completed prior to submission of building commissioning report and subsequent contract completion.

These activities may include but are not limited to: air and water balancing, building commissioning -functional performance testing, controls testing, and performance verification testing. Include other systems testing, if required. Include additional commissioning requirements noted elsewhere in the Contract as required in the schedule and discuss during the SEKO meeting, noted in paragraph SCHEDULING EXPECTATIONS KICKOFF MEETING (SEKO).

- k. Contractor's pre-final inspection.
- m. Correction of punch list from Contractor's pre-final inspection.
- n. Government's pre-final inspection.

- o. Correction of punch list from Government's pre-final inspection.
- p. Final inspection.

3.3.6 Government Activities

Show in schedule, Government and other agency activities that could impact progress. These activities include, but are not limited to approvals, acceptance, design reviews, environmental permit approvals by State regulators, inspections, utility tie-in, Government Furnished Equipment (GFE) and Notice to Proceed (NTP) for phasing requirements.

3.3.7 Standard Activity Coding Dictionary

Use the activity coding structure defined in the Standard Data Exchange Format (SDEF) in ER 1-1-11. This exact structure is mandatory. Develop and assign all Activity Codes to activities as detailed herein.

The SDEF format is as follows:

Field	Activity Code	Length	Description
1	WRKP	3	Workers per day
2	RESP	4	Responsible party
3	AREA	4	Area of work
4	MODF	6	Modification Number
5	BIDI	6	Bid Item (CLIN)
6	PHAS	2	Phase of work
7	CATW	1	Category of work
8	FOW	20	Feature of work*
*Some systems require that FEATURE OF WORK values be placed in several activity code fields. The notation shown is for Primavera P6. Refer to the specific software guidelines with respect to the FEATURE OF WORK field requirements.			

3.3.7.1 Workers Per Day (WRKP)

Assign Workers per Day for all field construction or direct work activities, if directed by the Contracting Officer. Workers per day is based on the average number of workers expected each day to perform a task for the duration of that activity.

3.3.7.2 Responsible Party Coding (RESP)

Assign responsibility code for all activities to the Prime Contractor, Subcontractor(s) or Government agency(ies) responsible for performing the activity.

- a. Activities coded with a Government Responsibility code include, but are not limited to: Government approvals, Government design reviews, environmental permit approvals by State regulators, Government Furnished Property/Equipment (GFP) and Notice to Proceed (NTP) for phasing requirements.
- b. Activities cannot have more than one Responsibility Code. Examples of acceptable activity code values are: DOR (for the designer of record); ELEC (for the electrical subcontractor); MECH (for the mechanical subcontractor); and GOVT (for USACE).

3.3.7.3 Area of Work Coding (AREA)

Assign Work Area code to activities based upon the work area in which the activity occurs. Define work areas based on resource constraints or space constraints that would preclude a resource, such as a particular trade or craft work crew from working in more than one work area at a time due to restraints on resources or space. Examples of Work Area Coding include different areas within a floor of a building, different floors within a building, and different buildings within a complex of buildings. Activities cannot have more than one Work Area Code.

3.3.7.4 Modification Number (MODF)

Assign a Modification Number Code to any activity or sequence of activities added to the schedule as a result of a Contract Modification, when approved by Contracting Officer. Key all Code values to the Government's modification numbering system. An activity can have only one Modification Number Code.

3.3.7.5 Bid Item Coding (BIDI)

Assign a Bid Item Code to all activities using the Contract Line Item Number (CLIN) to which the activity belongs, even when an activity is not cost loaded. An activity can have only one BIDI Code.

3.3.7.6 Phase of Work Coding (PHAS)

Assign Phase of Work Code to all activities. Examples of phase of work are design phase, procurement phase and construction phase. Each activity can have only one Phase of Work code.

- a. Code proposed fast track design and construction phases proposed to allow filtering and organizing the schedule by fast track design and construction packages.
- b. If the contract specifies phasing with separately defined performance periods, identify a Phase Code to allow filtering and organizing the schedule accordingly.

3.3.7.7 Category of Work Coding (CATW)

Assign a Category of Work Code to all activities. Category of Work Codes include, but are not limited to design, design submittal, design reviews, review conferences, permits, construction submittal, procurement, fabrication, weather sensitive installation, non-weather sensitive installation, start-up, and testing activities. Each activity can have no more than one Category of Work Code.

3.3.7.8 Feature of Work Coding (FOW)

Assign a Feature of Work Code to appropriate activities based on the Definable Feature of Work to which the activity belongs based on the approved QC plan.

Definable Feature of Work is defined in Section 01 45 00 QUALITY CONTROL. An activity can have only one Feature of Work Code.

3.3.8 Contract Milestones and Constraints

Milestone activities are to be used for significant project events including, but not limited to, project phasing, project start and end activities, or interim completion dates. The use of artificial float constraints such as "zero free float" or "zero total float" are prohibited.

Mandatory constraints that ignore or effect network logic are prohibited. No constrained dates are allowed in the schedule other than those specified herein. Submit additional constraints to the Contracting Officer for approval on a case by case basis.

3.3.8.1 Project Start Date Milestone and Constraint

The first activity in the project schedule must be a start milestone titled "NTP Acknowledged," which must have a "Start On" constraint date equal to the date that the NTP is acknowledged.

3.3.8.2 End Project Finish Milestone and Constraint

The last activity in the schedule must be a finish milestone titled "End Project."

Constrain the project schedule to the Contract Completion Date in such a way that if the schedule calculates an early finish, then the float calculation for "End Project" milestone reflects positive float on the longest path. If the project schedule calculates a late finish, then the "End Project" milestone float calculation reflects negative float on the longest path.

3.3.8.3 Interim Completion Dates and Constraints

Constrain contractually specified interim completion dates to show negative float when the calculated late finish date of the last activity in that phase is later than the specified interim completion date.

3.3.8.3.1 Start Phase

Use a start milestone as the first activity for a project phase. Call the start milestone "Start Phase X" where "X" refers to the phase of work.

3.3.8.3.2 End Phase

Use a finish milestone as the last activity for a project phase. Call the finish milestone "End Phase X" where "X" refers to the phase of work.

3.3.9 Adverse Weather

Ensure anticipated adverse weather is accounted for in the Project

Schedule. AACE 84R-13 provides methodologies for techniques in planning for adverse weather. The preferred methodology is the use of a weather calendar. If a method other than a weather calendar is proposed the discuss the reason during the SEKO meeting.

3.3.10 Calendars

Schedule activities on a Calendar to which the activity logically belongs. Develop calendars to accommodate any contract defined work period such as a 7-day calendar for Government Acceptance activities, concrete cure times, etc. Develop the default Calendar to match the physical work plan with non-work periods identified including weekends and holidays. Develop Seasonal Calendar(s) and assign to seasonally affected activities as applicable.

If a weather calendar is used its implementation and execution is to be in accordance with AACE 84R-13

3.3.11 Open Ended Logic

Only two open ended activities are allowed: the first activity "NTP Acknowledged" is to have no predecessor logic, and the last activity -"End Project" is to have no successor logic. Except as noted above all activities are to have at least a start-to-start or finish-to-start relationship with its predecessor(s) and a finish-to-finish or finish-to-start relationship with its successor(s).

Predecessor open ended logic may be allowed in a time impact analyses upon the Contracting Officer's approval.

3.3.12 Default Progress Data Disallowed

Actual Start and Finish dates must not automatically update with default mechanisms included in the scheduling software. Updating of the percent complete and the remaining duration of any activity must be independent functions. Disable program features that calculate one of these parameters from the other. Activity Actual Start (AS) and Actual Finish (AF) dates assigned during the updating process must match those dates provided in the Contractor Quality Control Reports. Failure to document the AS and AF dates in the Daily Quality Control report will result in disapproval of the Contractor's schedule.

3.3.13 Out-of-Sequence Progress

Activities that have progressed before all preceding logic has been satisfied (Out-of-Sequence Progress) will be allowed only on a case-by-case basis subject to approval by the Contracting Officer. Propose logic corrections to eliminate out of sequence progress or justify not changing the sequencing for approval prior to submitting an updated project schedule. Address out of sequence progress or logic changes in the Narrative Report and in the periodic schedule update meetings.

3.3.14 Added and Deleted Activities

Do not delete activities from the project schedule or add new activities to the schedule without approval from the Contracting Officer. Activity ID and description changes are considered new activities and cannot be changed without Contracting Officer approval.

3.3.15 Original Durations

Activity Original Durations (OD) must be reasonable to perform the work item. OD changes are prohibited unless justification is provided and approved by the Contracting Officer.

3.3.16 Leads, Lags, and Start to Finish Relationships

Lags must be reasonable as determined by the Government and not used in place of realistic original durations, must not be in place to artificially absorb or create float, or to replace proper schedule logic.

- a. Leads (negative lags) are prohibited.
- b. Start to Finish (SF) relationships are prohibited.

3.3.17 Retained Logic

Schedule calculations must retain the logic between predecessors and successors ("retained logic" mode) even when the successor activity(s) starts and the predecessor activity(s) has not finished (out-of-sequence progress). Software features that in effect sever the tie between predecessor and successor activities when the successor has started and the predecessor logic is not satisfied ("progress override") are not be allowed.

3.3.18 Percent Complete

Update the percent complete for each activity started, based on the realistic assessment of earned value. Activities which are complete but for remaining minor punch list work and which do not restrain the initiation of successor activities may be declared 100 percent complete to allow for proper schedule management. Budgeted cost of activity is to be reduced by the amount needed to correct deficiency. The activity referenced in paragraph COST LOADING OF CLOSEOUT ACTIVITIES must have its budgeted cost increased by this amount.

3.3.19 Remaining Duration

Update the remaining duration for each activity based on the number of estimated workdays it will take to complete the activity. Remaining duration may not mathematically correlate with percentage found under paragraph entitled Percent Complete. The remaining duration for unstarted activities are not to be less than its original duration.

3.3.20 Cost Loading of Closeout Activities

Cost load the "Correction of punch list from Government pre-final inspection" activity(ies) not less than 1 percent of the present contract value. Activity(ies) may be declared 100 percent complete upon the Government's verification of completion and correction of all punch list work identified during Government pre-final inspection(s).

3.3.20.1 As-Built Drawings

If there is no separate contract line item (CLIN) for as-built drawings, cost load the "Submission and approval of as-built drawings" activity not less than \$35,000 or 1 percent of the present contract value, which ever is greater, up to \$200,000. Activity will be declared 100 percent

complete upon the Government's approval.

3.3.20.2 O & M Manuals

Cost load the "Submission and approval of O & M manuals" activity not less than \$20,000. Activity will be declared 100 percent complete upon the Government's approval of all O & M manuals.

3.3.21 Early Completion Schedule and the Right to Finish Early

An Early Completion Schedule is an Initial Project Schedule (IPS) that indicates all scope of the required contract work will be completed before the contractually required completion date.

- a. No IPS indicating an Early Completion will be accepted without being fully resource-loaded (including crew sizes and manhours) and the Government agreeing that the schedule is reasonable and achievable.
- b. The Government is under no obligation to accelerate work items it is responsible for to ensure that the early completion is met nor is it responsible to modify incremental funding (if applicable) for the project to meet the contractor's accelerated work.

3.4 PROJECT SCHEDULE SUBMISSIONS

Provide the submissions as described below. The data reports and network diagrams required for each submission are contained in paragraph SUBMISSION REQUIREMENTS. If the Contractor fails or refuses to furnish the information and schedule updates as set forth herein, the Contractor will be deemed not to have provided an estimate upon which a progress payment can be made.

Review comments made by the Government on the schedule(s) do not relieve the Contractor from compliance with requirements of the Contract Documents.

3.4.1 Preliminary Project Schedule Submission

Within 15 calendar days after the NTP is acknowledged, submit the Preliminary Project Schedule defining the planned operations detailed for the first 90 calendar days for approval. The approved Preliminary Project Schedule will be used for payment purposes not to exceed 90 calendar days after NTP. Completely cost load the Preliminary Project Schedule to balance the contract award CLINS shown on the Price Schedule. The Preliminary Project Schedule may be summary in nature for the remaining performance period. It must be early start and late finish constrained and logically tied as specified. The Preliminary Project Schedule forms the basis for the Initial Project Schedule specified herein and must include all of the required plan and program preparations, submissions and approvals identified in the contract (for example, Quality Control Plan, Safety Plan, and Environmental Protection Plan) as well as design activities, planned submissions of all early design packages, permitting activities, design review conference activities, and other non-construction activities intended to occur within the first 90 calendar days. Government acceptance of the associated design package(s) and all other specified Program and Plan approvals must occur prior to any planned construction activities. Activity code any activities that are summary in nature after the first 90 calendar days with Bid Item (CLIN) code (BIDI), Responsibility Code (RESP) and Feature of Work code (FOW).

3.4.2 Initial Project Schedule Submission

Submit the Initial Project Schedule for approval within 42 calendar days after notice to proceed is issued. The schedule must demonstrate a reasonable and realistic sequence of activities which represent all work through the entire contract performance period. Include in the design-build schedule detailed design and permitting activities, including but not limited to identification of individual design packages, design submission, reviews and conferences; permit submissions and any required Government actions; and long lead item acquisition prior to design completion. Also cover in the initial design-build schedule the entire construction effort with as much detail as is known at the time but, as a minimum, include all construction start and completion milestones, and detailed construction activities through the dry-in, including all activity coding and cost loading. Include the remaining construction, including cost loading, but it may be scheduled summary in nature. As the design proceeds and design packages are developed, fully detail the remaining construction activities concurrent with the monthly schedule updating process. Constrain construction activities by Government acceptance of associated designs. When the design is complete, incorporate into the then approved schedule update all remaining detailed construction activities that are planned to occur after the dry-in milestone. No payment will be made for work items not fully detailed in the Project Schedule.

3.4.2.1 Design Package Schedule Submission

With each design package submitted to the Government, submit a fragment schedule extracted from the then current Preliminary, Initial or Updated schedule which covers the activities associated with that Design Package including construction, procurement and permitting activities.

3.4.3 Periodic Schedule Updates

Update the Project Schedule on a regular basis, monthly at a minimum. Provide a draft Periodic Schedule Update for review at the schedule update meetings as prescribed in the paragraph PERIODIC SCHEDULE UPDATE MEETINGS. These updates will enable the Government to assess Contractor's progress. Update the schedule to include detailed construction activities as the design progresses, but not later than the submission of the final un-reviewed design submission for each separate design package. The Contracting Officer may require submission of detailed schedule activities for any distinct construction that is started prior to submission of a final design submission if such activity is authorized.

- a. Update information including Actual Start Dates (AS), Actual Finish Dates (AF), Remaining Durations (RD), and Percent Complete. Updated information is subject to the approval of the Government at the meeting.
- b. AS and AF dates must match the date(s) reported on the Contractor's Quality Control Report for an activity start or finish.

3.5 SUBMISSION REQUIREMENTS

Submit the following items for the Preliminary Schedule, Initial Schedule, and every Periodic Schedule Update throughout the life of the project:

3.5.1 Electronic Scheduling Data

Provide electronic scheduling data containing the current project schedule and all previously submitted schedules in the format of the scheduling software (e.g. .xer). Also include the Narrative Report and all required Schedule Reports. The electronic file is to identify the type of schedule (Preliminary, Initial, Update), full contract number, Data Date and schedule file name. Each schedule must have a unique file name and use project specific settings. The file naming convention is to be determined at SEKO meeting.

3.5.2 Narrative Report

Provide a Narrative Report with each schedule submission. The Schedule Narrative Report is a "stand alone" report in PDF or Word format. Title the report "Schedule Narrative Report". Show the Report date, schedule name, and contract number on the first page. Indicate who is responsible for the schedule update and narrative report. Provide a brief description of the Project Scope. The Narrative Report is expected to communicate to the Government the thorough analysis of the schedule output and the plans to compensate for problems, either current or potential, which are revealed through that analysis. Include the following information at a minimum in the Narrative Report:

- a. Identify and discuss the work scheduled to start in the next update period.
- b. Describe activities along the critical path in addition to activities on the next float path after the critical path.
- c. Describe current and anticipated problem areas, delaying factors and their impact, and an explanation of corrective actions taken or required to be taken. Identify the party responsible for delay, if applicable.
- d. Identify and explain why activities, based on their calculated late dates, should have either started or finished during the update period but did not.
- e. Identify and discuss all schedule changes by activity ID and activity name including what specifically was changed and why the change was needed. Include at a minimum new and deleted activities, logic changes, duration changes, calendar changes, lag changes, resource changes, and actual start and finish date changes.
- f. Identify and discuss out-of-sequence work.
- g. If the longest path changes from a monthly update to the next, provide an explanation of the factors driving the change.
- h. Respond to USACE Schedule Review Comments from the previous update(s).

3.5.3 Schedule Reports

The format, filtering, organizing, and sorting for each schedule report will be as directed by the Contracting Officer. Typically, reports contain Activity Numbers, Activity Description, Original Duration, Remaining Duration, Early Start Date, Early Finish Date, Late Start Date, Late Finish Date, Total Float, Actual Start Date, Actual Finish Date, and

Percent Complete. Provide the reports electronically in .pdf format. Provide [_____] set(s) of hardcopy reports. The following lists typical reports that will be requested:

3.5.3.1 Activity Report

List of all activities sorted according to activity number.

3.5.3.2 Logic Report

List of detailed predecessor and successor activities for every activity in ascending order by activity number.

3.5.3.3 Total Float Report

A list of all incomplete activities sorted in ascending order of total float. List activities which have the same amount of total float in ascending order of Early Start Dates. Do not show completed activities on this report.

3.5.3.4 Earnings Report by CLIN

A compilation of the Total Earnings on the project from the NTP to the data date, which reflects the earnings of activities based on the agreements made in the schedule update meeting defined herein. Provided a complete schedule update has been furnished, this report serves as the basis of determining progress payments. Group activities by CLIN number and sort by activity number. Provide a total CLIN percent earned value, CLIN percent complete, and project percent complete. The printed report must contain the following for each activity: the Activity Number, Activity Description, Original Budgeted Amount, Earnings to Date, Earnings this period, Total Quantity, Quantity to Date, and Percent Complete (based on cost).

3.5.3.5 Scheduling and Leveling Report

Provide a Scheduling/Leveling Report generated from the current project schedule being submitted.

3.5.4 Network Diagram

The Network Diagram is required for the Preliminary, Initial and Periodic Updates. Depict and display the order and interdependence of activities and the sequence in which the work is to be accomplished. The Contracting Officer will use, but is not limited to, the following conditions to review compliance with this paragraph:

3.5.4.1 Continuous Flow

Show a continuous flow from left to right with no arrows from right to left. Show the activity number, description, duration, and estimated earned value on the diagram.

3.5.4.2 Project Milestone Dates

Show dates on the diagram for start of project, any contract required interim completion dates, and contract completion dates.

3.5.4.3 Critical Path

Show all activities on the critical path. The critical path is defined as the longest path.

3.5.4.4 Grouping and Sorting

Group and sort activities as directed to assist in the understanding of the activity sequence. Typically, this flow will group activities by major elements of work, category of work, work area and/or responsibility.

3.5.4.5 Cash Flow / Schedule Variance Control (SVC) Diagram

With each schedule submission, provide a SVC diagram showing 1) Cash Flow S-Curves indicating planned project cost based on projected early and late activity finish dates, and 2) Earned Value to-date.

3.6 PERIODIC SCHEDULE UPDATE

3.6.1 Periodic Schedule Update Meetings

Conduct periodic schedule update meetings for the purpose of reviewing the proposed Periodic Schedule Update, Narrative Report, Schedule Reports, and progress payment. Conduct meetings at least monthly within five days of the proposed schedule data date. Provide a computer with the scheduling software loaded and a projector which allows all meeting participants to view the proposed schedule during the meeting. The Contractor's authorized scheduler must organize, group, sort, filter, perform schedule revisions as needed and review functions as requested by the Contractor and/or Government. The Contractor scheduler must attend in person unless virtual attendance is approved by Contracting Officer. The meeting is a working interactive exchange which allows the Government and Contractor the opportunity to review the updated schedule on a real time and interactive basis. The meeting will last no longer than 8 hours. Submit schedule file and narrative to the Government a minimum of two workdays in advance of the meeting. The Contractor's Project Manager must attend the meeting with the authorized representative of the Contracting Officer. Superintendents, foremen and major subcontractors must attend the meeting as required to discuss the project schedule and work. Following the schedule update meeting, make corrections to the schedule and resubmit, following the required schedule naming convention. Include only those changes approved by the Government in the schedule review meeting. Only approved schedules may be used to generate invoices for payment.

3.6.2 Update Submission Following Progress Meeting

Submit the complete Periodic Schedule Update of the Project Schedule containing all approved progress, revisions, and adjustments, pursuant to paragraph SUBMISSION REQUIREMENTS not later than 4 workdays after the periodic schedule update meeting.

3.7 WEEKLY PROGRESS MEETINGS

Conduct a weekly meeting with the Government (or as otherwise mutually agreed to) between the meetings described in paragraph entitled PERIODIC SCHEDULE UPDATE MEETINGS for the purpose of jointly reviewing the actual progress of the project as compared to the as planned progress and to review planned activities for the upcoming two weeks. Use the current approved schedule update for the purposes of this meeting and for the

production and review of reports. At the weekly progress meeting, address the status of RFIs, RFPs and Submittals.

3.8 REQUESTS FOR TIME EXTENSIONS

ASCE 67-17 provides delay analysis guidelines. If there is a conflict between the contract and guidance provided in ASCE 67-17, the contract will govern. Provide justification of delay to the Contracting Officer in accordance with the contract provisions and clauses for approval, within 10 days of a delay occurring. Also prepare a time impact analysis for each Government request for proposal (RFP).

3.8.1 Justification of Delay

Provide a description of the event(s) that caused the delay and/or impact to the work. As part of the description, identify all schedule activities impacted. Show that the event that caused the delay/impact was the responsibility of the Government. Provide a time impact analysis that demonstrates the effects of the delay or impact on the project completion date, or interim completion date(s). Evaluate multiple impacts chronologically; each with its own justification of delay. With multiple impacts consider any concurrency of delay. A time extension and the schedule fragnet becomes part of the project schedule and all future schedule updates upon approval by the Contracting Officer.

3.8.2 Changes That Do Not Cause Delay

If it is determined that a change does not constitute a schedule time extension, a fragnet as described in paragraph FRAGMENTARY NETWORK, is to be created and inserted into the Project schedule.

3.8.3 Time Impact Analysis (Prospective Analysis)

Prepare a time impact analysis for approval by the Contracting Officer based on industry standard AACE 52R-06. Utilize a copy of the last approved schedule prior to the first day of the impact or delay for the time impact analysis. If Contracting Officer determines the time frame between the last approved schedule and the first day of impact is too great, prepare an interim updated schedule to perform the time impact analysis. Unless approved by the Contracting Officer, no other changes may be incorporated into the schedule being used to justify the time impact.

3.8.4 Forensic Schedule Analysis (Retrospective Analysis)

Prepare an analysis for approval by the Contracting Officer based on methodologies identified in industry standard AACE 29R-03. Identify which methodology was chosen and explain why it is more appropriate than other methodologies available.

3.8.5 Fragmentary Network (Fragnet)

Prepare a proposed fragnet for time impact analysis consisting of a sequence of new activities that are proposed to be added to the project schedule to demonstrate the influence of the delay or impact to the project's contractual dates. Clearly show how the proposed fragnet is to be tied into the project schedule including all predecessors and successors to the fragnet activities. The proposed fragnet must be approved by the Contracting Officer prior to incorporation into the

project schedule.

3.8.6 Time Extension

The Contracting Officer must approve the Justification of Delay including the time impact analysis before a time extension will be granted. No time extension will be granted unless the delay consumes all available Project Float and extends the projected finish date ("End Project" milestone) beyond the Contract Completion Date. The time extension will be in calendar days.

Actual delays that are found to be caused by the Contractor's own actions, which result in a calculated schedule delay, are not a cause for an extension to the performance period, completion date, or any interim milestone date.

3.8.7 Impact to Early Completion Schedule

No extended overhead will be paid for delay prior to the original Contract Completion Date for an Early Completion IPS unless the Contractor actually performed work in accordance with that Early Completion Schedule. The Contractor must show that an early completion was achievable had it not been for the impact.

3.9 FAILURE TO ACHIEVE PROGRESS

Should the progress fall behind the approved project schedule for reasons other than those that are excusable within the terms of the contract, the Contracting Officer may require provision of a written recovery plan for approval. The plan must detail how progress will be made-up to include which activities will be accelerated by adding additional crews, longer work hours, extra workdays, etc.

3.9.1 Artificially Improving Progress

Artificially improving progress by means such as, but not limited to, revising the schedule logic, modifying, or adding constraints, shortening activity durations, or changing calendars in the project schedule is prohibited. Indicate assumptions made and the basis for any logic, constraint, duration, and calendar changes used in the creation of the recovery plan. Any additional resources, manpower, or daily and weekly work hour changes proposed in the recovery plan must be evident at the work site and documented in the daily report along with the Schedule Narrative Report.

3.9.2 Failure to Perform

Failure to perform work and maintain progress in accordance with the supplemental recovery plan may result in an interim and final unsatisfactory performance rating and may result in corrective action directed by the Contracting Officer pursuant to FAR 52.236-15 Schedules for Construction Contracts, FAR 52.249-10 Default (Fixed-Price Construction), and other contract provisions.

3.9.3 Recovery Schedule

Should the Contracting Officer find it necessary, submit a recovery schedule pursuant to FAR 52.236-15 Schedules for Construction Contracts.

3.10 OWNERSHIP OF FLOAT

Except for the provision given in the paragraph IMPACT TO EARLY COMPLETION SCHEDULE, float available in the schedule, at any time, belongs to the Project and is available for Contractor and Government use. This includes activity and project float. Activity float is the number of workdays that an activity can be delayed without causing a delay to the "End Project" finish milestone. Project float (if applicable) is the number of work days between the projected early finish and the "End Project" finish.

3.11 TRANSFER OF SCHEDULE DATA INTO RESIDENT MANAGEMENT SYSTEM

Ensure schedule data is uploaded to RMS. This data is additional supporting data in a form and detail required by the Contracting Officer pursuant to FAR 52.232-5 Payments under Fixed-Price Construction Contracts. The receipt of a proper payment request pursuant to FAR 52.232-27 Prompt Payment for Construction Contracts is contingent upon the Government receiving both acceptable and approvable hard copies and matching electronic versions of the application for progress payment.

3.12 PRIMAVERA P6 MANDATORY REQUIREMENTS

Ensure Primavera P6 settings provide a schedule capable of fulfilling the requirements of the contract. The following settings are mandatory and required in all schedule submissions to the Government:

- a. Activity Codes must be Project Level, not Global or EPS level.
- b. Calendars must be Project Level, not Global or Resource level.
- c. Activity Duration Types must be set to "Fixed Duration & Units".
- d. Percent Complete Types must be set to "Physical".
- e. Time Period Admin Preferences must remain the default "8.0 hr/day, 40 hr/week, 172 hr/month, 2000 hr/year". Set Calendar Work Hours/Day to 8.0 Hour days.
- f. Set Schedule Option for defining Critical Activities to "Longest Path".
- g. Set Schedule Option for defining progressed activities to "Retained Logic".
- h. Set up cost loading using a single lump sum non-labor resource. The Price/Unit must be \$1/hr, Default Units/Time must be "8h/d", and settings "Auto Compute Actuals" and "Calculate costs from units" selected.
- i. Activity ID's must not exceed 10 characters.
- j. Activity Names must have a verb-noun structure and contain the most defining and detailed description within the first 30 characters.
- k. The daily ending hour for all work calendars must be the same.

-- End of Section --

SECTION 01 33 00

SUBMITTAL PROCEDURES
08/18, CHG 4: 02/21

PART 1 GENERAL

1.1 SUMMARY

1.1.1 Submittal Information

The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections. Each submittal is to be complete and in sufficient detail to allow ready determination of compliance with contract requirements.

Units of weights and measures used on all submittals are to be the same as those used in the contract drawings.

1.1.2 Project Type

The Contractor's Quality Control (CQC) System Manager are to check and approve all items before submittal and stamp, sign, and date indicating action taken. Proposed deviations from the contract requirements are to be clearly identified. Include within submittals items such as: Contractor's, manufacturer's, or fabricator's drawings; descriptive literature including (but not limited to) catalog cuts, diagrams, operating charts or curves; test reports; test cylinders; samples; O&M manuals (including parts list); certifications; warranties; and other such required submittals.

The Contractor and the Designer of Record (DOR), if applicable, are to check and approve all items before submittal and stamp, sign, and date indicating action taken. Proposed deviations from the contract requirements are to be clearly identified. Include within submittals items such as: Contractor's, manufacturer's, or fabricator's drawings; descriptive literature including (but not limited to) catalog cuts, diagrams, operating charts or curves; test reports; test cylinders; samples; O&M manuals (including parts list); certifications; warranties; and other such required submittals.

1.1.3 Submission of Submittals

Schedule and provide submittals requiring Government approval before acquiring the material or equipment covered thereby. Pick up and dispose of samples not incorporated into the work in accordance with manufacturer's Safety Data Sheets (SDS) and in compliance with existing laws and regulations.

1.2 DEFINITIONS

1.2.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Submittals that are required prior to or commencing with the start of work on site. Submittals that are required prior to or at the start of construction (work) or the next major phase of the construction on a multiphase contract.

[For Government approved division 01 preconstruction submittals that are required prior to or commencing with the start of work shall be submitted within 30 calendar days of contract award unless specified elsewhere in the specifications. For contractor approved division 01 submittals that are required prior to or commencing with the start of work shall be submitted within 45 calendar days of contract award unless specified elsewhere in the specifications.]

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

Certificates Of Insurance

Surety Bonds

List Of Proposed Subcontractors

List Of Proposed Products

Baseline Network Analysis Schedule (NAS)

Submittal Register

Schedule Of Prices Or Earned Value Report

Accident Prevention Plan Health And Safety Plan

Work Plan

Quality Control (QC) plan

Environmental Protection Plan [Explosive Safety Submission ESS Work Plan]

SD-02

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-04 Samples

Fabricated or unfabricated physical examples of materials, equipment or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged.

Color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project.

Field samples and mock-ups constructed on the project site establish standards ensuring work can be judged. Includes assemblies or portions of assemblies that are to be incorporated into the project and those that will be removed at conclusion of the work.

SD-05 Design Data

Design calculations, mix designs, analyses or other data pertaining to a part of work.

Design submittals, design substantiation submittals and extensions of design submittals.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. Unless specified in another section, testing must have been within three years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily logs and checklists

Final acceptance test and operational test procedure

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that the product, system, or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits

Text of posted operating instructions

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system or material, including special notices and (SDS) concerning impedances, hazards and safety precautions.

SD-09 Manufacturer's Field Reports

Documentation of the testing and verification actions taken by manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or after installation, to confirm compliance with manufacturer's standards or instructions. The documentation must be signed by an authorized official of a testing laboratory or agency and state the test results; and indicate whether the material, product, or system has passed or failed the test.

Factory test reports.

SD-10 Operation and Maintenance Data

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel.

Data required by operating and maintenance personnel for the safe and efficient operation, maintenance and repair of the item.

Data incorporated in an operations and maintenance manual or control system.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principle Validation (GPV) or Third Party Certification (TPC).

Special requirements necessary to properly close out a construction contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase contract.

1.2.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.2.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are [for Contractor Quality Control approval.][for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.] Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Submittal Register; G

1.4 SUBMITTAL CLASSIFICATION

1.4.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, variations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government.

Government approval is required for any variations from the Solicitation or the Accepted Proposal and for other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

1.4.2 Design-Build Submittal Classifications

1.4.2.1 Designer of Record Approved (DA)

Designer of Record (DOR) approval is required for extensions of design; critical materials; any variations from the Solicitation, the Accepted Proposal, or the completed design; equipment whose compatibility with the entire system must be checked; and other items as designated by the Contracting Officer. Provide the Government with the number of copies designated hereinafter of all DOR approved submittals. The Government may review any or all Designer of Record approved submittals for conformance

with the Solicitation, the Accepted Proposal, and the completed design. The Government will review all submittals designated as varying from the Solicitation or Accepted Proposal, as described below. Provide design submittals in accordance with Section 01 33 16 DESIGN AFTER AWARD. Generally, list design submittals under SD-05 Design Data.

1.4.2.2 Government Conformance Review of Design (CR)

The Government will review all intermediate and final design submittals for conformance with the technical requirements of the Solicitation. Section 01 33 16 DESIGN AFTER AWARD covers the design submittal and review process in detail. Review will be only for conformance with the applicable codes, standards, and contract requirements. Design data includes the design documents described in Section 01 33 16 DESIGN AFTER AWARD, DESIGN DATA.

1.4.2.3 Designer of Record Approved/Government Conformance Review (DA/CR)

1.4.2.3.1 Variations from the Accepted Design

DOR approval and the Government's concurrence are required for any proposed variation from the accepted design that still complies with the contract before the Contractor is authorized to proceed with material acquisition or installation. If necessary to facilitate the project schedule, before official submission to the Government, the Contractor and the DOR may discuss with the Contracting Officer's Representative a submittal proposing a variation. However, the Government reserves the right to review the submittal before providing an opinion. In any case, the Government will not formally agree to or provide a preliminary opinion on any variation without the DOR's approval or recommended approval. The Government reserves the right to reject any design, variation that may affect furniture, furnishings, equipment selections, or operational decisions that were made, based on the reviewed and concurred design.

1.4.2.3.2 Substitutions

Unless prohibited or otherwise provided for elsewhere in the contract, where the Accepted Proposal named products, systems, materials or equipment by manufacturer, brand name, model number, or other specific identification, and the Contractor desires to substitute a manufacturer or model after award, submit a requested substitution for Government concurrence. Include substantiation, through identifying information and the DOR's approval, that the substitute meets the contract requirements and that it is equal in function, performance, quality, and salient features to that in the accepted contract proposal. If the contract otherwise prohibits substitutions of equal named products, systems, materials or equipment by manufacturer, brand name, model number or other specific identification, the request is considered a "variation" to the contract. Variations are discussed below in paragraphs: "DESIGNER OF RECORD APPROVED/GOVERNMENT APPROVED" and VARIATIONS.

1.4.2.4 Designer of Record Approved/Government Approved (DA/GA)

In addition to the above-stated requirements for proposed variations to the accepted design, both DOR and Government Approval and, where applicable, a contract modification are required before the Contractor is authorized to proceed with material acquisition or installation for any proposed variation to the contract (the Solicitation or the Accepted Proposal), that constitutes a change to the contract terms. The Government

reserves the right to accept or reject any such proposed variation.

1.4.3 For Information Only

Submittals not requiring Government approval will be for information only. For Design-build construction all submittals not requiring DOR or Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

1.4.4 Sustainability Reporting Submittals (S)

Submittals for Guiding Principle Validation (GPV) or Third Party Certification (TPC) are indicated with an "S" designation. These submittals are for information only and for use as specified in Section 01 33 29 SUSTAINABILITY REPORTING.

Schedule submittals for these items throughout the course of construction as provided; do not wait until closeout.

1.5 PREPARATION

1.5.1 Transmittal Form

Transmit each submittal, except sample installations and sample panels to the office of the approving authority using the transmittal form prescribed by the Contracting Officer. Include all information prescribed by the transmittal form and required in paragraph IDENTIFYING SUBMITTALS. Use the submittal transmittal forms to record actions regarding samples.

Use the ENG Form 4025-R transmittal form for submitting both Government-approved and information-only submittals. Submit in accordance with the instructions on the reverse side of the form. These forms [will be furnished to the Contractor][are included in the RMS CM software that the Contractor is required to use for this contract]. Properly complete this form by filling out all the heading blank spaces and identifying each item submitted. Exercise special care to ensure proper listing of the specification paragraph and sheet number of the contract drawings pertinent to the data submitted for each item.

1.5.2 Identifying Submittals

The Contractor's [Quality Control Manager] [approving authority] must prepare, review and stamp submittals, including those provided by a subcontractor, before submittal to the Government.

Identify submittals, except sample installations and sample panels, with the following information permanently adhered to or noted on each separate component of each submittal and noted on transmittal form. Mark each copy of each submittal identically, with the following:

- a. Project title and location
- b. Construction contract number
- c. Dates of the drawings and revisions
- d. Name, address, and telephone number of Subcontractor, supplier,

manufacturer, and any other Subcontractor associated with the submittal.

- e. Section number of the specification by which submittal is required
- f. Submittal description (SD) number of each component of submittal
- g. For a resubmission, add alphabetic suffix on submittal description, for example, submittal 18 would become 18A, to indicate resubmission
- h. Product identification and location in project.

1.5.3 Submittal Format

1.5.3.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.3.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 210 by 297 mm nor more than 1189 by 841 mm, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

- a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.
- b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

Present 210 by 297 mm shop drawings sized as part of the bound volume for submittals. Present larger drawings in sets. Submit an electronic copy of drawings in PDF format.

1.5.3.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than [_____] millimeter on the right-hand side of each sheet for the Government disposition stamp.

1.5.3.3 Format of SD-03 Product Data

Present product data submittals for each section as a complete, bound volume. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

1.5.3.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in metric dimensions. Where product data are included in preprinted catalogs with English units only, submit metric dimensions on separate sheet.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

1.5.3.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.3.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will [not] be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.5.3.4 Format of SD-04 Samples

1.5.3.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

- a. Sample of Equipment or Device: Full size.
- b. Sample of Materials Less Than 50 by 75 mm: Built up to 210 by 297 mm.
- c. Sample of Materials Exceeding 210 by 297 mm: Cut down to 210 by 297 mm and adequate to indicate color, texture, and material variations.
- d. Sample of Linear Devices or Materials: 250 mm length or length to be supplied, if less than 250 mm. Examples of linear devices or materials are conduit and handrails.
- e. Sample Volume of Nonsolid Materials: 750 ml. Examples of nonsolid materials are sand and paint.
- f. Color Selection Samples: 50 by 100 mm. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.
- g. Sample Panel: 1200 by 1200 mm.
- h. Sample Installation: 10 square meters.

1.5.3.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

1.5.3.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.5.3.5 Format of SD-05 Design Data

Provide design data and certificates on 210 by 297 mm paper. Provide a bound volume for submittals containing numerous pages.

1.5.3.6 Format of SD-06 Test Reports

Provide reports on 210 by 297 mm paper in a complete bound volume.

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.3.7 Format of SD-07 Certificates

Provide design data and certificates on 210 by 297 mm paper. Provide a bound volume for submittals containing numerous pages.

1.5.3.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section as a complete, bound volume. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.5.3.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.3.9 Format of SD-09 Manufacturer's Field Reports

Provide reports on 210 by 297 mm paper in a complete bound volume.

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.3.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format.

1.5.3.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

Provide all dimensions in administrative submittals in metric. Where data are included in preprinted material with English units only, submit metric dimensions on separate sheet.

1.5.4 Source Drawings for Shop Drawings

1.5.4.1 Source Drawings

The entire set of source drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

1.5.4.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

1.5.5 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. [In addition to the electronic submittal, provide [three] [_____] hard copies of the submittals.] Compile the submittal file as a single, complete document, to include the Transmittal Form described within. Name the electronic submittal file specifically according to its contents, and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of a signature.

E-mail electronic submittal documents smaller than 10MB to an e-mail address as directed by the Contracting Officer. Provide electronic documents over 10 MB on an optical disc or through an electronic file sharing system such as the AMRDEC SAFE Web Application located at the following website: <https://safe.amrdec.army.mil/safe/>.

1.6 QUANTITY OF SUBMITTALS

1.6.1 Number of SD-01 Preconstruction Submittal Copies

Unless otherwise specified, submit [two][three] sets of administrative submittals.

1.6.2 Number of SD-02 Shop Drawing Copies

Submit [six][_____] copies of submittals of shop drawings requiring review and approval by a QC organization. Submit [seven][_____] copies of shop drawings requiring review and approval by the Contracting Officer.

1.6.3 Number of SD-03 Product Data Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.4 Number of SD-04 Samples

- a. Submit [two] [_____] samples, or [two] [_____] sets of samples showing the range of variation, of each required item. One approved sample or set of samples will be retained by the approving authority and one will be returned to the Contractor.
- b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.
- c. Submit one sample installation, where directed.
- d. Submit one sample of nonsolid materials.

1.6.5 Number of SD-05 Design Data Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.6 Number of SD-06 Test Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings, other than field test results that will be submitted with QC reports.

1.6.7 Number of SD-07 Certificate Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.8 Number of SD-08 Manufacturer's Instructions Copies

Submit in compliance with quantity requirements specified for shop

drawings.

1.6.9 Number of SD-09 Manufacturer's Field Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings other than field test results that will be submitted with QC reports.

1.6.10 Number of SD-10 Operation and Maintenance Data Copies

Submit [five][three][_____] copies of O&M data to the Contracting Officer for review and approval.

1.6.11 Number of SD-11 Closeout Submittals Copies

Unless otherwise specified, submit [two][three] sets of administrative submittals.

1.7 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe. For Design-Build construction, the Government will retain [_____] copies of information-only submittals.

1.8 PROJECT SUBMITTAL REGISTER AND DATABASE

A sample Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as "Appendix A - Submittal Register."

1.8.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. Use an electronic submittal register program furnished by the Government. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required. Maintain a submittal register for the project in accordance with Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE(RMS CM).[The Government will provide the initial submittal register][in electronic format][with the following fields completed, to the extent that will be required by the Government during subsequent usage.]

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

The database and submittal management program will be furnished to the Contractor on a writable compact disk (CD-R), for operation on a Windows-based personal computer.

Thereafter, the Contractor is to track all submittals by maintaining a complete list, including completion of all data columns and all dates on which submittals are received by and returned by the Government.

1.8.2 Design-Build Submittal Register

The Designer of Record develops a complete list of submittals during design and identify required submittals in the specifications, and use the list to prepare the Submittal Register. The list may not be all inclusive and additional submittals may be required by other parts of the contract. Complete the submittal register and submit it to the Contracting Officer for approval within 30 calendar days after Notice to Proceed. The approved submittal register will serve as a scheduling document for submittals and will be used to control submittal actions throughout the contract period. Coordinate the submit dates and need dates with dates in the Contractor prepared progress schedule. Submit monthly or until all submittals have been satisfactorily completed, updates to the submittal register showing the Contractor action codes and actual dates with Government action codes. Revise the submittal register when the progress schedule is revised and submit both for approval.

1.8.3 Preconstruction Use of Submittal Register

Submit the submittal register as an electronic database, using the submittal management program furnished to Contractor. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register database submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs

material delivered to Contractor control.

1.8.4 Contractor Use of Submittal Register

Update the following fields in the Government-furnished submittal register program or equivalent fields in the program used by the Contractor with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.8.5 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.8.6 Action Codes

Entries for columns (j) and (o) are to be used as follows (others may be prescribed by the Transmittal Form):

1.8.6.1 Government Review Action Codes

"A" - "Approved as submitted"; "Completed"

"B" - "Approved, except as noted on drawings"; "Completed"

"C" - "Approved, except as noted on drawings; resubmission required"; "Resubmit"

"D" - "Returned by separate correspondence"; "Completed"

"E" - "Disapproved (See attached)"; "Resubmit"

"F" - "Receipt acknowledged"; "Completed"

"G" - "Other (Specify)"; "Resubmit"

"X" - "Receipt acknowledged, does not comply with contract requirements"; "Resubmit"

1.8.6.2 Contractor Action Codes

DESIGN BID BUILD SUBMITTALS			
Submittal Classifications shown in UFGS Sections	Submittal Classification	Corresponding SpecsIntact Submittal Register Code which is populated in the SI Submittal Register. Software Limitations: (The software shows one character delineation in the SpecsIntact Submittal Register)	RMS - The following Submittal Classifications are populated in RMS when the SpecsIntact Submittal Data File is pulled into RMS)
G	Submittal requires Government Approval	G	GA
BLANK	Submittal is For Information Only (FIO)	BLANK	FIO
S	Submittal is for documentation of Sustainable requirements	S	S/FIO

1.8.6.3 Contractor Action Codes

DESIGN BUILD SUBMITTALS			
Submittal Classifications shown in UFGS Sections	Submittal Classification	Corresponding SpecsIntact Submittal Register Code which is populated in the SI Submittal Register. Software Limitations: (The software shows one character delineation in the SpecsIntact Submittal Register)	RMS - The following Submittal Classifications are populated in RMS when the SpecsIntact Submittal Data File is pulled into RMS)
G	Submittal requires Government Approval	G	GA

DESIGN BUILD SUBMITTALS			
BLANK	Submittal is For Information Only(FIO)	BLANK	FIO
DA	Submittal requires Designer of Record Approval	D	DA
CR	Submittal requires Government Conformance Review	C	CR
DA/CR	Submittal requires Designer of Record Approval and Government Conformance Review	R	DA/CR
DA/GA	Submittal requires Designer of Record Approval and Government Approval	A	DA/GA

1.8.7 Delivery of Copies

Submit an updated electronic copy of the submittal register to the Contracting Officer with each invoice request, unless a paper copy is requested by the Contracting Officer. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

1.9 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

1.9.1 Considering Variations

Discussion of variations with the Contracting Officer before submission [of a variation submittal] will help ensure that functional and quality requirements are met and minimize rejections and resubmittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the contractor.

Specifically point out variations from contract requirements in a [transmittal letter][variation submittal]. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.9.2 Proposing Variations

[When proposing variation, deliver a submittal, clearly marked as a "VARIATION" to the Contracting Officer, with documentation illustrating the nature and features of the variation including any necessary technical submittals and why the variation is desirable and beneficial to Government. If lower cost is a benefit, also include an estimate of the cost savings. In addition to documentation required for variation, include the submittals required for the item. Clearly mark the proposed variation in all documentation.]

[The Contracting Officer will indicate an approval or disapproval of the variation request; and if not approved as submitted, will indicate the Government's reasons therefore. Any work done before such approval is received is performed at the Contractor's risk.]"

Specifically point out variations from contract requirements in a [transmittal letter][variation submittal]. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

Check the column "variation" of ENG Form 4025 for submittals that include variations proposed by the Contractor. Set forth in writing the reason for any variations and note such variations on the submittal. The Government reserves the right to rescind inadvertent approval of submittals containing unnoted variations.

1.9.3 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.9.4 Review Schedule Extension

In addition to the normal submittal review period, a period of [14] [____] calendar days will be allowed for the Government to consider submittals with variations.

1.10 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals. [Allow an additional [____] calendar days for review and approval of submittals for [food service equipment] [and] [refrigeration and HVAC control systems]].

- a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as

for the initial submittal.

- b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."
- c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.

Contracting Officer review will be completed within [____] calendar days after the date of submission.

[Within [30][15] calendar days of Notice To Proceed][At the Preconstruction conference], provide the following schedule of submittals for approval by the Contracting Officer:

- d. A schedule of shop drawings and technical submittals required by the specifications and drawings. Indicate the specification or drawing reference requiring the submittal; the material, item, or process for which the submittal is required; the "SD" number and identifying title of the submittal; the anticipated submission date, and the approval need date.
- e. A separate schedule of other submittals required under the contract but not listed in the specifications or drawings. Indicate the contract requirement reference, the type or title of the submittal, the anticipated submission date, and the approval need date (if approval is required).

1.10.1 Government Reviewed Design

The Government will review design submittals for conformance with the technical requirements of the Solicitation. Section 01 33 16.00 10 DESIGN DATA (DESIGN AFTER AWARD) covers the design submittal and review process in detail. Government review is required for variations from the completed design. Review will be only for conformance with the contract requirements. Included are only those construction submittals for which the DOR's design documents do not include enough detail to ascertain contract compliance. The Government may, but is not required to, review extensions of design such as structural steel or reinforcement shop drawings.

1.11 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

- a. Note the date on which the submittal was received.
- b. Review submittals for approval within the scheduling period specified and only for conformance with project design concepts and compliance with contract documents.

- c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. [_____] copies of the submittal will be retained by the Contracting Officer and [_____] copies of the submittal will be returned to the Contractor. If the Government performs a conformance review of other Designer of Record approved submittals, the submittals will be identified and returned, as described above.

1.11.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

- a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.
- b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.
- c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.
- d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.
- e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.12 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

1.13 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as

a complete check, and indicates only that the general method of construction, materials, detailing, and other information are satisfactory. the design, general method of construction, materials, detailing, and other information appear to meet the Solicitation and Accepted Proposal.

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained within each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.14 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not to be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements.

1.15 WITHHOLDING OF PAYMENT

Payment for materials incorporated in the work will not be made if required approvals have not been obtained. No payment for materials incorporated in the work will be made unless all required DOR approvals or required Government approvals have been obtained. No payment will be made for any materials incorporated into the work for any conformance review submittals or information-only submittals found to contain errors or deviations from the Solicitation or Accepted Proposal.

1.16 CERTIFICATION OF SUBMITTAL DATA

Certify the submittal data as follows on Form ENG 4025: "I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except

as otherwise stated.

____NAME OF CONTRACTOR _____ SIGNATURE OF CONTRACTOR

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

DRAFT

U.S. Army Corps of Engineers (USACE) TRANSMITTAL OF SHOP DRAWINGS, EQUIPMENT DATA, MATERIAL SAMPLES, OR MANUFACTURER'S CERTIFICATES OF COMPLIANCE For use of this form, see ER 415-1-10; the proponent agency is CECW-CE.				DATE		TRANSMITTAL NO.		
SECTION I - REQUEST FOR APPROVAL OF THE FOLLOWING ITEMS (This section will be initiated by the contractor)								
TO:		FROM:		CONTRACT NO.		CHECK ONE: ☐ THIS IS A NEW TRANSMITTAL ☐ THIS IS A RESUBMITTAL OF TRANSMITTAL _____		
SPECIFICATION SEC. NO. (Cover only one section with each transmittal)		PROJECT TITLE AND LOCATION		THIS TRANSMITTAL IS FOR: (Check one) ☐ FIO ☐ GA ☐ DA ☐ CR ☐ DA/CR ☐ DA/GA				
ITEM NO. (See Note 3)	DESCRIPTION OF SUBMITTAL ITEM (Type size, model number/etc.)	SUBMITTAL TYPE CODE (See Note 8)	NO. OF COPIES	CONTRACT DOCUMENT REFERENCE		CONTRACTOR REVIEW CODE	VARIATION Enter "y" if requesting a variation (See Note 6)	USACE ACTION CODE (Note 9)
				SPEC. PARA. NO.	DRAWING SHEET NO.			
a.	b.	c.	d.	e.	f.	g.	h.	i.
REMARKS		I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.						
ENCLOSURES RETURNED (List by item No.)				NAME OF CONTRACTOR		SIGNATURE OF CONTRACTOR		
NAME AND TITLE OF APPROVING AUTHORITY				SIGNATURE OF APPROVING AUTHORITY		DATE		
SECTION II - APPROVAL ACTION								
PREVIOUS EDITIONS ARE OBSOLETE.								

INSTRUCTIONS		
1. Section I will be initiated by the Contractor in the required number of copies.		
2. Each Transmittal shall be numbered consecutively. The Transmittal Number typically includes two parts separated by a dash (-). The first part is the specification section number. The second part is a sequential number for the submittals under that spec section. If the Transmittal is a resubmittal, then add a decimal point to the end of the original Transmittal Number and begin numbering the resubmittal packages sequentially after the decimal.		
3. The "Item No." for each entry on this form will be the same "Item No." as indicated on ENG FORM 4288-R.		
4. Submittals requiring expeditious handling will be submitted on a separate ENG Form 4025-R.		
5. Items transmitted on each transmittal form will be from the same specification section. Do not combine submittal information from different specification sections in a single transmittal.		
6. If the data submitted are intentionally in variance with the contract requirements, indicate a variation in column h, and enter a statement in the Remarks block describing the detailed reason for the variation.		
7. ENG Form 4025-R is self-transmitting - a letter of transmittal is not required.		
8. When submittal items are transmitted, indicate the "Submittal Type" (SD-01 through SD-11) in column c of Section I. Submittal types are the following:		
SD-01 - Preconstruction	SD-02 - Shop Drawings	SD-03 - Product Data
SD-07 - Certificates	SD-08 - Manufacturer's Instructions	SD-09 - Manufacturer's Field Reports
		SD-05 - Design Data
		SD-10 - O&M Data
		SD-06 - Test Reports
		SD-11 - Closeout
9. For each submittal item, the Contractor will assign Submittal Action Codes in column g of Section I. The U.S. Army Corps of Engineers approving authority will assign Submittal Action Codes in column i of Section I. The Submittal Action Codes are:		
A -- Approved as submitted.	F -- Receipt acknowledged.	
B -- Approved, except as noted on drawings. Resubmission not required.	X -- Receipt acknowledged, does not comply with contract requirements, as noted.	
C -- Approved, except as noted on drawings. Refer to attached comments. Resubmission required.	G -- Other action required (Specify)	
D -- Will be returned by separate correspondence.	K -- Government concurs with intermediate design. (For D-B contracts)	
E -- Disapproved. Refer to attached comments.	R -- Design submittal is acceptable for release for construction. (For D-B contracts)	
10. Approval of items does not relieve the contractor from complying with all the requirements of the contract.		

SUBMITTAL REGISTER

CONTRACT NO.
W912HV-19-D-0010

TITLE AND LOCATION
PN77833 VEHICLE MAINTENANCE SHOP

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 11 00.00 10	SD-01 Preconstruction Submittals														
			Initial Site Conditions Photos	1.11													
			Key Personnel Qualifications	1.7	G												
			Conformed Drawings Submittal	1.10.2													
			INDOOR AIR QUALITY (IAQ)	1.10.3	G												
			MANAGEMENT PLAN														
			SD-07 Certificates														
			Monthly Progress Photos	1.11													
			SD-11 Closeout Submittals														
			Construction Completion Photos	1.11													
		01 20 00.00 10	SD-03 Product Data														
			Weight Certificates	1.6.2.2													
		01 32 01.00 10	SD-01 Preconstruction Submittals														
			Project Scheduler Qualifications	1.3	G												
			Preliminary Project Schedule	3.4.1	G												
			Initial Project Schedule	3.4.2	G												
			Periodic Schedule Update	3.6.2	G												
		01 33 00	SD-01 Preconstruction Submittals														
			Submittal Register	1.8	G												
		01 35 13	SD-01 Preconstruction Submittals														
			Heavy Equipment and Vehicle	3.1.4													
			List														
			Existing Conditions Survey	3.1.3.2													
			FAA Form 7460-1	3.1.2													
			FAA Form 7460-2	3.1.2													
			Construction Operations Plan	3.1.3.1													

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 13	Watercraft List	3.2													
			SD-04 Samples														
			Model Unit	3.3.3													
		01 35 26	SD-01 Preconstruction Submittals														
			Accident Prevention Plan (APP)	1.8	G												
			Dive Operations Plan	1.17	G												
			Final IAQ Management Plan	1.16	S												
			Indoor Air Quality (IAQ)	1.16	G												
			Management Plan														
			SD-06 Test Reports														
			Accident Reports	1.12.2	G												
			LHE Inspection Reports	1.12.3													
			Monthly Exposure Reports	1.5	G												
			SD-07 Certificates														
			Crane Operators/Riggers	1.7.7													
			Activity Hazard Analysis (AHA)	1.9	G												
			Certificate of Compliance	1.12.4													
			Hot Work Permit	1.13													
			Standard Lift Plan	1.12.4	G												
			Certifications	3.10.2													
			Licenses	3.10.2													
		01 45 00	SD-01 Preconstruction Submittals														
			Contractor Quality Control (CQC)	1.5.2	G												
			Plan														
			Additional Requirements For	1.5.2.2	G												
			Design Quality Control (DQC) Plan														

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 45 00	SD-05 Design Data														
			Design Quality Control	1.12.2													
			Discipline-Specific Checklists	1.5.2.2													
			SD-06 Test Reports														
			Verification Statement	1.12.3													
			SD-07 Certificates														
			Certificate Of Readiness	1.6.7.1	G												
		01 45 35	SD-01 Preconstruction Submittals														
			SIOR Letter of Acceptance	3.1.1	G												
			Project Manual	3.1.1	G												
			Written Practices	3.1.1													
			NDT Procedures and Equipment	3.1.1													
			Calibration Records														
			SD-06 Test Reports														
			Daily Reports	3.1.1													
			Biweekly Reports	3.1.1													
			SD-07 Certificates														
			Fabrication Plant	2.1													
			Certificate of Compliance	2.1													
			Special Inspector of Record	1.5.18	G												
			Special Inspector	1.5	G												
			Qualification Records	3.1.1													
			SD-11 Closeout Submittals														
			Comprehensive Final Report	3.1.3	G												
		01 50 00	SD-01 Preconstruction Submittals														
			Construction Site Plan	1.3	G												

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 50 00	Traffic Control Plan	3.4.1	G												
			Temporary Utility Connections Plan	3.3.2	G												
		01 57 19	SD-01 Preconstruction Submittals														
			Environmental Protection Plan	1.9	G												
			Environmental Manager Qualifications	1.8.3	G												
			Qualifications	1.8.3													
			Training Program	1.6.1.2													
			Written Assessment Of Friable Asbestos Disturbance	3.11													
			SD-06 Test Reports														
			Solid Waste Management Report	3.8.2.2	G												
			Nonhazardous Solid Waste Diversion Report	3.7.3													
			Disposal Requirements	3.5.7.2.1	G												
			SD-07 Certificates														
			Certificate of Competency	1.8.4.1													
			Asbestos Certification	1.7.1													
			Lead Certification	1.7.2													
			SD-11 Closeout Submittals														
			Disposal Documentation for Hazardous and Regulated Waste	3.8.3.8	G												
			Assembled Employee Training Records	1.8.4	G												
			Solid Waste Management Report	3.8.2.2	G												

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 19	Hazardous Waste/Debris Management	3.8.3.1	G												
			Environmental Records Binder	1.11													
		01 57 19.00 30	SD-01 Preconstruction Submittals														
			Preconstruction Survey	1.6.1													
			Regulatory Notifications	1.6.2	G												
			Environmental Manager Qualifications	1.6.4	G												
			Employee Training Records	1.6.5	G												
			Environmental Protection Plan	1.7	G												
			Dirt and Dust Control Plan	1.7.9.1	G												
			Solid Waste Management Permit	1.11	G												
			Stormwater Pollution Prevention Plan	3.2.1.1	G												
			Stormwater Notice of Intent	3.2.1.2	G												
			Spill Prevention Control And Countermeasure (SPCC) Plan	3.15.2	G												
			SD-06 Test Reports														
			Monthly Solid Waste Disposal Report	1.11.1	G												
			Inspection Reports	3.2.1.3													
			Laboratory Analysis	3.7.1.1.2													
			SD-07 Certificates														
			Employee Training Records	1.6.5	G												
			Erosion and Sediment Control	1.6.5													
			Inspector														

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 19.00 30	Certificate of Competency	1.6.5.1													
			SD-11 Closeout Submittals														
			Regulatory Notifications	1.6.2	G												
			Assembled Employee Training	1.6.5	G												
			Records														
			Solid Waste Management Permit	1.11	G												
			Stormwater Pollution Prevention	3.2.1.4	G												
			Plan Compliance Notebook														
			Stormwater Notice of Termination	3.2.1.5	G												
			As-Built Topographic Survey	3.2.1.5													
			Waste Determination	3.7.1	G												
			Documentation														
			Project Solid Waste Disposal	3.7.2.1	G												
			Documentation Report														
			Sales Documentation	3.7.2.1	G												
			Contractor Certification	3.7.2.1													
			Hazardous Waste/Debris	3.7.3.1	G												
			Management														
			Disposal Documentation for	3.7.3.6	G												
			Hazardous and Regulated Waste														
		01 57 19.01	SD-01 Preconstruction Submittals														
			Preconstruction Survey	1.8.1													
			Regulatory Notifications	1.8.2													
			Environmental Protection Plan	1.9	G												
			Environmental Protection Plan	3.8.2.1	G												

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CONTRACTOR

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		01 57 19.01	Environmental Manager Qualifications	1.8.4	G												
			Temporary Fuel Storage Tank Approval Request	3.15.3	G												
			Underground Fuel Storage Tank Approval Request	3.15.5	G												
			Licenses/Permits/Notifications General Use Permitted Equipment	1.8.3 1.10	G												
			Fuel Storage Tank Installation/Removal Report	1.9.11.2	G												
			Written Assessment Of Friable Asbestos Disturbance	3.11													
			SD-06 Test Reports Laboratory Analysis	3.8.1.1.2													
			Nonhazardous Solid Waste Diversion Report	3.7.2													
			SD-07 Certificates EMS Certificate of Completion	1.6.1.1	G												
			Asbestos Certification	1.7.1													
			Lead Certification	1.7.2													
			SD-11 Closeout Submittals Environmental Records Binder	1.8.5													
			Environmental Records Binder	1.8.5													
			Environmental Records Binder	1.11													
			Environmental Records Binder	3.2.1													

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		01 57 19.01	Environmental Records Binder	3.8.1													
		01 57 19.11	SD-01 Preconstruction Submittals														
			Preconstruction Survey	1.6.1													
			Regulatory Notifications	1.6.2	G												
			Environmental Manager	1.6.4	G												
			Qualifications														
			Employee Training Records	1.6.5	G												
			Environmental Protection Plan	1.7	G												
			Dirt and Dust Control Plan	1.7.9.1	G												
			Solid Waste Management Permit	1.11	G												
			Stormwater Pollution Prevention	3.2.1.1	G												
			Plan														
			Stormwater Notice of Intent	3.2.1.2	G												
			Spill Prevention Control And	3.15.2	G												
			Countermeasure (SPCC) Plan														
			SD-06 Test Reports														
			Monthly Solid Waste Disposal	1.11.1	G												
			Report														
			Inspection Reports	3.2.1.3													
			Laboratory Analysis	3.7.1.1.2													
			SD-07 Certificates														
			Employee Training Records	1.6.5	G												
			Erosion and Sediment Control	1.6.5													
			Inspector														
			Certificate of Competency	1.6.5.1													
			SD-11 Closeout Submittals														

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		01 57 19.11	Regulatory Notifications	1.6.2	G												
			Assembled Employee Training Records	1.6.5	G												
			Solid Waste Management Permit	1.11	G												
			Stormwater Pollution Prevention Plan Compliance Notebook	3.2.1.4	G												
			Stormwater Notice of Termination	3.2.1.5	G												
			As-Built Topographic Survey	3.2.1.5													
			Waste Determination Documentation	3.7.1	G												
			Project Solid Waste Disposal Documentation Report	3.7.2.1	G												
			Sales Documentation	3.7.2.1	G												
			Contractor Certification	3.7.2.1													
			Hazardous Waste/Debris Management	3.7.3.1	G												
			Disposal Documentation for Hazardous and Regulated Waste	3.7.3.6	G												
		01 74 19	SD-01 Preconstruction Submittals														
			Waste Management Plan	1.6	G												
			SD-11 Closeout Submittals														
			Records	1.7	G												
		01 78 00	SD-03 Product Data														
			Warranty Management Plan	1.7.1	G												
			Warranty Tags	1.7.5													
			Final Cleaning	3.8													

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		01 78 00	Spare Parts Data	1.5													
			Spare Parts And Spare Parts Data	1.5													
			SD-08 Manufacturer's Instructions														
			Instructions	1.7.1													
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	3.7	G												
			SD-11 Closeout Submittals														
			As-Built Drawings	3.1	G												
			Record Drawings	3.3	G												
			As-Built Record of Equipment and Materials	1.7.1	G												
			As-Built Record of Equipment and Materials	3.6	G												
			Final Approved Shop Drawings	3.4	G												
			Construction Contract Specifications	3.5	G												
			Interim DD FORM 1354	3.9	G												
			Checklist for DD FORM 1354	3.9	G												
		01 78 23	SD-10 Operation and Maintenance Data														
			O&M Database	1.4	G												
			Training Plan	3.1.1	G												
			Training Outline	3.1.3	G												

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		01 78 23	Training Content	3.1.2	G												
			SD-11 Closeout Submittals														
			Training Video Recording	3.1.4	G												
			Validation of Training Completion	3.1.6	G												
		01 91 00.15	SD-05 Design Data														
			Design Phase Commissioning Plan	3.2	G												
			SD-06 Test Reports														
			Completed Construction Observation Checklists	3.9	G												
			Design Review Report	3.3	G												
			Interim Construction Phase Commissioning Plan	3.7	G												
			Final Construction Phase Commissioning Plan	3.7	G												
			Issues Log	1.13	G												
			SD-07 Certificates														
			Commissioning Firm	1.10	G												
			Certificate of Readiness	1.14	G												
			SD-10 Operation and Maintenance Data														
			Maintenance and Service Life Plan	3.13	G												
			Systems Manual	3.12	G												
			SD-11 Closeout Submittals														
			Final Commissioning Report	3.14	G												

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		01 91 00.15	Updated Final Commissioning Report	3.15	G												
			Final Commissioning Report (eNotebook)	3.14	S												
			Updated Final Commissioning Report (eNotebook)	3.15	S												
		01 91 00.15 10	SD-01 Preconstruction Submittals														
			Commissioning Firm	1.7	G												
			Lead Commissioning Specialist	1.7.1													
			Technical Commissioning Specialists	1.7.2													
			Commissioning Firm's Contract	1.7													
			SD-06 Test Reports														
			Design Review Report	3.1.3	G												
			Interim Construction Phase Commissioning Plan	3.1.2.1	G												
			Final Construction Phase Commissioning Plan	3.1.2.2	G												
			Building Envelope Inspection Checklists	3.1.5.2	G												
			Pre-Functional Checklists	3.1.5.3	G												
			Issues Log	1.9													
			Commissioning Report	3.2	G												
			Post-Construction Trend Log Report	3.3.1	G												
			SD-07 Certificates														

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		01 91 00.15 10	Certificate of Readiness	1.10	G												
			SD-10 Operation and Maintenance Data														
			Training Plan	3.1.6	G												
			Training Attendance Rosters	3.1.6													
			Systems Manual	3.1.7	G												
			Maintenance and Service Life Plans	3.1.8													
			SD-11 Closeout Submittals														
			Construction Phase	3.1.2.1	G												
			Commissioning Plan														
			Final Commissioning Report	3.2	G												
		02 41 00	SD-01 Preconstruction Submittals														
			Demolition Plan	1.2.2	G												
			Deconstruction Plan	1.2.2	G												
			Existing Conditions	1.11													
			SD-07 Certificates														
			Notification	1.7	G												
			SD-11 Closeout Submittals														
			Receipts	3.3.4													
		02 82 00	SD-03 Product Data														
			Amended Water	1.2.2													
			Safety Data Sheets (SDS) for All Materials	1.3.9													
			Encapsulants	2.1													
			Respirators	3.1.2.1													

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		02 82 00	Local Exhaust Equipment	3.1.7													
			Pressure Differential Automatic	3.1.7													
			Recording Instrument														
			Vacuums	3.1.8													
			Glovebags	3.1.10													
			SD-06 Test Reports														
			Air Sampling Results	1.5.6													
			Pressure Differential Recordings	1.5.7													
			for Local Exhaust System														
			Clearance Sampling	3.2.13.4													
			Asbestos Disposal Quantity	3.3.3.2													
			Report														
			SD-07 Certificates														
			Employee Training	1.3.4													
			Notifications	1.3.5													
			Respiratory Protection Program	1.3.7													
			Asbestos Hazard Abatement Plan	1.3.10													
			Testing Laboratory	1.3.11													
			Landfill Approval	1.3.12													
			Delivery Tickets	1.3.12													
			Waste Shipment Records	1.3.12													
			Transporter Certification	1.3.13													
			Medical Certification	1.3.14													
			Private Qualified Person	1.5.2													
			Documentation														
			Competent Person	1.5.3													

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		02 82 00	Worker's License	1.5.4													
			Contractor's License	1.5.5													
			Equipment Used to Contain Airborne Asbestos Fibers	3.1													
			Water Filtration Equipment	3.1.3.3													
			Vacuums	3.1.8													
			Ventilation Systems	3.1.8													
			SD-11 Closeout Submittals														
			Permits[and Licenses]	1.3.5													
			Notifications	1.3.5													
			Respirator Program Records	1.3.7.1													
			Rental Equipment	1.7.1													
		02 83 00	SD-01 Preconstruction Submittals														
			Occupational and Environmental Assessment Data Report	1.5.2.3													
			Medical Examinations	1.5.2.4													
			Lead, Cadmium, Chromium	1.5.2.8													
			Waste Management Plan														
			Licenses, Permits and Notifications	1.5.4													
			Initial Sample Results	3.4.1.1													
			Written Evidence of TSD Approval	3.5.2.1													
			Lead, Cadmium, Chromium Compliance Plan	1.5.2.2													

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		02 83 00	Lead, Cadmium, Chromium Compliance Plan	3.1.1.6													
			SD-03 Product Data														
			Respirators	1.6.1													
			Vacuum Filters	1.6.4													
			Negative Air Pressure System	1.6.7													
			Materials and Equipment	2.1													
			Expendable Supplies	2.1.1													
			Local Exhaust Equipment	3.1.1.5													
			Pressure Differential Automatic Recording Instrument	3.1.1.5													
			Pressure Differential Log	3.1.1.6													
			SD-06 Test Reports														
			Occupational and Environmental Assessment Data Report	1.5.2.3													
			Sampling Results	1.5.2.3													
			Pressure Differential Recordings For Local Exhaust System	1.5.3													
			SD-07 Certificates														
			Notification of the Commencement of [LBP] Hazard Abatement														
			Clearance Certification	3.5.1.1													
			SD-11 Closeout Submittals														
			Hazardous Waste Manifest	3.5.2.1													

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		02 83 00	Turn-In Documents or Weight Tickets	3.5.2.1													
		02 84 16	SD-01 Preconstruction Submittals Notification	1.4.7													
			PCB and Lamp Management and Disposal Plan	1.4.1.1	G												
			SD-11 Closeout Submittals														
			Transporter certification	3.5.4	G												
			Certification of Decontamination	3.2.4													
			Copy E of the Japan Hazardous Waste Manifest	3.5.3													
			Disposal Request Form														
			Signed Ballast Turn-in (at Bldg 3625) Sheet	3.3.1													
		03 30 00	SD-01 Preconstruction Submittals														
			Quality Control Plan	1.6.5	G												
			Laboratory Accreditation	1.6.8	G												
			SD-02 Shop Drawings														
			Reinforcing Steel	1.6.2.1	G												
			SD-03 Product Data														
			Formwork Materials	2.1													
			Reinforcement	2.6													
			Liquid Chemical Floor Hardeners and Sealers	2.4.3.1													
			Mechanical Reinforcing Bar	2.6.2													
			Connectors														

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		03 30 00	Pumping Concrete	1.6.3.1													
			SD-05 Design Data														
			Concrete Mix Design	1.6.1.1	G												
			SD-06 Test Reports														
			Concrete Mix Design	1.6.1.1	G												
			Fly Ash	1.6.4.1													
			Pozzolan	1.6.4.1													
			Slag Cement	1.6.4.2													
			Aggregates	1.6.4.3													
			Compressive Strength Tests	3.14.2.3	G												
			Air Content	3.14.2.4													
			Slump Tests	3.14.2.1													
			Water	2.3.2													
			SD-07 Certificates														
			Reinforcing Bars	2.6.1													
			Welder Qualifications	1.8													
			VOC Content for Form Release	1.6.3.2													
			Agents, Curing Compounds, and														
			Concrete Penetrating Sealers														
			Safety Data Sheets	1.6.3.3													
			Field Testing Technician and	1.6.6.2													
			Testing Agency														
			SD-08 Manufacturer's Instructions														
			Liquid Chemical Floor Hardeners	2.4.3.1													
			and Sealers														
			Joint Sealants	2.4.5													

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		03 30 00	Curing Compound	2.4.1													
		03 42 13.00 10	SD-01 Preconstruction Submittals														
			Quality Control Procedures	1.3.2.2													
			SD-02 Shop Drawings														
			Standard Precast Units	2.1.1	G												
			Custom-Made Precast Units	2.1.2	G												
			Special Finishes	3.2.4.3													
			SD-03 Product Data														
			Standard Precast Units	2.1.1													
			Proprietary Precast Units	2.1.3													
			Embedded Items	3.1.3													
			Accessories	2.2.6													
			SD-05 Design Data														
			Design Calculations	2.1.2	G												
			Concrete Mix Proportions	2.1.5.1													
			SD-06 Test Reports														
			Test Reports	1.3.2.4													
			SD-07 Certificates														
			Quality Control Procedures	1.3.2.2													
			SD-11 Closeout Submittals														
			Recycled content for fly ash and pozzolan	2.2.1	S												
			Recycled content for Ground Iron Blast-Furnace Slag	2.2.1	S												
			Recycled content for Silica Fume	2.2.1	S												

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		03 42 13.00 10	Recycled content for Synthetic Fiber Reinforcement	2.2.1	S												
			Recycled content for steel	2.2.1	S												
		05 05 20	SD-01 Preconstruction Submittals														
			Installer Qualifications	1.5.1.1	G												
			Post-Installed Anchor Special Inspector Qualifications	1.5.1.2	G												
			SD-03 Product Data														
			Mechanical Anchors in Concrete	2.1.1.3	G												
			Adhesive Anchor System in Concrete	2.1.1.4	G												
			Mechanical and Adhesive Anchors in Masonry	2.1.1.5	G												
			Non-Shrink, Non-Metallic Grout	3.1.3	G												
			SD-06 Test Reports														
			Post-Installed Anchor Special Inspections Report	3.3.4	G												
			SD-07 Certificates														
			Post-Installed Anchor Certification	2.1.1.1	G												
			SD-08 Manufacturer's Instructions														
			Manufacturer's Printed Installation Instructions	2.1.1.2	G												
			Non-Shrink, Non-Metallic Grout	3.1.3	G												
			Manufacturer's Safety Data Sheets	1.5													

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		05 12 00	SD-01 Preconstruction Submittals														
			Erection and Erection Bracing	1.5.1.1	G												
			Drawings														
			SD-02 Shop Drawings														
			Fabrication Drawings	1.5.2	G												
			SD-03 Product Data														
			Welding Electrodes and Rods	2.4.1													
			Direct Tension Indicator Washers	2.3.2.3													
			Non-Shrink Grout	2.4.2													
			SD-06 Test Reports														
			Weld Inspection Reports	3.6.1.2													
			Direct Tension Indicator Washer	3.6.2.1													
			Inspection Reports														
			Bolt Testing Reports	3.6.3.1													
			Embrittlement Test Reports	3.6.4													
			SD-07 Certificates														
			MLIT Structural Steel Fabricator	1.3													
			Quality Certification														
			MLIT Structural Steel Erector	1.3													
			Quality Certification														
			Welding Procedure Specifications	3.4													
			(WPS)														
			Overhead, Top Running Crane	1.5.3.2													
			Rail Beam														
		05 30 00	SD-02 Shop Drawings														
			Fabrication Drawings	1.3.4	G												

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		05 30 00	SD-04 Samples														
			Steel Roof Deck Units														
			SD-05 Design Data														
			Deck Units	2.1	G												
			SD-07 Certificates														
			Welder Qualifications	1.3.2													
			Welding Procedures	1.3.2													
			Fire Safety	1.3.3.1													
			Wind Storm Resistance	1.3.3.2													
			Manufacturer's Certificate	1.3.1													
			Stud Manufacture's Certification	2.2.10													
			Stud Manufacture's Test Reports	2.2.10													
		05 50 13	SD-02 Shop Drawings														
			Floor Gratings	2.4													
			Bollards/Pipe Guards	2.5	G												
			Angles and Plates	2.7	G												
			Roof Hatches	2.9	G												
			SD-03 Product Data														
			Corner Guards	2.3													
			Floor Gratings	2.4													
			Roof Hatches	2.9													
		05 50 15	SD-02 Shop Drawings														
			Shop Fabricated Metal Items	2.2	G												
			SD-03 Product Data														
			Expansion Anchors	2.1.17													
			Sleeve Anchors	2.1.17													

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		05 50 15	Adhesive Anchors	2.1.17													
			Powder Driven Fasteners	2.1.20													
			Wire Rope	2.1.23													
			Gratings	2.1.28													
			Stairs	2.2.3													
			Ladders	2.2.5													
			Shop Fabricated Metal Items	2.2	G												
			SD-04 Samples														
			Shop Fabricated Metal Items	2.2	G												
			SD-06 Test Reports														
			Hardness Check	2.1.5.4.2													
			Rotational Capacity	2.1.16.1													
			SD-07 Certificates														
			Welder Certifications	1.3													
		05 51 00	SD-02 Shop Drawings														
			Iron and Steel Hardware	2.1	G												
			Steel Shapes, Plates, Bars, and Strips	2.1	G												
			Metal Stair System	2.2.1	G												
			SD-03 Product Data														
			Structural Steel Plates, Shapes, and Bars	2.4.1	G												
			Structural Steel Tubing	2.4.2	G												
			Hot-Rolled Carbon Steel Sheets and Strips	2.4.5	G												
			Cold-Finished Steel Bars	2.4.4	G												

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		05 51 00	Hot-Rolled Carbon Steel Bars	2.4.3	G												
			Cold-Rolled Carbon Steel Sheets	2.4.6	G												
			Galvanized Carbon Steel Sheets	2.4.7	G												
			Cold-Drawn Steel Tubing	2.4.8	G												
			Gray Iron Castings	2.4.9	G												
			Malleable Iron Castings	2.4.10	G												
			Concrete Inserts	2.3.4	G												
			Masonry Anchorage Devices	2.3.5	G												
			Protective Coating	2.2.4	G												
			Steel Pan Stairs	2.2.2	G												
			Steel Stairs	2.3.1	G												
			Steel Stairs, Circular	2.3.2	G												
			SD-07 Certificates														
			Welding Procedures	1.3.1	G												
			Welder Qualification	1.3.1	G												
			SD-08 Manufacturer's Instructions														
			Structural Steel Plates, Shapes, and Bars	2.4.1	G												
			Structural Steel Tubing	2.4.2	G												
			Hot-Rolled Carbon Steel Sheets and Strips	2.4.5	G												
			Cold-Finished Steel Bars	2.4.4	G												
			Hot-Rolled Carbon Steel Bars	2.4.3	G												
			Cold-Rolled Carbon Steel Sheets	2.4.6	G												
			Galvanized Carbon Steel Sheets	2.4.7	G												
			Cold-Drawn Steel Tubing	2.4.8	G												

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		05 51 00	Gray Iron Castings	2.4.9	G												
			Malleable Iron Castings	2.4.10	G												
			Protective Coating	2.2.4	G												
			Masonry Anchorage Devices	2.3.5	G												
		05 51 33	SD-02 Shop Drawings														
			Ladders	2.3													
			Ship's Ladder	2.3.3													
			SD-03 Product Data														
			Ladders	2.3													
			Ship's Ladder	2.3.3													
			Ladder Safety Devices	2.3.2													
			SD-07 Certificates														
			Fabricator Certification for Ladder Assembly	1.3													
			Fabricator Certification for Ships Ladder Assembly	1.3													
		05 52 00	SD-02 Shop Drawings														
			Fabrication Drawings	1.2.1	G												
			Iron and Steel Hardware	3.2													
			Steel Shapes, Plates, Bars and Strips	3.2													
			SD-07 Certificates														
			Welding Procedures	1.4.1	G												
			Welder Qualification	1.4.2	G												
			SD-08 Manufacturer's Instructions														
			Installation Instructions	3.2													

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		06 41 16.00 10	SD-02 Shop Drawings														
			Shop Drawings	2.9													
			Installation	3.1													
			SD-03 Product Data														
			Finish Schedule														
			SD-04 Samples														
			Plastic Laminates	2.3													
			Cabinet Hardware	2.6													
		06 61 16	SD-02 Shop Drawings														
			Detail Fabrication Drawings	1.4.2	G												
			Installation	3.1	G												
			SD-03 Product Data														
			Solid Polymer	2.1.1	G												
			Indoor air quality for solid surface seam and sealant products	2.2.2	S												
			SD-04 Samples														
			Material	2.1	G												
			Counter Tops	2.3.5	G												
			SD-06 Test Reports														
			Test Report Results	2.1.1													
			SD-07 Certificates														
			Qualifications	1.4.1													
			Indoor Air Quality for solid surface fabrication products	2.1.1	S												
			SD-10 Operation and Maintenance Data														

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		06 61 16	Solid Polymer	2.1.1	G												
		07 13 53	SD-01 Preconstruction Submittals														
			Draft Guarantee Certificate	1.7.1	G												
			Draft Warranty Certificate	1.7.2	G												
			SD-03 Product Data														
			Manufacturer's Standard Details	1.3	G												
			Elastomeric Waterproofing Sheet	2.1	G												
			Material														
			Protection Board	2.5	G												
			Primers, Adhesives, and Mastics	1.4	G												
			SD-06 Test Reports														
			Elastomeric Waterproofing Sheet	3.6	G												
			Material Test Reports														
			Protective Covering Inspection	3.7	G												
			Reports														
			SD-07 Certificates														
			Primers, Adhesives, and Mastics	1.4	G												
			SD-08 Manufacturer's Instructions														
			Primers, Adhesives, and Mastics	1.4	G												
			SD-11 Closeout Submittals														
			Final Guarantee Certificate	1.7.1	G												
			Final Warranty Certificate	1.7.2	G												
		07 14 00	SD-03 Product Data														
			Fluid-Applied Membrane	2.1													
			Membrane Primer	2.2													
			Bond Breaker	2.7													

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		07 14 00	SD-11 Closeout Submittals														
			Warranty	1.6													
			Information Card	3.6													
			Instructions To Government Personnel	3.5													
		07 21 16	SD-03 Product Data														
			Blanket Insulation	2.1													
			Recycled Content for Insulation Materials	2.1.2	S												
			Sill Sealer Insulation	2.2													
			Pressure Sensitive Tape	2.4													
			Accessories	2.5													
			SD-07 Certificates														
			Indoor Air Quality for Insulation Materials	2.1.4	S												
			Indoor Air Quality for Adhesives	2.5.1	S												
			SD-08 Manufacturer's Instructions														
			Insulation	3.3.1													
		07 24 00	SD-02 Shop Drawings														
			Shop Drawings	1.4.6	G												
			SD-03 Product Data														
			Sheathing Board	2.2													
			Thermal Insulation	2.6													
			Adhesive	2.3													
			Mechanical Fasteners	2.5													
			Accessories	2.12													

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		07 24 00	Base Coat	2.7													
			Portland Cement	2.8													
			Reinforcing Fabric	2.9													
			Finish Coat	2.10													
			Joint Sealant	2.13													
			Sealant Primer	2.11													
			Bond Breaker	2.14													
			Backer Rod	2.15													
			Insulation Board	1.4.5													
			Recycled Content for Insulation Materials	2.6.2	S												
			Warranty	1.7													
			SD-04 Samples														
			Sample Boards	1.2.3.7	G												
			Mock-up Installation of EIFS	1.2.1.4	G												
			SD-05 Design Data														
			Moisture Analysis	1.2.4													
			SD-06 Test Reports														
			Abrasion Resistance	1.2.3.1													
			Accelerated Weathering	1.2.3.2													
			Impact Resistance	1.2.2.3													
			Mildew Resistance	1.2.3.3													
			Salt Spray Resistance	1.2.3.4													
			Vapor Transmission	1.2.4													
			Absorption-Freeze-Thaw	1.2.3.6													
			Wall Fire Test	1.2.1.3													

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		07 24 00	Water Penetration	1.2.1.1													
			Water Resistance	1.2.3.5													
			Full Scale or Intermediate Scale	1.2.1.3													
			Fire Test														
			Surface Burning Characteristics	1.2.2.1													
			Radiant Heat	1.2.2.2													
			Substrate	3.1													
			Wind Load Test	1.2.1.2													
			SD-07 Certificates														
			Qualifications of EIFS	1.4.1													
			Manufacturer														
			Qualification of EIFS Installer	1.4.2													
			Qualification of Sealant Applicator	1.4.3													
			Qualifications of Third-Party	1.4.4													
			Inspector														
			Inspection Checklist	3.5.2	G												
			SD-08 Manufacturer's Instructions														
			Manufacturer's Instructions	3.3													
		07 27 36	SD-01 Preconstruction Submittals														
			Qualification of Manufacturer	1.10.1	G												
			Qualification of Installer	1.10.2	G												
			Quality Control Plan	1.11	G												
			Safety Plan	1.11	G												
			Fire Prevention Plan	1.9.1	G												
			Respiratory Protection Plan	1.9.2	G												
			SD-02 Shop Drawings														

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		07 27 36	Fire-Ratings Of	1.5.4	G												
			[Thermal][Ignition] Barrier														
			Materials														
			Flame Spread and Smoke	1.5.4	G												
			Developed Index Ratings of SPF														
			Products														
			Flame Propagation of Wall	1.5.4	G												
			Assemblies														
			Site Inspections	3.4.1	G												
			SD-07 Certificates														
			Indoor Air Quality for Spray Foam	2.1.5	S												
			Air Barrier														
			SD-08 Manufacturer's Instructions														
			SPF Handling, Storage, and	1.6.1	G												
			Spray Procedures														
			Substrate Preparation	3.2.1	G												
			Thermal Barrier	1.5.2	G												
			Ignition Barrier	1.5.2	G												
			Transition Membrane	3.3.2	G												
			Manufacturer's Installation														
			Instructions														
			Manufacturer's Printed	1.5.2	G												
			Installation Instructions for														
			Primers, Adhesives, and Mastics														
			SD-09 Manufacturer's Field														
			Reports														

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		07 27 36	Core Samples	1.11													
			Daily Work Record	3.3.3													
			Visual Inspection and Thermal Scanning	3.4.5													
		07 41 13	SD-01 Preconstruction Submittals														
			Sample Warranty	1.6	G												
			SD-02 Shop Drawings														
			Roofing Panels	1.3.4	G												
			Flashing and Accessories	1.3.4	G												
			Gutter/Downspout Assembly	1.3.4	G												
			SD-03 Product Data														
			Roof Panels	2.4.1	G												
			Recycled Content for Aluminum Roof Panels	2.4.1.1	S												
			Recycled Content for Steel Roof Panels		S												
			Heat Island Reduction	2.1.4.2	S												
			Factory-Applied Color Finish	1.3.4	G												
			Accessories	2.5	G												
			Fasteners	1.3.4	G												
			Pressure Sensitive Tape	1.3.4	G												
			Underlayments	2.4.2	G												
			Gaskets and Sealing/Insulating Compounds	2.5.6	G												
			Coil Stock	1.3.4	G												
			Enamel Repair Paint	1.3.4	G												

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		07 41 13	SD-04 Samples														
			Roof Panels	2.4.1	G												
			Factory-applied Color Finish	1.3.4	G												
			Accessories	2.5	G												
			Fasteners	1.3.4	G												
			Gaskets and Sealant/Insulating Compounds	1.3.4	G												
			SD-05 Design Data														
			Wind Uplift Resistance	2.1.2.3	G												
			SD-06 Test Reports														
			Leakage Test Report	2.1.2.1	G												
			Wind Uplift Test Report	2.1.2.3	G												
			Factory Finish and Color	2.1.4	G												
			Performance Requirements														
			Hydrostatic Join Performance Testing		G												
			SD-07 Certificates														
			Roof Panels	2.4.1	G												
			Coil Stock Compatibility	1.3.4	G												
			Self-Adhering Modified Bitumen Underlayment	2.4.2.1	G												
			Qualification of Manufacturer	1.3.1	G												
			Qualification of Applicator	1.3.2	G												
			SD-08 Manufacturer's Instructions														
			Installation Manual	1.3.4	G												

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		07 41 13	SD-09 Manufacturer's Field Reports														
			Manufacturer's Field Inspection Report	3.4.3.2	G												
			SD-11 Closeout Submittals														
			Warranties	1.6	G												
			Information Card	3.6.1	G												
		07 42 13	SD-01 Preconstruction Submittals														
			Qualification of Manufacturer	1.4.3	G												
			Qualification of Installation Contractor	1.4.4	G												
			Warranty	1.7	G												
			SD-02 Shop Drawings														
			Installation Drawings	1.4.1.1	G												
			SD-03 Product Data														
			Recycled Content;	2.2	S												
			Aluminum Sheet Panels	2.3.1	G												
			Steel Sheet Panels	2.3.2	G												
			Factory Color Finish	2.3.3													
			Closure Materials	1.4.5													
			Pressure Sensitive Tape	2.3.6.6													
			Sealants and Caulking	2.3.6.3													
			Galvanizing Repair Paint	1.4.3.1													
			Enamel Repair Paint	1.4.3.1													
			Aluminized Steel Repair Paint	2.3.8													
			Accessories	2.3.6													

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		07 42 13	SD-04 Samples														
			Aluminum Sheet Panels	2.3.1	G												
			Steel Sheet Panels	2.3.2	G												
			Fasteners	1.4.3.1	G												
			Metal Closure Strips	2.3.6.2	G												
			Color Chart	2.3.3.5	G												
			SD-05 Design Data														
			Wind Load Design Analysis	1.4.1.2	G												
			SD-06 Test Reports														
			Leakage Tests	3.5.1	G												
			Wind Load Tests	2.1.2	G												
			Coating	2.3.3.6	G												
			Chalking	2.3.3.6	G												
			Seismic Tests	2.1.2	G												
			SD-07 Certificates														
			Coil Stock	1.4.3.1	G												
			Fasteners	1.4.3.1	G												
			Galvanizing Repair Paint	1.4.3.1	G												
			Enamel Repair Paint	1.4.3.1	G												
			SD-08 Manufacturer's Instructions														
			Installation	3.3.1	G												
			SD-09 Manufacturer's Field														
			Reports														
			Manufacturer's Field Inspection	3.5.4.1.2	G												
			Reports														
			SD-11 Closeout Submittals														

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		07 42 13	Warranty	1.7	G												
			Maintenance Instructions	1.4.6	G												
			20 Year 'No Dollar Limit'	1.7.1													
			Warranty for Labor and Material														
		07 42 63	SD-01 Preconstruction Submittals														
			Qualification of Manufacturer	1.5.3													
			Qualification of Installer	1.5.4													
			Qualifications for Welding Work	1.5.4.1													
			SD-02 Shop Drawings														
			Fabrication and Installation	1.5.1													
			Drawings														
			Wall Panel Assemblies	1.5.1													
			Flashing and Accessories	1.5.1													
			Anchorage Systems	1.5.1													
			SD-03 Product Data														
			Certification	1.5.10													
			Recycled Content for Insulation	2.1.4	S												
			Manufacturer's Catalog Data	1.5.1													
			Closure Materials	2.4.2													
			Insulation	2.1.4													
			Tape Sealant	2.4.3.4													
			Sealants and Caulking	2.4.3.1													
			Rated Wall Assembly	1.5.1													
			Accessories	2.4													
			SD-04 Samples														
			Wall Panel Assemblies	1.5.1													

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		07 42 63	Fasteners	1.5.1													
			Metal Closure Strips	1.5.1													
			Insulation	2.1.4													
			Manufacturer's Color Charts and Chips	1.5.1													
			SD-05 Design Data														
			Wind Design Analysis	1.5.1													
			SD-06 Test Reports														
			Leakage Tests	3.7.2													
			Wind Load Tests	1.3.2													
			Seismic Tests	1.3.2													
			Factory Color Finish	1.5.1													
			SD-07 Certificates														
			Fasteners	1.5.1													
			Galvanizing Repair Paint	1.5.1													
			Enamel Repair Paint	1.5.1													
			Aluminized Steel Repair Paint	1.5.1													
			Wall System Assembly Wind	1.5.1													
			Load and Fire Rating Classification														
			Listings														
			SD-08 Manufacturer's Instructions														
			Installation of Wall Panels	1.5.1													
			SD-09 Manufacturer's Field														
			Reports														
			Manufacturer's Field Inspection	3.8.2	G												
			Reports														

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		07 42 63	SD-11 Closeout Submittals														
			Warranty	1.5.1													
			Safety Data Sheets	1.5.1													
		07 56 00	SD-02 Shop Drawings														
			Roof Assembly	1.2.1													
			SD-07 Certificates														
			Material and Equipment	1.4.1													
			Energy Efficiency	1.4.2													
			Qualifications	1.4.3													
		07 60 00	SD-02 Shop Drawings														
			Exposed Sheet Metal	2.2.1	G												
			Gutters	3.1.17	G												
			Downspouts	3.1.18	G												
			Base Flashing	3.1.11	G												
			Counterflashing	3.1.12	G												
			Flashing at Roof Penetrations	3.1.25													
			Scuppers	3.1.20	G												
			Drip Edges	3.1.16	G												
			Open Valley Flashing	3.1.22	G												
			Eave Flashing	3.1.23	G												
			SD-08 Manufacturer's Instructions														
			Instructions for Installation	1.4.3	G												
			Quality Control Plan	3.5	G												
			SD-10 Operation and Maintenance														
			Data														
			Cleaning and Maintenance	1.4.3	G												

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		07 61 14.00 20	SD-02 Shop Drawings														
			Roofing	1.2.5	G												
			SD-03 Product Data														
			Roofing Panels	2.1	G												
			Energy Star Label for Steel	2.1	S												
			Roofing Product														
			Recycled Content for Steel	2.1.1	S												
			Roofing Product														
			Heat Island Reduction	2.1	S												
			Attachment Clips	2.3													
			Closures	2.4.1													
			Accessories	2.4													
			Fasteners	2.4.2													
			Sealants	2.4.3													
			Insulation	2.5													
			Warranty	1.7	G												
			SD-04 Samples														
			Panel	2.1													
			Accessories	2.4													
			Sealants	2.4.3													
			Intermediate Support	2.2													
			SD-05 Design Data														
			Design Calculations	1.5													
			SD-06 Test Reports														
			Field Inspection	3.6													
			Structural Performance	1.3.3													

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		07 61 14.00 20	Finish	1.6.6													
			SD-07 Certificates														
			Manufacturer's Technical Representative	1.6.3													
			Installer's Qualifications	1.6.4													
			Coil Stock	2.1	G												
			SD-08 Manufacturer's Instructions														
			Installation	3.3	G												
			SD-11 Closeout Submittals														
			Information Card	3.8													
			Warranty	1.7													
		07 84 00	SD-02 Shop Drawings														
			Firestopping System	2.1	G												
			SD-03 Product Data														
			Firestopping Materials	2.2	G												
			SD-06 Test Reports														
			Inspection	3.3	G												
			SD-07 Certificates														
			Inspector Qualifications	1.5.2													
			Firestopping Materials	2.2													
			Installer Qualifications	1.5.1	G												
		08 11 13	SD-02 Shop Drawings														
			Doors	2.1	G												
			Doors	2.1	G												
			Frames	2.4	G												
			Frames	2.4	G												

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		08 11 16	SD-02 Shop Drawings														
			Door and Frame Assembly	1.5.1													
			SD-04 Samples														
			Finish Samples	1.5.2	G												
			SD-06 Test Reports														
			Air Infiltration	1.2.1	G												
			Water Penetration	1.2.2	G												
			SD-10 Operation and Maintenance														
			Data														
			Adjustments, Cleaning, and	1.5.3	G												
			Maintenance														
		08 14 00	SD-02 Shop Drawings														
			Doors	2.1	G												
			SD-04 Samples														
			Doors	2.1													
			SD-11 Closeout Submittals														
			Warranty	1.5													
		08 31 00	SD-02 Shop Drawings														
			Access Doors And Panels	1.3	G												
			SD-03 Product Data														
			Access Doors And Panels	1.3	G												
			Hardware	1.3.2	G												
			Accessories	2.2.8	G												
			Recycled Content	2.1	S												
			SD-04 Samples														
			Finishes	2.5	G												

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		08 31 00	SD-06 Test Reports														
			Fire-rating(s) of Assemblies	1.3.1	G												
			Acoustical Ratings of Assemblies	1.3.1	G												
		08 33 23	SD-02 Shop Drawings														
			Overhead Coiling Doors	2.2.1	G												
			Counterbalancing Mechanism	2.2.3	G												
			Manual Door Operators	2.2.4	G												
			Electric Door Operators	2.2.5	G												
			Bottom Bars	2.2.1.4	G												
			Guides	2.1.1.1	G												
			Mounting Brackets	2.2.3.1	G												
			Hood	2.2.2.2	G												
			Installation Drawings	2.1.1.1	G												
			SD-03 Product Data														
			Overhead Coiling Doors	2.2.1	G												
			Hardware	2.2.2	G												
			Counterbalancing Mechanism	2.2.3	G												
			Manual Door Operators	2.2.4	G												
			Electric Door Operators	2.2.5	G												
			Recycled content for steel curtain slats	2.2.1.1	S												
			SD-05 Design Data														
			Overhead Coiling Doors	2.2.1	G												
			Hardware	2.2.2	G												
			Counterbalancing Mechanism	2.2.3	G												
			Manual Door Operators	2.2.4	G												

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		08 33 23	Electric Door Operators	2.2.5	G												
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	1.3.2	G												
			SD-11 Closeout Submittals														
			Warranty	1.3.1	G												
		08 34 59	SD-02 Shop Drawings														
			Vault Door Units	2.1	G												
			Day Gates	2.2	G												
			SD-03 Product Data														
			Vault Door And Frame	1.4													
			Day Gates	2.2	G												
			Installation	3.2													
			SD-07 Certificates														
			Vault Door Units	2.1	G												
			SD-10 Operation and Maintenance Data														
			Vault Door Units	2.1	G												
			Day Gates	2.2	G												
		08 36 13	SD-02 Shop Drawings														
			Doors	2.2	G												
			SD-03 Product Data														
			Doors	2.2	G												
			Electric Operators	2.6	G												
			SD-08 Manufacturer's Instructions														

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		08 36 13	Doors	2.2													
			SD-10 Operation and Maintenance Data														
			Doors	2.2	G												
		08 36 19	SD-02 Shop Drawings														
			Doors	2.2	G												
			SD-03 Product Data														
			Doors	2.2	G												
			Motors	2.4.1	G												
			Controls	2.4.2	G												
			SD-08 Manufacturer's Instructions														
			Doors	2.2													
			SD-10 Operation and Maintenance Data														
			Doors	2.2	G												
		08 41 13	SD-01 Preconstruction Submittals														
			Sample Warranty	1.2.1	G												
			List of Product Installations	1.2.1	G												
			SD-02 Shop Drawings														
			Installation Drawings	3.3	G												
			Fabrication Drawings	2.2	G												
			SD-03 Product Data														
			Manufacturer's Catalog Data	1.2.1	G												
			Finish	2.2.3	G												
			Recycled Content of Aluminum Material	2.1.1.2	S												

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		08 41 13	SD-04 Samples														
			Finish and Color Samples	1.2.1	G												
			SD-06 Test Reports														
			Certified Test Reports	1.2.1	G												
			Deflection	3.4.3													
			Air Infiltration	3.4.1													
			Condensation Resistance and Thermal Transmittance	3.4.4													
			Water Infiltration	3.4.5													
			SD-08 Manufacturer's Instructions														
			Manufacturer's Instructions	3.3													
			SD-11 Closeout Submittals														
			Manufacturer's Product Warranty	3.6													
		08 51 13	SD-02 Shop Drawings														
			Windows	2.1	G												
			Fabrication Drawings	1.6													
			SD-06 Test Reports														
			Windborne-Debris-Impact Performance	1.7.6													
			SD-10 Operation and Maintenance Data														
			Windows	2.1	G												
		08 60 45	SD-02 Shop Drawings														
			Shop Drawings	3.2	G												
			SD-03 Product Data														
			Translucent Panels	2.1	G												

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		08 60 45	Recycled Content for Aluminum Framing Materials	2.2.3	S												
			Warranty	1.6													
			SD-06 Test Reports														
			Test Reports	2.1													
			SD-07 Certificates														
			Systems	2.4													
			Qualifications	1.4													
		08 71 00	SD-02 Shop Drawings														
			Manufacturer's Detail Drawings	1.3	G												
			Verification of Existing Conditions	1.3	G												
			Hardware Schedule	1.5	G												
			Keying System	2.3.8	G												
			SD-08 Manufacturer's Instructions														
			Installation	3.1													
			SD-10 Operation and Maintenance Data														
			Hardware Schedule	1.5	G												
			SD-11 Closeout Submittals														
			Key Bitting	1.6.1													
		08 81 00	SD-03 Product Data														
			Insulating Glass	1.7.1													
			Sealants	2.4.3.1													
			Joint Backer	2.4.4													
			SD-08 Manufacturer's Instructions														
			Setting and Sealing Materials	2.4													

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		08 81 00	Glass Setting	3.2													
			SD-11 Closeout Submittals														
			Insulated Glass Units	1.7.1													
		08 91 00	SD-02 Shop Drawings														
			Wall Louvers	1.4													
			SD-03 Product Data														
			Metal Wall Louvers	2.2													
			Door Louvers	2.3													
			SD-04 Samples														
			Wall Louver Samples	1.5	G												
			Door Louver Samples	1.5	G												
		09 22 00	SD-02 Shop Drawings														
			Metal Support Systems	2.1	G												
			SD-03 Product Data														
			Metal Support Systems	2.1													
		09 29 00	SD-08 Manufacturer's Instructions														
			Safety Data Sheets	2.1													
			SD-10 Operation and Maintenance														
			Data														
			Manufacturer Maintenance	2.1													
			Instructions														
		09 30 10	SD-02 Shop Drawings														
			Detail Drawings	3.2	G												
			SD-10 Operation and Maintenance														
			Data														
			Installation	3.2	G												

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		09 51 00	SD-02 Shop Drawings														
			Approved Detail Drawings	2.1	G												
		09 62 38	SD-03 Product Data														
			Static-Control Resilient Flooring	2.1	G												
			Recycled content for Conductive Vinyl Tile	2.1.1.1	S												
			Recycled content for Conductive Rubber Tile	2.1.1.2	S												
			Recycled content for Conductive Rubber Sheet Flooring	2.1.1.3	S												
			Recycled content for Static-Dissipative Vinyl Tile	2.1.2.1	S												
			Recycled content for Static-Control Carpet	2.2	S												
			Accessories	2.6	G												
			Adhesives	2.4	G												
			Warranty	1.9													
			SD-04 Samples														
			Static-Control Resilient Flooring	2.1	G												
			Static-Control Carpet	2.2	G												
			Moldings	2.5	G												
			Special Treatment Materials	1.3.1.4	G												
			Accessories	2.6	G												
			SD-06 Test Reports														
			Fire Resistance	2.9													
			Moisture, Alkalinity and Bond	3.2													

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		09 62 38	Testing	3.9													
			SD-07 Certificates														
			Indoor Air Quality for Conductive Vinyl Tile	2.1.1.1	S												
			Indoor Air Quality for Conductive Rubber Tile	2.1.1.2	S												
			Indoor Air Quality for Conductive Rubber Sheet Flooring	2.1.1.3	S												
			Indoor Air Quality for Static-Dissipative Vinyl Tile	2.1.2.1	S												
			Indoor Air Quality for Static-Dissipative Rubber Tile	2.1.2.2	S												
			Indoor Air Quality for Static-Control Carpet	2.2	S												
			Indoor Air Quality for Adhesives	2.4	S												
			Qualifications of Applicator	1.6													
			SD-08 Manufacturer's Instructions														
			Static-Control Resilient Flooring	2.1	G												
			Accessories	2.6	G												
			SD-10 Operation and Maintenance Data														
			Static-Control Resilient Flooring	2.1	G												
			Accessories	2.6	G												
		09 65 00	SD-02 Shop Drawings														
			Resilient Flooring and Accessories	2.7	G												

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		09 65 00	SD-10 Operation and Maintenance Data														
			Resilient Flooring and Accessories	2.7	G												
		09 67 23.13	SD-02 Shop Drawings														
			Installation Drawings	2.1	G												
			Fabrication Drawings		G												
			SD-03 Product Data														
			Manufacturer's Catalog Data	1.2.2	G												
			SD-04 Samples														
			Hardboard Mounted Epoxy Flooring	1.5.3	G												
			Floor Topping	3.1.4	G												
			Mockups	1.5.1	G												
			SD-05 Design Data														
			Design Mix Data	1.2.3	G												
			SD-07 Certificates														
			Listing of Product Installations	1.5.2													
			Referenced Standards	1.5													
			Certificates														
			SD-11 Closeout Submittals														
			Warranty	1.6	G												
		09 67 23.16	SD-03 Product Data														
			Joint Sealant	2.1	G												
			Epoxy Mortar Flooring System	2.2	G												

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		09 67 23.16	White Aluminum Oxide Non-Skid Grit	2.3	G												
			SD-05 Design Data														
			Environmental Control System	1.3.3.1													
			SD-06 Test Reports														
			Joint Sealant Test Report	1.3.4.1	G												
			Primer Coat	2.2.1	G												
			Epoxy Mortar Coat	2.2.2	G												
			Grout Coat	2.2.3	G												
			Urethane Topcoat	2.2.4	G												
			White Aluminum Oxide Non-Skid Grit	2.3	G												
			Patch Test Demonstration	1.7	G												
			Daily Inspection Report	1.3.4.2	G												
			Adhesion Testing	3.10.3	G												
			SD-07 Certificates														
			Coating Work Plan	1.3.2	G												
			Qualifications of Coating Contractors	1.3.5.5	G												
			Joint Sealant Certificates	1.3.5.6	G												
			Epoxy Mortar Flooring System Certificates	1.3.5.7	G												
			Qualifications of Certified Industrial Hygienist (CIH)	1.3.5.1													

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		09 67 23.16	Qualifications of Certified Protective Coatings Specialist (PCS)	1.3.5.2													
			Qualifications of Coating Inspection Company	1.3.5.3													
			Qualifications of QC Specialist Coating Inspector	1.3.5.4													
			Warranty	1.8	G												
			SD-08 Manufacturer's Instructions														
			Joint Sealant Manufacturer's Instructions	1.3.6.1	G												
			Epoxy Mortar Flooring System Manufacturer's Instructions	1.3.6.2	G												
			Water-Based Alkaline Degreaser	1.3.6.3	G												
			SD-11 Closeout Submittals														
			Inspection Logbook	3.10.2.2	G												
		09 90 00	SD-02 Shop Drawings														
			Piping Identification	3.10													
			SD-08 Manufacturer's Instructions														
			Mixing	3.6.2													
			SD-10 Operation and Maintenance Data														
			Coatings	1.3.1.1													
		09 96 00	SD-01 Preconstruction Submittals														
			Equipment List	1.3	G												
			SD-03 Product Data														

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		09 96 00	Polyurethane Coatings	2.2.1	G												
			SD-04 Samples														
			Color Chips	1.3	G												
			SD-07 Certificates														
			Polyurethane Coatings	2.2.1	G												
			Manufacturer's Printed	3.1.4	G												
			Instructions														
		10 11 00	SD-02 Shop Drawings														
			Placement Schedule	3.1	G												
			SD-03 Product Data														
			Visual Display Unit	1.2	G												
			Visual Display Unit	2.1	G												
			Projection Screen	2.10	G												
			SD-04 Samples														
			Aluminum	2.1.6	G												
			Hardwood	2.1.7	G												
			Porcelain Enamel	2.1.1	G												
			Cork	2.1.2	G												
			Fabric	2.1.3	G												
			Vinyl Wall Covering	2.1.5	G												
			Glass	2.1.8	G												
			SD-07 Certificates														
			Indoor air quality for	2.3.1	S												
			markerboards														
			Indoor air quality for tackboards	2.4	S												

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		10 11 00	Indoor air quality for projection screen	2.10	S												
			Certificate of Compliance	1.2													
			SD-08 Manufacturer's Instructions														
			Manufacturer's Cleaning Instructions	3.3													
			Manufacturer's Printed Installation Instructions	3.2													
			SD-10 Operation and Maintenance Data														
			Visual Display Unit[s]	2.1	G												
		10 14 00.10	SD-02 Shop Drawings														
			Approved Detail Drawings	3.1	G												
			SD-10 Operation and Maintenance Data														
			Protection and Cleaning	3.1.2	G												
		10 14 00.20	SD-02 Shop Drawings														
			Detail Drawings	1.4.2	G												
			SD-10 Operation and Maintenance Data														
			Approved Manufacturer's Instructions	3.1	G												
			Protection and Cleaning	3.1.2	G												
		10 14 53	SD-03 Product Data														
			Traffic Sign Posts	2.1													
			FHWA Acceptance Letter	2.1.3													

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		10 14 53	Traffic Sign Retroreflective Sheeting	2.4													
			SD-04 Samples														
			Flexible Posts	2.6.2													
		10 21 13	SD-02 Shop Drawings														
			Fabrication Drawings	2.1													
			Installation Drawings	3.2	G												
			SD-07 Certificates														
			Warranty	1.5													
			SD-10 Operation and Maintenance Data														
			Plastic Identification	2.1.1	G												
		10 22 13	SD-02 Shop Drawings														
			Wire Mesh Partitions	1.4													
			SD-03 Product Data														
			Wire Mesh Partitions	1.4													
			Recycled Content for Metal Post and Framing Materials	2.1	S												
			Recycled Content for Wire Materials	2.1	S												
			SD-04 Samples														
			Samples	2.1	G												
		10 22 39	SD-02 Shop Drawings														
			Coordination Drawings	1.4.1	G												
			Layouts	3.1.1	G												
			Installation Drawings	3.1	G												

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		10 22 39	SD-03 Product Data														
			Folding Panel Partitions	2.3	G												
			Recycled Content for Steel Components														
			Recycled Content for Aluminum Components	2.3.1													
			SD-04 Samples														
			Partition System Samples	2.6	G												
			SD-06 Test Reports														
			Acoustical Test Reports	2.1.3.2	G												
			Flame and Smoke Development Tests	2.1.3.1	G												
			SD-07 Certificates														
			Indoor Air Quality for Finish Covering	2.3.2													
			Installer Qualifications	1.4.2													
			Manufacturer's Qualifications	1.4.3													
			SD-08 Manufacturer's Instructions														
			Installation Instructions	3.1													
			SD-10 Operation and Maintenance Data														
			Folding Panel Partitions	2.3	G												
		10 26 00	SD-02 Shop Drawings														
			Corner Guards	2.2	G												
			SD-03 Product Data														
			Corner Guards	2.2	G												

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		10 26 00	SD-04 Samples														
			Finish	2.4	G												
		10 28 13	SD-04 Samples														
			Finishes	2.1.2	G												
			Accessory Items	2.2													
			SD-10 Operation and Maintenance														
			Data														
			Electric Hand Dryer		G												
		10 44 16	SD-02 Shop Drawings														
			Fire Extinguishers	2.1.1	G												
			Accessories	Part 2	G												
			Cabinets	Part 2	G												
			Wall Brackets	2.2.2	G												
			Schedule	1.5	G												
			SD-03 Product Data														
			Fire Extinguishers	2.1.1	G												
			Accessories	Part 2	G												
			Cabinets	Part 2	G												
			Wall Brackets	2.2.2	G												
			Replacement Parts List	3.2.1	G												
			SD-04 Samples														
			Equipment Samples	1.3.1	G												
			SD-07 Certificates														
			Fire Extinguishers Certifications	2.1.1	G												
			Manufacturer's Warranty with	1.4	G												
			Inspection Tag														

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		10 51 13	SD-02 Shop Drawings														
			Types	2.2	G												
			Location	1.4	G												
			Installation	3.1													
			Numbering System	3.2													
			SD-04 Samples														
			Color Chips	1.5.1	G												
		11 13 19.13	SD-02 Shop Drawings														
			Detail Drawings	2.1.1.1	G												
			SD-03 Product Data														
			Loading Dock Levelers	2.3	G												
			Dock Bumpers	2.3.5.2	G												
			Restraining Device	2.8.1	G												
			SD-04 Samples														
			Dock Bumpers	2.3.5.2													
			Rubber	2.3.5.2													
			SD-07 Certificates														
			Factory Test	2.9													
			SD-10 Operation and Maintenance Data														
			Loading Dock Levelers	2.3	G												
			Restraining Device	2.8.1	G												
			SD-11 Closeout Submittals														
			Record Drawings	3.6.1	G												
		12 00 01.00 20	SD-01 Preconstruction Submittals														

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		12 00 01.00 20	FF&E Schedule and Schedule Updates	3.1.1.3	G												
			Contractor's Interior Designer's Qualifications	3.1.1.1	G												
			SD-04 Samples														
			BVD FF&E Mock-Up	3.1.2.7	G												
			Final FF&E Mock-Up	3.1.2.10	G												
			Interior Finish Construction	3.1.2.15	G												
			Submittals														
			Post Option Award FF&E Finish	3.2.3	G												
			Submittals														
			SD-05 Design Data														
			FF&E Basis of Design Package	3.1.2.3.2	G												
			FF&E BVD Request for Quotation (RFQ) Package	3.1.2.6	G												
			BVD Package	3.1.2.8	G												
			Pre-Final FF&E Package 'Over the Shoulder'	3.1.2.11	G												
			Pre-Final FF&E Package	3.1.2.12	G												
			Final FF&E Package	3.1.2.13	G												
			Final FF&E Package Price Validation	3.1.2.14	G												
			SD-10 Operation and Maintenance Data														
			Warranty and Maintenance Information	3.5.1	G												

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		12 00 01.00 20	SD-11 Closeout Submittals														
			Punch List(s) Submittal	3.3.2	G												
			Interior Photography	3.5.2	G												
		12 24 13	SD-02 Shop Drawings														
			Detailed Drawings	3.2	G												
			Location Schedule	2.1	G												
			SD-03 Product Data														
			Window Shades	2.1	G												
			SD-04 Samples														
			Window Shades	2.1	G												
			SD-06 Test Reports														
			Flammability Requirements	1.3.2	G												
			SD-10 Operation and Maintenance														
			Data														
			Window Shades	2.1	G												
		12 48 13	SD-02 Shop Drawings														
			Installation Drawings	3.2	G												
			Detail Drawings	3.2	G												
			SD-03 Product Data														
			Entrance Floor Mats and Frames	2.1.1	G												
			Adhesives and Concrete Primers	2.1.2	G												
			SD-10 Operation and Maintenance														
			Data														
			Protection, Maintenance, and	3.2													
			Repair Information														
		12 50 00.13 10	SD-01 Preconstruction Submittals														

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		12 50 00.13 10	Storage Location	1.7.3	G												
			SD-02 Shop Drawings														
			Installation Drawings	3.3.1	G												
			Grommet[, Power and	3.3.1	G												
			Communication Units][, and Wire														
			Management] Locations														
			SD-03 Product Data														
			Product Data	2.3	G												
			Product Style Options	2.3	G												
			SD-04 Samples														
			Fabric and Finishes	2.3.6	G												
			SD-07 Certificates														
			Authorized Dealer	1.6	G												
			Certified Furniture Installers	1.6	G												
			Licensed Electrician	1.6	G												
			Certified Telecommunications	1.6	G												
			Installer														
			Manufacturer's Certification	2.3	G												
			Warranty	1.8	G												
			SD-10 Operation and Maintenance														
			Data														
			Furniture, Data Package 1	3.5	G												
			SD-11 Closeout Submittals														
			Energy Efficient Equipment	2.1.1	S												
			Reduced VOC's for Furniture	2.1.2	S												
			Recycled Content of Furniture	2.1.3	S												

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		12 50 00.13 10	Bio-Based Content of Furniture	2.1.4	S												
		12 59 00	SD-02 Shop Drawings														
			Detail Drawings	1.4.4	G												
			SD-03 Product Data														
			Warranty	1.6	G												
			Workstations	2.2.1													
			Power and Communications	2.11													
			Communications	2.11.8													
			Recycled Content for system furniture components	2.1	S												
			Energy Star Label for Task Lighting	2.11.7	S												
			SD-04 Samples														
			Workstations	2.2.1	G												
			Mock-up	2.2.3	G												
			Samples	2.2.2													
			SD-06 Test Reports														
			Selected Components	2.2.5.1	G												
			Panel Acoustics	2.2.5.2	G												
			Fire Safety	1.4.2	G												
			Electrical System	1.4.3	G												
			SD-07 Certificates														
			Workstations	2.2.1													
			door panels	2.3.8	S												
			SD-10 Operation and Maintenance Data														

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		12 59 00	Assembly Manuals	2.3.1	G												
			Maintenance Manuals	3.2	G												
			Cleaning	3.2	G												
			Electrical System	1.4.3	G												
			Installation	3.1	G												
		13 48 73	SD-01 Preconstruction Submittals														
			Independent Reviewer	2.1.1	G												
			Qualifications														
			SD-02 Shop Drawings														
			Supports and Attachments Shop	2.1	G												
			Drawings														
			Resilient Vibration Isolation	3.4	G												
			Devices														
			SD-03 Product Data														
			Material Supports	2.2.1	G												
			Common Bolts	2.2.2	G												
			High-Strength Bolts	2.2.2	G												
			Anchors Certified Manufacturer's	2.2.3	G												
			Mill Reports														
			Vibration Isolators	2.1.3	G												
			Isolation and Energy Dissipation	2.1	G												
			Systems														
			Isolation and Energy Dissipation	2.1	G												
			Systems														
			Snubbers	2.1.3	G												
			Welding Electrodes And Rods	2.2.4													

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		13 48 73	SD-05 Design Data														
			Design Calculations and Drawings	2.1	G												
			Details of Equipment Supports and Attachments	3.5.1	G												
			Independent Design Review Report	2.1.1	G												
			SD-06 Test Reports														
			Anchor Test Results	3.3.1	G												
			Shake Table Test	3.6	G												
			SD-07 Certificates														
			Certificate Of Completion	3.7	G												
			Welder Qualifications	3.1	G												
		14 21 23	SD-02 Shop Drawings														
			Elevator	2.1	G												
			Elevator Components	1.2.1	G												
			Elevator Components	1.2.2	G												
			Elevator Machine	1.2.1	G												
			Elevator Controller	1.2.1	G												
			Wiring Diagrams	1.3.4	G												
			SD-03 Product Data														
			Elevator	2.1	G												
			Elevator Components	1.2.1	G												
			Elevator Components	1.2.2	G												
			Data Sheets	1.2.2	G												

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		14 21 23	Elevator Microprocessor Controller	2.5.2	G												
			SD-05 Design Data														
			Heat Loads	1.2.3.2													
			Reaction Loads	1.2.3.1													
			SD-07 Certificates														
			Warranty	1.4													
			Endorsement Letter	1.3.1.1													
			Welders' Qualifications	1.2.4													
			SD-10 Operation and Maintenance Data														
			Elevator	2.1	G												
			Maintenance Control Program (MCP)	1.2.5	G												
		21 13 13	SD-01 Preconstruction Submittals														
			Qualified Fire Protection Engineer (QFPE)	1.2.3	G												
			Sprinkler System Designer	1.4.2.1	G												
			Sprinkler System Installer	1.4.2.2	G												
			SD-02 Shop Drawings														
			Shop Drawing	1.2.1.1	G												
			SD-03 Product Data														
			Pipe	2.2.1	G												
			Fittings	2.3.1.2	G												
			Valves	2.3.4	G												
			Relief Valves	2.8.5	G												

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		21 13 13	Sprinklers	2.7	G												
			Pipe Hangers and Supports	2.3.3	G												
			Sprinkler Alarm Switch	2.4.1	G												
			Valve Supervisory (Tamper) Switch	2.4.2	G												
			Fire Department Connection	2.6	G												
			Backflow Prevention Assembly	2.5	G												
			Air Vent	2.8.6	G												
			Hose Valve	2.5.1	G												
			Seismic Bracing	2.3.3	G												
			Nameplates	2.1.2	G												
			SD-05 Design Data														
			Seismic Bracing	2.3.3	G												
			Hydraulic Calculations	1.2.1.2	G												
			SD-06 Test Reports														
			Test Procedures	3.7.1	G												
			SD-07 Certificates														
			Verification of Compliant Installation	3.7.2.1	G												
			Request for Government Final Test	3.7.2.2	G												
			SD-10 Operation and Maintenance Data														
			Operating and Maintenance (O&M) Instructions	3.9	G												
			Spare Parts	1.6	G												

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		21 13 13	SD-11 Closeout Submittals														
			As-built drawings	3.9													
		21 30 00	SD-01 Preconstruction Submittals														
			Fire Pump Installation Related Submittals	1.3													
			Fire Protection Specialist	1.7.2	G												
			SD-02 Shop Drawings														
			Installation Drawings	3.3.1	G												
			As-Built Drawings	3.11.2	G												
			Piping Layout	3.3.2	G												
			Pump Room	3.3.2	G												
			SD-03 Product Data														
			Catalog Data	2.1	G												
			Spare Parts	1.6													
			Preliminary Tests	3.8.2													
			Field Tests	3.8	G												
			Manufacturer's Representative	1.7.7	G												
			Field Training	3.11.1	G												
			Final Acceptance Test	3.8.3													
			SD-06 Test Reports														
			Preliminary Tests	3.8.2													
			SD-07 Certificates														
			Fire Protection Specialist	1.7.2													
			Qualifications of Welders	1.7.3													
			Qualifications of Installer	1.7.4													
			Preliminary Test Certification	1.7.5													

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		21 30 00	Final Test Certification	1.7.6													
			SD-10 Operation and Maintenance Data														
			Operating and Maintenance Instructions	3.11.1	G												
			Flow Meter	2.14													
		22 00 00	SD-02 Shop Drawings														
			Plumbing System	3.9.1	G												
			SD-03 Product Data														
			Pipe Hangers, Supports and Anchors	2.15.3													
			Backflow Prevention Assemblies	3.9.1.1	G												
			Shower Faucets	2.6.2	G												
			Fixtures	2.4													
			Flush Valve Water Closets	2.4.2													
			Flush Valve Urinals	2.4.3													
			Kitchen Sinks	2.4.5													
			Service Sinks	2.4.6													
			Drinking-Water Coolers	2.4.7													
			Shower Stalls														
			Air Source Heat Pump Hot Water Heater	2.9.1													
			Hot-Water Storage Tanks	2.10													
			Expansion Tank	3.2.4													
			Air Compressors	2.12.1													
			Air Receivers	2.12.3													

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		22 00 00	Water Heaters	2.9													
			Pumps	2.11													
			SD-06 Test Reports														
			Tests, Flushing and Disinfection	3.9													
			Test of Backflow Prevention	3.9.1.1	G												
			Assemblies														
			SD-10 Operation and Maintenance														
			Data														
			Plumbing System	3.9.1													
		22 05 48.00 20	SD-02 Shop Drawings														
			Inertia Bases	2.7													
			Machinery Bases	2.6													
			Platforms	2.6													
			Rails	2.6													
			Saddles	2.6													
			SD-03 Product Data														
			Isolators	2.3													
			Flexible Connectors	2.8													
			Flexible Duct Connectors	2.9													
			Pipe Guides	2.11													
			Seismic Snubbers	2.10													
			Vertical Stops	3.1.3													
			Thrust Restraints	2.12													
			Inertia Bases	2.7													
			Machinery Bases	2.6													

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		22 05 48.00 20	Machinery Foundations and Subbases	3.1.13													
			Platforms	2.6													
			Rails	2.6													
			Saddles	2.6													
			Machinery Manufacturer's Sound Data	1.4.2													
			SD-05 Design Data														
			Inertia Bases	2.7													
			Machinery Bases	2.6													
			Platforms	2.6													
			Rails	2.6													
			Saddles	2.6													
			Machinery	1.4.3													
			Machinery over 136 Kilograms	1.4.4													
			SD-06 Test Reports														
			Seismic Snubbers	2.10													
			Equipment Vibration Tests	3.2.3.1													
			Equipment Sound Level Tests	3.2.3.2													
			Protected Spring Isolators	2.4													
			SD-08 Manufacturer's Instructions														
			Vibration and Noise Isolation	3.1.1													
			Components														
			Seismic Protection Components	2.13													
		23 05 48.19	SD-02 Shop Drawings														
			Coupling and Bracing	3.1													

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		23 05 48.19	Flexible Couplings or Joints	3.3													
			Equipment Restraint	2.2													
			Contractor Designed Bracing	1.2.4													
			SD-03 Product Data														
			Coupling and Bracing	3.1													
			Flexible Couplings Or Joints	3.3													
			Equipment Restraint	2.2													
			Contractor Designed Bracing	1.2.4													
			Anchor Bolts	3.9													
			Vibration Isolators	2.2.2													
			SD-05 Design Data														
			Design Calculations	1.2.4													
			SD-06 Test Reports														
			Anchor Bolts	3.9													
		23 05 93	SD-01 Preconstruction Submittals														
			TAB Firm And Personnel	1.6													
			Qualifications														
			Design Review Report	1.9													
			TAB Pre-Field Engineering Plan	1.10													
			Pre-Final DALT Report	3.1.6													
			Certified Final DALT Report	3.1.8													
			Certified TAB Report	3.2.8													
		23 07 00	SD-02 Shop Drawings														
			Pipe Insulation Systems	2.3													
			Duct Insulation Systems	3.3													
			Equipment Insulation Systems	3.4													

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		23 08 00	SD-03 Product Data														
			Test Equipment	2.1													
			SD-06 Test Reports														
			Pipe Flushing, Testing, And Water Treatment Reports	1.5													
			Completed Pre-Functional Checklists	3.3													
			Seasonal Test Report	3.12													
			Full-Load Test Report	3.13													
			Post-Construction Trend Log Report	3.16													
			SD-07 Certificates														
			Certificate Of Readiness	1.6													
		23 08 01.00 20	SD-06 Test Reports														
			Preliminary review report	1.4.5.1	G												
			Smoke tests report	1.4.5.2	G												
			Fan operating points report	1.4.5.3	G												
			Static pressure report	1.4.5.4	G												
			Volume and velocity flow rates report	1.4.5.5	G												
			Pitot traverse report	1.4.5.6	G												
			SD-07 Certificates														
			Test agency Qualifications	1.4.2.1	G												
			Record of Document Submittal to Testing Agency	1.4.2.2	G												
			Work plan	1.4.2.3	G												

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		23 08 01.00 20	List of test instruments	1.4.2.4	G												
		23 09 00	SD-02 Shop Drawings														
			DDC Contractor Design Drawings	3.2													
			Draft As-Built Drawings	3.2													
			Final As-Built Drawings	3.2													
			SD-03 Product Data														
			Programming Software	1.8.3													
			Controller Application Programs	1.8.4													
			Configuration Software	1.8.1													
			Controller Configuration Settings	1.8.2													
			Proprietary Multi-Split	1.1.1.3													
			Engineering Tool Software														
			Manufacturer's Product Data	2.2													
			SD-05 Design Data														
			Chiller Plant Gateway Request	1.9													
			SD-06 Test Reports														
			Pre-Construction Quality Control	1.10.1													
			(QC) Checklist														
			Post-Construction Quality Control	1.10.2													
			(QC) Checklist														
			Start-Up Testing Report	3.4.2													
			PVT Procedures	3.5.1													
			PVT Report	3.5.3													
			SD-10 Operation and Maintenance														
			Data														

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		23 09 00	Operation and Maintenance (O&M) Instructions	3.6													
			Training Documentation	3.8.1													
			SD-11 Closeout Submittals														
			Enclosure Keys	2.5													
			Password Summary Report	3.1.6.1													
			Closeout Quality Control (QC) Checklist	1.10.3													
		23 21 23	SD-02 Shop Drawings														
			System Coordination	2.1.2	G												
			SD-03 Product Data														
			Base-Mounted, Close Coupled, End Suction Pump[s]	2.3.5	G												
			Base-Mounted, Flexible Coupled, End suction Pump[s]	2.3.4	G												
			Base-Mounted, Flexible Coupled, Double Suction[,] [Horizontally Split][and][Vertically Split] Pump[s]	2.3.6	G												
			Circulator Pump[s]	2.3.1	G												
			Cooling Coil Condensate Pump Units	2.3.8	G												
			Equipment Data	2.2.5	G												
			Installation Instructions	2.2.5	G												
			Large In-Line Pump[s]	2.3.3	G												
			Packaged Pump System	2.6	G												

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		23 21 23	Small In-Line Pump[s]	2.3.2	G												
			Vertical Lineshaft Turbine Pump[s]	2.3.7	G												
			Training Period	3.5.2	G												
			Framed Instructions	2.2.2	G												
			SD-06 Test Reports														
			Factory Tests	2.9													
			Field Quality Control	3.3													
			Pump Motor/Alignment Report	3.5.3	G												
			SD-07 Certificates														
			Manufacturer's Representative	1.3.1													
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	3.5.1	G												
			Training	3.5.2	G												
		23 23 00	SD-02 Shop Drawings														
			Refrigerant Piping System	2.2													
			SD-06 Test Reports														
			Refrigerant Piping Tests	3.4													
			SD-10 Operation and Maintenance Data														
			Maintenance	1.5													
			Operation and Maintenance Manuals	3.2.4.3													
		23 25 00	SD-03 Product Data														

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		23 25 00	Water Analysis	2.1.3													
			Spare Parts	1.3													
			Field Instructions	3.4.1													
			Tests	3.3.1													
			Training Course	3.4.1													
		23 30 00	SD-02 Shop Drawings														
			Detail Drawings	1.4.4													
			SD-03 Product Data														
			Fire Dampers	2.8.3													
			Air Handling Units	2.10													
			Variable Volume, Single Duct	2.11.1.1													
			Terminal Units														
			Test Procedures	1.4.5													
			Diagrams	1.2.1.2													
			SD-06 Test Reports														
			Performance Tests	3.11													
			Damper Acceptance Test	3.9													
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.13.1													
			Manuals														
			Fire Dampers	2.8.3													
			Manual Balancing Dampers	2.8.4													
			Centrifugal Fans	2.9.1.1													
			In-Line Centrifugal Fans	2.9.1.2													
			Ceiling Exhaust Fans	2.9.1.3													

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		23 30 00	Air Handling Units	2.10													
			Variable Volume, Single Duct	2.11.1.1													
			Terminal Units														
		23 35 16	SD-02 Shop Drawings														
			Detail Drawings	1.4.1													
			SD-03 Product Data														
			Related Submittals	1.4.2													
			Ductwork Components	2.4													
			Materials and Equipment	2.1													
			Spare Parts	1.6													
			Field Instructions	3.6													
			Onsite Training	3.6													
			Exhaust System Specialist	1.4.2													
			SD-06 Test Reports														
			Final Acceptance Tests	3.7													
			SD-07 Certificates														
			Inspection	3.3													
			SD-10 Operation and Maintenance														
			Data														
			Exhaust System	1.2													
			Operation and Maintenance	3.6													
			Manuals														
		23 35 19.00 20	SD-02 Shop Drawings														
			Industrial Ventilation and Exhaust	1.2.4	G												
			Systems														
			SD-03 Product Data														

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		23 35 19.00 20	Fans	2.1	G												
			Dampers	2.7	G												
			Flexible Connectors	2.8.3													
			Flexible Duct	2.8.4	G												
			Gaskets	2.8.5													
			Protective Coating Materials	2.8.6													
			Sealants	2.8.7													
			Access Ports	2.9.1	G												
			Damper Regulators	2.9.2	G												
			Blast Gates	2.9.3	G												
			Vibration Isolators	2.10.5	G												
			Ductwork, Dust [and Fume] Collection	2.11													
			Steel Ducts	2.12	G												
			Fiberglass Ductwork	2.14	G												
			Thermoplastic Ductwork	2.13	G												
			Vehicle Tail Pipe Exhaust System	2.15	G												
			Welding Fume Exhaust System	2.16	G												
			Recycled Content of Ductwork	2.11.1	S												
			Steel Components														
			Recycled Content of Protectively Coated Steel Ducts	2.12.1	S												
			Indoor Air Quality for Duct	2.8.7.1	S												
			Sealants														
			SD-06 Test Reports														
			Fan Tests	2.1.1	G												

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		23 35 19.00 20	Start-Up Tests	1.2.5	G												
			Sound Level Tests	3.2.6	G												
			SD-07 Certificates														
			Welding Procedures	1.4.4	G												
			Welding Test Agenda	3.1.11	G												
			Welding Test Procedures	1.4.4	G												
			Welders' Identification	1.4.1	G												
			Fiberglass Fan Servicer	1.4.2	G												
			Experience Information														
			SD-10 Operation and Maintenance														
			Data														
			Fans	2.1	G												
			Vehicle Tail Pipe Exhaust System	2.15	G												
			Welding Fume Exhaust System	2.16	G												
			Industrial Ventilation and Exhaust	1.2.4	G												
			Systems														
			SD-11 Closeout Submittals														
			Posted Operating Instructions	1.5													
		23 64 10	SD-03 Product Data														
			Posted Instructions	3.1.3													
			Verification of Dimensions	1.6.1													
			System Performance Tests	3.6													
			Demonstrations	3.7													
			Refrigerant	2.5.1													
			Water Chiller - Field Acceptance	3.5.1													
			Test Plan														

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		23 64 10	Scroll Type Water Chillers	2.4.1													
			SD-06 Test Reports														
			Field Acceptance Testing	3.5													
			Water Chiller - Field Acceptance Test Report	3.5.2													
			Factory Tests	2.8													
			System Performance Tests	3.6													
			Factory Test Schedule	3.2													
			SD-07 Certificates														
			Refrigeration Technician Certification	1.3.1													
			SD-08 Manufacturer's Instructions														
			Water Chiller - Installation Instructions	3.1													
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	3.7													
			Training Course Schedule	3.7													
			SD-11 Closeout Submittals														
			Indoor Air Quality During Construction	3.4													
		23 64 26	SD-03 Product Data														
			Pumps	2.6													
			SD-06 Test Reports														
			Piping Welds NDE Report	3.1.1.3													

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		23 64 26	Pressure Tests Reports	3.4.2													
			SD-10 Operation and Maintenance Data														
			Calibrated Balancing Valves	2.4.8													
			Automatic Flow Control Valves	2.4.9													
			Pump Discharge Valve	2.4.10													
			Water Temperature Mixing Valve	2.4.11													
			Water Temperature Regulating Valves	2.4.12													
			Water Pressure Reducing Valve	2.4.13													
			Pressure Relief Valve	2.4.14													
			Combination Pressure and Temperature Relief Valves	2.4.15													
			Expansion Joints	2.5.8													
			Pumps	2.6													
			Combination Strainer and Pump Suction Diffuser	2.5.2													
			Expansion Tanks	2.7													
			Air Separator Tanks	2.8													
		23 81 00	SD-02 Shop Drawings														
			Control System Wiring Diagrams	1.3.2													
			SD-03 Product Data														
			Air Conditioners	2.1													
			Training	3.8													
			Posted Instructions	3.8													
			Spare Parts	3.9.1													

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		23 81 00	SD-06 Test Reports														
			Start-Up and Initial Operational Tests	3.7.3													
			SD-07 Certificates														
			Service Organizations	3.9.2													
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	3.8													
			Filters	2.2													
			Thermostats	2.1.11													
		25 05 11	SD-01 Preconstruction Submittals														
			Wireless Communication Request	3.1.5.3	G												
			Device Account Lock Exception Request	3.1.2.2	G												
			Multiple IP Connection Device Request	3.9	G												
			Contractor Computer Cybersecurity Compliance Statements	1.5.1.4	G												
			Contractor Temporary Network Cybersecurity Compliance Statements	1.5.6	G												
			SD-02 Shop Drawings														
			User Interface Banner Schedule	3.1.3.1	G												

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		25 05 11	Network Communication Report	1.3.2	G												
			Cybersecurity Riser Diagram	1.3.4	G												
			Control System Inventory Report	1.3.3	G												
			Cybersecurity Interconnection	1.3.1	G												
			Schedule														
			SD-03 Product Data														
			Control System Cybersecurity	1.5	G												
			SD-11 Closeout Submittals														
			Password Summary Report	3.5.2.2.5	G												
			Device Audit Record Upload	3.2.2.1	G												
			Software														
		26 05 13	SD-01 Preconstruction Submittals														
			List of Splices and Terminations	1.4.1	G												
			to be Installed by														
			Splicer/Terminator														
			SD-02 Shop Drawings														
			Pulling Plan	3.1.3	G												
			SD-03 Product Data														
			Multiple-Conductor Shielded		G												
			Cables														
			Multiple-Conductor Nonshielded		G												
			Cables														
			Single-Conductor Shielded	2.1.2	G												
			Cables														
			Single-Conductor Nonshielded		G												
			Cables														

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		26 05 13	Ground Conductors	2.3.1	G												
			Cable Cleats		G												
			Flexible and Portable Cable		G												
			Cable Supports and Fittings	2.5.1	G												
			Cable Grips	3.1.5.3	G												
			Cable Tags	2.5.2	G												
			Fireproof Tape	2.5.3	G												
			Splices	2.4.2	G												
			Terminations	2.4.3	G												
			Metal-Clad and Armored Cable		G												
			Jumper Cables		G												
			Cable Identification	3.1.8	G												
			SD-05 Design Data														
			Cable Pulling Calculations	3.1.2	G												
			Cleat Manufacturer's Calculations		G												
			SD-06 Test Reports														
			Factory Inspection and Tests	1.6	G												
			Field Acceptance Testing	3.3	G												
			Wire And Cable Test Record	3.3.1	G												
			Form														
			Phase Rotation Test Record	3.1.1	G												
			Form														
			SD-07 Certificates														
			Splicer/Terminator Certifications	1.5.3	G												
			SD-08 Manufacturer's Instructions														
			Terminations	2.4.3	G												

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		26 05 13	Splices	2.4.2	G												
		26 05 48	SD-02 Shop Drawings														
			Equipment Requirements	2.1	G												
			Lighting Fixtures	3.4	G												
			SD-03 Product Data														
			Supports and Attachments	1.2.3	G												
			Equipment Requirements	2.1	G												
			Lighting Fixtures	3.4	G												
			Flexible Fittings	2.4	G												
			Anchors	3.5	G												
			Isolation And Energy Dissipation Systems	2.1	G												
			SD-05 Design Data														
			Design Calculations	1.2.3	G												
			Design Drawings	1.2.3	G												
			Independent Design Review Report	1.2.3	G												
			SD-06 Test Reports														
			Anchors	3.5	G												
			SD-07 Certificates														
			ICC ES AC156 Shake Table Test	3.7	G												
		26 05 48.00 10	SD-02 Shop Drawings														
			Lighting Fixtures in Buildings	3.2													
			Equipment Requirements	1.3													
			SD-03 Product Data														
			Lighting Fixtures in Buildings	3.2	G												

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		26 05 48.00 10	Equipment Requirements	1.3	G												
			Contractor Designed Bracing	1.2.4	G												
		26 08 00	SD-06 Test Reports														
			Acceptance Tests and Inspections	3.1	G												
			SD-07 Certificates														
			Qualifications	1.4.1	G												
			Acceptance Test and Inspections Procedure	1.4.3	G												
		26 12 19	SD-02 Shop Drawings														
			Pad-mounted Transformer Drawings	1.5.1	G												
			SD-03 Product Data														
			Pad-mounted Transformers	2.2	G												
			SD-06 Test Reports														
			Acceptance Checks and Tests	3.7.1	G												
			SD-07 Certificates														
			Transformer Efficiencies	2.2.2.1	G												
			SD-09 Manufacturer's Field Reports														
			Transformer Test Schedule	2.8.1	G												
			Design Tests	2.8.2	G												
			Routine and Other Tests	2.8.3	G												
			SD-10 Operation and Maintenance Data														
			Transformer(s)	1.6.1	G												

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		26 12 19.10	SD-02 Shop Drawings														
			Pad-mounted Transformer	1.5.1	G												
			Drawings														
			SD-03 Product Data														
			Pad-mounted Transformers	2.2	G												
			SD-06 Test Reports														
			Acceptance Checks and Tests	3.7.1													
			SD-07 Certificates														
			Transformer Efficiencies	2.2.2.1													
			SD-10 Operation and Maintenance														
			Data														
			Transformer(s)	1.6.1													
		26 13 01	SD-02 Shop Drawings														
			Switchgear Drawings	1.5.1	G												
			SD-03 Product Data														
			Fuse Time-Current Characteristic	1.5.3	G												
			Curves														
			Air Insulated Pad-mounted	2.1	G												
			Switchgear														
			Insulated High-Voltage	2.2	G												
			Connectors														
			Surge Arresters	2.3	G												
			SD-06 Test Reports														
			Acceptance Checks and Tests	3.4.1	G												
			SD-07 Certificates														
			Paint Coating System	1.5.2	G												

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		26 13 01	SD-09 Manufacturer's Field Reports														
			Switchgear Design and Production Tests	2.6.1	G												
			SD-10 Operation and Maintenance Data														
			Air Insulated Pad-Mounted Switchgear Operation and Maintenance	1.6.1	G												
		26 20 00	SD-02 Shop Drawings														
			Panelboards	2.12	G												
			Transformers	2.14	G												
			Cable Trays	2.3	G												
			Wireways	2.25	G												
			Marking Strips	3.1.10.1	G												
			SD-03 Product Data														
			Switches	2.9	G												
			Receptacles	2.11	G												
			Circuit Breakers	2.12.3	G												
			Fuses	2.10	G												
			Enclosed Circuit Breakers	2.13	G												
			Transformers	2.14	G												
			Motor Controllers	2.16	G												
			Combination Motor Controllers		G												
			Manual Motor Starters	2.17	G												
			CATV Outlets	2.20.1.1	G												

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		26 20 00	Secondary Bonding Busbar	2.20.2	G												
			Surge Protective Devices	2.28	G												
			SD-06 Test Reports														
			600-volt Wiring Test	3.5.2	G												
			Grounding System Test	3.5.6	G												
			Transformer Tests	3.5.3	G												
			Ground-fault Receptacle Test	3.5.4	G												
			SD-07 Certificates														
			Fuses	2.10	G												
			SD-09 Manufacturer's Field Reports														
			Transformer Factory Tests	2.30.1													
			SD-10 Operation and Maintenance Data														
			Electrical Systems	1.6.1	G												
			Metering	2.26	G												
		26 24 13	SD-02 Shop Drawings														
			Switchboard Drawings	1.5.2	G												
			SD-03 Product Data														
			Switchboard	2.2	G												
			SD-06 Test Reports														
			Switchboard Design Tests	2.5.2	G												
			Switchboard Production Tests	2.5.3	G												
			Acceptance Checks and Tests	3.5.1	G												
			SD-07 Certificates														

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		26 24 13	Cybersecurity Installation Certification	3.5.1.7	G												
			SD-10 Operation and Maintenance Data														
			Switchboard Operation and Maintenance	1.6.1	G												
			SD-11 Closeout Submittals Assembled Operation and Maintenance Manuals	1.6.2	G												
			Equipment Test Schedule	2.5.1	G												
			Request for Settings	3.5	G												
			Required Settings	3.5	G												
			Service Entrance Available Fault Current Label	2.8	G												
		26 29 23	SD-02 Shop Drawings														
			Schematic Diagrams	1.5.1	G												
			Interconnecting Diagrams	1.5.2	G												
			Installation Drawings	1.5.3	G												
			As-Built Drawings	1.5.3	G												
			SD-03 Product Data														
			Adjustable Speed Drives	2.1	G												
			Wires and Cables	2.3													
			Equipment Schedule	1.5.4													
			SD-06 Test Reports														
			ASD Test	3.3.1													
			Performance Verification Tests	3.3.2													

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		26 29 23	Endurance Test	3.3.3													
			SD-07 Certificates														
			Testing Agency's Field Supervisor	3.3.1	G												
			SD-08 Manufacturer's Instructions														
			Installation instructions	1.5.5													
			SD-09 Manufacturer's Field Reports														
			ASD Test Plan	2.5.1	G												
			Standard Products	1.5.6													
			SD-10 Operation and Maintenance Data														
			Adjustable Speed Drives	2.1													
		26 41 00	SD-06 Test Reports														
			Lightning Protection and Grounding System Test Plan	1.4.2	G												
			Grounding Systems Testing	3.5.2	G												
			SD-07 Certificates														
			Component UL Listed and Labeled	1.4.1	G												
			Lightning Protection System Inspection Certificate	1.4.3	G												
			Roof Manufacturer's Warranty	3.1.1.1	G												
		26 51 00	SD-02 Shop Drawings														
			Luminaire Drawings	1.5.1	G												

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		26 51 00	Occupancy/Vacancy Sensor Coverage Layout	1.5.9	G												
			Lighting Control System One-Line Diagram	1.7.2	G												
			Sequence of Operation for Lighting Control System	2.5.1	G												
			SD-03 Product Data														
			Luminaires	2.2	G												
			Light Sources	2.3	G												
				03													
			Luminaire Warranty	1.6.1	G												
			Lighting Controls Warranty	1.6.2	G												
			Local Area Controller	2.5.1.1.1	G												
			Lighting Relay Panel	2.5.1.2.1	G												
			Lighting Control Panel	2.5.1.2.2	G												
			Gateway	2.5.1.2.3	G												
			Lighting Contactor	2.5.1.2.4	G												
			Switches	2.5.2.1	G												
			Digital Switch Timers	2.5.2.2													
			Wall Box Dimmers	2.5.2.3	G												
			Scene Wallstations	2.5.2.4	G												
			Occupancy/Vacancy Sensors	2.5.2.5	G												
			Photosensors	2.5.2.6	G												
			Power Packs	2.5.2.5.6	G												
			Power Hook Luminaire Hangers	2.7.4	G												
			Mini Inverters	2.6.5	G												

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		26 51 00	Exit Signs	2.6.1	G												
			Emergency Drivers	2.6.3	G												
			Energy Star Label For Residential Luminaires	2.2	S												
			Linear LED Lamps	2.3.1.1	G												
			SD-05 Design Data														
			Luminaire Design Data	1.5.2	G												
			Photometric Plan	1.5.8	G												
			SD-06 Test Reports														
			ANSI/IES LM-79 Test Report	1.5.3	G												
			ANSI/IES LM-80 Test Report	1.5.4	G												
			ANSI/IES TM-21 Test Report	1.5.5	G												
			ANSI/IES TM-30 Test Report	1.5.6	G												
			Occupancy/Vacancy Sensor Verification Test	3.2.1.1	G												
			Photosensor Verification Test	3.2.1.1	G												
			SD-07 Certificates														
			LED Driver and Dimming Switch Compatibility Certificate	1.5.7	G												
			SD-10 Operation and Maintenance Data														
			Lighting System	1.7.1	G												
			Lighting Control System	1.7.2	G												
			Maintenance Staff Training Plan	3.3.2.1	G												
			End-User Training Plan	3.3.2.2	G												
		26 56 00	SD-02 Shop Drawings														

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		26 56 00	Luminaire Drawings	1.5.1.1	G												
			Pole Drawings	1.5.1.2	G												
			Control System One-Line Diagram	1.8.2	G												
			Sequence of Operation for Exterior Lighting Control System	2.5.1	G												
			SD-03 Product Data														
			Luminaires	2.2	G												
			Light Sources	2.3	G												
			LED Drivers	2.4	G												
			Luminaire Warranty	1.7.1	G												
			Lighting Controls Warranty	1.7.2	G												
			Pole Warranty	1.7.3	G												
			Dimming Panel		G												
			Motion Sensors	2.5.2.3	G												
			Photosensors	2.5.2.2	G												
			Time Clock	2.5.2.1	G												
			Lighting Contactor	2.5.2.4	G												
			Poles	2.6	G												
			Brackets	2.6.2													
			Obstruction Marker Luminaires	2.2.2	G												
			SD-05 Design Data														
			Luminaire Design Data	1.5.2	G												
			Photometric Plan	1.5.8	G												
			SD-06 Test Reports														
			ANSI/IES LM-79 Test Report	1.5.3	G												

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		26 56 00	ANSI/IES LM-80 Test Report	1.5.4	G												
			ANSI/IES TM-21 Test Report	1.5.5	G												
			Pressure Treated Wood Pole	1.5.7	G												
			Quality														
			Tests for Fiberglass Poles	1.5.6	G												
			SD-08 Manufacturer's Instructions														
			Poles	2.6													
			SD-10 Operation and Maintenance														
			Data														
			Lighting System	1.8.1	G												
			Exterior Lighting Control System	1.8.2	G												
			Maintenance Staff Training Plan	3.3.1.1	G												
			End-User Training Plan	3.3.1.2	G												
		27 10 00	SD-02 Shop Drawings														
			T1-T6 Telecommunications	1.6.1.1	G												
			Drawings														
			SD-03 Product Data														
			Telecommunications Cabling	2.4													
			(Backbone and Horizontal)														
			Cable Trays	2.3	G												
			Non-continuous Cable Supports	2.3.6	G												
			Patch Panels	2.2.5	G												
			Telecommunications	2.5	G												
			Outlet/Connector Assemblies														
			Equipment Support Frames	2.2.2	G												
			Connector Blocks	2.2.3	G												

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		27 10 00	Multi-User Telecommunications Outlet Assembly	2.6.1	G												
			Spare Parts	1.11.3	G												
			Floor Outlet And Poke-Thru Boxes	2.3.8	G												
			Plywood Backboards	2.2.1	G												
			Consolidation Point	2.6.2	G												
			SD-05 Design Data														
			Fiber Link Loss Budget	1.6.3	G												
			SD-06 Test Reports														
			Telecommunications Cabling Testing	3.11.1	G												
			SD-07 Certificates														
			Telecommunications Contractor	1.6.2.1	G												
			Key Personnel	1.6.2.2	G												
			Manufacturer Qualifications	1.6.2.3	G												
			Test Plan	1.6.3	G												
			Factory Reel Tests	2.11.1	G												
			SD-10 Operation and Maintenance Data														
			Telecommunications Cabling and Pathway System	1.11.1	G												
			SD-11 Closeout Submittals														
			Record Documentation	1.11.2	G												
		28 08 10	SD-05 Design Data														
			Test Plan	3.1	G												

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		28 08 10	SD-06 Test Reports														
			Final Test Report	3.4	G												
			Pre-Acceptance Test Certification	3.2	G												
			SD-07 Certificates														
			Qualifications	1.4.1													
		28 10 05	SD-02 Shop Drawings														
			ESS Components	1.3.3.1	G												
			Access Control System (ACS)	2.4	G												
			Security Hardware Door	1.3.3.1	G												
			Schedule														
			Overall System Schematic	1.3.3.1	G												
			Video Surveillance System (VSS)	2.5.7	G												
			Schedule														
			SD-03 Product Data														
			Premise Control Unit	2.3.6	G												
			Detection Sensors	2.3.7	G												
			Access Control Unit	2.4.5	G												
			Access Control Devices	2.4.6	G												
			Cameras	2.5.1	G												
			Camera Lenses	2.5.1.2	G												
			Camera Housing and Mounts	2.5.1.3	G												
			Thermal Imaging System	2.5.2	G												
			Video Recording	2.5.5.3	G												
			Printers	2.6.4	G												
			Communications Interface	2.7	G												
			Devices														

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		28 10 05	Radio Frequency Link	2.7.3	G												
			Network Switch	2.7.5	G												
			Video and ESS Transmission	2.7.6	G												
			Infant Protection Alarm System (IPAS) Major Components	2.9.2	G												
			Uninterruptible Power Supply (UPS)	2.11.1	G												
			Batteries	2.11.2	G												
			Component Enclosure	2.13	G												
			Equipment Rack	2.14	G												
			SD-05 Design Data														
			Backup Battery Capacity Calculations	1.5.1	G												
			Video Surveillance System (VSS) Storage Calculations	1.5.2	G												
			Error and Throughput Rates	2.4.3	G												
			SD-07 Certificates														
			Contractor Qualifications	1.3.4.1	G												
			Instructor Qualifications	1.3.4.2	G												
			Contact Card Readers	2.4.6.1.1	G												
			Contactless Card Readers	2.4.6.1.2	G												
			Data Encryption	2.7.4	G												
			Wire And Cable	2.7.7	G												
			Bonding, Grounding, And Shielding	3.1.12	G												

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		28 10 05	SD-10 Operation and Maintenance Data														
			ESS Components	1.3.3.1	G												
			ESS Software, Data Package 4	1.6	G												
			ESS Training	3.6	G												
			ESS Training Outline	3.6.1	G												
			SD-11 Closeout Submittals														
			As-Built Drawings	1.7	G												
			Warranty	2.4.6.3.6	G												
		28 31 70	SD-01 Preconstruction Submittals														
			Qualified Fire Protection Engineer (QFPE)	1.3.2	G												
			Fire alarm system designer	1.8.2.1	G												
			Supervisor	1.8.2.2	G												
			Technician	1.8.2.3	G												
			Installer	1.8.2.4	G												
			Test Technician	1.8.2.5	G												
			Fire Alarm System Site-Specific Software Acknowledgement	1.7	G												
			SD-02 Shop Drawings														
			Nameplates	1.8.1.3	G												
			Instructions	2.2.4	G												
			Wiring Diagrams	1.8.1.4	G												
			System Layout	1.8.1.5	G												
			Notification Appliances	1.8.1.6	G												
			Initiating devices	1.8.1.7	G												

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		28 31 70	Battery Power	1.8.1.8	G												
			Voltage Drop Calculations	1.8.1.9	G												
			SD-03 Product Data														
			Fire Alarm Control Unit (FACU)	2.3	G												
			Manual Stations	2.4	G												
			Smoke Detectors	2.5	G												
			Duct Smoke Detectors	2.5.2	G												
			Addressable Interface Devices	2.6	G												
			Addressable Control Modules	2.7	G												
			Notification Appliances	1.8.1.6	G												
			Batteries	2.10.1	G												
			Battery Chargers	2.10.2	G												
			Supplemental Notification	2.10.1.1	G												
			Appliance Circuit Panels														
			Auxiliary Power Supply Panels	2.10.1.1	G												
			Surge Protective Devices	2.11	G												
			Alarm Wiring	2.11	G												
			Back Boxes and Conduit	3.3.4	G												
			Ceiling Bridges	3.2.6	G												
			Terminal Cabinets	3.3.2	G												
			Automatic Fire Alarm	2.13	G												
			Transmitters														
			Radio Transmitter and Interface	2.13.1	G												
			Panels														
			Document Storage Cabinet	3.10.3	G												
			SD-06 Test Reports														

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		28 31 70	Test Procedures	3.6.1	G												
			SD-07 Certificates														
			Verification of Compliant	3.6.2.1	G												
			Installation														
			Request for Government Final	3.6.2.2	G												
			Test														
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.8	G												
			(O&M) Instructions														
			Instruction of Government	3.9	G												
			Employees														
			SD-11 Closeout Submittals														
			As-Built Drawings	1.8.1.12													
			Spare Parts	1.10.1													
		28 31 76	SD-01 Preconstruction Submittals														
			Qualified Fire Protection Engineer	1.3.2	G												
			(QFPE)														
			Fire alarm system designer	1.8.2.1	G												
			Supervisor	1.8.2.2	G												
			Technician	1.8.2.3	G												
			Installer	1.8.2.4	G												
			Test Technician	1.8.2.5	G												
			Fire Alarm System Site-Specific	1.7	G												
			Software Acknowledgement														
			SD-02 Shop Drawings														

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		28 31 76	Nameplates	1.8.1.3	G												
			Instructions	2.2.4	G												
			Wiring Diagrams	1.8.1.4	G												
			System Layout	1.8.1.5	G												
			Notification Appliances	1.8.1.6	G												
			Initiating devices	1.8.1.7	G												
			Amplifiers	1.8.1.8	G												
			Battery Power	1.8.1.9	G												
			Voltage Drop Calculations	1.8.1.10	G												
			SD-03 Product Data														
			Fire Alarm and Mass Notification Control Unit (FMCU)	2.3	G												
			Local Operating Console (LOC)	1.4.4	G												
			Amplifiers	1.8.1.8	G												
			Tone Generators	2.5	G												
			Digitalized voice generators	2.5	G												
			LCD Annunciator	2.6.1	G												
			Manual Stations	2.7	G												
			Smoke Detectors	2.8	G												
			Duct Smoke Detectors	2.8.2	G												
			Addressable Interface Devices	2.9	G												
			Addressable Control Modules	2.10	G												
			Notification Appliances	1.8.1.6	G												
			Textual Display Signs	2.11.3	G												
			Batteries	2.13.1	G												
			Battery Chargers	2.13.2	G												

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		28 31 76	Supplemental Notification	2.13.1.1	G												
			Appliance Circuit Panels														
			Auxiliary Power Supply Panels	2.13.1.1	G												
			Surge Protective Devices	2.14	G												
			Alarm Wiring	2.14	G												
			Back Boxes and Conduit	3.3.4	G												
			Ceiling Bridges	3.2.8	G												
			Terminal Cabinets	3.3.2	G												
			Automatic Fire Alarm	2.16	G												
			Transmitters														
			Radio Transmitter and Interface Panels	2.16.1	G												
			Document Storage Cabinet	3.10.3	G												
			SD-06 Test Reports														
			Test Procedures	3.6.1	G												
			SD-07 Certificates														
			Verification of Compliant Installation	3.6.2.1	G												
			Request for Government Final Test	3.6.2.2	G												
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance (O&M) Instructions	3.8	G												
			Instruction of Government Employees	3.9	G												

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		28 31 76	SD-11 Closeout Submittals														
			As-Built Drawings	1.8.1.13													
			Spare Parts	1.10.1													
		31 00 00	SD-01 Preconstruction Submittals														
			Shoring	3.4	G												
			Dewatering Work Plan	1.3.2	G												
			SD-06 Test Reports														
			Testing	3.17													
			Borrow Site Testing	2.1													
		31 11 00	SD-01 Preconstruction Submittals														
			Herbicide Application Plan	3.1.1													
			SD-03 Product Data														
			Herbicides	1.3.2	G												
			SD-07 Certificates														
			Qualifications	1.3.2	G												
			SD-11 Closeout Submittals														
			Pest Management Report	3.5.1													
		31 21 13	SD-03 Product Data														
			Respirators	1.8.3.1													
			Radon Mitigation Systems	2.1.2													
			Components														
			Radon Mitigation Systems	2.2													
			Enclosure Components														
			Radon Mitigation Systems	2.2													
			Enclosure Components														
			SD-06 Test Reports														

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		31 21 13	Post Mitigation Testing	3.4.2													
			SD-07 Certificates														
			Worker Protection Plan	1.7													
			Medical Certification	1.7.3													
			Worker Notification	1.7.4													
			Respiratory Protection Program	1.8													
			Contractor Qualifications	1.9.1.1													
			Contractor Experience	1.9.1.2													
			Testing Laboratory Certification	1.9.2													
			Proof Of Current Calibration For Testing Devices	1.9.3													
			Radon Mitigation System Inspection	3.4.1													
			SD-08 Manufacturer's Instructions														
			Radon Mitigation Systems Components	2.1.2													
			Radon Mitigation Systems Enclosure Components	2.2													
			Radon Mitigation Systems Enclosure Components	2.2													
			SD-10 Operation and Maintenance Data														
			Radon Mitigation Systems	2.1													
			SD-11 Closeout Submittals														
			Radon Detector Location Log	1.6													
			Respirator Program Records	1.8.1													

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		31 31 16.13	SD-01 Preconstruction Submittals														
			Termiticide Application Plan	3.1.5	G												
			SD-03 Product Data														
			Termiticides	2.2.1													
			SD-05 Design Data														
			Mixing Formulation	3.2.2													
			SD-06 Test Reports														
			Soil Moisture	1.6.1													
			Calibration Test	3.2.1													
			SD-07 Certificates														
			Qualifications	1.4.2	G												
			Foundation Exterior	3.1.2													
			Utilities and Vents	3.1.3													
			Crawl and Plenum Air Spaces	3.1.4													
			List of Equipment	3.2.1													
			SD-08 Manufacturer's Instructions														
			Termiticides	2.2.1													
			SD-11 Closeout Submittals														
			Verification of Measurement	3.3.1													
			Warranty	1.7													
			Pest Management Report	3.4													
		31 31 16.19	SD-02 Shop Drawings														
			Steel Mesh Shop Drawings	3.3.1	G												
			Basaltic Sand Shop Drawings	3.3	G												
			SD-03 Product Data														
			Steel Mesh Materials	2.2.2.1													

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		31 31 16.19	Accessories	2.2.3													
			Steel Mesh System	2.1.1													
			Written Warranty	1.7													
			SD-04 Samples														
			Steel Mesh Materials	2.2.2.1	G												
			SD-06 Test Reports														
			Basaltic Sand	2.2.4	G												
			SD-07 Certificates														
			System Installers	1.4.1.1													
			Steel Mesh Materials	2.2.2.1													
			Written Verification	3.1													
			SD-08 Manufacturer's Instructions														
			Manufacturer's Installation	3.3.1													
			Instruction Manual														
			Manufacturer's Guidance	3.4.1													
			Manufacturer's Installation	3.3													
			Instructions														
			SD-09 Manufacturer's Field														
			Reports														
			Site Conditions	3.4.2													
			SD-11 Closeout Submittals														
			Written Warranty	1.7													
		31 62 13.24	SD-01 Preconstruction Submittals														
			Installation Procedures	1.10.3	G												
			Contractor's Geotechnical	1.10.4	G												
			Consultant Documentation														

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		31 62 13.24	Wave Equation Analysis	3.1.1	G												
			Quality Control Procedures	1.10.2	G												
			Instrumentation and Monitoring	3.8	G												
			Program Report														
			Testing Agency Qualifications	3.3.4	G												
			SD-02 Shop Drawings														
			Piles	1.10.1	G												
			Pile Splices	3.2.7	G												
			Pile Placement	3.2.3	G												
			As-Driven Survey	3.2.10	G												
			Load Tests	3.3.2	G												
			Pile Shoe	3.2.12	G												
			SD-03 Product Data														
			Pile Driving Equipment	2.11	G												
			SD-05 Design Data														
			Concrete Mix Design	1.10.5	G												
			Grout	2.2.7	G												
			Joint Sealing Material	2.2.8	G												
			Prestressing Tendons	2.5.5	G												
			SD-06 Test Reports														
			Aggregates	2.2.3	G												
			Silica Fume	2.2.1.4	G												
			Compressive Strength	2.9.3	G												
			Test Piles	3.3.1	G												
			Load Tests	3.3.2	G												
			Dynamic Pile Analysis	3.3.1.1	G												

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		31 62 13.24	Destructive Pile Testing	2.9.7	G												
			Instrumentation and Monitoring	3.8	G												
			Program Report														
			SD-07 Certificates														
			Aggregates	2.2.3	G												
			Admixtures	2.2.4	G												
			Silica Fume Manufacturer's	1.10.7	G												
			Representative														
			Prestressing Steel	2.2.5	G												
			Cement	2.2.1.1	G												
			Fly Ash and Pozzolan	2.2.1.2	G												
			Slag	2.2.1.3	G												
			Epoxy Coating	2.2.9	G												
			Supporting Data	1.10.6	G												
			SD-11 Closeout Submittals														
			Pile Records	3.3.3	G												
		32 01 19.61	SD-03 Product Data														
			Sealants	2.1													
			Manufacturer's	3.5.2													
			Recommendations														
			SD-04 Samples														
			Sealants	2.1	G												
			Blocking Media/Backup Materials	2.3.1													
			Backer Rod	3.3.4.1													
			Bond Breaking Tapes	2.3.2													
			SD-06 Test Reports														

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		32 01 19.61	Sealants	2.1	G												
			SD-07 Certificates														
			Equipment List	3.1													
			SD-08 Manufacturer's Instructions														
			Sealants	2.1													
		32 11 20	SD-03 Product Data														
			Plant, Equipment, and Tools	1.4	G												
			SD-06 Test Reports														
			Initial Tests	2.2.1	G												
			In-Place Tests	3.12.1	G												
		32 11 23	SD-03 Product Data														
			Plant, Equipment, and Tools	1.4	G												
			SD-06 Test Reports														
			Initial Tests	2.3.1	G												
			In-Place Tests	3.12.1	G												
		32 12 13	SD-06 Test Reports														
			Sampling and Testing	3.7	G												
		32 12 16.16	SD-03 Product Data														
			Mix Design	2.4	G												
			Quality Control	3.11	G												
			Material Acceptance	3.12	G												
			SD-06 Test Reports														
			Aggregates	2.2	G												
			QC Monitoring	3.11.3.2													
		32 13 14.13	SD-01 Preconstruction Submittals														
			Proposed Techniques	3.1.2	G RO												

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		32 13 14.13	Preliminary Proposed Proportioning	2.13.2	G DO												
			Pavement Quality Control Plan	1.4.3.3	G RO												
			Stringless Technology	3.5.6.3	G RO												
			SD-03 Product Data														
			Diamond Grinding Plan	2.1.7	G RO												
			Dowels	2.9.1	G AE												
			Dowel Bar Assemblies	2.9.2	G AE												
			Equipment	2.11	G RO												
			Concrete Patching Mixture	2.1.5	G AE												
			SD-05 Design Data														
			Proportioning Studies	2.13.2	G DO												
			SD-06 Test Reports														
			Batch Plant Manufacturer's Inspection Report	1.4.1	G RO												
			Batch Plant Manufacturer's Inspection Report	1.4.1	G RO												
			Slipform Paver Manufacturer's Inspection Report	1.4.1	G RO												
			Sampling and Testing	2.1.4.1	G RO												
			Diamond Grinding of PCC Surfaces	2.1.7	G RO												
			Mixer Performance (Uniformity) Testing	2.11.2.1	G RO												
			Repair Recommendations Plan	3.9.1	G AE												
			Paving Lot Report	3.12.5	G RO												

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		32 13 14.13	Test Section Paving Lot Report	1.4.6.1	G RO												
			Test Section Lot Report	1.4.6.4	G RO												
			SD-07 Certificates														
			Contractor Quality Control (CQC) Staff	1.4.1	G AE												
			Laboratory Accreditation and Validation	1.4.3	G AE												
			Commercial Laboratory	1.4.3	G AE												
			NRMCA Certificate of Conformance	2.11													
			Petrographer Resume	1.4.2	G AE												
			Licensed Surveyor	1.4.2	G RO												
			Concrete Batch Plant Operator	1.4.2	G RO												
		32 15 00	SD-03 Product Data														
			Plant, Equipment, and Tools	1.4	G												
			SD-06 Test Reports														
			Initial Tests	2.3.1	G												
			In-Place Tests	3.9.1	G												
		32 16 19	SD-03 Product Data														
			Concrete	2.1	G												
			SD-06 Test Reports														
			Field Quality Control	3.8													
		32 17 23.16	SD-01 Preconstruction Submittals														
			Quality Control Plan	1.4.1	G												
			Qualifications	1.4.2	G												

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		32 17 23.16	Safety Data Sheets For Each Paint Type	2.2													
			Data Sheets For Paint Removal Equipment	2.1.1													
			Surface Preparation Equipment List	2.1													
			Marking Applications Equipment List	2.1.2													
			Detour Plans	1.6.2	G												
			SD-03 Product Data														
			Manufacturer Data Sheets for all Marking Materials	2.2													
			Manufacturer Data Sheets for all Reflective Media	2.2.8													
			SD-04 Samples														
			Samples of Marking Materials	3.5.2													
			Samples of each Reflective Media	3.5.2													
			SD-06 Test Reports														
			Marking Application Wet Film Thickness Test	3.1.2													
			Marking Application Wet Film Thickness Test	3.5.4.1													
			Marking Application Wet Film Thickness Test	3.5.4.2													
			Reflective Media Reflectivity Test	3.1.3													

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		32 17 23.16	Reflective Media Reflectivity Test	3.5.4.1													
			SD-07 Certificates														
			Manufacturer Certificate of	2.2													
			Compliance for Marking Materials														
			Manufacturer Certificate of	2.2													
			Compliance for Reflective														
			Materials														
			Manufacturer Certificate of	1.6.1.3													
			Conformance for Volatile Organic														
			Compliance														
			SD-08 Manufacturer's Instructions														
			Marking Materials Storage and	1.5	G												
			Application														
			Reflective Media Storage and	1.5	G												
			Application														
			Chemicals Used in Surface	1.5	G												
			Preparation														
		32 31 13	SD-02 Shop Drawings														
			Fence Assembly	2.1	G												
			Location of Gate, Corner, End,	3.2.2.1	G												
			and Pull Posts														
			Gate Assembly	2.2.9.1	G												
			Gate Hardware and Accessories	2.2.9.3	G												
			Erection/Installation Drawings	Part 3	G												
			SD-03 Product Data														
			Fence Assembly	2.1	G												

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		32 31 13	Gate Assembly	2.2.9.1	G												
			Gate Hardware and Accessories	2.2.9.3	G												
			Zinc Coating	2.3.1	G												
			PVC Coating		G												
			Fabric	2.2.1	G												
			Stretcher Bars	2.2.6	G												
			Barbed Wire	2.2.4	G												
			Precast Concrete Posts	2.2.2.1	G												
			Padlocks	2.2.12	G												
			Turnbuckles	2.2.9.4	G												
			Truss Rod	2.2.9.5	G												
			Tension Wires	2.2.9.6	G												
			Wire Ties	2.2.10	G												
			Concrete	2.3.2	G												
			SD-07 Certificates														
			Certificates of Compliance	1.3.1													
			SD-08 Manufacturer's Instructions														
			Fence Assembly	2.1													
			Gate Assembly	2.2.9.1													
			Hardware Assembly	2.1													
			Accessories	2.1													
			SD-11 Closeout Submittals														
			Recycled Material Content	3.3													
		33 01 50.31	SD-02 Shop Drawings														
			Leak Detection System	2.3	G												

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		33 01 50.31	Electronic Monitoring/Alarm Panel	2.4													
			Computational Pipeline Monitoring System	2.5													
			SD-03 Product Data														
			Leak Detection System	2.3	G												
			Electronic Monitoring/Alarm Panel	2.4													
			Computational Pipeline Monitoring System	2.5													
			SD-06 Test Reports														
			Leak Detection System Test	3.2.1													
			SD-07 Certificates														
			Demonstrations	3.3													
			SD-08 Manufacturer's Instructions														
			Leak Detection System	2.3													
			SD-10 Operation and Maintenance Data														
			Leak Detection System	2.3	G												
			Electronic Monitoring/Alarm Panel	2.4	G												
			Computational Pipeline Monitoring System	2.5	G												
		33 11 00	SD-03 Product Data														
			Fire Hydrants	2.1.3.1													
			Meters	2.1.4													

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		33 11 00	Backflow Preventer	2.1.5													
			Disinfection Procedures	3.2.4	G												
			SD-06 Test Reports														
			Backflow Preventer Tests	3.3.1.4	G												
			Bacteriological Samples	3.3.1.3	G												
			Leakage Test	3.3.1.2	G												
			Hydrostatic Test	3.3.1.1	G												
			SD-07 Certificates														
			Fire Hydrants	2.1.3.1													
			Backflow Certificate	2.1.5													
			SD-08 Manufacturer's Instructions														
			PVC Piping	2.1.1.2.1.1													
		33 30 00	SD-06 Test Reports														
			Hydrostatic Sewer Test														
			Infiltration Tests or Exfiltration Tests	3.2.1.1													
			Deflection Testing	3.2.1.2													
			SD-07 Certificates														
			Gaskets														
		33 40 00	SD-07 Certificates														
			Crushed Stone for Perforated Pipes	2.4.7													
			Geotextile Filter Fabric for Perforated Pipes	2.7													
			Determination of Density	3.10.1.1													

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		33 40 00	Post-Installation Inspection Report	3.10.2.1.2G													
			Placing Pipe	3.3													
		33 52 00	SD-02 Shop Drawings														
			POL System	1.4.1	G												
			Pipe Hangers and Supports	2.3.7													
			SD-03 Product Data														
			Stainless Steel Pipe	2.2.1	G												
			Pressure Gauge	2.3.6													
			Swing Type Check Valves	2.4.1													
			Globe Valve	2.4.2													
			Pressure Relief Valve	2.4.3													
			Foot Valve	2.4.4													
			Air Operated Lube-Oil Pumps	2.5.1	G												
			Air Operated Anti-Freeze Pumps	2.5.2	G												
			Lube-Oil Hose Reels	2.6.1	G												
			Anti-Freeze Hose Reels	2.6.2	G												
			Prefabricated Hose Reel Service	2.7.1	G												
			Columns (Single)														
			Prefabricated Hose Reel Service	2.7.2	G												
			Columns (Bilateral)														
			Lube-Oil Dispensing Handles	2.8.1	G												
			Anti-freeze Dispensing Handles	2.8.2	G												
			SD-06 Test Reports														
			Preliminary Pneumatic Test	3.3.3.1													
			Final Pneumatic Test	3.3.3.2													

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		33 52 00	Hydrostatic Test	3.3.3.3													
			Non-Destructive Examination (NDE)	3.1.2.2													
			SD-07 Certificates														
			Contractor Qualifications	1.4.2	G												
			Licensed Personnel	1.4.3.1													
			Demonstrations	3.4													
			SD-10 Operation and Maintenance Data														
			Swing Type Check Valves	2.4.1													
			Globe Valve	2.4.2	G												
			Pressure Relief Valve	2.4.3	G												
			Foot Valve	2.4.4	G												
			Air Operated Lube-Oil Pumps	2.5.1	G												
			Air Operated Anti-Freeze Pumps	2.5.2	G												
			Lube-Oil Hose Reels	2.6.1	G												
			Anti-Freeze Hose Reels	2.6.2	G												
			Prefabricated Hose Reel Service Columns (Single)	2.7.1	G												
			Prefabricated Hose Reel Service Columns (Bilateral)	2.7.2	G												
			Lube-Oil Dispensing Handles	2.8.1	G												
			Anti-freeze Dispensing Handles	2.8.2	G												
		33 52 10	SD-02 Shop Drawings														
			Grounding and Bonding	2.1.2													
			Pipe Supports	2.5.9													

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		33 52 10	SD-03 Product Data														
			Insulating Flange Kits	2.3.2	G												
			Flange Protectors	2.3.3	G												
			Fuel Piping Flange Bolts, Nuts, and Washers	2.3.4	G												
			Carbon Steel Pipe	2.4.1	G												
			Stainless Steel Pipe	2.4.2	G												
			Flexible Non-Metallic Pipe	2.4.3	G												
			Double Wall Carbon Steel Piping	2.4.4	G												
			Steel Reinforced Flexible Pipe	2.4.5	G												
			Copper Piping	2.4.6	G												
			Joint Compound	2.5.4	G												
			Flexible Connector	2.5.5	G												
			Strainer	2.5.6	G												
			Thermometers	2.5.7	G												
			Pressure Gauge	2.5.8	G												
			Flexible Ball Joint	2.5.11	G												
			Bellows Expansion Joint	2.5.12	G												
			Flow Meter	2.5.13	G												
			Ball Valves	2.6.1	G												
			Plug (Double Block and Bleed) Valves	2.6.2	G												
			Swing Type Check Valves	2.6.3	G												
			Wafer Type Check Valve	2.6.4	G												
			Globe Valve	2.6.5	G												
			Thermal Relief Valve	2.6.6	G												

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		33 52 10	Pressure/Vacuum Relief Valve	2.6.7	G												
			Foot Valve	2.6.8	G												
			Tank Overfill Prevention Valve (Gravity Fill)	2.6.9	G												
			Tank Overfill Prevention Valve (Pumped Fuel Receipt)	2.6.10	G												
			Anti-Siphon Valves	2.6.11	G												
			Submersible Pump	2.7.1	G												
			ANSI Type Centrifugal Pump	2.7.2	G												
			Sliding Vane Rotary Pump	2.7.3	G												
			Self-Priming Centrifugal Pump	2.7.4	G												
			Pump Control Panel	2.7.5	G												
			FRP Containment Sump	2.8	G												
			Pipeline Markers	2.9.8	G												
			SD-06 Test Reports														
			Exterior Coating Holiday Test	3.3.2.1													
			Preliminary Pneumatic Test	3.3.2.2													
			Final Pneumatic Test	3.3.2.3													
			Hydrostatic Test	3.3.2.4													
			Exterior Containment Piping Tests	3.3.2.5													
			SD-07 Certificates														
			Contractor Qualifications	1.4.1	G												
			Licensed Personnel	1.4.2.1													
			Stage II Vapor Recovery System	1.4.2.2													
			Pipeline Inventory	1.4.3.1	G												

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		33 52 10	Demonstrations	3.5													
			SD-08 Manufacturer's Instructions														
			Flexible Ball Joint	2.5.11													
			Bellows Expansion Joint	2.5.12													
			SD-10 Operation and Maintenance Data														
			Insulating Flange Kits	2.3.2	G												
			Flange Protectors	2.3.3	G												
			Strainer	2.5.6	G												
			Thermometers	2.5.7	G												
			Flexible Ball Joint	2.5.11	G												
			Bellows Expansion Joint	2.5.12	G												
			Flow Meter	2.5.13	G												
			Ball Valves	2.6.1	G												
			Plug (Double Block and Bleed) Valves	2.6.2	G												
			Swing Type Check Valves	2.6.3	G												
			Wafer Type Check Valve	2.6.4	G												
			Globe Valve	2.6.5	G												
			Thermal Relief Valve	2.6.6	G												
			Pressure/Vacuum Relief Valve	2.6.7	G												
			Foot Valve	2.6.8	G												
			Tank Overfill Prevention Valve (Gravity Fill)	2.6.9	G												
			Tank Overfill Prevention Valve (Pumped Fuel Receipt)	2.6.10	G												

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		33 52 10	Anti-Siphon Valves	2.6.11	G												
			Submersible Pump	2.7.1	G												
			ANSI Type Centrifugal Pump	2.7.2	G												
			Sliding Vane Rotary Pump	2.7.3	G												
			Self-Priming Centrifugal Pump	2.7.4	G												
			Pump Control Panel	2.7.5	G												
		33 52 23.15	SD-01 Preconstruction Submittals														
			Welding Procedure Specification	1.9.1	G												
			Procedure Qualification Record	1.9.2	G												
			Welder Performance Qualification	1.9.3.1	G												
			Weld Plan	1.7.5	G												
			SD-06 Test Reports														
			Weld Log	1.8.2	G												
			SD-07 Certificates														
			Weld Inspector Certificate	1.9.6.2	G												
			Independent Test Organization	1.9.5	G												
			In-Process Weld Inspector	1.9.6.3	G												
			Certificates														
			Weld Examiner Certificate	1.9.6.4	G												
		33 56 10	SD-02 Shop Drawings														
			Grounding and Bonding	2.1.1													
			SD-03 Product Data														
			Aboveground Storage Tank	2.4.1	G												
			(Double Wall, Concrete Encased)														
			Tank Protective Coatings	2.5	G												
			Atmospheric Vent	2.6.5.1	G												

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		33 56 10	Pressure Vent	2.6.5.2	G												
			Emergency Vent	2.6.5.3	G												
			Independent Level Alarm System	2.7	G												
			Tank Gauges	2.8	G												
			Tank Anchorage System	3.1	G												
			Materials														
			SD-05 Design Data														
			Tank Anchorage Calculations	3.1	G												
			SD-06 Test Reports														
			Aboveground Storage Tank	3.2.1	G												
			Tightness Tests														
			Tank Manufacturer's Tests	3.2.2													
			Tank Fill Tests	3.4													
			Tank Inspection Reports	3.2.4	G												
			SD-07 Certificates														
			Contractor Qualifications	1.4.1	G												
			Manufacturer	1.4.1	G												
			Pollution Liability Insurance	1.4.1													
			Permitting	1.4.2.1													
			Registration	1.4.2.2													
			Licensed Personnel	1.4.2.3													
			Demonstrations	3.3													
			STI SP001 Inspector's	3.2.4	G												
			Certification														
			SD-08 Manufacturer's Instructions														
			Aboveground Storage Tank	2.4													

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		33 56 10	Independent Level Alarm System	2.7													
			Tank Gauges	2.8													
			SD-10 Operation and Maintenance Data														
			Aboveground Storage Tank	2.4	G												
			Independent Level Alarm System	2.7	G												
			Tank Gauges	2.8	G												
		33 57 55	SD-02 Shop Drawings														
			Grounding and Bonding	2.3.1	G												
			Tightness Monitoring System	2.6.4	G												
			Truck Fillstand Overfill Protection and Ground Verification Unit	2.6.5	G												
			Venturi Tubes	2.6.7	G												
			Meters	2.8	G												
			Jockey Pump	2.10.5	G												
			Packaged Truck Offload System	2.11	G												
			High Point Vent and Low Point Drain Pits	2.19	G												
			Water Draw-Off System	2.26	G												
			Operating Tank Vent	2.28	G												
			SD-03 Product Data														
			Pressure Gages	2.4	G												
			Differential Pressure Gauge	2.5	G												
			Automatic Pump Controls	2.6	G												
			Tightness Monitoring System	2.6.4	G												

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		33 57 55	Truck Fillstand Overfill Protection and Ground Verification Unit	2.6.5	G												
			Flow Switches	2.6.6	G												
			Venturi Tubes	2.6.7	G												
			Differential Pressure Transmitter	2.6.8	G												
			Pressure Sensor	2.6.9	G												
			Relaxation Tank	2.7	G												
			Meters	2.8	G												
			Submersible Pump	2.10.1	G												
			ANSI Type Centrifugal Pump	2.10.2	G												
			Sliding Vane Rotary Pump	2.10.3	G												
			Self-Priming Centrifugal Pump	2.10.4	G												
			Jockey Pump	2.10.5	G												
			Packaged Truck Offload System	2.11	G												
			Deaerator Tank	2.12	G												
			Truck Fillstand Hose	2.14.1	G												
			Truck Fillstand Swivel Joints	2.14.2	G												
			Tank Truck Bottom Loading Arm	2.15	G												
			Top Loading Arm	2.16	G												
			Filter/Separator	2.18	G												
			High Point Vent and Low Point Drain Pits	2.19	G												
			FRP Containment Sump	2.20	G												
			Liquid Level Gauge	2.21	G												
			Operating Tank Level Indicator	2.22	G												
			Operating Tank Level Switches	2.23	G												

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		33 57 55	Operating Tank Level Switches	2.24	G												
			Operating Tank Level Switches	2.25	G												
			Water Draw-Off System	2.26	G												
			Operating Tank Vent	2.28	G												
			Product Dispensing Unit	2.29.1	G												
			SD-06 Test Reports														
			Tightness Monitoring System	2.6.4	G												
			Coating Testing	3.2.1	G												
			SD-07 Certificates														
			System Supplier	1.5	G												
			Tightness Monitoring System	2.6.4	G												
			SD-10 Operation and Maintenance														
			Data														
			Automatic Pump Controls	2.6	G												
			Truck Fillstand Overfill Protection and Ground Verification Unit	2.6.5	G												
			Relaxation Tank	2.7	G												
			Meters	2.8	G												
			Submersible Pump	2.10.1	G												
			ANSI Type Centrifugal Pump	2.10.2	G												
			Sliding Vane Rotary Pump	2.10.3	G												
			Self-Priming Centrifugal Pump	2.10.4	G												
			Jockey Pump	2.10.5	G												
			Packaged Truck Offload System	2.11	G												
			Deaerator Tank	2.12	G												
			Filter/Separator	2.18	G												

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		33 57 55	Operating Tank Level Indicator	2.22	G												
			Water Draw-off System	2.26	G												
			Operating Tank Vent	2.28	G												
			Product Dispensing Unit	2.29.1	G												
		33 61 14	SD-02 Shop Drawings														
			Factory-prefabricated	2.1													
			preinsulated water piping system														
			field joints	3.2													
			SD-03 Product Data														
			Pipe, fittings, and end connections	2.1													
			Factory-prefabricated	2.1													
			preinsulated water piping system														
			Plastic reinforced thermosetting resin (RTR) piping	2.2.3													
			SD-07 Certificates														
			Certification of welders' qualifications	1.4.1													
			SD-08 Manufacturer's Instructions														
			factory-prefabricated preinsulated	2.1													
			water piping system														
		33 71 02	SD-02 Shop Drawings														
			Precast Underground Structures	1.6.1	G												
			SD-03 Product Data														
			Medium Voltage Cable	2.5	G												
			Medium Voltage Cable Joints	2.7	G												

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		33 71 02	Medium Voltage Cable Terminations	2.6	G												
			Live End Caps	2.9	G												
			Precast Concrete Structures	2.14.2.1	G												
			Sealing Material	2.14.2.4													
			Pulling-In Irons	3.5.3													
			Manhole Frames and Covers	2.14.3	G												
			Handhole Frames and Covers	2.14.4	G												
			Composite/Fiberglass Handholes	2.14.7	G												
			Cable Supports	2.15	G												
			Protective Devices and Coordination	2.19	G												
			SD-06 Test Reports														
			Medium Voltage Cable Qualification and Production Tests	2.20.2	G												
			Field Acceptance Checks and Tests	3.20.1	G												
			Arc-proofing Test	2.20.1	G												
			Cable Installation Plan and Procedure	3.3	G												
			SD-07 Certificates														
			Cable splicer/terminator	1.6.2	G												
			Cable Installer Qualifications	1.6.3	G												
		33 82 00	SD-02 Shop Drawings														
			Telecommunications Outside Plant	1.6.1.1	G												

SUBMITTAL REGISTER

CONTRACT NO.
W912HV-19-D-0010

TITLE AND LOCATION
PN77833 VEHICLE MAINTENANCE SHOP

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION REVIEWER	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		33 82 00	Telecommunications Entrance Facility Drawings	1.6.1.2	G												
			SD-03 Product Data														
			Wire and Cable	2.7	G												
			Cable Splices, and Connectors	2.4	G												
			Closures	2.3	G												
			Building Protector Assemblies	2.2.1	G												
			Protector Modules	2.2.2	G												
			Cross-Connect Terminal Cabinets		G												
			Spare Parts	1.8.2	G												
			SD-06 Test Reports														
			Pre-installation Tests	3.5.1	G												
			Acceptance Tests	3.5.2	G												
			Outside Plant Test Plan	1.6.3	G												
			SD-07 Certificates														
			Telecommunications Contractor	1.6.2.1	G												
			Key Personnel	1.6.2.2	G												
			Manufacturer's Qualifications	1.6.2.3	G												
			SD-08 Manufacturer's Instructions														
			Building Protector Assembly Installation	2.2.1	G												
			Cable Tensions	3.1.7.1	G												
			Fiber Optic Splices	3.1.8.2	G												
			SD-09 Manufacturer's Field Reports														

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CONTRACTOR

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		33 82 00	Factory Reel Test Data	2.15.1	G												
			SD-10 Operation and Maintenance Data														
			Telecommunications Outside Plant (OSP)	1.6.1.1	G												
			SD-11 Closeout Submittals Record Documentation	1.8.1	G												
		41 22 13.14	SD-02 Shop Drawings														
			Overhead Electric Traveling Crane	1.6.4	G												
			Complete Schematic Wiring Diagram	3.3.2	G												
			SD-03 Product Data														
			Gear Reducers	2.3.7.1	G												
			Hoist Brakes	2.3.10	G												
			Travel Brakes	2.3.9	G												
			Load Blocks and Hooks	2.3.3	G												
			Wheels	2.3.8	G												
			Hoists	2.3.1	G												
			Sheaves	2.3.5	G												
			End Trucks	2.2.5	G												
			Bridge Rails	2.2.3	G												
			End Stops	2.2.6	G												
			Bumpers	2.2.6	G												
			Variable Frequency Drives	2.4.1	G												
			Motors	2.4.1	G												

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CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION REVIEW	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		41 22 13.14	Runway Conductor System	2.4.8.1	G												
			Runway Conductor System	2.4.8.1	G												
			Bridge Conductor System	2.4.8.2	G												
			Bridge Conductor System	2.4.8.2	G												
			Limit Switches	2.4.6	G												
			Pendant Pushbutton Station	2.4.7.1	G												
			Pendant Conductor System	2.4.7.1.1	G												
			Controls	2.4.2	G												
			Control Parameter Settings	3.3.2	G												
			Capacity Overload Protective Device	2.4.9	G												
			Painting System	2.5	G												
			SD-05 Design Data														
			Load and Sizing Calculations	1.6.5	G												
			SD-06 Test Reports														
			Hook Non-destructive Test (NDT)	1.6.2.2.1	G												
			Post-erection Inspection	3.4.1	G												
			Operational Tests	3.4.2	G												
			Hook Tram Measurement	3.4.4	G												
			Load Tests	3.4.5	G												
			SD-07 Certificates														
			Wire Ropes	1.6.3	G												
			Crane Runway System	1.6.3	G												
			Hazardous Material	1.6.3	G												
			Loss of Power Test	1.6.3	G												

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CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		APPROVING AUTHORITY						MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION			
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	
		41 22 13.14	Coupling Alignment Verification Record	1.6.3	G													
			Overload Test		G													
			Brake Adjustment Record	1.6.3	G													
			Compliance with Listed Standards	1.6.3	G													
			Contractor Hazardous Environment	1.6.3	G													
			SD-10 Operation and Maintenance Data															
			Operation and Maintenance Manuals	3.6	G													
		46 72 11	SD-02 Shop Drawings															
			Installation	3.4.1														
			SD-03 Product Data															
			Inclined Plate Clarifier Tank	2.2.3														
			Submersible Redelivery Pump	2.2.4														
			Spare Parts	2.1.4														
			Framed Instructions	3.3														
			Qualifications	1.3														
			Water Cannon	2.2.5														
			Cold Water Pressure Washer	2.2.6														
			SD-06 Test Reports															
			Testing	3.4.3														
			SD-10 Operation and Maintenance Data															

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PN77833 VEHICLE MAINTENANCE SHOP

CONTRACTOR

[illegible]

SECTION 01 33 23.33

AVIATION FUEL SYSTEM SPECIFIC SUBMITTAL REQUIREMENTS
08/18, CHG 1: 02/21

PART 1 GENERAL

1.1 SUBMITTALS

Provide submittals as specified in each individual section and obtain approval as specified before procurement, fabrication, or delivery to the job site. Partial submittals are not acceptable and will be returned without review. Subject Matter Expert (SME) is defined as Service Headquarters Subject Matter Experts. SME for this project is [Air Force - The Air Force Fuels Facilities Subject Matter Expert (HQ AFCEC/COS)] [Army - Headquarters, U.S. Army Corps of Engineers, POL-MCX Facilities Proponent (CECW-EC) through the Army Petroleum Center (APC)] [Navy/Marine Corps - NAVFAC POL Facility Subject Matter Expert (NAVFAC EXWC, CI11)]]]. In addition to the definitions in Section 01 33 00 SUBMITTAL PROCEDURES, the following provisions apply:

1.1.1 SD-02 Shop Drawings

Drawings must be a minimum of A3 sheet in size, with a minimum scale of 1:100, except as specified otherwise. Include floor plans, sectional views, wiring diagrams, and installation details of equipment; and equipment spaces identifying and indicating proposed location, layout and arrangement of items of equipment, control panels, accessories, piping, ductwork, and other items that must be shown to assure a coordinated installation. Wiring diagrams must identify circuit terminals, and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Drawings must indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices. If equipment is disapproved, drawings must be revised to show acceptable equipment and be resubmitted. (Prior to the completion of the contract, on A1 sheet size Adobe.pdf reproducible drawings, for each system, wiring/control diagram and approved system layout drawing must be provided to the Contracting Officer with the operation and maintenance manuals specified herein).

1.1.2 SD-03 Product Data

Submittals for each manufactured item must be manufacturer's descriptive literature of cataloged products, equipment drawings, diagrams, performance and characteristic curves, and catalog cuts. The submittals must also include the manufacturer's name, trade name, catalog model or number, nameplate data, size, layout dimensions, capacity, project specification and paragraph reference, applicable Government, industry, and technical society publication references, years of satisfactory service, and other information necessary to establish contract compliance of each item the Contractor proposes to provide. Photographs of existing installations and data submitted in lieu of catalog data are not acceptable and will be returned without approval.

1.1.2.1 Standards Compliance

When materials or equipment are required to conform to the standards of organizations such as the American National Standards Institute (ANSI),

ASTM International (ASTM), National Electrical Manufacturers Association (NEMA), ASME International (ASME), American Gas Association (AGA), American Petroleum Institute (API), Air-Conditioning and Refrigeration Institute (ARI), and Underwriters Laboratories (UL) or equivalent, submit proof of such conformance to the Contracting Officer. Factory Mutual (FM) listing or CSA International (CSA) listing will be acceptable in lieu of any UL listing requirements. If an organization uses a label or listing to indicate compliance with a particular standard, the label or listing will be acceptable evidence, unless otherwise specified in the individual sections. In lieu of the label or listing, submit a certificate from an independent testing organization, which is competent to perform acceptable test and is approved by the Contracting Officer. The certificate must state that the item has been tested in accordance with the specified organization's test methods and that the item conforms to the specified organization's standard. For materials and equipment whose compliance with organizational - standards or specifications is not regulated by an organization using its own listing or label as proof of compliance, submit a certificate of compliance from the manufacturer. The certificate must identify the manufacturer, the product, and the referenced standard and must simply state that the manufacturer certifies that the product conforms to all requirements of the project specification and of the referenced standards listed.

1.1.2.2 Manufacturer's Installation Instructions

Where installation procedures or any part thereof are required to be in accordance with the recommendations of the manufacturer of the material being installed, furnish printed copies of these recommendations prior to installation. Installation of the item will not be allowed to proceed until the recommendations are received. Failure to furnish these recommendations can be cause for rejection of the material.

PART 2 PRODUCTS

2.1 STANDARD PRODUCTS/SERVICE AVAILABILITY

2.1.1 Materials and Equipment

Provide materials and equipment that are standard products of a manufacturer regularly engaged in the manufacture of such products, which are of a similar material, design and workmanship. The standard products must have been in satisfactory commercial or industrial use for two years prior to start of construction. The two-year use must include applications of equipment and materials under similar circumstances and of similar size. Contracting Officer approval of materials with less than two years experience is allowed if acceptable by the design agency and SME.

2.1.1.1 Experience Required

The two-year experience requirement must be satisfactorily completed for a product which has been sold or is offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures.

2.1.1.2 Alternative Service Record

Products having less than a two-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory test, can be shown. Contracting Officer approval of materials

with less than two years experience is allowed if acceptable by the design agency and SME.

2.1.2 Service Support

The equipment items must be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. These service organizations must be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

2.1.3 Manufacturer's Nameplate

Provide a nameplate, on each item of equipment, bearing the manufacturer's name, address, model number, and serial number securely permanently affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION 01 35 13

SPECIAL PROJECT PROCEDURES
11/20, CHG 2: 08/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. Code (USC)

49 USC 44718 Structures Interfering with Air Commerce
or National Security

49 USC 46301 Civil Penalties

U.S. FEDERAL AVIATION ADMINISTRATION (FAA)

FAA AC 70/7460-1 (2016; Rev L; Change 2) Obstruction
Marking and Lighting

FAA AC 150/5300-13 (2022; Rev B) Airport Design

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

14 CFR 77 Safe, Efficient Use, and Preservation of
the Navigable Airspace

1.2 DEFINITIONS

1.2.1 Landing Areas

"Landing Areas" means:

- a. The primary surfaces, comprising the surface of the runway, runway shoulders, and lateral safety zones. The length of each primary surface is the same as the runway length. The width of each primary surface is 609.6 meters (304.8 meters on each side of the runway centerline). [Exceptions: Some airfields are based on a primary width of 457.2 meters (228.6 meters on each side of the runway centerline). In such instances, substitute the proper width in the applicable statements.]
- b. The "clear zone" beyond the ends of each runway is the extension of the primary surface for a distance of 914.4 meters in length for fixed wing aircraft and 121.92 meters in length for helicopter only runways beyond each end of each runway.
- c. All taxiways, plus the lateral clearance zones along each side for the length of the taxiways (the outer edge of each lateral clearance zone is laterally 76.2 meters from the far or opposite edge of the taxiway (example: a 22.86 meter taxiway must have a combined width and lateral clearance zone of 129.54 meters.)

- d. All aircraft parking aprons, plus the area 38.1 meters in width extending beyond each edge around the aprons.

1.2.2 Safety Precaution Areas

"Safety Precaution Areas" means those portions of approach-departure clearance zones and transitional zones where placement of objects incident to Contract performance might result in vertical projections at or above the approach-departure clearance, or the transitional surface.

- a. The "approach-departure clearance surface" is an extension of the primary surface and the clear zone at each end of each runway, for a distance of 15240 meters, first along an inclined (glide angle) and then along a horizontal plane, both flaring symmetrically about the runway centerline extended.
 - (1) The inclined plane (glide angle) begins in the clear zone 61 meters past the end of the runway (and primary surface) at the same elevation as the end of the runway. It continues upward at a slope of 50:1 (.305 meters vertically for each 15.24 meters horizontally) to an elevation of 152.4 meters above the established airfield elevation. At that point the plane becomes horizontal, continuing at that same uniform elevation to a point 15240 meters longitudinally from the beginning of the inclined plane (glide angle) and ending there.
 - (2) The width of the surface at the beginning of the inclined plane (glide angle) is the same as the width of the clear zone. It then flares uniformly, reaching the maximum width of 4876.8 meters at the end.
- b. The "approach-departure clearance zone" is the ground area under the approach-departure clearance surface.
- c. The "transitional surface" is a sideways extension of all primary surfaces, clear zones, and approach-departure clearance surfaces along inclined planes.
 - (1) The inclined plane in each case begins at the edge of the surface.
 - (2) The slope of the incline plane is 7:1 (.305 meters vertically for each 2.13 meters horizontally). It continues to the point of intersection with the:
 - (a) Inner horizontal surface (which is the horizontal plane 45.72 meters above the established airfield elevation); or
 - (b) Outer horizontal surface (which is the horizontal plane 152.4 meters above the established airfield elevation), whichever is applicable.
- d. The "transitional zone" is the ground area under the transitional surface. (It adjoins the primary surface, clear zone, and approach-departure clearance zone.)

1.2.3 Federal Aviation Administration (FAA) Notice of Proposed Construction or Alteration

- a. FAA Notice of Proposed Construction or Alteration may be required in

accordance with 49 USC 44718 and 14 CFR 77, depending on height of construction equipment on site, height of temporary structures, proximity to an airport or heliport, and specific location of equipment or temporary structure. For the purpose of notifying the FAA, proximity is defined as within 5 nautical miles of a Government or civilian airfield, including landing areas, taxiways, runways and helicopter pads.

- b. In order to determine if a FAA Notice of Proposed Construction or Alteration is required, refer to 14 CFR 77 Subpart B. Alternately, utilize the FAA's Notice Criteria Tool located at: <https://oeaaa.faa.gov/oeaaa/external/portal.jsp>. The FAA will determine if the equipment or temporary structure exceeds obstruction standards and may pose a hazard to air navigation.
- c. Failure to comply with the provisions of 14 CFR 77 are subject to Civil Penalty under Section 902 of the Federal Aviation Act of 1958, as amended and pursuant to 49 USC 46301 Subpart (a).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

[Heavy Equipment and Vehicle List

][Existing Conditions Survey

] FAA Form 7460-1

FAA Form 7460-2

Construction Operations Plan

[Watercraft List

][SD-04 Samples

Model Unit

]PART 2 PRODUCTS

2.1 AIRFIELD OBSTRUCTION LIGHTS

Airfield obstruction lights must be in accordance with FAA AC 70/7460-1 and have red or white lenses.

PART 3 EXECUTION

[3.1 HAZARDS TO [AIRFIELD][HELIPORT] OPERATION

In addition to DFARS 252.236-7005 Airfield Safety Precautions, the

following paragraphs apply.

3.1.1 Work in Proximity to [Landing Areas] [Landing Strips] [Landing Pad(s)]

Place nothing upon the landing area or applicable portions of safety precaution areas without authority of the Contracting Officer.

Use of [landing areas] [landing strips] [landing pads] for purposes other than aircraft operation, is prohibited without permission of the Contracting Officer, and the [landing area] [landing strip] [landing pad] is closed by order of the Contracting Officer and marked as indicated herein.

Accomplish all construction work on [the runways, taxiways, and parking aprons and in the end zones of the runways and 23 m to each side of the runways and taxiways][the landing strip, 23 m to each side thereof, and on the taxiways and parking aprons][the landing pad(s)]with extreme care regarding the operation of aircraft. Cooperate closely, and coordinate with [the Operations Officer and]the Contracting Officer. Park equipment in an area designated by the Contracting Officer. Parking of equipment, vehicles, or any storage features overnight or for any extended period of time in the proximity of the [landing areas or taxiways][landing strip][landing pad]is strictly prohibited. Leave no material in areas where extreme care is to be taken regarding the operation of aircraft.

During periods of active performance of work on the airfield by the Contractor, govern all operations of mobile equipment in accordance with the safety provisions.

3.1.2 Contractor FAA Notification

When required in accordance with 49 USC 44718 and 14 CFR 77, submit FAA Form 7460-1 and attachments directly to the FAA a minimum of 60 calendar days prior to the start date of the operations that may affect air traffic. Submit supplemental notification FAA Form 7460-2 to the FAA within 48 hours prior to start of the construction. Simultaneous with submission to the FAA, submit both forms to the Contracting Officer for information. It is the Contractor's responsibility to notify the FAA when required.

3.1.3 Schedule of Work/Aircraft Operating Schedules

Schedule work to conform to aircraft operating schedules. The Government will exert every effort to schedule aircraft operations so as to permit the maximum amount of time for the Contractor's activities; however, in the event of emergency, intense operational demands, adverse wind conditions, and other such unforeseen difficulties, the Contractor must cease operations at the specified locations in the aircraft operational area for the safety of the Contractor and military personnel and Government property. Submit a schedule of the work to the Contracting Officer [for transmittal to the Operations Officer]describing the work to be accomplished; the location of the work, noting distances from the ends of [landing areas, taxiways][landing strips][landing pads]and buildings and other structures as necessary; and dates and hours during which the work is to be accomplished. Keep the approved schedule of work current, and notify the Contracting Officer of changes prior to beginning each day's work.

Where flying is controlled, obtain permission from the control tower operator to enter a landing area unless such area is marked as hazardous to aircraft.

3.1.3.1 Construction Operations Plan

Submit a Construction Operations Plan prior to the start of work that includes a description of the airfield work to be accomplished; the exact location of the work, noting distances from the ends of [landing areas, taxiways][landing strips][landing pads]and buildings and other structures as necessary; and dates and hours during which the work is to be accomplished. Keep the approved schedule of work current and notify the Contracting Officer of changes prior to beginning each day's work.

[3.1.3.2 Existing Conditions Survey

Perform an Existing Conditions Survey and submit results to the Government prior to the start of work. As part of this survey, inspect the work site to identify existing conditions located within the vicinity of the contract work. Utilize original as-builts if available showing possible utilities, type of pavement, thickness, airfield lighting, equipment, vehicles, structures, and parked aircraft. Unless already specified in the Contract documents, utilize the RFI process to clarify responsibility for relocating appurtenances that will interfere with work. If the site is opened up (in/under construction), update the existing survey condition to include the actual condition of the underlying conditions such as sub base, base, and pavement thickness.

]3.1.4 Daytime Markings

During daylight, mark stationary and mobile equipment with international orange and white checkered flags, mark the material, and work with yellow flags.

Submit a Heavy Equipment and Vehicle List identifying all stationary and mobile equipment and vehicles that will be operating on the airfield. All equipment and vehicles must be identified by means of a flag on a staff attached to and flying above the vehicle. Flag size must be not less than one meter square and consist of a checkered pattern of international orange and white squares not less than 300 millimeter on each side. Flags varying in any dimension by not more than 10 percent of the specified dimensions are considered to comply with the stated requirements.

3.1.5 Nighttime Markings

During nighttime, which begins 2-hours before sundown and ends 2-hours after sunrise, mark stationary and mobile equipment and material, and work with [red lanterns][battery-operated, low-intensity, red flasher lights]. Where the [Operations Officer][Contracting Officer]determines that the [red lanterns][red flasher lights]may confuse pilots approaching for landings, the [Operations Officer][Contracting Officer]may direct that the [red lanterns][red flasher lights]be left off or that the color of the [globes][lights]be changed.

Provide lighting in accordance with FAA AC 70/7460-1. Provide a minimum of two aviation red or high intensity white obstruction lights on temporary structures (including cranes) over 30 meter above ground level. Lights must be operational during periods of reduced visibility, darkness, and as directed by the Contracting Officer.

No separate payments will be made for lighting and protection necessitated by the safety provisions.

[3.1.6 Excavation

Open only those trenches for which material is on hand and ready for placing therein. As soon as possible after the material has been placed and work approved, backfill and compact the trenches as specified.

Maintain [landing areas][landings strips][landing pads]at all times free from hazards, holes, material piles, or projecting shoulders that might damage tires or landing gear. Keep paved surfaces clean at all times and free from small stones or other objects which could cause damage to propellers, craft, and personnel.

][3.1.7 Contractor Safety Precautions

The Contractor is advised that aircraft operations will produce extremely high noise levels and will induce vibrations in pavements, structures, and equipment in the vicinity, and may result in high velocity flying debris in the area. These anticipated hazards must be appropriately addressed in the Activity Hazard Analysis (AHA) associated with airfield work - the Activity Hazard Analysis must be in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. The Contractor is responsible for providing personal protective equipment (PPE) and other safety devices required to ensure protection of contractor personnel and equipment. Schedule the work to eliminate hazards to personnel and equipment and to prevent damage to work performed.

Boundary areas for hazardous work locations and restrictions are defined in FAA AC 150/5300-13. Construction activity within the limits of the boundary areas without approval of the Contracting Officer is prohibited.

]3.1.8 Base Civil Engineering (BCE) Work Clearance Request

Obtain an approved BCE Work Clearance Request, AF Form 103, prior to the start of excavation, digging work, or work that disrupts aircraft or vehicular traffic flow, base utility services, fire and intrusion alarm system, or routine activities of the Activity.

3.1.9 Radio Contact

Provide necessary battery powered portable radios, including one radio for the tower. During work within the landing area, have an operator (who speaks fluent English) available for radio contact with the tower at all times. Obtain approval of radio frequency from the tower.

][3.2 HARBOR WATERWAYS

In addition to DFARS 252.236-7002 Obstruction of Navigable Waterways, obtain from the [Operations Officer at the Naval Base, Pearl Harbor via the] [_____] Contracting Officer, permission to use [Pearl Harbor] [_____] waterways and the regulations and directives governing such usage. Submit a watercraft list with a description of crafts, including sizes, types, numbers and boat crew for approval.

[3.2.1 Hazards to Navigation

Maintain complete control of the movement of floating equipment and material. Loose floating equipment and material are not permitted. Keep in readiness at all times a powered craft capable of moving, securing and disposing of floating equipment which may get loose and become a hazard to navigation.

][3.2.2 Submarine Cables or Underwater Utilities

Physically locate submarine cables or underwater utilities as indicated prior to performance of work.

]][3.3 MODEL UNIT

Prior to placement of material orders for components of the [BEQ living suites] [BOQ living suites] [family housing units] [_____] and the installation of those components throughout the project, the Contractor must complete the construction of the components of the model unit, and gain the approval of the Contracting Officer for the materials and workmanship therein.

3.3.1 Model Unit Description

Provide a model unit made up of enough [living suites consisting of two bedrooms and an interconnecting bathroom] [single family housing units] [_____] to show all of the different levels of completion required below. [The model unit must have enough [suites] [housing units] to show all level of completion simultaneously[, except for foundation and floor slab levels which are approved by the Contracting Officer and covered by following levels].] The model unit must be [living suites] [family housing units] [_____] constructed at the location agreed upon with the Contracting Officer and [a part of the permanent structure as indicated] [a temporary structure constructed to full scale at the location indicated, and demolished at the completion of the Contract].

3.3.2 Model Unit Requirements

Adhere to the following requirements for model construction:

- a. Provide materials in model unit identical to those actually approved in submittals. The QC Manager and Organization must approve and process submittals in accordance with Section 01 33 00 SUBMITTAL PROCEDURES. Do not procure, fabricate, or install materials prior to approval by the Contracting Officer.
- b. The Contracting Officer retains the right to disapprove and reject materials or items which are not in compliance with the project drawings and specifications.
- c. After approval of initial product submittals, provide full size samples of these components in the required color, and in sufficient quantity to completely outfit the model unit. Provide full size samples as dictated by the construction schedule for installation of those components in the model unit. Do not order or approve production of materials beyond those required for the model unit, until the full size samples have been installed in the model unit and approved by the Contracting Officer.

- d. Proceed immediately with construction of the model unit after the Contracting Officer acknowledges the required product submittals are approved.
- e. Provide temporary utilities and climate control necessary to construct, inspect, approve, and maintain the model unit.
- f. Provide and maintain the following throughout the construction period of the entire project: Temporary parking at the model unit; signage from the temporary parking to the model unit; the visual approach to the unit including trash removal, lawn [and landscaping]maintenance.
- g. Locate the model unit in an easily accessible location for public tours. The Government retains the right to use the model unit for supervised tours and public relations activities following each level of completion. Tours will take place during normal working hours.
- [h. Prior to completion and approval of the model unit, and with written permission to start from the Contracting Officer, the Contractor may perform general sitework and run major utility lines for the rest of the project.
-]
- [i. As each model unit's level of completion is approved by the Contracting Officer, construction of that level's components throughout the remainder of the project may proceed.

]3.3.3 Model Unit Levels of Completion

[The completed model unit must be exhibited for approval. Upon completion, the model unit must be available to the Government for a period of 15 working days for inspection after notification is received by Contracting Officer.

][Exhibit and secure approval of model unit at the following levels of completion:

- a. Level 1 (Pre-Finished Level) - Model unit construction within a completed building exterior must exhibit plumbing, ductwork, electrical locations, exposed framing, and windows. Leave walls and floors unfinished but ready for application of final finish. Upon completion of this level, model unit must be available to the Government for a period of five working days for inspection.
- b. Level 2 (Finished Level) - Model unit construction must exhibit all materials, completely finished as specified, including fixtures, wall and floor finishes, cabinets, and shelving. Upon completion, the model unit must be available to the Government for a period of 15 working days for inspection.

][Model unit must be exhibited and approved at the following [7] [_____] levels of completion:

- c. Level 1, model must be available for Government inspection for 28 calendar days.
- d. Levels 2 through 7, model unit must be available for Government inspection for 10 calendar days per level.
- a. Level 1 (Foundation & Floor Slab Level) - Clearly demonstrate

treatment of earth excavation, testing of underslab mechanical piping, earth compaction, assembly and support of concrete reinforcing, technique of concrete movement and placement, framing of foundation steps, placement of insulation, final exposed concrete finished surface treatment, location outlines of all walls, all slab penetrations, and structural support points.

- b. Level 2 (Framing Level) - Clearly exhibit all load bearing and non-load bearing framing, sheathing, anchorage for exterior shell materials, and pre-trim frame-out.
- c. Level 3 (Exterior Shell Level) - Exhibit all exterior finishes, windows and doors with temporary hardware, caulking, sealant, sheathing paper, wall ties, locations of mechanical equipment, roofing, louvers, vents and trim.
- d. Level 4 (Plumbing, Mechanical and Electrical rough-in Level) - Exhibit electrical conduit and box location for switches, outlets fixtures, cable and data collection, ductwork, built-in plumbing fixtures, plumbing piping, and water pressure test of all plumbing pipe.
- e. Level 5 (Interior Finishes-rough Level) - Exhibit non-veneer plaster, unspackled and uncaulked wall and ceiling finishes, insulation, interior doors, final locations of all outlets and fixtures.
- f. Level 6 (Interior Finishes & Fixtures-final Level) - Exhibit final interior finishes, veneer plaster, painting, caulking, built-in shelving, cabinetwork, cabinet supported sinks, final door hardware, mechanical louvers, final electrical equipment connections, electrical fixtures, freestanding bathroom fixtures, properly functioning appliances, and operation of mechanical and electrical systems.
- [g. Level 7 (Site Finishes Level) - Exhibit final driveway, walks, exterior stairs, final concrete and asphalt finished surface treatment, topsoil quality and application, sodding/seeding of lawn, tree and shrub planting, landscaping beds, playground fall areas and playground equipment. Contracting Officer retains the right to inspect all preparation and installation of pavings. Provide adequate notice to the Contracting Officer prior to commencing paving work.

]]] -- End of Section --

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 52.2 (2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP A10.34 (2021) Protection of the Public on or Adjacent to Construction Sites

ANSI/ASSP A10.44 (2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations

ANSI/ASSP A10.8 (2019) Scaffolding Safety Requirements

ASTM INTERNATIONAL (ASTM)

ASTM D6245 (2012) Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation

ASTM D6345 (2010) Standard Guide for Selection of Methods for Active, Integrative Sampling of Volatile Organic Compounds in Air

ASTM F855 (2020) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048 (2016) Guide for Protective Grounding of Power Lines

IEEE C2 (2023) National Electrical Safety Code

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z89.1 (2014; R 2019) American National Standard for Industrial Head Protection

Japanese Standards Association (JSA)

JIS A 8951 (1995) Tubular Steel Scaffolds

MINISTRY OF HEALTH, LABOUR AND WELFARE, GOVERNMENT OF JAPAN (MHLW)

MHLW (1972) Amendment No. 57 - 2006 Industrial Safety and Health Law

ORDINANCE NO. 34 (1972) The Safety Ordinance for Cranes

ORDINANCE NO. 47 (2007) Ordinance on Industrial Safety and Health

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 51B (2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work

NFPA 70 (2023; ERTA 7 2023; TIA 23-15) National Electrical Code

NFPA 70E (2024) Standard for Electrical Safety in the Workplace

NFPA 241 (2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

NFPA 306 (2024) Standard for the Control of Gas Hazards on Vessels

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

ANSI/SMACNA 008 (2007) IAQ Guidelines for Occupied Buildings Under Construction, 2nd Edition

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 Safety -- Safety and Health Requirements Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910 Occupational Safety and Health Standards

29 CFR 1910.1000 Air Contaminants

29 CFR 1915 Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment

29 CFR 1926 Safety and Health Regulations for Construction

CPL 2.100 (1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146

1.2 DEFINITIONS

The following definitions are for the convenience of the reader. If there is a referenced document in the text of this specification section, that is the document that should define terms for that paragraph. If further clarification is needed, contact the Contracting Officer.

1.2.1 Site Safety and Health Officer (SSHO)

A Contractor Employee that is responsible for overseeing and ensuring implementation of the prime Contractor's Safety and Occupational Health (SOH) program according to the Contract, EM 385-1-1, Chapter 2 applicable federal, state, and local requirements.

1.2.1.1 Level One SSHO

A designated employee with full-time SOH responsibility that meets and follows the requirements of EM 385-1-1, Chapter 2-3. In conjunction with the local Safety and Occupational Health Office (SOHO), the Contracting Officer will evaluate proposed Japanese equivalent training for applicability to the contract scope of work being performed.

1.2.1.2 Level Two SSHO

A designated employee with Level Two SSHO responsibility that meets and follows the requirements of EM 385-1-1, Chapter 2-3. Level Two SSHOs cannot be assigned to projects that have a residual Risk Assessment Code (RAC) of high or extremely high.

1.2.1.3 Level Three SSHO

A designated Qualified Person or Competent Person with SOH responsibility that meets and follows the requirements of EM 385-1-1, Chapter 2-3. Level 3 SSHOs cannot be assigned to projects that have a residual RAC of high or extremely high.

1.2.1.4 Alternate SSHO

An employee that meets the definition of the contract-required level SSHO, but is not the primary SSHO.

1.2.2 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate them.

1.2.3 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.3 SUBMITTALS

Government Acceptance or Approval does not remove responsibility from the Contractors for their actions or liability.

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. Submittals not having a "G" classification are for Contractor Quality Control or Designer of Record approval. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

Dive Operations Plan; G

[Final IAQ Management Plan; S

] Indoor Air Quality (IAQ) Management Plan; G

SD-06 Test Reports

Accident Reports; G

LHE Inspection Reports

Monthly Exposure Reports; G

SD-07 Certificates

Crane Operators/Riggers

Activity Hazard Analysis (AHA); G

Certificate of Compliance

Hot Work Permit

Standard Lift Plan; G

Certifications

Licenses

1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by installation emergency management requirement.

1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report and attach to the monthly billing request. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the voucher. Monthly exposure reports are to be tracked and generated through RMS.

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, MHLW Laws and follow host nation laws, ordinances, criteria, rules and regulations at the date of the Solicitation for this Contract. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Site Safety and Health Officer (SSHO)

1.7.1.1 Qualifications of SSHO

All SSHOs will have met the training, experience requirements identified in the EM 385-1-1, Chapter 2 and this Contract. In conjunction with the local Safety and Occupational Health Office (SOHO), the Contracting Officer will evaluate proposed Japanese equivalent training for applicability to the contract scope of work being performed.

1.7.1.2 Duties of SSHO

All SSHOs will carry out the roles and responsibilities as identified in this Contract and the EM 385-1-1, Chapter 2. All SSHOs will be designated on an ENG Form 6282, as approved by the Contracting Officer. Superintendent, QC Manager, and SSHO are subject to dismissal if their required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.7.1.3 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, or foremen must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.7.2 Roles and Responsibilities of Prime Contractor and SSHO

The Prime Contractor and SSHO must ensure that the requirements of all applicable OSHA and EM 385-1-1, Chapter 2 are met for the project. The Prime Contractor must ensure an SSHO or an equally qualified Alternate SSHO(s) is at the worksite at all times to implement and administer the Contractor's safety program and Government accepted Accident Prevention Plan. If the required SSHO has to temporarily (that is, up to 24 hours / 1 day) leave the site of work due to unforeseen or emergency situations, a Level One, Two, or Three SSHO may be used in the interim and must be on

the site of work at all times when work is being performed. For features of work with multiple high or extremely high RAC codes, a Level One SSHO or Alternate Level One SSHO presence is required on site at all times the work is being performed.

If the SSHO must be off-site for a period longer than 24 hours / 1 day, a qualified alternate that meets the contract requirements must be onsite.

a. Prime contractor must ensure all SSHOs will:

- (1) Are designated on an ENG Form 6282.
- (2) Meet minimum training and experience requirements identified in EM 385-1-1, Chapter 2.
- (3) Execute roles and responsibilities identified in EM 385-1-1, Chapter 2.

[1.7.3 Contract Site Safety And Health Officer(s)(SSHOs) Minimum Requirements

Provide a minimum of one Level One SSHO that meets the requirements of EM 385-1-1, Chapter 2 for this project.

]1.7.4 Contract Site Safety and Health Officer(s)(SSHOs) Minimum Requirements for Projects with[Multiple Work Sites,][Multiple Shifts,][Limited Scope,][or Maintenance, or Service Contracts].

Provide a separate full-time Level [One][Two][Three] SSHO at each of the following worksites:

- a. [INDICATE WORKSITE LOCATION] Level [One][Two][Three] SSHO
- b. [INDICATE WORKSITE LOCATION] Level [One][Two][Three] SSHO
- c. [INDICATE WORKSITE LOCATION] Level [One][Two][Three] SSHO

The SSHOs for the worksites listed above must each have the required training, experience, and qualifications in accordance with EM 385-1-1, Chapter 2.

Each SSHO is responsible for implementing and managing the Safety and Occupational Health (SOH) program at the worksite indicated, while ensuring that the 29 CFR 1926, EM 385-1-1, Contracts, and all applicable federal, state, and Local requirements are met.

]1.7.5 Dredging Contract Site Safety and Health Officer(s)(SSHOs) Requirements

Since dredging typically involves underwater diving, refer to EM 385-1-1, Chapter 30 and Section 01 11 30.00 25 DIVING for additional guidance.

1.7.5.1 Dredging SSHO Personnel Requirements

- a. Provide a minimum of one primary Level One SSHO assigned for the primary shift.

Note: Hopper Dredges with U.S. Coast Guard, credentialed crews the prime contractor may designate a Level Two SSHO in lieu of having a

Level one SSHO onboard.

- b. For a project involving multiple work shifts, provide a minimum of a Level[One][Two] SSHO for each additional shift.
- c. For individual dredging projects the prime contractor will designate additional Level Three SSHOs at locations where the primary Level One SSHO is not located (example: on dredge, tug, material placement site).

Examples of one dredging project site is reflected in each of the following:

- (1) a mechanical dredge, tug(s) and scow(s), scow route, and material placement site; or
- (2) a hydraulic pipeline dredge, attendant plant, and material placement site; or,
- (3) a hopper dredge (include land-based material placement site - if applicable.)
- d. Designated SSHOs must be present at the project site, located so that they have full mobility and reasonable access to all major work operations and must be available during their shift for immediate verbal consultation and notification.
- e. Designated Level One and Two SSHOs must have direct report authority to a senior project (or corporate) management official.
- f. Designated Level Three SSHOs must report potential safety and occupational health hazards, incidents, and concerns to the Level One or Two SSHO on shift.
- g. Level One and Level Two SSHOs for dredging must have a minimum of 3 years experience in one of the following areas:
 - (1) Supervising/managing dredging activities.
 - (2) Supervising/managing marine construction activities.
 - (3) Supervising/managing land-based construction activities.
 - (4) Work managing safety programs or processes.
 - (5) Conducting hazard analyses and developing controls in activities or environments with similar hazards.

1.7.6 Competent Person for Confined Space Entry

Provide a CP for Confined Space Entry who meets the requirements of EM 385-1-1, Chapter 34 and herein. The CP for Confined Space Entry must supervise the entry into each confined space.

- [Since this work involves operations that handle combustible or hazardous materials, this person must have the ability to understand and follow through on the air sampling, Personal Protective Equipment (PPE), and instructions of a Marine Chemist, Coast Guard authorized persons, or Certified Industrial Hygienist. Confined space and enclosed space work must comply with NFPA 306, 29 CFR 1915, "Confined and Enclosed Spaces and

Other Dangerous Atmospheres in Shipyard Employment," or as applicable in 29 CFR 1910 for general industry, 29 CFR 1926 for construction.]

1.7.7 Crane Operators/Riggers

Provide Operators meeting the requirements in EM 385-1-1, Chapter 15 for Riggers and Chapter 16 for Crane Operators. In addition, for mobile cranes with Original Equipment Manufacturer (OEM) rated capacities of 22,680 kg or greater, designate crane operators qualified by a source that qualifies crane operators (i.e., union, a Government agency, or an organization that tests and qualifies crane operators). Provide proof of current qualification.

1.7.7.1 Load Handling Equipment Operator Accepted Alternative

The Commander, POJ, has determined that there are adequate provisions provided under the Government of Japan (GOJ) Ordinance to achieve the required protections for crane operator licensing and practical training requirements employed as part of contracted activities provided for USACE within the locality of Japan. Reference POJ-SO decision memorandum, subject, Alternate Requirement Analysis - Load Handling Equipment, Qualifications, dated 8 January 2020.

Provisions within the JISHA (ORDINANCE NO. 34) Articles 223-228 provide sufficient confirmation of an operator's certification/qualification and practical training to achieve the intent of EM 385-1-1, Chapter 16-3.

JISHA (ORDINANCE NO. 34) Articles 223 establishes the Director of the Prefectural Labour Bureau as the licensing authority. Based upon this Article the Ministry of Labour is the recognized auditor, established for local accreditation of certification and training programs.

1.7.8 Preconstruction Conference

- a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).
- b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures, and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays. The creation of the APP and Schedule will be created after being given Notice to Proceed.
- c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.8 ACCIDENT PREVENTION PLAN (APP)

1.8.1 Design Accident Prevention Plan (APP)

Provide a site-specific Accident Prevention Plan (APP), including Activity Hazard Analyses (AHA), in accordance with EM 385-1-1, ENG Form 6293, for the design team to follow during site visits and investigations. For subsequent visits, update the plan if there are changes in the personnel who will be attending, or the tasks to be performed. Submit the APP for review and acceptance by the Government at least 15 calendar days prior to the start of the design field work after being given Notice to Proceed. Field work must not begin until the design APP is accepted by the Contracting Officer. Prior to the start of construction incorporate the Design APP into the Construction APP so that one site specific APP exists for the project and submit to the Contracting Officer for acceptance.

If the design scope includes borings or other subsurface investigations, include in the APP the type of field investigation and verification techniques, such as visual, local utility locating service scanning and third party subcontractor scanning, potholing, or hand digging within two feet of a known utility that will be required. Mark underground utilities before starting any ground-disturbing actions. Notify the Contracting Officer 15 days prior to the start of soil borings or sub-surface investigations.

1.8.2 [Construction] Accident Prevention Plan (APP)

Submit the Accident Prevention Plan (APP) for review and acceptance by the Government at least 15 calendar days prior to the start, after being given Notice to Proceed. A competent person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, ENG Form 6293, and herein. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and occupational health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling employer" for all worksite safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed in accordance with ENG Form 6293 Accident Prevention Plan Worksheet. The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer as attachment 01 35 26-A "ACCIDENT PREVENTION PLAN (APP) WORKSHEET; ENG Form 6293" 15 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of

the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e., imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ANSI/ASSP A10.34), and the environment.

Copies of the accepted APP shall be maintained at the resident engineer's office and at the job site. Continuously review and amend the APP, as necessary, throughout the life of the contract. Incorporate unusual or high-hazard activities not identified in the original APP as discovered.

1.8.3 Names and Qualifications

Provide plans in accordance with the requirements outlined in EM 385-1-1, including the following:

- a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.
- b. As a minimum, designate and submit qualifications of Competent Persons (CP) for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance. Designate and submit qualifications for additional CPs as applicable to the work performed under this Contract.

1.8.4 Plans

Provide plans in the APP in accordance with the requirements outlined in EM 385-1-1, including the following:

[1.8.4.1 Lead, Cadmium, and Chromium Compliance Plan

Identify the safety and health aspects of work involving lead, cadmium and chromium, and prepare in accordance with Section 02 83 00 LEAD REMEDIATION.

] [1.8.4.2 Asbestos Hazard Abatement Plan

Identify the safety and health aspects of asbestos work, and prepare in accordance with Section 02 82 00 ASBESTOS REMEDIATION.

] [1.8.4.3 Site Safety and Health Plan

Identify the safety and health aspects, and prepare in accordance with

Section 01 35 29.13 HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES.

][1.8.4.4 Polychlorinated Biphenyls (PCB) Plan

Identify the safety and health aspects of Polychlorinated Biphenyls work, and prepare in accordance with Sections 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBs) and 02 61 23 REMOVAL AND DISPOSAL OF PCB CONTAMINATED SOILS.

][1.8.4.5 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section 02 41 00 [DEMOLITION] [AND] [DECONSTRUCTION] and referenced sources.

]1.9 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1, Chapter 2 or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOW must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.10 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including

those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.11 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. The Government has no responsibility to provide emergency medical treatment.

1.12 NOTIFICATIONS AND REPORTS

1.12.1 Accident Notification

Notify the Contracting Officer in accordance with the EM 385-1-1 Accident Reporting Timeline.

Table Accident Reporting Required Timeline		
Accident Type	Notify KO or COR	Complete Final Accident Report on ENG 3394 and provide to KO or COR
Fatality, in-patient hospitalization, amputation, eye loss, or property damage over \$600,000.	Immediately, no later than (NLT) 8 Hours	Within 7 Days
All other accidents and near misses	Immediately, no later than (NLT) 24 Hours	Within 7 Days

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any accident or near miss.

1.12.2 Accident Reports

- a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1 to establish the root cause(s) of the accident. All accidents are reportable, regardless of whether or not it is recordable. Complete the applicable USACE Accident Report, ENG Form 3394, and provide the report to the Contracting Officer within 7 calendar days of the accident. The Contracting Officer will provide copies of any required or special forms. All accidents are reportable, regardless of whether or not it is recordable.
- b. Near Misses: Report all "Near Misses" to the Contracting Officer, using local accident reporting procedures, within 24 hours. The Contracting Officer will provide the Contractor the required forms.

Near miss reports are considered positive and proactive Contractor safety management actions.

1.12.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.12.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a Certificate of Compliance for LHE entering an activity under this Contract and in accordance with EM 385-1-1, Chapter 16-10, paragraph b (16-10.b). Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1, Chapter 16-10, paragraph a (16-10.a) and using Standard Pre-Lift Crane Plan/Checklist for each lift planned. Submit SLP to the Contracting Officer for approval within 15 calendar days in advance of planned lift.

1.13 HOT WORK PERMIT

1.13.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e. welding or cutting) or operating other flame-producing/spark producing devices, from the Installation Fire Department. Contractors are required to meet all criteria before a permit is issued. Provide at least two (2) 9kg 4A:20 BC rated extinguishers for normal "Hot Work". Extinguishers shall be current inspection tagged, and have approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch shall be trained in accordance with NFPA 51B and remain on-site for a minimum of one hour after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency Base Fire Department phone number. Report any fire, no matter how small, to the responsible Base Fire Department immediately.

1.13.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "Hot Work" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1, Chapter 6.

1.14 RADIATION SAFETY REQUIREMENTS

Notwithstanding any other hazardous material used in this Contract,

radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. The Contractor shall request approval in writing to the Contracting Officer not less than thirty (30) calendar days in advance of any request for exception. Following Contracting Officer approval of any radiation related exception, the Contractor shall coordinate with the Contracting Officer's Representative at least fifteen (15) calendar days in advance to notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

1.15 CONFINED SPACE ENTRY REQUIREMENTS

Confined space entry must comply with EM 385-1-1, Chapter 34, 29 CFR 1926, 29 CFR 1910, Host Nation, local regulatory requirements and Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used. Obtain written permission from the Contracting Officer prior to confined space entry by personnel.

1.15.1 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.16 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN

Submit an IAQ Management Plan [within [15][_____] calendar days after design[Contract award][notice to proceed] and not less than 10 calendar days before the preconstruction conference.][not less than 10 calendar days before the preconstruction conference.] Revise and resubmit Plan as required by the Contracting Officer. Make copies of the final plan available to all workers on site. Include provisions in the Plan to meet the requirements specified below and to ensure safe, healthy air for construction workers and building occupants.[Submit Final IAQ Management Plan for inclusion in the Sustainability eNotebook, in accordance with Section 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING.]

1.16.1 Requirements During Construction

Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with ASTM D6245. Provide for evaluation of volatile organic compounds (VOCs) in indoor air in accordance with ASTM D6345. Use filters with a Minimum Efficiency Reporting Value (MERV) of 8 in permanently installed air handlers during construction.

1.16.1.1 Control Measures

Meet or exceed the requirements of ANSI/SMACNA 008 to help minimize contamination of the building from construction activities. The five requirements of this manual which must be adhered to are described below:

- a. HVAC protection: Isolate return side of HVAC system from surrounding environment to prevent construction dust and debris from entering the duct work and spaces.
- b. Source control: Use low emitting paints and other finishes, sealants, adhesives, and other materials as specified. When available, cleaning products must have a low VOC content and be non-toxic to minimize building contamination. Utilize cleaning techniques that minimize dust generation. Cycle equipment off when not needed. Prohibit idling motor vehicles where emissions could be drawn into building. Designate receiving/storage areas for incoming material that minimize IAQ impacts.
- c. Pathway interruption: When pollutants are generated use strategies such as 100 percent outside air ventilation or erection of physical barriers between work and non-work areas to prevent contamination.
- d. Housekeeping: Clean frequently to remove construction dust and debris. Promptly clean up spills. Remove accumulated water and keep work areas dry to discourage the growth of mold and bacteria. Take extra measures when hazardous materials are involved.
- e. Scheduling: Control the sequence of construction to minimize the absorption of VOCs by other building materials.

1.16.1.2 Moisture Contamination

- a. Remove accumulated water and keep work dry.
- b. Use dehumidification to remove moist, humid air from a work area.
- c. Do not use combustion heaters or generators inside the building.
- d. Protect porous materials from exposure to moisture.
- e. Remove and replace items which remain damp for more than a few hours.

1.16.2 Requirements After Construction

After construction ends and prior to occupancy, conduct a building flush-out or test the indoor air contaminant levels. Flush-out must be a minimum 2 weeks with MERV-13 filtration media as determined by ASHRAE 52.2 at 100 percent outside air. Air contamination testing must be consistent with EPA's current Compendium of Methods for the Determination of Air Pollutants in Indoor Air. After building flush-out or testing and prior to occupancy, replace filtration media. Filtration media must have a MERV of 13 as determined by ASHRAE 52.2.

1.17 DIVE SAFETY REQUIREMENTS

Develop a Dive Operations Plan, AHA, emergency management plan, and personnel list that includes qualifications, for each separate diving operation in accordance with Section 01 11 30.00 25 Diving. Submit these documents to the District Dive Coordinator (DDC) via the Contracting Officer, for review and approval at least 30 working days prior to commencement of diving operations. These documents must be at the diving location at all times. Provide each of these documents as a part of the project file.

1.18 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

- a. Secure outside equipment and materials and place materials that could be damaged in protected areas.
- b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.
- c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

- a. Head Protection that meets ANSI/ISEA Z89.1
- b. Long Pants
- c. Appropriate Safety Footwear
- d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear

density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. The Contractor shall request approval in writing to the Contracting Officer not less than thirty (30) calendar days in advance of any request for exception. Following Contracting Officer approval of any radiation related exception, the Contractor shall coordinate with the Contracting Officer's Representative at least fifteen (15) calendar days in advance to notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

3.1.3 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e., 29 CFR 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during demolition, repair, renovation, or construction operations. Stop that portion of work and notify the Contracting Officer immediately. Within [14][____] calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages at least [sixty (60)][____] days in advance. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1, Chapter 12. In accordance with EM 385-1-1, Chapter 12 where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy.

These activities include, but are not limited to, a review of Hazardous Energy Control Program (HECP) and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

3.2.1 Okinawa Utility Outages

Outages (power, water, sewer, and communications) for the following listed dates, times, and locations shall not be permitted during the performance of this Contract, unless approved by the authorized agent of the Installation Commander and Contracting Officer's Representative:

- a. DoDEA Schools: No outages from 01 September through 31 July, Monday through Friday between the hours of 0700 and 1800.

- b. Family Housing Areas: No outages will be allowed on Saturdays when DoDEA schools are not in session. No more than two (2) outages for the same utility service shall be allowed in a one-month period (30 calendar days), and no more than one (1) outage in a two-week period (14 calendar days).
- c. U.S. Government Holidays: No outages during the following periods:
 - (1) New Year's Day - to include seven (7) calendar days before and one (1) calendar day after this date (contiguous with the Christmas Day outage requirement).
 - (2) Martin Luther King's Birthday - to include the following Tuesday.
 - (3) President's Day - to include the following Tuesday.
 - (4) Memorial Day - to include the following Tuesday.
 - (5) Juneteenth - to include one (1) calendar day before and after this date.
 - (6) Independence Day - to include one (1) calendar day before and after this date.
 - (7) Labor Day - to include the following Tuesday.
 - (8) Columbus Day - to include the preceding Friday.
 - (9) Veteran's Day - to include two (2) calendar days before and after this date.
 - (10) Thanksgiving Day - to include one (1) calendar day before and after this date.
 - (11) Christmas Day - to include seven (7) calendar days before and after this date (contiguous with the New Year's Day outage requirement).
- d. Child Development Centers (CDC): Outages allowed only when the CDC is closed or on Sundays.
- e. Marine Corps Bases: No outages during Marine Corps Balls or Change of Commands which are normally scheduled between 20 October and 21 November. Specific dates and locations shall be provided during or subsequent to the Preconstruction Conference.
- f. Kadena Air Base: No outages during Change of Commands. Specific dates and locations shall be provided during or subsequent to the Preconstruction Conference.
- g. Torii Station: No outages during Change of Commands. Specific dates and locations shall be provided during or subsequent to the Preconstruction Conference.

3.2.2 Reimbursement for Public Works Support of Electrical Outages

The Contractor shall reimburse the NAF Atsugi Public Works Department for its costs to prepare for, support, and recover from Contractor requested

utility outages associated with this work. Public Works efforts include setting up generators, securing elevators and securing utilities before the outage, and restoring these systems after the outage. The work includes resetting steam systems, air conditioning systems, hot water systems and electrical systems. Outages during the heating and cooling season cost \$1,369 to support during normal working hours, and \$3,200 during overtime hours. Outages in the spring and fall, between heating and cooling seasons, cost \$1,150 to support during normal working hours, and \$2,696 during overtime hours.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1, Chapter 12. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, and the Installation representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HEC training in accordance with EM 385-1-1, Chapter 12. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1, Chapter 12 29 CFR 1910, 29 CFR 1915, ANSI/ASSP A10.44, NFPA 70E.

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1, Chapter 12. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1, Chapter 12. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1, Chapter 12.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECF. In accordance with EM 385-1-1, Chapter 12, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with EM 385-1-1, Chapter 21.

3.5.1 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1, Chapter 21-4.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 and 29 CFR 1926.

3.5.1.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1, Chapter 21-8.t. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

3.5.1.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 1633 kg in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 1.8 m, unless the proper

energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1, Chapter 21-8.t.

3.6 SCAFFOLDING ACCEPTED ALTERNATIVE

The Commander, POJ, has determined that there are adequate provisions provided under the Government of Japan Ordinance to achieve the required protections for supported metal scaffolding employed as part of contracted activities provided for U.S. Army Corps of Engineers (USACE) within the locality of Japan. Reference POJ-SO decision memorandum, subject, Alternate Requirement Analysis - Scaffolding Standards Review, dated 24 September 2019.

This determination only includes supported metal scaffolding structures and does not include other forms of scaffolding methods included in EM 385-1-1, Chapter 22.

JISHA requirements for steel scaffolding structures to conform to the Japanese Industrial Standard JIS A 8951 are comparable to those of ANSI/ASSP A10.8, providing sufficient protection required to achieve the provisions of EM 385-1-1, Chapter 22.

3.7 RIGGING ACCEPTED ALTERNATIVE

The Commander, POJ, has determined that there are adequate provisions provided under the Government of Japan (GOJ) Ordinance, which warrant a partial acceptance for rigging procedures and rigging equipment, employed as part of contracted activities provided for USACE within the locality of Japan. Reference POJ-SO decision memorandum, subject, Alternate Requirement Analysis - Rigging Requirements and Procedures, dated 15 January 2020.

This acceptance excludes Multiple Lift Rigging (MLR) procedures as well as standards for all hooks used for lifting or load handling purposes. MLR and hooks used for load handling purposes will adhere to the respective sections of the EM 385-1-1, Chapter 15 without exception.

Accepted standards include, JISHA ORDINANCE NO. 34, "The Safety Ordinance for Cranes", Chapter VIII Sling Work, Section 1 Slings Equipment Articles 213-220. Also referenced, is the JISHA's "Guideline for sling work safety (2000)".

3.8 EXCAVATIONS

Soil classification shall be performed by a competent person in accordance with EM 385-1-1, Chapter 25.

3.8.1 Excavations and Trenching Accepted Alternative

The Commander, POJ, has determined that there are adequate provisions provided under the Government of Japan Ordinance to achieve the required protections for excavation and trenching systems including shoring, shielding, and engineered designed systems employed as part of contracted activities provided for USACE within the locality of Japan. Reference

POJ-SO decision memorandum, subject, Alternate Requirement Analysis - Excavation and Trenching Review, dated 7 November 2019.

This determination only includes excavation and trenching systems including shoring, shielding, and engineered designed systems where the manufactures or a registered professional engineer's tabulated data has been employed.

Provisions within JISHA ORDINANCE NO. 47, Articles 368-371 provide sufficient protection required to achieve the provisions of EM 385-1-1, Chapter 25-2, paragraph j (25-2.j). JISHA ORDINANCE NO. 47 instructions included in Articles 368-371 conform with the provisions for manufactured shoring/shielding and engineered designed support systems including material serviceability, structural design, and erection/installation. These provisions provided sufficient protection required to achieve the provisions of EM 385-1-1, Chapter 25-2, paragraph j (25-2.j).

3.8.2 Utility Locations

The Contractor shall positively identify underground utilities in the work area and coordinate with the installation utility department. The Contractor shall obtain digging permits prior to start of excavation by contacting the Contracting Officer at least fifteen (15) calendar days in advance.

3.8.3 Utility Location Verification

The Contractor shall scan the construction site with electromagnetic or sonic equipment, and mark the surface of the ground where existing underground utilities are discovered. Verify the elevations of existing piping, utilities, and any type of underground or encased obstruction not indicated to be specified or removed, but indicated or discovered during scanning in locations to be traversed by piping, ducts, and other work to be conducted or installed. Verify elevations before installing new work closer than nearest manhole or other structure at which an adjustment in grade can be made. Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within one meter of the underground system.

3.8.4 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. Whenever Contract work involves chipping, saw cutting, or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with installation utility departments in addition to location and depth verification by the Contractor. The Contractor shall locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement.

3.8.5 Munitions and Explosives of Concern (MEC)

Munitions and Explosives of Concern (MEC): Unexploded Ordinance (UXO),

Material Presenting a Potential Explosive Hazard (MPPEH), Chemical Agents (CA), or Discarded Military Munitions (DMM) on jobsites shall be treated as extremely dangerous and must be reported immediately. Follow the 3Rs: RECOGNIZE, RETREAT, and REPORT.

3.8.5.1 Identification and Notification

If you encounter or suspect you have encountered MEC, the on-site supervisor shall immediately suspend all operations put at risk due to the suspected MEC identified. STOP WORK, and DO NOT TOUCH, mark the location, keep people out of the area, and report to the Government Designated Authority. Project personnel will withdraw along cleared path upwind 300 meters from the discovery or as determined by emergency responders.

The on-site supervisor shall contact the installation emergency services who will intern contact the appropriate Explosives Ordinance Disposal (EOD) unit. EOD will determine the threat and mitigate any immediate explosives hazard(s).

Where suspected MEC are encountered during construction operations and other activities, notify the Contracting Officer's Representative/Contracting Officer immediately. If no hazard is present or poses no danger, the Government will direct the Contractor to proceed without change after given the all clear from the emergency responders. If MEC are found the condition or probability of encountering MEC may also change. The responsible government authority will conduct a new probability assessment, this determination will be used to plan the level of support required and determine further requirements. Pursuant to FAR 52.243-4, "Changes" the Government may issue a modification.

3.8.6 Training

All workers on site involved in the earthwork, in any manner, must attend a UXO brief. Coordinate with Contracting Officer for scheduling of this Briefing. Upon completion of the brief, the worker will receive a laminated card verifying that they attended the brief. The card does not expire and is valid for all USACE construction contracts on Okinawa, but is non-transferrable between workers. The USACE team will help coordinate the training and assist with processing base access requests as needed. The brief is free of charge, but the contractor is responsible for all other costs associated with transport to Camp Foster, labor, and all other costs related to the training. The expected duration of this training is two hours.

3.9 EQUIPMENT

3.9.1 Use of Explosives

Explosives must not be used or brought to the project site without prior written approval from the Contracting Officer. Such approval does not relieve the Contractor of responsibility for injury to persons or for damage to property due to blasting operations.

Storage of explosives, when permitted on Government property, must be only where directed and in approved storage facilities. These facilities must be kept locked at all times except for inspection, delivery, and withdrawal of explosives.

3.10 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1, Chapters 11 and 12.

3.10.1 Electrical Work

As described in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety footwear, insulating gloves and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910 and 29 CFR 1926.

3.10.1.1 Energized Electrical Work Permit (EERP)

All equipment and circuits must be placed into an Establishing an Electrically Safe Working Condition (ESWC) must be de-energized before work is started unless energized work can be justified. A QP must complete and verify the following to establish electrically safe working condition in accordance with EM 385-1-1, paragraph 11-8.d unless the work meets the EERP exemption located under EM 385-1-1, paragraph 11-8.e(5) or the host employer or system owner can show that there are additional hazards or an increased risk from de-energizing.

Hazardous Energy: Any energy including but not limited to mechanical (for example, power transmission apparatus, counterbalances, springs, pressure, gravity), pneumatic, hydraulic, electrical, stored electrical energy exists such as Uninterrupted Power Systems (UPS), chemical, nuclear, and thermal (for example, high or low temperature) energies, which could cause injury to employees.

Once it has been determined that equipment must be worked on in an energized condition, an EERP must be submitted to the USACE supervisor/KO or COR and AHJ for acceptance. Do not perform energized work without prior authorization. All permits must be prepared, signed, and authorized in advance of performing any electrical work. A non-mandatory ENG Form 6277 (Energized Electrical Work Permit) is provided in EM 385-1-1, Chapter 11-10.

3.10.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with Host Nation requirements applicable to where work is being performed.

3.10.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.10.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

3.10.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

-- End of Section --

U.S. Army Corps of Engineers (USACE)
ACCIDENT PREVENTION PLAN (APP) WORKSHEET

For use of this form, see EM 385-1-1; the proponent agency is CESO.

This form serves as a guide. It does not replace or eliminate the need to comply with the requirements set forth in Engineering Manual 385-1-1, Safety and Occupational Health Requirements Manual. The references included in this form correspond to the applicable chapters of EM 385-1-1. No work will begin onsite until the accident prevention plan (APP) is developed according to the EM 385-1-1, reviewed to ensure minimum requirements are met, and approved by the appropriate personnel identified in "Part 1 -Signatures"

PART 1 - SIGNATURES

Plan Preparer - For example: Qualified Person (QP), Competent Person (CP), or project Quality Control (QC).

1. Name:

2. Title:

3. Email Address:

4. Phone Number:

5. Signature

Plan Approver - Corporate Safety Official

1. Name:

2. Title:

3. Email Address:

4. Phone Number:

5. Signature

Plan Concurrence - For example: Industrial hygienist, Project Manager, Superintendent, Site Safety & Health Officer, Quality Control, etc.

1. Name:

2. Title:

3. Email Address:

4. Phone Number:

5. Signature:

PART 2: PROJECT INFORMATION

1. Project Name:

2. Project Address (attach map in Appendix 1):

3. Estimated Project Start Date:

4. Estimated Project Completion Date:

5. Project Description / Description of work to be performed:

PART 3: PRIME CONTRACTOR INFORMATION

1. Prime Contractor Name:

2. Contract Number:

3. Project Manager Name:

4. Quality Control (QC) Manager Name:

5. Contractor Corporate Safety Official Name:

6. Primary Site Safety & Health Officer (SSHO) Name:

7. Alternate Site Safety & Health Officer (SSHO) Name(s):

PART 4: SOH COMMITMENT AND POLICY

1. Provide a statement of safety and health policy detailing commitment to providing a safe and healthful workplace for all employees on this project.

2. Provide a statement outlining the Contractor's safety program goals and objectives.

3. Identify policies and procedure regarding non-compliance with safety requirements.

4. Provide written procedures for holding managers and supervisors accountable for safety.

5. Prime contractor is responsible for the implementation and enforcement of the SOH program for contractor employees, subcontractors, suppliers, and visitors at the site of work?

☐ Yes☐ No

6. Prime contractor will require subcontractors, suppliers, and visitors to comply with the APP and EM 385-1-1?

☐ Yes☐ No

7. No work will be performed unless a designated SSHO is present on the site of work.

☐ Yes☐ No

8. A list of all known subcontractor(s) and supplier(s) are listed in Appendix 2.

☐ Yes☐ No

9. If all subcontractor(s) and supplier(s) are not known at the time of initial APP submittal, Appendix 2 will be updated and resubmitted to the KO/COR for acceptance prior to the start of any work activities.

☐ Yes☐ No

10. All subcontractors will have a copy of the APP, applicable AHAs and will be required to comply with them.

☐ Yes☐ No

11. Subcontractors and suppliers at the site of work will abide by the EM 385-1-1, this APP, and all AHAs.

☐ Yes☐ No

12. This APP and AHAs will be available to all workers at the site of work.

☐ Yes☐ No

PART 5: TRAINING

1. All employees will be trained according to applicable EM 385-1-1, federal, state, and local requirements prior to the commencement of work.

☐ Yes☐ No

2. The following personnel are identified and designated as personnel responsible for SOH on this project. Add the following documentation to Appendix 3 of this worksheet: 1) SSHO designation letter (ENG Form 6282), 2) instructor signed OSHA 30-hour card (or course completion certificate if within 90 days of having completed the OSHA 30-hour training), and 3) proof of 24 hours of competency of SOH training within the last 3 years for the personnel identified below. (EM 385-1-1, Chapter 2, 2-3 Personnel Qualification/Training):
(Check and list names of personnel being designated)

☐ Primary SSHO Name:

☐ Alternate SSHO Name(s):

3. QP(s) and CP(s) required by EM 385-1-1 (for example, excavation, confined space, scaffolding, fall protection, etc.) for this project will be included in Appendix 3. Include 1) List of names or designation letters and 2) proof of qualification/competency.

PART 6: SOH INSPECTIONS

1. All applicable inspections required by EM 385-1-1, contract, federal, state, and local requirements will be completed.

☐ Yes☐ No

2. The following personnel are assigned to conduct inspections, the type of inspection and frequency for the worksite. At minimum the SSHO must conduct daily site inspections.

Who will conduct inspection?	Frequency?	What type of inspection?
+ -		

3. Employees that conduct inspections will have technical proficiency needed to perform inspections. Proof of inspector's training/qualification is included in Appendix 3.

☐ Yes☐ No

4. Deficiencies will be documented and tracked as required by this EM 385-1-1?

☐ Yes☐ No

6. Will any external inspections be required (for example, USCG, OSHA, etc.)?
If yes, describe:

☐ Yes ☐ No
PART 7: ACCIDENT REPORTING AND INVESTIGATION

1. Exposure data (man-hours worked) will be reported to KO/COR as required by the contract?

☐ Yes☐ No

2. All accidents will be reported as soon as possible according to EM 385-1-1?

☐ Yes☐ No

3. All accidents occurring incidentally to an operation, project, or facility for which the EM 385-1-1 is applicable will be thoroughly investigated and analyzed. Corrective actions to prevent reoccurrence will be implemented as soon as reasonably possible. The KO/COR will be notified when corrective actions are completed.

☐ Yes☐ No

4. Accident notification will be provided to the KO/COR as stated below:

☐ Yes☐ No

ACCIDENT TYPE	KO/COR NOTIFICATION TIME
Fatality, permanent and partial disability, or property damage over \$600,000	Immediately, no later than (NLT) 8 hours
All other accidents and near misses	Immediately, NLT 24 hours

PART 8: SOH OVERSIGHT

- | | | |
|--|------------------------------|-----------------------------|
| 1. A risk management process that meets or exceeds the requirement of the EM 385-1-1 will be implemented on this project site? | ☐ Yes | ☐ No |
| 2. Hazardous conditions and unsafe acts will be immediately corrected? | ☐ Yes | ☐ No |
| 3. APP and AHAs will be prepared by a QP or CP according to the EM 385-1-1 prior to the start of any work activities on the site of work (as much as the information can be known at that point in time). | ☐ Yes | ☐ No |
| 4. The plans and AHAs will be updated throughout the life of the project to include changes in personnel, equipment, conditions, etc. Additional revisions must be incorporated as necessary to reflect changing site conditions, construction methods, personnel roles and responsibilities and construction schedules. | ☐ Yes | ☐ No |
| 5. No activity/work will be started at the site of work until the APP is reviewed and submitted to the KO/COR for acceptance, with the site-specific plans and procedures required by the EM 385-1-1 to complete the project? | ☐ Yes | ☐ No |
| 6. APP and AHAs will be provided to and reviewed by all workers involved in the activity prior to work? | ☐ Yes | ☐ No |
| 7. Workers will sign AHA(s) to concur with identified job steps and controls needed to mitigate hazards? | ☐ Yes | ☐ No |

PART 9: SEVERE WEATHER PLAN

- | | | |
|--|------------------------------|-----------------------------|
| 1. The SSHO or designated onsite personnel is responsible for checking the weather conditions at a minimum of twice a day? | ☐ Yes | ☐ No |
| 2. What severe weather forecast will require the SSHO to monitor weather conditions continuously? Describe. | | |

- | |
|---|
| 3. What are the training requirements for severe weather precautions and actions? |
|---|

- | |
|---|
| 4. Identify area of retreat or other actions to be taken such as evacuation, shelter in place, etc. |
|---|

- | |
|---|
| 5. Identify procedures to stop outdoor activities if extreme weather conditions to include thunderstorms are in the area. |
|---|

PART 10: ACTIVITY HAZARD ANALYSIS (RISK MANAGEMENT)

1. Major phases of work anticipated:

2. Definable Features of Work (DFOW)

Note. An activity-specific AHA will be submitted to the KO/COR for each DFOW prior to any work being performed.

3. List equipment to be used:

4. Anticipated High-Risk Activities: (See examples below, check all that apply):

☐ Electrical and/or Hazardous Energy

☐ Confined Space

☐ Working at Heights

☐ Excavations & Trenching

☐ Scaffolding

☐ Other: _____

☐ Diving

☐ Other: _____

☐ Load Handling Equipment

☐ Other: _____

☐ Occupational Exposure

☐ Other: _____

5. Detailed project-specific hazards and controls will be provided utilizing AHAs for each activity/DFOW.

☐ Yes

☐ No

6. No work will begin on an activity/DFOW until the initial AHA addressing the project-specific hazards has been developed, reviewed, and accepted by the KO/COR. (See EM 385-1-1 Chapter 2 section 2-6).

☐ Yes

☐ No

7. AHAs are living documents and are intended to be created, utilized in the field, and updated (by the workers) as necessary.

☐ Yes

☐ No

8. Workers/crews will have accepted AHA(s) that reflects current site conditions, personnel, equipment, control measures, etc. while the work is being performed?

☐ Yes

☐ No

9. AHA will be used by the contractor personnel to assure work is being performed consistent with the AHA?

☐ Yes

☐ No

10. Once the activity has been completed, the AHA will be available and kept onsite for the length of the contract.

☐ Yes

☐ No

11. AHAs for unknown work activities will be submitted to the KO/COR prior to work and attached to Appendix 4 of this document.

☐ Yes

☐ No

12. Who has the authority to accept residual risk on AHA(s)? (List below)

Category of Risk	Approval Authority (Name and Title)
Extremely High Risk	
High Risk	
Medium Risk	
Low Risk	

PART 11: REQUIRED PROJECT SITE-SPECIFIC PLANS AND PROCEDURES

- | | | |
|---|------------------------------|-----------------------------|
| 1. Based on a risk assessment of contracted activities and on mandatory OSHA compliance programs, the contractor will ensure that all applicable SOH risks and associated compliance plans are in place. | ☐ Yes | ☐ No |
| 2. Include all project-specific compliance plans, as applicable to the work being performed, and as identified below. The plans will incorporate project-wide procedures to control hazards to which the employees of all project employers may be exposed. | ☐ Yes | ☐ No |
| 3. These plans and procedures will be coordinated with all project employers and be included in project-specific emergency response and evacuation procedures. | ☐ Yes | ☐ No |
| 4. Address each of the plans in the outline below in the order that they are listed.
Option 1: If a plan is applicable, develop and attach it to Appendix 5. Submit to the KO/COR with this worksheet for acceptance prior to work.
Option 2: If a plan is applicable, but all of the minimum required information for the plan is not yet known, provide justification next to each plan title, and provide an expected date the plan will be created, reviewed, and attached to Appendix 5. Submit to the KO/COR with this worksheet for acceptance prior to work.
Option 3: If a plan is not applicable because of the nature of the work to be performed, provide justification next to each plan title. | | |

Minimum plans required by EM 385-1-1	Reference EM 385-1-1 paragraph:	Select Option	Justification for Option 2 and Option 3
Exposure Control Plan	3-7.a		
AED Program	3-7.b		
Construction Site Plan	4-7.a		
Access/Haul Road Plan	4-7.b		
Hearing Conservation Program	5-7.a		
Respiratory Protection Program	5-7.b		
Hazard Communication (HazCom) Program	6-7.a		
Process Safety Management (PSM) Program	6-7.b		
Lead Compliance Plan	6-7.c		
Asbestos Management Plan	6-7.d		
Ionizing Radiation Safety Plan	6-7.e		
Radiofrequency Plan	6-7.e		
Abrasive Blasting Plan	6-7.g		
Thermal Stress Management Plan	6-7.h		
Mold Remediation Plan	6-7.i		
Silica Compliance Plan	6-7.j		
Fatigue Management Plan	6-7.k		
Night Operations Lighting Plan	7-7		
Traffic Control Plan	8-7		
Fire Prevention Plan	9-7.a		
Wildland Fire Management Plan	9-7.b		
Housekeeping Plan	10-7		
Electrical Safety Plan	11-7		
Assured Equipment Grounding Conductor Program	11-7.b		
Energy Control Plan	12-7.a		
Hazard Isolation Procedures	12-7.b		

Demolition Plan	17-7		
Emergency Plan for Marine Activities	19-7.a		
Severe Weather Plan for Marine Activities	19-7.b		
Float Plans	19-7.c		
Tow Plans	19-7.d		
Fall Protection and Prevention Plan	21-7.a		
Rescue Plan	21-7.b		
Scaffold Work Plan	21-7.b		
Rope Access Work Plan	23-7		
Excavation and Trenching Plan	25-7		
Fire Prevention and Protection Plan	26-7 a & f		
Compressed Air Work Plan	26-7 a & b		
Contingency Response Plan	26-7 a & c		
Emergency Rescue Plans and Equipment	26-7 a & d		
Ventilation and Dust Control Plan	26-7 a & e		
Formwork and Shoring Plan	27-7.a		
Masonry Bracing Plan	27-7.b		
Steel Erection Plan	28-7.a		
Site Layout Plan	28-7.b		
Explosive Site Safety Plan	29-7.a		
Master Blasting Plan	29-7.b		
Vibrations Monitoring Plan	29-7.c		
Dive Operations Plan	30-7.a & d		
Emergency Management Plan	30-7.a & e		
Safe Practices Manual	30-7.b		
Snorkeling Plan	30-7.f		
Tree Maintenance and Removal Plan	31-7		
Construction Safety and Phasing Plan	32-7.a		
Safety Plan Compliance Document	32-7.b		
Aviation Pre-Accident Plan	33-7		
Standard Lift Plan	34-7.b		
Confined Space Plan	34-7.b & c		
Standard Lift Plan - Floating Plan	34-7.c		
Critical Lift Plan	34-7.d		
Pile Driving	34-7.e		
Housekeeping Plan	35-7.a		
Extermination Plan	35-7.b		
Site Safety and Health Plan	36-7.a		
Comprehensive Work Plan	36-7.b		
Emergency Response Plan	36-7.c		

PART 12: PROJECT SITE COORDINATION

1. Provide procedures for coordinating SOH activities with other employers on the site of work:

PART 13: REQUIRED APPENDICES

Indicate if each of the following items are, or are not, attached to this plan.

Appendix 1. Project Map	☐ Yes	☐ No
Appendix 2. Subcontractor/Suppliers	☐ Yes	☐ No
Appendix 3. Personal Qualifications and Designations	☐ Yes	☐ No
Appendix 4. Activity Hazard Analysis	☐ Yes	☐ No
Appendix 5. Site-specific plans and procedures	☐ Yes	☐ No
Appendix 6. Any additional plans, procedures, etc. can be added to this appendix and will be coordinated with the KO/COR.	☐ Yes	☐ No

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS
02/19

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

AACE INTERNATIONAL (AACE)
1265 Suncrest Towne Centre Drive
Morgantown, WV 26505-1876 USA
Ph: 304-296-8444
Fax: 304-291-5728
Internet: <https://web.aacei.org/>

AIR BARRIER ASSOCIATION OF AMERICA (ABAA)
1600 Boston-Providence Hwy
Walpole, MA 02081
Ph: 1-866-956-5888
Fax: 1-866-956-5819
Internet: <https://www.airbarrier.org/>

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)
30 West University Drive
Arlington Heights, IL 60004 U.S.A.
Ph: +1 847-394-0150
E-mail: communications@amca.org
Internet: <https://www.amca.org/>

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)
2111 Wilson Blvd, Suite 400
Arlington, VA 22201
Ph: 703-524-8800
Internet: <http://www.ahrinet.org>

ALUMINUM ASSOCIATION (AA)
1400 Crystal Drive
Suite 430
Arlington, VA 22202
Ph: 703-358-2960

E-Mail: info@aluminum.org
Internet: <https://www.aluminum.org/>

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)
1900 E Golf Rd, Suite 1250
Schaumburg, IL 60173
Ph: 847-303-5664
E-mail: customerservice@aamanet.org
Internet: <https://aamanet.org/>

AMERICAN ASSOCIATION OF RADON SCIENTISTS AND TECHNOLOGISTS (AARST)
475 South Church Street - Suite 600
Hendersonville, NC 28792
Ph: 800-269-4174
Fax: 828-214-6299
E-mail: info@aarst.org
Internet: <http://aarst-nrpp.com/wp/>

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)
444 North Capital Street, NW, Suite 249
Washington, DC 20001
Ph: 202-624-5800
Fax: 202-624-5806
E-Mail: info@aaashto.org
Internet: <https://www.transportation.org/>

AMERICAN CONCRETE INSTITUTE (ACI)
ACI World Headquarters
38800 Country Club Drive
Farmington Hills, MI
48331-3439 USA
Ph: 248-848-3800
Internet: <https://www.concrete.org/>

AMERICAN FOREST FOUNDATION (AFF)
American Tree Farm System
2000 M Street, NW, Suite 550
Washington, DC 20036
Ph: 202-765-3660
Fax: 202-827-7924
Email: info@forestfoundation.org
Internet: <https://www.treefarmssystem.org>

AMERICAN HARDBOARD ASSOCIATION (AHA)
1210 West Northwest Highway
Palatine, IL 60067
Ph: 847-934-8800
Fax: 847-934-8803
E-mail: aha@hardboard.org
Internet: <http://domensino.com/AHA/>

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)
130 East Randolph, Suite 2000
Chicago, IL 60601
Ph: 312-670-5444
Fax: 312-670-5403
Steel Solutions Center: 866-275-2472
E-mail: solutions@aisc.org

Internet: <https://www.aisc.org/>

AMERICAN INSTITUTE OF TIMBER CONSTRUCTION (AITC)
7012 South Revere Parkway, Suite 140
Centennial, CO 80112
Ph: 503-639-0651
Fax: 503-684-8928
E-mail: mschoen@wclib.org
Internet: <http://www.aitc-glulam.org>

AMERICAN IRON AND STEEL INSTITUTE (AISI)
25 Massachusetts Avenue, NW Suite 800
Washington, DC 20001
Ph: 202-452-7100
Internet: <https://www.steel.org/>

AMERICAN LUMBER STANDARDS COMMITTEE (ALSC)
7470 New Technology Way, Suite F
Frederick, MD 21703
Ph: 301-972-1700
Fax: 301-540-8004
E-mail: alasc@alasc.org
Internet: <http://www.alasc.org>

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
1899 L Street, NW, 11th Floor
Washington, DC 20036
Ph: 202-293-8020
Fax: 202-293-9287
E-mail: storemanager@ansi.org
Internet: <https://www.ansi.org/>

AMERICAN PETROLEUM INSTITUTE (API)
1220 L Street, NW
Washington, DC 20005-4070
Ph: 202-682-8000
Internet: <https://www.api.org/>

AMERICAN RAILWAY ENGINEERING AND MAINTENANCE-OF-WAY ASSOCIATION
(AREMA)
4501 Forbes Blvd., Suite 130
Lanham, MD 20706
Ph: 301-459-3200
E-mail: info@arema.org
Internet: <https://www.arema.org>

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)
P.O. Box 28518
1711 Arlingate Lane
Columbus, OH 43228-0518
Ph: 800-222-2768 or 614-274-6003
Fax: 614-274-6899
E-mail: tjones@asnt.org
Internet: <https://www.asnt.org/>

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)
1801 Alexander Bell Drive
Reston, VA 20191
Ph: 800-548-2723; 703-295-6300

Internet: <https://www.asce.org/>

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)
1791 Tullie Circle, NE
Atlanta, GA 30329
Ph: 404-636-8400 or 800-527-4723
Fax: 404-321-5478
E-mail: ashrae@ashrae.org
Internet: <https://www.ashrae.org/>

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Two Park Avenue
New York, NY 10016-5990
Ph: 800-843-2763
Fax: 973-882-1717
E-mail: customercare@asme.org
Internet: <https://www.asme.org/>

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)
520 N. Northwest Highway
Park Ridge, IL 60068
Ph: 847-699-2929
E-mail: customerservice@assp.org
Internet: <https://www.assp.org/>

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)
18927 Hickory Creek Drive, Suite 220
Mokena, IL 60448
Ph: 708-995-3019
Fax: 708-479-6139
Internet: <http://www.asse-plumbing.org>

AMERICAN WATER WORKS ASSOCIATION (AWWA)
6666 W. Quincy Avenue
Denver, CO 80235 USA
Ph: 303-794-7711 or 800-926-7337
Fax: 303-347-0804
Internet: <https://www.awwa.org/>

AMERICAN WELDING SOCIETY (AWS)
8669 NW 36 Street, #130
Miami, FL 33166-6672
Ph: 800-443-9353
Internet: <https://www.aws.org/>

AMERICAN WOOD COUNCIL (AWC)
222 Catoctin Circle SE, Suite 201
Leesburg, VA 20175
Ph: 800-890-7732
Fax: 412-741-0609
E-mail: publications@awc.org
Internet: <https://www.awc.org/>

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)
P.O. Box 361784
Birmingham, AL 35236-1784
Ph: 205-733-4077
Fax: 205-733-4075

Internet: <http://www.awpa.com>

APA - THE ENGINEERED WOOD ASSOCIATION (APA)
7011 South 19th St.
Tacoma, WA 98466-5333
Ph: 253-565-6600
Fax: 253-565-7265
Internet: <https://www.apawood.org/>

ARCHITECTURAL INSTITUTE OF JAPAN (AIJ)
26-20 Shiba 5-chome
Minato-ku, Tokyo 108-8414
PH: +81-3-3456-2051
E-mail: info@aij.or.jp
Internet: <https://www.aij.or.jp/aijhome.htm>

ASSOCIATED AIR BALANCE COUNCIL (AABC)
1220 19th St NW, Suite 410
Washington, DC 20036
Ph: 202-737-0202
Fax: 202-315-0285
E-mail: info@aabc.com
Internet: <https://www.aabc.com/>

ASSOCIATION FOR MATERIALS PROTECTION AND PERFORMANCE (AMPP)
800 Trumbull Drive
Pittsburgh, PA 15205
Ph (US): +1-877-281-7772
Ph (International): +1-412-281-2331
Internet: <https://www.ampp.org/home>

ASSOCIATION OF DIVING CONTRACTORS INTERNATIONAL (ADCI)
5206 Cypress Creek Parkway Ste. 202
Houston, TX 77069
Ph: 281-891-8388 or 281-893-6438
Fax: 281-893-5118
Internet: <https://www.adc-int.org/>

ASTM INTERNATIONAL (ASTM)
100 Barr Harbor Drive, P.O. Box C700
West Conshohocken, PA 19428-2959
Ph: 610-832-9500
Fax: 610-832-9555
E-mail: service@astm.org
Internet: <https://www.astm.org/>

BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)
529 14th Street NW Suite 1280
Washington, DC 20045
Ph: 212-297-2122
Fax: 212-370-9047
Internet: <https://www.buildershardware.com/>

CALIFORNIA AIR RESOURCES BOARD (CARB)
1001 I Street
Sacramento, CA 95814
Ph: 800-242-4450
Email: helpline@arb.ca.gov
Internet: <https://ww2.arb.ca.gov/>

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)
PO Box 997377, MS 0500
Sacramento, CA 95899-7377
Ph: 916-558-1784
Internet: <https://www.cdph.ca.gov/>

COMPOSITE PANEL ASSOCIATION (CPA)
19465 Deerfield Avenue, Suite 306
Leesburg, VA 20176
Ph: 703-724-1128
Fax: 703-724-1588
Internet: <https://www.compositepanel.org/>

COMPRESSED GAS ASSOCIATION (CGA)
14501 George Carter Way, Suite 103
Chantilly, VA 20151-1788
Ph: 703-788-2700
Fax: 703-961-1831
E-mail: cga@cganet.com
Internet: <https://www.cganet.com/>

COUNCIL ON ENVIRONMENTAL QUALITY (CEQ) (WHITE HOUSE)
722 Jackson Place
Washington DC 20506
Internet: <https://www.whitehouse.gov/administration/eop/ceq>

CRANE MANUFACTURERS ASSOCIATION OF AMERICA (CMAA)
8720 Red Oak Boulevard, Suite 201
Charlotte, NC 28217-3992
Ph: 704-676-1190
Fax: 704-676-1199
Internet: <http://www.mhi.org/cmaa>

CSA GROUP (CSA)
178 Rexdale Blvd.
Toronto, ON, Canada M9W 1R3
Ph: 416-747-4044
Fax: 416-747-2510
E-mail: member@csagroup.org
Internet: <https://www.csagroup.org/>

EUROPEAN COMMITTEE FOR STANDARDIZATION (CEN/CENELEC)
CEN-CENELEC Management Centre
Rue de la Science 23
B - 1040 Brussels, Belgium
Ph: 32-2-550-08-11
Fax: 32-2-550-08-19
Internet: <https://www.cencenelec.eu/>

ELECTRICAL SAFETY INSPECTION ASSOCIATIONS
Shintoranomon Jitsugyokaikan 9F
1-1-21 Toranomon
Minato-ku, Tokyo 105-0011, JAPAN
E-mail: info@denki-hoan.org
Internet: <https://denkihoan.org/en/>

FEDERAL EMERGENCY MANAGEMENT AGENCY (FEMA)
615 Chestnut Street

One Independence Mall, Sixth Floor
Philadelphia, PA 19106-4404
Ph: 215-931-5597
E-mail: femar3newsdesk@fema.dhs.gov
Internet: <https://www.fema.gov/>

FM GLOBAL (FM)
270 Central Avenue
Johnston, RI 02919-4949
Ph: 401-275-3000
Fax: 401-275-3029
Internet: <https://www.fmglobal.com/>

FOREST STEWARDSHIP COUNCIL (FSC)
708 First Street North, Suite 235
Minneapolis, MN 55401
Ph: 612-353-4511
E-mail: info@us.fcs.org
Internet: <https://us.fsc.org/>

FOUNDATION FOR CROSS-CONNECTION CONTROL AND HYDRAULIC RESEARCH
(FCCCHR)
USC Foundation Office
Research Annex 219
Los Angeles, CA 90089-7700
Ph: 866-545-6340
Fax: 213-740-8399
E-mail: fccchr@usc.edu
Internet: <https://fccchr.usc.edu/>

GREEN BUSINESS CERTIFICATION INC. (GBCI)
Washington, D.C.
Ph: 1-800-795-1747
Internet: <https://www.gbci.org>

GREEN BUILDING INITIATIVE (GBI)
7805 SW 40th Ave. #80010
Portland, Oregon 97219
Ph: 503-274-0448
Email: info@thegbi.org
Internet: <https://www.thegbi.org/>

GREEN SEAL (GS)
1001 Connecticut Avenue, NW
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Ph: + 33 1 45 24 82 00
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U. S. GREEN BUILDING COUNCIL (USGBC)
2101 L St NW, Suite 500
Washington, DC 20037
Ph: 202-828-7422
Internet: <https://new.usgbc.org/>

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)
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College Park, MD 20740-6001
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Internet: <https://www.archives.gov/>
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U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)
1322 Patterson Ave. SE, Suite 1000
Washington Navy Yard, DC 20374-5065
Ph: 202-685-9387
Internet: <http://www.navfac.navy.mil>

U.S. NAVAL SEA SYSTEMS COMMAND (NAVSEA)
Commander Naval Sea Systems Command
1333 Isaac Hull Ave., SE
Washington Navy Yard, DC 20376-1080
Ph: 202-781-0000
E-mail: navsea_publicqueries@navy.mil
Internet: <http://www.navsea.navy.mil/PublicInquiries.aspx>

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UL Directories available through IHS at <https://ihsmarkit.com/>

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7 Underwriters Road
Toronto, Ontario, Canada M1R 3A9
Ph: 866-937-3852
Fax: 416.757.8727
E-mail: cec@ul.com
Internet: <https://canada.ul.com/>

VIBRATION ISOLATION AND SEISMIC CONTROL MANUFACTURERS ASSOCIATION
(VISCMA)
994 Old Eagle School Road
Suite 1019
Wayne, PA 19087-1866
Ph: 610-971-4850
E-mail: info@viscma.com
Internet: <http://www.viscma.com>

WEST COAST LUMBER INSPECTION BUREAU (WCLIB)
6980 S.W. Varns
Tigard, OR 97223
Ph: 503-639-0651
Fax: 503-684-8928
E-mail: info@wclib.org
Internet: <http://www.wclib.org>

WINDOW AND DOOR MANUFACTURERS ASSOCIATION (WDMA)
2001 K Street NW
3rd Floor North
Washington, DC 20006
Ph: 202-367-1157
or
330 N Wabash Avenue, Suite 2000
Chicago, IL 60611
Ph: 312-321-6802
E-mail: membersupport@wdma.com
Internet: <https://www.wdma.com/>

WOOD MOULDING AND MILLWORK PRODUCERS ASSOCIATION (WMMPA)
1282 Stabler Lane, STE 630 #146
Yuba City, CA95993-2625
Ph: 530-661-9591
Fax: 530-661-9586
E-mail: info@wmmpa.com
Internet: <https://www.wmmpa.com/>

WOODWORK INSTITUTE (WI)
1455 Response Road, Suite 110
Sacramento, CA 95815
Ph: 916-372-9943
Fax: 916-372-9950
E-mail: info@woodinst.com
Internet: <https://woodworkinstitute.com>

WESTERN WOOD PRODUCTS ASSOCIATION (WWPA)
1500 SW First Ave., Suite 870
Portland, OR 97201
Ph: 503-224-3930
E-mail: info@wwpa.org
Internet: <http://www.wwpa.org>

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

Not used

-- End of Section --

SECTION 01 42 15

METRIC MEASUREMENTS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM SI10	(2010) American National Standard for Use of the International System of Units (SI): The Modern Metric System
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1.2 GENERAL

This project includes metric units of measurements. The metric units used are the International System of Units (SI) developed and maintained by the General Conference on Weights and Measures (CGPM); the name International System of Units and the international abbreviation SI were adopted by the 11th CGPM in 1960. The following criteria allow both metric and English inch-pound (I-P) measurements to be shown for a product:

- a. To substitute products that are manufactured to an industry recognized rounded metric (hard metric) dimension with I-P products.
- b. To indicate industry and/or Government standards, test values or other controlling factors (such as the code requirements where I-P values are needed for clarity).
- c. To trace back to the referenced standard(s), test value(s) or code(s).

1.3 USE OF MEASUREMENTS IN DRAWINGS AND SPECIFICATIONS

Measurements in drawings and specifications shall be in SI units, with I-P units in parenthesis in cases as indicated in Paragraph GENERAL and as otherwise authorized by the Contracting Officer. Procure the product in SI units unless otherwise authorized by the Contracting Officer. The Contractor is responsible for all associated labor and materials when authorized to substitute one system of units for another and for the final assembly and performance of the specified work and/or products.

1.3.1 Hard Metric

Hard metric measurements are often used for field data such as distance from one point to another or distance above the floor. Products are considered to be hard metric when they are manufactured to metric dimensions or have an industry recognized metric designation.

1.3.2 Soft Metric

- a. A soft metric measurement is a non-mathematical, industry related conversion. Soft metric measurements are used for measurements

pertaining to products, test values, and other situations where the I-P units are the standard for manufacture, verification, or other controlling factor.

- b. A soft metric measurement is also indicated for products that are manufactured in industry designated metric dimensions but are required by law to allow substitute I-P products.

1.3.3 Neutral

A neutral measurement is indicated by an identifier which has no expressed relation to either an SI or an I-P value (e.g., American Wire Gage (AWG) which indicates thickness but in itself is neither SI nor I-P).

1.4 COORDINATION

Bring discrepancies, such as mismatches or product unavailability, arising from use of both metric and non-metric measurements and discrepancies between the measurements in the specifications and the measurements in the drawings to the attention of the Contracting Officer for resolution.

1.5 RELATIONSHIP TO SUBMITTALS

Submittals for Government approval or for information only covers the SI products actually being furnished for the project. Submit the required drawings and calculations in the same units used in the Contract documents describing the product or requirement unless otherwise instructed or approved. Use ASTM SI10 as the basis for establishing metric measurements required to be used in submittals.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION 01 45 00

QUALITY CONTROL
08/23, CHG 3: 08/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D3740	(2019) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
ASTM E90	(2023) Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements
ASTM E329	(2023) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements
ER 1110-3-12	(2021) Military Engineering and Design Quality Management

1.2 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program. Include all associated costs in the applicable Bid Pricing Schedule item.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G, [_____]

Additional Requirements For Design Quality Control (DQC) Plan; G, [_____]

SD-05 Design Data

Design Quality Control

Discipline-Specific Checklists

SD-06 Test Reports

Verification Statement

SD-07 Certificates

Certificate Of Readiness; G, [____]

1.4 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system that complies with FAR 52.246-12 Inspection of Construction. QC is comprised of plans, procedures, and organization necessary to produce an end product that complies with the Contract requirements. The QC system covers all design and construction operations, both onsite and offsite, and must be keyed to the proposed design and construction sequence. The Quality Control Manager, Superintendent, Site Safety and Health Officer (SSHO), and all on-site supervisors are responsible for the quality of work and are subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the Contract. The Quality Control Manager must maintain a physical presence at the work site at all times and is the primary individual responsible for all quality control.

1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this section. This QC program is a key element in meeting the objectives of the Commissioning Process (Cx). The QC program consists of a QC Organization, QC Plan, QC Plan Meeting(s), a Coordination and Mutual Understanding Meeting, QC meetings, three phases of control, submittal review and approval, testing, completion inspections, QC certifications, independent Special Inspections in accordance with Section 01 45 35 SPECIAL INSPECTIONS, and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations that comply with the requirements of this Contract. The QC program must cover on-site and off-site work and be keyed to the work sequence. No construction work or testing may be performed unless the QC Manager is on the work site. The QC Manager must report to an officer of the firm and not be subordinate to the Project Superintendent or the Project Manager. The QC Manager, Project Superintendent and Project Manager must work together effectively. Although the QC Manager is the primary individual responsible for quality control, all individuals will be held responsible for the quality of work on the job.

1.5.1 Meetings

1.5.1.1 Quality Control Plan Meeting

Prior to submission of the QC Plan, the Contractor may request a meeting with the Contracting Officer to discuss the QC Plan requirements of this Contract.

The purpose of this meeting is to develop a mutual understanding of the QC Plan requirements prior to plan development and submission and to agree on the Contractor's list of Definable Feature of Work (DFOW).

1.5.1.2 Coordination and Mutual Understanding Meeting

After the Preconstruction Conference, Post Award Conference, before start of design or construction, and prior to acceptance by the Government of the CQC Plan, meet with the Contracting Officer and discuss the Contractor's quality control system. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations, design activities, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the QC Manager and signed by the Contractor, the Architect/Engineer (A/E), and the Government. Provide a copy of the signed minutes to all attendees[and incorporate into the relevant QC Plan]. At a minimum the Coordination and Mutual Understanding Meeting must be repeated when a new QC Manager is appointed. There can be other occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

1.5.1.2.1 Purpose

The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site work, design intent, Cx in accordance with Section 01 91 00.15 BUILDING COMMISSIONING, environmental requirements and procedures, coordination of activities to be performed, Special Inspections, and the coordination of the Contractor's management, production, and QC personnel. At the meeting, the Contractor must explain in detail how three phases of control will be implemented for each DFOW, as well as how each DFOW will be affected by each management plan or requirement as listed below:

- a. Waste Management Plan.
- b. Procedures for noise and acoustics management.
- c. Environmental Protection Plan.
- d. Environmental regulatory requirements.
- e. Cx Plan requirements in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.
- f. Special Inspections.
- g. Indoor Air Quality (IAQ) Management Plan.

1.5.1.2.2 Coordination of Activities

Coordinate activities included in various sections to assure efficient and orderly installation of each component. Coordinate operations included under different sections that are dependent on each other for proper

installation and operation. Schedule construction operations with consideration for indoor air quality as specified in the IAQ Management Plan. Coordinate Special Inspections.

1.5.1.2.3 Attendees

As a minimum, the Contractor's personnel required to attend include an officer of the firm, the Project Manager, Project Superintendent, QC Manager, Alternate QC Manager, [Assistant QC Manager,] [QC Specialists,] Special Inspector, [Special Inspector of Record,] Commissioning Provider (Cx/C), [Environmental Manager,] and subcontractor representatives. Each subcontractor who will be assigned QC responsibilities must have a principal of the firm at the meeting.

1.5.1.3 Quality Control (QC) Meetings

After the start of construction, conduct weekly QC meetings led by the QC Manager at the work site with the Project Superintendent, [the QC Specialists,] the Special Inspector, [the Special Inspector of Record,] CxC, and the other personnel as necessary. The QC Manager is to prepare the minutes of the meeting and provide a copy to the Contracting Officer within 2 working days after the meeting. The Contracting Officer may attend these meetings. As a minimum, accomplish the following at each meeting:

- a. Review the minutes of the previous meeting.
- b. Review the schedule and the status of work and deficiencies/rework. Review the most current approved schedule (in accordance with schedule specification) and the status of work and deficiencies/rework.
- c. Review the status of submittals and Request For Information (RFIs).
- d. Review the work to be accomplished in the next 3 weeks as defined by the schedule section paragraph WEEKLY PROGRESS MEETINGS in Section 01 32 01.00 10 PROJECT SCHEDULE and all quality control documentation required for that work.
- e. Review Testing Plan and Log including status of tests performed since last QC Meeting.
- f. Resolve QC and production problems. Discuss status of pending change orders.
- g. Address items that may require revising the QC Plan.
- h. Review Accident Prevention Plan (APP) and effectiveness of the safety program.
- i. Review environmental requirements and procedures.
- j. Review Environmental Management Plan.
- k. Review Waste Management Plan.
- l. Review the status of training completion.
- m. Review Cx Plan and progress. Review Issues Log and resolution.

- n. Review IAQ Management Plan.

1.5.2 Contractor Quality Control (CQC) Plan

Submit no later than [30][_____] days after receipt of notice to proceed, the CQC Plan proposed to implement the requirements FAR 52.246-12 Inspection of Construction. The Government will consider an interim plan for the first [_____] days of operation. Design and Construction will be permitted to begin only after acceptance of the CQC Plan and other Contract requirements or acceptance of an interim plan applicable to the particular feature of work to be started. Work outside of the accepted interim plan will not be permitted to begin until acceptance of a CQC Plan or another interim plan containing the additional work. Design work is not permitted to start prior to approval of a Design Quality Control Plan.

1.5.2.1 Content of Contractor Quality Control (CQC) Plan

Provide a CQC Plan, prior to start of construction that includes a table of contents, with major sections identified, pages numbered sequentially, and that documents the proposed methods and responsibilities for accomplishing quality control during the construction of the project. The CQC Plan must at a minimum include the following sections:

- a. A description of the quality control organization and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified.
- b. An organizational chart showing the quality control organization with individual names and job titles and lines of authority up to an executive of the company at the home office.
- c. NAMES AND QUALIFICATIONS: Names and qualifications, in resume format, (including position titles and durations for qualifying experiences) for each person in the QC organization. Include the Construction Quality Management (CQM) for Contractors course certifications for the QC personnel as required by the paragraph CONSTRUCTION QUALITY MANAGEMENT TRAINING.
- d. DUTIES, RESPONSIBILITY AND AUTHORITY OF QC PERSONNEL: Duties, responsibilities, and authorities of each person in the QC organization.
- e. OUTSIDE ORGANIZATIONS: A listing of outside organizations, such as architectural and consulting engineering firms, that will be employed by the Contractor and a description of the services these firms will provide.
- f. APPOINTMENT LETTERS: Letters signed by an officer of the firm appointing the QC Manager and Alternate QC Manager, Design Quality Control Manager, CxC, and stating that they are responsible for implementing and managing the QC program as described in this Contract. Include in this letter the responsibility of the QC Manager and Alternate QC Manager to implement and manage the three phases of control, and their authority to stop work that is not in compliance with the Contract. Letters of direction are to be issued by the QC Manager to[the Assistant QC Manager and] all other QC Specialists or quality control representatives outlining their duties, authorities,

and responsibilities. Include copies of the letters in the QC Plan.

- g. SUBMITTAL PROCEDURES AND INITIAL SUBMITTAL REGISTER: Procedures for reviewing, approving, scheduling, and managing submittals, including those of subcontractors, designers of record, consultants, architect-engineers (AE), offsite fabricators, suppliers, and purchasing agents. Provide the name(s) of the person(s) in the QC organization authorized to review and certify submittals prior to approval. Provide the initial submittal of the Submittal Register as specified in Section 01 33 00 SUBMITTAL PROCEDURES.
- h. TESTING LABORATORY INFORMATION: Testing laboratory information required by the paragraph ACCREDITATION REQUIREMENTS, as applicable.
- i. TESTING PLAN AND LOG: A Testing Plan and Log that includes the tests required, associated feature of work required, referenced by the specification paragraph number requiring the test, the frequency, and the person responsible for each test.
- j. Procedures to complete design and construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected. This phase is performed prior to beginning work on each definable feature of work, after all required plans, documents, materials are approved, and after copies are at the work site.
- k. Reporting procedures, including proposed reporting formats.
- l. Procedures for submitting and reviewing design changes/variations prior to submission to the Contracting Officer.
- m. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task that is separate and distinct from other tasks and has control requirements and work crews unique to that task. A DFOW is identified by different trades or disciplines, or it is work by the same trade in a different environment. A DFOW is by definition any item or activity on the construction schedule, and the schedule specification provides direction regarding how the DFOWs are to be structured. Include in the list of DFOWs for all activities on the Construction Schedule. Provide separate DFOWs in the Network Analysis Schedule for each design development stage and submittal package. Although each section of the specifications can generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section. Identify the specification section number and schedule activity ID for each DFOW listed. The DFOW list will be reviewed in coordination with the construction schedule and agreed upon during the Coordination and Mutual Understanding Meeting.
- n. PROCEDURES FOR PERFORMING AND TRACKING THE THREE PHASES OF CONTROL: Identify procedures used to ensure the three phases of control to manage the quality on this project. For each Definable Feature of Work (DFOW), a Preparatory and Initial phase checklist will be filled out during the Preparatory and Initial phase meetings. Conduct the Preparatory and Initial Phases and meetings with a view towards obtaining quality construction by planning ahead and identifying potential problems for each DFOW.
- o. Coordinate scheduled work with Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and

the Special Inspections Project Manual. Where the applicable code issued by the International Code Council (ICC) calls for inspections by the Building Official, the Contractor must include the inspections in the Quality Control Plan and must perform the inspections required by the applicable ICC. The Contractor must perform these inspections using independent qualified inspectors. Include the Special Inspections Project Manual requirements in the QC Plan.

- [p. [o][p]. PROCEDURES FOR COMPLETION INSPECTION: Procedures for identifying and documenting the completion inspection process. Include in these procedures the responsible party for punch out inspection, pre-final inspection, and final acceptance inspection.
-] [q. [p][q]. TRAINING PROCEDURES AND TRAINING LOG: Procedures for coordinating and documenting the training of personnel required by the Contract.
-] [r. [q][r]. ORGANIZATION AND PERSONNEL CERTIFICATIONS LOG: Procedures for coordinating, tracking and documenting all certifications required for entities such as subcontractors, testing laboratories, suppliers, and personnel. The QC Manager will ensure that certifications are current, appropriate for the work being performed, and will not lapse during any period of the Contract that the work is being performed.

]1.5.2.2 Additional Requirements for Design Quality Control (DQC) Plan

The following additional requirements apply to the DQC Plan:

- a. Submit and maintain a DQC Plan as an effective quality control program which assures that all services required by this Contract are performed and provided in a manner that meets professional architectural and engineering quality standards. As a minimum, all documents must be technically reviewed by competent, independent reviewers identified in the DQC Plan. The same element that produced the product may not perform the [Independent Technical Review (ITR)][Agency Technical Review (ATR)]. Correct errors and deficiencies in the design documents prior to submitting them to the Government.
- b. Include the design schedule in the master project schedule, showing the sequence of events involved in carrying out the project design tasks within the specific Contract period. Schedule must include sufficient detail to identify all major design tasks, including those that control the flow of work. Include review and correction periods associated with each item. Schedule must be a forward planning as well as a project monitoring tool. The schedule reflects calendar days and not dates for each activity. If the schedule is changed, submit a revised schedule reflecting the change within 7 calendar days.
- c. Include in the DQC Plan the discipline-specific checklists to be used during the design and quality control of each submittal. Submit at each design phase as part of the project documentation these completed discipline-specific checklists. ER 1110-3-12 provides some useful information in developing checklists.
- d. Implement the DQC Plan by a Design Quality Control Manager who has the responsibility of being cognizant of and assuring that all documents on the project have been coordinated and who coordinates work with and reports to the Contracting Officer via the QC Manager. Notify the Contracting Officer, in writing, of the name of the individual and the

name of an alternate person assigned to the position.

- e. Provide Quality Control Documentation procedures such as QC review sets and QC comments to demonstrate that cross checking of all engineering discipline's design drawings and specifications has taken place. The QC review documentation must exhibit a checking process of the design documents for completeness, accuracy, and constructability.

1.5.3 Acceptance of the Quality Control (QC) and Design Quality Control (DQC) Plan

The Contracting Officer's acceptance of the Contractor QC Plan, or interim plan applicable to the particular feature of work to be started, and Design Quality Control Plan is required prior to the start of design and construction. Work outside of the accepted interim plan will not be permitted to begin until acceptance of a CQC Plan or another interim plan containing the additional work. The Contracting Officer reserves the right to require changes in the QC and DQC Plan and operations as necessary, including removal or addition of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC and DQC organization at any time to verify the submitted qualifications. All QC and DQC organization personnel are subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the Contract.

1.5.4 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed changes in the QC Plan or changes to the QC organization personnel. Proposed changes are subject to acceptance by the Contracting Officer.

1.5.5 Special Inspections

Perform all required Special Inspections per Section 01 45 35 SPECIAL INSPECTIONS, the statement of Special Inspections and the Schedule of Special Inspections.

1.6 QUALITY CONTROL (QC) ORGANIZATION

1.6.1 Personnel Requirements

The requirements for the CQC organization are a Site Safety and Health Officer (SSHO), QC Manager, a Design Quality Manager, and enough qualified personnel to ensure safety and Contract compliance. The SSHO reports directly to a senior project (or corporate) official independent from the QC Manager. The SSHO will also serve as a member of the CQC Staff. Personnel identified in the technical provisions as requiring specialized skills to assure the required work is being performed properly. The CQC staff always maintains a presence at the site during progress of the work and have complete authority and responsibility to take any action necessary to ensure Contract compliance. The CQC staff will be subject to acceptance by the Contracting Officer. Provide adequate office space, filing systems and other resources as necessary to maintain an effective and fully functional CQC organization. Promptly complete and furnish all letters, material submittals, shop drawing submittals, schedules, and all other project documentation to the CQC organization. The CQC organization is responsible for always maintaining these documents and records at the site, except as otherwise acceptable to the Contracting Officer.

1.6.2 Quality Control (QC) Manager

1.6.2.1 Duties

Provide a QC Manager at the work site to implement and manage the QC program. The QC Manager must be employed by the Prime Contractor. [In addition to implementing and managing the QC program, the QC Manager may perform the duties of Project Superintendent.] [The only duties and responsibilities of the QC Manager are to manage and implement the QC program on this Contract.] The QC Manager must attend the Coordination and Mutual Understanding Meeting, perform the three phases of control [except for those phases of control designated to be performed by QC Specialists], perform submittal review and approval, ensure testing is performed and provide QC certifications and documentation required in this Contract. The QC Manager is responsible for managing and coordinating the three phases of control and documentation performed by [the QC Specialists,] testing laboratory personnel and any other inspection and testing personnel required by this Contract. The QC Manager is the manager of all QC activities. The QC manager is responsible for notifying the Special Inspector [and Special Inspector of Record] of activities which require their review. The QC manager is responsible for coordinating the Special Inspection activities, see paragraph CONTRACTOR'S QUALITY CONTROL (QC) MANAGER, in Section 01 45 35 SPECIAL INSPECTIONS. The QC manager is responsible for the quality control for [Secure Space] [Controlled Area] [Sound Rated] perimeter construction.

1.6.2.2 Qualifications

[The QC Manager must be an individual with a minimum of [5][10][_____] years combined experience in the following positions: Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction Contracts which included the major trades that are part of this Contract. The individual must have at least [2][4][_____] years experience as a QC Manager. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification, safety compliance, and sustainability.

] [The QC Manager must possess at least a Bachelor's degree in engineering, architecture, or construction management, with [a current professional engineer registration [in the state of [_____]] [in a US State or Territory] or a current licensed architect [in the state of [_____]] [in a US State or Territory] and] a minimum of [10][_____] years construction experience as a Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction Contracts which included the major trades that are part of this Contract. The individual must have at least [2][4][_____] years experience as a QC Manager. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification, safety compliance, and sustainability.

] The QC Manager and all members of the QC organization must be capable of reading, writing, and conversing fluently in the English language.

1.6.2.3 Construction Quality Management Training

In addition to the above experience and education requirements, the QC Manager Alternate QC Manager, Superintendent, QC Specialist, and DQCM must have completed the CQM for Contractors course. If the QC Manager does not

have a current certification, obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Systems Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

The Construction Quality Management Training certificate expires after 5 years. If the QC Manager's certificate has expired, retake the course to remain current.

[1.6.2.4 Quality Control Requirements for Multiple Work Sites

Provide separate full-time QC Managers at the work sites listed below. Provide a separate full-time QC Manager at the following work sites:

a. [INDICATE WORK SITE LOCATION]

b. [INDICATE WORK SITE LOCATION]

[c. [INDICATE WORK SITE LOCATION]

] The QC Managers for the work sites listed above are each responsible for implementing and managing the QC Program at the designated work site. They must perform the three phases of control, perform submittal review, ensure testing is performed, and prepare QC certifications and documentation required by this Contract. The qualification requirements for the QC Managers at each site are listed in the paragraph QUALIFICATIONS.

]1.6.3 Organizational Changes

Maintain the QC staff with personnel as required by the specification section at all times. When it is necessary to make changes to the QC staff, revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

1.6.4 Design Quality Control (DQC) Manager

The DQC Manager must be a member of the QC organization, must coordinate actions with the QC Manager, and must not be subordinate to the Project Superintendent or the Project Manager. The DQC Manager may also act as the CxC if all CxC qualifications are met.

1.6.4.1 Qualifications

- a. Must be a person who has verifiable engineering or architectural design experience and is a registered professional engineer or architect.

1.6.4.2 Responsibilities

- a. Be responsible for the design integrity, professional design standards, and all design services required.

1.6.5 Alternate Quality Control (QC) Manager Duties and Qualifications

Designate an alternate for the QC Manager at the work site to serve in the event of the designated QC Manager's absence. The qualification

requirements for the Alternate QC Manager must be the same as for the QC Manager.

[1.6.6 Assistant Quality Control (QC) Manager Duties and Qualifications

[Provide a full-time assistant to the QC Manager at the work site to perform the three phases of control, perform submittal review, ensure testing is performed, and prepare QC certifications and documentation required by this Contract. The qualification requirements for the Assistant QC Manager must be [____]. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance.

][Provide a full-time assistant to the QC Manager at the work site to perform the three phases of control, perform submittal review, ensure testing is performed, and prepare QC certifications and documentation required by this Contract. The Assistant QC Manager must be on the work site during supplemental work shifts[beyond the regular shift] and perform the duties of the QC Manager during such supplemental shift work. The qualification requirements for the Assistant QC Manager must be [FILL IN BASED ON NATURE AND COMPLEXITY OF JOB]. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance.

] The Assistant QC Manager must be capable of reading, writing, and conversing fluently in the English language.

]1.6.7 Commissioning

Commissioning (Cx) is a systematic, quality-focused process for delivery of a project focusing on verifying and documenting all commissioned systems and assemblies are installed, tested, and operating as they were planned and designed to meet the project requirements. The Quality Control requirements outlined in this specification section are key in supporting the objectives of the Cx process, specifically coordinating testing, documenting, and verifying proper system operation. Properly executed the Quality Control support of Cx ensures timely execution of necessary tasks to deliver the fully commissioned and operating systems in coordination with the overall construction and project schedule.

Provide Cx in addition to the quality control requirements of this section and not as a substitute for quality control requirements. The QC Manager is responsible for carrying out the three phases of control while ensuring the functional performance and integrated systems tests are coordinated with the Cx provider as required for each system to be commissioned.

1.6.7.1 Certificate of Readiness

The QC Manager must issue a Certificate of Readiness for Government approval for each system to be commissioned. Schedule Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. The Certificate of Readiness certifies that all required inspections have been completed and deficiencies that were identified through any prior review, inspection, or test activity have been corrected before the start of Functional Performance Tests. Refer to Cx requirements in Section 01 91 00.15 BUILDING COMMISSIONING for a list of systems to be commissioned and detailed requirements for the Cx provider.

1.6.8 Quality Control (QC) Specialists

Provide a separate QC Specialist at the work site for each of the areas as listed in the Matrix listed below, who must assist and report to the QC Manager and who[may perform production related duties but must be allowed sufficient time to perform][must have no duties other than] their assigned quality control duties. These individuals or specialized technical companies[are directly employed by the Prime Contractor and cannot be employed by a supplier or subcontractor on this project][are employees of the Prime or subcontractor including Designer of Record (DOR)]. A single person can cover more than one area provided that the single person is qualified to perform quality control activities in each designated and that workload allows. QC Specialists must be physically present at the work site with frequency as indicated in the Experience Matrix below, to participate in the QC Meetings, perform the three phases of control, including participation in Preparatory and Initial Phase meetings, and to perform and document Follow-up inspections as an extension of the QC Manager for each definable feature of work in their area of responsibility. QC Specialist must assist and be present for training events, and Critical System Acceptance inspections by the Government. Qualification, experience, Area of Responsibility, and frequency of QC surveillance are provided in Matrix listed herein.

Experience Matrix		
1. Area	2-1. Qualification 2-2. Experience	3-1. Area of Responsibility 3-2. Frequency
[Civil]	[2-1. Graduate Civil Engineer or Construction Manager with [a professional engineer registration [in the state of [____]], and] 2-2. 2 years experience in the type of work being performed on this project or technician with 5 years related experience]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]
[Mechanical]	[2-1. Graduate Mechanical Engineer with [a professional engineer registration [in the state of [____]], and] 2-2. 2 years experience or person with 5 years of experience supervising mechanical features of work in the field with a construction company]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]

[Electrical]	[2-1. Graduate Electrical Engineer with [a professional engineer registration [in the state of [____]], and] 2-2. 2 years related experience or person 5 years of experience supervising electrical features of work in the field with a construction company]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]
[Structural]	[2-1. Graduate Civil Engineer (with Structural Track or Focus) or Construction Manager with [a professional engineer registration [in the state of [____]], and] 2-2. 2 years experience or person 5 years of experience supervising structural features of work in the field with a construction company]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]
[Architectural]	[2-1. Graduate Architect with [a registered architect [in the state of [____]], and] 2-2. 2 years experience or person with 5 years related experience]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]
[Environmental]	[2-1. Graduate Environmental Engineer with [a professional engineer registration [in the state of [____]], and] 2-2. 3 years experience]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]
[Submittals]	[2-1. Submittal Clerk 2-2. 1 year experience]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]

[Occupied Family Housing]	[2-1. Person, customer relations type, 2-2. Coordinator experience]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]
[Concrete, Pavements and Soils]	[2-1. Materials Technician 2-2. 2 years experience for the appropriate area]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]
[Testing, Adjusting and Balancing (TAB) Personnel]	[2.1. TAB Team Field Leader must be a member of AABC or an experienced technician of the firm certified by the NEBB 2-2. 3 years experience immediately preceding this Contract]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]
[Roofing Manufacturer's Technical Representative]	[2-1. Installation and testing of roofing systems, Section 07 53 23 ETHYLENE-PROPYLENE-DIENE-MONOMER ROOFING 2-2. 5 years experience minimum]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [Full-time] during installation and including final inspection.]
[Mechanical Inspector, International Code Council (ICC) Certified]	[2-1. Installation and testing of boilers, Section 23 52 49.00 20 STEAM BOILERS AND EQUIPMENT (500,000 - 18,000,000 BTU/HR) 2-2. 5 years experience minimum]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [Minimum three times a week during installation and full-time during testing] during installation and including final inspection.]
[Geotechnical]	[2-1. [____] 2-2. [____] years of experience]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]

[Telecommunications]	[2-1. [____] 2-2. [____] years of experience]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]
[Low-Voltage Specialties/ESS]	[2-1. [____] 2-2. [____] years of experience]	[3-1. Area of Responsibility: [____] 3-2. Frequency of QC surveillance and inspection will be [____] during installation and including final inspection.]
[Underwater QC Team]	[Note: See paragraph UNDERWATER QC TEAM]	[Note: See paragraph UNDERWATER QC TEAM]
[Fire Protection QC Specialist (FPQC)]	[Note: See paragraph FIRE PROTECTION QC SPECIALIST (FPQC)]	[Note: See paragraph FIRE PROTECTION QC SPECIALIST (FPQC)]
[Building Envelope QC Specialist]	[2-1. Roofing Manufacturer's Technical Representative 2-2. 5 years minimum with roofing system used]	[3-1. Area of Responsibility: Installation and testing of roofing. 3-2. Frequency of QC surveillance and inspection will be once a week during installation, once a week during flashing installation and full-time during roof testing.]
[____]	[____]	[____]

[1.6.8.1 Underwater QC Team

Provide Underwater QC (UWQC) Team at the work site to perform underwater surveillance and inspection for the Contractor. The UWQC Team divers must have current commercial diver's license, with a minimum of 5 years experience with underwater inspection. The personnel make-up of the UWQC team must comply with EM 385-1-1, OSHA, and local requirements for Contract diving operations. Comply with all the applicable safety requirements of EM 385-1-1, OSHA, and local requirements for Contract diving operations. The UWQC lead diver must be thoroughly familiar with the design plans and specifications to sufficiently understand the engineering aspects of the underwater construction and to be able to recognize and document potential problem areas, such as improperly constructed or defective areas. Provide all necessary equipment to conduct surveillance and inspection services, including diver's equipment,

dive boat, communication equipment, and photographic or video equipment or both. Diver(s) must be equipped to maintain two-way communication with QC personnel during diving operations. Prepare and submit a report including photographs or videos or both with the QC report after each dive. Frequency of underwater surveillance and inspection will be [_____] during installation and including final inspection. The UWQC Team must be an independent third party hired directly by the Prime Contractor, and must not be involved with the design, preparation of Contract, or installation of work.

][1.6.8.2 Fire Protection QC Specialist (FPQC)

Provide a Fire Protection Quality Control Specialist (FPQC) within the QC organization to perform quality control related activities as specified herein on fire protection and life safety systems installed under this Contract.

1.6.8.2.1 Qualifications

The FPQC must have the following qualifications:

- a. Be a registered Professional Engineer (P.E.) licensed by a Licensing Board in the United States, the District of Columbia, Guam or Puerto Rico, having passed the National Council of Examiners for Engineering and Surveying (NCEES) examination specifically in the discipline of Fire Protection Engineering.
- b. Have a minimum of 5 years of Fire Protection Engineering experience on projects of similar relevance and complexity to the fire protection work specified under this Contract.
- c. Other than the contractual obligations with the Prime Contractor, the FPQC must have no other business relationship (i.e., employee, owner, partner, operating officer, distributor, salesman, technical representative, family relationship, or financial investment) with the Prime Contractor or subcontractors.
- d. Be employed by an independent engineering firm or company or be the Fire Protection Engineer on the Prime Contractor's Design-Build team. The firm may identify multiple, to a maximum of five, licensed Fire Protection Engineers for the performance of the duties under this Contract but must submit the names and qualifications for Government approval for all individuals identified prior to them performing any work under this Contract. These individuals may not be substituted without prior approval from the Contracting Officer.

1.6.8.2.2 Responsibilities

FPQC duties and responsibilities:

- a. Assist in the development of the QC Plan including the Testing Plan and Log and executing the three phases of control for work involving the installation and testing of fire protection and life safety systems as an extension of the QC Manager.
- b. Participate in project QC Meetings. Participate in Preparatory and Initial Phase meetings and perform and Follow-up inspections for work involving the installation and testing of fire protection and life safety systems.

- c. Review and certify that all submittals pertaining to fire protection and life safety systems are complete and accurate prior to submission to the Government for surveillance approval. Forward each submittal reviewed by the FPQC to the Government for surveillance within 10 working days of FPQC certification. The FPQC Specialist is responsible for ensuring submittals are complete and accurate and all corrections have been made prior to submission to the Government. The Government reserves the right to reject any submittal that has not first been reviewed and certified by the FPQC and so marked, in writing, attesting to such review and completeness of the submittal.
- d. The Government reserves the right to reject any submittal or construction that is not in compliance to Contract. Government reviews do not relieve the Contractor responsibility for providing adequate quality control measures and do not constitute or imply acceptance of Contract variation.
- e. Perform construction surveillance in accordance with the Schedule of Fire Protection System Inspections. Construction surveillance includes but is not limited to performing periodic on-site inspections during construction at specified milestones, performing a pre-final inspection of installed systems and witnessing functional testing; and participating and documenting in an on-site final acceptance inspection of fire protection and life safety systems with the Government FPE.
- f. Document inspection results on a FPQC report prepared each day inspections are performed. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract documents, and signature of the FPQC Specialist. [In person inspection][and][Remote inspections] must be documented via [video (.mp4)][or][photo (.jpeg)][or both]. Video/photographic documentation must include before and after conditions and physical measurements.] Forward the FPQC daily report to the QC Manager who must include the report with the submission of their daily QC Report to the Government each day. Every site visit by the FPQC must be documented on a FPQC daily report.

1.6.8.2.3 Schedule of Fire Protection System Inspections

A schedule, prepared by the Fire Protection DOR, which lists each of the required visual inspections and observations required by the FPQC. The schedule is included at the end of this UFGS section.

[1.6.8.3 Quality Control (QC) for[Secure Space][Controlled Area][Sound Rated] Perimeter Construction

1.6.8.3.1 Periodic (Follow-Up Phase) Inspections

Once construction begins, perform periodic inspections of[Secure Space][Controlled Area][Sound Rated Area] identified in the Contract drawings at least once every 2 weeks. Increase frequency to weekly inspections within 30 days of planned acceptance testing. Coordinate periodic inspections with the appointed Government Site Security Manager (SSM) responsible for ensuring the assembly meets the requirements for accreditation.] Inspections must verify that construction and materials comply with the Contract documents, the description of the assembly in the ASTM E90

Factory Report for acoustical testing, and the approved submittals. Focus inspections on the construction of the sound rated assemblies, perimeter penetrations, perimeter doors, [electronic security system,][man-bar installation,][inspection ports,][and TEMPEST countermeasures]. Document periodic inspections in Daily QC Reports.

1.6.8.3.2 Preliminary Inspection

The Government and QC Manager will perform a joint preliminary inspection of the [Secure Space][Controlled Area][Sound Rated Area] after construction of the assembly is complete to verify compliance with the design requirements and other Contract documents. The Contracting Officer [and the appointed Government SSM] will participate in the preliminary inspection. Provide the Contracting Officer a minimum [14][] calendar days notification in advance of the preliminary inspection.

As a result of the preliminary inspection, prepare a [Secure Space][Controlled Area][Sound Rated Area] punch list with deficiencies identified. Include with the punch list the estimated date by which the deficiencies will be corrected. Document the preliminary inspection in the Daily QC Report and attach the punch list. Notify the Contracting Officer when deficiencies are corrected. Deficiencies from the Preliminary Inspection must be corrected prior to scheduling the Final Acceptance Inspection.

1.6.8.3.3 Acceptance Testing for Sound Attenuation

Perform acceptance testing for sound transmission loss of sound rated door assemblies as required in Section 08 34 73 SOUND CONTROL DOOR ASSEMBLIES and Section 09 29 00 GYPSUM BOARD for sound rated assemblies. Acceptance testing must be performed during the preliminary inspection. The Contracting Officer [and the appointed Government SSM] must witness acceptance testing. Deficiencies identified during acceptance testing must be included in the [Secure Space][Controlled Area][Sound Rated Area] punch list and corrected prior to the final acceptance inspection.

1.6.8.3.4 Acceptance Testing for Electronic Security Systems

Perform acceptance testing for Electronic Security Systems in accordance with Section 28 08 10 ELECTRONIC SECURITY SYSTEM ACCEPTANCE TESTING. Acceptance testing must be performed during the preliminary inspection. The Contracting Officer [and the appointed Government SSM] must witness acceptance testing. Deficiencies identified during acceptance testing must be included in the [Secure Space][Controlled Area][Sound Rated Area] punch list and corrected prior to the Final Inspection.

1.6.8.3.5 Final Inspection

Perform a final inspection of the [Secure Space][Controlled Area][Sound Rated Area] after required testing has been successfully completed as part of the preliminary inspection and all punch list items corrected. Testing is not permitted during the final inspection. QC Manager and Superintendent must attend the final inspection and Government attendees will include the Contracting Officer [and appointed Government SSM]. Request a final inspection by the Contracting Officer a minimum of 14 calendar days in advance.

1.6.9 Special Inspector [and Special Inspector of Record]

The Special Inspector (SI)[and Special Inspector of Record (SIOR)] must be an independent third party hired directly by the Prime Contractor. The SI[and SIOR] must not be a company employee of the Contractor or any subcontractor performing the work to be inspected. The qualifications of the SI[and SIOR] are defined in Section 01 45 35 SPECIAL INSPECTION.

[1.6.10 Submittal Reviewer[s] Duties and Qualifications

Provide[a] Submittal Reviewer[s], other than the QC Manager or CxC, qualified in the discipline[s] being reviewed, to review and certify that the submittals meet the requirements of this Contract prior to certification or approval by the QC Manager.

[Each submittal must be reviewed by an individual with 10 years of construction experience.

] [Each submittal must be reviewed by a registered architect or professional engineer.

] [Each of the following submittals must be reviewed by[an] individual[s] meeting the qualifications/experience specified below:

Qualification / Experience in Submittal Discipline	Submittals to be reviewed:	
	Section No	Submittal
[_____]	[_____]	[_____]

]]1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL

Procedures for submission, review and approval of submittals are described in Section 01 33 00 SUBMITTAL PROCEDURES. Procedures must include field verification of relevant dimensions and component characteristics by the QC organization prior to submittal being sent to the Contracting Officer. The CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the Contract. When Section 01 91 00.15 BUILDING COMMISSIONING are included in the Contract, the submittals required by those sections have to be coordinated with Section 01 33 00 SUBMITTAL PROCEDURES to ensure adequate time is allowed for each type of submittal required.

1.8 THREE PHASES OF CONTROL

CQC enables the Contractor to ensure that the construction, including that of subcontractors and suppliers, complies with the requirements of the Contract. At least three phases of control must be conducted by the QC Manager to adequately cover both on-site and off-site work for each definable feature of the construction work as follows:

1.8.1 Preparatory Phase

Document the results of the preparatory phase actions by separate minutes prepared by the QC Manager and attach to the daily CQC report. Instruct applicable workers as to the acceptable level of workmanship required to

meet Contract specifications.

Notify the Contracting Officer at least 2 business days in advance of each preparatory phase meeting. The meeting will be conducted by the QC Manager and attended by[the QC Specialists,] the Project Superintendent, the CxC, the Special Inspector,[the Special Inspector of Record,] and the foreman responsible for the DFOW. When the DFOW will be accomplished by a subcontractor, that subcontractor's foreman must attend the preparatory phase meeting. This phase is performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. Perform the following prior to beginning work on each DFOW:

- a. Review each paragraph of the applicable specification sections, reference codes, and standards. Make available during the preparatory inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.
- b. Review the Contract drawings.
- c. Verify that field measurements are as indicated on construction or shop drawings or both before confirming product orders, to minimize waste due to excessive materials.
- d. Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.
- e. Review the testing plan and ensure that provisions have been made to provide the required QC testing.
- f. Examine the work area to ensure that the required preliminary work has been completed and complies with the Contract and ensure any deficiencies/rework items in the preliminary work have been corrected and confirmed by the Contracting Officer.
- g. Review coordination of product/material delivery to designated prepared areas to execute the work.
- h. Examine the required materials, equipment and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data and are properly stored.
- i. Check to assure that all materials and equipment have been tested, submitted, and approved.
- j. Discuss specific controls to be used, construction methods, construction tolerances, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each DFOW. Ensure any portion of the plan requiring separate Contracting Officer acceptance has been approved.
- k. Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.

- l. Review the Cx requirements in accordance with Section 01 91 00.15 BUILDING COMMISSIONING and ensure all preliminary work items have been completed and documented.
- m. Discussion of procedures for controlling quality of the work including repetitive deficiencies. Document construction tolerances and workmanship standards for that feature of work.
- n. Discuss schedule and execution of the initial control phase and confirmation or construction quality compliance.
- o. Review Special Inspections required by Section 01 45 35 SPECIAL INSPECTION, the Statement of Special Inspections and the Schedule of Special Inspections.

1.8.2 Initial Phase

Notify the Contracting Officer at least 2 business days in advance of each initial phase. When construction crews are ready to start work on a DFW, conduct the initial phase with[the QC Specialists,] the Project Superintendent, the Special Inspector,[the Special Inspector of Record,] and the foreman responsible for that DFW. Observe the initial segment of the DFW to ensure that the work complies with Contract requirements. Document the results of the initial phase in the[daily CQC Report and in the] Initial Phase Checklist. Repeat the initial phase for each new crew to work on-site when acceptable levels of specified quality are not being met. Indicate the exact location of initial phase for definable feature of work for future reference and comparison with follow-up phases. Perform the following for each DFW:

- a. Check work to ensure that it is in full compliance with Contract requirements. Review minutes of the preparatory meeting.
- b. Verify adequacy of controls to ensure full Contract compliance. Verify required control inspection and testing comply with the Contract.
- c. Establish level of workmanship and verify that it meets the minimum acceptable workmanship standards. Compare with required sample panels as appropriate.
- d. Resolve any workmanship issues.
- e. Ensure that testing is performed by the approved laboratory.
- f. Check work procedures for compliance with the APP and the appropriate AHA to ensure that applicable safety requirements are met.
- g. Review project specific work plans (i.e., Cx, HAZMAT Abatement, Stormwater Management) to ensure all preparatory work items have been completed and documented.
- h. Coordinate scheduled work with Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Schedule of Special Inspections.

1.8.3 Follow-Up Phase

Perform the following for on-going DFW daily, or more frequently as necessary, until the completion of each DFW. The Final Follow-Up for any DFW will clearly note in the daily report the DFW is completed, and all deficiencies/rework items have been completed in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Each DFW that has completed the Initial Phase and has not completed the Final Follow-up must be included on each daily report. If no work was performed on that DFW for the period of that daily report, it must be so noted. Document all Follow-Up activities for DFWs in the daily CQC Report:

- a. Ensure the work including control testing complies with Contract requirements until completion of that particular work feature. Record checks in the CQC documentation.
- b. Maintain the quality of workmanship required.
- c. Ensure that testing is performed by the approved laboratory.
- d. Ensure that deficiencies/rework items are being corrected. Conduct final follow-up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work.
- e. Do not build upon nor conceal non-conforming work.
- f. Assure manufacturers' representatives have performed necessary inspections if required and perform safety inspections.
- g. Review the Cx requirements in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.
- h. Coordinate scheduled work with Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Schedule of Special Inspections.

1.8.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same DFW if the quality of on-going work is unacceptable, if there are changes in the applicable QC organization, if there are changes in the on-site production supervision or work crew, if work on a DFW has not started within 45 days of the initial preparatory meeting or has resumed after 45 days of inactivity, or if other problems develop.

1.8.5 Notification of Three Phases of Control for Off-Site Work

Notify the Contracting Officer at least 2 weeks prior to the start of the preparatory and initial phases.

1.8.6 Deficiency/Rework Items List

The QC Manager must maintain a list of work that does not comply with the Contract, identifying what items need to be corrected, the activity ID number associated with the item, the date the item was originally

discovered, the date the item will be corrected by, and the date the item was corrected.

The QC Manager reviews the list at each weekly QC Meeting:

- a. There is no requirement to report a deficiency/rework item that is corrected the same day it is discovered.
- b. No successor task may be advanced beyond the preparatory phase meeting until all deficiencies/rework items have been cleared by the QC Manager and concurred with by the Contracting Officer. This must be confirmed as part of the Preparatory Phase activities.
- c. Attach a copy of the "Deficiency/Rework Items List" to the last daily CQC Report of each month.
- d. The Contractor is responsible for including those items identified by the Contracting Officer.
- e. All deficiencies/rework items must be confirmed as corrected by the QC Manager, and concurred by the Contracting Officer, prior to commencement of any completion inspections per paragraph COMPLETION INSPECTIONS unless specifically exempted by the Contracting Officer.
- f. Non-Compliance with these requirements is grounds for removal in accordance with paragraph ACCEPTANCE OF THE QUALITY CONTROL (QC) and DESIGN QUALITY CONTROL (DQC) PLAN.
- g. All delays, concurrent or related to failure to manage, monitor, control, and correct deficiencies/rework items are entirely the responsibility of the Contractor and can not be made the subject, or any component of any request for additional time or compensation.

1.9 TESTING

Perform specified or required tests to verify that control measures are adequate to provide a product which conforms to Contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance tests when specified. Procure the services of an U.S. Army Corps of Engineers approved testing laboratory or establish an approved testing laboratory at the project site or within [_____] miles. Perform the following activities and record and provide the following data:

- a. Verify that testing procedures comply with Contract requirements.
- b. Verify that facilities and testing equipment are available and comply with testing standards.
- c. Check test instrument calibration data against certified standards.
- d. Verify that recording forms and test identification control number system, including all test documentation requirements, have been prepared.
- e. Record results of all tests taken, both passing and failing on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. If approved by the Contracting Officer, actual

test reports are submitted later with a reference to the test number and date taken. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this Contract.

1.9.1 Laboratory Accreditation Authorities

All testing laboratories must be validated by the USACE Material Testing Center (MTC) for the tests to be performed. Information on the USACE MTC with web-links to both a list of validated testing laboratories and for the laboratory inspection request for can be found at:
<https://mtc.erdc.dren.mil>

1.9.2 Capability Check

The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract. Laboratories utilized for testing soils, concrete, asphalt, and steel must meet criteria detailed in ASTM D3740 and ASTM E329.

1.9.2.1 Capability Recheck

If the selected laboratory fails the capability check, the Contractor will be assessed a charge of [] to reimburse the Government for each succeeding recheck of the laboratory or the checking of a subsequently selected laboratory. Such costs will be deducted from the Contract amount due the Contractor.

1.9.2.2 Onsite Laboratory

The Government reserves the right to utilize the Contractor's control testing laboratory and equipment to make assurance tests, and to check the Contractor's testing procedures, techniques, and test results at no additional cost to the Government.

1.9.3 Test Results

Cite applicable Contract requirements, tests or analytical procedures used. Provide actual results and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify the Contracting Officer immediately. Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results must be signed by a testing laboratory representative authorized to sign certified test reports. Furnish the signed reports, certifications, and other documentation to the Contracting Officer via the QC Manager.

1.10 COMPLETION INSPECTIONS

1.10.1 Punch-Out Inspection

Near the completion of all work or any increment thereof, established by a completion time stated in the Contract Clause entitled "Commencement, Prosecution, and Completion of Work," or stated elsewhere in the specifications, the QC Manager must conduct an inspection of the work and

develop a "punch list" of items which do not conform to the approved drawings, specifications, and Contract. Include in the punch list any remaining items on the "Deficiency/Rework Items List", that were not corrected prior to the Punch-Out Inspection as approved by the Contracting Officer in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Include within the punch list the estimated date by which the deficiencies will be corrected. Provide a copy of the punch list to the Contracting Officer.

The QC Manager, or staff, must make follow-on inspections to ascertain that all deficiencies have been corrected. All punch list items must be confirmed as corrected by the QC Manager and concurred by the Contracting Officer. Once this is accomplished, notify the Government that the facility is ready for the Government "Pre-Final Inspection".

1.10.2 Pre-Final Inspection

The Government and QC Manager will perform this inspection to verify that the facility is complete and ready to be occupied. A Government "Pre-Final Punch List" will be documented by the QC Manager as a result of this inspection. The QC Manager will ensure that all items on this list are corrected and concurred by the Contracting Officer prior to notifying the Government that a "Final" inspection with the Client can be scheduled. All items noted on the "Pre-Final" inspection must be corrected and concurred by the Contracting Officer in a timely manner and be accomplished before the Contract completion date for the work, or any increment thereof, if the project is divided into increments by separate completion dates unless exceptions are directed by the Contracting Officer.

1.10.3 Final Acceptance Inspection

Notify the Contracting Officer at least 14 calendar days prior to the date a final acceptance inspection can be held. State within the notice that all items previously identified on the pre-final punch list will be corrected and acceptable, along with any other unfinished Contract work, by the date of the final acceptance inspection. The Contractor must be represented by the QC Manager, the Project Superintendent, and others deemed necessary. Attendees for the Government will include the Contracting Officer, other Government QA personnel, and personnel representing the Client. Failure of the Contractor to have all Contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with the Contract Clause entitled "Inspection of Construction."

1.11 QUALITY CONTROL (QC) CERTIFICATIONS

1.11.1 Contractor Quality Control (CQC) Report Certification

Contain the following statement within the CQC Report: "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used, and work performed during this reporting period is in compliance with the Contract drawings and specifications to the best of my knowledge, except as noted in this report."

1.11.2 Completion Certification

Upon completion of work under this Contract, the QC Manager must furnish a certificate to the Contracting Officer attesting that "the work has been

completed, inspected, tested and is in compliance with the Contract." Provide a copy of this final QC Certification for completion to the preparer of the Operation & Maintenance (O&M) documentation.

1.12 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER

Maintain current and complete records of on-site and off-site QC program operations and activities.

Contact the Contracting Officer for sample forms or print from RMS-QCS as needed. Prior to commencing work on construction, the Contractor must obtain a copy set of the current report forms. The report forms will consist of the Contractor Quality Control (CQC) Report, CQC Report (Continuation Sheet), Contractor Production Report, Contractor Production Report (Continuation Sheet), Preparatory Phase Checklist, Initial Phase Checklist, Testing Plan and Log, and Rework Items List. Unless otherwise provided by the Contracting Officer, Contractor may use the forms provided as related material located at www.wbdg.org/dod/ufgs/ufgs-01-45-00.

1.12.1 Construction Documentation

Prepare a Construction Production Report and a Contractor Quality Control Report for each day that work is performed. Attach the Construction Production Report to the Contractor Quality Control Report prepared for the same day. Maintain current and complete records of on-site and off-site QC program operations and activities.

Account for each calendar day throughout the life of the Contract.

The Project Superintendent and the QC Manager must prepare and sign the Contractor Production and CQC Reports, respectively. Every space on the forms must be filled in. Use N/A if nothing can be reported in one of the spaces. The reporting of work must be identified by terminology consistent with the construction schedule. In the "Remarks" sections of the reports, enter pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, delays encountered, a record of visitors to the work site, quality control problem areas, deviations from the QC Plan, construction deficiencies encountered, and meetings held. For each entry in the report(s), identify the Schedule Activity No. that is associated with the entered remark.

1.12.2 Quality Control Activities

CQC and Contractor Production reports will be prepared daily to maintain current records providing factual evidence that required quality control activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

- a. The name and area of responsibility of the Contractors and any subcontractors.
- b. Operating plant/equipment with hours worked, idle, or down for repair.
- c. Work performed each day, giving location, description, and by whom. When a Network Analysis Schedule (NAS) is used, identify each item of work performed each day by NAS activity number.

- d. Control phase activities performed. Preparatory, and Initial phase Checklists associated with the DFOW referenced to the construction schedule. Follow-up phase activities identified to the DFOW. If testing or specific QC Specialist activities are associated with the Follow-up phase activities for a specific DFOW note this and include those reports.
- e. Test and control activities performed with results and references to specifications and drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST.
- f. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications and drawings requirements.
- g. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.
- h. Offsite surveillance activities, including actions taken.
- i. Job safety evaluations stating what was checked, results, and instructions or corrective actions.
- j. Instructions given/received and conflicts in plans and specifications.
- k. Provide documentation of design quality control activities. For independent design reviews, provide, as a minimum, identification of the[Independent Technical Review (ITR)][Agency Technical Review (ATR)] team and their review comments, responses and the record of resolution of the comments.

1.12.3 Verification Statement

Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract.

Furnish the original and one copy of these records in report form to the Government by 10:00 AM the next working day after the date covered by the report. As a minimum, prepare and submit one report for every 7 days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the Contract. The first report following a day of no work will be for that day only. Reports need to be signed and dated by the QC Manager. Include copies of test reports and copies of reports prepared by all subordinate quality control personnel within the QC Manager Report.

[1.12.4 Reports from the Quality Control (QC) Specialist(s)

Document inspection results on a QC specialist report prepared each day work is performed in their area of responsibility. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract

documents, and signature of the QC Specialist. In person inspections must be documented with Video/photographs. Video/photographic documentation of deficiencies must include before and after conditions and physical measurements, as necessary. Forward the QC daily report to the QC Manager who must include the report with the submission of their daily QC Report to the Government each day. Every site visit by the QC Specialist must be documented on a QC Specialist daily report.

]1.12.5 Quality Control Validation

Establish and maintain the following in an electronic folder. Divide folder into a series of tabbed sections as shown below. Ensure folder is updated at each required progress meeting.

- a. CQC Meeting minutes in accordance with paragraph QUALITY CONTROL (QC) MEETINGS.
- b. All completed Preparatory and Initial Phase Checklists, arranged by specification section, further sorted by DFOV referenced to the construction schedule. Submit each individual Phase Checklist the day the phase event occurs as part of the CQC daily report.
- c. All milestone inspections, arranged by Activity Number referenced to the construction schedule.
- d. An up-to-date copy of the Testing Plan and Log with supporting field test reports, arranged by specification section referenced to the DFOV to which individual reports results are associated. Individual field test reports will be submitted within 2 working days after the test is performed in accordance with the paragraph QUALITY CONTROL ACTIVITIES.
- e. Copies of all Contract modifications, arranged in numerical order. Also include documentation that modified work was accomplished.
- f. An up-to-date copy of the paragraph DEFICIENCY/REWORK ITEMS LIST.
- g. Cx documentation in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.
- h. [g][h]. Special Inspection reports.
- i. [g][h][i]. Upon commencement of Completion Inspections of the entire project or any defined portion, maintain up-to-date copies of all punch lists issued by the QC staff to the Contractor and subcontractors and all punch lists issued by the Government in accordance with the paragraph COMPLETION INSPECTIONS.

1.12.6 Testing Plan and Log

As tests are performed, the CxC and the QC Manager will record on the "Testing Plan and Log" the date the test was performed and the date the test results were forwarded to the Contracting Officer. Attach a copy of the updated "Testing Plan and Log" to the last daily CQC Report of each month. Provide a copy of the final "Testing Plan and Log" to the preparer of the Operation & Maintenance (O&M) documentation.

1.12.7 As-Built Drawings

The QC Manager must ensure the as-built drawings, required by Section 01 78 00 CLOSEOUT SUBMITTALS are kept current on a daily basis and marked to show deviations which have been made from the Contract drawings. The as-built drawings document commences with the QC Manager ensuring all amendments, or changes to the Contract prior to Contract award are accurately noted in the initial document set creating the accurate baseline of the Contract prior to any work starting. Ensure each deviation has been identified with the appropriate modifying documentation (e.g., PC No., Modification No., Request for Information No.). The QC Manager[or QC Specialist assigned to an area of responsibility] must initial each revision. Upon completion of work, the QC Manager will furnish a certificate attesting to the accuracy of the as-built drawings prior to submission to the Contracting Officer.

1.13 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected non-compliance with the Contract. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, is deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of a claim for extension of time for excess costs or damages by the Contractor.

1.14 DELIVERY, STORAGE, AND HANDLING

Designate receiving/storage areas for incoming material to be delivered according to installation schedule and to be placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. Store and handle materials in a manner as to prevent loss from weather and other damage. Keep materials, products, and accessories covered and off the ground, and store in a dry, secure area. Prevent contact with material that may cause corrosion, discoloration, or staining. Protect all materials and installations from damage by the activities of other trades.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)
11/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1

Safety -- Safety and Health Requirements
Manual

1.2 MEASUREMENT AND PAYMENT

The work of this section is not measured for payment. The Contractor is responsible for the work of this section, without any direct compensation other than the payment received for contract items.

1.3 CONTRACT ADMINISTRATION

The Government will use the Resident Management System (RMS) to assist in its monitoring and administration of this contract. The Government accesses the system using the Government Mode of RMS (RMS GM) and the Contractor accesses the system using the Contractor Mode (RMS CM). The term RMS will be used in the remainder of this section for both RMS GM and RMS CM. The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of the contract. The Contractor accesses RMS to record, maintain, input, track, and electronically share information with the Government throughout the contract period in the following areas:

- Administration
- Finances
- Quality Control
- Submittal Monitoring
- Scheduling
- Closeout
- Import/Export of Data

1.3.1 Correspondence and Electronic Communications

For ease and speed of communications, exchange correspondence and other documents in electronic format to the maximum extent feasible. Some correspondence, including pay requests and payrolls, are also to be provided in paper format with original signatures. Paper documents will govern, in the event of discrepancy with the electronic version.

1.3.2 Other Factors

Other portions of this document have a direct relationship to the reporting accomplished through RMS. Particular attention is directed to

FAR 52.236-15 Schedules for Construction Contracts; FAR 52.232-27 Prompt Payment for Construction Contracts; FAR 52.232-5 Payments Under Fixed-Price Construction Contracts; [Section 01 32 01.00 10 PROJECT SCHEDULE][Section 01 32 16.00 20 SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES]; Section 01 33 00 SUBMITTAL PROCEDURES; Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS; and Section 01 45 00.00 10 QUALITY CONTROL.

1.4 RMS SOFTWARE

RMS is a Windows-based program that can be run on a Windows-based PC meeting the requirements as specified in paragraph SYSTEM REQUIREMENTS. Download, install and be able to utilize the latest version of the RMS software within 7 calendar days of receipt of the Notice to Proceed. RMS software, user manuals, access and installation instructions, program updates and training information are available from the RMS website (<https://rms.usace.army.mil>). The Government and the Contractor will have different access authorities to the same contract database through RMS. The common database will be updated automatically each time a user finalizes an entry or change.

1.5 SYSTEM REQUIREMENTS

The following is the recommended system configuration to run the Contractor Mode RMS for full utilization of all features for all types and sizes of contracts. Smaller, less complicated, projects may not require the configuration levels described below. Required configuration also noted below.

Recommended RMS System Requirements	
Hardware	
Windows-based PC	1.7 GHz i3; AMD A6 3650 GHz or higher processor (REQUIRED)
RAM	8 GB
Hard drive disk	100 GB space for sole use by RMS system
Monitor	Screen resolution 1366 x 768
Mouse or other pointing device	
Windows compatible printer	Laser printer must have 4 MB+ of RAM
Connection to the Internet	minimum 4 Mbs per user
Software	
MS Windows	Windows 10 x 64 bit (RMS requires 64 bit O/S) or newer (REQUIRED)
Word Processing software	Viewer for MS Word 2013, MS Excel 2013 or newer (REQUIRED)

Recommended RMS System Requirements	
E-mail	MAPI compatible (REQUIRED)
Virus protection software	Regularly upgraded with all issued Manufacturer's updates and is able to detect most zero day viruses (REQUIRED)

1.6 CONTRACT DATABASE - GOVERNMENT

The Government will enter the basic contract award data in RMS prior to granting the Contractor access. The Government entries into RMS will generally be related to submittal reviews, correspondence status, and Quality Assurance(QA)comments, as well as other miscellaneous administrative information.

1.7 CONTRACT DATABASE - CONTRACTOR

Contractor entries into RMS establish, maintain, and update data throughout the duration of the contract. Contractor entries generally include prime and subcontractor information, daily reports, submittals, RFI's, schedule updates and payment requests. RMS includes the ability to import attachments and export reports in many of the modules, including submittals. The Contractor responsibilities for entries in RMS typically include the following items:

1.7.1 Administration

1.7.1.1 Contractor Information

Enter all current Contractor administrative data and information into RMS within 7 calendar days of receiving access to the contract in RMS. This includes, but is not limited to, Contractor's name, address, telephone numbers, management staff, and other required items.

1.7.1.2 Subcontractor Information

Enter all missing subcontractor administrative data and information into RMS CM within 7 calendar days of receiving access to the contract in RMS or within 7 calendar days of the signing of the subcontractor agreement for agreements signed at a later date. This includes name, trade, address, phone numbers, and other required information for all subcontractors. A subcontractor is listed separately for each trade to be performed.

1.7.1.3 Correspondence

Identify all Contractor correspondence to the Government with a serial number. Prefix correspondence initiated by the Contractor's site office with "S". Prefix letters initiated by the Contractor's home (main) office with "H". Letters are numbered starting from 0001. (e.g., H-0001 or S-0001). The Government's letters to the Contractor will be prefixed with "C" or "RFP".

1.7.1.4 Equipment

Enter and maintain a current list of equipment planned for use or being

used on the jobsite, including the most recent and planned equipment inspection dates.

1.7.1.5 Reports

Track the status of the project utilizing the reports available in RMS. The value of these reports is reflective of the quality of the data input. These reports include the Progress Payment Request worksheet, Quality Control (QC) comments, Submittal Register Status, and Three-Phase Control worksheets.

1.7.1.6 Request For Information (RFI)

Create and track all Requests For Information (RFI) in the RMS Administration Module for Government review and response.

1.7.2 Finances

1.7.2.1 Pay Activity Data

Develop and enter a list of pay activities in conjunction with the project schedule. The sum of pay activities equals the total contract amount, including modifications. Each pay activity must be assigned to a Contract Line Item Number (CLIN). The sum of the activities assigned to a CLIN equals the amount of each CLIN.

1.7.2.2 Payment Requests

Prepare all progress payment requests using RMS. Update the work completed under the contract at least monthly, measured as percent or as specific quantities. After the update, generate a payment request and prompt payment certification using RMS. Submit the signed prompt payment certification and payment request as well as supporting data either electronically or by hard copy. Unless waived by the Contracting Officer, a signed paper copy of the approved payment certification and request is also required and will govern in the event of discrepancy with the electronic version.

1.7.3 Quality Control (QC)

Enter and track implementation of the 3-phase QC Control System, QC testing, transferred and installed property and warranties in RMS. Prepare daily reports, identify and track deficiencies, document progress of work, and support other Contractor QC requirements in RMS. Maintain all data on a daily basis. Insure that RMS reflects all quality control methods, tests and actions contained within the Contractor Quality Control (CQC) Plan and Government review comments of same within 7 calendar days of Government acceptance of the QC Plan.

1.7.3.1 Quality Control (QC) Reports

The Contractor's Quality Control (QC) Daily Report in RMS is the official report. The Contractor can use other supplemental formats to record QC data, but information from any supplemental formats are to be consolidated and entered into the RMS QC Daily Report. Any supplemental information may be entered into RMS as an attachment to the report. QC Daily Reports must be finalized and signed in RMS within 24 hours after the date covered by the report. Provide the Government a printed signed copy of the QC Daily Report, unless waived by the Contracting Officer.

1.7.3.2 Deficiency Tracking.

Use the QC Daily Report Module to enter and track deficiencies. Deficiencies identified and entered into RMS by the Contractor or the Government will be sequentially numbered with a QC or QA prefix for tracking purposes. Enter each deficiency into RMS the same day that the deficiency is identified. Monitor, track and resolve all QC and QA entered deficiencies. A deficiency is not considered to be corrected until the Government indicates concurrence in RMS.

1.7.3.3 Three-Phase Control Meetings

Maintain scheduled and actual dates and times of preparatory and initial control meetings in RMS. Worksheets for the three-phase control meetings are generated within RMS.

1.7.3.4 Labor and Equipment Hours

Enter labor and equipment exposure hours on a daily basis. Roll up the labor and equipment exposure data into a monthly exposure report.

1.7.3.5 Accident/Safety Reporting

Both the Contractor and the Government enter safety related comments in RMS as a deficiency. The Contractor must monitor, track and show resolution for safety issues in the QC Daily Report area of the RMS QC Module. In addition, follow all reporting requirements for accidents and incidents as required in EM 385-1-1, Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and as required by any other applicable Federal, State or local agencies.

1.7.3.6 Definable Features of Work

Enter each feature of work, as defined in the approved CQC Plan, into the RMS QC Module. A feature of work may be associated with a single or multiple pay activities, however a pay activity is only to be linked to a single feature of work.

1.7.3.7 Activity Hazard Analysis

Import activity hazard analysis electronic document files into the RMS QC Module utilizing the document package manager.

1.7.4 Submittal Management

Enter all current submittal register data and information into RMS within 7 calendar days of receiving access to the contract in RMS. The information shown on the submittal register following the specification Section 01 33 00 SUBMITTAL PROCEDURES will already be entered into the RMS database when access is granted. Group electronic submittal documents into transmittal packages to send to the Government, except very large electronic files, samples, mock ups, color boards, or where hard copies are specifically required. Track transmittals and update the submittal register in RMS on a daily basis throughout the duration of the contract. Submit hard copies of all submittals unless waived by the Contracting Officer.

1.7.5 Schedule

Enter and update the contract project schedule in RMS by either manually entering all schedule data or by importing the Standard Data Exchange Format (SDEF) file, based on the requirements in [Section 01 32 01.00 10 PROJECT SCHEDULE][Section 01 32 16.00 20 SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES].

1.7.6 Closeout

Closeout documents, processes and forms are managed and tracked in RMS by both the Contractor and the Government. Ensure that all closeout documents are entered, completed and documented within RMS.

1.8 IMPLEMENTATION

Use of RMS as described in the preceding paragraphs is mandatory. Ensure that sufficient resources are available to maintain contract data within the RMS system. RMS is an integral part of the Contractor's required management of quality control.

1.9 MONTHLY COORDINATION MEETING

Update the RMS CM database each workday. At least monthly, generate and submit one schedule update. At least one week prior to submittal, meet with the Government representative to review the planned progress payment data submission for errors and omissions.

Make required corrections prior to Government acceptance of the export file and progress payment request. Payment requests accompanied by incomplete or incorrect data submittals will not be accepted. The Government will not process progress payments until all required corrections are processed.

1.10 NOTIFICATION OF NONCOMPLIANCE

Take corrective action within 7 calendar days after receipt of notice of RMS non-compliance by the Contracting Officer.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION 01 45 35

SPECIAL INSPECTIONS
02/15

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7 (2017) Minimum Design Loads for Buildings and Other Structures

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2021) International Building Code

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS Z2305 (2013) Non-destructive Testing - Qualification and Certification of Personnel

1.2 GENERAL REQUIREMENTS

Perform Special Inspections in accordance with the Statement of Special Inspections, Schedule of Special Inspections and Chapter 17 of ICC IBC. The Statement of Special Inspections and Schedule of Special Inspections are included as an attachment to this specification. Special Inspections are to be performed by an independent third party and are intended to ensure that the work of the prime contractor is in accordance with the Contract Documents and applicable building codes. Special inspections do not take the place of the three phases of control inspections performed by the Contractor's QC Manager or any testing and inspections required by other sections of the specifications.

Structural observations will be performed by the Government. The contractor must provide notification to the Contracting Officer 14 days prior to the following points of construction:

- a. Reinforcing Steel Placement
- b. Concrete Placement
- c. Welding

1.3 DEFINITIONS

1.3.1 Continuous Special Inspections

Continuous Special Inspections is the constant monitoring of specific tasks by a special inspector. These inspections must be carried out continuously over the duration of the particular tasks.

1.3.2 Periodic Special Inspections

Periodic Special Inspections is Special Inspections by the special inspector who is intermittently present where the work to be inspected has been or is being performed.

1.3.3 Perform

Perform these Special Inspections tasks for each welded joint or member, bolted or fastener connection, and required verification.

1.3.4 Observe

Observe these Special Inspections items on a random daily basis. Operations need not be delayed pending these inspections.

1.3.5 Special Inspector (SI)

A qualified person retained by the contractor and approved by the Contracting Officer as having the competence necessary to inspect a particular type of construction requiring Special Inspections. The SI must be an independent third party hired directly by the Prime Contractor.

1.3.6 Associate Special Inspector (ASI)

A qualified person who assists the SI in performing Special Inspections but must perform inspection under the direct supervision of the SI and cannot perform inspections without the SI on site.

1.3.7 Third Party

A third party inspector must not be company employee of the Contractor or any Sub-Contractor performing the work to be inspected.

1.3.8 Special Inspector of Record (SIOR)

A licensed engineer in responsible charge of supervision all special inspectors for the project and approved by the Contracting officer. The SIOR must be an independent third party hired directly by the Prime Contractor.

1.3.9 Contracting Officer

The Government official having overall authority for administrative contracting actions. Certain contracting actions may be delegated to the Contracting Officer's Representative (COR).

1.3.10 Contractor's Quality Control (QC) Manager

An individual retained by the prime contractor and qualified in accordance with the Section 01 45 00.00 10 QUALITY CONTROL having the overall responsibility for the contractor's QC organization.

1.3.11 Designer of Record (DOR)

A registered design professional contracted by the Government as an A/E responsible for the overall design and review of submittal documents prepared by others. The DOR is registered or licensed to practice their respective design profession as defined by the statutory requirements of

the professional registration laws in state in which the design professional works. The DOR is also referred to as the Engineer of Record (EOR) in design code documents.

1.3.12 Statement of Special Inspections (SSI)

A document developed by the DOR identifying the material, systems, components and work required to have Special Inspections.

1.3.13 Schedule of Special Inspections

A schedule which lists each of the required Special Inspections, the extent to which each Special Inspections is to be performed, and the required frequency for each in accordance with ICC IBC Chapter 17.

1.3.14 Designated Seismic System

Those nonstructural components that require design in accordance with ASCE 7 Chapter 13 and for which the component importance factor, I_p , is greater than 1.0. This designation applies to systems that are required to be operational following the Design Earthquake for RC I - IV structures.

1.3.15 Component Certification and O&M Manual

For any electrical or mechanical component required by ASCE 7 Section 13.2.2 to be certified, evidence demonstrating compliance with the requirement shall be maintained in a file identified as "Equipment Certification Documentation." This file shall be a part of the Operations & Maintenance Manual that is turned over to the Contracting Officer. The project specifications shall require the Operations & Maintenance Manual state that replaced or modified components need to be certified per the original certification criteria.

1.3.16 Component Identification Nameplate

Any electrical or mechanical component required by ASCE 7 SSection 13.2.2 to be certified shall bear permanent marking or nameplates constructed of a durable heat and water resistant material. Nameplates shall be mechanically attached to such nonstructural components and placed on each component for clear identification. The nameplate shall not be less than 125 x 180 with red letters 25 in height on a white background stating "Certified Equipment." The following statement shall be on the nameplate: "This equipment/component is certified. No modifications are allowed unless authorized in advance and documented in the Equipment Certification Documentation file." The nameplate shall also contain the component identification number in accordance with the drawings/specifications and the O&M manuals.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

SIOR Letter of Acceptance; G

Special Inspections Project Manual; G

Special Inspections Agency's Written Practices

NDT Procedures and Equipment Calibration Records

SD-06 Test Reports

Special Inspections Daily Reports

Special Inspections Biweekly Reports

SD-07 Certificates

Fabrication Plant

Certificate of Compliance

Special Inspector of Record Qualifications; G

Special Inspector Qualifications; G

Qualification Records for NDT technicians

SD-11 Closeout Submittals

Comprehensive Final Report of Special Inspections; G

1.5 SPECIAL INSPECTOR QUALIFICATIONS

Submit qualifications for each special inspector and the special inspector of record.

Certifying Associations	
FM	Factory Mutual
ICC	International Code Council
JCI	Japan Concrete Institute
JSFA	Japan Steel Fabricators Association
JWES	Japan Welding Engineering Society
UL	Underwriters Laboratories
Architect Under Bldg Standard Law of Japan	First Class Architect (Kenchikushi) Registered Under Standard Law of Japan

1.5.1 Steel Construction and High Strength Bolting

1.5.1.1 Special Inspector

- a. Structural Steel Work Manager (Tekkotsu Koji Kanri Sekininsha), or

- b. Japan Welding Society Certified Inspector (Nihon Yosetsu Kyokai Kensa-in), or
- c. Non-Destructive Test Technician (Hihakai Kensa Gijutsusha), or
- d. Architectural or Civil Construction Managing Engineer (Kenchiku/Doboku Koji Sekou Kanrigishi), or
- e. JSFA Certified Architectural High Strength Bolt Joint Management Engineer

1.5.2 Welding Structural Steel

1.5.2.1 Special Inspector

- a. JWES Certified Welding Management Engineer
- b. JSFA Certified Architectural Structural Steel Products Inspection Engineer

1.5.3 Nondestructive Testing of Welds

1.5.3.1 Special Inspector

JIS Z2305 NDT Level III Certificate.

1.5.3.2 Associate Special Inspector

JIS Z2305 NDT Level II Certificate plus one year of related experience.

1.5.4 Cold Formed Steel Framing

1.5.4.1 Special Inspector

- a. JSFA Certified Architectural Structural Steel Products Inspection Engineer, or

1.5.5 Concrete Construction

1.5.5.1 Special Inspector

- a. Japan Concrete Institute Concrete Engineer (JCI Concrete Gishi), or
- b. Registered Professional Engineer with related experience, or
- c. First Class Kenchikushi with four years of related experience.
- d. Architectural or Civil Construction Managing Engineer (Kenchiku/Doboku Sekou Kanrigishi)

1.5.6 Prestressed Concrete Construction

1.5.6.1 Special Inspector

- a. Registered Professional Engineer with related experience, or
- b. First Class Kenchikushi with four years of related experience

1.5.7 Post-tensioned Concrete Construction

1.5.7.1 Special Inspector

- a. Registered Professional Engineer with related experience, or
- b. First Class Kenchikushi with four years of related experience.

1.5.8 Verification of Site Soil Condition, Fill Placement and Load-Bearing Requirements

1.5.8.1 Special Inspector

- a. First Class Kenchikushi, or
- b. First Class Doboku Sekou Kanrigishi (1st class Civil Engineering Works Management Engineer), or
- c. Geotechnical Survey Technician (Chishitsu Chosa Gishi), or
- d. Soil Investigation Technician (Jiban Kensa Gishi)

1.5.9 Deep Foundations

1.5.9.1 Special Inspector

Chishitsu Chosa Gishi (Professional Geotechnical Engineer)

1.5.10 Sprayed Fire Resistant Material

1.5.10.1 Special Inspector

- a. ICC Spray-applied Fireproofing Special Inspector Certificate, or
- b. ICC Fire Inspector I Certificate with one year of related experience, or
- c. Registered Professional Engineer with related experience, or
- d. First Class Kenchikushi with four years related experience.

1.5.11 Mastic and Intumescent Fire Resistant Coatings

1.5.11.1 Special Inspector

- a. ICC Spray-applied Fireproofing Special Inspector Certificate, or
- b. ICC Fire Inspector I Certificate with one year of related experience, or
- c. Registered Professional Engineer with related experience, or
- d. First Class Kenchikushi with four years related experience

1.5.12 Exterior Insulation and Finish System (EIFS)

1.5.12.1 Special Inspector

- a. Registered Professional Engineer with related experience, or

- b. First Class Kenchikushi with four years related experience.

1.5.13 Fire-Resistant Penetrations and Joints

1.5.13.1 Special Inspector

- a. Passed the UL Firestop Exam with one year of related experience, or
- b. Passed the FM Firestop Exam with one year of related experience, or
- c. Registered Professional Engineer with related experience, or
- d. First Class Kenchikushi with four years related experience.

1.5.14 Smoke Control

1.5.14.1 Special Inspector

- a. AABC Technician Certification with one year of related experience, or
- b. Registered Professional Engineer with related experience, or
- c. First Class Kenchikushi with four years related experience

1.5.15 Architectural Components

1.5.15.1 Special Inspector

- a. Registered Professional Architect with related experience, or
- b. Registered Professional Engineer with related experience, or
- c. First Class Kenchikushi with four years related experience.

1.5.16 Mechanical, Electrical, and Plumbing Designated Seismic Systems

1.5.16.1 Special Inspector

- a. Registered Professional Engineer with related experience, or
- b. First Class Kenchikushi with four years related experience.

1.5.17 Verification of Electrical Systems

- a. Chief Electrical Engineer (Denki Shunin Gijutusha), or
- b. Electrical Construction Managing Engineer (Denki Sekou Kanrigishi), or
- c. Japanese Licensed Electrician (Denki Koujishi)

1.5.18 Special Inspector of Record (SIOR)

Registered Professional Engineer or First Class Kenchikushi with four years of related experience.

PART 2 PRODUCTS

2.1 FABRICATOR SPECIAL INSPECTIONS

Special Inspections of fabricator's work performed in the fabricator's shop is required to be inspected in accordance with the Statement of Special Inspections and the Schedule of Special Inspections unless the fabricator is certified by the approved agency to perform such work without Special Inspections. Submit the following certification [certifications] to the Contracting Officer for information to allow work performed in the fabricator's shop to not be subjected to Special Inspections.

Minister of Land, Infrastructure and Transportation (MLIT) Certified Fabrication Plant, Category as specified

At the completion of fabrication, submit a certificate of compliance, to be included with the comprehensive final report of Special Inspections, stating that the materials supplied and work performed by the fabricator are in accordance the construction documents.

PART 3 EXECUTION

3.1 RESPONSIBILITIES

3.1.1 Special Inspector of Record

- a. Supervise all Special Inspections required by the contract documents and the IBC.
- b. Submit a SIOR Letter of Acceptance to the Contracting Officer attesting to acceptance of the duties of SIOR, signed and sealed by the SIOR.
- c. Verify the qualifications of all of the Special Inspectors.
- d. Verify the qualifications of fabricators.
- e. Submit Special Inspections agency's written practices for the monitoring and control of the agency's operations to include the following:
 - (1) The agency's procedures for the selection and administration of inspection personnel, describing the training, experience and examination requirements for qualifications and certification of inspection personnel.
 - (2) The agency's inspection procedures, including general inspection, material controls, and visual welding inspection.
- f. Submit qualification records for nondestructive testing (NDT) technicians designated for the project.
- g. Submit NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.
- h. Prepare a Special Inspections Project Manual, which will cover the following:

- (1) Roles and responsibilities of the following individuals during Special Inspections: SIOR, SI, General Contractor, Subcontractors, QC Manager, and DOR.
- (2) Organizational chart and/or communication plan, indicating lines of communication.
- (3) Contractor's internal plan for scheduling inspections. Address items such as timeliness of inspection requests, who to contact for inspection requests, and availability of alternate inspectors.
- (4) Indicate the government reporting procedures.
- (5) Propose forms or templates to be used by SI and SIOR to document inspections.
- (6) Indicate procedures for tracking nonconforming work and verification that corrective work is complete.
- (7) Indicate how the SIOR and/or SI will participate in weekly QC meetings.
- (8) Indicate how Special Inspections of shop fabricated items will be handled when the fabricator's shop is not certified per paragraph FABRICATOR SPECIAL INSPECTIONS.
- (9) Include a section in the manual that covers each specific item requiring Special Inspections that is indicated on the Schedule of Special Inspections. Provide names and qualifications of each special inspector who will be performing the Special Inspections for each specific item. Provide detail on how the Special Inspections are to be carried out for each item so that the expectations are clear for the General Contractor and the Subcontractor performing the work.

Make a copy of the Special Inspections Project Manual available on the job site during construction. Submit a copy of the Special Inspections Project Manual for approval.

- i. Attend coordination and mutual understanding meeting where the information in the Special Inspections Project Manual will be reviewed to verify that all parties have a clear understanding of the Special Inspections provisions and the individual duties and responsibilities of each party.
- j. Maintain a 3- ring binder for the Special Inspector's daily and biweekly reports and the Special Inspections Project Manual. This file must be located in a conspicuous place in the project trailer/office to allow review by the Contracting Officer and the DOR.
- k. Submit a copy of the Special Inspector's daily reports to the QC Manager.
- l. Discrepancies that are observed during Special Inspections must be reported to the QC Manager for correction. If discrepancies are not corrected before the special inspector leaves the site the observed discrepancies must be documented in the daily report.
- m. Submit a biweekly Special Inspections report until all work requiring

Special Inspections is complete. A report is required for each biweekly period in which Special Inspections activity occurs, and must include the following:

- (1) A brief summary of the work performed during the reporting time frame.
 - (2) Changes and/or discrepancies with the drawings, specifications and mechanical or electrical component certification, that were observed during the reporting period.
 - (3) Discrepancies which were resolved or corrected.
 - (4) A list of nonconforming items requiring resolution.
 - (5) All applicable test results including nondestructive testing reports.
- n. At the completion of each Definable Feature of Work (DFOW) requiring Special Inspections, submit an interim final report of Special Inspections that documents the Special Inspections completed for that DFOW and corrections of all discrepancies noted in the daily reports. The interim final report of Special Inspections must be signed, dated and bear the seal of the SIOR.
- o. At the completion of the project submit a comprehensive final report of Special Inspections that documents the Special Inspections completed for the project and corrections of all discrepancies noted in the daily reports. The comprehensive final report of Special Inspections must be signed, dated and bear the seal of the SIOR.
- 3.1.2 Quality Control Manager
- a. Supervise all Special Inspectors required by the contract documents and the IBC.
 - b. Verify the qualifications of all of the Special Inspectors.
 - c. Verify the qualifications of fabricators.
 - d. Maintain a 3- ring binder for the Special Inspector's daily and biweekly reports. This file must be located in a conspicuous place in the project trailer/office to allow review by the Contracting Officer and the DOR.
 - e. Maintain a rework items list that includes discrepancies noted on the Special Inspectors daily report.

3.1.3 Special Inspectors

- a. Inspect all elements of the project for which the special inspector is qualified to inspect and are identified in the Schedule of Special Inspections.
- b. Attend preparatory phase meetings related to the Definable Feature of Work (DFOW) for which the special inspector is qualified to inspect.
- c. Submit a copy of the daily reports to the QC Manager.

- d. Discrepancies that are observed during Special Inspections must be reported to the QC Manager for correction. If discrepancies are not corrected before the special inspector leaves the site the observed discrepancies must be documented in the daily report.
- e. Submit a biweekly Special Inspection Report until all inspections are complete. A report is required for each biweekly period in which Special Inspections activity occurs, and must include the following:
 - (1) A brief summary of the work performed during the reporting time frame.
 - (2) Changes and/or discrepancies with the drawings, specifications and mechanical or electrical component certification, that were observed during the reporting period.
 - (3) Discrepancies which were resolved or corrected.
 - (4) A list of nonconforming items requiring resolution.
 - 5) All applicable test result including nondestructive testing reports.
- f. At the completion of the project submit a comprehensive final report of Special Inspections that documents the Special Inspections completed for the project and corrections of all discrepancies noted in the daily reports. The comprehensive final report of Special Inspections must be signed, dated and indicate the certification of the special inspector qualifying them to conduct the inspection.
- g. Submit daily reports to the SIOR.

3.2 DEFECTIVE WORK

Check work as it progresses, but failure to detect any defective work or materials must in no way prevent later rejection if defective work or materials are discovered, nor obligate the Contracting Officer to accept such work.

-- End of Section --

Project: P234 - Maintenance Hangar
Location: Anytown, VA
Project #: 12345678
Date: 4/10/2017

STATEMENT OF SPECIAL INSPECTIONS

Project Seismic Design Category: E
Project Risk Category: IV
Project Design Wind Speed (km/h): 392
Number of Stories: 2
Structure Height Above Grade (M): 21
Hazardous Occupancy or attached to such? No Group H Occupancies (2018 IBC, Section 415)

Special Inspector of Record (SIOR)

A Special Inspector of Record (SIOR) IS required (per UFGS 01 45 35, Section 1.3.8)

SIOR Name (Registered Professional): John Doe P.E.
Professional Registration Number: C222222
Consulting Firm Name (if any): ABC Structural Consulting
SIOR Office AND Mobile Phone Number: 1234567895 (Mobile), 123456789 (Office)

This Statement of Special Inspections is in accordance with Section 1704.3 of the 2018 IBC. It includes a Schedule of Special Inspections applicable to the above referenced project.

2018 IBC 1704.3.2 and 1704.3.3

Following is a listing of critical main wind/seismic force resisting systems for this structure. Carefully inspect these elements as part of the roles and responsibilities of the Special Inspector (reference the Schedule of Special Inspections for inspection checklists).

Vertical LFRS Elements	Notes
Ordinary Concentric Braced Frames	North - South Direction Only, See Plan
Special Concentric Braced Frames	Lettered Gridline only, see plan for call out
Ordinary Steel Moment resisting Frames (SMRF)	Both orthogonal Directions, See plan for call out
Ordinary Reinforced Concrete Shearwalls	See Key Notes
Concrete Grade Beams Resisting Lateral Loads	Office Portion Only, see plan key notes
Ordinary Reinforced Masonry Shear Walls	High Bay, Numbered Gridlines only, see plan
OSB Sheathed Shear Walls (nailing, sill bolting, Etc)	See schedule on plan
Shear Wall Hold Downs	Identified on Plan & in Detail Sheet X.XX
Horizontal LFRS Elements	Notes
Continuous Roof Ties	See Key Notes on Roof Plan
Collector Elements	Identified on plan with key notes
Concrete over metal deck	2nd floor and roof
Metal Roof Deck & Related Fastening System	See Roof Plan
Out of Plane Wall Connections	See Structural Details XX & XX
Diaphragm Cross Rod Bracing	See Roof Plan for locations
Cast in Place Concrete Floor and Roof Diaphragms	See sheet XX for details and rebar schedule
Special Force Transfer Connection	See Detail X on Sheet X.XX

Project: P234 - Maintenance Hangar
Location: Anytown, VA
Project #: 12345678
Date: 4/10/2017

Designated Seismic Systems (DSS)

(2018 IBC 1705.13.3.4) (ASCE 7-16, 13.2.2, C13.2.2) (UFC 3-301-04, 2-11.2 & 2-13.2.2)

Non-structural 'Designated Seismic Systems' (DSS) have an I_p greater than 1.0, and must be certified by the manufacturer to remain operable after a design earthquake. The Prime Contractor shall collect the manufacturers Certificates of Compliance for incorporation into the Operations and Maintenance manual per Section 01 78 23 paragraph 1.3. The SIOR shall review the Certificates of Compliance as part of the special inspection. Additionally, the below listed Designated Seismic Systems must be carefully inspected by the Special Inspector according to the requirements noted in the Schedule of Special Inspections.

ELECTRICAL Designated Seismic Systems (DSS) Requiring a Certificate of Compliance	
1.	DSS Emergency or Standby Power System
2.	DSS Lighting Inverters and Associated Conduits
3.	DSS Electrical System (including conduits) and Panels Serving Fire Pump Controller and Fire Pump
4.	DSS Light Fixture with Emergency Battery Backup
5.	DSS Component XX

If additional space is required, append an additional sheet listing the remaining DSS

MECHANICAL/PLUMBING Designated Seismic Systems (DSS) Requiring a Certificate of Compliance	
1.	DSS Gas lines and associated fittings, anchorage, & flexible Connections
2.	DSS Component XX
3.	DSS Component XX
4.	DSS Component XX
5.	DSS Component XX

If additional space is required, append an additional sheet listing the remaining DSS

OTHER Designated Seismic Systems (DSS) Requiring a Certificate of Compliance	
1.	DSS Building egress stair systems
2.	DSS Building fire sprinkler systems
3.	DSS Elevator System, see specification section 14 21 23
4.	DSS Fire pump, Fire pump controller, fire alarm control panel and mass notification
5.	DSS Component XX

Final Walk Down Inspection and Report

(UFC 3-301-01 SECTION 2-2.4.3)

Designated Seismic Systems shall receive a final walk-down inspection by the Registered Design Professional in Responsible Charge.

Final Walk Down Report, Prepared by the Registered Design Professional in Responsible Charge, Must Include:

1. Record observations of Final Walk Down Inspection
2. Document that Inspections were performed in accordance with the Schedule of Special Inspections
3. Document that all Designated Seismic Systems are installed according to construction/manufacture document requirements, and that Compliance Certificates have been collected (UFC 3- 301-01, 01, 2-13.2.2.1).

Statement of Special Inspections Page 2 of 2

Japan Edited Specifications, Feb 2020

SCHEDULE OF SPECIAL INSPECTIONS

Reference UFGS 01 45 35 for all requirements not noted as part of this schedule.

INSPECTION DEFINITIONS:

- PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and noted verification.
- OBSERVE:** Observe these items randomly during the course of each work day to ensure that applicable requirements are being met. Operations need not be delayed pending these inspections at contractor's risk.
- DOCUMENT:** Document, with a report, that the work has been performed in accordance with the contract documents. This is in addition to any other reports required in the Special Inspections guide specification.
- CONTINUOUS:** Constant monitoring by a special inspector of identified tasks over the duration of performance of said tasks.
- PERIODIC:** Intermittent monitoring by a special inspector of identified tasks that have been or are being performed.

DESIGNER NOTES (to be deleted after reviewing):

1. This schedule contains minimum requirements. Do not delete applicable inspection tasks unless notes in blue indicate it is acceptable to do so.
2. Blue text = designers notes. The designer must review and edit all blue text in this schedule prior to inserting this schedule into the special inspections spec (UFGS 01 45 35).
3. Check section boxes with ANY inspection tasks applicable to your project. You may choose to delete unchecked sections or leave them in the scheduled unchecked.
4. Individual rows/tasks that are not applicable to the project may be left in the section, as the inspector can determine whether they occur/apply (e.g. metal trusses in the light gauge framing section for example).
5. Design discipline sections are color coded for easier reference by designers. This schedule does NOT need to be printed in color.
6. When finished editing, delete this note box and save this schedule as a PDF and insert into the project specifications (special inspections section).

A. STRUCTURAL - STEEL – WELDING SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

STEEL INSPECTION <u>PRIOR TO WELDING</u> – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-16: Table C-N5.4-1		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Verify that the welding procedures specification (WPS) is available	PERFORM	
2. Verify manufacturer certifications for welding consumables are available	PERFORM	
3. Verify material identification	PERFORM	Type and grade.
4. Welder Identification System	PERFORM	The fabricator or erector, as applicable, shall maintain a system by which a welder who has welded a joint or member can be identified. Stamps, if used, shall be the low-stress type.
5. Fit-up of groove welds (including joint geometry)	OBSERVE	✓ Joint preparation ✓ Dimensions (alignment, root opening, root face, bevel) ✓ Cleanliness (condition of steel surfaces) ✓ Tacking (tack weld quality and location) ✓ Backing type and fit (if applicable)
6. Configuration and finish of access holes	OBSERVE	
7. Fit-up of fillet welds	OBSERVE	✓ Dimensions (alignment, gaps at root) ✓ Cleanliness (condition of steel surfaces) ✓ Tacking (tack weld quality and location)
8. Check welding equipment	OBSERVE	
STEEL INSPECTION <u>DURING WELDING</u> – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-16: Table C-N5.4-2		
TASK	INSPECTION TYPE ²	DESCRIPTION
9. Use of qualified welders	PERFORM	Welding by welders, welding operators, and tack welders who are qualified in conformance with requirements.
10. Control and handling of welding consumables	OBSERVE	✓ Packaging ✓ Electrode atmospheric exposure control
11. No welding over cracked tack welds	OBSERVE	
12. Environmental conditions	OBSERVE	✓ Wind speed within limits ✓ Precipitation and temperature
13. Welding Procedures Specification followed	OBSERVE	✓ Settings on welding equipment ✓ Travel speed ✓ Selected welding materials ✓ Shielding gas type/flow rate ✓ Preheat applied ✓ Interpass temperature maintained (min./max.)

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

² **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

STEEL INSPECTION <u>DURING</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-16: Table C-N5.4-2		
TASK	INSPECTION TYPE ¹	DESCRIPTION
14. Welding techniques	OBSERVE	✓ Interpass and final cleaning ✓ Each pass within profile limitations ✓ Each pass meets quality requirements
		✓ Proper position (F, V, H, OH) ✓ Intermix of filler metals avoided

CONTINUED ON FOLLOWING PAGE

A. STRUCTURAL - STEEL – WELDING SECTION (CONTINUED)

STEEL INSPECTION <u>AFTER</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 2018 1705.2.1, AISC 360-16: Table C-N5.4-3		
TASK	INSPECTION TYPE ²	DESCRIPTION
15. Welds cleaned	OBSERVE	
16. Size, length, and location of all welds	PERFORM	Size, length, and location of all welds conform to the requirements of the detail drawings.
17. Welds meet visual acceptance criteria	PERFORM AND DOCUMENT	✓ Crack prohibition ✓ Weld/base-metal fusion ✓ Crater cross section ✓ Weld profiles ✓ Weld size ✓ Undercut ✓ Porosity
18. Arc strikes	PERFORM	
19. k-area	PERFORM	When welding of doubler plates, continuity plates or stiffeners has been performed in the k-area, visually inspect the web k-area for cracks.
20. Backing removed, weld tabs removed and finished, and fillet welds added where required	PERFORM	
21. Repair activities	PERFORM AND DOCUMENT	
22. Document acceptance or rejection of welded joint or member	PERFORM	
23. No prohibited welds have been added without the approval of the Contracting Officer	OBSERVE	

END SECTION

¹ **OBSERVE:** Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

² **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.

OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

B. STRUCTURAL - STEEL – BOLTING SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

STEEL INSPECTION TASKS <u>PRIOR TO</u> BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-16: Table C-N5.6-1		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Manufacture's certifications available for fastener materials	PERFORM	
2. Fasteners marked in accordance with ASTM or JIS requirements	OBSERVE	
3. Proper fasteners selected for joint detail (grade, type, bolt length if threads are to be excluded from shear plane)	OBSERVE	
4. Proper bolting procedure selected for joint detail	OBSERVE	
5. Connecting elements, including appropriate faying surface condition and hole preparation, if specified, meet applicable requirements	OBSERVE	
6. Pre-installation verification testing by installation personnel observed and documented for fastener assemblies and methods used	PERFORM	
7. Proper storage provided for bolts, nuts, washers, and other fastener components	OBSERVE	
STEEL INSPECTION TASKS <u>DURING</u> BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-16: Table C-N5.6-2		
TASK	INSPECTION TYPE ¹	DESCRIPTION
8. Fastener assemblies of suitable condition, placed in all holes and washers (if required) are positioned as required	OBSERVE	
9. Joint brought to the snug-tight condition prior to pretensioning operation	OBSERVE	
10. Fastener component not turned by the wrench prevented from rotating	OBSERVE	
11. Bolts are pretensioned in accordance with RCSC Specification, progressing systematically from the most rigid point toward the free edges	OBSERVE	
STEEL INSPECTION TASKS <u>AFTER</u> BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-16: Table C-N5.6-3		
TASK	INSPECTION TYPE ¹	DESCRIPTION
12. Document acceptance or rejection of all bolted connections	DOCUMENT	

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

C. STRUCTURAL - STEEL - NON DESTRUCTIVE TESTING SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

NONDESTRUCTIVE TESTING OF WELDED JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-16: Section N5.5		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Use of qualified nondestructive testing personnel	PERFORM	Visual weld inspection and nondestructive testing (NDT) shall be conducted by personnel qualified in accordance with AWS D1.8 clause 7.2
2. CJP groove welds	OBSERVE	[NOTE: DOR must delete this row if section D (SEISMIC PROVISIONS SECTION) is checked] Dye penetrant testing (DT) and ultrasonic testing (UT) shall be performed on 20% of CJP groove welds for materials greater than 8mm thick. Testing rate must be increased to 100% if greater than 5% of welds tested have unacceptable defects.
3. Welded joints subject to fatigue	OBSERVE	Dye penetrant testing (DT) and Ultrasonic testing (UT) shall be performed on 100% of welded joints identified on contract drawings as being subject to fatigue.
4. Weld tab removal sites	OBSERVE	At the end of welds where weld tabs have been removed, magnetic particle testing shall be performed on the same beam-to-column joints receiving UT

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

D. STRUCTURAL - STEEL – AISC 341 REQUIREMENTS (SEISMIC PROVISIONS) SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

NONDESTRUCTIVE TESTING OF WELDED JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 341-16: Section J6.2		
TASK	INSPECTION TYPE ¹	DESCRIPTION
[NOTE: DOR may uncheck this section for projects NOT designed in accordance with AISC 341 (Seismic Provisions) or for projects designed according to AISC 341, but using an R value equal to 3]		
1. CJP groove welds	OBSERVE	Dye penetrant testing (DT) and ultrasonic testing (UT) shall be performed on 100% of CJP groove welds for materials greater than 5/16" thick (8mm).
2. Beam cope and access hole.	OBSERVE	At welded splices and connections, thermally cut surfaces of beam copes and access holes shall be tested using magnetic particle testing (MT) or dye penetrant testing (DT), when the flange thickness exceeds 38mm for rolled shapes, or when the web thickness exceeds 38mm for built-up shapes.
3. K-area NDT (AISC 341)	PERFORM	Where welding of doubler plates, continuity plates or stiffeners has been performed in the k-area, the web shall be tested for cracks using magnetic particle testing (MT). The MT inspection area shall include the k-area base metal within 75mm of the weld. The MT shall be performed no sooner than 48 hours following completion of the welding.
4. Placement of reinforcing or contouring fillet welds	DOCUMENT	

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

E. STRUCTURAL - STEEL - COMPOSITE CONSTRUCTION ¹

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

COMPOSITE CONSTRUCTION <u>PRIOR TO</u> PLACING CONCRETE – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-16: Table N6.1, AISC 341-16: Table J9-1		
TASK	INSPECTION TYPE ²	DESCRIPTION
1. Placement and installation of steel headed stud anchors	PERFORM	
2. Material identification of reinforcing steel (Type/Grade)	OBSERVE	
3. Determination of carbon equivalent for reinforcing steel other than ASTM A706	OBSERVE	
4. Proper reinforcing steel size, spacing, clearances, support, and orientation	OBSERVE	
5. Reinforcing steel has been tied and supported as required	OBSERVE	

END SECTION

F. STRUCTURAL - STEEL - OTHER INSPECTIONS

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

OTHER STEEL INSPECTIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 341-16: Tables J8-1 & J10-1		
TASK	INSPECTION TYPE ²	DESCRIPTION
1. Anchor rods and other embedments supporting structural steel	PERFORM	Verify the diameter, grade, type, and length of the anchor rod or embedded item, and the extent or depth of embedment prior to placement of concrete.
2. Fabricated steel or erected steel frame	OBSERVE	Verify compliance with the details shown on the construction documents, such as braces, stiffeners, member locations and proper application of joint details at each connection.
3. Reduced beam sections (RBS) where/if occurs	DOCUMENT	✓ Contour and finish ✓ Dimensional tolerances
4. Protected zones	DOCUMENT	No holes or unapproved attachments made by fabricator or erector
5. H-piles where/if occurs	DOCUMENT	No holes or unapproved attachments made by the responsible contractor

END SECTION

¹ See Concrete Construction Section for all concrete related inspection of composite steel construction.

² **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

G. STRUCTURAL - COLD-FORMED METAL DECK - PLACEMENT SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

METAL DECK INSPECTION <u>PRIOR TO</u> DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2017, Appendix 1, Table 1.1		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Verify compliance of materials (deck and all deck accessories) with construction documents, including profiles, material properties, and base metal thickness	PERFORM	
2. Document acceptance or rejection of deck and deck accessories	DOCUMENT	
METAL DECK INSPECTION <u>DURING</u> DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2017, Appendix 1, Table 1.2		
TASK	INSPECTION TYPE ¹	DESCRIPTION
3. Verify compliance of deck and all deck accessories installation with construction documents	PERFORM	
4. Verify deck materials are represented by the mill certifications that comply with the construction documents	PERFORM	
5. Document acceptance or rejection of installation of deck and deck accessories	DOCUMENT	
METAL DECK INSPECTION <u>AFTER</u> DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2017, Appendix 1, Table 1.3		
TASK	INSPECTION TYPE ¹	DESCRIPTION
6. Welding procedure specification (WPS) available	PERFORM	
7. Manufactures certifications for welding consumables available	OBSERVE	
8. Material identification (type/grade)	OBSERVE	
9. Check welding equipment	OBSERVE	

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

H. STRUCTURAL - COLD-FORMED METAL DECK – WELDING SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

METAL DECK INSPECTION <u>DURING</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2017, Appendix 1, Table 1.4 or JASS 6		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Use of qualified welders	OBSERVE	
2. Control and handling of welding consumables	OBSERVE	
3. Environmental conditions (wind speed, moisture, temperature)	OBSERVE	
4. WPS followed	OBSERVE	
METAL DECK INSPECTION <u>AFTER</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2017, Appendix 1, Table 1.5 or JASS 6		
TASK	INSPECTION TYPE ¹	DESCRIPTION
5. Verify size and location of welds, including support, sidelap, and perimeter welds.	PERFORM	
6. Welds meet visual acceptance criteria	PERFORM	
7. Verify repair activities	PERFORM	
8. Document acceptance or rejection of welds	DOCUMENT	

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

I. STRUCTURAL - COLD-FORMED METAL DECK – FASTENING SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☐

METAL DECK INSPECTION <u>BEFORE</u> MECHANICAL FASTENING – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2017, Appendix 1, Table 1.6		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Manufacturer installation instructions available for mechanical fasteners	OBSERVE	
2. Proper tools available for fastener installation	OBSERVE	
METAL DECK INSPECTION <u>DURING</u> MECHANICAL FASTENING – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2017, Appendix 1, Table 1.7		
TASK	INSPECTION TYPE ¹	DESCRIPTION
3. Fasteners are positioned as required	OBSERVE	
4. Fasteners are installed in accordance with manufacturer's instructions	OBSERVE	
METAL DECK INSPECTION <u>AFTER</u> MECHANICAL FASTENING – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2017, Appendix 1, Table 1.8		
TASK	INSPECTION TYPE ¹	DESCRIPTION
5. Check spacing, type, and installation of support fasteners	PERFORM	
6. Check spacing, type, and installation of sidelap fasteners	PERFORM	
7. Check spacing, type, and installation of perimeter fasteners	PERFORM	
8. Verify repair activities	PERFORM	
9. Document acceptance or rejection of mechanical fasteners	DOCUMENT	

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

J. STRUCTURAL - LIGHT GAUGE STEEL FRAMING AND/OR LIGHT GAUGE TRUSSES SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☐

LIGHT GAUGE STEEL CONSTRUCTION AND CONNECTIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.2, 1705.11.2, 1705.11.3, UFC 4 023 03		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Trusses spanning 18.3m or greater where/if applies	PERFORM	Verify that temporary and permanent truss restraint/bracing is installed in accordance with approved truss submittal package.
2. Welded connections (seismic and/or wind resisting system)	OBSERVE	Visually inspect all welds composing part of the main wind or seismic force resisting system, including shearwalls, braces, collectors (drag struts), and hold-downs. [NOTE: DOR must identify critical wind and/or seismic force resisting welds in the contract drawings so that the special inspector can confirm compliance.]
3. Connections (seismic and/or wind resisting system)	OBSERVE	Visually inspect all screw attachment, bolting, anchoring and other fastening of components within the main wind or seismic force resisting system, including roof deck, roof framing, exterior wall covering, wall to roof/floor connections, braces, collectors (drag struts) and hold-downs. [NOTE: DOR must identify critical wind and/or seismic force resisting connection/fastener components in the contract drawings so that the special inspector can confirm compliance.]
4. Cold-formed steel (progressive collapse resisting system where/if applies)	OBSERVE	Verify proper welding operations, screw attachment, bolting, anchoring and other fastening of components within the progressive collapse resisting system, including horizontal tie force elements, vertical tie force elements and bridging elements (UFC 4 023 03). [NOTE: DOR must identify critical progressive collapse resisting connection/fastener components in the contract drawings so that the special inspector can confirm compliance.]

END SECTION

K. STRUCTURAL - OPEN-WEB STEEL JOISTS SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☐

OPEN-WEB STEEL JOISTS AND JOIST GIRDERS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC TABLE 1705.2.3		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Installation of open-web steel joists and joist girders	OBSERVE	✓ End connections – welded or bolted ✓ All Bridging – horizontal and diagonal, per Steel Joist Institute specifications listed in Section 2207.1.

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

L. STRUCTURAL - CONCRETE CONSTRUCTION SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

CONCRETE CONSTRUCTION, INCLUDING COMPOSITE DECK – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC TABLE 1705.3 (ACI 318 REFERENCES NOTED IN IBC TABLE)		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Inspect reinforcement, including prestressing tendons, and verify placement.	PERIODIC	Verify prior to placing concrete that reinforcing is of specified type, grade and size; that it is free of oil, dirt and unacceptable rust; that it is located and spaced properly; that hooks, bends, ties, stirrups and supplemental reinforcement are placed correctly; that lap lengths, stagger and offsets are provided; and that all mechanical connections are installed per the manufacturer's instructions and/or evaluation report. [NOTE: DOR must identify critical progressive collapse resisting connection/fastener components in the contract drawings so that the special inspector can confirm compliance.]
2. Reinforcing bar welding	PERIODIC	✓ Verify weldability of reinforcing bars other than ASTM A 706 ✓ Inspect single-pass fillet welds, maximum 8mm in accordance with AWS D1.4 or JASS 6
3. All other welding	CONTINUOUS	Visually inspect all welds in accordance with AWS D1.4 or JASS 6
4. Gas Pressure Welding	CONTINUOUS	Visually inspect all welds for compliance with construction documents and workmanship and check all welds for internal flaws by ultrasonic inspection
5. Cast in place anchors and post installed drilled anchors (downward inclined)	PERIODIC	Verify prior to placing concrete that cast in place anchors and post installed drilled anchors have proper embedment, spacing and edge distance.
6. Post-installed adhesive anchors in horizontal or upward inclined orientations	CONTINUOUS AND DOCUMENT	✓ Inspect as required per approved ICC-ES report ✓ Verify that installer is certified for installation of horizontal and overhead installation applications ✓ Inspect proof loading as required by the contract documents
7. Verify use of required mix design	PERIODIC	Verify that all mixes used comply with the approved construction documents
8. Prior to concrete placement, fabricate specimens for strength tests, perform slump and air content tests, and determine the temperature of the concrete	CONTINUOUS	At the time fresh concrete is sampled to fabricate specimens for strength test verify these tests are performed by qualified technicians.

CONTINUED ON FOLLOWING PAGE

¹ **PERIODIC:** Intermittent monitoring by a special inspector of identified tasks that have been or are being performed.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.
CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

STRUCTURAL - CONCRETE CONSTRUCTION (CONTINUED)

CONCRETE CONSTRUCTION, INCLUDING COMPOSITE DECK – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC TABLE 1705.3 (ACI 318 REFERENCES NOTED IN IBC TABLE)		
TASK	INSPECTION TYPE ¹	DESCRIPTION
9. Inspect concrete and/or shotcrete placement for proper application techniques	CONTINUOUS	Verify proper application techniques are used during concrete conveyance and depositing avoids segregation or contamination. Verify that concrete is properly consolidated.
10. Verify maintenance of specified curing temperature and technique	PERIODIC	Inspect curing, cold weather protection, and hot weather protection procedures.
11. Pre-stressed concrete	CONTINUOUS	Verify application of prestressing forces and grouting of bonded prestressing tendons.
12. Inspect erection of precast concrete members	PERIODIC	
13. Verify in-situ concrete strength, prior to stressing of tendons in post-tensioned concrete and prior to removal of shores and forms from beams and structural slabs.	PERIODIC	
14. Inspect formwork for shape, location and dimensions of the concrete member being formed.	PERIODIC	

END SECTION

¹ **PERIODIC:** Intermittent monitoring by a special inspector of identified tasks that have been or are being performed.
CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

M. STRUCTURAL - MASONRY CONSTRUCTION SECTION (ALL RISK CATEGORIES)

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE <u>AT START</u> OF CONSTRUCTION IBC 1705.4 (ACI 530-13 TABLE 3.1.2 & 3.1.3)		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Compliance with approved submittals prior to start	PERIODIC	
2. Proportions of site-mixed mortar.	PERIODIC	
3. Grade and type of reinforcement, anchor bolts, and prestressing tendons and anchorages	PERIODIC	
4. Prestressing technique	PERIODIC	
5. Properties of thin bed mortar for AAC masonry	PERIODIC	
MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE <u>PRIOR TO</u> GROUTING IBC 1705.4 (ACI 530-13 TABLE 3.1.2 & 3.1.3)		
TASK	INSPECTION TYPE ¹	DESCRIPTION
6. Grout space	PERIODIC CONTINUOUS	[NOTE: DOR must either delete 'OBSERVE' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/ III]
7. Proportions of site-prepared grout and prestressing grout for bonded tendons	PERIODIC	
8. Proportions of site-mixed grout and prestressing grout for bonded tendons	PERIODIC	
9. Placement of masonry units and mortar joints	PERIODIC	
10. Welding of reinforcement	CONTINUOUS	
MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE <u>DURING</u> CONSTRUCTION IBC 1705.4 (ACI 530-13 TABLE 3.1.2 & 3.1.3)		
TASK	INSPECTION TYPE ¹	DESCRIPTION
11. Size and location of structural elements is in compliance	PERIODIC	
12. Preparation, construction, and protection of masonry during cold weather (temperature below 40°F (4.4°C) or hot weather (temp above 90°F (32.2°C))	PERIODIC	
13. Application and measurement of prestressing force	CONTINUOUS	
14. Placement of grout and prestressing grout for bonded tendons	CONTINUOUS	
15. Placement of AAC masonry units and construction of thin bed mortar joints	CONTINUOUS	Continuous for first 465 square meters only.
16. Observe preparation of grout specimens, mortar specimens, and/or prisms	PERIODIC	
17. Type, size and placement of reinforcement, connectors, anchor bolts and prestressing tendons and anchorages, including details of anchorage of masonry to structural members, frames, or other construction	PERIODIC CONTINUOUS	[NOTE: DOR must either delete 'OBSERVE' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/III]

END SECTION

¹ **PERIODIC:** Intermittent monitoring by a special inspector of identified tasks that have been or are being performed.
CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

N. STRUCTURAL - WOOD CONSTRUCTION – SPECIALTY ITEMS SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

WOOD CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.5		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. High-load diaphragms where applicable	PERIODIC	Verify thickness and grade of sheathing, size of framing members at panel edges, nail diameters and length, and the number of fastener lines and that fastener spacing is per approved contract documents.
2. Metal-plate connected wood trusses spanning 18.3m or greater	PERIODIC	Verify that the temporary installation restraint/bracing and the permanent individual truss member restraint/bracing are installed in accordance with the approved truss submittal package

END SECTION

O. STRUCTURAL - WOOD CONSTRUCTION - SEISMIC & WIND SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

WOOD CONSTRUCTION SEISMIC AND WIND – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.5		
TASK	INSPECTION TYPE ¹	DESCRIPTION
[NOTE: DOR may uncheck this section where sheathing nailing/fasteners (both shearwall and roof) are consistently greater than 100mm on center, or if the design wind speed is less than 49 meters/sec AND the seismic design category is A or B]		
1. Field gluing operation of elements of the main wind/seismic force-resisting system	CONTINUOUS	
2. Nailing, bolting, anchoring and other fastening of elements of the main wind/seismic force-resisting system	PERIODIC	Includes connectors for: shearwall sheathing, roof/floor sheathing, drag struts/collectors, braces, hold downs, roof and floor framing connections to exterior walls.

END SECTION

P. STRUCTURAL – ISOLATION AND ENERGY DISSIPATION SYSTEMS SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

ISOLATION AND ENERGY DISSIPATION SYSTEMS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC TABLE 1705.2.3		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Fabrication and installation	PERIODIC	Verify that fabrication and installation of isolator units and energy dissipation devices conform to manufacturer's recommendations and approved construction documents

END SECTION

¹ **PERIODIC:** Intermittent monitoring by a special inspector of identified tasks that have been or are being performed.
CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

Q. GEOTECHNICAL - SOILS INSPECTION SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

SOILS INSPECTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.6		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Materials below shallow foundations are adequate to achieve the design bearing capacity.	PERIODIC	
2. Excavations are extended to proper depth and have reached proper material	PERIODIC	
3. Perform classification and testing of compacted fill materials	PERIODIC	
4. Verify use of proper materials, densities and lift thicknesses during placement and compaction of compacted fill	CONTINUOUS	
5. Prior to placement of compacted fill, inspect subgrade and verify that site has been prepared properly.	PERIODIC	During fill placement, the special inspector shall verify that proper materials and procedures are used in accordance with the provisions of the approved geotechnical report

END SECTION

R. GEOTECHNICAL - DRIVEN DEEP FOUNDATION ELEMENTS SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

DEEP DRIVEN FOUNDATION CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.5		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Verify element materials, sizes and lengths comply with requirements	CONTINUOUS	
2. Determine capacities of test elements and conduct additional load tests as required	CONTINUOUS	
3. Inspect driving operations and maintain complete and accurate records for each element	CONTINUOUS	
4. Verify placement locations and plumbness, confirm type and size of hammer, record number of blows per foot of penetration, determine required penetrations to achieve design capacity, record tip and butt elevations and document any damage to foundation element	CONTINUOUS	

END SECTION

¹ **PERIODIC:** Intermittent monitoring by a special inspector of identified tasks that have been or are being performed.
CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

S. GEOTECHNICAL - HELICAL PILE FOUNDATIONS SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

HELICAL PILE FOUNDATIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.9		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Record installation equipment used, pile dimensions, tip elevations, final depth, final installation torque and other pertinent installation data. The approved geotechnical report (provided upon request) and the contract documents shall be used to determine compliance	CONTINUOUS	

END SECTION

T. GEOTECHNICAL - CAST IN PLACE DEEP FOUNDATION ELEMENTS SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

CAST IN PLACE DEEP FOUNDATION ELEMENTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.8		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Inspect drilling operations and maintain complete and accurate records for each element.	CONTINUOUS	
2. Verify placement locations and plumbness, confirm element diameters, bell diameters (if applicable), lengths, embedment into bedrock (if applicable and adequate end-bearing strata capacity. Record concrete or grout volumes	CONTINUOUS	

END SECTION

¹ **CONTINUOUS:** Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

U. FIRE PROTECTION - SPRAYED FIRE-RESISTANT MATERIALS SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

SPRAYED FIRE RESISTANT MATERIALS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.14		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Surface condition	PERIODIC	Prior to application confirm that surface has been prepared per the approved fire-resistance design and manufacturer's instructions.
2. Application	PERIODIC	Prior to application confirm that the substrate meets the minimum ambient temperature per the approved fire-resistance design and manufacturer's instructions.
3. Material thickness	PERIODIC	Verify that the thickness of the SFRM to structural elements is not less than the thickness require by the fire-resistant design in more that 10 percent of the measurement, but in no case less than minimum allowable thickness required by 1705.14.4.
4. Material density	PERIODIC	Verify that the density of the SFRM is not less than the density required by the fire-resistant design and according to IBC 1705.14.5.
5. Bond strength	PERIODIC	Verify cohesive/adhesive bond strength of the cured SFRM applied to the structural element is not less than 150psf and according to IBC 1705.14.6

END SECTION

V. FIRE PROTECTION - MASTIC AND INTUMESCENT COATINGS SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

MASTIC AND INTUMESCENT FIRE-RESISTANT COATINGS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.15		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Surface preparation	PERIODIC	Inspections shall be performed in accordance with AWCI 12-B and the contract documents

END SECTION

W. FIRE PROTECTION – FIRE RESISTANT PENETRATIONS AND JOINTS SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

FIRE RESISTANT PENETRATIONS AND JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.17		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Inspections of penetration firestop systems conducted in accordance with ASTM E 2174.	PERFORM	[NOTE: This section applies to Risk Category III, IV, & V only. DOR may choose to uncheck this section where project is assigned to Risk Category I or II. Confirm Risk Category with Structural Engineer]
2. Inspections of fire-resistant joint systems conducted in accordance with ASTM E 2393	PERFORM	

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
PERIODIC: Intermittent monitoring by a special inspector of identified tasks that have been or are being performed.

X. FIRE PROTECTION – SMOKE CONTROL SECTION
THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

SMOKE CONTROL – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.17		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Verify device locations and perform leakage testing	PERIODIC	Perform during erection of ductwork and prior to concealment
2. Pressure difference testing, flow measurements and detection and control verification	PERIODIC	Perform prior to occupancy and after sufficient completion

END SECTION

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¹ **PERIODIC:** Intermittent monitoring by a special inspector of identified tasks that have been or are being performed.

Y. ARCHITECTURAL - EXTERIOR INSULATION AND FINISH SYSTEMS SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

EXTERIOR INSULATION AND FINISH SYSTEMS (EIFS) – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.16		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Water resistive barrier coating applied over a sheathing substrate.	PERIODIC	Verify that water resistive barrier coating complies with ASTM E 2570. [NOTE: not applicable to masonry or concrete wall applications. Uncheck this section in those cases]

END SECTION

Z. ARCHITECTURAL – ARCHITECTURAL COMPONENTS

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☐

ARCHITECTURAL COMPONENTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.12.4, 1705.12.7		
TASK	INSPECTION TYPE ¹	DESCRIPTION
[NOTE: This section is not applicable to Seismic Design Categories A, B, & C. Uncheck this section if one of those categories applies. Confirm Seismic Design Category with the structural engineer]		
1. Erection and fastening of exterior cladding and interior and exterior veneer.	PERIODIC	Verify appropriate materials, fasteners and attachment at commencement of work and at completion. Inspector Note: Inspection not required if height is less than 9.15m or weight is less than 0.24kPa
2. Interior and exterior non-load bearing walls	PERIODIC	Verify appropriate materials, fasteners and attachment at commencement of work and at completion. Inspector Note: Inspection not required if height is less than 9.15m. Also, Interior non-load bearing walls need not be inspected if weighing less than 0.72kPa
3. Access floors	PERIODIC	Verify that anchorage complies with approved construction documents. Inspection of post-installed anchors shall comply with approved ICC-ES report
4. Storage racks	PERIODIC	Verify that anchorage complies with approved construction documents. Inspection of post-installed anchors shall comply with approved ICC-ES report. Inspector Note: Not required for racks less than 8 feet in height
5. Acoustical ceiling with 22mm wall molding and clip	PERIODIC	Verify ceiling system is installed per manufacturer details. Conduct special inspection per IBC 2018, 1705.1.1 and provide certification from manufacturer based on the Seismic Design Category of the project.
6. Acoustical ceiling with 50mm wall molding and no clip	PERIODIC	Verify ceiling system is installed per ASTM E580.
7. Elevator equipment, supports and seismic switch	PERIODIC	Verify anchors have been prequalified per ACI 355.2 and designed per ACI 318 Appendix D. Locate switch on loadbearing wall or column. Do not attach to metal stud wall.

END SECTION

¹ PERIODIC: Intermittent monitoring by a special inspector of identified tasks that have been or are being performed.

AA. PLUMBING/MECHANICAL DESIGNATED SEISMIC SYSTEMS SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☐

PLUMBING AND MECHANICAL - DESIGNATED SEISMIC SYSTEMS IBC 1705.12.4, 1705.13.3		
TASK	INSPECTION TYPE ¹	DESCRIPTION
[NOTE: This section is not applicable to Seismic Design Categories A or B. Uncheck this section if one of those categories applies. Confirm Seismic Design Category with structural engineer]		
1. Designated Seismic Systems equipment verification	PERIODIC	✓ Verify model number and serial number are in conformance with project specific seismic qualification (PSSQ) ✓ Verify Tag ID is correct and installed per specifications
2. Designated Seismic Systems equipment Mounting	PERIODIC	✓ Verify that Anchor Base Bolting is installed per PSSQ ✓ Verify that Equipment Bracing is Installed per PSSQ ✓ Verify that Bracing Attachments are installed per PSSQ
3. Designated Seismic Systems utility Conduit/Piping	PERIODIC	✓ Verify that Conduit/Piping is connected to the equipment per PSSQ (flex or rigid) ✓ Verify that Conduit/Piping is seismically supported independently of equipment and in accordance with PSSQ support requirements
4. Designated Seismic Systems clearance	PERIODIC	✓ Adjacent Equipment – Verify that there is adequate gap to eliminate possibility of pounding ✓ Conduit/Piping - Verify that there is adequate gap to eliminate possibility of pounding
5. Installation of Vibration Isolation System (Snubber)	PERIODIC	✓ Verify Anchors have been pre-qualified per ACI 355.2 and designed per ACI 318 Appendix D.

END SECTION

¹ **PERIODIC:** Intermittent monitoring by a special inspector of identified tasks that have been or are being performed.

BB. ELECTRICAL DESIGNATED SEISMIC SYSTEMS SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

ELECTRICAL - DESIGNATED SEISMIC SYSTEMS IBC 1705.12.4		
TASK	INSPECTION TYPE ¹	DESCRIPTION
[NOTE: This section is not applicable to Seismic Design Categories A or B. Uncheck this section if one of those categories applies. Confirm Seismic Design Category with structural engineer]		
1. Generator	PERIODIC	✓ Verify certification of emergency generator, ATS, switchgear, and control panelboard by NIPPON Engine Generator Association (NEGA) ✓ Verify conformance with the Fire Protection Law of Standby Generator
2. Automatic Transfer Switch (ATS)	PERIODIC	✓ Verify certification of emergency generator, ATS, switchgear, and control panelboard by NIPPON Engine Generator Association (NEGA) ✓ Verify conformance with the Fire Protection Law of Standby Generator
3. Uninterrupted Power Source (UPS)	PERIODIC	✓ Verify anchors have been pre-qualified per ACI 355.2 and designed per ACI 318 Appendix D.
4. Electrical Panels	PERIODIC	✓ Verify anchors have been pre-qualified per ACI 355.2 and designed per ACI 318 Appendix D.
5. Switchgear	PERIODIC	✓ Verify Anchors have been pre-qualified per ACI 355.2 and designed per ACI 318 Appendix D.
6. Emergency Fixtures	PERIODIC	✓ Verify light fixtures have been installed per ASTM E580.

END SECTION

¹ **PERIODIC:** Intermittent monitoring by a special inspector of identified tasks that have been or are being performed.

SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C511 (2007) Standard for Reduced-Pressure
Principle Backflow Prevention Assembly

FOUNDATION FOR CROSS-CONNECTION CONTROL AND HYDRAULIC RESEARCH
(FCCCHR)

FCCCHR List (continuously updated) List of Approved
Backflow Prevention Assemblies

FCCCHR Manual (10th Edition) Manual of Cross-Connection
Control

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 241 (2022) Standard for Safeguarding
Construction, Alteration, and Demolition
Operations

NFPA 70 (2023; ERTA 7 2023; TIA 23-15) National
Electrical Code

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 Safety -- Safety and Health Requirements
Manual

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

PL 93-523 (1974; A 1999) Safe Drinking Water Act

U.S. FEDERAL AVIATION ADMINISTRATION (FAA)

FAA AC 70/7460-1 (2016; Rev L; Change 2) Obstruction
Marking and Lighting

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submittals not having a "G" designation are for Contractor Quality Control or Designer of Record approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Site Plan; G
Traffic Control Plan; G
Temporary Utility Connections Plan; G

1.3 CONSTRUCTION SITE PLAN

Prior to the start of work, submit a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage area (onsite and offsite), and access and haul routes, avenues of ingress/egress to the fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.4 BACKFLOW PREVENTERS CERTIFICATE

Certificate of Full Approval from FCCCHR List, attesting that the design, size and make of each backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. Certificate of Provisional Approval will not be acceptable.

1.4.1 Backflow Tester Certificate

Certification issued by the local regulatory agency attesting that the backflow tester has successfully completed a certification course sponsored by the regulatory agency. Tester must not be affiliated with any company participating in any other phase of this Contract.

1.4.2 Backflow Prevention Training Certificate

Certification recognized by the local authority that states the Contractor has completed at least 10 hours of training in backflow preventer installations. The certificate must be current. Training and certification can be obtained from the following sources:

- a. Kyuusuikouji Gijutsu Koshin Zaidan; <https://www.kyuukou.or.jp/>
- b. Zenkoku Kensetsu Kenshu Center; <https://www.jctc.jp/>
- c. Tester Training Course, Foundation for Cross-Connection Control and Hydraulic Research, University of Southern California;
<https://fccchr.usc.edu/tester.html>

1.5 TYPHOON AND STORM PREPARATIONS

In order to minimize damage to public properties and prevent personal injury, the following actions shall be taken upon declaration of the typhoon conditions described below. Ensure that the construction sites are well-prepared for protection from the damages of heavy rain and strong winds. Develop and establish the necessary procedures to be taken for storm preparation. In Part 1 of the Accident Prevention Plan, provide emergency phone numbers and addresses where at least three Contractor officials may be reached and notified in the event that an immediate typhoon alert is declared.

1.5.1 Tropical Cyclone Conditions of Readiness (TCCOR)

- a. TCCOR STORM WATCH: The winds are not forecast to exceed 50 knots (58 mph/26 m/s) sustained (the criteria for "destructive winds") but there still exists a probability of high winds due to the proximity of the storm. High winds may include gusts exceeding 50 knots and/or sustained winds meeting TCCOR 1 Caution criteria. The storm is also close enough to the area that a heightened alert status is necessary in order to rapidly establish elevated TCCOR conditions should the storm deviate from the forecast track. Personnel should follow Standard Operating Procedures for TCCOR Storm Watch and stay alert for any changes to TCCOR status.

When wind gusts of 30 knots (35 mph/16 m/s) or greater are forecast, the Contractor shall be required to perform general area cleanup and monitor TCCOR levels.

When a local storm warning has been issued, the Contractor competent person and/or SSHO shall determine whether it is necessary to implement manufacturer recommendations for securing the equipment to include Weight Handling Equipment (WHE).

Contractor reviews project site storm preparation plans and discuss plan of action with Contracting Officer's Representative.

- b. TCCOR 4: Winds of 50 knots (58 mph/26 m/s) sustained or greater are possible within 72 hours.

When wind gusts of 30 knots (35 mph/16 m/s) or greater are forecast, the Contractor will be required to perform general area cleanup and monitor TCCOR levels.

Cranes shall not be operated when wind speeds at the site attain the maximum wind velocity based on the surface/load ratio recommendations of the manufacturer.

At winds greater than 20 mph (17 knots/9 m/s), the operator, rigger, lift supervisor, and SSHO shall cease all crane operations, evaluate conditions and determine if the lift can proceed with Contracting Officer's Representative acceptance.

All elevated work above 6 ft (1.8 m) shall be risk assessed by the Contractor's competent person and/or SSHO and discussed with the Contracting Officer's Representative for acceptance.

- c. TCCOR 3: Winds of 50 knots (58 mph/26 m/s) sustained or greater are possible within 48 hours.

(1) Clean and remove all loose debris and waste including scrap wood and metal, empty barrels, and construction form materials no longer required on the job site, to a safe area for proper disposition.

(2) Secure, tie down, and anchor construction field office and storage facilities, scaffolding, concrete forms and supports, doors, windows, opening covers, stored lumber and other materials, mechanized construction equipment, temporary power lines and supports, and other items that may be blown away or that might

cause injury or damage.

- (3) Inspect all excavation and trenching work in process, and provide necessary temporary drainage and proper protection and shoring for excavation sides and openings, in order to prevent damage to public roads and facilities by slides or flooding. Accumulation of water in the excavation of structure foundation work shall be controlled and minimized.
- (4) Inspect scaffolding or work platforms for loose materials, planking, etc. that could become airborne projectile hazards and secure scaffold netting, tarps, etc. from wind loads.

- d. TCCOR 2: Winds of 50 knots (58 mph/ 26 m/s) sustained or greater are possible within 24 hours.

Work required at remote areas such as off-shore facilities or high elevations shall cease immediately and the workers shall be evacuated to a safe area. During TCCOR 2, the Contractor shall continue the actions described in TCCOR 3 above and the construction site shall be inspected for storm preparation by the Contracting Officer's Representative. The Contractor shall request an inspection by calling the Contracting Officer's Representative at the appropriate Japan Engineer District field office.

- e. TCCOR 1: Winds of 50 knots (58 mph/26 m/s) sustained or greater are possible within 12 hours.

All work shall cease immediately and the Contractor's representative shall insure that all necessary storm preparations, including the items listed below, are completed.

- (1) All electrical circuits and equipment including temporary power lines are cut off and secured against unauthorized use.
- (2) Gas cylinders, hot work equipment, and flammable materials properly stored at a safe area.
- (3) No igniting source is present.
- (4) All workers have been evacuated from the construction site. When TCCOR 1 is declared without the normal progression through TCCOR 3 and/or 2, the Contractor shall take the actions listed in TCCOR 3 above, and also follow the procedures described herein.

- f. TCCOR 1 Caution: Winds of 35 to 49 knots (39/17 to 56/25 mph/ m/s) sustained are occurring.

Construction Sites Secured.

- g. TCCOR 1 Emergency: Winds of 50 knots (58 mph/26 m/s) sustained or greater are occurring.

Construction Sites Secured.

- h. TCCOR 1 Recovery: Winds of 50 knots (58 mph/26 m/s) sustained or greater are no longer forecast to occur. Strong winds may still exist.

Installation personnel begin storm damage assessments and clean up.

- i. TCCOR Storm Clear: The storm is over and not forecast to return. Used to inform personnel that the threat of the storm is over, but the storm damage could still present a danger.

Contractor may resume normal activities. The construction site shall be investigated for all damage caused by the typhoon or high winds, and the result of the investigation shall be furnished in verbal or written form to the Contracting Officer's Representative as soon as practicable.

Complete and submit PODWP 134 Typhoon Damage Report for storm damages or negative report of damage to Emergency Management, Construction Division, and Safety Office.

- j. TCCOR All Clear: The storm is over and not forecast to return, and recovery efforts are complete.

Resume normal activities.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Immediately upon beginning of work, provide a weatherproof glass-covered bulletin board not less than 915 by 1220 mm in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, and other information approved by the Contracting Officer. Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, as approved by the Contracting Officer.

2.1.2 Project and Safety Signs

Provide [one (1)] project sign and [one (1)] safety sign fabricated to size and design as shown in Attachment 01 50 00-A. The signs shall be rigidly formed and erected at location designated by the Contracting Officer prior to commencement of work. A blue-line drawing of different letter sizes and style shall be made available by the Contracting Officer upon request. Prior to painting the sign, submit for approval a sketch, similar to diagram shown in Attachment 01 50 00-A, indicating the actual information for the project. The sketch shall indicate lettering dimensions and locations. The Corps of Engineers castle logo shall be painted red per Attachment 01 50 00-A. Correct the data required by the safety sign daily, with light-colored metallic or non-metallic numerals. No separate payment shall be made for the sign, and all costs in connection therewith shall be included in the Contract price for the project. The sign shall be subject to the approval of the Contracting Officer. Erect signs within 15 days after receipt of the Notice to Proceed. Correct the data required by the safety sign daily, with light-colored metallic or non-metallic numerals. Upon completion of work under this Contract, the signs shall be removed from the job site and shall remain the property of the Contractor.

2.2 TEMPORARY TRAFFIC CONTROL

2.2.1 Haul Roads

Construct access and haul roads necessary for proper prosecution of the work under this Contract. Construct with suitable grades and widths; sharp curves, blind corners, and dangerous cross traffic are to be avoided. Provide necessary lighting, signs, barricades, and distinctive markings for the safe movement of traffic. The method of dust control must be adequate to ensure safe operation at all times. Location, grade, width, and alignment of construction and hauling roads are subject to approval by the Contracting Officer. Lighting must be adequate to assure full and clear visibility for full width of haul road and work areas during any night work operations.

2.2.2 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Whenever safe public access to paved areas such as roads, parking areas or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic barricades will be required. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.2.3 Temporary Wiring

Provide temporary wiring in accordance with NFPA 241 and NFPA 70. Include frequent inspection of all equipment and apparatus.

2.2.4 Backflow Preventers

Reduced pressure principle type conforming to the applicable requirements AWWA C511. Provide backflow preventers complete with 65 kg flanged, mounted gate valve and strainer, stainless steel or bronze, internal parts. After installation conduct Backflow Preventer Tests and provide test reports verifying that the installation meets the FCCCHR Manual Standards.

PART 3 EXECUTION

3.1 EMPLOYEE PARKING

Contractor employees will park privately owned vehicles in an area designated by the Contracting Officer. This area shall be within reasonable walking distance of the construction site. Contractor employee parking must not interfere with existing and established parking requirements of the government installation.

3.2 TEMPORARY BULLETIN BOARD

Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, as approved by the Contracting Officer.

3.3 AVAILABILITY AND USE OF UTILITY SERVICES

3.3.1 Payment for Utility Services

The Government shall make all reasonably required utilities available to

the Contractor[, without charge,] from existing outlets and supplies, as specified in the Contract. Carefully conserve any utilities.

3.3.2 Meters and Temporary Connections

At the Contractor's expense and in a manner satisfactory to the Contracting Officer, provide and maintain necessary temporary connections, distribution lines, and meter bases required to measure the amount of each utility used. Materials may be new or used, must be adequate for the required usage, not create unsafe conditions, and not violate applicable codes and standards. Submit a Temporary Utility Connections Plan for approval to the Contracting Officer, fifteen (15) working days before making temporary utility connections.

- [The Government shall
 - [Furnish the water meter]
 - [Furnish the electric meter]
 - [Furnish the backflow preventer]
 - [Install the water meter]
 - [Install the electric meter]
 - [Install the backflow preventer]
-][The Contractor shall
 - [Furnish the water meter]
 - [Furnish the electric meter]
 - [Furnish the backflow preventer]
 - [Install the water meter]
 - [Install the electric meter]
 - [Install the backflow preventer]

[3.3.3 Advance Deposit

An advance deposit for utilities consisting of an estimated month's usage or a minimum of [6,000] JPY shall be required. Adjustments to the monthly bills shall be settled at the end of the fiscal year ending on 30 September. Services to be rendered for the next fiscal year, beginning 1 October, shall require a new deposit. Notification of the due date for the new deposit shall be mailed to the Contractor prior to the end of the previous fiscal year.

]3.3.4 Final Meter Reading

Before completion of the work and final acceptance of the work by the Government, notify the Contracting Officer, in writing, five working days before termination is desired. A final reading shall be taken by the Government. Afterwards, remove all temporary distribution lines, meter bases, meters, and associated appurtenances.[Pay all outstanding utility bills before final acceptance of the work by the Government.]

3.3.5 Sanitation

Provide and properly maintain temporary toilet facility for use of Contractor personnel. Government toilet facilities are not available to Contractor personnel. Toilet facility shall be constructed as required in EM 385-1-1 and by local installation regulations, where directed. Include provisions for pest control and elimination of odors. Upon completion of the Contract, dispose facility outside the limits of Government-controlled land and at Contractor's expense.

3.3.6 Electricity

The Contractor shall be responsible for all temporary connections required, as well as the removal of these temporary connections upon completion of the project. The Contractor's plan for temporary connections shall be submitted to the Contracting Officer Representative for approval prior to making any temporary connections.

Electrical power system frequencies for different installations are as follows:

- a. 50 Hertz (Hz): Camp Zama, Sagami Depot, Sagamihara Housing Area, Yokota Air Base, Commander Fleet Activities Yokosuka, Naval Air Facility Atsugi, Misawa Air Base, Camp Fuji, and Naval Support Facility Kamiseya.
- b. 60 Hertz (Hz): Marine Corps Air Station Iwakuni, Commander Fleet Activities Sasebo, and all of Okinawa.

3.3.7 Telephone and Data

Make arrangements and pay all costs for telephone facilities and data connections.

3.3.8 Obstruction Lighting of Cranes

Provide a minimum of 2 aviation red or high intensity white obstruction lights on temporary structures (including cranes) over 30 meter above ground level. Light construction and installation shall comply with FAA AC 70/7460-1. Lights shall be operational during periods of reduced visibility, darkness, and as directed by the Contracting Officer.

3.3.9 Fire Protection

Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials daily to minimize potential hazards.

3.4 TRAFFIC PROVISIONS

3.4.1 Maintenance of Traffic

- a. Conduct operations in a manner that will not close any thoroughfare or interfere in any way with traffic except with written permission of the Contracting Officer at least 30 calendar days prior to the proposed modification date, and provide a Traffic Control Plan detailing the proposed controls to traffic movement for approval. The plan must be in accordance with installation regulations. Coordinate with installation officials. Contractor may move oversized and slow-moving vehicles to the worksite once the requirements of the highway authority have been met. Include a schedule of planned road closures in the Initial Project Schedule.
- b. Conduct work so [there is no obstruction of traffic, and maintain traffic on the full roadway width at all times][as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times]. Obtain approval from the Contracting Officer prior to starting any activity that will obstruct traffic.

- c. At the Contractor's expense, provide, erect, and maintain: lights, barriers, signals, passageways, detours, and other items, that may be required by the Life Safety Signage overhead protection authority having jurisdiction.

3.4.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction period except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment, the work, and the erection and maintenance of adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Protect the traveling public from damage to person and property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. Contractor is responsible for the repair of any damage to roads caused by construction operations.

3.4.3 Dust Control

Dust control methods and procedures must be approved by the Contracting Officer. Treat dust abatement on access roads with applications of calcium chloride, water sprinklers, or similar methods or treatment.

3.5 CONTRACTOR'S TEMPORARY FACILITIES

3.5.1 Safety

Protect the integrity of any installed safety systems or personnel safety devices. If entrance into systems serving safety devices is required, the Contractor must obtain prior approval from the Contracting Officer. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish contract requirements, provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and obtain approval from the Contracting Officer.

3.5.2 Administrative Field Offices

Provide and maintain administrative field office facilities within the construction area at the designated site. Government office and warehouse facilities will not be available to the Contractor's personnel.

3.5.3 Temporary Project Safety and Storage Area Fencing

As soon as practicable, but not later than 15 days after the date established for commencement of work, furnish and erect temporary project safety and storage area fencing to control access by unauthorized people. Fencing shall be able to restrain a force of at least 114 kg. Safety and storage area fencing shall be [1.8 m high wire mesh fence with lockable gates and visibility screening (similar to tennis court screening)][2.5 to 3.0m high sheet metal fence (similar to construction fencing used throughout Japan)] [located around the perimeter of project work areas and the perimeter of Contractor lay-down areas]. [Use plastic panels or perforated steel sheets to improve visibility for vehicular traffic when erecting fencing within 3 meters of street corners.] The intent is to block (screen) public view of the construction, and prevent unauthorized

access to the site. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Maintain the safety and storage area fencing during the life of the Contract and, upon completion and acceptance of the work, shall become the property of the Contractor and be removed from the work site.

3.5.4 Storage Area

Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the installation boundaries. Trailers, equipment, or materials must not be open to public view with the exception of those items which are in support of ongoing work on any given day. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

3.5.5 Supplemental Storage Area

Upon Contractor's request, the Contracting Officer will designate another or supplemental area for the Contractor's use and storage of trailers, equipment, and materials. This area may not be in close proximity of the construction site but will be within the installation boundaries. Fencing of materials or equipment will be required at this site. The Contractor is responsible for cleanliness and orderliness of the area used and for the security of any material or equipment stored in this area. Utilities will not be provided to this area by the Government.

3.5.6 Appearance of Trailers

Trailers utilized by the Contractor for administrative or material storage purposes must present a clean and neat exterior appearance and be in a state of good repair. Trailers which, in the opinion of the Contracting Officer, require exterior painting or maintenance will not be allowed on installation property.

3.5.7 Maintenance of Storage Area

Keep fencing in a state of good repair and proper alignment. Grassed or unpaved areas, which are not established roadways, will be covered with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways, should the Contractor elect to traverse them with construction equipment or other vehicles; gravel gradation will be at the Contractor's discretion. Mow and maintain grass to a maximum height of 100 mm located within the boundaries of the construction site for the duration of the project. Grass and vegetation along fences, buildings, under trailers, and in areas not accessible to mowers will be edged or trimmed neatly.

3.5.8 Security Provisions

Provide adequate outside security lighting at the Contractor's temporary facilities. The Contractor will be responsible for the security of its own equipment.

3.5.9 Weather Protection of Temporary Facilities and Stored Materials

Take necessary precautions to ensure that roof openings and other critical openings in the building are monitored carefully. Take immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the building from damage.

3.5.9.1 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions must include, but are not limited to, closing openings; removing loose materials, tools and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

3.6 GOVERNMENT FIELD OFFICE

The Contractor shall be responsible for providing and maintaining, in good condition, an office for the sole use of Government personnel, in a structure physically separated from the Contractor's site offices. Facility shall be furnished at a location designated and approved by the Contracting Officer, and available for use concurrent with availability of Contractor's site office or within 15 calendar days of Notice to Proceed, whichever occurs first. The office facility, including all furniture, equipment, and connections, shall remain the property of the Contractor and, upon completion of work under this Contract, shall be removed from the site by the Contractor and the site returned as nearly as practical to the original conditions.

Contractor shall provide two designated USACE parking spaces as close to the Government Field Office as practicable. The spaces shall not exceed 50 meters from the field office without written approval from the Contracting Officer.

3.6.1 Facility Requirements

Provide the Government an office with a minimum of [67 square meters (720 square feet)] in floor area, windows, a door with lockset individually keyed with two keys, and a battery operated smoke detector in each room. Provide batteries for the smoke detectors, and replace batteries in each smoke detector at least twice annually.

The walls and ceiling of the office shall be insulated. All windows shall be screened. Windows shall be openable and be securely fastened from the inside. Glass panels in windows shall be protected by bars or heavy mesh screens to prevent easy access to the facility through these panels. The office shall contain a minimum of eight (8) standard electrical outlets meeting local requirements and additionally provide for delivery of a minimum 3,000 Watts.

The office shall contain an integral toilet facility with western-type (with a push-type flush button) toilet fixture (bowl), which shall hold water in it at all times, and a wash basin.

The office facility shall contain a kitchenette type facility with a

countertop/work area with a minimum surface area of 55 cm (W) by 150 cm (L). A minimum of two electrical outlets (local current) shall be located adjacent to the countertop.

Hot water for all faucets shall be provided.

3.6.2 Furniture and Equipment

Furnish the office with the following furniture and equipment. All furnishing shall be new and manufactured by a recognized supplier of office equipment. All items shall be assembled, installed, and maintained by the Contractor.

- a. Two desks (minimum 80 cm by 150 cm) double pedestal with lockable drawers.
- b. Two desk chairs with arms: swivel tilt, hydraulic height-adjustable, medium back, fabric covered seat cushion, castored.
- c. One conference table (minimum 120 cm by 250 cm), with a minimum of six new arm-chairs, castored.
- d. Three file cabinets, capable of holding US letter-size documents, with a minimum five drawers each.
- e. Two bookcases with a minimum of five shelves each. Minimum dimensions are 120 cm (W) by 200 cm (H) by 30 cm (D).
- f. One clock: wall hung, quartz, 12 dial, battery powered.
- g. Plan racks capable of holding two full-size set of Contract plans.
- h. One dry-erase board with a minimum dimension of 1.5 m by 1.0 m. Dry erase markers (minimum four different colors) shall be provided and replaced as depleted.
- i. One microwave (minimum 0.028 cubic meter).
- j. One refrigerator (minimum 0.70 cubic meter).

3.6.3 Lighting Requirements

The office facility shall include suitable electric lighting to meet the minimum lighting requirements (luminance) of administrative areas as indicated in EM 385-1-1.

3.6.4 Heating/Cooling Requirements

Provide heating and cooling capable of maintaining an interior temperature of between 16 degrees Celsius and 25 degrees Celsius in all seasons. If applicable, use a fluorocarbon gas refrigerant with an Ozone Depletion Potential (ODP) of less than or equal to 0.05.

3.6.5 Drinking Water

Provide potable drinking water for the facility that meets or exceeds PL 93-523.

3.6.6 Data Connection

Provide network connectivity (ADSL, Cable or Fiber), one USACE-dedicated router and one switching hub, eight-5 m network cables, and eight power strips.

3.6.7 Maintenance

Provide maintenance and janitorial services, to include:

- a. Emptying of trash at least twice a week.
- b. Cleaning of the office, including the toilet facilities, weekly.
- c. Replenishment of liquid soap, toilet paper, and paper towels for daily use. Adequate quantity of these supplies shall be provided on-site and replaced as needed.

3.6.8 Security Requirements

The physical security of the Government personnel, property, and field office shall be the responsibility of the Contractor throughout the duration of the Contract.

If the project site is outside of the controlled area of an existing military installation, the Government field office shall have 30 meters of standoff from a fence enclosing the office and all parking or roads. Additional fencing, earth berms, concrete barriers, controlled access points, and similar security features shall be provided to limit access to the site and provide additional building standoff or security.

3.7 PLANT COMMUNICATION

Whenever the Contractor has the individual elements of its plant so located that operation by normal voice between these elements is not satisfactory, the Contractor must install a satisfactory means of communication, such as telephone or other suitable devices and made available for use by Government personnel.

3.8 CLEANUP

Remove construction debris, waste materials, packaging material and the like from the work site daily. Any dirt or mud which is tracked onto paved or surfaced roadways must be cleaned away. Store any salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

3.9 RESTORATION OF STORAGE AREA

Upon completion of the project remove the bulletin board, signs, barricades, haul roads, and any other temporary products from the site. All tools, equipment, and materials not the property of the Government shall be removed from the premises and properly disposed of off Government property. After removal of trailers, materials, and equipment from within the fenced area, remove the temporary project safety and storage area fencing. Restore areas used by the Contractor for the storage of equipment or material, or other use to the original or better condition.

Remove gravel used to traverse grassed areas and restore the area to its original condition, including top soil and seeding as necessary. Upon completion of the work, notify the Contracting Officer for final inspection.

-- End of Section --

DRAFT

Introduction: Construction Project Signs

EP 310-1-6a
01 Jun 06

The use of signs to identify Corps managed or supervised design, construction, and rehabilitation projects for military is an important part of efforts to keep the public informed of Corps work. For this purpose, a construction project sign package has been adopted. This package consists of two signs: one for project identification and the other to show on-the-job safety performance of the contractor.

These two signs are to be displayed side by side and mounted for reading by passing viewers. Exact placement location will be designated by the contracting officer's representative.

The panel sizes and graphic formats have been standardized for visual consistency throughout all Corps operations.

Panels are fabricated using HDO plywood or aluminum with dimensional lumber uprights and bracing. The sign faces are nonreflective vinyl.

All legends are to be die-cut or computer-cut in the sizes and typefaces specified and applied to the white panel background following the graphic formats shown on pages 16-2 and 16-3. The Communication Red panel on the left side of the construction project sign with Corps Signature (reverse version) is screen-printed onto the white background.

A display of these two signs is shown on the following two pages. Mounting and fabrication details are provided on page 16-4.

Special applications or situations not covered in these guidelines should be referred to the district Sign Program Manager.

EP 310-1-6a (Modified per ECB 2020-1)
01 Jun 06 (ECB modified 1/31/2020)

Construction Project Identification Sign

Below is one sample of the Construction Project Identification sign showing how this panel is to identify military projects. The graphic format for this 4'x 6' sign panel follows the legend guidelines and layout as specified below. The large 4'x 4' section of the panel on the right is to be white with black legend.

The 2'x 4' section of the sign on the left with the full Corps Signature (reverse version) is to be screen-printed Communication Red on the white background. The designation of a sponsor in the area indicated is optional with Military construction signs.

This sign is to be placed with the Safety Performance sign shown on the following page. Mounting and fabrication details are provided on page 16-4.

Special applications or situations not covered in these guidelines should be referred to the district Sign Program Manager.

Post size is 4"x4".
Mounting height is 48".

Legend Group 1: One- to two-line description of Corps relationship to project.
Color: White
Typeface: 1.25" Helvetica Regular
Maximum line length: 19"

Legend Group 2: Division or District Name (optional). Placed below 10.5" reverse Signature (6" Castle).
Color: White
Typeface: 1.25" Helvetica Regular

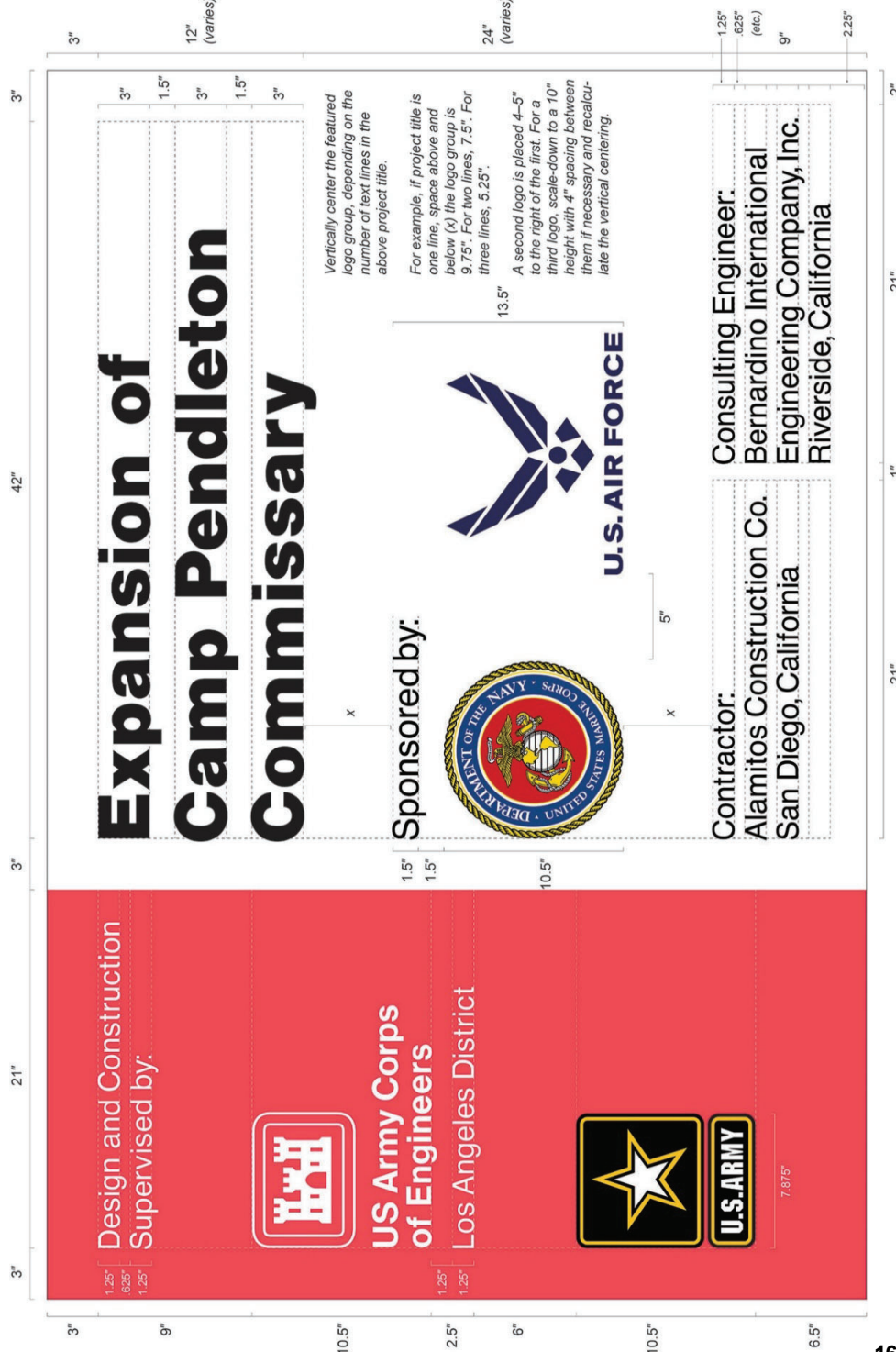
Legend Group 2a: Army Star emblem.
Color: Various

Legend Group 3: One- to three-line project title legend describes the work being done under this contract.
Color: Black
Typeface: 3" Helvetica Bold
Maximum line length: 42"

Legend Group 4: Logo of sponsoring department (military).
Color: Various

Legend Groups 5a-b: One- to five-line identification of prime contractors including: type (architect, general contractor, etc.), corporate or firm name, city, state. Use of Legend Group 5 is optional.
Color: Black
Typeface: 1.25" Helvetica Regular
Maximum line length: 21"

All typography is flush left and rag right, upper and lower case with initial capitals only as shown. Letter- and word-spacing to follow Corps standards as specified in Appendix D.



16-2

Safety Performance Sign

EP 310-1-6a
01 Jun 06

Each contractor's safety record is to be posted on Corps managed or supervised construction projects and mounted with the Construction Project Identification sign specified on page 16-2.

The graphic format, color, size and typefaces used on the sign are to be reproduced exactly as specified below. The

title with First Aid logo in the top section of the sign, and the performance record captions are standard for all signs of this type. Legend groups 2 and 3 below identify the project and the contractor and are to be placed on the sign as shown.

Safety record numbers are mounted on individual metal plates and are screw-

mounted to the background to allow for daily revisions to posted safety performance record.

Special applications or situations not covered in these guidelines should be referred to the district Sign Program Manager.

Legend Group 1: Standard two-line title "Safety is a Job Requirement" with 8" (outside diameter) Safety Green first aid logo.
Color: To match Pantone system 347
Typeface: 3" Helvetica Bold
Color: Black

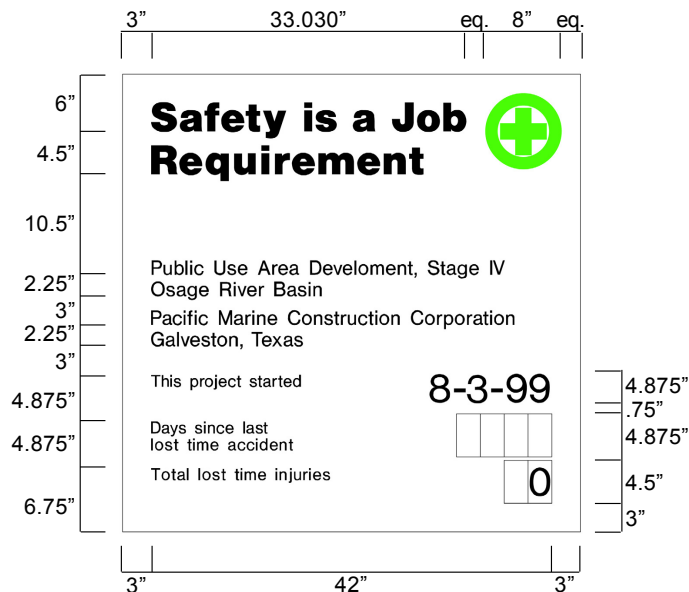
Legend Group 2: One- to two-line project title legend describes the work being done under this contract and name of host project.
Color: Black
Typeface: 1.5" Helvetica Regular
Maximum line length: 42"

Legend Group 3: One- to two-line identification: name of prime contractor and city, state address. Color: Black
Typeface: 1.5" Helvetica Regular
Maximum line length: 42"

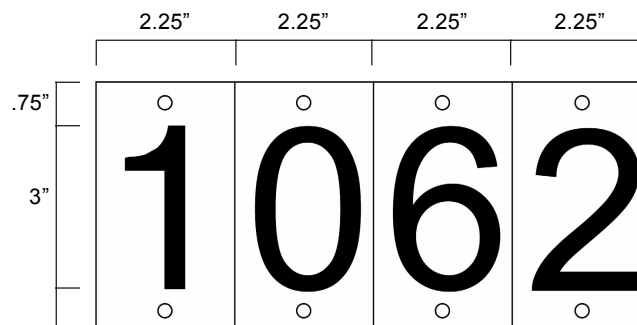
Legend Group 4: Standard safety record captions as shown.
Color: Black
Typeface: 1.25" Helvetica Regular

Replaceable numbers are to be mounted on white .060 aluminum plates and screw-mounted to background.
Color: Black
Typeface: 3" Helvetica Regular
Plate size: 2.5" x 4.5"

All typography is flush left and rag right, upper and lower case with initial capitals only as shown. Letter- and word-spacing to follow Corps standards as specified in Appendix D.



Sign Type	Legend Size (A)	Panel Size	Post Size	Specification Code	Mounting Height	Color Bkg/Lgd
CID-02	various	4'x4'	4"x4"	HDO-3	48"	WH/BK-SG



16-3

EP 310-1-6a
01 Jun 06

Fabrication and Mounting Guidelines

All Construction Project Identification signs and Safety Performance signs are to be fabricated and installed as described below. The signs are to be erected at a location designated by the contracting officer representative and shall conform to the size, format, and typographic standards shown on pages 16-2 and 16-3. Detailed specifications

for HDO plywood panel preparation are provided in Appendix B.

For additional information on the proper method to prepare sign panel graphics, contact the district Sign Program Manager.

Shown below the mounting diagram is a panel layout grid with spaces provided for project information. Photocopy this page and use as a worksheet when preparing sign legend orders.

The sign panels are to be fabricated from .75" High Density Overlay Plywood. Panel preparation to follow HDO specifications provided in Appendix B.

Sign graphics to be prepared on a white nonreflective vinyl film with positionable adhesive backing.

All graphics except for the Communication Red background with Corps Signature on the project sign are to be die-cut or computer-cut nonreflective vinyl, prespaced legends prepared in the sizes and typefaces specified and applied to the background panel following the graphic formats shown on pages 16-2 and 16-3.

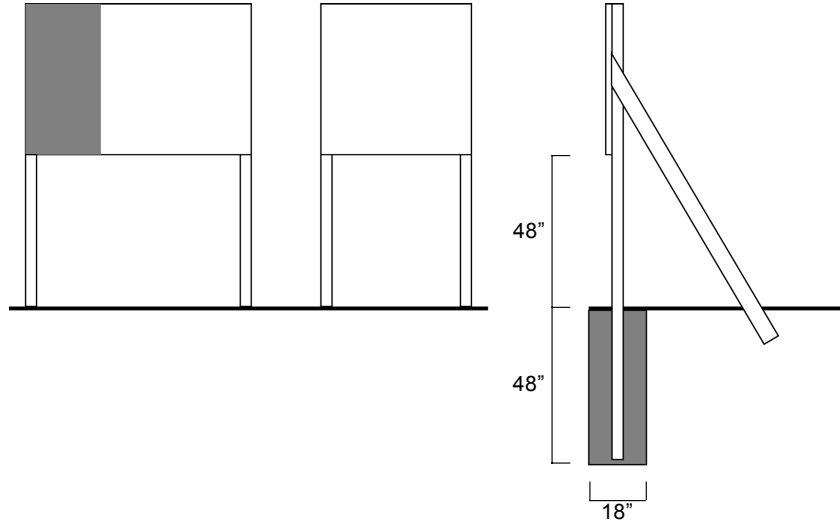
The 2'x 4' Communication Red panel (to match Pantone system 032) with full Corps Signature (reverse version) is to be screen-printed on the white background. Identification of the district or division may be applied under the signature with white cut vinyl letters prepared to Corps standards.

Drill and insert six (6) .375" T-nuts from the front face of the HDO sign panel. Position holes as shown. Flange of T-nut to be flush with sign face.

Apply graphic panel to prepared HDO plywood panel following manufacturers' instructions.

Sign uprights to be structural grade 4" x 4" treated Douglas Fir or Southern Yellow Pine, No.1 or better. Post to be 12' long. Drill six (6) .375" mounting holes in uprights to align with T-nuts in sign panel. Countersink (.5") back of hole to accept socket head cap screw (4" x .375").

Assemble sign panel and uprights. Imbed assembled sign panel and uprights in 4' hole. Local soil conditions and/or wind loading may require bolting additional 2" x 4" struts on inside face of uprights to reinforce installation as shown.



Construction Project Identification Sign
Legend Group 1: Corps Relationship

1. _____
2. _____

Legend Group 2: Division/District Name

1. _____
2. _____

Legend Group 2a: Military/Civil Works Sponsor

1. _____
2. _____

Legend Group 3: Project Title

1. _____
2. _____
3. _____

Legend Group 4: Facility Name

1. _____
2. _____

Legend Group 5: Contractor/A&E

1. _____
2. _____
3. _____
4. _____
5. _____

Legend Group 5b: Contractor/A&E

1. _____
2. _____
3. _____
4. _____
5. _____

Safety Performance Sign
Legend Group 2: Project Title

1. _____
2. _____

Legend Group 3: Contractor/A&E

1. _____
2. _____

SECTION 01 57 19

TEMPORARY ENVIRONMENTAL CONTROLS

01/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. The Contractor shall comply with applicable U.S. Government and Government of Japan (GOJ) laws and regulations, however denominated, including those applicable political subdivision, departments and other entities, to include but not limited to the following:

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS (Apr 2024) Japan Environmental Governing Standards

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910 Occupational Safety and Health Standards

29 CFR 1926.1101 Asbestos

40 CFR 261 Identification and Listing of Hazardous Waste

1.2 DEFINITIONS

In some cases, definitions are given only for illustrative purposes. Prevailing JEGS definitions shall be used for environmental compliance requirements.

1.2.1 Aboveground Storage Tank

A portable or fixed POL aboveground storage container as defined in the JEGS with a capacity greater than 55 gallons.

1.2.2 Asbestos Containing Material (ACM)

Any material containing more than one tenth of one percent (0.1%) asbestos by weight.

1.2.3 Bulky Waste

Large items of solid waste such as household appliances, furniture, large auto parts, trees, branches, stumps, and other oversize wastes whose large size precludes or complicates their handling by normal solid waste collection, processing, or disposal methods.

1.2.4 Chemical Wastes

Salts, acids, alkalis, herbicides, pesticides, organic chemicals, and

spent products, which serve no purpose.

1.2.5 Class I and II Ozone Depleting Substance (ODS)

Class I and II ODS are listed in the JEGS.

1.2.6 Construction and Demolition Waste

The waste building materials, packaging, and rubble resulting from construction, remodeling, repair and demolition operations on pavements, houses, commercial buildings, and other structures.

1.2.7 Contractor Generated Hazardous Waste

Contractor generated hazardous waste is materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e. methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment rinse water.

1.2.8 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, or disposal.

1.2.9 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

1.2.10 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.11 Food Waste

Organic residues generated by the handling, storage, sale, preparation, cooking, and serving of foods (commonly called garbage).

1.2.12 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with the JEGS. Hazardous debris also includes debris that exhibits a characteristic of hazardous waste in accordance with the JEGS.

1.2.13 Hazardous Materials

Hazardous material is any material that is capable of posing an unreasonable risk to health, safety, or the environment if improperly handled, stored, issued, transported, labeled, or disposed because it displays a characteristic listed in the JEGS, "Typical Hazardous Materials Characteristics," or the material is listed in the "List of Hazardous Waste/Substances/Materials" or is regulated as a hazardous material in accordance with the JEGS or GOJ or local installation regulations, or requires a Safety Data Sheet (SDS), or during end use, treatment, handling, packaging, storage, transpiration, or disposal meets or has components that meet or have potential to meet the definition of a hazardous waste as defined by the JEGS or prefectural regulations. Munitions are excluded.

Designation of a material by this definition, when separately regulated or controlled by other instructions or directives, does not eliminate the need for adherence to that hazard-specific guidance which takes precedence over this instruction for "control" purposes. Such material include ammunition, weapons, explosive actuated devices, propellants, pyrotechnics, chemical and biological warfare materials, medical and pharmaceutical supplies, medical waste and infectious materials, bulk fuels, radioactive materials, and other materials such as asbestos, mercury, and polychlorinated biphenyls (PCBs). Nonetheless, the exposure may occur incident to manufacture, storage, use and demilitarization of these items.

1.2.14 Hazardous Substances

Any substance having the potential to do serious harm to human health or the environment if spilled or released in reportable quantity. A list of these substances and the corresponding reportable quantities is contained in the JEGS, "Characteristics of Hazardous Waste and Lists of Hazardous Waste and Hazardous Material."

1.2.15 Hazardous Waste

Hazardous Waste is discarded material that may be solid, semi-solid, liquid, or contained gas, and either exhibits a characteristic of a hazardous waste as defined in the JEGS. Excluded from this definition are domestic sewage sludge, household wastes, and medical wastes.

1.2.16 Installation Pest Management Coordinator

Installation Pest Management Coordinator (IPMC) is the individual officially designated by the Installation Commander to oversee the Installation Pest Management Program and the Installation Pest Management Plan.

1.2.17 Installation Pest Management Consultant

Installation Pest Management Consultant (IPMC) is the professional DoD pest management personnel located at component headquarters, field operating agencies, major commands, facilities engineering field divisions or activities, or area support activities who provide technical and management guidance for the conduct of installation pest management operations. Some pest management consultants may be designated by their component as certifying officials.

1.2.18 Land Application

Land Application means spreading or spraying discharge water at a rate that allows the water to percolate into the soil. No sheeting action, soil erosion, discharge into storm sewers, or discharge into defined drainage areas (includes drainage ditches, streams, rivers, ocean, etc.) must occur. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

1.2.19 Lead-based Paint (LBP)

Paint or other surface coatings that contain lead greater than or equal to 1.0 milligram per square centimeter, or 0.5 percent by weight, or 5,000 ppm by weight.

1.2.20 Oily Wastes

Oily wastes are those materials that are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from those POLs. Oily wastes also means materials, including wastewaters, centrifuge solids, filter residues or sludges, bottom sediments, tank bottoms, and sorbents which have come into contact with and have been contaminated by, POLs and may be appropriately tested and discarded in a manner which is in compliance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements..

This definition includes materials such as oily rags, "kitty litter" sorbent clay and organic sorbent material. These materials may be land filled provided that: It is not prohibited in the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements; the amount generated is "de minimus" (a small amount); it is the result of minor leaks or spills resulting from normal process operations; and free-flowing oil has been removed to the practicable extent possible. Large quantities of this material, generated as a result of a major spill or in lieu of proper maintenance of the processing equipment, are a solid waste. As a solid waste, perform a hazardous waste determination prior to disposal. As this can be an expensive process, it is recommended that this type of waste be minimized through good housekeeping practices and employee education.

1.2.21 Pesticide

Pesticide is any substance or mixture of substances, including biological control agents that may prevent, destroy, repel, or mitigate pests.

1.2.22 Pesticide Treatment Plan

A plan for the prevention, monitoring, and control to eliminate pest infestation.

1.2.23 Pests

Pests are arthropods, birds, rodents, nematodes, fungi, bacteria, viruses, algae, snails, marine borers, snakes, weeds and other organisms (except for human or animal disease-causing organisms) that adversely affect readiness, military operations, or the well-being of personnel and animals; attack or damage real property, supplies, equipment, or vegetation; or are otherwise undesirable.

1.2.24 Project Pesticide Coordinator

The Project Pesticide Coordinator (PPC) is an individual who resides at a Civil Works Project office and who is responsible overseeing of pesticide application on project grounds.

1.2.25 Petroleum, Oil, and Lubricants

Refined petroleum, oils, and lubricants, including, but not limited to, petroleum, fuel, lubricant oils, synthetic oils, mineral oils, animal fats, vegetable oil, sludge, and POL mixed with wastes other than dredged spoil.

1.2.26 Polychlorinated Biphenyl (PCB)

Any PCB article, PCB article container, PCB container, or PCB equipment that deliberately or unintentionally contains or has as a part of it any detectable concentration of PCB.

1.2.27 Regulated Waste

Regulated waste are solid wastes that have specific additional Federal, GOJ national, and prefectural controls for handling, storage, or disposal.

1.2.28 Rubbish

A general term for solid waste, excluding food wastes and ashes, taken from residences, commercial establishments, and institutions.

1.2.29 Sanitary Waste

a. Sewage: Wastes characterized as domestic sanitary sewage.

b. Garbage: Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

1.2.30 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.31 Solid Waste

Solid waste is garbage, refuse, sludge, and other discarded materials, including solid, semi-solid, liquid, and contained gaseous materials resulting from industrial and commercial operations and from community activities. It does not include solids or dissolved material in domestic sewage or their significant pollutants in water resources, such as silt, dissolved or suspended solids in industrial wastewater effluent, dissolved materials in irrigation return flows, or other common water pollutants. Material not regulated as solid waste are: nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production. Types of solid waste typically generated at construction sites may include:

1.2.31.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure that exceeds 60 mm particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials [may][may not] be reinforced with or contain ferrous wire, rods, accessories and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.31.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps and plant roots. Marketable trees, grasses and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.31.3 Hazardous Waste

By definition, to be a hazardous waste a material must first meet the definition of a solid waste. Hazardous waste and hazardous debris are special cases of solid waste. They have additional regulatory controls and must be handled separately. They are thus defined separately in this document.

1.2.31.4 Material Not Regulated as Solid Waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.31.5 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, characteristic of a hazardous waste as defined in the JEGS or is listed as a hazardous waste in the JEGS. Excluded from this definition is medical wastes.

1.2.31.6 Paint Cans

Metal cans that are empty of paints, solvents, thinners and adhesives. If permitted by the paint can label, a thin dry film may remain in the can.

1.2.31.7 Recyclables

Recyclables are materials, equipment and assemblies such as doors, windows, door and window frames, plumbing fixtures, glazing and mirrors that are recovered and sold as recyclable, wiring, insulated/non-insulated copper wire cable, wire rope, and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade

paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Metal meeting the definition of lead contaminated or lead based paint contaminated may not be included as recyclable if sold to a scrap metal company.

1.2.31.8 Surplus Soil

Surplus soil is existing soil that is in excess of what is required for this work, including aggregates intended, but not used, for on-site mixing of concrete, mortars, and paving. Contaminated soil meeting the definition of hazardous material or hazardous waste is not included and must be managed in accordance with paragraph HAZARDOUS MATERIAL MANAGEMENT.

1.2.31.9 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included.

1.2.31.10 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, hardboard. Treated or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.2.32 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, or creeks, or waters of Japan. Surface discharges are discrete, identifiable sources and require a permit from the governing agency. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

1.2.33 Wastewater

Wastewater is the used water and solids from a community that flow to a treatment plant.

1.2.33.1 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.34 Waters of Japan

Waters of Japan, as defined by the JEGS, means surface water including the territorial seas recognized under customary international law.

1.2.35 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

1.3 PROHIBITED PRODUCTS

The following items are forbidden for use by the JEGS or other criteria. Details of each are included in the text of each chapter of the Environmental Protection Plan (EPP) described in this section.

- a. Asbestos Containing Materials (ACM).
- b. Lead-Containing Paint. Paint containing greater than 0.009 percent lead by weight.
- c. Polychlorinated Biphenyls (PCBs). Materials containing PCBs greater than 0.5mg/kg shall not be used.
- d. Class I Ozone Depleting Substances (ODS). Class 1 ODS listed in the JEGS shall not be used.
- e. Lead Drinking-Water Pipes, Solders, Flux, and Fittings.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are [for Contractor Quality Control approval.][for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.] Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Environmental Protection Plan; G[, [____]]

Environmental Manager Qualifications; G[, [____]]

Qualifications

Training Program

Written Assessment Of Friable Asbestos Disturbance

SD-06 Test Reports

Solid Waste Management Report; G[, [____]]

Nonhazardous Solid Waste Diversion Report

Disposal Requirements; G

SD-07 Certificates

Certificate of Competency

Asbestos Certification

Lead Certification

SD-11 Closeout Submittals

Disposal Documentation for Hazardous and Regulated Waste; G[, [____]](Disposal documentation includes certificates of landfill facility, treatment facility, and transportation of hazardous/regulated waste, and completed signed hazardous/regulated waste manifest.)

[Assembled Employee Training Records; G[, [____]]]

Solid Waste Management Report; G[, [____]]

Hazardous Waste/Debris Management; G[, [____]]

Environmental Records Binder

1.5 PAYMENT

No separate payment shall be made for work covered under this section. Payment of fees associated with environmental permits, application, and/or notices obtained by the Contractor, and payment of all fines/fees for violation or non-compliance with GOJ, Federal, and local laws and regulations are the Contractor's responsibility. All costs associated with this section shall be included in the Contract price.

1.6 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the Contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements and management plans pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

Tests and procedures assessing whether construction operations comply with applicable Environmental Laws may be required. Laboratory analyses necessary to implement the JEGS shall be conducted in a laboratory certified by a U.S. or GOJ regulatory authority for the applicable test method, and where required by law, the laboratories shall be certified. In the absence of a certified laboratory, contact the Contracting Officer for further guidance.

Contractor shall be responsible to ensure that subcontractors comply with all environmental protection requirements of this section.

The Contractor shall record any problems in complying with laws, regulations, permit requirements, ordinances, and corrective actions taken. The Contractor shall immediately inform the Contracting Officer of any environmental problems.

1.6.1 Conformance with the Environmental Management System

Perform work under this Contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In the case of a noncompliance, the Contractor shall assume legal and financial liability for the noncompliance and immediately take corrective action and document the root cause. In the case of a nonconformance, the Contractor shall respond and take corrective action based on the time schedule established by the Contracting Officer. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities affect work performed under the Contract.

The Contractor shall perform work under this Contract consistent with the following EMS goals and policy.

1.6.1.1 Goals

1. Reduce purchase and use of toxic and hazardous materials.
2. Expand purchase of green products and services; increase recycling.
3. Reduce energy and water use.
4. Increase use of alternative fuels and renewable energy.
5. Integrate green building concepts in major renovations and new construction.
6. Prevent pollution at the source.
7. Continual improvement.

1.6.1.2 Policy

1. Protect public health and the environment by being an environmentally responsible member of Sasebo's community.
2. Preserve natural, historic and cultural resources.
3. Conserve natural resources by reducing what we discard, reusing items, and recycling materials, which includes purchasing products made from recycled materials.
4. Integrate sound environmental practices into all operations and business decisions; Integrate environmental protection requirements and pollution prevention initiatives into the early planning, design and procurement of facilities, equipment and

material.

5. Prevent or minimize pollution at its source and seek out ways to eliminate or further minimize use of hazardous materials and generation of hazardous waste.

6. Maintain a sound partnership with regulatory agencies to sustain compliance with existing and new environmental laws and regulations.

7. Enhance our program as we develop and implement an Environmental Management System.

8. Adhere to this policy, remind one another to do so, and ensure that our entire community knows this is our policy by our actions as well as our words

Coordinate with the installation's EMS coordinator to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Office will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

The Contractor is responsible for ensuring that their employees receive applicable environmental and occupational health and safety training and remains current on regulatory required specific training for the type of work to be conducted onsite. All on-site Contractor personnel, and their subcontractor personnel, performing tasks that have the potential to cause a significant environmental impact shall be competent on the basis of appropriate education, training or experience.

The Contractor shall provide a list of employees who are expected to perform work or services on CFAS property to the Contracting Officer before the commencement of construction. Provide a copy of the written Training Program to be implemented on this project, including, but not limited to project site-specific training material as indicated in 29 CFR 1926.1101 and comply with applicable laws and regulations of Government of the United States and Japan, however denominated including those applicable political subdivisions, departments, and other entities that will be used to train on-site employees (if required).

1.7 SPECIAL ENVIRONMENTAL REQUIREMENTS

Comply with the special environmental requirements listed here [_____] and attached at the end of this section.

1.7.1 Asbestos Prohibition and Certification

- a. Materials or products containing more than one-tenth of one percent (0.1 percent) by total weight, of the material or product, of asbestos shall not be used in this project. The Contracting Officer, at any time prior to acceptance of the work, or during the period designated for warranty of the work, if any, may reject materials and products that contain asbestos in excess of one-tenth of one percent by weight, and direct the removal of such materials and products from the job site, at the sole expense of the Contractor, and without additional

time granted for performance of the work. After completion of this Contract, if asbestos (exceeding 0.1 percent by weight) is discovered in the products or materials (excluding items permitted by the exception) installed by the Contractor, the Government reserves the right to direct the Contractor to perform asbestos abatement and restoration work, as required, at the Contractor's sole cost. Asbestos abatement work (removal and disposal of asbestos-containing materials and products) shall be accomplished in accordance with currently applicable Government standards for such work.

Exception: Where suitable asbestos-free substitutes do not exist for a material or product, the Contractor may use a material or product containing asbestos in the excess of 0.1 percent by weight, with prior written approval of the Contracting Officer. Submit a written request for substitution, accompanied by a certification from the manufacturer of the material or product that shall set forth, in specific detail, the amount of asbestos present in the material or product. When available, laboratory analysis of the material or product for asbestos content shall be included with the submittal.

- b. The Government may conduct asbestos testing on suspected asbestos-containing materials and products excluding items permitted by the "exception", and such testing shall be conducted at the expense of the Government. However, wherever destructive testing is required, or a material or product must be utilized by the Government for testing, the Contractor shall, at its own expense, repair or replace the material or product, or the item of work that has been disturbed by testing, if the results confirm presence of asbestos exceeding 0.1 percent by weight. In the event test results indicate 0.1 percent or less asbestos content or complete absence of asbestos, the Contractor shall restore the test site to its original condition and the cost of restoration work, as approved by the Contracting Officer, shall be borne by the Government.
- c. As a minimum, furnish manufacturer's certification for the items listed below, excluding items permitted by the "exception", certifying that the items are asbestos-free and do not contain asbestos in excess of 0.1 percent by weight, as applicable. However, when presence of asbestos is suspected in other products and materials used in this project, the Contractor shall be required to provide such certification for those additional items when so directed by the Contracting Officer. Asbestos certification shall be required for the items applicable to this project only.

- (1) Vinyl sheet/vinyl tile flooring, including accessories and adhesives.
- (2) Insulation materials, including facing.
- (3) Gaskets for piping and duct work.
- (4) Acoustical tiles.
- (5) Firestopping materials.
- (6) Fireproofing materials.
- (7) Special coating, including factory applied coatings, on sheet metal roofing and siding.

- (8) Wallboard for all interior and exterior applications, including joint compounds.
- (9) Adhesives (other than item 1) used in the project.
- (10) Tape materials used in the project.
- (11) Roofing and siding, nonmetallic.
- (12) Felt materials and cushion materials.
- (13) Pre-mixed mortars, grouts, leveling compounds, fillers, and other cementitious materials.
- (14) Caulking and sealing materials.

- d. All submittals shall be accompanied by a certification from the manufacturer of the material or product that the material or product is asbestos-free; or shall set forth, in specific detail, the amount of asbestos present in the material or product. Documentary evidence of laboratory analysis of the material or product for asbestos content, conducted by a qualified independent testing laboratory, shall be included with the submittal.

1.7.2 Lead Prohibition and Certification

- a. Paint or product coating containing more than 0.009 percent) by total weight of lead shall not be used in this project. The Contracting Officer, at any time prior to acceptance of the work, or during the period designated for warranty of the work, if any, may reject materials and products that contain lead in excess of 0.009 percent by weight, and direct the removal of such materials and products from the job site, at the sole expense of the Contractor, and without additional time granted for performance of the work. After completion of this Contract, if lead (exceeding 0.009 percent by weight) is discovered in the products or materials (excluding items permitted by the exception) installed by the Contractor, the Government reserves the right to direct the Contractor to perform lead abatement and restoration work, as required, at the Contractor's sole cost. Lead abatement work (removal and disposal of lead-containing materials and products) shall be accomplished in accordance with currently applicable Government standards for such work.

Exception: Where suitable lead-free substitutes do not exist for a paint or product coating, the Contractor may use a material or product containing lead in the excess of 0.009 percent by weight, with prior written approval of the Contracting Officer. Submit a written request for substitution, accompanied by a certification from the manufacturer of the material or product that shall set forth, in specific detail, the amount of lead present in the material or product. When available, laboratory analysis of the material or product for lead content shall be included with the submittal.

- b. The Government may conduct lead testing on suspected lead-containing materials and products excluding items permitted by the "exception", and such testing shall be conducted at the expense of the Government. However, wherever destructive testing is required, or a material or product must be utilized by the Government for testing, the Contractor

shall, at its own expense, repair or replace the material or product, or the item of work that has been disturbed by testing, if the results confirm presence of lead exceeding 0.009 percent by weight. In the event test results indicate 0.009 percent or less lead content or complete absence of lead, the Contractor shall restore the test site to its original condition and the cost of restoration work, as approved by the Contracting Officer, shall be borne by the Government.

- c. As a minimum, furnish manufacturer's certification for the items listed below, excluding items permitted by the "exception", certifying that the items are lead-free and do not contain lead in excess of 0.009 percent by weight, as applicable. However, when presence of lead is suspected in other products and materials used in this project, the Contractor shall be required to provide such certification for those additional items when so directed by the Contracting Officer. Lead certification shall be required for the items applicable to this project only.

1. Paints and Coatings

2. Any product or material with a factory applied coating

1.7.3 Class I and Class II Ozone Depleting Chemicals (ODC) or Substances (ODS)

Class I and II Ozone Depleting Substances listed in the JEGS are prohibited from being used. Contractor must provide certifications that materials utilized do not contain Class I and Class II ODC/ODS.

1.7.4 Polychlorinated Biphenyls (PCB)

Materials (ballasts, capacitors, transformers, dielectric fluid, switches, etc.) that contain PCBs are prohibited. Contractor must provide certifications that materials utilized do not contain PCBs, in accordance with the JEGS.

1.7.5 Hazardous Material Survey

[The Contractor shall review the [title of survey] Survey by [Company that performed Survey] dated [date of survey], attached at the end of this section, to familiarize themselves with the materials that have been sampled and tested for this project. The Contractor shall utilize the information contained within the report to develop their work and compliance plans with regards to Hazardous Materials. During construction, should potentially hazardous material be discovered, which has not been previously tested, the Contractor shall take action to assure that the untested material is not disturbed and contact the Contracting Officer. The quantities provided in the report represents a rough order of magnitude estimate of materials and shall not be used for bidding purposes as conditions may have changed from the date of the survey and the present existence of the material(s).]

[[No existing Hazardous Material Survey for the material to be disturbed under this project, exists. The Contractor shall assume that all potential ACM is ACM and all paints and coatings contain lead. A Hazardous Material Survey is not required for this project.] [The Contractor shall perform a Hazardous Material Survey prior to initiating any action that could disturb suspect Hazardous Materials as required by this Contract. The

Contractor shall notify the Contracting Officer upon finding any material that will be disturbed under this Contract and is suspected to contain Hazardous Materials. The Contracting Officer will handle the identified materials under, Unexpected Discovery of Hazardous Materials.]]

1.7.6 Cultural/Historic Resources

The Contractor must receive approval from the Contracting Officer prior to working in or near historic buildings and for any excavation work. The Contractor is responsible for fully complying with the JEGS and the CFAS ICRMP which includes both prior work approval and stoppage of work if Cultural Resource (CR) artifacts are discovered. The Contracting Officer must be notified immediately in the case of the discovery of undocumented CR and the location must be secured and protected until investigation by the CFAS Environmental Office.

Prior to any construction, repair, or maintenance work affecting the ground and/or any building or structure, the proposed project must be reviewed by the Installation Cultural Resources Manager (CRM) to determine whether it is located within a culturally sensitive (high probability) area or the building or structure is a cultural property.

If a potential impact to cultural resources exists, CFAS Environmental Office will consult with the Host Nation CR representatives and provide recommendations for minimizing any negative impacts.

1.7.7 Natural Resources

Blasting or use of explosives are not permitted without written permission from the Contracting Officer. Construction activities must be kept under surveillance and control to minimize environment damage by noise. Comply with the provisions of any applicable host nation regulations.

1.7.8 Water Quality Protection

1.7.8.1 Water Supply Systems

Contractors are required to take Drinking Water for Contractors- Japan available through an on-line ECATTS course. Provide proof of training to the Contracting Officer prior to work commencement. File course certificates in Environmental Manager's Binder.

New pipelines and pipelines that will be repaired shall be performed in accordance with AWWA C651 procedures and the JEGS for disinfection of the potable water system.

1.7.8.2 Cross Connection and Backflow Prevention

Personnel or Contractors who use hose bibbs shall keep on hand hose bibb vacuum breakers to connect to hose bibbs that do not already have a backflow preventer installed. The personnel or Contractor shall also ensure that an air gap is maintained between the hose and water surface. When use of the hose bibb is finished, the backflow preventer may be disconnected and put aside for subsequent use on hose bibbs that do not have a backflow preventer.

Construction projects are to be designed without cross-connections. The design must provide adequate backflow protection through the use of approved backflow preventers (BFPs). BFP selection should be based on the

Degree of Hazard associated with the cross-connection. All newly installed BFPs will be tested and certified before being placed into service.

Consult with CFAS PWD Facilities Engineering or Environmental Division to determine if cross connections and backflow protection exist for construction projects based on cross connection surveys and corrective action records.

Document corrective actions taken to correct breaches of cross connection and backflow prevention criteria and report these corrective actions to the Contracting Officer and CFAS PWD Environmental Division within 10 working days after the corrective actions are made.

Installing temporary backflow devices (i.e. check valves) may be required when conducting necessary hydrostatic tests of equipment or piping.

1.7.9 Waste Management

Identify construction activities that will generate hazardous waste or debris. Provide a documented waste determination for resultant waste streams. Identify, label, handle, store, and dispose of hazardous waste or debris in accordance with CFA Sasebo Hazardous Waste Management Plan.

Manage hazardous waste in accordance with the approved Hazardous Waste Management Section of the EPP. Store hazardous wastes in approved containers in accordance with CFA Sasebo Environmental Division Guidelines. Hazardous waste generated within the confines of Government facilities is identified as being generated by the Government. Prior to removal of any hazardous waste from Government property, removal must be coordinated with the installation Environmental Office. Do not bring hazardous waste onto Government property. Provide the Contracting Officer with a copy of waste determination documentation for any solid waste streams that have any potential to be hazardous waste or contain any chemical constituents listed in Japanese Environmental Government Regulations.

1.8 QUALITY ASSURANCE

1.8.1 Regulatory Notifications

Provide regulatory notification requirements in accordance with GOJ national and prefectural laws and regulations and installation requirements.. Include copies of regulatory notifications in the approved EPP prior to commencement of work activities translated to English.

1.8.2 Environmental Brief

Attend an environmental brief to be included in the preconstruction meeting. Provide the following information: types, quantities, and use of hazardous materials that will be brought onto the installation; and types and quantities of wastes/wastewater that may be generated during the Contract. Discuss the results of the Preconstruction Survey at this time.

Prior to initiating any work on site, meet with the Contracting Officer and installation Environmental Office to discuss the proposed Environmental Protection Plan (EPP). Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural and cultural resources, required reports, required permits, permit requirements (such as mitigation measures), and other measures to be taken.

Permits, licenses, or other forms of official approvals are not required by DoD activities and installations. Permits, licenses, or other forms of official approvals may, however, be required under GOJ law for certain contracted activities. When required, all such permits, licenses and other forms of official approval shall be obtained by the Contractor from the appropriate GOJ authorities and included in the Environmental Protection Plan. DoD Components shall assist Contractors when they are applying for a required permit, license or other form of official approval by providing necessary information only.

1.8.3 Environmental Manager

Appoint in writing an Environmental Manager for the project site. The Environmental Manager is directly responsible for Environmental Quality Control and coordinating Contractor compliance with the JEGS, Federal, GOJ national and prefectural laws and regulations, and installation requirements. The Environmental Manager must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, manifesting, and disposal); implement the EPP; ensure environmental permits are obtained, maintained, and closed out; ensure compliance with Stormwater Program requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; and coordinate any remediation of regulated substances (lead, asbestos, PCB transformers). This can be a collateral position; however, the person in this position must be trained to adequately accomplish the following duties: ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure Contractor personnel are trained in JEGS requirements in accordance with their position requirements; coordinate removal of waste containers; and maintain the Environmental Records binder and required documentation, including environmental permits compliance and close-out. Insert Environmental Manager Qualifications into the Environmental Records Binder to indicate the training and past experience which meets the requirements of this position as described in this section.

Contractor must also provide a written report providing evidence of Qualifications for personnel, facilities, and equipment assigned to the work.

1.8.4 Employee Training Records

Prepare and maintain Employee Training Records throughout the term of the Contract meeting applicable JEGS requirements. Provide Employee Training Records in the Environmental Records Binder. Ensure every employee completes a program of classroom instruction or on-the-job training that teaches them to perform their duties in a way that ensures compliance with the JEGS and all applicable Federal, GOJ national or prefectural laws or regulations and installation requirements. All Contractor personnel and their supervisors who are assigned duties involving actual or potential exposure to hazardous waste must successfully complete an appropriate training program prior to assuming those duties and must successfully complete all training as stipulated in the JEGS prior to commencement of project work. [Submit these Assembled Employee Training Records to the Contracting Officer at the conclusion of the project, unless otherwise directed.]

Train personnel to meet the JEGS and all applicable Federal, GOJ national

or prefectural laws or regulations, and Installation requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of Japan, and endangered species and their habitat that are known to be in the area.

[1.8.4.1 Pest Control Training

Trained personnel in pest control. Conduct a pest control meeting for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting pest infestation; familiarization with statutory and contractual pest control standards; installation and care of devices, and instruments, if required, for monitoring purposes to ensure adequate and continuous pest control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of waters of the United States, and endangered species and their habitat that are known to be in the area. Provide a Certificate of Competency for the personnel who will be conducting the pesticide application and management of pest control.

]1.8.5 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with the JEGS, applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements, and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions will be granted or equitable adjustments allowed for any such suspensions. This is in addition to any other actions the Contracting Officer may take under the Contract, or in accordance with the Federal Acquisition Regulation or Federal Law.

1.9 ENVIRONMENTAL PROTECTION PLAN

The purpose of the EPP is to present an overview of known or potential environmental issues that must be considered and addressed during construction. Incorporate construction related objectives and targets from the installation's EMS into the EPP. Include in the EPP measures for protecting natural and cultural resources, required reports, and other measures to be taken. Meet with the Contracting Officer or Contracting Officer Representative to discuss the EPP and develop a mutual understanding relative to the details for environmental protection including measures for protecting natural resources, required reports, and other measures to be taken. Submit the EPP within 15 days after receiving notice to proceed. The format of the EPP shall follow the CFAY Environmental template provided as Attachment 01 57 19-A. Commencement of work will not begin until the Environmental Protection Plan has been approved.

Revise the EPP throughout the project to include any reporting requirements, changes in site conditions, or Contract modifications that change the project scope of work in a way that could have an environmental impact. No requirement in this section will relieve the Contractor of the JEGS, and applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. During Construction, identify, implement, and submit for approval any additional requirements to be included in the EPP. Maintain the current version of the EPP onsite. In addition, the Contractor shall complete a CFAY Environmental Brief Checklist for Contractor and Sub-contractor(s) form and Roster Sheet (Attachment 01 57 19-B). The checklist and roster sheet shall be included as an appendix to the EPP submittal.

The EPP includes, but is not limited to, the following elements as they apply to this project:

1.9.1 General Overview and Purpose

1.9.1.1 Descriptions

A brief description of each specific plan required by environmental permit or elsewhere in this Contract such as[stormwater pollution prevention plan,][spill control plan,][solid waste management plan,][wastewater management plan,][air pollution control plan,][contaminant prevention plan,][pesticide treatment plan,][a historical, archaeological, cultural resources, biological resources and wetlands plan,][traffic control plan][Hazardous, Toxic and Radioactive Waste (HTRW) Plan][Non-Hazardous Solid Waste Disposal Plan][borrowing material plan][_____].

Include a list of applicable Federal, GOJ, JEGS, prefectural laws, regulations, and permits concerning environmental protection, pollution control, and abatement that are applicable to the Contractor's proposed operations and requirements imposed by those laws, regulations, and permits. Whenever there is conflict between Federal, GOJ, JEGS, or prefectural laws, regulations, and permit requirements, the strictest applicable rule applies.

List what notifications, permit applications and licenses must be made or secured. Some permits may take a considerable amount of time to obtain. Demonstrate that those permits have been obtained or applied for by including copies of applicable environmental permits. The EPP will not be approved until the permits have been obtained.

1.9.1.2 Duties

The duties and level of authority assigned to the person(s) on the job site who oversee environmental compliance, such as who is responsible for adherence to the EPP, who is responsible for spill cleanup and training personnel on spill response procedures, who is responsible for manifesting hazardous waste to be removed from the site (if applicable), and who is responsible for training the Contractor's environmental protection personnel.

Provide the name, telephone number, and address of a Primary and Alternate Environmental Representative.

1.9.1.3 Procedures

A copy of any standard or project-specific operating procedures that will be used to effectively manage and protect the environment on the project site.

Provide schedule of digging or trenching actions to include date/time, location, purpose, method and depth.

1.9.1.4 Communications

Communication and training procedures that will be used to convey environmental management requirements to Contractor employees and subcontractors.

1.9.1.5 Contact Information

Emergency contact information contact information (office phone number, cell phone number, and e-mail address).

1.9.2 General Site Information

1.9.2.1 Drawings

Drawings showing locations of proposed temporary excavations or embankments for haul roads, stream crossings, jurisdictional wetlands, material storage areas, structures, sanitary facilities, storm drains and conveyances, and stockpiles of excess soil.

1.9.2.2 Work Area

Work area plan showing the proposed activity in each portion of the area and identify the areas of limited use or nonuse. Include measures for marking the limits of use areas, including methods for protection of features to be preserved within authorized work areas and methods to control runoff and to contain materials on site, and a traffic control plan.

1.9.2.3 Documentation

A letter signed by an officer of the firm appointing the Environmental Manager and stating that person is responsible for managing and implementing the Environmental Program as described in this Contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work.

1.9.3 Management of Natural Resources

- a. Land resources
- b. Tree protection
- c. Replacement of damaged landscape features
- d. Temporary construction
- e. Stream crossings
- f. Fish and wildlife resources

- g. Wetland areas
- h. Waters of Japan

1.9.4 Protection of Historical and Archaeological Resources

- a. Objectives
- b. Methods

1.9.5 Stormwater Management and Control

- a. Ground cover
- b. Erodible soils
- c. Temporary measures
 - (1) Structural Practices
 - (2) Temporary and permanent stabilization
- d. Effective selection, implementation and maintenance of Best Management Practices (BMPs).
- e. Erosion Control Plan. Prepared in accordance with the JEGS.
- f. Storm Water Pollution Prevention Plan. Prepared in accordance with paragraph Stormwater Pollution Prevention Plan and the JEGS (if required)

1.9.6 Hazardous Material List

Prepared in accordance with paragraph Hazardous Material and the JEGS (if required). Safety Data Sheets (SDS): SDS in English for all Hazardous materials to be used in accordance with the JEGS (if required).

List shall include all pesticides and herbicides that will be brought onto the Installation.

1.9.7 Protection of the Environment from Waste Derived from Contractor Operations

Control and disposal of solid and sanitary waste. Control and disposal of hazardous waste.

Include a Hazardous Waste Management and Disposal Plan prepared in accordance with paragraph Control and Management of Hazardous Waste and the JEGS (if required). This item consists of the management procedures for hazardous waste to be generated. The elements of those procedures will coincide with the Installation Hazardous Waste Management Plan. The Contracting Officer will provide a copy of the Installation Hazardous Waste Management Plan. As a minimum, include the following:

- a. List of the types of hazardous wastes expected to be generated
- b. Procedures to ensure a written waste determination is made for appropriate wastes that are to be generated

- c. Sampling/analysis plan, including laboratory method(s) that will be used for waste determinations and copies of relevant laboratory certifications
- d. Methods and proposed locations for hazardous waste accumulation/storage (that is, in tanks or containers)
- e. Management procedures for storage, labeling, transportation, and disposal of waste (treatment of waste is not allowed unless specifically noted)
- f. Management procedures and regulatory documentation ensuring disposal of hazardous waste complies with Land Disposal Restrictions, applicable Federal, GOJ and prefectural laws and regulations, JEGS, and Installation Hazardous Waste Management Plan. .
- g. Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar
- h. Used oil management procedures in accordance with the JEGS; Hazardous waste minimization procedures, and Installation Hazardous Waste Management Plan.
- i. Plans for the disposal of hazardous waste by permitted facilities; and Procedures to be employed to ensure required employee training records are maintained.
- j. Procedures to be employed to ensure required employee training records are maintained.
- k. Hazardous Waste Disposal local permits or licenses for hazardous waste disposal in accordance with the JEGS (if required).
- l. Hazardous Waste Disposal Statement of Agreement. From a treatment, storage, or disposal (TSD) facility that will accept the waste from the Contractor and also a statement from a certified hazardous waste transporter who will transport the waste to the TSD facility in accordance with the JEGS (if required).

1.9.8 Solid Waste Management Plan

Prepared in accordance with the Waste Management Plan requirements in paragraph Section 01 74 19 Solid Waste Management Plan and the JEGS.

1.9.9 Site Specific Spill Prevention Plan

Prepared in accordance with paragraph SPILLS and the JEGS (if required).

1.9.10 Prevention of Releases to the Environment

- a. Procedures to prevent releases to the environment
- b. Notifications in the event of a release to the environment

1.9.11 Regulatory Notification and Permits

List what notifications and permit applications must be made. Some permits require up to 180 days to obtain. Demonstrate that those permits

have been obtained or applied for by including copies of applicable environmental permits. The EPP will not be approved until the permits have been obtained.

1.9.12 Air Quality Compliance

1.9.12.1 Haul Route

Include the truck and material haul routes along with a Dirt and Dust Control Plan for controlling dirt, debris, and dust on Installation roadways as a part of the EPP. As a minimum, identify in the plan the subcontractor and equipment for cleaning along the haul route and measures to reduce dirt, dust, and debris from roadways.

1.9.12.2 Pollution Generating Equipment

Identify air pollution generating equipment or processes that may generate air emissions and require compliance with the JEGS or GOJ national or prefectural laws and regulations. Determine requirements based on any current JEGS and installation requirements and the impacts of the project. Provide a list of all fixed or mobile equipment, machinery or operations that could generate air emissions during the project to the Installation Environmental Office (Air Program Manager) in the EPP. Air emissions must comply with the JEGS.

1.9.12.3 Stationary Internal Combustion Engines

Identify portable and stationary internal combustion engines that will be supplied, used or serviced in the EPP. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. At minimum, include the make, model, serial number, manufacture date, size (engine brake horsepower), and EPA emission certification status of each engine. Maintain applicable records and log hours of operation and fuel use. Logs must include reasons for operation and delineate between emergency and non-emergency operation.

1.9.12.4 Refrigerants

Identify management practices to ensure that heating, ventilation, and air conditioning (HVAC) work involving refrigerants and ozone depleting substances comply with the JEGS and installation requirements. Technicians [must be certified, maintain copies of certification on site, use certified equipment and log work that requires the addition or removal of refrigerant][shall be trained in proper recovery/recycling procedures, leak detection, safety, shipping, and disposal in accordance with recognized industry standards or Japanese equivalent]. Any refrigerant reclaimed is the property of the Government, coordinate with the Installation Environmental Office to determine the appropriate turn in location.

1.9.12.5 Air Pollution-engineering Processes

Identify planned air pollution-generating processes and management control measures (including, but not limited to, spray painting, abrasive blasting, demolition, material handling, fugitive dust, and fugitive emissions). Log hours of operations and track quantities of materials used.

1.9.12.6 Monitoring

For the protection of public health, monitor and control contaminant emissions to the air from Hazardous, Toxic, and Radioactive Waste remedial action area sources to minimize short-term risks that might be posed to the community during implementation of the remedial alternative in accordance with the following.

- a. Perimeter Air Contaminant of Concern [____].
- b. Time Averaged Perimeter Action Levels [____].

Concentration	[____]
Time	[____]

- c. Perimeter Sampling/Monitoring Location[s] [____].
- d. Monitoring Instruments/Sampling and Analysis Methods [____].
- e. Staffing [____].

1.9.12.7 Compliant Materials

Provide the Government a list of and SDSs for all hazardous materials proposed for use on site. Materials must be compliant with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements for emissions including solvent and volatile organic compound contents, and applicable National Emission Standards for Hazardous Air Pollutants requirements and either already be included on the CFAS approved Authorized Use List (AUL) or be approved for inclusion on the AUL prior to the project start. The Government may alter or limit use of specific materials as needed to meet installation permit requirements for emissions.

1.10 LICENSES AND PERMITS

Obtain licenses and permits required for the construction of the project and in accordance with FAR 52.236-7. Notify the Government of all general use permitted equipment the Contractor plans to use on site. This paragraph supplements the Contractor's responsibility under FAR 52.236-7. A copy of all Licenses and Permits shall be included in the Environmental Protection Plan.

[a. The following permits have been obtained by the Government:

[(1) [____]

][(2) [____]

][(3) [____]

]]

[b. The following permits will be obtained by the Government:

[(1) [____]

][(2) [____]

][(3) [____]

]]1.11 ENVIRONMENTAL RECORDS BINDER

Maintain on-site a separate three-ring Environmental Records Binder and submit at the completion of the project. Make separate parts within the binder that correspond to each submittal listed under paragraph CLOSEOUT SUBMITTALS in this section.

[1.12 PESTICIDE DELIVERY, STORAGE, AND HANDLING

1.12.1 Delivery and Storage

Deliver pesticides to the site in the original, unopened containers bearing legible labels indicating the EPA registration number and the manufacturer's registered uses. Store pesticides according to manufacturer's instructions and under lock and key when unattended.

1.12.2 Handling Requirements

Formulate, treat with, and dispose of pesticides and associated containers in accordance with label directions and use the clothing and personal protective equipment specified on the labeling for use during each phases of the application. Furnish SDSs for pesticide products.

]1.13 SOLID WASTE MANAGEMENT PERMIT

Include as a part of the Environmental Protection Plan (EPP) written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Include in the report the locations where various types of waste will be disposed or recycled. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location and local Solid Waste Management Permit or license showing such agency's approval of the disposal plan before transporting wastes off Government property.

1.13.1 Solid Waste Management Report

Monthly, submit a solid waste disposal report to the Contracting Officer. For each waste, the report will state the classification (using the definitions provided in this section), amount, location, and name of the business receiving the solid waste.

The Contractor shall submit all waste disposal manifests. For each solid waste retained by the Contractor for his own use, the Contractor will include in the solid waste disposal report the information previously described in this paragraph. Prices paid or received will not be reported to the Contracting Officer unless required by other provisions or specifications of this Contract or public law.

PART 2 PRODUCTS

2.1 GREEN PROCUREMENT

The Contractor is required to buy items using recovered, or bio-based, materials listed at the following website:
<http://www.epa.gov/epp/pubs/products/index.htm> provided, however, the cost of buying and transporting these materials does not exceed the cost of buying the listed recovered, or bio-based, materials, and providing that

these materials are competitively available and available within reasonable period. A reasonable period is the time required to buy, ship, receive, and use the materials that does not exceed the time allowed for completion of a delivery order. Recovered and bio-based materials include items listed by the EPA in the following categories:

- a. Bio-based.
- b. Construction Products.
- c. Non-Paper Office Products.
- d. Miscellaneous Products.
- e. Transportation Products.
- f. Vehicular Products.
- g. Park and Recreation Products.
- h. Landscape Products.
- i. Paper and Paper Products.

The Contractor shall provide a Recovered Materials Certification and estimate of Percentage of Recovered Material Content for EPA designated items. 2.1.1 Recycled Paper The Contractor shall use, as a minimum, paper that is Processed Chlorine Free (PCF) and contains at least 30 percent recycled content. All printed or copied products shall be double-sided and on recycled paper.

2.1.2 Energy Star and Energy Efficient Products

All applicable products delivered, used in the performance of the work, furnished for Government use, or specified in the design of a building or work shall be Energy Star and Energy Efficient Products.

2.1.3 Electronic Product Environmental Assessment Tool (EPEAT)

Purchased computers, notebook and monitors shall, as a minimum, meet the Electronic Product Environmental Assessment Tool (EPEAT) Bronze level of certification.

2.1.4 Leadership in Energy and Environmental Design (LEED) All new construction Contracts or those performing major renovations shall incorporate the Leadership in Energy and Environmental Design (LEED) System guidelines with Silver being the goal, or shall use Comprehensive Assessment System for Building Environmental Efficiency (CASBEE).

PART 3 EXECUTION

3.1 PROTECTION OF NATURAL RESOURCES

Minimize interference with, disturbance to, and damage to fish, wildlife, and plants, including their habitats. Prior to the commencement of activities, consult with the Installation Environmental Office, regarding rare species or sensitive habitats that need to be protected. The protection of rare, threatened, and endangered animal and plant species identified, including their habitats, is the Contractor's

responsibility.[The following species are known and could be affected within the construction area: [____].]

Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work that is consistent with the requirements of the Installation Environmental Office or as otherwise specified. Confine construction activities to within the limits of the work indicated or specified.

The Contractor shall take precautions to preserve all such resources as they existed at the time they were first pointed out. The Contractor shall provide and install protection for these resources and be responsible for their preservation during the life of the Contract. Environmental protection shall be conducted as follows:

- a. Except in areas indicated on the drawing or specified to be cleared, the Contractor shall not remove, cut, deface, injure, or destroy land resources, including trees, shrubs, vines, grasses, topsoil, and land forms without the Contracting Officer's permission. Vegetation resources, land forms and other landscape features indicated and defined on the drawings to be preserved shall be clearly identified by marking, fencing, or wrapping with boards, or any other approved techniques. Any anticipated vegetation disturbance needs to be coordinated with the Government before it occurs.

3.1.1 Flow Ways

Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as specified and permitted.

3.1.2 Vegetation

Except in areas to be cleared, do not remove, cut, deface, injure, or destroy trees or shrubs without the Contracting Officer's permission. Do not fasten or attach ropes, cables, or guys to existing nearby trees for anchorages unless authorized by the Contracting Officer. Where such use of attached ropes, cables, or guys is authorized, the Contractor is responsible for any resultant damage.

Protect existing trees that are to remain to ensure they are not injured, bruised, defaced, or otherwise damaged by construction operations. Remove displaced rocks from uncleared areas. By approved excavation, remove trees with 30 percent or more of their root systems destroyed. Coordinate with the Contracting Officer and Installation Environmental Office to determine appropriate action for trees and other landscape features scarred or damaged by equipment operations. Obtain Contracting Officer's approval prior to replacement.

3.1.3 G0J-Protected Species

G0J-Protected species are typically found in undeveloped and unmaintained portions of the base. Projects that may affect protected species shall include mitigation measures to eliminate or minimize effects.

3.1.4 Streams

Stream crossings must allow movement of materials or equipment without

violating water pollution control standards of the JEGS and applicable, Federal, and GOJ national and prefectural laws and regulations, and installation requirements. Construction of stream crossing structures must be in compliance with the JEGS, applicable Federal and GOJ national and prefectural laws and regulations, and installation requirements, including the Storm Water Pollution Prevention Plan.

The Contracting Officer's approval and appropriate permits are required before any equipment will be permitted to ford live streams. In areas where frequent crossings are required, install temporary culverts or bridges. Obtain Contracting Officer's approval prior to installation. Remove temporary culverts or bridges upon completion of work, and repair the area to its original condition unless otherwise required by the Contracting Officer. A copy of all secured permits shall be included in the EPP.

3.1.5 Banyan Trees

All Banyan trees must be preserved and protected. Protect existing trees that are to remain and which may be injured, bruised, defaced, or otherwise damaged by construction operations. Remove displaced rocks from uncleared areas. Remove trees with 30 percent or more of their root systems destroyed. Any banyan tree requiring destruction or removal must first be coordinated with the installation environmental office or designated representative.

3.1.6 Endangered Species

At any time an endangered or threatened species, flora or fauna, to include sea turtle nests are encountered, all activities shall stop and the Contracting Officer shall be notified.

3.1.7 Indigenous/Native Flora and Fauna

The Contractor shall place emphasis on the protection of habitats favorable to the reproduction and survival of indigenous flora and fauna. The Contractor shall use indigenous flora for planting/sodding.

3.1.8 Invasive Species

Invasive Species are prohibited to be raised, planted, stored or possessed on DoD installations. The Contractor shall not bring in any invasive species to DoD installations.

3.2 STORMWATER

Obtain authorization in advance from the Installation Environmental Office for any release of contaminated water.

3.2.1 Construction Operations and Management

Maintain construction operations and management as required by the Installation Storm Water Pollution Prevention Plan, which addresses pollution prevention for storm water discharges from construction activities.

[3.2.2 Stormwater Pollution Prevention Plan

If applicable, submit a project-specific Stormwater Pollution Prevention

Plan (SWPPP) to the Contracting Officer as a part of the EPP, for approval, prior to the commencement of work. The SWPPP shall meet the requirements of the Japanese Laws, local government regulations and the Installation SWPPP for pollution prevention to storm water discharges from construction sites. The SWPPP along with any required Notice of Intents, Notice of Termination, and appropriate permit fees, shall be incorporated into the EPP. The Contractor shall maintain an approved copy of the SWPPP at the construction on-site office, and continually update as regulations require, to reflect current site conditions. The Contractor may be required to install, inspect, maintain best management practices (BMPs), and submit storm water BMPs inspection reports and storm water pollution prevention plan inspection reports.

The SWPPP must meet the requirements of the JEGS and the Installation's SWPPP.

Include the following:

- a. Identify potential sources of pollution which may be reasonably expected to affect the quality of storm water discharge from the site.
- b. Comply with terms of the Installation SWPPP for stormwater discharges from construction activities. Prepare SWPPP in accordance with the JEGS and installation requirements.
- c. Select applicable BMPs from the JEGS and/or the Installation SWPPP. Applicable Best Management Practices in the JEGS shall be incorporated into the site-specific SWPPP and implemented.
- d. The SWPPP shall also address erosion and sediment control measures and stormwater management and control including, but not limited to ground cover, erodible soils, temporary measures - structural practices, temporary and permanent stabilization.
- e. Describe and ensure implementation of practices which will be used to reduce the pollutants in storm water discharge from the site.
- f. Ensure compliance with terms of the Japanese Law and local government general permit for storm water discharge.
- g. Include a completed copy of the Registration Statement, BMP Inspection Report Template and Notice of Termination except for the effective date.

3.2.2.1 Inspection Reports

Insert "Inspection Reports" into the Environmental Records Binder in accordance with the JEGS and applicable GOJ or local laws and regulations.

]3.2.3 Erosion and Sediment Control Measures

Provide erosion and sediment control measures in accordance with the JEGS and applicable GOJ or local laws and regulations. Preserve vegetation to the maximum extent practicable.

Erosion control inspection reports may be compiled as part of a stormwater pollution prevention plan inspection reports. Insert "Erosion and Sediment Control Inspection Reports" (E&S) into the Environmental Records Binder once every seven (7) calendar days and within 24 hours of a storm

event that produces 12 mm or more of rain.

[3.2.3.1 Erosion Control

Prevent erosion by[mulching,][Compost Blankets,][Geotextiles,][temporary slope drains,][_____]. Stabilize slopes by[chemical stabilization,][sodding,][seeding,][_____] or such combination of these methods necessary for effective erosion control. Use of hay bales is prohibited.

[Provide seeding in accordance with Section 32 92 1932 92 19 SEEDING.]

Any disturbed area with exposed soil that is not being worked on must be seeded or sodded no later than two weeks from the last disturbance regardless of the installation of any other erosion control measures in place. Remove any temporary measures after the area has been stabilized. No "red soil" shall be allowed to leave the site through runoff.

]

3.2.3.2 Protection of Erodible Soils

Immediately finish the earthwork brought to a final grade, as indicated or specified. Immediately protect the side slopes and back slopes upon completion of rough grading. Plan and conduct earthwork to minimize the duration of exposure of unprotected soils.

3.2.3.3 Temporary Protection of Erodible Soils

Use the following methods to prevent erosion and control sedimentation:

a. Mechanical Retardation and Control of Runoff

- (1) Mechanically retard and control the rate of runoff from the construction site. This includes construction of diversion ditches, benches, berms, and use of silt fences and straw bales to retard and divert runoff to protected drainage courses.

b. Sediment Basins

- (1) Trap sediment in temporary sediment basins. Select a basin size to accommodate the runoff of a local 10-year storm ("kou no kakuritsu nen"). Pump dry and remove the accumulated sediment, after each storm. Use a paved weir or vertical overflow pipe for overflow. Remove collected sediment from the site. Institute effluent quality monitoring programs.
- (2) Install, inspect, and maintain best management practices (BMPs) as required by the general permit. Prepare BMP Inspection Reports as required by the general permit. If required by the permit, include those inspection reports.

c. Vegetation and Mulch

- (1) Provide temporary protection on sides and back slopes as soon as rough grading is completed or sufficient soil is exposed to require erosion protection. Protect slopes by accelerated growth of permanent vegetation, temporary vegetation, mulching, or netting. Stabilize slopes by hydroseeding, anchoring mulch in place, covering with anchored netting, sodding, or such combination of these and other methods necessary for effective

erosion control.

- (2) Seeding: Provide new seeding where ground is disturbed. Include topsoil or nutriment during the seeding operation necessary to reestablish a suitable stand of grass. The seeding operation will be as specified.

[3.2.3.4 Sediment Control Practices

Implement sediment control practices to divert flows from exposed soils, temporarily store flows, or otherwise limit runoff and the discharge of pollutants from exposed areas of the site. Implement sediment control practices prior to soil disturbance and prior to creating areas with concentrated flow, during the construction process to minimize erosion and sediment laden runoff. Best Management Practices (BMPs) may include, but not limited to, vegetation cover, stream bank stabilization, slope stabilization, silt fences, construction of terraces, interceptor channels, sediment traps, inlet and outfall protection, diversion channels, and sedimentation basins. [Location and details of installation and construction are as indicated on the drawings.] Include the following devices: [silt fence,] [temporary diversion dikes,] [storm drain inlet protection,] [_____,] [Location and details of installation and construction are indicated on the drawings.]

]3.2.4 Work Area Limits

Mark the areas that need not be disturbed under this Contract prior to commencing construction activities. Mark or fence isolated areas within the general work area that are not to be disturbed. Protect monuments and markers before construction operations commence. Where construction operations are to be conducted during darkness, any markers must be visible in the dark. Personnel must be knowledgeable of the purpose for marking and protecting particular objects.

3.2.5 Contractor Facilities and Work Areas

Place field offices, staging areas, stockpile storage, and temporary buildings in areas designated on the drawings or as directed by the Contracting Officer. Move or relocate the Contractor facilities only when approved by the Government. Provide erosion and sediment controls for onsite borrow and spoil areas to prevent sediment from entering nearby waters. Control temporary excavation and embankments for plant or work areas to protect adjacent areas.

3.2.6 Silt Screens

Silt screens must be installed prior to start of construction. Silt screens and/or other erosion control devices shall be installed on construction sites that are in or near water. Silt screens shall consist of trenched and staked filter fabric and trenched and staked hay bales. Filter fabric must be toed 8 inches into the soil to avoid sediments that would be transported via water under the screen. Hay bales must be placed end to end on the downstream side of the screen and be trenched and staked firmly into the ground. Chinking is usually required to fill gaps between the bales. Silt screens must be maintained properly. Screens and other control devices must be inspected once a week and after any rainfall event totaling 1.27 cm (0.5 in.) or more to ensure they are in good repair and functioning properly. In areas that experience high flow rates, extra precautions will be necessary to stabilize screens. Trenching of hay bale

barriers is required to adequately control runoff. A series of screens may have to be installed in waters that are especially turbid to properly filter out sediments. Silt screens will remain in place and properly maintained until the site is properly stabilized with sod or seeding.

3.2.7 Protection of Excavated Soil

During the temporary storing of excavated soil until off base disposal or backfill, the excavated soil/rocks shall be completely segregated with materials from other locations, and covered with vinyl sheets to prevent the soil from run-off by rain, and label indicates "Project Name", "Contract Number", "Contractor's Name", "POC phone number" shall be posted on the vinyl sheet.

3.3 SURFACE AND GROUNDWATER

3.3.1 Cofferdams, Diversions, and Dewatering

Construction operations for dewatering, removal of cofferdams, tailrace excavation, and tunnel closure must be constantly controlled to maintain compliance with existing GOJ, Prefectural, installation water quality standards and designated uses of the surface water body. Comply with water quality standards and anti-degradation provisions. Do not discharge excavation ground water to the sanitary sewer, storm drains, or to surface waters without prior specific authorization in writing from the Installation Environmental Office. Discharge of hazardous substances will not be permitted under any circumstances. Use sediment control BMPs to prevent construction site runoff from directly entering any storm drain or surface waters.

Construction site runoff will be prevented from entering any storm drain or the river directly by the use of silt fence, straw bales, or other method suitable to the Environmental Division. Contractor will provide erosion protection of the surrounding soils.

Construction Dewatering shall not be discharged to the sanitary sewer. If the construction dewatering is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is specifically permitted. Obtain authorization for any contaminated groundwater release in advance from the Installation Environmental Officer and the GOJ, Prefectural, installation, or local authority, as applicable. Discharge of hazardous substances will not be permitted under any circumstances.

Water pumped from the trenches shall be filtered using an appropriate dewatering devices, such as a weir tank or other suitable devices, before discharging into any storm drain/ditch. Filtration shall be capable of treating the water to meet the allowable limit of total suspended solids (TSS) as listed in the JEGS. Discharging and dumping of wastewaters and waste materials into storm drains, ditches, canals, and local bodies of water are forbidden; this includes mop water, cleaning/stripping solutions and solvents, paint related wastes (including water-borne paints), pesticides (insecticide, herbicide, and/or fungicide), herbicides, fire extinguishing agents, fuels, oils, hazardous materials, hazardous substances, and hazardous wastes.

3.3.2 Waters of Japan

Do not enter, disturb, destroy, or allow discharge of contaminants into

waters of Japan[.][except as authorized herein. The protection of waters of Japan shown on the drawings in accordance with paragraph LICENSES AND PERMITS is the Contractor's responsibility. Authorization to enter specific waters of Japan identified does not relieve the Contractor from any obligation to protect other waters of Japan within, adjacent to, or in the vicinity of the construction site and associated boundaries.]

3.3.3 Wetlands

All wetlands on the installation (currently only Harioshima Ordnance area) must be identified on the project drawings. If wetlands on the project site must be disturbed, coordinate with the CFAS Environmental Office before start of project. The Contractor is required to be cognizant of their responsibility to protect wetlands regardless of whether they are identified on drawings or in the event site conditions have changed since design (JEGS).

3.3.4 Turbidity

In some cases where severe erosion results in waters becoming turbid despite control measures, regular turbidity monitoring and documentation shall be necessary. Any such documentation shall be forwarded to USAG-O Environmental Division for review via the Contracting Officer.

3.3.5 Water Protection

Contractor shall prevent oily wastes or other hazardous substances from entering the ground, drainage areas, or local bodies of water.

3.3.6 Sewage

Contractor is not authorized sewage holding tanks on base and must procure a porta-potty service Contract. The porta-potty Contract must include the correct removal of sewage and maintenance of the facilities.

3.3.7 Wastewater

If the project generates wastewater from rinsing tanks, dewatering sites, etc., contact the Contracting Officer on proper disposal.

3.3.7.1 Wastewater Generated by concrete/asphalt cutting

Dispose wastewater generated from the concrete/asphalt cutting event as industrial waste. If wastewater generated from concrete/asphalt cutting activities will be discharged to the storm/sewer system, then the discharge must satisfy the following conditions in accordance with COMFLEACTINST 5090.13_Direct Discharge Standard_Revised_03022020.

- 1)pH must be between 6.0 - 9.0
- 2)TSS must be below 45
- 3)The concentration of Hexavalent Chromium must be below 0.5 mg/L

Notify UEM and PRY4 before discharging the wastewater into sewer line.
UEM: DSN: 243-5232 and 243-5701
PRY4:DSN: 243-9578

3.4 PROTECTION OF CULTURAL RESOURCES

3.4.1 Archaeological Resources

[Existing archaeological resources within the work area are shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract.]If, during excavation or other construction activities, any previously unidentified or unanticipated historical, archaeological, and cultural resources are discovered or found, activities that may damage or alter such resources will be suspended. Resources covered by this paragraph include, but are not limited to: any human skeletal remains or burials; artifacts; shell, midden, bone, charcoal, or other deposits; rock or coral alignments, pavings, wall, or other constructed features; and any indication of agricultural or other human activities. Upon such discovery or find, immediately notify the Installation Cultural Resources Manager via the Contracting Officer so that the appropriate authorities may be notified and a determination made as to their significance and what, if any, special disposition of the finds should be made. Cease all activities that may result in impact to or the destruction of these resources. Secure the area and prevent employees or other persons from trespassing on, removing, or otherwise disturbing such resources. The Government retains ownership and control over archaeological resources.

3.4.1.1 Restrictions

Project work may be restricted to be performed only during certain periods or times as specified.

3.4.1.2 Protection Measures

The Contractor shall implement and adhere to all protection measures specified to ensure protection of archaeologically sensitive areas and to avoid impact to known properties and artifacts as identified

3.4.1.3 Training and Awareness - Site Walk

When specified, the Contractor shall arrange to have site personnel receive appropriate awareness training to be provided by or under the guidance of the Environmental Office. In addition, a site walk may also be requested in order to ensure that the Contractor understands and is prepared to implement required protection measures.

[3.4.2 Historical Resources

Existing historical resources within the work area are shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract.]

3.4.3 Historical and Archaeological Resources

If the items need to be temporarily removed, receive approval from the Contracting Officer and PWD Environmental Cultural Resources Manager (CRM) then carefully protect them from disturbance, including weather conditions, unforeseen traffic, and pilfering. The CRM PWD Environmental Cultural Resources Manager through the Contracting Officer will provide further direction/guidance on how to protect the items. The Government retains ownership and control over historical and archaeological resources.

3.5 AIR RESOURCES

Equipment operation, activities, or processes will be in accordance with the JEGS and all applicable Federal, GOJ national or prefectural air emission and performance laws and regulations.

3.5.1 Oil or Dual-fuel Boilers and Furnaces

Provide product data and details for new, replacement, or relocated fuel fired boilers, heaters, or furnaces to the Installation Environmental Office (Air Program Manager) through the Contracting Officer. Data to be reported include: equipment purpose (water heater, building heat, process), manufacturer, model number, serial number, fuel type (oil type, gas type) size (MMBTU heat input). Product data shall be provided prior to equipment installation.

3.5.2 Burning

[Burning is prohibited on the Government premises.] [Burning is allowed on Government premises[; confine fires to a closed vessel that is guarded and under constant surveillance until contents have burned out or have been extinguished]. [Burning must completely reduce the materials to ashes.]]

3.5.3 Class I and II ODS Prohibition

The Contractor shall be responsible to turn in all of recyclable Class I and II ODS (R22) to DLA Yokosuka. The Contractor shall follow Feb 2017 Turn-in Procedures prepared by DLA, and coordinate with Torii Environmental Division prior to recovering ODS. The Contractor shall conduct necessary sampling, marking, labeling, and attach other required information. For replacement of air conditioning systems/units or in new construction, the Contractor shall provide quantity of refrigerant recovered from the existing equipment as well as the quantity of refrigerant charged in new equipment to the Contracting Officer. Under no circumstances shall the Contractor design or use ODS products for new systems to be installed on the installation.

Class I and II ODS are Government property and must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Office to determine the required recovery cylinders and the appropriate location for turn in of all reclaimed refrigerant. Class I and II ODS as defined and identified herein shall not be used in the performance of this Contract.

The Contractor shall coordinate with the [Refrigerant, AC and Heating Shop, (DSN 652-4302)][] on the quantity of ODS recovered from system for the purpose of recycle, reuse or dilution. In the event the ODS can be recycled, reused or diluted, the entire quantity will be turned over to the [Refrigerant, AC and Heating shop][].

In cases where the ODS is sufficiently contaminated and cannot be recycled, reused and/or diluted, the Contractor shall dispose of the ODS according to the JEGS and local regulations.

In the event of a release of ODS, base personnel and/or Contractor shall inform [Fire and Emergency's Service Branch and Environmental Division][].

3.5.3.1 Recycling Requirements

Recycle used refrigerants and ozone depleting substances generated during the performance of this Contract to the maximum extent practicable to minimize used refrigerant and ozone depleting substance disposal as regulated waste in accordance with the JEGS and local laws and regulations. Test, collect, transfer, recycle, and/or arrange for shipping and proper disposal of used refrigerants and ozone depleting substances generated during the performance of work under this Contract. The Contractor is responsible for all associated costs. Any and all Class I ODS and R-22 recovered by the Contractor as part of this Contract shall be packaged and turned in to the Government for recycling upon the completion of the work covered by this Contract. Turn-in of ODS shall be performed in accordance with "Department of Defense Ozone Depleting Substances Turn-In Procedures". The Contractor shall arrange for recycling of used refrigerants not turned in to the government, at a licensed refrigerant recycler approved by the Contracting Officer.

3.5.3.2 ODS Storage

The Contractor is prohibited from storing or accumulating cylinders/containers of ODS, refrigerant or fire suppression chemicals on all installations. The Contractor shall store or accumulate the ODS cylinders/containers on Contractor's premises.

3.5.4 Accidental Venting of Refrigerant

Accidental venting of a refrigerant is a release and must be reported immediately to the Contracting Officer. Comply with the JEGS.

3.5.5 Halon Fire Extinguishing Equipment

Remove fire extinguishing equipment without releasing Halons to the atmosphere.

3.5.6 Training/Certification Requirements

Heating and air conditioning technicians must be trained to meet requirements in the JEGS. Maintain copies of certifications at the employees' places of business; technicians must carry certification wallet cards, as provided by GOJ national or prefectural laws and regulations.

3.5.7 Dust Control

Keep dust down at all times, including during nonworking periods.[Sprinkle or treat, with dust suppressants, the soil at the site, haul roads, and other areas disturbed by operations.] Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Do not unnecessarily shake bags of cement, concrete mortar, or plaster. Do not discharge any construction waste water or surface cleaning water into the storm drain or on-base sewer line directly without authorization of NAVFAC Utilities Services (US) Branch and NAVFAC PWD Environmental Office via the Contracting Officer. Keep sediment leaves, and construction debris away from storm drains (use barriers).

3.5.7.1 Particulates

Dust particles, aerosols and gaseous by-products from construction activities, and processing and preparation of materials (such as from asphaltic batch plants) must be controlled at all times, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed the JEGS and applicable US Federal, GOJ national or prefectural air pollution standards or that would cause a hazard or a nuisance. Sprinkling, chemical treatment of an approved type, baghouse, scrubbers, electrostatic precipitators, or other methods will be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp. Provide sufficient, competent equipment available to accomplish these tasks. Perform particulate control as the work proceeds and whenever a particulate nuisance or hazard occurs. Comply with the JEGS, GOJ national or prefectural laws and regulations, and installation requirements.

3.5.7.2 Abrasive Blasting

Blasting operations cannot be performed without prior approval of the Installation Air Program Manager. The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive agent, paint chips, and other debris.[Perform work involving removal of hazardous material in accordance with the JEGS, GOJ national or prefectural laws and regulations, and installation requirements.]

3.5.7.2.1 Disposal Requirements

Submit analytical results of the debris generated from abrasive blasting operations per paragraph entitled Laboratory Analysis of this section. Hazardous waste generated from blasting operations will be managed in accordance with paragraph entitled "HAZARDOUS WASTE AND DEBRIS MANAGEMENT" of this section and with the approved HWMP. Disposal of non-hazardous abrasive blasting debris will be in accordance with paragraph entitled "CONTROL AND MANAGEMENT OF SOLID WASTES".

3.5.8 Odors

Control odors from construction activities. The odors must be in compliance with applicable GOJ national or prefectural laws and regulations and may not constitute a health hazard.

3.6 WATER RESOURCES

Keep construction activities under surveillance, management, and control, and monitor all water areas affected by construction activities in order to prevent pollution of surface and ground waters. Toxic or hazardous chemicals shall not be applied to soil or vegetation when such application may cause contamination. Prevent oily wastes or other hazardous substances from entering the ground, drainage areas, or local bodies of water.

3.6.1 Lead-Free Drinking Water Pipe, Solders, Flux and Fittings

The maximum allowable lead content for pipes, fittings, and fixtures intended to convey or dispense water for human consumption and cooking shall be a weighted average of 0.25 percent with respect to the wetted surfaces of pipes and pipe fittings, plumbing fittings, and fixtures in accordance with NSF 61.

3.7 WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of waste. Include procedures for pollution prevention/ hazardous waste minimization in the Hazardous Waste Management Section of the EPP. Obtain a copy of the installation's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EPP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in the construction when requesting information.

3.7.1 Salvage, Reuse and Recycle

Identify anticipated materials and waste for salvage, reuse, and recycling. Describe actions to promote material reuse, resale or recycling. The following items shall be recycled: scrap metal, Styrofoam, cardboard, glass, scrap wood, wood pallets, concrete, and asphalt. Other waste minimization strategies include composting or mulching green wastes. To the extent practicable, all scrap metal must be sent [for reuse or recycling][to the Installation's Recycling Yard][] and will not be disposed of in a landfill.

Include the name, physical address, and telephone number of the hauler, if transported by a franchised solid waste hauler. Include the destination and, unless exempted, provide a copy of the GOJ, Prefectural, installation or local permit (cover) or license for recycling.

All solid wastes and materials that have been separated for the purpose of recycling shall be stored in such a manner that they do not constitute a fire, health or safety hazard, or provide food or harborage for vectors, and shall be contained or bundled so as to not result in spillage. Containers must be: leak proof, water proof, and vermin proof including sides, seams, tops and bottoms, durable enough to withstand anticipated usage, and stored on a firm, level, well-drained surface).

The Contractor shall complete and submit COMFLEACTY 5090.8A (attachment 4 at the end of this specification section), and CFAY PWD Env Jun 2022 (attachment 5, at the end of this specification section) each time material is recycled and turned into the installation recycling center or each time solid waste is disposed of off post, respectively.

3.7.1.1 Prohibitions for Hazardous Waste (CFAS Requirement)

The Contractor shall minimize the amount of hazardous materials brought onto the project work site and only bring onto Government property HM directly associated and necessary to perform work assigned. The Contractor shall also minimize the amount of hazardous waste generated through the use of effective construction management practices. The Contractor is fully responsible to properly handle, segregate and containerize all HW as required by the CFAS Hazardous Waste Management Plan (HWMP) and guidance provided by the Environmental Office HW personnel. Failure by the

Contractor to properly manage HW as described above and that results in increased costs to the Government shall be billed back to the Contractor.

3.7.1.2 Prohibitions for Nonhazardous Waste (CFAS Requirement)

No hazardous material shall be brought onto the government property that are not directly related to requirements for the performance of this Contract. The government is not responsible for the disposals of Contractor's wastes/materials brought on the job site which are not required in the performance of this Contract. The intent of this provision is to only dispose of hazardous wastes that is defined in this section, is generated as part of this Contract, and exist within the boundary of the Contract limits and has not been brought in from offsite by the Contractor.

[3.7.1.3 Qualified Recycling Program (QRP) at U.S. Marine Corps Air Station (MCAS), Iwakuni

NOTE: These paragraphs only apply to projects located at U.S. Marine Corps Air Station (MCAS), Iwakuni. Iwakuni Marine Corps Air Station Order (MCASO) 5090.2 "Environmental Compliance and Protection Standard Operating Procedures (ECPSOP)", Chapter 17 "Solid Waste Management and Resource Recovery" applies to existing materials and equipment to be removed but to remain the property of the Government. All scrap metals generated from the project shall be salvaged and remain the property of the Government. The salvaged property shall be segregated, cleaned, itemized, delivered from the project site to the following storage area, and off-loaded, under the direction of the Contracting Officer.

SALVAGED PROPERTY
Scrap Metal

STORAGE AREA
Bldg.7725

All "metal items" to be removed shall be vested in the Government Property and delivered to Bldg.7725 (Facilities Recycle Center) in accordance with the following regulations:

- a. Metal items that are to be "turned-in" shall be pre-cut to 3 meters (10 feet) in length or less. Steel decking, steel plates, deck grating or plate metal will be accepted in any size.
- b. Segregate copper, brass and other metal items from steel items as much as possible.
- c. Before delivery, metal items shall be free of all lagging, clamps, straps, fittings or any other type of encumbrances, and metal items being used for fuel and oil handling system shall be completely purged.
- d. Follow the directions of the Contracting Officer when off-loading and positioning of "salvage metal items" in this Recycle Center.
- e. Make an appointment prior to turn-ins on Monday, Tuesday, Wednesday, and Friday between the hours of 0800-1500, and on Thursday between the hours of 0900-1500.
- f. Provide document describing item description, quantity, project number and date delivered to recycle center.

]3.7.2 Rubbish and Debris Disposal and Reporting

Remove rubbish and debris from Government property and dispose in

accordance with Japanese law. The Contractor shall provide to the DPW project manager the weight of construction and demolition debris in metric tons within 30 days or sooner of each disposal and detail how it has been disposed of to include landfill, recycle, reuse, composted, or mulched. These metrics are required by Department of the Army 27 Feb 2006 Memorandum for Commander, U.S. Army Garrison Japan.

Segregation measures shall be employed so that no hazardous or toxic waste is co-mingled with solid waste. The Contractor shall transport solid waste off Government property and dispose of it in compliance with applicable laws and regulations of U.S. Government and GOJ, however denominated, including those applicable political subdivisions, departments, and other entities for solid waste disposal. Vehicles used in transporting refuse shall be covered and enclosed to prevent spillage. The Contractor shall be responsible for all expense and cleanup of any spills on- or off-base.

3.7.3 Nonhazardous Solid Waste Diversion Report

Maintain an inventory of nonhazardous solid waste diversion and disposal of construction and demolition debris. Submit a report to the Installation Environmental Office [____ through] the Contracting Officer on the first working day after each fiscal year quarter, starting the first quarter that nonhazardous solid waste has been generated. Include the following in the report:

Construction and Demolition (C&D) Debris Disposed	(____) kilograms or metric tons as appropriate
C&D Debris Recycled	(____) kilograms or metric tons as appropriate
Total C&D Debris Generated	(____) kilograms or metric tons as appropriate
Waste Sent to Waste-To-Energy Incineration Plant (This amount should not be included in the recycled amount)	(____) kilograms or metric tons as appropriate

3.8 WASTE MANAGEMENT AND DISPOSAL

3.8.1 Waste Determination Documentation

Incorporate the Waste Determination Documentation for Contractor-derived wastes to be generated into the Environmental Records Binder. All potentially hazardous solid waste streams that are not subject to a specific exclusion or exemption from the hazardous waste regulations (e.g. scrap metal, domestic sewage) or subject to special rules, (lead-acid batteries and precious metals) must be characterized in accordance with the requirements of the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations. Base waste determination on user knowledge of the processes and materials used, and analytical data when necessary. Consult with the Installation environmental staff for guidance on specific requirements. Attach support documentation to the Waste Determination Documentation. As a minimum, provide Waste Determination Documentation for the following waste (this listing is not inclusive):

oil- and latex -based painting and caulking products, solvents, adhesives, aerosols, petroleum products, and containers of the original materials.

NOTE: The Contractor should include waste determination/categorization in the EPP for all common construction wastes to be generated under this Contract. If a waste is generated but not included in the approved EPP, additional waste determination may be required by the Government prior to accepting the waste for disposal. Complete Waste Determination Documentation for all Contractor building demolition related derived wastes to be generated. Base the waste determination upon laboratory analysis and attach all support documentation to the Waste Determination. Safety Data Sheets (SDS) by themselves are not adequate other than for disposal of remaining unusable portions of HM.

[3.8.1.1 Sampling and Analysis of Waste

3.8.1.1.1 Waste Sampling

Sample waste in accordance with the JEGS, Installation Hazardous Waste Management Plan, and appropriate Japanese or US EPA testing protocols that meet the purpose of the testing. Clearly mark each sampled drum or container with the Contractor's identification number, and cross reference to the chemical analysis performed.

Excess excavated soil shall become the property of the Contractor and shall be disposed of off base at a properly licensed facility in accordance with the latest version of the JEGS and applicable GOJ federal and local laws and regulations. Any excess soil shall be sampled and tested according to the applicable GOJ federal and local laws and regulations and disposal facility requirements prior to disposal.

3.8.1.1.2 Laboratory Analysis

Follow the analytical procedure and methods in accordance with the JEGS and all applicable GOJ national or prefectural laws and regulations and the Installation Hazardous Waste Management Plan. Provide analytical results and reports performed in the Environmental Records Binder.

3.8.1.1.3 Analysis Type

Identify hazardous waste by analyzing for the following characteristics:[ignitability,][corrosivity,][reactivity,][toxicity based on TCLP results,] [_____].

]3.8.2 Solid Waste Management

3.8.2.1 Solid Waste Management Plan

The Solid Waste Management Plan in the EPP shall include, but not limited to, the following:

- a. Description and estimated quantities of the proposed job-site waste to be generated.
- b. Landfill Options: The name of the landfill(s) where trash will be disposed of, applicable landfill tipping fee(s), and the projected cost of disposing of all project waste in the landfill(s).
- c. Waste Diversion: A list of the waste materials from the project that

will be separated for reuse, salvage, or recycling, associated weights and estimated cost savings shall be reported to USAG-O by the 25th day of each month via the Contracting Officer.

d. Handling Procedures: A description of the means by which any waste materials identified in (c) above will be stored and a description of the means to be employed in recycling the above materials consistent with requirements for acceptance by designated facilities.

e. Transportation: A description of the means of transportation of the waste and recycled materials (whether materials will be site-separated and self-hauled to designated center, or whether mixed materials will be collected by a waste hauler and removed from the site). Request manufacturers to use the minimum packaging required for protection and identification of project products, and to use packaging materials with recycled content where economically feasible.

f. Submit cost information on the wastes diverted from the landfill to the Contracting Officer or USAG-O by the 25th day of each month via the Contracting Officer.

3.8.2.2 Solid Waste Management Report

Provide copies of the waste handling facilities' weight tickets, receipts, bills of sale, Contractor Certification and other sales documentation. In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. The sales documentation[Contractor certification] must include the receiver's tax identification number and business, GOJ or prefectural registration number, along with the receiver's delivery and business addresses and telephone numbers. Submit a copy of the applicable Japanese local permits and licenses for transportation, treatment, storage and disposal of solid waste ("Sangyoku Haikibutsu") by permitted facilities.

3.8.2.3 Control and Management of Solid Wastes

Pick up solid wastes, and place in covered containers that are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Recycling is encouraged and can be coordinated with the Contracting Officer and the activity recycling coordinator. Employ segregation measures so that no hazardous or toxic waste will become co-mingled with non-hazardous solid waste.[Transport solid waste off Government property and dispose of it in compliance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements for solid waste disposal. Solid waste disposal offsite must comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. Verify that the selected transporters and disposal facilities have the necessary permits and licenses to operate. Include a copy of all licenses and permits secured in the Environmental Protection Plan]

Manage hazardous material used in construction, including but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

Submit Disposal manifests as specified in Section 01 74 19 CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT.

3.8.2.4 Solid Waste Management Plan Implementation

a. The Contractor shall designate an on-site party (or parties) responsible for instructing workers and overseeing and documenting results of the Solid Waste Management Plan for the project.

b. The Contractor shall distribute copies of the Solid Waste Management Plan to key personnel and submit the plan to the Contracting Officer as part of the Environmental Protection Plan.

c. The Contractor shall provide on-site instruction of appropriate separation, handling, and recycling, salvage, reuse, and return methods to be used by all parties.

d. The Contractor shall lay out and label a specific area to facilitate separation of materials for potential recycling, salvage, reuse, and return. Recycling and waste bin areas are to be kept neat and clean and clearly marked in order to avoid contamination of materials.

e. The Contractor shall submit a Summary of Solid Waste Generated by the project to USAG-O via the Contracting Officer with each application for progress payment. Failure to submit this information shall render the application for progress payment incomplete and shall delay progress payment. The Summary shall be submitted on a form acceptable to the Owner and shall contain the following information:

(1) The amount (in tons) of material land-filled from the project, the identity of the landfill, the total amount of tipping fees paid at the landfill, and the total disposal cost. Include manifests, weight tickets, receipt, and invoices.

(2) For each material recycled, reused or salvaged from the project, the amount (in tons), the date removed from the jobsite, the receiving party, the transportation cost, the amount of any money paid or received for the recycled or salvaged material, and the net total cost or savings of salvage or recycling each material. Attach manifests, weight tickets, receipts, and invoices.

3.8.3 Control and Management of Hazardous Waste

The Contractor shall be considered the primary generator for all hazardous wastes generated throughout the duration of the Contract. However, all hazardous waste management activities shall be coordinated and approved by USAG-O Environmental Division via the Contracting Officer.

All hazardous waste must be managed in accordance with applicable laws and regulations of U.S. Government and GOJ, however denominated, including those applicable political subdivisions, department, and other entities. The Contracting Officer shall contact USAG-O Environmental Division if handling procedures for hazardous wastes and materials are unclear.

Hazardous wastes, including excavated contaminated soil, shall be stored, transported, and disposed in accordance with applicable laws and regulations of U.S. Government and GOJ, however denominated, including those applicable political subdivisions, departments, and other entities,

including obtaining necessary local permits, licenses, and approvals. The Contractor shall identify what wastes are hazardous using specific and technical knowledge and/or sampling and analysis. This responsibility also includes preparation of waste profile sheets, packaging, marking and labeling of wastes in accordance with Federal, JEGS, GOJ, prefectural, and local requirements. All persons involved in the hazardous waste management process or potential exposure to hazardous waste must have hazardous waste training required by the JEGS.

All costs for labor, equipment, materials, transportation, and other services required to comply with applicable laws and regulations of U.S. Government and GOJ, however denominated, including those applicable political subdivisions, departments, and other entities, governing hazardous waste generations are the responsibility of the Contractor. This requirement extends to personnel training and the identification, initial accumulation and transportation of hazardous waste generated during the project.

Contractor must package hazardous waste in a United Nations (UN) approved container, labeled in English and Japanese, furnish analytical results and submit the container(s) and information to the USAG-O Environmental Division via the Contracting Officer. The Contractor shall be responsible for collection and disposal of hazardous wastes. When the waste is ready for shipment, the USAG-O Environmental Division personnel will inspect the Contractor's material if necessary. The Contractor's representative will be present for the inspection and will correct any discrepancies on the spot or remove the unacceptable containers and correct the deficiencies. For further information, contact the USAG-O Hazardous Waste Manager at DSN 652-4910 via the Contracting Officer.

Do not dispose of hazardous waste on Government property. Do not discharge any waste to a sanitary sewer, storm drain, or to surface waters or conduct waste treatment or disposal on Government property without written approval of the Contracting Officer.

3.8.3.1 Hazardous Waste/Debris Management

Identify construction activities that will generate hazardous waste or debris. Provide a documented waste determination for resultant waste streams. Identify, label, handle, store, and dispose of hazardous waste or debris in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements

Manage hazardous waste in accordance with the approved Hazardous Waste Management Section of the EPP. Store hazardous wastes in approved containers in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. Prior to removal of any hazardous waste from Government property, all hazardous waste manifests must be signed by waste transporter and disposal facility. If hazardous wastes are turned into the Government, submit Hazardous Waste turn-in document instead of manifest. Do not bring hazardous waste onto Government property. Provide the Contracting Officer with a copy of waste determination documentation for any solid waste streams that have any potential to be hazardous waste or contain any chemical constituents listed in the JEGS. For hazardous wastes spills, verbally notify the Contracting Officer immediately. Hazardous waste generated within the confines of Government facilities will be identified as being generated by the Government and will be labeled, handled, stored, and transport to designated Government facility specified

in Section 01 74 19 CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT.

3.8.3.2 Contractor Hazardous Waste Accumulation Point (HWAP)

If the work requires the temporary storage/collection of hazardous wastes on site overnight, the Contractor shall request the establishment of a Contractor HWAP at the point of generation. The Contractor must submit a request form with other required documentation/information to the Installation Environmental Office via the Contracting Officer. The Contractor shall not store/collect hazardous waste on site overnight unless the requested Contractor HWAP is approved and managed properly. When hazardous waste will be removed from the job site properly daily, the Contractor HWAP is not required.

Accumulate hazardous waste at HWAP and in compliance with the JEGS and CFAS HWMP. Individual waste streams will be limited to 208 liter (55 gallons) of accumulation, or 0.95 liter (1 quart) for acutely hazardous wastes. If the Contractor expects to generate hazardous waste at a rate and quantity that makes a HWAP impractical, the Contractor must request assistance from the Installation Hazardous Waste Manager via the Contracting Officer. Submit a request for assistance in writing to the Contracting Officer and provide the following information:

- a. Contract Number
- b. Contractor
- c. Haz Waste or Regulated Waste POC
- d. Phone Number
- e. Type of Waste
- f. Source of Waste
- g. Emergency POC
- h. Phone Number
- i. Location of the Site (Attach Site Plan to the Request)

Attach a waste determination form for the expected waste streams. Allow (10) ten working days for processing this request. Additional compliance requirements (e.g. training and contingency planning) that may be required and are the responsibility of the Contractor.

3.8.3.3 Contractor Disposal Turn-In Requirements

Handling and disposal of HW shall be performed in accordance with the JEGS. Unless otherwise specified or directed, all hazardous waste disposal will be coordinated with the CFAS Environmental Office and transferred to (usually picked up by) CFAS Hazardous Waste personnel to process final disposal. THE CONTRACTOR SHALL NOT DISPOSE OF ANY HAZARDOUS WASTES INTO ANY AREA OUTSIDE THE GOVERNMENT'S CONTROL.

To coordinate hazardous waste disposal needs, the Contractor shall inform the Contracting Officer in writing prior to generation of the hazardous waste by submitting a Hazardous Waste Disposal Disposition form provided by the Contracting Officer. The Government will provide HW accumulation containers for all HW disposed of through the CFAS Environmental Office

Hazardous waste will also be managed in accordance with the approved Hazardous Waste Management Section of the EPP. Store hazardous wastes in approved containers in accordance with the JEGS. Prior to removal of any hazardous waste from Government property, all hazardous waste manifests must be signed by waste transporter and disposal facility. If hazardous

wastes are turned into the Government, submit Hazardous Waste turn-in document instead of manifest. No hazardous waste will be brought onto Government property. Provide to the Contracting Officer a copy of waste determination documentation for any solid waste streams that have any potential to be hazardous waste or contain any chemical constituents listed in the JEGS. For hazardous wastes spills, call base 911 and verbally notify the Contracting Officer immediately.

Prior to shipping any hazardous Waste off the installation, the Contractor must provide the CFAS Environmental Office via the Contracting Officer a copy of the applicable manifest for transportation, treatment, storage, and disposal of hazardous, regulated, or toxic waste.

Incidental wastes used to support this Contract including, but not limited to aerosol cans, waste paints, cleaning solvents, contaminated brushes, rags, clothing, Personal Protection Equipment (PPE), etc. are considered as hazardous wastes. Chemical paint removers shall also be disposed of as Hazardous Waste.

HWAP - For projects that result in generation of HW over a period of time and in such quantity as to need a temporary HWAP to ensure proper, efficient and safe storage of HW at the job site, the CFAS Environmental Office will coordinate with the Contractor and provide guidance to set up a temporary HWAP for the duration of the project task order. The Contractor shall set-up and manage the temporary HWAP to meet the CFAS HWMP and guidance provided by the Environmental Office.

3.8.3.4 Hazardous Waste Records

The Contractor shall maintain sampling, analysis, and turn-in records for all hazardous waste generated during the project. These records shall include, but not be limited to: [waste profile sheets provided by HWSF personnel for wastes streams turned in to the HWSF, manifests (DD forms 1348-1A) for all wastes turned over to the HWSF,] logs of sample locations or container identification data (including time and date of sample collection), analytical results, and quality control data provided by the analytical lab pertaining to the samples analyzed. Copies of this data shall be submitted to [][374 CES/CEIE][35 CES/CEIE][INSERT ORGANIZATION] via the COR after the work is completed.

3.8.3.5 Excavated Soil or Other Excavated Materials

If excavated soil or other excavated materials, not indicated, that may be hazardous waste or hazardous to human health during construction operations is encountered, stop that portion of work and notify the Contracting Officer immediately as specified in paragraph entitled "EXCAVATION REQUIREMENTS", Section 01 11 00 GENERAL PARAGRAPHS. Any test results from the excavated soil shall be forwarded to the environment office. Case by case situation for special soil excavations i.e. soil with asbestos, PCB and heavy-metal. For further soil excavation/disposal guidance, contact the Installation Environmental Office.

3.8.3.6 Mercury Containing Items

Mercury is prohibited in the construction of this facility, unless specified otherwise, and with the exception of mercury vapor lamps and fluorescent lamps. Dumping of mercury-containing materials and devices such as mercury vapor lamps, fluorescent lamps, and mercury switches, etc., in Base trash enclosures/rubbish containers is strictly prohibited.

Such material shall be carefully removed without breaking, appropriately packaged to prevent breakage, and transferred to the Government, coordinate turn-in with the Hazardous Waste Program Manager of Installation's Environmental Office.

In case of breakage or any mercury spillage, the Contactor shall call base 911 and report immediately to the contracting Officer. The Contactor shall cleanup mercury spill area to the satisfaction of the Contracting Officer. Do not recycle a mercury spill cleanup; manage it as a hazardous waste for disposal.

3.8.3.7 Electronics End-of-Life Management

Recycle or dispose of electronics waste, including, but not limited to, used electronic devices such as computers, monitors, hard-copy devices, televisions, mobile devices, in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

3.8.3.8 Disposal Documentation for Hazardous and Regulated Waste

[If the Contractor is disposing of Hazardous waste off-base, Contact the Contracting Officer for the facility unique identification number (if applicable) that is to be used on each manifest.

Submit a copy of the applicable local permit(s), manifest(s), or license(s) for transportation, treatment, storage, and disposal of hazardous and regulated waste by permitted facilities. Hazardous or toxic waste manifests must be reviewed, signed, and approved by the Contracting Officer before the Contractor may ship waste. To obtain specific disposal instructions, coordinate with the Installation Environmental Office.]The hazardous wastes shall be taken to a TSD facility which has been properly licensed by the prefecture government or the mayor of a city and complies with all the provisions of applicable disposal regulations. An acceptance statement of agreement from the proposed TSD and certified transporters shall be submitted to the Contracting Office not less than 14 days before transporting of wastes. The original return copy of the manifest, signed by the owner or operator of a permitted TSD facility, shall be provided to the Contracting Officer no later than 5 working days following the delivery of the wastes to the TSD facility. Upon completion of waste disposal, a copy of the disposal certificate from TSD facility must also be provided to the Contracting Officer.

The Contractor shall maintain records of all waste sampling and test data. The data includes the results of analyses performed, sample locations, the time of collection, and other pertinent data during the life of the Contract. Transportation, treatment, and disposal methods and dates; the quantities of wastes; and the names and addresses of each transporter and the disposal or reclamation facility, shall also be recorded and available for inspection, as well as the following documents:

- (1) Manifests
- (2) Waste analyses or waste profile sheets
- (3) Certification of final treatment or disposal will be signed by the licensed TSD facility official.

3.8.4 Releases/Spills of Oil and Hazardous Substances

3.8.4.1 Response and Notifications

Exercise due diligence to prevent, contain, and respond to spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with the JEGS and Installation plans. Maintain spill cleanup equipment and materials at the work site. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of any releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the [Installation][Activity] Fire Department [dial 911 on base], the [Installation][Activity] Command Duty Officer, the [Installation][Activity] Environmental Office, the Contracting Officer. The Contractor shall immediately report all POL or Hazardous Substances spills to the Contracting Officer. Submit verbal and written notifications as required by the JEGS and installation and service component instructions and plans, and local regulations. Provide copies of the written notification and documentation that a verbal notification was made within [the timeframes required by the JEGS and installation instructions][five working days]. Spill response must be in accordance with the JEGS and installation requirements. Contain and clean up these spills without cost to the Government. If the Contractor's response is inadequate, the Navy may respond. If this should occur, the Contractor will be required to reimburse the government for spill response assistance and analysis. Spill response shall be in accordance with the JEGS, CFAS SPRP, GOJ and local Prefecture regulations.

3.8.4.2 Clean Up

The Contractor is responsible for spill cleanup. Determine, as quickly as possible, the nature of the spilled substance and implement necessary safety precautions to protect both human health and the environment. Cleanup shall be in accordance with applicable local laws and regulations as determined by the Environmental Division at no additional cost to the Government, and shall only be performed by personnel adequately trained in spill response and cleanup techniques for the severity of the spill incident.

a. The Contractor shall provide and maintain spill equipment, sufficient in both type and quantity, at all sites involving the storage, use or handling of hazardous materials and/or hazardous wastes. The type of equipment and quantity required will be identified in the Contractor's site specific contingency plans. Equipment must be adequate to contain any release.

b. Cost incurred from any Contractor spills is the responsibility of the Contractor, if the Government has to perform emergency spill response and cleanup, due to the non-availability of designated personnel or if spill is beyond the capability of designated personnel, the Contractor shall be held liable for all costs associated with performing said work. The costs of clean-up will be subject to Contract price offsets.

If remaining Contract payments are insufficient, the government reserves the right to pursue other offsets, administrative or civil actions to satisfy this. (JEGS)

c. The Contractor shall train employees in the use of the above equipment

and document training.

d. The Contractor shall take sufficient measures to prevent spillage of hazardous and toxic materials during dispensing and shall collect waste in suitable containers observing chemical compatibility requirements. If the Contractor is required to stockpile contaminated soil for testing prior to disposal, the Contracting Officer will inform the Contractor of the stockpile location after consultation with USAG-O.

e. The Contractor waives all claims to damage or loss that arise from Government actions after thirty (30) days.

Clean up hazardous and non-hazardous waste spills in accordance with the JEGS, local regulations and the installation's spill response procedures. Reimburse the Government for costs incurred including sample analysis materials, clothing, equipment, and labor if the Government will initiate its own spill cleanup procedures, for Contractor-responsible spills, when: Spill cleanup procedures have not begun within one hour of spill discovery/occurrence; or, in the Government's judgment, spill cleanup is inadequate and the spill remains a threat to human health or the environment.

3.8.4.3 Reporting Requirements

Submit a written follow-up report to the Contracting Officer not later than 7 days after the incident. The written report shall conform to the requirements detailed in the Installation Spill Response Plan and to the site specific spill prevention plan, originally submitted. At a minimum, the report shall provide the information necessary to complete an Army Environmental Pollution Incident Report Form. Copies may be obtained from the Environmental Division.

The Emergency Response Plan as required in EPP section shall include, but not be limited to the following:

a. Information for review by the Contracting Officer to show that the Contractor has successfully performed hazardous waste management (as applicable).

b. The "Site Specific Spill Prevention/Response Plan" shall be adapted to specific sites where the Contractor handles chemicals or hazardous materials. Site-specific spill plans shall be revised as necessary to reflect current hazardous material storage and usage. The plan shall detail the Contractor representative and alternate that will be contacted in the event of an emergency and the equipment to be used to adequately contain and cleanup spills. Revisions to the plan shall be coordinated with Fire Department and USAG-O Environmental Division through the Contracting Officer.

3.8.5 Mercury Materials

Immediately report to the Environmental Office and the Contracting Officer instances of breakage or mercury spillage. Clean mercury spill area to the satisfaction of the Contracting Officer.

Do not recycle a mercury spill cleanup; manage it as a hazardous waste for disposal.

3.8.6 Wastewater

3.8.6.1 Disposal of wastewater must be as specified below.

3.8.6.1.1 Treatment

Do not allow wastewater from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, and forms to enter water ways or to be discharged prior to being treated to remove pollutants. Dispose of the construction- related waste water[off-Government property in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.][by collecting and placing it in a retention pond where suspended material can be settled out or the water can evaporate to separate pollutants from the water. The site for the retention pond must be coordinated and approved with the Contracting Officer. The residue left in the pond prior to completion of the project must be removed, tested, and disposed of off- Government property in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. Backfill the area to the original grade, top-soiled, and seeded or sodded.[Consult with the Installation Environmental Office on specific testing requirements for water in the retention pond. If required, test the water in the retention pond for [_____] and have the results reviewed and approved by the Installation Environmental Office via the Contracting Officer prior to being discharged or disposed of off-Government property].]

3.8.6.1.2 Surface Discharge

For discharge of ground water,[if applicable, obtain a GOJ or local prefectural permit specific for pumping and discharging ground water prior to surface discharging.][Surface discharge shall be done in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations and installation requirements.]

3.8.6.1.3 Land Application

Water generated from the flushing of lines after[disinfection or disinfection in conjunction with hydrostatic testing][hydrostatic testing] must be[land- applied in accordance with the JEGS and applicable Federal, GOJ, prefectural and local laws and installation requirements for land application][discharged into the sanitary sewer with prior approval and notification to the owner of the sanitary sewer system].

3.9 HAZARDOUS MATERIAL MANAGEMENT for the performance of this Contract.

Include hazardous material control procedures in the Safety Plan, in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. Do not bring hazardous material onto Government property that does not directly relate to requirements for the performance of this Contract. Submit an Safety Data Sheet in both English and Japanese and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the installation. Typical materials requiring SDS and quantity reporting include, but are not limited to, oil and latex based painting and caulking products, solvents, adhesives, aerosol, and petroleum products. Use hazardous materials in a manner that minimizes

the amount of hazardous waste generated. Containers of hazardous materials must have National Fire Protection Association labels or their equivalent.

Storage and handling of hazardous materials will adhere to the DoD Component policies, including Joint Service Publication on Storage and Handling of Hazardous Materials. Defense Logistics Agency Instruction (DLAI) 4145.11, Army Technical Manual (TM) 38-410, Naval Supply Publication (NAVSUP PUB) 573, Air Force Joint Manual (AFJMAN) 23-209, and Marine Corps Order (MCO) 4450.12A, "Storage and Handling of Hazardous Materials," January 13, 1999 provide additional guidance on the storage and handling of hazardous materials. The International Maritime Dangerous Goods (IMDG) Code and appropriate DoD and Component instructions provide requirements for international maritime transport of hazardous materials originating from DoD installations. International air shipments of hazardous materials originating from DoD installations are subject to International Civil Aviation Organization Technical Instructions or DoD Component guidance, including Air Force Manual 24-204, (Interservice) TM 38-250, NAVSUP PUB 505, MCO P4030.19J, and DLAI 4145.3, DCMAD1, Ch3.4 (HM24), "Preparing Hazardous Materials for Military Air Shipments," 3 December 2012. At the end of the project, provide the Contracting Officer with the maximum quantity of each material that was present at the site at any one time, the dates the material was present, the amount of each material that was used during the project, and how the material was used. Ensure that hazardous materials are utilized in a manner that will minimize the amount of hazardous waste that is generated. Ensure that all containers of hazardous materials have NFPA labels or their Japanese equivalent. Keep copies of the SDS for hazardous materials on site at all times and provide them to the Contracting Officer at the end of the project. Certify that hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste, in accordance with the 40 CFR 261, JEGS, and installation requirements.

3.9.1 Hazardous Material

Contractors bringing hazardous material onto the installation must have a list of the hazmats, along with an SDS in both English and Japanese provided in the EPP submittal documents. These documents must be coordinated and routed through Fire Department and Environmental Division. All excess materials and empty containers are the responsibility of the Contractor and shall be removed at the end of the Contract. Should HM requirements change during the performance period, the Contractor shall immediately notify the Environmental Division and Fire Department of such changes in writing.

Hazardous materials must be stored, handled, used, and disposed of, in accordance with 29 CFR 1910, JEGS, and local regulations.

A list of HM's, and the associated SDS in both English and Japanese must be presented to the Contracting Officer before bringing HM's on the base. All containers with hazardous materials shall be correctly marked/labeled with DD Form 2522 (DD-2522) or an equivalent label with the container contents and protected from damage (compressed gases, fuels, acids, and refrigerants (ODS) are also included).

At the end of the project all Hazardous Materials will be removed from the installation, unless approval is granted by the Environmental Division, Safety and Fire Department via the Contracting Officer.

The Contracting Officer may request documentation for any spills or releases, environmental reports, or off-site transfers.

3.9.2 HAZARDOUS MATERIAL AND HAZARDOUS WASTE PROHIBITIONS

Unless otherwise specified in task order, do not dispose hazardous materials and hazardous waste into any area outside the Government control area. The Contractor shall inform of hazardous waste to be salvaged to the Contracting Officer for the notification purpose to Environmental Hazardous Waste Disposal Br., Environmental Division, PWD Sasebo, NAVFAC FE for CFAS projects in writing and coordinate setting up the boxes with the Contracting Officer. The Contractor shall submit Hazardous Waste Disposal Disposition form provided by the Contracting Officer. No hazardous material shall be brought onto the government property that are not directly related to requirements for the performance of this Contract. The government is not responsible for the disposals of Contractor's wastes/materials brought on the job site which are not required in the performance of this Contract. The intent of "hazardous wastes that exist in the Government facility" is hazardous wastes that is generated as part of this Contract, and is existed within the boundary of the Contract limits and not brought in from offsite by the Contractor. The disposal of incidental materials used to accomplish the work including, but not limited to aerosol cans, waste paint, cleaning solvents, contaminated brushes, rags, clothing, etc. are the responsibility of the Contractor, except the incidental hazardous waste specified in elsewhere of this Contract.

3.10 PREVIOUSLY USED EQUIPMENT

Clean previously used construction equipment prior to bringing it onto the project site. Equipment must be free from soil residuals, egg deposits from plant pests, noxious weeds, and plant seeds.

[3.11 CONTROL AND MANAGEMENT OF ASBESTOS-CONTAINING MATERIAL (ACM)]

Manage and dispose of asbestos- containing waste in accordance with the JEGS and applicable Federal, GOJ national or prefectural laws and regulations. Contractor must double-bag ACM waste, package in a UN approved Tri-wall container, weigh with a certified scale, label in English and Japanese, furnish analytical result and submit information to the USAG-O Environmental Division. The Contractor shall be responsible for collection and disposal of ACM waste. The ACM waste shall be subjected to pre-shipment inspections by the Environmental Division if necessary. [Contact JED Environmental for the most current version of][Refer to] Section 02 82 00 ASBESTOS REMEDIATION. Manifest asbestos-containing waste and provide the manifest to the Contracting Officer. Notifications to the Contracting Officer and Installation Air Program Manager are required before starting any asbestos work.

Removal of mastics, necessary to achieve the Contract's objectives, shall be performed in a manner as if the material contains asbestos.

When applicable, a minimum of 14 calendar days prior to the demolition or renovation of a facility that involves removing or disturbing friable ACM, the Contractor shall prepare a Written Assessment Of Friable Asbestos Disturbance and submit to the COR, who will, in turn submit to the Installation Commander in accordance with the JEGS.

[For any previously untested material suspected to contain asbestos and located in areas impacted by the work, notify the Contracting Officer (CO) who will order up to [] bulk samples to be obtained and analyzed at the Contractor's expense. Samples shall be collected in sufficient amounts such that a follow-on test utilizing TEM or 1000 point count analysis can be performed, should it be necessary. The Contractor shall deliver the sample(s) to a laboratory accredited under the National Institute of Standards and Technology (NIST) "National Voluntary Laboratory Accreditation Program (NVLAP)" and analyzed by PLM. The laboratory shall have a working definition of "Trace" amounts of asbestos, and the laboratory shall report any detectable amount of asbestos in a bulk sample that is less than the PLM Limit of Quantification of 1% as a "Trace" concentration. If PLM does not detect the presence of asbestos (e.g. "non-detect"), the material shall be considered <0.1% asbestos. If PLM analysis detects asbestos in any discernible amount (to include "trace" or "less than 1%"), the material shall be considered >0.1% asbestos unless proven to be non-ACM by the use of quantification methods capable of achieving an analytical sensitivity of less than 0.1%, such as Transmission Electron Microscopy (TEM) or 1000 point counting.

The Contracting Officer (CO) will order, testing by TEM or 1000 point counting, for up to [] samples, to be obtained at the Contractor's expense and delivered to a laboratory accredited under the National Institute of Standards and Technology (NIST) "National Voluntary Laboratory Accreditation Program (NVLAP)", for those samples with results of < 1% ACM as analyzed by PLM.

Any additional components identified as ACM that have been approved by the CO for removal shall be removed and will be paid for by an equitable adjustment to the Contract price under the CONTRACT CLAUSE titled "changes". Sampling shall be conducted by personnel who have successfully completed the EPA Model Accreditation Plan (MAP) "Building Inspector" training course and is AHERA certified as a "Building Inspector".]

][3.12 CONTROL AND MANAGEMENT OF LEAD-BASED PAINT (LBP)

Manage and dispose of lead-contaminated waste in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations and installation requirements. [Contact JED Environmental for the most current version of][Refer to] Section 02 83 00 LEAD REMEDIATION. The Contractor shall assume all paints and coatings that will be disturbed during the course of meeting the contractual requirements, contain detectable levels of lead, requiring the Contractor to incorporate OSHA regulations for worker and occupant protection. Contractor must package lead contaminated waste in a UN approved container, weigh with a certified scale, label in English and Japanese, furnish analytical result and submit information to USAG-O Environmental Division. The Contractor shall be responsible for collection and disposal of LBP waste. The LBP waste shall be subjected to pre-shipment inspections by the Environmental Division if necessary. The Contractor is to assume that all existing paint encountered in this Contract contain detectable levels of lead, and the OSHA regulations referenced in these specification applies. Any attached lead material survey information is provided for disposal purposes only. Manifest any lead-contaminated waste and provide the manifest to the Contracting Officer. No paint containing lead shall be used during the course of this Contract. Work with LBP must comply with 29 CFR 1926.62, 29 CFR 1910.134, JEGS, GOJ, Prefectural, and local requirements. The Contractor must maintain all documentation regarding lead exposure by either historical data or project data. Conduct in accordance with Section

[].

a. Prior to the commencement of construction, the prime Contractor, each subcontractor and material/equipment supplier shall provide to the CO and USAG-O Environmental Division the most current Safety Data Sheets proving that the paint does not have any lead content.

b. The Contractor shall be responsible for collection and disposal of all lead paint chips and lead paint-contaminated materials, and for accumulation of these chips/materials on site. The Contractor shall test the paint materials, provide containers for proper disposal, transport and dispose of any resulting hazardous waste as described in paragraphs [].

c. Construction and demolition waste suspected of, or reported to contain LBP contamination must be profiled through the performance of a toxicity analysis. A copy of the toxicity analysis must be provided to USAG-O Environmental Division and the CO. LBP wastes are specially controlled industrial waste having the potential to have lead and cadmium levels above the regulatory levels. Lead and cadmium levels above the 0.3 mg/L regulatory levels will not be disposed of in a solid waste landfill. These types of wastes shall be disposed of as hazardous waste. LBP wastes below the lead and cadmium regulatory levels may be disposed of in a permitted solid waste landfill.

d. Copies of all lead paint-related documentation generated from this project, including lead testing, air monitoring and hazardous waste manifests, shall be provided by the Contractor to the Contracting Officer. A copy shall be forwarded to USAG-O Environmental Division via the Contracting Officer within 10 working days of task completion.

][3.13 CONTROL AND MANAGEMENT OF POLYCHLORINATED BIPHENYLS (PCBS)

Manage and dispose of PCB-contaminated waste in accordance with the JEGS, installation requirements, and Section 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCB). Purchase of electrical equipment and transformers containing PCBs is prohibited.

3.13.1 Removal Or Installation

Work on PCB equipment is forbidden using this supplement alone. If the project involves the disturbance, relocation, installation, or removal of products that have historically contained PCBs, follow the instructions in the supplement to the specification that pertains to that product's installation or removal (Transformer Replacement, Lighting Replacement, etc.).

a. Transformers

(1) Contractor must provide a manufacturer certification stating that the transformers used in the project are PCB-free (no detectable PCBs). The phrase "No PCBs" must appear on the label of the transformer. (JEGS)

(2) Contractor shall sample the transformer for the presence of PCBs using samples analyzed at a local Japanese laboratory. Results are to be bilingual. Recommend that the laboratory technician pull the sample. Contractor is required to submit bilingual copies of the lab result to USAG-O Environmental Division for approval. Contractor will also provide a copy of the lab result to the CO.

(3) If lab results indicate that the PCBs content is more than 0.5 mg/Kg contact USAG-O Environmental Division for guidance on turn-in procedures of transformers.

(4) If lab results indicate that the transformer is non-PCB (i.e. less than 0.5 mg/Kg) then it may be possible to recycle the transformer and dielectric fluid.

b. Paper Insulated Lead Covered (PILC) Cable

(1) Contractor must provide a manufacturer certification stating that the PILC used in the project is PCB-free (no detectable PCBs) and non-Asbestos. (JEGS).

(2) Contractor shall sample each run of cable for the presence of PCBs and Asbestos using a local Japanese laboratory in accordance with Japanese disposal methods. PCB results are to be bilingual. Asbestos test method is to be in accordance to JIS A 1481 to 0.1% by weight. Recommend that the laboratory technician pull the sample. Contractor is required to submit bilingual copies of the lab result to USAG-O Environmental Division for approval. Contractor will also provide a copy of the lab result to the CO.

(3) If the lab results indicate that there are PCBs present in the PILC in concentrations above 0.5 mg/Kg, contact USAG-O Environmental Division for guidance on turn-in procedures.

(4) Contractor shall take care when handling PILC. Prior to removal, the Contractor shall lay down tarps (blue poly sheets are acceptable) at the manhole and surrounding area. Cable shall be pulled in long runs, minimizing cuts, and placed in a spool and identified as to which manhole they originated.

(5) Contractor shall seal ends of cable, and any area where sheathing is compromised with splicing tape. A cover, such as a blue poly sheet, is to be placed over the spool of cable to prevent rainfall from contacting the cable. Cable is to be staged away from storm drains while awaiting results.

(6) Contractor shall place splices along with tarps and rags used in pulling cable in UN certified open-head drums and identified as to which manhole they originated. Debris is to be cleaned up and containerized daily.

(7) If the lab results indicate Asbestos is present in the PILC in concentrations above 0.1 % by weight, abate Asbestos in accordance with Section 13280A, Asbestos Hazard Control Activities.

(8) If lab results indicate that the PILC is non-PCB (i.e. less than 0.5 mg/Kg) and non- Asbestos (less than 0.1% by weight) then the material may be managed as a scrap metal and disposed of off-base by the Contractor.

] [3.14 CONTROL AND MANAGEMENT OF LIGHTING BALLAST AND LAMPS CONTAINING PCBS

Manage and dispose of contaminated waste in accordance with the JEGS and installation requirements. [Contact JED Environmental for the most current version of] [Refer to] Section 02 84 16 HANDLING OF LIGHTING BALLASTS AND LAMPS CONTAINING PCBS AND MERCURY.]

If ballasts without a "No PCBs" label are encountered, the manufacturer shall be contacted to determine the PCB-content of the ballast.

Furthermore, there is a possibility that ballasts labeled "No PCBs" may still contain PCBs above the Japan-established level of 0.5 mg/Kg. Because the U.S. definition for PCBs is 50 mg/Kg and over, it is possible that some U.S. manufacturers may consider "No PCBs" as 50 mg/Kg or below. Therefore, if a project involves the removal, disposal and installation of ballasts, the manufacturer shall be contacted and an MSDS or a written statement indicating that the ballasts contain less than 0.5 mg/Kg of PCBs shall be requested, even if the existing ballasts are labeled "No PCBs". However, it is best if the manufacturer is able to state that there is absolutely 0 PCBs in their product. If the Contractor cannot obtain any PCB information for the ballasts, the Contractor shall take a representative sample of each model and have the samples analyzed at a local Japanese laboratory. The Contractor is required to submit bilingual copies of the lab results to USAG-O Environmental Division. If the lab results indicate that the PCB-content is more than 0.5 mg/Kg, USAG-O Environmental Division shall be contacted for guidance on the turn-in procedures for the wastes containing PCBs. Segregate PCBs free ballasts from PCBs containing ballasts and further segregated according to model. The ballasts shall be placed in a united nation certified drum container and in separate boxes per brand as well as per US and Japanese products. Contractor must package PCBs ballast in a UN approved container, weigh with a certified scale, label in English and Japanese, furnish analytical result and submit information to USAG-O Environmental Division. For further information, contact the PCBs Manager at 652-4382 or 652-4910.

Light Fixtures waste including, but not limited to mercury containing equipment such as florescent lamps, mercury vapor lamps, and high pressure sodium lamps shall be disposed as Hazardous Waste and turned in to the Government in accordance with the JEGS. Non hazardous waste shall be disposed off-base in accordance with all applicable JEGS, Japanese laws and local regulations. Lighting fixture ballasts may contain PCB as specified in Paragraph "PCB Contained Waste Materials and Equipment". When the light ballast does not contain PCBs, submit the ballast manufacturer's non-PCB certificate to the NAVFAC PWD Environment Office via the Contracting Officer.

a. Light Ballasts/Capacitors.

(1) Contractor must provide a manufacturer certification stating that the ballasts/capacitors used in the project are PCB-free (no detectable PCBs). The phrase "No PCBs " must appear on the label of the ballast/capacitor. (JEGS).

(2) All ballasts/capacitors must be removed from the fixtures. Ballasts/capacitors are to be segregated based on PCB-free from PCB-containing, U.S. and Japanese-made and according to model. Refer to Section 3.73 of this part for turn-in and disposal.

]3.15 CONTROL AND DISPOSAL OF IONIZATION SMOKE DETECTORS AND TRITIUM EXIT SIGNS

3.15.1 Material Bagging

Remove existing ionization smoke detectors and tritium exit signs, and place like types together; i.e. same manufacturer and model number, in a plastic bag. Provide a label on the bag with the following data:

Manufacturer
Model Number

Isotope/Quantity (if known)
Activity
Contract Number

3.15.2 Material Storage

Do not open smoke detectors due to a small amount of radioactive (Americium 241) material. Store plastic bags in 55-gallon covered drum(s). Do not seal the drum(s). Apply a label entitled "RADIOACTIVE" and the storage inventory form to exterior surface of the cover and side of the drum(s). Provide a record copy, for each drum storage inventory to the Contracting Officer.

3.15.2.1 Storage Site and Disposal for U.S. Products

Deliver drums to designated facility for storage and disposal of U.S made ionization smoke detectors and tritium exit signs as directed by the Contracting Officer.

3.15.3 Storage and Disposal for Japanese Products

The Contractor shall be responsible for storage of Japanese ionization smoke detectors and delivery to each manufacturer in accordance with Japanese law, "Houshasai Douigenso Tou Ni Yoru Houshasen Shougai Boushi Ni Kannsuru Hourishu". If unable to identify the manufacturer, items may be sent to the "Japan Radioisotope Association".

3.16 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING

POL products include flammable or combustible liquids, such as gasoline, diesel, lubricating oil, used engine oil, hydraulic oil, mineral oil, and cooking oil. Store POL products and fuel equipment and motor vehicles in a manner that affords the maximum protection against spills into the environment. Manage and store POL products in accordance with the JEGS and installation requirements and applicable GOJ national or prefectural laws and regulations. Use secondary containments, dikes, curbs, and other barriers, to prevent POL products from spilling and entering the ground, storm or sewer drains, stormwater ditches or canals, or navigable waters of Japan. Describe in the EPP (see paragraph ENVIRONMENTAL PROTECTION PLAN) how POL tanks and containers must be stored, managed, and inspected and what protections must be provided.[Storage of oil, including fuel, on the project site is not allowed. Fuel must be brought to the project site each day that work is performed.][Storage of fuel on the project site must be in accordance with the JEGS and installation requirements, and local laws and regulations and paragraph OIL STORAGE INCLUDING FUEL TANKS.]

If temporary petroleum products and lubricants containing tanks/containers are to be stored at a job site, spill preventive measures shall be taken. A secondary containment system shall be applied. Appropriate and compatible spill kits shall be readily available at a job site. A sign larger than 0.3 meters wide and 0.6 meters long or clearly visible from 16 meters shall be provided containing following information in English and Japanese: identification number, maximum storage capacity, the title and telephone number of the point of contact, and state that "DANGER: NO FLAMMABLE OR IGNITION SOURCES WITHIN 50 FEET OR 16 METERS" in red letters. Further guidance on spill prevention measures is available; contact Installation Environmental Office.

3.16.1 Used Oil Management

"Used oil," means any oil or other waste petroleum, oil, or lubricant (POL) product that has been refined from crude oil, or is synthetic oil, has been used and as a result of such use, is contaminated by physical or chemical impurities, or is off specification and cannot be used as intended. Although used oil may exhibit the characteristics of reactivity, toxicity, ignitability, or corrosivity, it is still considered used oil, unless it has been mixed with hazardous waste. Manage used oil generated on site in accordance with the JEGS and installation requirements. Determine if any used oil generated while onsite exhibits a characteristic of hazardous waste. Used oil containing 1,000 parts per million of solvents is considered a hazardous waste and disposed of at the Contractor's expense. Used oil mixed with a hazardous waste is also considered a hazardous waste. Dispose in accordance with paragraph HAZARDOUS WASTE DISPOSAL.

Petroleum Product Used or Remaining In Existing Facility - Turn-in petroleum products generated from existing facilities which is defined as a hazardous waste or recyclable. Turn-in to Government's storage facility in accordance with Section 01 74 19 CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT.

3.16.2 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection for oil storage tanks. A berm used to provide secondary containment must be of sufficient size and strength to contain the contents of the tanks plus 12 centimeters freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks. If tanks and containers with an aggregate aboveground capacity greater than 5000 liter will be used onsite (only containers with a capacity of 208 liter or greater are counted), provide and implement a SPCC plan meeting the requirements of 40 CFR 112. Do not bring underground storage tanks to the installation for Contractor use during a project. Submit the SPCC plan to the Contracting Officer for approval.

Monitor and remove any rainwater that accumulates in open containment dikes or berms. Inspect the accumulated rainwater prior to draining from a containment dike to the environment, to determine there is no oil sheen present. If a sheen is present, remove the sheen in accordance with CFAS Environmental Management Plans, prior to draining containment dike. Maintain a written record of containment dike inspections, sheen removal, and drainage by date and time.

3.16.3 Secondary Containment

A secondary containment capable of holding the entire contents of the largest single tank plus sufficient freeboard to allow for precipitation and expansion of product shall be provided. Alternatively, a double wall container equipped with adequate technical spill and leak prevention options such as overfill alarm and flow shutoff may provide as the secondary containment.

3.16.4 FUEL STORAGE TANKS

US Army Garrison Fire Department and USAG-O Environmental Division must approve the use of fuel storage tanks on base, and the Contractor must ensure adequate spill containment (spill kits) for any tanks approved for use. The Contractor must have written spill procedures for tanks and heavy equipment that they use on base. Temporary gasoline storage is not permitted on base.

Provide secondary containment such as spill pans or impervious berms where necessary when bringing fuel storage tanks on base. Contact the Contracting Officer for inspection to ensure proper setup of fuel storage prior to dispensing of fuel.

All tanks regardless of size must have secondary containment or double wall capable of holding the entire contents of the largest single tank plus sufficient freeboard to allow for precipitation and expansion of product.

When a facility with fuel storage tanks is demolished, demolition of the tank(s) shall be included in the scope of the project. The following standards shall be applied when demolishing an existing tank.

a. Training: Personnel involved in removing tanks must have training on UST/AST removal and confined space entry.

b. Environmental, Health and Safety Plan: Prepare environmental, health and safety plan and keep it at the site during tank removal. Coordinate plan with USAG-O Environmental Division via the Contracting Officer prior to start of construction.

Plan shall include:

- (1) All planned safety measures
- (2) Protective equipment to be used by workers
- (3) Documentation of worker training
- (4) Lab analysis of the material in the tank
- (5) Disposal method for tank contents and any contamination in accordance with local regulations
- (6) Location of post-excavation soil sampling points, analytical methods to be used on samples, and clean up levels. At least 5 sampling points will be used after excavation or soil known to be contaminated has been removed. Consult with USAG-O Environmental Division via the Contracting Officer for assistance in planning sampling.

c. Before Removing Tank: Notify Environmental Division, at least 24 hours before removing tank. Clean the tank (triple rinse) and render tank inert (vapor must not to exceed 10 percent Lower Explosive Limit).

d. Tank Cutting: Cut a 2' by 2' opening on each side of the tank to prevent reuse, paint the Bldg. Number and date of removal on the tank, take photos for documentation purpose.

e. Soil Removal: Segregate clean soil from the contaminated soil. The soil

shall be visually inspected for staining after removal of all obviously contaminated soil, then screened for the presence of volatile and/or semi-volatile contamination using an approved vapor monitoring instrument. If the soil is contaminated with petroleum, it shall be disposed of in accordance with local regulations. If it is contaminated with hazardous substance, the Contractor shall dispose of it in accordance with section 3.08 of this part. If additional information is needed, consult with USAG-O Environmental Division via the Contracting Officer.

f. Closure Report: Prepare closure report, including documentation of all of the requirements of this section and the following information as a minimum, and submit to USAG-O Environmental Division via the Contracting Officer.

(1) A cover letter signed by a responsible company official certifying that all services involved have been performed in accordance with the terms and conditions of this section.

(2) A narrative report describing what was encountered at each site, including: condition of the UST, any visible evidence of leaks or stained soils, results of vapor monitoring readings, actions taken including quantities of materials treated or removed, reasons for selecting sample locations, sample locations, whether or not groundwater was encountered.

(3) Copies of all analyses.

(4) Information on who transported and accepted all wastes encountered, including copies of manifests.

(5) Scaled one-line drawings showing tank locations, limits of excavation, limits of contamination, underground utilities within fifty feet, sample locations, and sample identification numbers.

(6) The Contractor shall take site photos including: site condition before work begins, soil stockpile area, tank, excavation area, site condition after the construction.

3.17 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS WASTES

If petroleum-contaminated soil, or suspected hazardous waste is found during construction that was not identified in the Contract documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.18 Contaminated Soil

In case the Government assumes that soil on project site is contaminated soil, the Contractor shall store excavated soil and take necessary measure to protect ground from contamination. Excavation and handling soil shall be performed in accordance with direction from Activity environmental office.

In accordance with CNFJ Policy Letter "CONTAMINATED SOIL EXCAVATION AND MANAGEMENT POLICY", back filling of existing contaminated soil to its original place will be allowed and provide covering for the surface. Transport and dispose of excess contaminated soil to off-base facility as specified Paragraph "Disposal of Contaminated Soil" in this section.

3.18.1 Soil Sampling and Testing

If soil contamination is unknown, (i.e. sample results do not exist), previous samples indicating contamination exist, but are over one-year old or soil contamination is suspected then sampling and analysis is required in accordance with Soil Contamination Countermeasures Law Soil Testing Standards. As a minimum, sampling will consist of a composite sample composed of five separate samples per 100 square meters. A sampling plan will be submitted to the Installation Environmental Office for approval.

Sample analysis will be performed utilizing an independent accredited laboratory. Testing shall consist of those contaminants listed in the JEGS. Samples shall be analyzed in accordance with Soil Contamination Countermeasures Law Soil Testing Standards. Results shall be provided to the Activity Environmental Soil Manager for determination of soil disposal methodology. For constituent(s) exceeding the limits of the soil primary leachate standard (JEGS), the Installation Environmental Office will be the determining authority for the need for soil contents test(s).

Soil awaiting analysis or disposal maybe temporarily stored within the installation. Temporary soil staging areas shall be located and maintained in such a way as to prevent damage to cultural / natural resources, impact waterways or drinking water sources, create an air or odorous nuisance or endanger public health. Map(s) indicating temporary storage area location and a plan describing pollution prevention measures shall be submitted to the Installation Environmental Office for approval.

Contaminated soil piles shall be on an impermeable surface or membrane. Soils shall be completely and securely covered, for the duration of the storage period, with an impermeable material of sufficient strength, thickness, anchoring or weighting to prevent tearing or lifting of the cover, infiltration of precipitation or runoff, and exposure of the soil to the atmosphere.

Appropriate steps shall be taken to deter public access to the storage area. This may include fencing, similar barriers, security patrols or warning signs.

Soils relocated to a Temporary Staging, Transfer, and/or Temporary Storage Area (i.e. an area away from the project site) to await off-site disposition shall have a sign posted identifying in indelible ink:

- a. "Temporary Staged Soil"
- b. Contract Number
- c. POC Phone Number
- d. Contaminants
- e. Date Soil Staged

3.18.2 Disposal of Contaminated Soil

The Contractor shall dispose of contaminated soil off-base in accordance with Japanese laws, local prefectural regulations, and the JEGS.

[3.19 PEST MANAGEMENT

In order to minimize impacts to existing fauna and flora, coordinate with the Installation Pest Management Coordinator (IPMC) or the Natural Resources Manager in the DPW Environmental Division, through the Contracting Officer, at the earliest possible time prior to pesticide application. Discuss integrated pest management strategies with the [IPMC][PPC] and receive concurrence from the [IPMC][PPC] through the Contracting Officer prior to the application of any pesticide associated with these specifications. Provide Installation Project Office Pest Management personnel the opportunity to be present at meetings concerning treatment measures for pest or disease control and during application of the pesticide.[For termiticide requirements, see[Section 31 31 16.13 CHEMICAL TERMITE CONTROL][Section 31 31 16.19 TERMITE CONTROL BARRIERS]] The use and management of pesticides are regulated under the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations and installation requirements.

3.19.1 Application

Apply pesticides using a DoD-certified or equivalent Japanese-certified pesticide applicator in accordance with guidance given on the pesticide label. The certified applicator must wear clothing and personal protective equipment as specified on the pesticide label. If local nationals will be using the pesticides, the precautionary messages and use instructions shall be in English and Japanese. The Contracting Officer will designate locations for water used in formulating. Do not allow the equipment to overflow. Inspect equipment for leaks, clogging, wear, or damage and repair prior to application of pesticide.

3.19.2 Pesticide Treatment Plan

Include and update a pesticide treatment plan, as information becomes available. Include in the plan the sequence of treatment, dates, times, locations, pesticide trade name, EPA registration numbers if applicable, authorized uses, chemical composition, formulation, original and applied concentration, application rates of active ingredient (that is, pounds of active ingredient applied), equipment used for application and calibration of equipment. Comply with the JEGS or any applicable Federal GOJ national or prefectural and installation pest management record-keeping and reporting requirements as well as any additional Installation Project Office specific requirements in conformance with [DA AR 200-1 Chapter 5, Pest Management, Section 5-4 "Program requirements"] for data required to be reported to the Installation.

]3.20 POST CONSTRUCTION CLEANUP

Clean up areas used for construction in accordance with Contract Clause: "Cleaning Up". Unless otherwise instructed in writing by the Contracting Officer, remove traces of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. Grade parking area and similar temporarily used areas to conform with surrounding contours.

-- End of Section --

SECTION 01 57 19.00 30

TEMPORARY ENVIRONMENTAL CONTROLS KAB SUPPLEMENTAL
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA SW-846 (Third Edition; Update IV) Test Methods
for Evaluating Solid Waste:
Physical/Chemical Methods

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1053 Respirable Crystalline Silica
29 CFR 1910.1200 Hazard Communication
29 CFR 1926.1153 Respirable Crystalline Silica
40 CFR 50 National Primary and Secondary Ambient Air
Quality Standards
40 CFR 60 Standards of Performance for New
Stationary Sources
40 CFR 63 National Emission Standards for Hazardous
Air Pollutants for Source Categories
40 CFR 64 Compliance Assurance Monitoring
40 CFR 82 Protection of Stratospheric Ozone
40 CFR 112 Oil Pollution Prevention
40 CFR 122.26 Storm Water Discharges (Applicable to
State NPDES Programs, see section 123.25)
40 CFR 152 Pesticide Registration and Classification
Procedures
40 CFR 152 - 186 Pesticide Programs
40 CFR 241 Guidelines for Disposal of Solid Waste
40 CFR 243 Guidelines for the Storage and Collection
of Residential, Commercial, and
Institutional Solid Waste
40 CFR 258 Subtitle D Landfill Requirements

40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 261.7	Residues of Hazardous Waste in Empty Containers
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 262.11	Hazardous Waste Determination and Recordkeeping
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 266	Standards for the Management of Specific Hazardous Wastes and Specific Types of Hazardous Waste Management Facilities
40 CFR 268	Land Disposal Restrictions
40 CFR 273	Standards for Universal Waste Management
40 CFR 273.2	Standards for Universal Waste Management - Batteries
40 CFR 273.3	Standards for Universal Waste Management - Pesticides
40 CFR 273.4	Standards for Universal Waste Management - Mercury Containing Equipment
40 CFR 273.5	Standards for Universal Waste Management - Lamps
40 CFR 273.6	Applicability - Aerosol Cans
40 CFR 279	Standards for the Management of Used Oil
40 CFR 300	National Oil and Hazardous Substances Pollution Contingency Plan
40 CFR 300.125	National Oil and Hazardous Substances Pollution Contingency Plan - Notification and Communications
40 CFR 355	Emergency Planning and Notification
40 CFR 403	General Pretreatment Regulations for

Existing and New Sources of Pollution

40 CFR 745	Lead-Based Paint Poisoning Prevention in Certain Residential Structures
40 CFR 761	Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions
49 CFR 171	General Information, Regulations, and Definitions
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 172.101	Hazardous Material Regulation-Purpose and Use of Hazardous Material Table
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
49 CFR 178	Specifications for Packagings

WASHINGTON STATE ADMINISTRATIVE CODE (WAC)

WAC-173-303-573	Standards for Universal Waste Management
WAC-173-303-573(2)	Standards for Universal Waste Management - Batteries
WAC-173-303-573(3)	Standards for Universal Waste Management - Mercury-containing Equipment
WAC-173-303-573(5)	Standards for Universal Waste Management - Lamps

1.2 DEFINITIONS

1.2.1 Class I and II Ozone Depleting Substance (ODS)

Class I ODS is defined in Section 602(a) of The Clean Air Act. A list of Class I ODS can be found on the EPA website at the following weblink.
<https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances>.

Class II ODS is defined in Section 602(s) of The Clean Air Act. A list of Class II ODS can be found on the EPA website at the following weblink.
<https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances>.

1.2.2 Contractor Generated Hazardous Waste

Contractor generated hazardous waste is materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e., methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and

contaminated pesticide equipment rinse water.

1.2.3 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, or disposal.

1.2.4 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

1.2.5 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.6 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with 40 CFR 261. Hazardous debris also includes debris that exhibits a characteristic of hazardous waste in accordance with 40 CFR 261.

1.2.7 Hazardous Materials

Hazardous material is any material that: Is defined in 49 CFR 171, listed in 49 CFR 172, regulated as a hazardous material in accordance with 49 CFR 173; or requires a Safety Data Sheet (SDS) in accordance with 29 CFR 1910.1200; or during end use, treatment, handling, packaging, storage, transportation, or disposal meets or has components that meet or have potential to meet the definition of a hazardous waste as defined by 40 CFR 261 Subparts A, B, C, or D. Designation of a material by this definition, when separately regulated or controlled by other sections or directives, does not eliminate the need for adherence to that hazard-specific guidance which takes precedence over this section for "control" purposes. Such material includes ammunition, weapons, explosive actuated devices, propellants, pyrotechnics, chemical and biological warfare materials, medical and pharmaceutical supplies, medical waste and infectious materials, bulk fuels, radioactive materials, and other materials such as asbestos, mercury, and polychlorinated biphenyls (PCBs).

1.2.8 Hazardous Waste

Hazardous Waste is any material that meets the definition of a solid waste and exhibits a hazardous characteristic (ignitability, corrosivity, reactivity, or toxicity) as specified in 40 CFR 261, Subpart C, or contains a listed hazardous waste as identified in 40 CFR 261, Subpart D, or meets a state, local, or host nation definition of a hazardous waste.

1.2.9 Installation Pest Management Coordinator

Installation Pest Management Coordinator (IPMC) is the individual officially designated by the Installation Commander to oversee the Installation Pest Management Program and the Installation Pest Management Plan.

1.2.10 Land Application

Land Application means spreading or spraying discharge water at a rate that allows the water to percolate into the soil. No sheeting action, soil erosion, discharge into storm sewers, discharge into defined drainage areas, or discharge into the "waters of the United States" must occur. Comply with federal, state, and local laws and regulations.

1.2.11 Municipal Separate Storm Sewer System (MS4) Permit

MS4 permits are those held by municipalities or installations to obtain NPDES permit coverage for their stormwater discharges.

1.2.12 National Pollutant Discharge Elimination System (NPDES)

The NPDES permit program controls water pollution by regulating point sources that discharge pollutants into waters of the United States.

1.2.13 Oily Waste

Oily waste are those materials that are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from that POLs. Oily wastes also means materials, including wastewaters, centrifuge solids, filter residues or sludges, bottom sediments, tank bottoms, and sorbents which have come into contact with and have been contaminated by, POLs and may be appropriately tested and discarded in a manner which is in compliance with other state and local requirements.

This definition includes materials such as oily rags, "kitty litter" sorbent clay and organic sorbent material. These materials may be land filled provided that: It is not prohibited in other state regulations or local ordinances; the amount generated is "de minimus" (a small amount); it is the result of minor leaks or spills resulting from normal process operations; and free-flowing oil has been removed to the practicable extent possible. Large quantities of this material, generated as a result of a major spill or in lieu of proper maintenance of the processing equipment, are a solid waste. As a solid waste, perform a hazardous waste determination prior to disposal. As this can be an expensive process, it is recommended that this type of waste be minimized through good housekeeping practices and employee education.

1.2.14 Pesticide

Pesticide is any substance or mixture of substances intended for preventing, destroying, repelling, or mitigating any pest, or intended for use as a plant regulator, defoliant or desiccant.

1.2.15 Pesticide Treatment Plan

A plan for the prevention, monitoring, and control to eliminate pest infestation.

1.2.16 Pests

Pests are arthropods, birds, rodents, nematodes, fungi, bacteria, viruses, algae, snails, marine borers, snakes, weeds and other organisms (except for human or animal disease-causing organisms) that adversely affect readiness, military operations, or the well-being of personnel and animals; attack or damage real property, supplies, equipment, or vegetation; or are otherwise undesirable.

1.2.17 Project Pesticide Coordinator

The Project Pesticide Coordinator (PPC) is an individual who resides at a Civil Works Project office and who is responsible overseeing of pesticide application on project grounds.

1.2.18 Regulated Waste

Regulated waste are solid wastes that have specific additional federal, state, or local controls for handling, storage, or disposal.

1.2.19 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.20 Solid Waste

Solid waste is a solid, liquid, semi-solid or contained gaseous waste. A solid waste can be a hazardous waste, non-hazardous waste, or non-Resource Conservation and Recovery Act (RCRA) regulated waste. Types of solid waste typically generated at construction sites may include:

1.2.20.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure that exceeds 60 mm particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials [may][may not] be reinforced with or contain ferrous wire, rods, accessories and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.20.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps and plant roots. Marketable trees, grasses and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.20.3 Material Not Regulated As Solid Waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges;

regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.20.4 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, hazardous waste criteria in accordance with 40 CFR 261.

1.2.20.5 Recyclables

Recyclables are materials, equipment and assemblies such as doors, windows, door and window frames, plumbing fixtures, glazing and mirrors that are recovered and sold as recyclable, [wiring,] [insulated/non-insulated copper wire cable,] [wire rope,] and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Metal meeting the definition of lead contaminated or lead based paint contaminated [may][may not] be included as recyclable if sold to a scrap metal company. Paint cans that meet the definition of empty containers in accordance with 40 CFR 261.7 may be included as recyclable if sold to a scrap metal company.

1.2.20.6 Surplus Soil

Surplus soil is existing soil that is in excess of what is required for this work, including aggregates intended, but not used, for on-site mixing of concrete, mortars, and paving. Contaminated soil meeting the definition of hazardous material or hazardous waste is not included and must be managed in accordance with paragraph HAZARDOUS MATERIAL MANAGEMENT.

1.2.20.7 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included.

1.2.20.8 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, hardboard. Treated or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.2.21 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, or creeks meeting the definition of "waters of the United States". Surface discharges from construction sites are discrete, identifiable sources and require a permit from the governing agency. Comply with federal, state, and local laws and regulations.

1.2.22 Wastewater

Wastewater is the used water and solids that flow through a sanitary sewer

to a treatment plant.

1.2.22.1 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.23 Waters of the United States

Waters of the United States means Federally jurisdictional waters, including wetlands, that are subject to regulation under Section 404 of the Clean Water Act or navigable waters, as defined under the Rivers and Harbors Act.

1.2.24 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

1.2.25 Universal Waste

The universal waste regulations streamline collection requirements for certain hazardous wastes in the following categories: batteries, pesticides, mercury-containing equipment (for example, thermostats), and lamps (for example, fluorescent bulbs). The rule is designed to reduce hazardous waste in the municipal solid waste (MSW) stream by making it easier for universal waste handlers to collect these items and send them for recycling or proper disposal. These regulations can be found at 40 CFR 273.

1.2.26 Location Specific Universal Waste

Any of the following dangerous waste that are subject to the universal waste requirements of WAC-173-303-573: Batteries as described in WAC-173-303-573(2)); Lamps as described in WAC-173-303-573(5); Mercury-containing equipment as described in WAC-173-303-573(3).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are [for Contractor Quality Control approval.][for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.] Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Preconstruction Survey

Regulatory Notifications; G[, [____]]

Environmental Manager Qualifications; G[, [____]]

Employee Training Records; G[, [____]]

Environmental Protection Plan; G[, [____]]
Dirt and Dust Control Plan; G[, [____]]
Solid Waste Management Permit; G[, [____]]
[Stormwater Pollution Prevention Plan (SWPPP); G[, [____]]
][Stormwater Notice of Intent (for NPDES coverage under the general
permit for construction activities); G[, [____]]
] Spill Prevention Control And Countermeasure (SPCC) Plan; G[,
[____]]

SD-06 Test Reports

Monthly Solid Waste Disposal Report; G[, [____]]
[Inspection Reports
][Laboratory Analysis

] SD-07 Certificates

Employee Training Records; G[, [____]]
[Erosion and Sediment Control Inspector Qualifications
] Certificate of Competency

SD-11 Closeout Submittals

Regulatory Notifications; G[, [____]]
Assembled Employee Training Records; G[, [____]]
Solid Waste Management Permit; G[, [____]]
[Stormwater Pollution Prevention Plan Compliance Notebook; G[,
[____]]
][Stormwater Notice of Termination (for NPDES coverage under the
general permit for construction activities); G[, [____]]
][As-Built Topographic Survey
] Waste Determination Documentation; G[, [____]]
Project Solid Waste Disposal Documentation Report; G[, [____]]
Sales Documentation; G[, [____]]
[Contractor Certification
] Hazardous Waste/Debris Management; G[, [____]]
Disposal Documentation for Hazardous and Regulated Waste; G[,
[____]]

1.4 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract. Comply with federal, state, and local regulations pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

Tests and procedures assessing whether construction operations comply with Applicable Environmental Laws may be required. Analytical work must be performed by qualified laboratories; and where required by law, the laboratories must be certified.

1.4.1 Conformance with the Environmental Management System

Perform work under this contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities affect work performed under the contract.

Coordinate with the installation's EMS coordinator to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Office will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

1.5 SPECIAL ENVIRONMENTAL REQUIREMENTS

Comply with the special environmental requirements listed here [_____] and attached at the end of this section.

1.6 QUALITY ASSURANCE

1.6.1 Preconstruction Survey and Protection of Features

This paragraph supplements the Contract Clause PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS. Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer, and take

photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record. Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the Contractor's assigned storage area and access route(s), as applicable. The Contractor and the Contracting Officer will sign this survey report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract.

1.6.2 Regulatory Notifications

Provide regulatory notification requirements in accordance with federal, state and local regulations. In cases where the Government will also provide public notification (such as stormwater permitting), coordinate with the Contracting Officer. Submit copies of regulatory notifications to the Contracting Officer at least [____] days prior to commencement of work activities. Typically, regulatory notifications must be provided for the following (this listing is not all-inclusive): demolition, renovation, NPDES defined site work, construction, removal or use of a permitted air emissions source, and remediation of controlled substances (asbestos, hazardous waste, lead paint).

1.6.3 Environmental Brief

Attend an environmental brief to be included in the preconstruction meeting. Provide the following information: types, quantities, and use of hazardous materials that will be brought onto the installation; and types and quantities of wastes/wastewater that may be generated during the Contract. Discuss the results of the Preconstruction Survey at this time.

Prior to initiating any work on site, meet with the Contracting Officer and installation Environmental Office to discuss the proposed Environmental Protection Plan (EPP) or equipment local requirement. Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural and cultural resources, required reports, required permits, permit requirements (such as mitigation measures), and other measures to be taken.

[1.6.4 Environmental Manager

Appoint in writing an Environmental Manager for the project site. The Environmental Manager is directly responsible for coordinating contractor compliance with federal, state, local, and installation requirements. The Environmental Manager must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, manifesting, and disposal); implement the EPP; ensure environmental permits are obtained, maintained, and closed out; ensure compliance with Stormwater Program requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; and coordinate any remediation of regulated substances (lead, asbestos, PCB transformers). This can be a collateral position; however, the person in this position must be trained to adequately accomplish the following duties: ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure Contractor personnel are trained in 40 CFR

requirements in accordance with their position requirements; coordinate removal of waste containers; and maintain the Environmental Records binder and required documentation, including environmental permits compliance and close-out. Submit Environmental Manager Qualifications to the Contracting Officer.

]1.6.5 Employee Training Records

Prepare and maintain Employee Training Records throughout the term of the contract meeting applicable 40 CFR requirements. Provide Employee Training Records in the Environmental Records Binder.[Ensure every employee completes a program of classroom instruction or on-the-job training that teaches them to perform their duties in a way that ensures compliance with federal, state and local regulatory requirements for RCRA Large Quantity Generator. Provide a Position Description for each employee, by subcontractor, based on the Davis-Bacon Wage Rate designation or other equivalent method, evaluating the employee's association with hazardous and regulated wastes. This Position Description will include training requirements as defined in 40 CFR 265 for a Large Quantity Generator facility.] Submit these Assembled Employee Training Records to the Contracting Officer at the conclusion of the project, unless otherwise directed.

Train personnel to meet [EPA][state] requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of the United States, and endangered species and their habitat that are known to be in the area.[Provide copy of the Erosion and Sediment Control Inspector[Qualifications as defined by EPA][Certification as required by[state]].]

1.6.5.1 Pest Control Training

Trained personnel in pest control. Conduct a pest control meeting for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and pest infestation; familiarization with statutory and contractual pest control standards; installation and care of devices, and instruments, if required, for monitoring purposes to ensure adequate and continuous pest control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of waters of the United States, and endangered species and their habitat that are known to be in the area. Provide a Certificate of Competency for the personnel who will be conducting the pesticide application and management of pest control.

1.6.6 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with federal, state or local environmental laws or regulations, permits, and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed

corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. FAR 52.242-14 Suspension of Work provides that a suspension, delay, or interruption of work due to the fault or negligence of the Contractor allows for no adjustments to the contract for time extensions or equitable adjustments. In addition to a suspension of work, the Contracting Officer may use additional authorities under the contract or law.

1.7 ENVIRONMENTAL PROTECTION PLAN

The purpose of the EPP is to present an overview of known or potential environmental issues that must be considered and addressed during construction. Incorporate construction related objectives and targets from the installation's EMS into the EPP. Include in the EPP measures for protecting natural and cultural resources, required reports, and other measures to be taken. Meet with the Contracting Officer or Contracting Officer Representative to discuss the EPP and develop a mutual understanding relative to the details for environmental protection including measures for protecting natural resources, required reports, and other measures to be taken. [Submit the EPP within [15][____] days after [Contractor award][notice to proceed] and not less than [____][10] days before the[preconstruction] meeting.][Submit the EPP not less than 60 calendar days before scheduled final site or building design approval.] Revise the EPP throughout the project to include any reporting requirements, changes in site conditions, or contract modifications that change the project scope of work in a way that could have an environmental impact. No requirement in this section will relieve the Contractor of any applicable federal, state, and local environmental protection laws and regulations. During Construction, identify, implement, and submit for approval any additional requirements to be included in the EPP. Maintain the current version onsite.

The EPP includes, but is not limited to, the following elements:

1.7.1 General Overview and Purpose

1.7.1.1 Descriptions

A brief description of each specific plan required by environmental permit or elsewhere in this Contract such as[stormwater pollution prevention plan,][spill control plan,][solid waste management plan,][wastewater management plan,][air pollution control plan,][contaminant prevention plan,][pesticide treatment plan,][a historical, archaeological, cultural resources, biological resources and wetlands plan,][traffic control plan][Hazardous, Toxic and Radioactive Waste (HTRW) Plan][Non-Hazardous Solid Waste Disposal Plan][borrowing material plan][____].

1.7.1.2 Duties

The duties and level of authority assigned to the person(s) on the job site who oversee environmental compliance, such as who is responsible for adherence to the EPP, who is responsible for spill cleanup and training personnel on spill response procedures, who is responsible for manifesting hazardous waste to be removed from the site (if applicable), and who is responsible for training the Contractor's environmental protection personnel.

1.7.1.3 Procedures

A copy of any standard or project-specific operating procedures that will be used to effectively manage and protect the environment on the project site.

1.7.1.4 Communications

Communication and training procedures that will be used to convey environmental management requirements to Contractor employees and subcontractors.

1.7.1.5 Contact Information

Emergency contact information contact information (office phone number, cell phone number, and e-mail address).

1.7.2 General Site Information

1.7.2.1 Drawings

Drawings showing locations of proposed temporary excavations or embankments for haul roads, stream crossings, jurisdictional wetlands, material storage areas, structures, sanitary facilities, storm drains and conveyances, and stockpiles of excess soil.

1.7.2.2 Work Area

Work area plan showing the proposed activity in each portion of the area and identify the areas of limited use or nonuse. Include measures for marking the limits of use areas, including methods for protection of features to be preserved within authorized work areas and methods to control runoff and to contain materials on site, and a traffic control plan.

Show where any fuels, hazardous substances, solvents, or lubricants will be stored. Provide a spill plan to address any releases of those materials.

1.7.2.3 Documentation

A letter signed by an officer of the firm appointing the Environmental Manager and stating that person is responsible for managing and implementing the Environmental Program as described in this contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work.

1.7.3 Management of Natural Resources

- a. Land resources
- b. Tree protection
- c. Replacement of damaged landscape features
- d. Temporary construction
- e. Stream crossings

- f. Fish and wildlife resources
- g. Wetland areas

1.7.4 Protection of Historical and Archaeological Resources

- a. Objectives
- b. Methods

1.7.5 Stormwater Management and Control

- a. Ground cover
- b. Erodible soils
- c. Temporary measures
 - (1) Structural Practices
 - (2) Temporary and permanent stabilization
- d. Effective selection, implementation and maintenance of Best Management Practices (BMPs).
- e. Stormwater Pollution Prevention Plan (SWPPP).

1.7.6 Protection of the Environment from Waste Derived from Contractor Operations

Control and disposal of solid and sanitary waste.

Control and disposal of hazardous waste.

This item consist of the management procedures for hazardous waste to be generated. The elements of those procedures will coincide with the Installation Hazardous Waste Management Plan when within an installation. The Contracting Officer will provide a copy of the Installation Hazardous Waste Management Plan as applicable.

As a minimum, include the following:

- a. List of the types of hazardous wastes expected to be generated
- b. Procedures to ensure a written waste determination is made for appropriate wastes that are to be generated
- c. Sampling/analysis plan, including laboratory method(s) that will be used for waste determinations and copies of relevant laboratory certifications
- d. Methods and proposed locations for hazardous waste accumulation/storage (that is, in tanks or containers)
- e. Management procedures for storage, labeling, transportation, and disposal of waste (treatment of waste is not allowed unless specifically noted)
- f. Management procedures and regulatory documentation ensuring disposal

of hazardous waste complies with Land Disposal Restrictions (40 CFR 268)

- g. Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar
- h. Used oil management procedures in accordance with 40 CFR 279; Hazardous waste minimization procedures
- i. Plans for the disposal of hazardous waste by permitted facilities; and Procedures to be employed to ensure required employee training records are maintained.

1.7.7 Prevention of Releases to the Environment

Procedures to prevent releases to the environment

Notifications in the event of a release to the environment

1.7.8 Regulatory Notification and Permits

List what notifications and permit applications must be made. Some permits require up to 180 days to obtain. Demonstrate that those permits have been obtained or applied for by including copies of applicable environmental permits. The EPP will not be approved until the permits have been obtained.

1.7.9 Clean Air Act Compliance

1.7.9.1 Haul Route

Submit truck and material haul routes along with a Dirt and Dust Control Plan for controlling dirt, debris, and dust on Installation roadways. As a minimum, identify in the plan the subcontractor and equipment for cleaning along the haul route and measures to reduce dirt, dust, and debris from roadways.

1.7.9.2 Pollution Generating Equipment

Identify air pollution generating equipment or processes that may require federal, state, or local permits under the Clean Air Act. Determine requirements based on any current installation permits and the impacts of the project. Provide a list of all fixed or mobile equipment, machinery or operations that could generate air emissions during the project to the Installation Environmental Office (Air Program Manager). Ensure required permits are obtained prior to installing and operating applicable equipment/processes.

1.7.9.3 Stationary Internal Combustion Engines

Identify portable and stationary internal combustion engines that will be supplied, used or serviced. Comply with 40 CFR 60 Subpart IIII, 40 CFR 60 Subpart JJJJ, 40 CFR 63 Subpart ZZZZ, and local regulations as applicable. At minimum, include the make, model, serial number, manufacture date, size (engine brake horsepower), and EPA emission certification status of each engine. Maintain applicable records and log hours of operation and fuel use. Logs must include reasons for operation and delineate between maintenance/testing, emergency, and non-emergency operation.

1.7.9.4 Refrigerants

Identify management practices to ensure that heating, ventilation, and air conditioning (HVAC) work involving refrigerants complies with 40 CFR 82 requirements. Technicians must be certified, maintain copies of certification on site, use certified equipment and log work that requires the addition or removal of refrigerant. Any refrigerant reclaimed is the property of the Government, coordinate with the Installation Environmental Office to determine the appropriate turn in location.

1.7.9.5 Air Pollution-engineering Processes

Identify planned air pollution-generating processes and management control measures (including, but not limited to, spray painting, abrasive blasting, demolition, material handling, fugitive dust, and fugitive emissions). Log hours of operations and track quantities of materials used.

1.7.9.6 Monitoring

For the protection of public health, monitor and control contaminant emissions to the air from Hazardous, Toxic, and Radioactive Waste remedial action area sources to minimize short-term risks that might be posed to the community during implementation of the remedial alternative in accordance with the following.

- a. Perimeter Air Contaminant of Concern [____].
- b. Time Averaged Perimeter Action Levels [____].

Concentration	[____]
Time	[____]

- c. Perimeter Sampling/Monitoring Location[s] [____].
- d. Monitoring Instruments/Sampling and Analysis Methods [____].
- e. Staffing [____].

1.7.9.7 Compliant Materials

Provide the Government a list of SDSs for all hazardous materials proposed for use on site. Materials must be compliant with all Clean Air Act regulations for emissions including solvent and volatile organic compound contents, and applicable National Emission Standards for Hazardous Air Pollutants requirements. The Government may alter or limit use of specific materials as needed to meet installation permit requirements for emissions.

1.8 LICENSES AND PERMITS

Obtain licenses and permits required for the construction of the project and in accordance with FAR 52.236-7 Permits and Responsibilities. Notify the Government of all equipment that may require permits or special approvals that the Contractor plans to use on site. This paragraph

supplements the Contractor's responsibility under FAR 52.236-7 Permits and Responsibilities.

[a. The following permits have been obtained by the Government:

[(1) [____]

][(2) [____]

][(3) [____]

]]

[b. The following permits will be obtained by the Government:

[(1) [____]

][(2) [____]

][(3) [____]

]]1.9 ENVIRONMENTAL RECORDS BINDER

Maintain on-site a separate three-ring Environmental Records Binder and submit at the completion of the project. Make separate parts within the binder that correspond to each submittal listed under paragraph CLOSEOUT SUBMITTALS in this section.

1.10 PESTICIDE DELIVERY, STORAGE, AND HANDLING

1.10.1 Delivery and Storage

Deliver pesticides to the site in the original, unopened containers bearing legible labels indicating the EPA registration number and the manufacturer's registered uses. Store pesticides according to manufacturer's instructions and under lock and key when unattended.

1.10.2 Handling Requirements

Formulate, treat with, and dispose of pesticides and associated containers in accordance with label directions and use the clothing and personal protective equipment specified on the labeling for use during each phases of the application. Furnish SDSs for pesticide products.

1.11 SOLID WASTE MANAGEMENT PERMIT

Provide the Contracting Officer with written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Include in the report the locations where various types of waste will be disposed or recycled. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location state and local Solid Waste Management Permit or license showing such agency's approval of the disposal plan before transporting wastes off Government property.

1.11.1 Monthly Solid Waste Disposal Report

Monthly, submit a solid waste disposal report to the Contracting Officer. For each waste, the report will state the classification (using the definitions provided in this section), amount, location, and name of the business receiving the solid waste.

1.12 FACILITY HAZARDOUS WASTE GENERATOR STATUS

[_____] is designated as a[Large Quantity Generator][Small Quantity Generator][Conditionally Exempt-Small Quantity Generator]. Meet the regulatory requirements of this generator designation for any work conducted within the boundaries of this Installation. Comply with provisions of federal, state, and local regulatory requirements applicable to this generator status regarding training and storage, handling, and disposal of construction derived wastes.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 PROTECTION OF NATURAL RESOURCES

Minimize interference with, disturbance to, and damage to fish, wildlife, and plants, including their habitats. Prior to the commencement of activities, consult with the Installation Environmental Office as applicable, regarding rare species or sensitive habitats that need to be protected. The protection of rare, threatened, and endangered animal and plant species identified, including their habitats, is the Contractor's responsibility.[The following species are known and could be affected within the construction area: [_____]].]

Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work that is consistent with the requirements of the Installation Environmental Office or as otherwise specified. Confine construction activities to within the limits of the work indicated or specified.

3.1.1 Flow Ways

Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as specified and permitted.

3.1.2 Vegetation

Except in areas to be cleared, do not remove, cut, deface, injure, or destroy trees or shrubs without the Contracting Officer's permission. Do not fasten or attach ropes, cables, or guys to existing nearby trees for anchorages unless authorized by the Contracting Officer. Where such use of attached ropes, cables, or guys is authorized, the Contractor is responsible for any resultant damage.

Protect existing trees that are to remain to ensure they are not injured, bruised, defaced, or otherwise damaged by construction operations. Remove displaced rocks from uncleared areas. Coordinate with the Contracting Officer and Installation Environmental Office to determine appropriate action for trees and other landscape features scarred or damaged by equipment operations.

3.1.3 Streams

Stream crossings must allow movement of materials or equipment without violating water pollution control standards of the federal, state, and local governments. Construction of stream crossing structures must be in compliance with all required permits including, but not limited to, Clean Water Act Section 404, and Section 401 Water Quality.

The Contracting Officer's approval and appropriate permits are required before any equipment will be permitted to ford live streams. In areas where frequent crossings are required, install temporary culverts or bridges. Obtain Contracting Officer's approval prior to installation. Remove temporary culverts or bridges upon completion of work, and repair the area to its original condition unless otherwise required by the Contracting Officer.

3.2 STORMWATER

Do not discharge stormwater from construction sites to the sanitary sewer. If the water is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is specifically permitted. Obtain authorization in advance from the Installation Environmental Office for any release of contaminated water.

[3.2.1 Construction General Permit

Provide a Construction General Permit as required by 40 CFR 122.26 or [EPA][the State of [____]] General Permit. Under the terms and conditions of the permit, install, inspect, maintain BMPs, prepare stormwater erosion and sediment control inspection reports, and submit SWPPP inspection reports. Maintain construction operations and management in compliance with the terms and conditions of the general permit for stormwater discharges from construction activities.

3.2.1.1 Stormwater Pollution Prevention Plan

Submit a project-specific Stormwater Pollution Prevention Plan (SWPPP) to the Contracting Officer for approval, [within 30 days of Contract Award and]prior to the commencement of work. The SWPPP must meet the requirements of 40 CFR 122.26 and [the EPA General Permit][the [____] State General Permit] for stormwater discharges from construction sites.

Include the following:

- a. Comply with terms of the [EPA][state] general permit for stormwater discharges from construction activities. Prepare SWPPP in accordance with [state][EPA] requirements. Use [state][EPA guide Developing your Stormwater Pollution Prevention Plan located at <https://www.epa.gov/npdes/developing-stormwater-pollution-prevention-plan-swppp> to prepare the SWPPP.]
- b. Select applicable BMPs from EPA Fact Sheets located at <https://www.epa.gov/npdes/national-menu-best-management-practices-bmps-stormwater#constr> or in accordance with applicable state or local requirements.
- c. Include a completed copy of the Notice of Intent, BMP Inspection Report Template, and Stormwater Notice of Termination, except for the effective date.

3.2.1.2 Stormwater Notice of Intent for Construction Activities

[Prepare and submit the Notice of Intent for NPDES coverage under the general permit for construction activities to the Contracting Officer for review.

][Prepare and submit the Notice of Intent for NPDES coverage under the general permit for construction activities to the Contracting Officer for review and approval.

][Prepare and submit a Notice of Intent as a co-permittee to the Contracting Officer, for review and approval.

] Submit the approved NOI and appropriate permit fees onto the appropriate federal or state agency for approval. No land disturbing activities may commence without permit coverage. Maintain an approved copy of the SWPPP at the onsite construction office, and continually update as regulations require, reflecting current site conditions.

3.2.1.3 Inspection Reports

Submit "Inspection Reports" to the Contracting Officer in accordance with [EPA][the State of [____]]Construction General Permit.

3.2.1.4 Stormwater Pollution Prevention Plan Compliance Notebook

Create and maintain a three ring binder of documents that demonstrate compliance with the Construction General Permit. Include a copy of the permit Notice of Intent, proof of permit fee payment, SWPPP and SWPPP update amendments, inspection reports and related corrective action records, copies of correspondence with the [EPA][the [____]State Permitting Agency], and a copy of the permit Notice of Termination in the binder. At project completion, the notebook becomes property of the Government. Provide the compliance notebook to the Contracting Officer.

3.2.1.5 Stormwater Notice of Termination for Construction Activities

Submit a Notice of Termination to the Contracting Officer for approval once construction is complete and final stabilization has been achieved on all portions of the site for which the permittee is responsible. Once approved, submit the Notice of Termination to the appropriate state or federal agency.[Prepare as-built topographic survey information required by the permitting agency for certification of the stormwater management system, and provide to the Contracting Officer.]

3.2.2 Erosion and Sediment Control Measures

Provide erosion and sediment control measures in accordance with state and local laws and regulations. Preserve vegetation to the maximum extent practicable.

Erosion control inspection reports may be compiled as part of a stormwater pollution prevention plan inspection reports.

[3.2.2.1 Erosion Control

Prevent erosion by[mulching,][Compost Blankets,][Geotextiles,][temporary slope drains,][____]. Stabilize slopes by[chemical

stabilization,][sodding,][seeding,][_____] or such combination of these methods necessary for effective erosion control. Use of hay bales is prohibited.

[Provide seeding in accordance with Section 32 92 19 SEEDING.

]][3.2.2.2 Sediment Control Practices

Implement sediment control practices to divert flows from exposed soils, temporarily store flows, or otherwise limit runoff and the discharge of pollutants from exposed areas of the site. Implement sediment control practices prior to soil disturbance and prior to creating areas with concentrated flow, during the construction process to minimize erosion and sediment laden runoff. Include the following devices:[silt fence,][temporary diversion dikes,][storm drain inlet protection,][_____,][Location and details of installation and construction are indicated on the drawings].

]3.2.3 Work Area Limits

Mark the areas that need not be disturbed under this Contract prior to commencing construction activities. Mark or fence isolated areas within the general work area that are not to be disturbed. Protect monuments and markers before construction operations commence. Where construction operations are to be conducted during darkness, all markers must be visible in the dark. Personnel must be knowledgeable of the purpose for marking and protecting particular objects.

3.2.4 Contractor Facilities and Work Areas

Place field offices, staging areas, stockpile storage, and temporary buildings in areas designated on the drawings or as directed by the Contracting Officer. Move or relocate the Contractor facilities only when approved by the Government. Provide erosion and sediment controls for onsite borrow and spoil areas to prevent sediment from entering nearby waters. Control temporary excavation and embankments for plant or work areas to protect adjacent areas.

3.2.5 Municipal Separate Storm Sewer System (MS4) Management

Comply with the Installation's MS4 permit requirements.

3.3 SURFACE AND GROUNDWATER

3.3.1 Cofferdams, Diversions, and Dewatering

Construction operations for dewatering, removal of cofferdams, tailrace excavation, and tunnel closure must be constantly controlled to maintain compliance with existing state water quality standards and designated uses of the surface water body. Comply with[the State of [_____] water quality standards and anti-degradation provisions] [and][the Clean Water Act Section 404, Nation Wide Permit No. [_____]]. Do not discharge excavation ground water to the sanitary sewer, storm drains, or to surface waters without prior specific authorization in writing from the Installation Environmental Office or Contracting Officer. Discharge of hazardous substances will not be permitted under any circumstances. Use sediment control BMPs to prevent construction site runoff from directly entering any storm drain or surface waters.

If the construction dewatering is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is specifically permitted. Obtain authorization for any contaminated groundwater release in advance from the Installation Environmental Officer and the federal or state authority, as applicable. Discharge of hazardous substances will not be permitted under any circumstances.

3.3.2 Waters of the United States

Do not enter, disturb, destroy, or allow discharge of contaminants into waters of the United States[.] except as authorized herein. The protection of waters of the United States shown on the drawings in accordance with paragraph LICENSES AND PERMITS is the Contractor's responsibility. Authorization to enter specific waters of the United States identified does not relieve the Contractor from any obligation to protect other waters of the United States within, adjacent to, or in the vicinity of the construction site and associated boundaries.]

3.4 PROTECTION OF CULTURAL RESOURCES

3.4.1 Archaeological Resources

[Existing archaeological resources within the work area are shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract.]If, during excavation or other construction activities, any previously unidentified or unanticipated historical, archaeological, and cultural resources are discovered or found, activities that may damage or alter such resources will be suspended. Resources covered by this paragraph include, but are not limited to: any human skeletal remains or burials; artifacts; shell, midden, bone, charcoal, or other deposits; rock or coral alignments, pavings, wall, or other constructed features; and any indication of agricultural or other human activities. Upon such discovery or find, immediately notify the Contracting Officer so that the appropriate authorities may be notified and a determination made as to their significance and what, if any, special disposition of the finds should be made. Cease all activities that may result in impact to or the destruction of these resources. Secure the area and prevent employees or other persons from trespassing on, removing, or otherwise disturbing such resources. The Government retains ownership and control over archaeological resources.

[3.4.2 Historical Resources

Existing historical resources within the work area are shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract.

]3.5 AIR RESOURCES

Equipment operation, activities, or processes will be in accordance with 40 CFR 64 and state air emission and performance laws and standards.

3.5.1 Preconstruction Air Permits

Notify the Air Program Manager, through the Contracting Officer, at least 6 months prior to bringing equipment, assembled or unassembled, onto the Installation, so that air permits can be secured. Necessary permitting

time must be considered in regard to construction activities. Clean Air Act (CAA) permits must be obtained prior to bringing equipment, assembled or unassembled, onto the Installation.

[Permits will be provided by the Government.] [Confirm that these permits have been obtained.]

3.5.2 Oil or Dual-fuel Boilers and Furnaces

Provide product data and details for new, replacement, or relocated fuel fired boilers, heaters, or furnaces to the Installation Environmental Office (Air Program Manager) through the Contracting Officer. Data to be reported include: equipment purpose (water heater, building heat, process), manufacturer, model number, serial number, fuel type (oil type, gas type) size (MMBTU heat input). Provide in accordance with paragraph PRECONSTRUCTION AIR PERMITS.

3.5.3 Burning

[Burning is prohibited on the Government premises.] [Burning is allowed on Government premises[; confine fires to a closed vessel that is guarded and under constant surveillance until contents have burned out or have been extinguished]. [Burning must completely reduce the materials to ashes.]]

3.5.4 Class I [and II]ODS Prohibition

Class I [and II]ODS are Government property and must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Office to determine the appropriate location for turn in of all reclaimed refrigerant.

3.5.5 Venting of Refrigerant

Accidental venting of a refrigerant is a release and must be reported immediately to the Contracting Officer. Intentional venting of refrigerants (including most Non-ODS substitute refrigerants) is prohibited per 40 CFR 82.

3.5.6 EPA Certification Requirements

Heating and air conditioning technicians must be certified through an EPA-approved program. Maintain copies of certifications at the employees' places of business; technicians must carry certification wallet cards, as provided by environmental law.

3.5.7 Dust Control

Keep dust down at all times, including during nonworking periods.[Sprinkle or treat, with dust suppressants, the soil at the site, haul roads, and other areas disturbed by operations.] Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Do not unnecessarily shake bags of cement, concrete mortar, or plaster. Since these products contain Crystalline Silica, comply with the applicable OSHA standard, 29 CFR 1910.1053 or 29 CFR 1926.1153 for controlling exposure to Crystalline Silica Dust.

3.5.7.1 Particulates

Dust particles, aerosols and gaseous by-products from construction activities, and processing and preparation of materials (such as from asphaltic batch plants) must be controlled at all times, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed 40 CFR 50, state, and local air pollution standards or that would cause a hazard or a nuisance. Sprinkling, chemical treatment of an approved type, baghouse, scrubbers, electrostatic precipitators, or other methods will be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp. Provide sufficient, competent equipment available to accomplish these tasks. Perform particulate control as the work proceeds and whenever a particulate nuisance or hazard occurs. Comply with state and local visibility regulations.

3.5.7.2 Abrasive Blasting

Blasting operations cannot be performed without prior approval of the Installation Air Program Manager. The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive agent, paint chips, and other debris.[Perform work involving removal of hazardous material in accordance with 29 CFR 1910.]

3.5.8 Odors

Control odors from construction activities. The odors must be in compliance with state regulations and local ordinances and may not constitute a health hazard.

3.6 WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of waste. Include procedures for pollution prevention/ hazardous waste minimization in the Hazardous Waste Management Section of the EPP. Obtain a copy of the installation's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EPP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in the construction when requesting information.

3.6.1 Salvage, Reuse and Recycle

Identify anticipated materials and waste for salvage, reuse, and recycling. Describe actions to promote material reuse, resale or recycling. To the extent practicable, all scrap metal must be sent for reuse or recycling and will not be disposed of in a landfill.

Include the name, physical address, and telephone number of the hauler, if transported by a franchised solid waste hauler. Include the destination and, unless exempted, provide a copy of the state or local permit (cover) or license for recycling.

3.6.2 Nonhazardous Solid Waste Diversion Report

Maintain an inventory of nonhazardous solid waste diversion and disposal of construction and demolition debris. Submit a report to [____] through the Contracting Officer on the first working day after each fiscal year quarter, starting the first quarter that nonhazardous solid waste has been generated. Include the following in the report:

Construction and Demolition (C&D) Debris Disposed	[____] [cubic yards][tons],[cubic meters] as appropriate
C&D Debris Recycled	[____] [cubic yards][tons],[cubic meters] as appropriate
C&D Debris Composted	[____] [cubic yards][tons],[cubic meters] as appropriate
Total C&D Debris Generated	[____] [cubic yards][tons],[cubic meters] as appropriate
Waste Sent to Waste-To-Energy Incineration Plant (This amount should not be included in the recycled amount)	[____] [cubic yards][tons],[cubic meters] as appropriate

3.7 WASTE MANAGEMENT AND DISPOSAL

3.7.1 Waste Determination Documentation

Complete a Waste Determination form (provided at the pre-construction conference) for Contractor-derived wastes to be generated. All potentially hazardous solid waste streams that are not subject to a specific exclusion or exemption from the hazardous waste regulations (e.g., scrap metal, domestic sewage) or subject to special rules, (lead-acid batteries and precious metals) must be characterized in accordance with the requirements of 40 CFR 262.11 or corresponding applicable state or local regulations. Base waste determination on user knowledge of the processes and materials used, and analytical data when necessary. Consult with the Installation environmental staff for guidance on specific requirements. Attach support documentation to the Waste Determination form. As a minimum, provide a Waste Determination form for the following waste (this listing is not exhaustive): oil- and latex-based painting and caulking products, solvents, adhesives, aerosols, petroleum products, and containers of the original materials.

[3.7.1.1 Sampling and Analysis of Waste

3.7.1.1.1 Waste Sampling

Sample waste in accordance with EPA SW-846. Clearly mark each sampled drum or container with the Contractor's identification number, and cross reference to the chemical analysis performed.

3.7.1.1.2 Laboratory Analysis

Follow the analytical procedure and methods in accordance with the 40 CFR 261. Provide analytical results and reports performed to the Contracting Officer. Coordinate all activities with Installation Hazardous Waste Manager.

3.7.1.1.3 Analysis Type

Identify hazardous waste by analyzing for the following characteristics:[
ignitability,][corrosivity,][reactivity,][toxicity based on TCLP
results,] [_____].

]3.7.2 Solid Waste Management

3.7.2.1 Project Solid Waste Disposal Documentation Report

Provide copies of the waste handling facilities' weight tickets, receipts, bills of sale, and other sales documentation. In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. The sales documentation[Contractor certification] must include the receiver's tax identification number and business, EPA or state registration number, along with the receiver's delivery and business addresses and telephone numbers. For each solid waste retained for the Contractor's own use, submit the information previously described in this paragraph on the solid waste disposal report. Prices paid or received do not have to be reported to the Contracting Officer unless required by other provisions or specifications of this Contract or public law.

3.7.2.2 Control and Management of Solid Wastes

Pick up solid wastes, and place in covered containers that are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Employ segregation measures so that no hazardous or toxic waste will become co-mingled with non-hazardous solid waste.[Transport solid waste off Government property and dispose of it in compliance with 40 CFR 260, state, and local requirements for solid waste disposal. A Subtitle D RCRA permitted landfill is the minimum acceptable offsite solid waste disposal option. Verify that the selected transporters and disposal facilities have the necessary permits and licenses to operate.][Haul waste materials to the Government landfill site[shown on the drawings][designated by the Contracting Officer].][Comply with site procedures.][Segregate and separate treated wood components disposed at a lined landfill approved to accept this waste in accordance with local and state regulations] Solid waste disposal offsite must comply with most stringent local, state, and federal requirements, including 40 CFR 241, 40 CFR 243, and 40 CFR 258.

Manage hazardous material used in construction, including but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, in accordance with 49 CFR 173.

3.7.3 Control and Management of Hazardous Waste

Do not dispose of hazardous waste on Government property. Do not discharge any waste to a sanitary sewer, storm drain, or to surface waters or conduct waste treatment or disposal on Government property without written approval of the Contracting Officer and Installation Hazardous Waste Manager.

3.7.3.1 Hazardous Waste/Debris Management

Identify construction activities that will generate hazardous waste or debris. Provide a documented waste determination for resultant waste streams. Identify, label, handle, store, and dispose of hazardous waste or debris in accordance with federal, state, and local regulations, including 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, 40 CFR 266, and 40 CFR 268.

Manage hazardous waste in accordance with the approved Hazardous Waste Management Section of the EPP. Store hazardous wastes in approved containers in accordance with 49 CFR 173 and 49 CFR 178. Hazardous waste generated within the confines of Government facilities is identified as being generated by the Government. Prior to removal of any hazardous waste from Government property, hazardous waste manifests must be signed by personnel from the Installation Environmental Office. Do not bring hazardous waste onto Government property. Provide the Contracting Officer with a copy of waste determination documentation for any solid waste streams that have any potential to be hazardous waste or contain any chemical constituents listed in 40 CFR 372-SUBPART D.

3.7.3.2 Waste Storage/Satellite Accumulation/90 Day Storage Areas

Accumulate hazardous waste at satellite accumulation points and in compliance with 40 CFR 262 and applicable state or local regulations. Individual waste streams will be limited to 208 liter of accumulation (or 0.95 liter for acutely hazardous wastes). If the Contractor expects to generate hazardous waste at a rate and quantity that makes satellite accumulation impractical, the Contractor may request a temporary 90-day or 180-day, as appropriate, accumulation point be established. Submit a request in writing to the Contracting Officer and provide the following information (Attach Site Plan to the Request):

Contract Number	[_____]
Contractor	[_____]
Haz/Waste or Regulated Waste POC	[_____]
Phone Number	[_____]
Type of Waste	[_____]
Source of Waste	[_____]
Emergency POC	[_____]
Phone Number	[_____]
Location of the Site	[_____]

Attach a Waste Determination form for the expected waste streams. Allow 10 working days for processing this request. Additional compliance requirements (e.g., training and contingency planning) that may be required are the responsibility of the Contractor. Barricade the designated area where waste is being stored and post a sign identifying as follows:

"DANGER - UNAUTHORIZED PERSONNEL KEEP OUT"

3.7.3.3 Hazardous Waste Disposal

[3.7.3.3.1 Responsibilities for Contractor's Disposal

Provide hazardous waste manifest to the Installations Environmental Office for review, approval, and signature prior to shipping waste off Government property.

3.7.3.3.1.1 Services

Provide service necessary for the final treatment or disposal of the hazardous material or waste in accordance with 40 CFR 260 - 40 CFR 279, local, and state, laws and regulations, and the terms and conditions of the Contract within 60 days after the materials have been generated. These services include necessary personnel, labor, transportation, packaging, detailed analysis (if required for disposal or transportation, include manifesting or complete waste profile sheets, equipment, and compile documentation).

3.7.3.3.1.2 Samples

Obtain a representative sample of the material generated for each job done to provide waste stream determination.

3.7.3.3.1.3 Analysis

Analyze each sample taken and provide analytical results to the Contracting Officer. See paragraph WASTE DETERMINATION DOCUMENTATION.

3.7.3.3.1.4 Labeling

During waste accumulation label all containers in accordance with 40 CFR 262. Prior to offering a waste for off-site transport, determine the Department of Transportation's (DOT's) proper shipping names for waste in accordance with 49 CFR 172 (each container requiring disposal) and demonstrate to the Contracting Officer how this determination is developed and supported by the sampling and analysis requirements contained herein. Label all containers of hazardous waste with the words "Hazardous Waste" or other words to describe the contents of the container in accordance with 40 CFR 262 and applicable state or local regulations.

]3.7.3.3.2 Contractor Disposal Turn-In Requirements

Hazardous waste generated must be disposed of in accordance with the following conditions to meet installation requirements:

- a. Drums must be compatible with waste contents and drums must meet DOT requirements for 49 CFR 173 for transportation of materials.
- b. Band drums to wooden pallets.
- c. No more than three 208 liter drums or two 321 liter over packs are to be banded to a pallet.
- d. Band using 32 millimeters minimum band on upper third of drum.
- e. Provide label in accordance with 49 CFR 172.101.
- f. Leave 7 to 12 centimeters of empty space above volume of material.

]3.7.3.4 Universal Waste Management

Manage the following categories of universal waste in accordance with federal, state, and local requirements and installation instructions:

- a. Batteries as described in 40 CFR 273.2
- b. Lamps as described in 40 CFR 273.5
- c. Mercury-containing equipment as described in 40 CFR 273.4
- d. Aerosol cans as described in 40 CFR 273.6
- e. Pesticides as described in 40 CFR 273.3

Mercury is prohibited in the construction of this facility, unless specified otherwise, and with the exception of mercury vapor lamps and fluorescent lamps. Dumping of mercury-containing materials and devices such as mercury vapor lamps, fluorescent lamps, and mercury switches, in rubbish containers is prohibited. Remove without breaking, pack to prevent breakage, and transport out of the activity in an unbroken condition for disposal as directed.

3.7.3.5 Electronics End-of-Life Management

Recycle or dispose of electronics waste, including, but not limited to, used electronic devices such computers, monitors, hard-copy devices, televisions, mobile devices, in accordance with 40 CFR 260-262, state, and local requirements, and installation instructions.

3.7.3.6 Disposal Documentation for Hazardous and Regulated Waste

Contact the Contracting Officer [or designated representative]for the facility RCRA identification number that is to be used on each manifest.

- [Submit a copy of the applicable EPA and or state permit(s), manifest(s), or license(s) for transportation, treatment, storage, and disposal of hazardous and regulated waste by permitted facilities. Hazardous or toxic waste manifests must be reviewed, signed, and approved by the Contracting Officer before the Contractor may ship waste. To obtain specific disposal instructions, coordinate with the Installation Environmental Office. Refer to location special requirements for the Installation Point of Contact information.

]3.7.4 Releases/Spills of Oil and Hazardous Substances

3.7.4.1 Response and Notifications

Exercise due diligence to prevent, contain, and respond to spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with 40 CFR 300. Maintain spill cleanup equipment and materials at the work site. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of any releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the Installation Fire Department, the Installation Command Duty Officer, the Installation Environmental Office, the

Contracting Officer[and the state or local authority].

Submit verbal and written notifications as required by the federal (40 CFR 300.125 and 40 CFR 355), state, local regulations and instructions. Provide copies of the written notification and documentation that a verbal notification was made within 20 days. Spill response must be in accordance with 40 CFR 300 and applicable state and local regulations. Contain and clean up these spills without cost to the Government.

3.7.4.2 Clean Up

Clean up hazardous and non-hazardous waste spills. Reimburse the Government for costs incurred including sample analysis materials, clothing, equipment, and labor if the Government will initiate its own spill cleanup procedures, for Contractor- responsible spills, when: Spill cleanup procedures have not begun within one hour of spill discovery/occurrence; or, in the Government's judgment, spill cleanup is inadequate and the spill remains a threat to human health or the environment.

3.7.5 Mercury Materials

Immediately report to the Environmental Office and the Contracting Officer instances of breakage or mercury spillage. Clean mercury spill area to the satisfaction of the Contracting Officer.

Do not recycle a mercury spill cleanup; manage it as a hazardous waste for disposal.

3.7.6 Wastewater

3.7.6.1 Disposal of Wastewater

Disposal of wastewater must be as specified below.

3.7.6.1.1 Treatment

Do not allow wastewater from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, and forms to enter water ways or to be discharged prior to being treated to remove pollutants. Dispose of the construction- related waste water[off-Government property in accordance with 40 CFR 403, state, regional, and local laws and regulations.][by collecting and placing it in a retention pond where suspended material can be settled out or the water can evaporate to separate pollutants from the water. The site for the retention pond must be coordinated and approved with the Contracting Officer. The residue left in the pond prior to completion of the project must be removed, tested, and disposed of off-Government property in accordance with federal, state, and local laws and regulations. Backfill the area to the original grade, top-soiled, and seeded or sodded.[Test the water in the retention pond for [_____] and have the results reviewed and approved by the Contracting Officer prior to being discharged or disposed of off- Government property].]

3.7.6.1.2 Surface Discharge

For discharge of ground water,[obtain a state or federal permit specific for pumping and discharging ground water prior to surface discharging.][

Surface discharge in accordance with federal, state, and local laws and regulations.][Surface discharge in accordance with the requirements of the NPDES or state STORMWATER DISCHARGES FROM CONSTRUCTION SITES permit.]

3.7.6.1.3 Land Application

Water generated from the flushing of lines after[disinfection or disinfection in conjunction with hydrostatic testing][hydrostatic testing] must be[land- applied in accordance with federal, state, and local laws and regulations for land application][discharged into the sanitary sewer with prior approval and notification to the Wastewater Treatment Plant's Operator].

3.8 HAZARDOUS MATERIAL MANAGEMENT

Include hazardous material control procedures in the Safety Plan, in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. Do not bring hazardous material onto Government property that does not directly relate to requirements for the performance of this contract. Submit an SDS and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the installation. Typical materials requiring SDS and quantity reporting include, but are not limited to, oil and latex based painting and caulking products, solvents, adhesives, aerosol, and petroleum products. Use hazardous materials in a manner that minimizes the amount of hazardous waste generated. Containers of hazardous materials must have National Fire Protection Association labels or their equivalent. Certify that hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste, in accordance with 40 CFR 261 and state[and installation] requirements.

3.9 PREVIOUSLY USED EQUIPMENT

Clean previously used construction equipment prior to bringing it onto the project site. Equipment must be free from soil residuals, egg deposits from plant pests, noxious weeds, and plant seeds. Consult with the U.S. Department of Agriculture jurisdictional office for additional cleaning requirements.

[3.10 CONTROL AND MANAGEMENT OF ASBESTOS-CONTAINING MATERIAL (ACM)

Manage and dispose of asbestos- containing waste in accordance with all applicable federal, state, and local (or Host Nation) requirements. Refer to Section 02 82 00 ASBESTOS REMEDIATION. Manifest asbestos-containing waste and provide the manifest to the Contracting Officer. Notifications to the regulatory authorities and Installation Air Program Manager are required before starting any asbestos work.

][3.11 CONTROL AND MANAGEMENT OF LEAD-BASED PAINT (LBP)

Manage and dispose of lead-contaminated waste in accordance with 40 CFR 745 and Section 02 83 00 LEAD REMEDIATION. Manifest any lead-contaminated waste and provide the manifest to the Contracting Officer.

][3.12 CONTROL AND MANAGEMENT OF POLYCHLORINATED BIPHENYLS (PCBs)

Manage and dispose of PCB-contaminated waste in accordance with 40 CFR 761

and Section 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBs).

] [3.13 CONTROL AND MANAGEMENT OF LIGHTING BALLAST AND LAMPS CONTAINING PCBs

Manage and dispose of contaminated waste in accordance with 40 CFR 761. [Refer to Section 02 84 16 HANDLING OF LIGHTING BALLASTS AND LAMPS CONTAINING PCBs AND MERCURY.]

] [3.14 MILITARY MUNITIONS

In the event military munitions, as defined in 40 CFR 260, are discovered or uncovered, immediately stop work in that area and immediately inform the Contracting Officer.

] 3.15 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING

POL products include flammable or combustible liquids, such as gasoline, diesel, lubricating oil, used engine oil, hydraulic oil, mineral oil, and cooking oil. Store POL products and fuel equipment and motor vehicles in a manner that affords the maximum protection against spills into the environment. Manage and store POL products in accordance with EPA 40 CFR 112, and other federal, state, regional, and local laws and regulations. Use secondary containments, dikes, curbs, and other barriers, to prevent POL products from spilling and entering the ground, storm or sewer drains, stormwater ditches or canals, or navigable waters of the United States. Describe in the EPP (see paragraph ENVIRONMENTAL PROTECTION PLAN) how POL tanks and containers must be stored, managed, and inspected and what protections must be provided. [Storage of oil, including fuel, on the project site is not allowed. Fuel must be brought to the project site each day that work is performed.] [Storage of fuel on the project site must be in accordance with EPA, state, and local laws and regulations and paragraph OIL STORAGE INCLUDING FUEL TANKS.]

3.15.1 Used Oil Management

Manage used oil generated on site in accordance with 40 CFR 279. Determine if any used oil generated while onsite exhibits a characteristic of hazardous waste. Used oil containing 1,000 parts per million of solvents is considered a hazardous waste and disposed of at the Contractor's expense. Used oil mixed with a hazardous waste is also considered a hazardous waste. Dispose in accordance with paragraph HAZARDOUS WASTE DISPOSAL.

3.15.2 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection for oil storage tanks. A berm used to provide secondary containment must be of sufficient size and strength to contain the contents of the tanks plus 12 centimeters freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks. If tanks and containers with an aggregate aboveground capacity greater than 5000 liter will be used onsite (only containers with a capacity of 208 liter or greater are counted), provide and implement a Spill Prevention Control and Countermeasure (SPCC) plan

meeting the requirements of 40 CFR 112. Do not bring underground storage tanks to the installation for Contractor use during a project. Submit the SPCC plan to the Contracting Officer for approval.

Monitor and remove any rainwater that accumulates in open containment dikes or berms. Inspect the accumulated rainwater prior to draining from a containment dike to the environment, to determine there is no oil sheen present.

3.16 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS WASTES

If petroleum-contaminated soil, or suspected hazardous waste is found during construction that was not identified in the Contract documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.17 PEST MANAGEMENT

In order to minimize impacts to existing fauna and flora, coordinate with the Installation Pest Management Coordinator (IPMC) or Project Pesticide Coordinator (PPC), through the Contracting Officer, at the earliest possible time prior to pesticide application. Discuss integrated pest management strategies including prevention of invasive species as applicable with the [IPMC][PPC] and receive concurrence from the [IPMC][PPC] through the Contracting Officer prior to the application of any pesticide associated with these specifications. Provide Installation Project Office Pest Management personnel the opportunity to be present at meetings concerning treatment measures for pest or disease control and during application of the pesticide. [For termiticide requirements, see[Section 31 31 16.13 CHEMICAL TERMITE CONTROL][Section 31 31 16.19 TERMITE CONTROL BARRIERS]] The use and management of pesticides are regulated under 40 CFR 152 - 186.

3.17.1 Application

Apply pesticides using a state-certified pesticide applicator in accordance with EPA label restrictions and recommendation. The certified applicator must wear clothing and personal protective equipment as specified on the pesticide label. The Contracting Officer will designate locations for water used in formulating. Do not allow the equipment to overflow. Inspect equipment for leaks, clogging, wear, or damage and repair prior to application of pesticide.

3.17.2 Pesticide Treatment Plan

Include and update a pesticide treatment plan, as information becomes available. Include in the plan the sequence of treatment, dates, times, locations, pesticide trade name, EPA registration numbers, authorized uses, chemical composition, formulation, original and applied concentration, application rates of active ingredient (that is, pounds of active ingredient applied), equipment used for application and calibration of equipment. Comply with 40 CFR 152-189, state, regional, and local pest management record-keeping and reporting requirements as well as any additional Installation Project Office specific requirements in conformance with [DA AR 200-1 Chapter 5, Pest Management, Section 5-4 "Program requirements"] for data required to be reported to the Installation.

3.18 CHLORDANE

Evaluate excess soils and concrete foundation debris generated during the demolition of housing units or other wooden structures for the presence of chlordane or other pesticides prior to reuse or final disposal.

3.19 SOUND INTRUSION

Make the maximum use of low-noise emission products, as certified by the EPA. Blasting or use of explosives are not permitted without written permission from the Contracting Officer, and then only during the designated times. Confine pile-driving operations to the period between [_____] [8 a.m.] and [_____] [4 p.m.], [_____] [Monday through Friday], exclusive of holidays, unless otherwise specified.

Keep construction activities under surveillance and control to minimize environment damage by noise. Comply with the provisions of the State of [_____] rules.

3.20 POST CONSTRUCTION CLEANUP

Clean up areas used for construction in accordance with Contract Clause: "Cleaning Up". Unless otherwise instructed in writing by the Contracting Officer, remove traces of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. Grade parking area and similar temporarily used areas to conform with surrounding contours.

-- End of Section --

SECTION 01 57 19.01

TEMPORARY ENVIRONMENTAL CONTROLS - AIR FORCE INSTALLATIONS
01/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 14001 (2015) Environmental Management Systems - Requirements with Guidance for Use

U.S. AIR FORCE (USAF)

AFI 23-204 (2012) Organizational Fuel Tanks

AFMAN 32-1067 (4 Aug 2020) Water and Fuel Systems

AFI 32-7001 (04 Oct 2019) Environmental Management

AFMAN 32-7002 (4 Feb 2020) Environmental Compliance and Pollution Prevention

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS (Apr 2024) Japan Environmental Governing Standards

UFC 3-460-01 (2015) Design: Petroleum Fuel Facilities

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910 Occupational Safety and Health Standards

1.2 DEFINITIONS

In some cases, definitions are given only for illustrative purposes. Prevailing JEGS definitions shall be used for environmental compliance requirements.

1.2.1 Aboveground Storage Tank

[A portable or fixed POL aboveground storage container as defined in the JEGS with a capacity greater than 55 gallons.][A container with a capacity greater than 55 gallons which is a fixed installation or is portable and is either an aboveground storage container as defined in the JEGS or a below ground storage container as defined in the JEGS which contains POL, hazardous materials, hazardous substances or hazardous waste.]

1.2.2 Asbestos Containing Material (ACM)

Any material containing greater than or equal to one-tenth of one percent (0.1 percent) asbestos, by weight.

1.2.3 Bulky Waste

Large items of solid waste such as household appliances, furniture, large auto parts, trees, branches, stumps, and other oversize wastes whose large size precludes or complicates their handling by normal solid waste collection, processing, or disposal methods.

1.2.4 Chemical Wastes

Salts, acids, alkalis, herbicides, pesticides, organic chemicals, and spent products, which serve no purpose.

1.2.5 Class I and II Ozone Depleting Substance (ODS)

Class I and II ODS are listed in the JEGS.

1.2.6 Construction and Demolition Waste

The waste building materials, packaging, and rubble resulting from construction, remodeling, repair and demolition operations on pavements, houses, commercial buildings, and other structures.

1.2.7 Contractor Generated Hazardous Waste

Contractor generated hazardous waste is materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e. methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment rinse water.

1.2.8 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, or disposal.

1.2.9 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

1.2.10 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other

pollutants.

1.2.11 Food Waste

Organic residues generated by the handling, storage, sale, preparation, cooking, and serving of foods (commonly called garbage).

1.2.12 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with the JEGS. Hazardous debris also includes debris that exhibits a characteristic of hazardous waste in accordance with the JEGS.

1.2.13 Hazardous Materials

Hazardous material is any material that is capable of posing an unreasonable risk to health, safety, or the environment if improperly handled, stored, issued, transported, labeled, or disposed because it displays a characteristic listed in the JEGS. Munitions are excluded.

1.2.14 Hazardous Substances

Any substance having the potential to do serious harm to human health or the environment if spilled or released in reportable quantity. A list of these substances and the corresponding reportable quantities is contained in the JEGS.

1.2.15 Hazardous Waste

Hazardous Waste is discarded material that may be solid, semi-solid, liquid, or contained gas, and either exhibits a characteristic of a hazardous waste as defined in the JEGS. Excluded from this definition are domestic sewage sludge, household wastes, and medical wastes.

1.2.16 Hazardous Waste Accumulation Point

A shop, site, or other work center where hazardous wastes are accumulated until removed to a Hazardous Waste Storage Area (HWSA) or shipped for treatment or disposal. A HWAP may be used to accumulate no more than 208 liters (55 gallons) of hazardous waste, or 1 liter (1 quart) of acute hazardous waste, from each waste stream. The HWAP must be at or near the point of generation and under the control of the operator.

1.2.17 Installation Pest Management Consultant

Installation Pest Management Consultant (IPMC) is the professional DoD pest management personnel located at component headquarters, field operating agencies, major commands, facilities engineering field divisions or activities, or area support activities who provide technical and management guidance for the conduct of installation pest management operations. Some pest management consultants may be designated by their component as certifying officials.

1.2.18 Land Application

Land Application means spreading or spraying discharge water at a rate that allows the water to percolate into the soil. No sheeting action,

soil erosion, discharge into storm sewers, or discharge into defined drainage areas (includes drainage ditches, streams, rivers, ocean, etc.) must occur. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

1.2.19 Lead-Based Paint (LBP)

Paint or other surface coatings that contain greater than or equal to 1.0 milligram per square centimeter (cm), or 0.5 percent by weight, or 5,000 ppm of lead by weight.

1.2.20 Oily Wastes

Oily wastes are those materials that are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from those POLs. Oily wastes also means materials, including wastewaters, centrifuge solids, filter residues or sludge, bottom sediments, tank bottoms, and sorbents which have come into contact with and have been contaminated by, POLs and may be appropriately tested and discarded in a manner which is in compliance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

This definition includes materials such as oily rags, "kitty litter" sorbent clay and organic sorbent material. These materials may be land filled provided that: It is not prohibited in the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements; the amount generated is "de minimus" (a small amount); it is the result of minor leaks or spills resulting from normal process operations; and free-flowing oil has been removed to the practicable extent possible. Large quantities of this material, generated as a result of a major spill or in lieu of proper maintenance of the processing equipment, are a solid waste. As a solid waste, perform a hazardous waste determination prior to disposal. As this can be an expensive process, it is recommended that this type of waste be minimized through good housekeeping practices and employee education.

1.2.21 Pesticide

Pesticide is any substance or mixture of substances, including biological control agents that may prevent, destroy, repel, or mitigate pests.

1.2.22 Pesticide Treatment Plan

A plan for the prevention, monitoring, and control to eliminate pest infestation.

1.2.23 Pests

Anthropods, birds, rodents, nematodes, fungi, bacteria, viruses, algae, snails, marine borers, snakes, weeds, undesirable vegetation, and other organisms (except for microorganisms that cause human or animal disease) that adversely affect the well-being of humans or animals; attack real property, supplies, equipment, or vegetation; or are otherwise undesirable.

1.2.24 Petroleum, Oil, and Lubricants

Refined petroleum, oils, and lubricants, including, but not limited to, petroleum, fuel, lubricant oils, synthetic oils, mineral oils, animal fats, vegetable oil, sludge, and POL mixed with wastes other than dredged

spoil.

1.2.25 Polychlorinated Biphenyl (PCB)

Any PCB article, PCB article container, PCB container, or PCB equipment that deliberately or unintentionally contains or has as a part of it any detectable concentration of PCB.

1.2.26 Regulated Waste

Regulated waste are solid wastes that have specific additional Misawa AB, JEGS, Federal, or GOJ national and prefectural controls for handling, storage, or disposal.

1.2.27 Rubbish

A general term for solid waste, excluding food wastes and ashes, taken from residences, commercial establishments, and institutions.

1.2.28 Sanitary Waste

a. Sewage: Wastes characterized as domestic sanitary sewage.

b. Garbage: Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

1.2.29 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.30 Solid Waste

Solid waste is garbage, refuse, sludge, and other discarded materials, including solid, semi-solid, liquid, and contained gaseous materials resulting from industrial and commercial operations and from community activities. It does not include solids or dissolved material in domestic sewage or their significant pollutants in water resources, such as silt, dissolved or suspended solids in industrial wastewater effluent, dissolved materials in irrigation return flows, or other common water pollutants.

Types of solid waste typically generated at construction sites may include:

1.2.30.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure that exceeds 60 mm particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials [may][may not] be reinforced with or contain ferrous wire, rods, accessories and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.30.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and

grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps and plant roots. Marketable trees, grasses and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.30.3 Material Not Regulated as Solid Waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.30.4 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, characteristic of a hazardous waste as defined in the JEGS or is listed as a hazardous waste in the JEGS. Excluded from this definition is medical wastes.

1.2.30.5 Recyclables

Recyclables are materials, equipment and assemblies such as doors, windows, door and window frames, plumbing fixtures, glazing and mirrors that are recovered and sold as recyclable, wiring, insulated/non-insulated copper wire cable, wire rope, and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Metal meeting the definition of lead contaminated or lead based paint contaminated may not be included as recyclable if sold to a scrap metal company.

1.2.30.6 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included.

1.2.30.7 Surplus Soil

Surplus soil is existing soil that is in excess of what is required for this work, including aggregates intended, but not used, for on-site mixing of concrete, mortars, and paving. Contaminated soil meeting the definition of hazardous material or hazardous waste is not included and must be managed in accordance with paragraph HAZARDOUS MATERIAL MANAGEMENT.

1.2.30.8 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, hardboard. Treated or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.2.30.9 Universal Waste

The universal waste regulations streamline collection requirements for certain hazardous wastes in the following categories: batteries, pesticides, mercury-containing equipment (for example, thermostats), and lamps (for example, fluorescent bulbs). The rule is designed to reduce hazardous waste in the municipal solid waste (MSW) stream by making it easier for universal waste handlers to collect these items and send them for recycling or proper disposal.

1.2.31 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, or creeks, or waters of Japan. Surface discharges are discrete, identifiable sources and require a permit from the governing agency. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

1.2.32 Wastewater

Wastewater is the used water and solids from a community that flow to a treatment plant.

1.2.32.1 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.33 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

1.3 PROHIBITED PRODUCTS

The following items are forbidden for use by the JEGS or other criteria. Details of each are included in the text of each chapter of the Environmental Protection Plan (EPP) described in this section.

- a. Asbestos Containing Materials (ACM). Do not use materials containing asbestos unless deemed absolutely necessary. If necessary, asbestos concentration shall be limited to less than 0.1 percent by weight. The Contractor shall provide manufacturer certifications that state that materials utilized do not contain asbestos or justification for ACM and relevant certification.
- b. Lead-Containing Paint. Paint containing greater than 0.009 percent lead by weight. Contractor must provide manufacturer certifications that materials utilized are lead free per the JEGS. The contractor shall furnish a certificate for paint and paint related materials proposed for use attesting that the paint is lead free unless deemed necessary.
- c. Polychlorinated Biphenyls (PCBs). Materials containing PCBs greater than 0.5mg/kg shall not be used.

- d. Class I Ozone Depleting Substances (ODS). Class 1 ODS listed in the JEGS shall not be used.
- e. Lead-containing Drinking-Water Pipes, Solders, Flux, and Fittings. Comply with JEGS requirements where applicable

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Preconstruction Survey

Regulatory Notifications; G

Environmental Protection Plan; G

Environmental Manager Qualifications; G

Temporary Fuel Storage Tank Approval Request; G

Underground Fuel Storage Tank Approval Request; G

Licenses/Permits/Notifications; G

General Use Permitted Equipment

Fuel Storage Tank Installation/Removal Report; G[, [____]]

Written Assessment Of Friable Asbestos Disturbance

SD-06 Test Reports

Laboratory Analysis

Nonhazardous Solid Waste Diversion Report

SD-07 Certificates

EMS Certificate of Completion; G

Asbestos Certification; G

Lead Certification; G

SD-11 Closeout Submittals

Environmental Records Binder

1.5 PAYMENT

No separate payment shall be made for work covered under this section. Payment of fees associated with environmental permits, application, and/or notices obtained by the Contractor, and payment of all fines/fees for

violation or non-compliance with GOJ, Federal, and local laws and regulations are the Contractor's responsibility. All costs associated with this section shall be included in the Contract price.

1.6 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the Contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements pertaining to the environment, including but not limited to water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

Tests and procedures assessing whether construction operations comply with Applicable Environmental Laws may be required. Laboratory analyses necessary to implement the JEGS shall be conducted in a laboratory certified by a U.S. or GOJ regulatory authority for the applicable test method. In the absence of a certified laboratory, contact the Contracting Officer for further guidance.

Contractor shall be responsible to ensure that subcontractors comply with all environmental protection requirements of this section.

The Contractor shall record any problems in complying with laws, regulations, permit requirements, ordinances, and corrective actions taken. The Contractor shall immediately inform the Contracting Officer of any environmental problems.

1.6.1 Environmental Management System (EMS) Training

1.6.1.1 Personnel Requirements

The Environmental Manager is responsible for environmental compliance on projects. The Environmental Manager [and other staff], must complete applicable EMS training prior to starting respective portions of on-site work under this Contract. Contact 35 CES/CEIE for obtaining training material. If personnel changes occur for any of these positions after starting work, replacement personnel must complete applicable EMS training within 14 days of assignment to the project. EMS Training is available in English and Japanese at <https://usaf.learningbuilder.com>. New users must create a new account. Search for EMS Training once logged in. Print out the EMS Certificate of Completion and submit with the EPP.

1.6.1.2 Certification

Submit a roster of individuals whom have completed the EMS training to 35 CES/CEIE. This training is a one time requirement; 35 CES/CEIE shall maintain a listing of Contractor personnel whom have completed the training, thereby eliminating the need for Contractors who frequently perform work at Misawa AB to take this training prior to each contract.

1.6.2 Conformance with the Environmental Management System

All contractor personnel shall be aware of and comply with the Yokota Air Base Environmental Commitment Statement. The Contractor shall also comply with the EMS program, including all base environmental plans, instructions and the Japan Environmental Governing Standards (JEGS). This also includes instructions/comments provided on planning documents such as 332's (Base Civil Engineer Work Request), and 813s (Request for Environmental Impact Analysis).

Communicate with Environmental Program Managers (EPMs) to be aware of any operational controls (permits, plans, requirements to purchase recycled content, bio-based, or energy efficient products).

Provide copies of environmental records as requested by EPMs to meet recordkeeping requirements.

Ensure that personnel are properly trained in accordance with applicable statutes and regulations.

Immediately report all hazardous waste or hazardous material releases to the installation emergency response activity, and fully cooperate with any emergency response.

Perform work under this Contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities affect work performed under the Contract.

Coordinate with the installation's EMS coordinator to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Office will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

1.7 SPECIAL ENVIRONMENTAL REQUIREMENTS

1.7.1 Asbestos Prohibition and Certification

- a. Materials or products containing more than one-tenth of one percent (0.1 percent) by total weight, of the material or product, of asbestos shall not be used in this project. The Contracting Officer, at any time prior to acceptance of the work, or during the period designated for warranty of the work, if any, may reject materials and products that contain asbestos in excess of one-tenth of one percent by weight,

and direct the removal of such materials and products from the job site, at the sole expense of the Contractor, and without additional time granted for performance of the work. After completion of this Contract, if asbestos (exceeding 0.1 percent by weight) is discovered in the products or materials (excluding items permitted by the exception) installed by the Contractor, the Government reserves the right to direct the Contractor to perform asbestos abatement and restoration work, as required, at the Contractor's sole cost. Asbestos abatement work (removal and disposal of asbestos-containing materials and products) shall be accomplished in accordance with currently applicable Government standards for such work.

Exception: Where suitable asbestos-free substitutes do not exist for a material or product, the Contractor may use a material or product containing asbestos in the excess of 0.1 percent by weight, with prior written approval of the Contracting Officer. Submit a written request for substitution, accompanied by a certification from the manufacturer of the material or product that shall set forth, in specific detail, the amount of asbestos present in the material or product. When available, laboratory analysis of the material or product for asbestos content shall be included with the submittal.

- b. The Government may conduct asbestos testing on suspected asbestos-containing materials and products excluding items permitted by the "exception", and such testing shall be conducted at the expense of the Government. However, wherever destructive testing is required, or a material or product must be utilized by the Government for testing, the Contractor shall, at its own expense, repair or replace the material or product, or the item of work that has been disturbed by testing, if the results confirm presence of asbestos exceeding 0.1 percent by weight. In the event test results indicate 0.1 percent or less asbestos content or complete absence of asbestos, the Contractor shall restore the test site to its original condition and the cost of restoration work, as approved by the Contracting Officer, shall be borne by the Government.
- c. As a minimum, furnish manufacturer's certification for the items listed below, excluding items permitted by the "exception", certifying that the items are asbestos-free or do not contain asbestos in excess of 0.1 percent by weight, as applicable. However, when presence of asbestos is suspected in other products and materials used in this project, the Contractor shall be required to provide such certification for those additional items when so directed by the Contracting Officer. Asbestos certification shall be required for the items applicable to this project only.
 - (1) Vinyl sheet/vinyl tile flooring, including accessories and adhesives.
 - (2) Insulation materials, including facing.
 - (3) Gaskets for piping and duct work.
 - (4) Acoustical tiles.
 - (5) Firestopping materials.
 - (6) Fireproofing materials.

- (7) Special coating, including factory applied coatings, on sheet metal roofing and siding.
- (8) Wallboard for all interior and exterior applications, including joint compounds.
- (9) Adhesives (other than item 1) used in the project.
- (10) Tape materials used in the project.
- (11) Roofing and siding, nonmetallic.
- (12) Felt materials and cushion materials.
- (13) Pre-mixed mortars, grouts, leveling compounds, fillers, and other cementitious materials.
- (14) Caulking and sealing materials.

- d. All submittals shall be accompanied by a certification from the manufacturer of the material or product that the material or product is asbestos-free; or shall set forth, in specific detail, the amount of asbestos present in the material or product. Documentary evidence of laboratory analysis of the material or product for asbestos content, conducted by a qualified independent testing laboratory, shall be included with the submittal.

1.7.2 Lead Prohibition and Certification

- a. Paint or product coating containing more than 0.009 percent) by total weight of lead shall not be used in this project. The Contracting Officer, at any time prior to acceptance of the work, or during the period designated for warranty of the work, if any, may reject materials and products that contain lead in excess of 0.009 percent by weight, and direct the removal of such materials and products from the job site, at the sole expense of the Contractor, and without additional time granted for performance of the work. After completion of this Contract, if lead (exceeding 0.009 percent by weight) is discovered in the products or materials (excluding items permitted by the exception) installed by the Contractor, the Government reserves the right to direct the Contractor to perform lead abatement and restoration work, as required, at the Contractor's sole cost. Lead abatement work (removal and disposal of lead-containing materials and products) shall be accomplished in accordance with currently applicable Government standards for such work.

Exception: Where suitable lead-free substitutes do not exist for a paint or product coating, the Contractor may use a material or product containing lead in the excess of 0.009 percent by weight, with prior written approval of the Contracting Officer. Submit a written request for substitution, accompanied by a certification from the manufacturer of the material or product that shall set forth, in specific detail, the amount of lead present in the material or product. When available, laboratory analysis of the material or product for lead content shall be included with the submittal.

- b. The Government may conduct lead testing on suspected lead-containing materials and products excluding items permitted by the "exception", and such testing shall be conducted at the expense of the Government.

However, wherever destructive testing is required, or a material or product must be utilized by the Government for testing, the Contractor shall, at its own expense, repair or replace the material or product, or the item of work that has been disturbed by testing, if the results confirm presence of lead exceeding 0.009 percent by weight. In the event test results indicate 0.009 percent or less lead content or complete absence of lead, the Contractor shall restore the test site to its original condition and the cost of restoration work, as approved by the Contracting Officer, shall be borne by the Government.

- c. As a minimum, furnish manufacturer's certification for the items listed below, excluding items permitted by the "exception", certifying that the items are lead-free and do not contain lead in excess of 0.009 percent by weight, as applicable. However, when presence of lead is suspected in other products and materials used in this project, the Contractor shall be required to provide such certification for those additional items when so directed by the Contracting Officer. Lead certification shall be required for the items applicable to this project only.

- 1. Paints and Coatings
- 2. Any product or material with a factory applied coating

1.7.3 Class I and Class II Ozone Depleting Chemicals (ODC) or Substances (ODS)

Class I and II Ozone Depleting Substances listed in the JEGS are prohibited from being used. Contractor must provide certifications that materials utilized do not contain Class I and Class II ODC/ODS.

1.7.4 Polychlorinated Biphenyls (PCB)

Materials (ballasts, capacitors, transformers, dielectric fluid, switches, etc.) that contain PCBs are prohibited. Contractor must provide certifications that materials utilized do not contain PCBs, in accordance with the JEGS.

1.7.5 Hazardous Material Survey

The Contractor shall review the Hazardous Materials Inspection Report, PN77833 Vehicle Maintenance Shop, Kadena Air Base, Okinawa, Japan by OYO Corporation, Pacific dated November 2025, attached at the end of this section, to familiarize themselves with the materials that have been sampled and tested for this project. The Contractor shall utilize the information contained within the report to develop their work and compliance plans with regards to Hazardous Materials. During construction, should potentially hazardous material be discovered, which has not been previously tested, the Contractor shall take action to assure that the untested material is not disturbed and contact the Contracting Officer. The quantities provided in the report represents a rough order of magnitude estimate of materials and shall not be used for bidding purposes as conditions may have changed from the date of the survey and the present existence of the material(s).

The Contractor shall notify the Contracting Officer upon finding any material that will be disturbed under this contract and is suspected to contain Hazardous Materials. The Contracting Officer will handle the

identified materials under, Unexpected Discovery of Hazardous Materials.

1.8 QUALITY ASSURANCE

1.8.1 Preconstruction Survey and Protection of Features

This paragraph supplements the Contract Clause PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS. Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer, and take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record. Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the Contractor's assigned storage area and access route(s), as applicable. The Contractor and the Contracting Officer will sign this survey report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract.

1.8.2 Regulatory Notifications

Provide regulatory notification requirements in accordance with GOJ national and prefectural laws and regulations and installation requirements. Submit copies of regulatory notifications to the Contracting Officer at least 14 days prior to commencement of work activities.

1.8.3 Environmental Brief

Attend an environmental brief to be included in the preconstruction meeting. Provide the following information: types, quantities, and use of hazardous materials that will be brought onto the installation; and types and quantities of wastes/wastewater that may be generated during the Contract. Discuss the results of the Preconstruction Survey at this time.

Prior to initiating any work on site, meet with the Contracting Officer and installation Environmental Office to discuss the proposed Environmental Protection Plan (EPP). Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural and cultural resources, required reports, required permits, permit requirements (such as mitigation measures), and other measures to be taken.

Permits, licenses, or other forms of official approvals are not required by DoD activities and installations. Permits, licenses, or other forms of official approvals may, however, be required under GOJ law for certain contracted activities. When required, all such permits, licenses and other forms of official approval shall be obtained by the Contractor from the appropriate GOJ authorities. DoD Components shall assist Contractors when they are applying for a required permit, license or other form of official approval by providing necessary information only. Submit copies of all Licenses/Permits/Notifications obtained.

1.8.4 Environmental Manager

Appoint in writing an Environmental Manager for the project site. The Environmental Manager is directly responsible for coordinating Contractor compliance with the JEGS, Federal, GOJ national and prefectural laws and regulations, and installation requirements. The Environmental Manager must ensure adherence to the EPP; ensure all appropriate documents are submitted to 718 CES/CEIE; appoint an emergency response manager for environmental emergencies; train personnel in accordance with the requirements of the EPP; ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, manifesting, and disposal); implement the EPP; ensure environmental permits are obtained, maintained, and closed out; ensure compliance with Stormwater Program requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; and coordinate any remediation of regulated substances (lead, asbestos, PCB transformers). This can be a collateral position; however, the person in this position must be trained to adequately accomplish the following duties: ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure Contractor personnel are trained in JEGS requirements in accordance with their position requirements; coordinate removal of waste containers; and maintain the Environmental Records binder and required documentation, including environmental permits compliance and close-out. Submit Environmental Manager Qualifications to the Contracting Officer to indicate the training and past experience which meets the requirements of this position as described in this section.

1.8.5 Employee Training Records

Prepare and maintain Employee Training Records throughout the term of the Contract meeting applicable JEGS requirements. Provide Employee Training Records in the Environmental Records Binder. Ensure every employee completes a program of classroom instruction or on-the-job training that teaches them to perform their duties in a way that ensures compliance with the JEGS and all applicable Federal, GOJ national or prefectural laws or regulations and installation requirements. Submit these Assembled Employee Training Records in the Environmental Records Binder to the Contracting Officer at the conclusion of the project, unless otherwise directed.

Train personnel to meet the JEGS and all applicable Federal, GOJ national or prefectural laws or regulations, and Installation requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of Japan, and endangered species and their habitat that are known to be in the area.

1.8.6 Non-Compliance Notifications

The Contracting Officer shall notify the Contractor in writing of any observed noncompliance with the JEGS, applicable Federal, GOJ national or

prefectural laws and regulations, and installation requirements, and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions will be granted or equitable adjustments allowed for any such suspensions. This is in addition to any other actions the Contracting Officer may take under the Contract, or in accordance with the Federal Acquisition Regulation or Federal Law.

1.9 ENVIRONMENTAL PROTECTION PLAN

The purpose of the EPP is to present an overview of known or potential environmental issues that must be considered and addressed during construction. The EPP consists of 15 chapters and requires an entry for each chapter. Incorporate construction related objectives and targets from the installation's EMS into the EPP. Include in the EPP measures for protecting natural and cultural resources, required reports, and other measures to be taken. Meet with the Contracting Officer or Contracting Officer Representative to discuss the EPP and develop a mutual understanding relative to the details for environmental protection including measures for protecting natural resources, required reports, and other measures to be taken. Submit the EPP within 15 days after receiving notice to proceed. Commencement of work will not begin until the Environmental Protection Plan has been approved.

Revise the EPP throughout the project to include any reporting requirements, changes in site conditions, or Contract modifications that change the project scope of work in a way that could have an environmental impact. No requirement in this section will relieve the Contractor of any JEGS, and applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. During Construction, identify, implement, and submit for approval any additional requirements to be included in the EPP. Provide an electronic copy of the EPP to 718 CES/CEIE via the Contracting Officer. Maintain the current version of the EPP onsite.

1.9.1 Submission Guidance

The EPP is a required submittal for all projects, and shall be submitted a minimum of 14 days prior to the scheduled start of construction activities or delivery of materials to the site. Work impacted by the contents of the EPP cannot start prior to approval of the EPP. Maintain a copy of the approved EPP on-site at all times.

If disapproved, the Contractor shall revise and resubmit the EPP to properly address reviewer comments. Delays resulting from disapproval are the responsibility of the Contractor.

1.9.2 EPP Drafting Instructions

Address the topic of each chapter at a level of detail commensurate with the environmental issue and the scope of work. The EPP shall explain how the Contractor will meet the requirements in each chapter.[Chapters that are not applicable to this project shall be replaced with a statement indicating as such.][Chapters that are not applicable to this project shall be replaced with the following:

"This chapter does not apply to the Contract because" followed by a brief explanation on why it does not apply.

The EPP shall not be merely a reiteration of the requirements or the text of instructions in this section, except for the "Not Applicable" chapters. The EPP shall explain how the Contractor will meet the requirements in each chapter.

Any site-specific environmental issues identified as necessary by the Government shall be addressed in the EPP even if this supplement does not specifically mention such a situation as a requirement.]

1.9.3 EPP General Overview and Purpose

1.9.3.1 Descriptions

A brief description of each specific plan required by environmental permit or elsewhere in this Contract such as stormwater pollution prevention plan, spill control plan, solid waste management plan, wastewater management plan, air pollution control plan, contaminant prevention plan, a historical, archaeological, cultural resources, biological resources and wetlands plan, traffic control plan, Hazardous, Toxic and Radioactive Waste (HTRW) Plan, Non-Hazardous Solid Waste Disposal Plan, and borrowing material plan.

Include a list of applicable Federal, GOJ, JEGS, prefectural laws, regulations, and permits concerning environmental protection, pollution control, and abatement that are applicable to the Contractor's proposed operations and requirements imposed by those laws, regulations, and permits. Whenever there is conflict between Federal, GOJ, JEGS, or prefectural laws, regulations, and permit requirements, the strictest applicable rule applies.

1.9.3.2 Duties

The duties and level of authority assigned to the person(s) on the job site who oversee environmental compliance, such as who is responsible for adherence to the EPP, who is responsible for spill cleanup and training personnel on spill response procedures, who is responsible for manifesting hazardous waste to be removed from the site (if applicable), and who is responsible for training the Contractor's environmental protection personnel.

Provide the name, telephone number, and address of a Primary and Alternate Environmental Representative.

1.9.3.3 Procedures

A copy of any standard or project-specific operating procedures that will be used to effectively manage and protect the environment on the project site.

Provide schedule of digging or trenching actions to include date/time, location, purpose, method and depth.

1.9.3.4 Communications

Communication and training procedures that will be used to convey environmental management requirements to Contractor employees and

subcontractors.

1.9.3.5 Contact Information

Emergency contact information contact information (office phone number, cell phone number, and e-mail address).

1.9.4 General Site Information

1.9.4.1 Drawings

Drawings showing locations of proposed temporary excavations or embankments for haul roads, stream crossings, jurisdictional wetlands, material storage areas, structures, sanitary facilities, storm drains and conveyances, and stockpiles of excess soil.

Drawings showing locations of any proposed temporary facilities, hazardous materials storage, hazardous and solid waste collection points, excavations, embankments, existing utility lines to include storm drains included in the EPP.

1.9.4.2 Work Area

Work area plan showing the proposed activity in each portion of the area and identify the areas of limited use or nonuse. Include measures for marking the limits of use areas, including methods for protection of features to be preserved within authorized work areas and methods to control runoff and to contain materials on site, and a traffic control plan.

1.9.4.3 Documentation

A letter signed by an officer of the firm appointing the Environmental Manager and stating that person is responsible for managing and implementing the Environmental Program as described in this contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work.

1.9.4.4 Regulatory Notification and Permits

List the required notifications and permit applications. Some permits require up to 180 days to obtain. Demonstrate that permits have been obtained or applied for by including hard copies. The EPP shall not be approved until the permits have been obtained.

1.9.5 CHAPTER 1: ENVIRONMENTAL REQUIREMENTS AND TRAINING

1.9.5.1 Criteria Summary

Provide and maintain environmental protective measures to control pollution that develops during normal construction practice during the life of the Contract.

1.9.5.2 Applicable Regulations

Comply with all laws and regulations referenced in the drawings and specifications, as well as applicable GOJ and prefectural laws or regulations, referenced or not.

1.9.5.3 EPP Mandatory Items

Include the following items in the EPP:

- a. Environmental Manager Appointment Letter: Include a letter signed by an officer of the firm designating a Primary and an Alternate Environmental Manager with telephone number (office phone number, cell phone number), e-mail address, and mailing address and stating that this person is responsible for managing and implementing the Environmental Program as described in this Contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work. The Environmental Manager shall be responsible for providing all project Environmental submittals and for coordinating with the Contracting Officer on any environmental issues encountered during work.
- b. Employee Training Letter: The Environmental Manager shall implement a training program to ensure that all Contractor personnel are trained in environmental protection requirements. Include a letter that lists employee's names, types of environmental training completed, and dates of completion.
- c. Employee Training Roster: Provide a roster of individuals involved with the project who have completed the required EMS training.
- d. Include in this section the requirements outlined in paragraphs 1.9.6.1 - Drawings and 1.9.6.2 - Work Area.

1.9.6 CHAPTER 2: AIR EMISSIONS AND OZONE DEPLETING SUBSTANCES (ODS)

1.9.6.1 Criteria Summary

ODSs, including Hydrofluorocarbons (HFC) and perfluorocarbons (PFC) from refrigeration and fire suppression equipment, are categorized into two specific classes. See the JEGS for proper classifications.

1.9.6.2 EPP Mandatory Items

If this project involves the servicing, relocation, removal, or installation of any ODS containing items, contractor shall provide, in the EPP, an explanation of how requirements of paragraph 3.5 will be achieved/maintained.

Include the following items in the EPP as applicable:

- a. Haul Route: Submit truck and material haul routes along with a Dirt and Dust Control Plan in the EPP for controlling dirt, debris, and dust on Installation roadways. As a minimum, identify in the plan the subcontractor and equipment for cleaning along the haul route and measures to reduce dirt, dust, and debris from roadways.
- b. Pollution Generating Equipment: Identify air pollution generating equipment or processes that may require compliance with the JEGS or GOJ national or prefectural laws-regulations. Determine requirements based on any current JEGS and installation requirements and the impacts of the project. Provide a list of all fixed or mobile equipment, machinery or operations that could generate air emissions during the project to the Installation Environmental Office (Air

Program Manager).

- c. Stationary Internal Combustion Engines: Identify portable and stationary internal combustion engines that will be supplied, used or serviced. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. At minimum, include the make, model, serial number, manufacture date, size (engine brake horsepower), and EPA emission certification status of each engine. Maintain applicable records and log hours of operation and fuel use. Logs must include reasons for operation and delineate between emergency and non-emergency operation.
- d. Refrigerants: Identify management practices to ensure that heating, ventilation, and air conditioning (HVAC) work involving refrigerants complies with the JEGS and installation requirements. Technicians shall be trained in proper recovery/recycling procedures, leak detection, safety, shipping, and disposal in accordance with recognized industry standards or Japanese equivalent. Any refrigerant reclaimed is the property of the Government; coordinate with the Installation Environmental Office to determine the appropriate turn in location.
- e. Training/Certification Requirements: Heating and air conditioning technicians must be trained to meet requirements in the JEGS. Maintain copies of certifications at the employees' places of business; technicians must carry certification wallet cards, as provided by GOJ national or prefectural laws and regulations. Submit training certificates as part of the EPP.
- f. Air Pollution-Engineering Processes: Identify planned air pollution-generating processes and management control measures (including, but not limited to, spray painting, abrasive blasting, demolition, material handling, fugitive dust, and fugitive emissions). Log hours of operations and track quantities of materials used.
- g. Compliance Materials: Provide the Government a list of and SDSs for all hazardous materials proposed for use on site. Materials must be compliant with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements for emissions including solvent and volatile organic compound contents, and applicable National Emission Standards for Hazardous Air Pollutants requirements. The Government may alter or limit use of specific materials as needed to meet installation permit requirements for emissions.

1.9.6.3 Other Required Submittals

For Contracts servicing, removing, or replacing ODS-containing items, such as refrigeration or fire suppression equipment, follow the guidance for submittals in accordance with the specifications under which the work is being performed (HVAC, Fire Suppression, etc.).

1.9.7 CHAPTER 3: WATER RESOURCES

1.9.7.1 EPP Mandatory Items

Include the following items in the EPP as applicable:

- a. Site-Specific Storm Water Pollution Prevention Plan (SWPPP): For any Contract where work may possibly lead to disruption of the stormwater flow, the Contractor shall develop and submit a project-specific SWPPP as a mandatory part of Chapter 3 of the EPP.

The SWPPP must include effective selection, implementation and maintenance of Best Management Practices (BMPs) as indicated in the JEGS, and include methods and drawings as necessary to prevent storm water pollution as specified in the Storm Water Pollution Prevention Measures section in Part 3. The SWPPP shall also address erosion and sediment control measures and stormwater management and control including, but not limited to ground cover, erodible soils, temporary measures - structural practices, temporary and permanent stabilization. The SWPPP must meet the requirements of the JEGS and the Installation's SWPPP.

If applicable, submit a project-specific Stormwater Pollution Prevention Plan (SWPPP) to the Contracting Officer for approval, prior to the commencement of work. The Contractor shall maintain an approved copy of the SWPPP at the construction on-site office, and continually update as regulations require, to reflect current site conditions. The Contractor may be required to install, inspect, maintain best management practices (BMPs), and submit storm water BMPs inspection reports and storm water pollution prevention plan inspection reports.

The SWPPP must meet the requirements of the JEGS and the Installation's SWPPP.

Include the following:

- a. Identify potential sources of pollution which may be reasonably expected to affect the quality of storm water discharge from the site.
 - b. Comply with terms of the Installation SWPPP for stormwater discharges from construction activities. Prepare SWPPP in accordance with the JEGS and installation requirements.
 - c. Select applicable BMPs from the JEGS and/or the Installation SWPPP. Applicable Best Management Practices in the JEGS shall be incorporated into the site-specific SWPPP and implemented.
 - d. The SWPPP shall also address erosion and sediment control measures and stormwater management and control including, but not limited to ground cover, erodible soils, temporary measures - structural practices, temporary and permanent stabilization.
- b. Wastewater Management Plan (WWMP): For any Contract that may generate wastewater, the Contractor shall submit a WWMP.
 - (1) The WWMP must identify the methods and procedures for management and/or discharge of wastewater which are directly derived from construction activities, such as concrete curing water, clean-up water, dewatering of ground water, disinfection water, hydrostatic test water, and water used in flushing of lines.
 - (2) If a settling/retention pond is required, the WWMP must include the design of the pond, including drawings, removal plan, and testing requirements for possible pollutants. If disposal is to a

sanitary sewer, the WWMP must include documentation that the Waste Water Treatment Plant Operator has approved the flow rate, volume, and type of discharge.

- (3) Contractor is not authorized sewage holding tanks on base and is not authorized to dispose of waste from chemical toilets/porta-potties via the installation sewer system.
- (4) When using chemical toilets, plan for procuring a porta-potty/chemical toilet service Contract. The porta-potty/chemical toilet service Contract must include the correct maintenance, waste collection, transportation, and disposal of all porta-potty/chemical toilets and content and include provisions for pest control and elimination of odors.
- (5) If the project generates wastewater from rinsing tanks, dewatering sites, etc., the Contractor shall ensure proper disposal with CEIE approval. The Contractor may be required to provide further analysis and treatment based on the condition of wastewater.

1.9.8 CHAPTER 4: HAZARDOUS MATERIALS

1.9.8.1 Criteria Summary

Hazardous Material Inventory, Material Safety Data Sheets, and Safety Data Sheets: Submit documentation and receive approval prior to bringing any hazardous materials (HAZMAT) onto the installation. Consult with the Hazardous Materials Program Manager 718 CES/CEIE at DSN 634-2600 for a determination of whether or not a product is a HAZMAT.

1.9.8.2 EPP Mandatory Items

For all Contracts where HAZMAT will be brought to the worksite, submit the following documents in Chapter 4 of the EPP:

- a. Hazardous Material Inventory: A table listing all hazardous materials brought onto the base, including the estimated amounts of each material that will be used. List shall include all pesticides and herbicides that will be brought onto the Installation.
- b. Hazmat SDS (in English and Japanese): An SDS sheet for each hazardous material on the inventory.
- c. AF Form 3000.
- d. If HAZMAT is suspected to contain lead or asbestos, such as paint, primer, caulking, a certificate from the manufacturer shall be provided to prove that the HAZMAT is free of asbestos/lead.

The AF Form 3000 package must be reviewed by Bioenvironmental (374 AMDS/SGPB), Safety (374 AW/SE) and Environmental (374 CES/CEIE) via the Contracting Officer.

1.9.8.3 Other Required Submittals

If after beginning work, the Contractor discovers previously undisclosed, unexpected, or suspected HAZMAT, substances, chemicals, or contaminated areas, the Contractor shall immediately report the discovery to the Contracting Officer and to 718 CES/CEIE (DSN 634-2600). Contractor

personnel shall immediately cease work in the area, unless the work is of an emergency nature and the risk of exposure can be mitigated by the use of personal protective equipment (PPE) or clothing. Submit, in writing, a Notification of Unexpected Hazardous Material Discovery to the Contracting Officer outlining the events leading to the discovery.

1.9.9 CHAPTER 5: HAZARDOUS WASTE

1.9.9.1 EPP Mandatory Items

For all Contracts where Hazardous Waste will be generated, submit the following information and documents in Chapter 5 of the EPP:

- a. Hazardous Waste Manager Letter: A letter designating the hazardous waste manager. The letter shall include:
 - (1) Manager's name.
 - (2) Training and experience.
 - (3) Contact information.
- b. Management Procedures for Hazardous Waste to be Generated: The elements of those procedures will coincide with the Installation Hazardous Waste Management Plan (HWMP). The Contracting Officer will provide a copy of the Installation Hazardous Waste Management Plan. As a minimum, include the following:
 - (1) List of the types of hazardous wastes expected to be generated.
 - (2) Procedures to ensure a written waste determination is made for appropriate wastes that are to be generated.
 - (3) Sampling/analysis plan, including laboratory method(s) that will be used for waste determinations and copies of relevant laboratory certifications. Follow the analytical procedure and methods in accordance with the JEGS and all applicable GOJ national or prefectural laws and regulations. Identify hazardous waste by analyzing for the following characteristics: ignitability, corrosivity, reactivity, and toxicity based on TCLP results. Provide analytical results and reports performed to the Contracting Officer. Sample waste in accordance with appropriate Japanese or U.S. EPA testing protocols that meet the purpose of the testing. Clearly mark each sampled drum or container with the Contractor's identification number, and cross reference to the chemical analysis performed.
 - (4) Methods and proposed locations for hazardous waste accumulation/storage (that is, in tanks or containers).
 - (5) Management procedures for storage, labeling, transportation, and disposal of waste (treatment of waste is not allowed unless specifically noted).
 - (6) Management procedures and regulatory documentation ensuring disposal of hazardous waste complies with Land Disposal Restrictions, applicable Federal, GOJ and prefectural laws and regulations, JEGS, and Installation Hazardous Waste Management Plan.

- (7) Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar.
- (8) Used oil management procedures in accordance with the JEGS; Hazardous waste minimization procedures. Obtain a copy of the installation's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EPP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in the construction when requesting information.
- (9) Plans for the disposal of hazardous waste by permitted facilities; and procedures to be employed to ensure required employee training records are maintained.
- (10) Procedures to be employed to ensure required employee training records are maintained.
- (11) Hazardous Waste Disposal local permits or licenses for hazardous waste disposal in accordance with the JEGS (if required).
- (12) Hazardous Waste Disposal Statement of Agreement. From a treatment, storage, or disposal (TSD) facility that will accept the waste from the contractor and also a statement from a certified hazardous waste transporter who will transport the waste to the TSD facility in accordance with the JEGS (if required).

1.9.9.2 Other Required Submittals

- a. Hazardous Waste Analysis Report: For any materials that were sampled and analyzed for hazardous waste content, submit a Hazardous Waste Analysis Report containing:
 - (1) Who the sample was taken by.
 - (2) What type of sample was taken (soil, water, chemical, etc.).
 - (3) Where the sample was taken (building number, room number, etc.).
 - (4) The sampling time and date.
 - (5) An explanation of why the material was sampled (disposal determination, final clearance, etc.).
 - (6) Photographs of the sampled location or material (if required to positively identify the exact location or material later).
 - (7) The laboratory's certified analysis results letter.

1.9.10 CHAPTER 6: SOLID WASTE

1.9.10.1 EPP Mandatory Items

Solid Waste Management Plan: To ensure the Contractor understands the solid waste reduction requirements, the Contractor is required to submit a Solid Waste Management Plan SWMP, as part of the EPP. The Solid Waste Management Plan shall be consistent with the Installation Integrated Solid

Waste Management Plan (ISWMP).

The SWMP has four (4) sections: Diversion, Procedures, Training, and Permits.

Provide the Contracting Officer with written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Include in the report the locations where various types of waste will be disposed or recycled. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location state and local Solid Waste Management Permit or Solid Waste Management License showing such agency's approval of the disposal plan before transporting wastes off Government property.

a. SWMP: Diversion Section

Provide the dates for the start and end of work, the types of wastes expected to be generated by the project, estimated weights for each type of waste, and the facility or facilities where each waste will be taken. Also include a list of which wastes will be salvaged, reused, and recycled, estimated weights for each recyclable commodity, and the facility or facilities where each recyclable will be taken.

To the extent practicable, all scrap metal must be sent for reuse or recycling and will not be disposed of in a landfill. Include the name, physical address, and telephone number of the hauler, if transported by a franchised solid waste hauler. Include the destination and, unless exempted, provide a copy of the state or local permit (cover) or license for recycling.

b. SWMP: Procedures Section: Describe the planned Separation, Contamination Prevention, and Transportation procedures.

- (1) Outline how recyclables will be separated from other wastes, how wastes will be collected, and where they will be stored at the project site until transported to recycling facilities. The Separation Procedures shall also detail how other wastes will be collected and where they will be stored at the project site until transported to disposal facilities. Containers for recyclables and other wastes must meet the requirements in the JEGS that storage of recyclables and other wastes must meet the JEGS and that storage of bulky wastes must meet the JEGS.
- (2) Contamination Prevention Procedures shall explain how recyclables will be kept free from contaminants (hazardous, toxic, or other wastes) and how other non-recyclable wastes will be kept free of contaminants (recyclables and hazardous or toxic wastes).
- (3) Transportation Procedures shall outline how spills of recyclables or other wastes will be prevented during collection, storage, loading, and transportation. They also detail how the recyclables and other wastes will be transported from the project site to the recycling or disposal facility. These procedures also must state how often recyclables and other wastes will be picked up by the transportation company.

c. SWMP: Training Section

Set forth how employees and subcontractors will be trained on solid

waste management policies and procedures. Include how training will be documented and who will conduct the training. Also outline any consequences or repercussions for employees or subcontractors that fail to follow the policies and procedures.

d. SWMP: Permits Section

Include copies of all permits, licenses, and agreements for the recycling and disposal facilities and for the transportation companies utilized during the project. Include English translations for all documents in this section.

1.9.10.2 Other Required Submittals

- a. Solid Waste Management Report: During project execution, the Contractor shall submit weight tickets or manifests, receipts, bills of sale, and other sales documentation (with English translations) for all materials removed from the project site. Weight tickets and manifests receipts, bills of sale, and other sales documentation shall be submitted during the project every month by the 5th day of the following month, (include this statement as part of the SWMP). Designate on each document whether the material was disposed of, recycled, or reused, and state the amounts, locations, and names of the businesses receiving the solid waste.

In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. Also include any additional or renewal permits, licenses, or agreements, with recycling or disposal landfills. In addition, a copy of Contract(s) for waste collection/transportation and disposal permits issued from metropolitan/prefectural governments shall be submitted with AF Form 3000 to 374 CES/CEIE via the Contracting Officer. After the disposal, waste manifests shall be submitted to 374 CES/CEIE via the Contracting Officer along with recycling report that shows the amount of waste recycled, incinerated/landfilled (Attachment 1).

1.9.11 CHAPTER 7: FUEL STORAGE TANKS

1.9.11.1 EPP Mandatory Item

Describe in the EPP how POL tanks and containers must be stored, managed, and inspected and what protections must be provided. POL tank work cannot be performed using this section alone. All UST tank removals shall be performed in accordance with SECTION 02 65 00. Tank installations shall be performed in accordance with SECTION 33 56 10, and AFI 23-204. Contact KB Environmental Section at to obtain copies of the requirements.

1.9.11.2 Other Required Submittals

Fuel Storage Tank Removal/Installation Report: If an AST/UST (to include temporary fuel storage tanks) is installed or removed as part of the project, the Contractor shall prepare a Fuel Storage Tank Installation/Removal Report and submit it to 718 CES/CEIE via the COR within 30 days of the completion of site work. At a minimum, the report shall include the following as applicable:

- a. Summary of work conducted.

- b. Installation/Removal date for each installed/removed storage tank.
- c. Date placed in service, capacity, manufacturer name, model number, and serial number for each new AST/UST.
- d. As-built drawings showing the removed AST/USTs and/or the newly installed AST/USTs and associated infrastructure.
- e. Copy of the Manufacturer's Operations and Maintenance Manuals and product information.
- f. Copy of the Manufacturer's Warranty Cards.
- g. Photo documentation of construction activities, including AST/UST removals and/or new AST/UST installations. Photos shall show the initial and completed site condition, each exposed UST, each tank's condition upon removal, the completed UST excavation(s) to show any contamination, and the contaminated soil pile, if any.
- h. If a tank was removed, include a photograph of the old tank with a hole cut into the side as described in paragraph UST REMOVAL. Provide a location map and tank ID (when available).
- i. Leak check report for buried piping associated with storage tank at time of installation, modification, construction, relocation or replacement.

1.9.12 CHAPTER 8: HISTORICAL, CULTURAL, AND NATURAL RESOURCES

1.9.12.1 EPP Mandatory Items

- a. Description of measures to manage and mitigate damage to natural resources including, but not limited to land resources, tree protection, replacement of damaged landscape features, temporary construction, stream crossings, fish and wildlife resources, and wetlands areas.
- b. Objectives and methods to protect identified historical and archeological resources.

1.9.12.2 Other Required Submittals

Notification Of Discovery Or Historical Or Cultural Items: The Contractor shall immediately report the discovery of any historical and/or archaeological items or human remains in the course of work to the Contracting Officer and the Kadena Air Base Environmental Office at DSN 634-2600.

Resources covered by this paragraph include, but are not limited to: any human skeletal remains or burials; artifacts; shell, midden, bone, charcoal, or other deposits; rock or coral alignments, pavings, wall, or other constructed features; and any indication of agricultural or other human activities.

Stop work immediately in the area of the discovery until the Environmental Office evaluates the site. The Contracting Officer will provide direction on how to proceed. Secure the area and prevent employees or other persons from trespassing on, removing, or otherwise disturbing such resources. The Government retains ownership and control over archaeological resources.

The Contractor shall be responsible for completing and submitting official paperwork on the discovery. The Environmental Office's Cultural Resources Manager will advise the Contractor concerning the content of the report.

1.9.13 CHAPTER 9: POLYCHLORINATED BIPHENYLS (PCB)

1.9.13.1 EPP Mandatory Items

[Information on the PCB equipment that will be removed, including, but not limited to:

- (a) Equipment type.
- (b) Month/year of manufacture.
- (c) Date removed from service.
- (d) Building number, circuit number or set number.
- (e) Base from which item was removed.

Items (c) - (e) shall also be written on the equipment chassis in permanent ink.]

Provide a list of any equipment which is known, or has potential to contain (contains dielectric fluid) PCBs removed under this project. List shall include equipment serial numbers and map indicating location of equipment. When listing light ballasts provide the quantity to be removed.

If the project involves the removal of an oil-filled transformer or switchgear manufactured prior to 1980, perform the work in accordance with the specification and supplement that applies to the work, such as: (Section 26 12 21 SINGLE-PHASE PAD-MOUNTED TRANSFORMERS, etc.) In Chapter 9 of the EPP, enter the following:

"CHAPTER 9: POLYCHLORINATED BIPHENYLS (PCB)

A separate PCB equipment removal plan will be submitted in accordance with the local supplement."

If the project involves the removal of an oil-filled transformer or switchgear that was manufactured after 1980, perform the work in accordance with the specification and supplement that applies to the work, such as: (Section 26 12 21 SINGLE-PHASE PAD-MOUNTED TRANSFORMERS, etc.)

In Chapter 9 of the EPP, enter the following:

- a. (A transformer) (and) (or) (Switchgear) manufactured (insert month/year of manufacture) will be removed in accordance with (insert spec section) and placed at Bldg 1465 for storage.
 - (1) Insert the item's month/year of manufacture in the spaces above.
 - (2) Write the specification used to perform the work in the spec spaces.
- b. The Contractor shall write the following information on the chassis with permanent ink.

- (1) "Date removed from service".
- (2) "Building number", "circuit-number", or "set-number".
- (3) "Base" where item was removed."]

1.9.14 CHAPTER 10: PESTICIDES

1.9.14.1 EPP Mandatory Items

Pesticide Treatment Plan: Include and update a pesticide treatment plan, as information becomes available. Include in the plan the sequence of treatment, dates, times, locations, pesticide trade name, EPA registration numbers if applicable, authorized uses, chemical composition, formulation, original and applied concentration, application rates of active ingredient (that is, pounds of active ingredient applied), equipment used for application, and calibration of equipment. Comply with the JECS or any applicable Federal GOJ national or prefectural and installation requirements.

1.9.15 CHAPTER 11: ASBESTOS CONTAINING MATERIALS (ACM)

1.9.15.1 EPP Mandatory Items

Indicate whether the project involves sampling, disturbance, and/or abatement of any ACM. Refer to section 02 82 00 for asbestos management and disposal requirements.

Insert the following text in Chapter 11 of the EPP:

"CHAPTER 11: ASBESTOS-CONTAINING MATERIALS (ACM)

1. An Asbestos Sampling and Analysis Report and Hazard Abatement and Disposal Plan will be submitted separately in accordance with the ACM specifications and supplements. The Contractor understands that the plan must be approved prior to beginning site work."

1.9.15.2 Other Mandatory Requirements

If ACM is unexpectedly discovered at the worksite, submit an Unexpected Discovery of Asbestos Report to the Contracting Officer, briefly explaining the location of discovery, the time and date.

Consult with the Environmental Representative in advance of drafting the EPP if asbestos applicability is uncertain.

1.9.16 CHAPTER 12: LEAD-BASED PAINT (LBP)

1.9.16.1 EPP Mandatory Items

Indicate whether the project involves sampling, disturbance, removal, and/or abatement of lead-containing and/or LBP. Refer to section 02 83 00 for lead paint management and disposal requirements.

Consult with the Environmental Representative in advance of drafting the EPP if lead applicability is uncertain.

1.9.17 CHAPTER 13: SPILL PREVENTION AND RESPONSE

1.9.17.1 EPP Mandatory Items

Site-specific Spill Contingency Plan (SSCP): The Contractor shall develop and submit a SSCP for any project at which POL or Hazardous Substances are used and/or stored at the project site. The SSCP shall be consistent with the Installation Spill Prevention and Response Plan (SPRP). The Contractor shall train employees on the contents of the SSCP and the use of spill response equipment and shall document such training. The SSCP shall be posted at the work site at all times.

The SSCP must be included in Chapter 13 of the EPP and be at least two pages in length:

- a. (Page 1) Description of the spill response procedures, an inventory of POL and/or hazardous substances, probable spill routes, and an inventory of spill response equipment.
- b. (Page 2) Project site layout diagram.
- c. An SSCP Plan template can be obtained from 718 CES Environmental Section by calling DSN 634-2600.

1.9.17.2 Other Mandatory Requirements

Prevention: Procedures to prevent releases to the environment.

Spill Notification: The Contractor is required to submit a written Spill Notification when a worksite spill occurs. The quantity of POL spilled determines the amount of information required on the report and the deadline for submission. Consult with Kadena Air Base Environmental Section on the requirements when initially reporting the spill.

1.9.18 CHAPTER 14: ENVIRONMENTAL MANAGEMENT SYSTEM (EMS)

1.9.18.1 Criteria Summary

In accordance with EO 13693, Kadena Air Base has implemented the ISO 14001 to recognize environmental impacts, reduce our environmental footprint, and to always look to improve the way business is conducted at Kadena Air Base.

1.9.18.2 Training

In accordance with AFI 32-7001, the Contractor shall ensure that all personnel and subcontractors complete the EMS Awareness training via the Environmental Occupational Safety Health-Training Network, (<https://esohtn.com/start>, password: esohtn) prior to commencing construction/project activities or delivery of materials to the site. For questions, please contact 718 CES/CEIE.

1.9.18.3 EPP Mandatory Items

The Contractor is required to understand and implement EMS and shall describe how this will be accomplished in this Chapter of the EPP. For chapter 14, submit a print-out of the EMS Training Certificate (from the website) for each employee showing completion of EMS Training courses.

1.9.19 CHAPTER 15: SUSTAINABLE PROCUREMENT PROGRAM

1.9.19.1 Criteria Summary

The Sustainable Procurement Program (SPP), establishes goals for all federal employees, to include contractors, to improve sustainable environment practices and procedures through the purchase and use of recycled content, bio-based products, energy and water efficient products, and environmentally preferable products. Recycled content products are made from or contain recycled materials. Bio-based products are made with biological, agriculture, or forestry materials. Energy-efficient products use less energy than comparable equipment. Water-efficient products use less water than comparable equipment. Environmentally preferable products are materials and items that have reduced effects on health and the environment compared to other products with the same purpose.

In the performance of this contract, the contractor shall make maximum effort to improve sustainable environmental practices and procedures as follows: reduce solid and hazardous waste generation, reduce greenhouse gas emissions, increase the use of renewable energy and bio-based products, reduce the use of ozone depleting substances and hazardous and toxic chemicals, reduce consumption of energy and natural resources, expand markets for green products and services, and reduce dependence on fossil fuel-based products. Make maximum use of products containing recovered materials that are EPA-designated items unless the product cannot be acquired:

- a. Competitively within a timeframe providing for compliance with the contract performance schedule,
- b. Meeting contract performance requirements, or
- c. At a reasonable price.

The U.S. Environmental Protection Agency (EPA) has developed the Comprehensive Procurement Guidelines (CPG) to help identify recycled content products. The CPG also provides a "Total Recovered Materials Content" percentage that is a range of how much recycled content should be in each item. CPG-listed items must be purchased at the amounts recommended by EPA (the Total Recovered Materials Content percentage).

The CPG categories are: Paper and Paper Products, Vehicular Products, Construction Products, Transportation Products, Park and Recreation Products, Landscaping Products, Non-paper Office Products, and Miscellaneous Products.

The U.S. Department of Agriculture (USDA) has developed a list of bio-based products, similar to that of EPA's CPG for recycled content products. The Contractor shall purchase bio-based products whenever possible.

The USDA's Biopreferred categories are: Construction and Road Maintenance, Furniture and Furnishings, Housewares and Cleaning, Industrial Supplies, Landscaping and Agriculture, Office Supplies, Personal Care and Toiletries, The Great Outdoors, and Utilities.

Information about this requirement and these products is available at:
<http://www.biopreferred.gov>.

Recycled content products not listed in the CPG may also be used in the execution of any contract. Look for opportunities to maximize the use of products containing recycled content. Green products can and should be used in every construction project, including new construction, addition, renovation, or repair projects, and every service contract.

1.9.19.2 EPP Mandatory Items

For Chapter 14, list products with recycled content, environmentally preferable, bio-based, and energy and water efficient products will be purchased or utilized, if any.

1.10 LICENSES AND PERMITS

Obtain licenses and permits required for the construction of the project and in accordance with FAR 52.236-7. Notify the Government of all general use permitted equipment the Contractor plans to use on site. This paragraph supplements the Contractor's responsibility under FAR 52.236-7.

1.11 ENVIRONMENTAL RECORDS BINDER

Maintain on-site a separate three-ring Environmental Records Binder and submit at the completion of the project. Maintain digital copies of records to submit. Make separate parts within the binder that correspond to each submittal listed below:

- a) Waste Determination Documentation
- b) Disposal Documentation for Hazardous and Regulated Waste
- c) Stormwater Inspection Reports and Stormwater Pollution Prevention Plan Compliance Notebook

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 PROTECTION OF NATURAL RESOURCES

Minimize interference with, disturbance to, and damage to fish, wildlife, and plants, including their habitats. Prior to the commencement of activities, consult with the Installation Environmental Office, regarding rare species or sensitive habitats that need to be protected. The protection of rare, threatened, and endangered animal and plant species identified, including their habitats, is the Contractor's responsibility.

Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work that is consistent with the requirements of the Installation Environmental Office or as otherwise specified. Confine construction activities to within the limits of the work indicated or specified.

The Contractor shall take precautions to preserve all such resources as they existed at the time they were first pointed out. The Contractor shall provide and install protection for these resources and be responsible for their preservation during the life of the Contract. Environmental protection shall be conducted as follows:

- a. Except in areas indicated on the drawing or specified to be cleared, the Contractor shall not remove, cut, deface, injure, or destroy land resources, including trees, shrubs, vines, grasses, topsoil, and land forms without the Contracting Officer's permission. Any anticipated vegetation disturbance needs to be coordinated with the Government before it occurs.
- b. All construction activities shall stay out of the drip line of trees. Damaging roots and/or compacting soil within the dripline of trees is prohibited. Trees, shrubs, and other vegetation not identified for removal shall be protected against removal, injury, defacing, and scarring - no ropes, cables, or guys shall be fastened to or attached to any trees for anchorage unless specifically authorized. Where such emergency use is permitted, the contractor shall provide effective protection for land and vegetation resources at all times as defined in the following subparagraphs. Stone, earth or other material displaced into uncleared areas shall be removed.
- c. Where an exception is permitted, the Contractor shall provide effective protection for land and vegetation resources at all times and shall be responsible for any resultant damage.

3.1.1 Flow Ways

Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as specified and permitted.

3.1.2 Vegetation

Except in areas to be cleared, do not remove, cut, deface, injure, or destroy trees or shrubs without the Contracting Officer's permission. Do not fasten or attach ropes, cables, or guys to existing nearby trees for anchorages unless authorized by the Contracting Officer. Where such use of attached ropes, cables, or guys is authorized, the Contractor is responsible for any resultant damage.

Protect existing trees that are to remain to ensure they are not injured, bruised, defaced, or otherwise damaged by construction operations. Remove displaced rocks from uncleared areas. Coordinate with the Contracting Officer and Installation Environmental Office to determine appropriate action for trees and other landscape features scarred or damaged by equipment operations.

3.1.3 G0J-Protected Species

G0J-Protected species are typically found in undeveloped and unmaintained portions of the base. Projects that may affect protected species shall include mitigation measures to eliminate or minimize effects.

3.1.4 Banyan Trees

All Banyan trees must be preserved and protected. Banyan trees mark the villages and are used as landmarks for the Okinawans. Any banyan tree requiring destruction or removal must first be coordinated with the 18 CEG/CC, or in his absence with the 18 CEG/CD (for example, trees with 30 percent or more of their root systems destroyed may need to be removed). Protect existing trees from injury, bruising, defacing, or other damage by

construction operations.

3.1.5 Post-Construction Site Restoration

Remove traces of temporary construction facilities such as haul roads, work areas, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other signs of construction. Grade temporary roads, parking areas, and similar temporarily used areas to conform to surrounding contours. Stabilize disturbed soils after construction to prevent erosion.

3.1.6 Streams

Stream crossings must allow movement of materials or equipment without violating water pollution control standards of the JEGS and applicable, Federal, and GOJ national and prefectural laws and regulations, and installation requirements. Construction of stream crossing structures must be in compliance with JEGS, applicable Federal and GOJ national and prefectural laws and regulations, and installation requirements, including the Storm Water Pollution Prevention Plan.

The Contracting Officer's approval and applicable local permits are required before any equipment will be permitted to ford live streams. In areas where frequent crossings are required, install temporary culverts or bridges. Obtain Contracting Officer's approval prior to installation. Remove temporary culverts or bridges upon completion of work, and repair the area to its original condition unless otherwise required by the Contracting Officer.

3.1.7 Endangered Species

At any time an endangered or threatened species, flora or fauna, to include sea turtle nests are encountered, all activities shall stop and the Contracting Officer shall be notified.

3.1.8 Indigenous/Native Flora and Fauna

The Contractor shall place emphasis on the protection of habitats favorable to the reproduction and survival of indigenous flora and fauna. The Contractor shall use indigenous flora for planting/sodding.

3.1.9 Invasive Species

Invasive Species are prohibited to be raised, planted, stored or possessed on DoD installations. The Contractor shall not bring in any invasive species to DoD installations.

3.2 STORMWATER

Do not discharge stormwater from construction sites to the sanitary sewer. If the water is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is specifically permitted. Obtain authorization in advance from the Installation Environmental Office for any release of contaminated water. Storm water management and control shall be detailed in the SWPPP submitted as part of Chapter 3 of the EPP submittal. Refer to paragraph 1.9.9.1.a of this section.

3.2.1 Inspection Reports

Submit Inspection Reports and Stormwater Pollution Prevention Plan Compliance Notebook in the Environmental Records Binder and provide to the Contracting Officer in accordance with the JEGS and applicable GOJ or Local Laws and Regulations.

- a. Runoff from the construction site or from storms shall be controlled, retarded, and diverted, as indicated on the SWPPP drawings, to protect drainage courses by means of diversion ditches, benches, and berms. Berms, dikes, drains, sedimentation basins, grassing, and mulching shall be maintained until permanent drainage and erosion control facilities are completed and operative.
- b. Silt screens must be installed prior to start of construction. Silt screens and/or other erosion control devices shall be installed on construction sites that are in or near water. Silt screens shall consist of trenched and staked filter fabric and trenched and staked hay bales. Filter fabric must be toed 8 inches into the soil to avoid sediments that would be transported via water under the screen. Hay bales must be placed end-to-end on the downstream side of the screen and be trenched and staked firmly into the ground. Chinking is usually required to fill gaps between the bales. Silt screens must be maintained properly. Screens and other control devices must be inspected once a week and after any rainfall event totaling one-half inch or more to ensure they are in good repair and functioning properly.
- c. In areas that experience high flow rates, extra precautions shall be necessary to stabilize screens. Trenching of hay bale barriers is required to adequately control runoff. A series of screens may have to be installed in waters that are especially turbid to properly filter out sediments. Silt screens shall remain in place and properly maintained until the site is properly stabilized with sod or seeding.
- d. Providing erosion and sediment control measures is the Contractor's responsibility. The erosion and sediment controls selected and maintained by the Contractor shall be such that water quality standards, listed in the JEGS are not violated as a result of construction activities. The area of bare soil exposed at any one time by construction operations shall be kept to a minimum. Construct or install temporary and permanent erosion and sediment control BMPs as indicated on the drawings and as specified in the SWPPP.
- e. BMPs may include, but are not limited to, vegetation cover, stream bank stabilization, slope stabilization, silt fences, construction of terraces, interceptor channels, sediment traps, inlet and outfall protection, diversion channels, and sedimentation basins. The Contractor's Best Management Practices must also be in accordance with the JEGS and the SWPPP. Remove any temporary measures after the area has been stabilized.
- f. All earthwork shall be planned and conducted to minimize the duration of exposure of unprotected soils. Side and back slopes shall be protected as soon as practicable upon completion of rough grading. Earthwork brought to final grade shall be finished as indicated.
- g. The Contractor's field offices, staging areas, stockpile storage, and

temporary buildings shall be placed in areas designated on the drawings, or as directed by the Contracting Officer. Temporary movement or relocation of the Contractor's facilities shall be made only when approved. Borrow areas shall be managed to minimize erosion and to prevent sediment from entering nearby waters.

- h. Spoil areas shall be managed and controlled to limit spoil intrusion into areas designated on the drawings and to prevent erosion of soil or sediment from entering nearby waters. Spoil areas shall be developed in accordance with the grading plan indicated on the drawings. Temporary excavation and embankments for plant and/or work areas shall be controlled to protect adjacent areas from despoilment.

3.2.2 Erosion and Sediment Control Measures

Provide erosion and sediment control measures in accordance with GOJ, Federal, and Local prefecture laws and regulations, and JEGS. Preserve vegetation to the maximum extent practicable.

Erosion control inspection reports may be compiled as part of a stormwater pollution prevention plan inspection reports.

3.2.2.1 Erosion Control

Prevent erosion by mulching, compost blankets, geotextiles, or temporary slope drains. Stabilize slopes by [chemical stabilization, sodding, seeding, or a combination of these methods necessary for effective erosion control. Provide seeding in accordance with Section 32 92 19 SEEDING. Use of hay bales is prohibited.

Any disturbed area with exposed soil that is not being worked on must be seeded or sodded no later than two weeks from the last disturbance regardless of the installation of any other erosion control measures in place. Remove any temporary measures after the area has been stabilized.

3.2.2.2 Sediment Control Practices

Implement sediment control practices to divert flows from exposed soils, temporarily store flows, or otherwise limit runoff and the discharge of pollutants from exposed areas of the site. Implement sediment control practices prior to soil disturbance and prior to creating areas with concentrated flow, during the construction process to minimize erosion and sediment laden runoff. Best Management Practices (BMPs) may include, but not limited to, vegetation cover, stream bank stabilization, slope stabilization, silt fences, construction of terraces, interceptor channels, sediment traps, inlet and outfall protection, diversion channels, and sedimentation basins. Include the following devices: silt fence, temporary diversion dikes, and/or storm drain inlet protection.

3.2.3 Work Area Limits

Mark the areas that need not be disturbed under this Contract prior to commencing construction activities. Mark or fence isolated areas within the general work area that are not to be disturbed. Protect monuments and markers before construction operations commence. Where construction operations are to be conducted during darkness, any markers must be visible in the dark. Personnel must be knowledgeable of the purpose for marking and protecting particular objects.

3.2.4 Contractor Facilities and Work Areas

Place field offices, staging areas, stockpile storage, and temporary buildings in areas designated on the drawings or as directed by the Contracting Officer. Move or relocate the Contractor facilities only when approved by the Government. Provide erosion and sediment controls for onsite borrow and spoil areas to prevent sediment from entering nearby waters. Control temporary excavation and embankments for plant or work areas to protect adjacent areas.

3.3 SURFACE AND GROUNDWATER

3.3.1 Cofferdams, Diversions, and Dewatering

Construction operations for dewatering, removal of cofferdams, tailrace excavation, and tunnel closure must be constantly controlled to maintain compliance with existing water quality standards and designated uses of the surface water body. Comply with water quality standards and anti-degradation provisions. Do not discharge excavation ground water to the sanitary sewer, storm drains, or to surface waters without prior specific authorization in writing from the Installation Environmental Office. Discharge of hazardous substances will not be permitted under any circumstances. Use sediment control BMPs to prevent construction site runoff from directly entering any storm drain or surface waters.

If the construction dewatering is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is specifically permitted. Obtain authorization for any contaminated groundwater release in advance from the Installation Environmental Officer. Discharge of hazardous substances will not be permitted under any circumstances.

3.3.2 Waters of Japan

Do not enter, disturb, destroy, or allow discharge of contaminants into waters of Japan. The protection of waters of Japan shown on the drawings in accordance with paragraph LICENSES AND PERMITS is the Contractor's responsibility. Authorization to enter specific waters of Japan identified does not relieve the Contractor from any obligation to protect other waters of Japan within, adjacent to, or in the vicinity of the construction site and associated boundaries.

3.3.3 Water Protection

Contractor shall prevent oily wastes or other hazardous substances from entering the ground, drainage areas, or local bodies of water.

3.3.4 Sewage

Contractor is not authorized sewage holding tanks on base and must procure a porta-potty service Contract. The porta-potty Contract must include the correct removal of sewage and maintenance of the facilities.

3.3.5 Wastewater

If the project generates wastewater from rinsing tanks, dewatering sites, etc., contact the Contracting Officer on proper disposal.

3.4 PROTECTION OF CULTURAL RESOURCES

3.4.1 Archaeological Resources

Existing historical, archaeological, and cultural resources within the Contractor's work area will be so designated by the Contracting Officer if any such areas have been identified. If, during excavation or other construction activities, any previously unidentified or unanticipated historical, archaeological, and cultural resources are discovered or found, activities that may damage or alter such resources will be suspended. Resources covered by this paragraph include, but are not limited to: any human skeletal remains or burials; artifacts; shell, midden, bone, charcoal, or other deposits; rock or coral alignments, pavings, wall, or other constructed features; and any indication of agricultural or other human activities. Upon such discovery or find, immediately notify the Installation Cultural Resources Manager via the Contracting Officer so that the appropriate authorities may be notified and a determination made as to their significance and what, if any, special disposition of the finds should be made. Cease all activities that may result in impact to or the destruction of these resources. Secure the area and prevent employees or other persons from trespassing on, removing, or otherwise disturbing such resources. The Government retains ownership and control over archaeological resources.

[3.4.2 Historical Resources

If applicable, existing historical resources within the work area will be shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract.

3.5 AIR RESOURCES

Equipment operation, activities, or processes will be in accordance with the JEGS and all applicable Federal, GOJ national or prefectural air emission and performance laws and regulations.

3.5.1 Oil or Dual-fuel Boilers and Furnaces

Provide product data and details for new, replacement, or relocated fuel fired boilers, heaters, or furnaces to the Installation Environmental Office (Air Program Manager) through the Contracting Officer. Data to be reported include: equipment purpose (water heater, building heat, process), manufacturer, model number, serial number, fuel type (oil type, gas type) size (MMBTU heat input). Product data shall be provided prior to equipment installation.

3.5.2 Burning

Burning is prohibited on the Government premises.

3.5.3 Class I and II ODS Prohibition

Class I and II ODS are Government property and must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Office to determine the required recovery cylinders and the appropriate location for turn in of all reclaimed refrigerant. Class I and II ODS as defined and identified herein shall not be used in the performance of this contract.

The contractor shall coordinate with the Kadena CES HVAC Shop on the quantity of ODS recovered from system for the purpose of recycle, reuse or dilution. In the event the ODS can be recycled, reused or diluted, the entire quantity will be turned over to the Refrigerant, AC and Heating shop.

In cases where the ODS is sufficiently contaminated and cannot be recycled, reused and/or diluted, the Contractor shall dispose of the ODS according to the JEGS and local regulations.

In the event of a release of ODS, base personnel and/or Contractor shall inform Fire and Emergency's Service Branch and Environmental Division.

3.5.4 Accidental Venting of Refrigerant

Accidental venting of a refrigerant is a release and must be reported immediately to the Contracting Officer. Comply with the JEGS.

3.5.5 Training/Certification Requirements

Heating and air conditioning technicians must be trained to meet requirements in the JEGS. Maintain copies of certifications at the employees' places of business; technicians must carry certification wallet cards, as provided by GOJ national or prefectural laws and regulations.

3.5.6 Dust Control

Keep dust down at all times, including during nonworking periods. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Do not unnecessarily shake bags of cement, concrete mortar, or plaster.

3.5.6.1 Particulates

Dust particles, aerosols and gaseous by-products from construction activities, and processing and preparation of materials (such as from asphaltic batch plants) must be controlled at all times, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed the JEGS and applicable U.S. Federal, GOJ national or prefectural air pollution standards or that would cause a hazard or a nuisance.

Sprinkling, chemical treatment of an approved type, baghouse, scrubbers, electrostatic precipitators or other methods shall be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp at all times. Provide sufficient, competent equipment available to accomplish these tasks. Perform particulate control as the work proceeds and whenever a particulate nuisance or hazard occurs. Comply with the JEGS, GOJ national or prefectural laws and regulations, and installation requirements.

Responsible units/activities/stakeholders/contractors must have sufficient, competent equipment available and shall use Best Management Practice (BMP) at all times to accomplish these tasks. Air emission/pollutant control shall be performed as the work proceeds and whenever potential hazards exist, emission standards are listed in the JEGS.

3.5.6.2 Abrasive Blasting

Blasting operations cannot be performed without prior approval of the Installation Air Program Manager. The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive agent, paint chips, and other debris. Perform work involving removal of hazardous material in accordance with the JEGS, GOJ national or prefectural laws and regulations, and installation requirements.

3.5.7 Odors

Control odors from construction activities. The odors must be in compliance with applicable GOJ national or prefectural laws and regulations, and may not constitute a health hazard.

3.5.8 All Refrigerant Types

Ensure a weatherproof data plate is permanently attached to refrigeration (or air conditioning) equipment in a location visible to maintenance workers, showing refrigerant type and the full charge quantity.

3.5.9 Other Pollutants

Dust particles, aerosols and gaseous by-products from construction activities, and processing and preparation of materials shall be controlled at all times, including weekends, holidays, and hours when work is not in progress. Responsible units/activities/stakeholders/Contractors must have sufficient, competent equipment available and shall use Best Management Practice (BMP) at all time while accomplishing tasks. Air emission/pollutant control shall be performed as the work proceeds and whenever potential hazards exist. Emission standards are listed in the JEGS.

3.5.10 Class I ODS

Class I ODS listed in the JEGS are prohibited from use in construction, remodeling, or maintenance. Provide certifications that materials utilized during the project do not contain Class I ODSs. The purchase of Class I ODS for fire suppression equipment, air conditioning, and refrigeration equipment for ground applications is prohibited by AFMAN 32-7002 on all new and/or refurbishing projects. Limited exceptions for using Class I ODS for airborne equipment exist, but shall require prior approval.

- a. Phase-out of Class I ODS: Installation to comply with the requirements of EO 13148, Sec 505(b), to cost effectively phase-out Class I ODS requirements as equipment using Class I ODS reach the end of their expected service life per AFMAN 32-7002. Class 1 ODS shall not be disposed of off the installation; Contractor shall coordinate with 718 CES/CEIE for procedures to ensure Class I ODSs are turned into the DLA ODS Reserve.

3.5.11 Class II ODS Approval Request

Class II ODS listed in the JEGS and AFMAN 32-7002 must first be approved by 718 CES/CEIE, Environmental Office prior to use. 718 CES/CEIE

recommends using non-ODS products as a substitute.

- a. Phase-out of Class II ODS: Installations, Contractors, and other using agencies shall comply with USAF phase-out goal by 01 Jan 2020 per AFMAN 32-7002.

3.5.12 Refrigerant Recovery and Recycling

Prior to recovery and recycling of ODS, the Contractor is required to coordinate with the using agencies (918 CES HVAC Shop, or 718 CES Environmental Branch) of the future intended use of the ODS. All repairs, including leak repairs or services to appliances, industrial process refrigeration units, air conditioning units, or motor vehicle air conditioners, must be performed using commercially available refrigerant recovery/recycling equipment operated by trained personnel. Refrigerant technicians shall be trained in proper recovery/recycling procedures, leak detection, safety, shipping, and disposal in accordance with recognized industry standards or Japanese equivalent.

3.5.13 Refrigerant Venting Prohibition

Any class I or class II ODS, HFC, and PFC refrigerant shall not be intentionally released in the course of maintaining, servicing, repairing, or disposing of appliances, industrial process refrigeration units, air conditioning units, or motor vehicle air conditioners. Minor releases associated with good faith attempts to recycle or recover ODS, HFC, and PFC refrigerants are not subject to this prohibition.

3.5.14 Halon Venting Prohibition

Halons shall not be intentionally released into the environment while testing, maintaining, servicing, repairing, or disposing of Halon-containing equipment or using such equipment for technician training. This venting prohibition does not apply to the Halon releases listed in the JEGS.

- a. Halon De minimis release associated with good faith attempts to recycle or recover halon (i.e. release of residual halon contained in fully discharged total flooding fire extinguishing systems).
 - b. Emergency release for the legitimate purpose of fire extinguishing, explosion inertion, or other emergency applications for which the equipment or systems were designed.
 - c. Releases during the testing of fire extinguishing systems if each of the following is true; systems or equipment employing suitable alternative fire extinguishing agents are not available; release of extinguishing agent is essential to demonstrate equipment functionality; failure of system equipment would pose a great risk to human safety or the environment; and simulant agent cannot be used.
- b. Hydrocarbons and Carbon Monoxide: Hydrocarbons and carbon monoxide emissions from equipment shall be controlled to allowable limits at all times.

3.5.15 Generators, Incinerators, and Boilers

Generators, Incinerators, and Boilers: If generators, incinerators, or boilers are to be installed or replaced as part of the project, they must meet standards listed in the JEGS. Provide 718 CES with the stack height,

Fuel Combustion Rate (L/hr) for generators; Grate area (m²), Rate capacity (t/day), Incineration rate (kg/h) for incinerators; and heating area (m²), burner combustion rate (L/hr), Heat input rating (MMbtu/hr) for boilers, manufacturer's results from their product emission testing or other emission testing results, the heat exchange surface area in square meters for boilers, and the fuel oil consumption rate in liters per hour for boilers and generators.

3.5.16 Hydrocarbons and Carbon Monoxide

Hydrocarbons and Carbon Monoxide: Hydrocarbons and carbon monoxide emissions from equipment shall be controlled to allowable limits at all times.

3.6 WATER RESOURCES

Keep construction activities under surveillance, management, and control, and monitor all water areas affected by construction activities in order to prevent pollution of surface and ground waters. Toxic or hazardous chemicals shall not be applied to soil or vegetation when such application may cause contamination. Prevent oily wastes or other hazardous substances from entering the ground, drainage areas, or local bodies of water.

Monitoring of water areas affected by construction shall be the Contractor's responsibility. All water areas affected by construction activities shall be monitored by the Contractor.

3.6.1 Lead-Free Drinking Water Pipe, Solders, Flux and Fittings

The maximum allowable lead content for pipes, fittings, and fixtures intended to convey or dispense water for human consumption and cooking shall be a weighted average of 0.25 percent with respect to the wetted surfaces of pipes and pipe fittings, plumbing fittings, and fixtures in accordance with NSF 61. Refer to section 22 00 00 PLUMBING, GENERAL PURPOSE.

3.7 WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of waste. Include procedures for pollution prevention/hazardous waste minimization in the Hazardous Waste Management Section of the EPP. Obtain a copy of the installation's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EPP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in the construction when requesting information.

3.7.1 Salvage, Reuse and Recycle

Identify anticipated materials and waste for salvage, reuse, and recycling. Describe actions to promote material reuse, resale or recycling. To the extent practicable, all scrap metal must be sent for reuse or recycling and will not be disposed of in a landfill.

Include the name, physical address, and telephone number of the hauler, if transported by a franchised solid waste hauler. Include the destination and, unless exempted, provide a copy of the state or local permit (cover) or license for recycling.

All solid wastes and materials that have been separated for the purpose of recycling shall be stored in such a manner that they do not constitute a fire, health or safety hazard, or provide food or harborage for vectors, and shall be contained or bundled so as to not result in spillage. Containers must be: leak proof, water proof, and vermin proof including sides, seams, tops and bottoms, durable enough to withstand anticipated usage, and stored on a firm, level, well-drained surface).

3.7.2 Nonhazardous Solid Waste Diversion Report

Maintain an inventory of nonhazardous solid waste diversion and disposal of construction and demolition debris. Submit a report to the Environmental Office through the Contracting Officer on the first working day after each fiscal year quarter, starting the first quarter that nonhazardous solid waste has been generated. Include the following in the report:

Construction and Demolition (C&D) Debris Disposed	(____) cubic meters, as appropriate
C&D Debris Recycled	(____) cubic meters, as appropriate
Total C&D Debris Generated	(____) cubic meters, as appropriate
Waste Sent to Waste-To-Energy Incineration Plant (This amount should not be included in the recycled amount)	(____) cubic meters, as appropriate

Include the completed form, Disposal and Recycling Data Sheet, Attachment A, attached at the end of the section.

3.8 WASTE MANAGEMENT AND DISPOSAL

3.8.1 Waste Determination Documentation

Complete Waste Determination Documentation for Contractor-derived wastes to be generated and submit in the Environmental Records Binder. All potentially hazardous solid waste streams that are not subject to a specific exclusion or exemption from the hazardous waste regulations (e.g. scrap metal, domestic sewage) or subject to special rules, (lead-acid batteries and precious metals) must be characterized in accordance with the requirements of the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. Base waste determination on user knowledge of the processes and materials used, and analytical data when necessary. Consult with the Installation environmental staff for guidance on specific requirements. Attach support documentation to the Waste Determination form. As a minimum, provide a Waste Determination form for the following waste (this listing is not inclusive): oil- and latex -based painting and caulking products, solvents, adhesives, aerosols, petroleum products, and containers of the original materials.

3.8.1.1 Sampling and Analysis of Waste

3.8.1.1.1 Waste Sampling

Sample waste in accordance with the JEGS, Installation Hazardous Waste Management Plan, appropriate Japanese or U.S. EPA testing protocols that meet the purpose of the testing. Clearly mark each sampled drum or container with the Contractor's identification number, and cross reference to the chemical analysis performed.

Excess excavated soil shall become the property of the Contractor and shall be disposed of off base at a properly licensed facility in accordance with the latest version of the JEGS and applicable GOJ federal and local laws and regulations. Any excess soil shall be sampled and tested according to the applicable GOJ federal and local laws and regulations and disposal facility requirements prior to disposal.

3.8.1.1.2 Laboratory Analysis

Follow the analytical procedure and methods in accordance with the JEGS and all applicable GOJ national or prefectural laws and regulations and Installation Hazardous Waste Management Plan. Provide analytical results and reports performed to the Contracting Officer.

3.8.1.1.3 Analysis Type

Identify hazardous waste by analyzing for the following characteristics: ignitability, corrosivity, reactivity, and toxicity based on TCLP results.

3.8.2 Solid Waste Management

3.8.2.1 Solid Waste Management Report

Provide copies of the waste handling facilities' weight tickets, receipts, bills of sale, Contractor Certification and other sales documentation in the Environmental Protection Plan. In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. The sales documentation or Contractor certification must include the receiver's tax identification number and business, GOJ or prefectural registration number, along with the receiver's delivery and business addresses and telephone numbers.

3.8.2.2 Control and Management of Solid Wastes

Pick up solid wastes, and place in covered containers that are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Employ segregation measures so that no hazardous or toxic waste will become co-mingled with non-hazardous solid waste. Transport solid waste off Government property and dispose of it in compliance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements for solid waste disposal. Solid waste disposal offsite must comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. Verify that the selected transporters and disposal facilities have the necessary permits and licenses to operate.

Manage hazardous material used in construction, including but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

3.8.3 Control and Management of Hazardous Waste

Do not dispose of hazardous waste on Government property. Do not discharge any waste to a sanitary sewer, storm drain, or to surface waters or conduct waste treatment or disposal on Government property without written approval of the Contracting Officer.

The Contractor is responsible for managing all hazardous wastes (HW) generated throughout the duration of the Contract. The Contractor shall designate a hazardous waste manager in writing who shall be responsible for all HW management activities. The Contractor may attend the Hazardous Waste Accumulation Point Management course (English only) offered quarterly by 718CES/CEIE.

All HW generated by the Contractor shall be disposed/recycled off base in accordance with Japanese regulation, Article 3 of "Law of Disposal and Clearing of Waste" Haikibutsu no shori oyobi seisou ni kansuru horitsu" and local regulations. U.S./foreign made ionization smoke detectors shall be turned in to 718 CES/CEIE HWSA. The Contractor shall not leave any hazardous waste at work site areas at the end of duty hours. If temporary hazardous waste storage is needed to accumulate hazardous waste on base, the Contractor shall submit spill response plan with site map, the list of all hazardous waste to be stored on-base and estimated quantities to 718 CES/CEIE, 718 CES/CEF, 718 AMDS/SGPB and 718 AW/SE for approval via the Contracting Officer.

3.8.3.1 Hazardous Waste/Debris Management

Identify construction activities that will generate hazardous waste or debris. Provide a documented waste determination for resultant waste streams. Identify, label, handle, store, and dispose of hazardous waste or debris in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

Manage hazardous waste in accordance with the approved Hazardous Waste Management Section of the EPP. Store hazardous wastes in approved containers in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. Do not bring hazardous waste onto Government property. Provide the Contracting Officer with a copy of waste determination documentation for any solid waste streams that have any potential to be hazardous waste or contain any chemical constituents listed in the JEGS. For hazardous wastes spills, verbally notify the Contracting Officer immediately.

3.8.3.2 Hazardous Waste Records

The Contractor shall maintain sampling, analysis, and turn-in records for all hazardous waste generated during the project. These records shall include, but not be limited to: waste profile sheets provided by HWSF personnel for wastes streams turned in to the HWSF, manifests (DD forms

1348-1A) for all wastes turned over to the HWSF, logs of sample locations or container identification data (including time and date of sample collection), analytical results, and quality control data provided by the analytical lab pertaining to the samples analyzed. Copies of this data shall be submitted to 718 CES/CEIE via the COR after the work is completed.

3.8.3.3 Electronics End-of-Life Management

Recycle or dispose of electronics waste, including, but not limited to, used electronic devices such as computers, monitors, hard-copy devices, televisions, mobile devices, in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, local requirements and installation instructions.

3.8.3.4 Disposal Documentation for Hazardous and Regulated Waste

If the Contract specifies that the Contractor is responsible for disposal at a commercial landfill, submit copies of the following documents:

- a. Copy E of the Hazardous Waste Manifest form used to transport from on-base to the off-base disposal facility (Sangyou Haikibutsu Manifesuto) within 60 days of transport.
- b. Transportation company's Industrial Waste Collection and Transport Business Permit (Sangyou Haikibutsu Shuushuu Unpangyou no Kyoka) as a part of a separate Hazardous Waste Abatement Plan.
- c. Disposal facility's Industrial Waste Business Permit (Sangyou Haikibutsu Shobungyou no Kyoka) as a part of a separate Hazardous Waste Abatement Plan.

3.8.4 Releases/Spills of Oil and Hazardous Substances

3.8.4.1 Response and Notifications

The Contractor shall conduct all operations in a manner that prevents spills of POL or other hazardous substances. The Contractor is required to familiarize personnel with spill prevention and response procedures, fire suppression systems, and SDSs for all materials used and/or stored on the project site. The Contractor shall provide and maintain spill equipment, sufficient in both type and quantity, at all sites involving the storage, use, or handling of POL and/or hazardous substances. Exercise due diligence to prevent, contain, and respond to spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with the JEGS and installation plans. Maintain spill cleanup equipment and materials at the work site.

Follow the spill prevention measures detailed in the Installation Spill Prevention and Response Plan. The Contractor is required to familiarize personnel with spill prevention and response procedures, fire suppression systems, and SDSs for all materials used and/or stored on the project site. Provide and maintain spill equipment, sufficient in both type and quantity, at all sites involving the storage, use, or handling of POL and/or hazardous substances.

In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the

spill/release. In the event of any releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the 18 CES/CEF, Installation Fire Department (at 911 from a landline or 098-934-5911 from a cell phone), the Installation Command Duty Officer, the 718 CES Installation Environmental Office at DSN 634-2600, and the Contracting Officer.

. Submit verbal and written notifications as required by the JEGS and installation and service component instructions and plans, and local regulations. Provide copies of the written notification and documentation that a verbal notification was made within the timeframes required by the JEGS and installation instructions. Spill response must be in accordance with the JEGS and installation requirements. Contain and clean up these spills without cost to the Government.

Submit verbal and written notifications as required by the JEGS and installations instructions. Provide copies of the written notification and documentation that a verbal notification was made within the timeframes required by the JEGS and installation instructions. Spill response must be in accordance with the JEGS and installation requirements. Contain and clean up these spills without cost to the Government.

3.8.4.2 Clean Up

The Contractor is responsible for the cleanup of POL and/or hazardous substance spills and disposal of clean-up material. The Contractor shall determine, as quickly as possible, the nature of the spilled substance and implement necessary safety precautions to protect both human health and the environment. Cleanup shall be in accordance with applicable local laws and regulations, as determined by the Fire Department and 718 CES/CEIE. Cleanup shall only be performed by personnel adequately trained in spill response and cleanup techniques for the severity of the spill incident. Clean up hazardous and non-hazardous waste spills. Reimburse the Government for costs incurred including sample analysis materials, clothing, equipment, and labor if the Government will initiate its own spill cleanup procedures, for Contractor-responsible spills, when: Spill cleanup procedures have not begun within one hour of spill discovery/occurrence; or, in the Government's judgment, spill cleanup is inadequate and the spill remains a threat to human health or the environment.

The Contractor shall be responsible for all costs incurred from any Contractor spills. The Contractor shall be held liable for all costs associated with performing emergency spill response and cleanup by the U.S. Government due to the non-availability of designated personnel or the limited capability of designated personnel. The U.S. Government's cost of cleanup shall be considered a Contract debt and collected in accordance with FAR subpart 32.6. If remaining Contract payments are insufficient, the Government reserves the right to pursue other offsets or administrative or civil actions to satisfy the debt.

3.8.5 Mercury Materials

Immediately report to the Environmental Office and the Contracting Officer instances of breakage or mercury spillage. Clean mercury spill area to the satisfaction of the Contracting Officer.

Do not recycle a mercury spill cleanup; manage it as a hazardous waste for

disposal.

3.8.6 Wastewater

Disposal of wastewater must be as specified below.

3.8.6.1 Treatment

Do not allow wastewater from construction activities, such as on-site material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, and forms to enter water ways[.] [or to be discharged without proper approval from the Contracting Officer.] Dispose of the construction-related waste water off-Government property in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements, or by collecting and placing it in a retention pond where suspended material can be settled out or the water can evaporate to separate pollutants from the water. The site for the retention pond as well as its use must be coordinated and approved with the Contracting Officer. The residue left in the pond prior to completion of the project must be removed and disposed of off-Government property in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. Backfill the area to the original grade, top-soiled, and seeded or sodded.

3.8.6.2 Surface Discharge

For discharge of ground water, if applicable, obtain a GOJ or local prefectural permit specific for pumping and discharging ground water prior to surface discharging. Surface discharge shall be done in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations and installation requirements.

3.8.6.3 Land Application

Water generated from the flushing of lines after[disinfection or disinfection in conjunction with hydrostatic testing][hydrostatic testing] must be[land- applied in accordance with the JEGS and applicable Federal, GOJ, prefectural and local laws and installation requirements for land application][discharged into the sanitary sewer with prior approval and notification to the owner of the sanitary sewer system].

3.8.7 Contaminated Soil Stockpile Request

The Contractor shall submit a Contaminated Soil Stockpile Request to 718 CES/CEIE for stockpiling contaminated soil for testing prior to disposal. Upon receipt of request, 718 CES/CEIE will inform the Contractor of an approved stockpile location. The Contractor shall be responsible for containment and monitoring of the contaminated soil stockpile at no additional cost to the U.S. Government.

3.9 HAZARDOUS MATERIAL MANAGEMENT

Include hazardous material control procedures in the Safety Plan, in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. Do not bring hazardous material onto Government property that does not directly relate to requirements for the performance of this Contract. Submit a Safety Data

Sheet and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the installation. Typical materials requiring SDS and quantity reporting include, but are not limited to, oil and latex based painting and caulking products, solvents, adhesives, aerosol, and petroleum products. Use hazardous materials in a manner that minimizes the amount of hazardous waste generated. Containers of hazardous materials must have National Fire Protection Association labels or their equivalent.

Storage and handling of hazardous materials will adhere to the DoD Component policies, including Joint Service Publication on Storage and Handling of Hazardous Materials. Defense Logistics Agency Instruction (DLAI) 4145.11, Army Technical Manual (TM) 38-410, Naval Supply Publication (NAVSUP PUB) 573, Air Force Joint Manual (AFJMAN) 23-209, and Marine Corps Order (MCO) 4450.12A, "Storage and Handling of Hazardous Materials," January 13, 1999 provide additional guidance on the storage and handling of hazardous materials. The International Maritime Dangerous Goods (IMDG) Code and appropriate DoD and Component instructions provide requirements for international maritime transport of hazardous materials originating from DoD installations. International air shipments of hazardous materials originating from DoD installations are subject to International Civil Aviation Organization Technical Instructions or DoD Component guidance, including Air Force Manual 24-204, (Interservice) TM 38-250, NAVSUP PUB 505, MCO P4030.19J, and DLAI 4145.3, DCMAD1, Ch3.4 (HM24), "Preparing Hazardous Materials for Military Air Shipments," 3 December 2012. Certify that hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste, in accordance with the JEGS and installation requirements.

3.9.1 Hazardous Material Approval

Contractors seeking approval to bring HAZMAT onto the installation must submit an AF Form 3952 or AF Form 3000. A Safety Data Sheet (SDS) or Safety Data Sheet (SDS), in English and Japanese, must accompany the AF Form 3952 or AF Form 3000. For specific details on information required for submitting an AF Form 3952 or AF Form 3000, contact the Hazardous Material Program Manager at DSN 634-2600. The Contractor must coordinate and route the AF Form 3952 or AF Form 3000 through 18th Wing Safety, Fire Department, Bioenvironmental, and 718 CES/CEIE. All excess materials and empty containers are the responsibility of the Contractor and shall be removed at the end of the Contract. If HAZMAT requirements change during the project, submit a new or revised USAF Form 3952 or AF Form 3000 and obtain approval before bringing the HAZMAT onto the installation. For Army projects, the AF Form 3952 shall be coordinated through the Directorate of Logistics (DOL) HAZMAT and approved by Fire Department, Safety, and Environmental. When occupants of the facility request that the contractor leave the excess HAZMAT, the Contractor shall coordinate with and obtain approval from 718 CES/CEIE.

3.9.2 Hazardous Material Storage, Handling and Marking

Storage of oils, greases, chemicals, fuels or other liquids will require spill prevention and security. Contractor is not allowed to store HAZMAT on base for over 24 hours without authorization. If on-base HAZMAT storage areas are needed, Contractor shall submit the list of all items that are to be stored on-base and estimated quantities to 718 CES/CEIE, 718 CES/CEF and 718 AW/SE for approval. Hazardous materials storage areas may be inspected by 718 CES/CEIE, 718 CES/CEF and 718 AW/SE. SDSs must be available at storage and on site at all times.

3.10 PREVIOUSLY USED EQUIPMENT

Clean previously used construction equipment prior to bringing it onto the project site. Equipment must be free from soil residuals, egg deposits from plant pests, noxious weeds, and plant seeds.

3.11 CONTROL AND MANAGEMENT OF ASBESTOS-CONTAINING MATERIAL (ACM)

Manage and dispose of asbestos-containing waste in accordance with the JEGS and applicable Federal, GOJ national or prefectural laws and regulations. Refer to Section 02 82 00 ASBESTOS REMEDIATION. Manifest asbestos-containing waste and provide the manifest to the Contracting Officer. Notifications to the Contracting Officer and Installation Air Program Manager are required before starting any asbestos work.

Removal of mastics, necessary to achieve the contract's objectives, shall be performed in a manner as if the material contains asbestos.

When applicable, a minimum of 14 calendar days prior to the demolition or renovation of a facility that involves removing or disturbing friable ACM, the Contractor shall prepare a written assessment of friable asbestos disturbance and submit to the COR, who will, in turn submit to the Installation Commander in accordance with the JEGS.

For any previously untested material suspected to contain asbestos and located in areas impacted by the work, notify the Contracting Officer, who will order up to 100 bulk samples to be obtained and analyzed at the Contractor's expense. Samples shall be collected in sufficient amounts such that a follow-on test utilizing TEM or 1000 point count analysis can be performed, should it be necessary. The Contractor shall deliver the sample(s) to a laboratory accredited under the National Institute of Standards and Technology (NIST) "National Voluntary Laboratory Accreditation Program (NVLAP)" and analyzed by PLM. The laboratory shall have a working definition of "Trace" amounts of asbestos, and the laboratory shall report any detectable amount of asbestos in a bulk sample that is less than the PLM Limit of Quantification of 1 percent as a "Trace" concentration. If PLM does not detect the presence of asbestos (e.g. "non-detect"), the material shall be considered less than 0.1 percent asbestos. If PLM analysis detects asbestos in any discernible amount (to include "trace" or "less than 1 percent"), the material shall be considered greater than 0.1 percent asbestos unless proven to be non-ACM by the use of quantification methods capable of achieving an analytical sensitivity of less than 0.1 percent, such as Transmission Electron Microscopy (TEM) or 1000 point counting.

The Contracting Officer (CO) will order, testing by TEM or 1000 point counting, for up to 20 samples, to be obtained at the Contractor's expense and delivered to a laboratory accredited under the National Institute of Standards and Technology (NIST) "National Voluntary Laboratory Accreditation Program (NVLAP)", for those samples with results of < 1% ACM as analyzed by PLM.

Any additional components identified as ACM that have been approved by the Contracting Officer for removal shall be removed and will be paid for by an equitable adjustment to the Contract price under the CONTRACT CLAUSE titled "changes". Sampling shall be conducted by personnel who have successfully completed the EPA Model Accreditation Plan (MAP) "Building Inspector" training course and is AHERA certified as a "Building

Inspector".]

3.11.1 Sample Result Reporting

The contractor shall submit to the Contracting Officer a "Suspected ACM Discovery Report" within 5 days of receiving the final laboratory analysis. The report shall contain, at a minimum, a text description of the location, building and room number of the discovery, and the condition of the material; a copy of the sampler's "Building Inspector" certificate; copies of the laboratory's accreditations; a color photograph of the sampled location; the chain of custody document; a certified copy of laboratory analysis sheet discerning the non-detect/detect threshold at 0.1 percent.

3.11.2 Contract Changes

ACM-related work beyond the scope of this specification, such as the removal of ACM identified by the contractor after beginning site work, must be approved by the Contracting Officer, in advance, and will be paid for by an equitable adjustment to the Contract price under the CONTRACT CLAUSE titled "changes".

3.11.3 New Materials

All materials provided by the Contractor shall contain less than 0.1% asbestos per weight in accordance with the JEGS. The COR may request the contractor to provide documentation from the manufacturer for any material suspected of containing asbestos to confirm asbestos content.

3.11.4 Flooring/Base Cove Mastic

Removal of mastics, necessary to achieve the contract's objectives, shall be performed in a manner as if the material contains asbestos.

3.11.5 Additional Requirements

All such work "changes" must be performed in accordance with Section ASBESTOS REMEDIATION 02 82 00 ASBESTOS REMEDIATION and is subject to the submittal requirements listed therein.

3.12 CONTROL AND MANAGEMENT OF LEAD-BASED PAINT (LBP)

Manage and dispose of lead-contaminated waste in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations and installation requirements. Refer to Section 02 83 00.01 LEAD REMEDIATION. The Contractor is to assume that all existing paint encountered in this Contract contain detectable levels of lead, and the OSHA regulations referenced in these specification applies. Any attached lead material survey information is provided for disposal purposes only. Manifest any lead-contaminated waste and provide the manifest to the Contracting Officer.

For LBP and other lead abatement projects, refer to the JEGS and Section 02 83 00.01 LEAD REMEDIATION.

3.13 CONTROL AND MANAGEMENT OF POLYCHLORINATED BIPHENYLS (PCBS)

Manage and dispose of PCB-contaminated waste in accordance with the JEGS,

installation requirements, and Section 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCB). Purchase of electrical equipment and transformers containing PCBs is prohibited.

The JEGS prohibits the acquisition of any new items containing any detectable PCBs. Items that historically contained PCB includes, but are not limited to: any dielectric oil-filled items such as transformers, switchgear, capacitors, rectifiers, high-voltage paper insulated lead cable (PILC), magnetic lighting ballasts (including some labeled No-PCB), high-voltage potheads, High Intensity Discharge (HID) lighting ballasts, hydraulic oil-filled machinery, etc.

Some items manufactured for the U.S. market that are labeled "No-PCB" may legally contain as high as 49ppm PCBs. However, these items are regulated under the JEGS and not suitable for use on USFJ projects.

All procurement of transformers or any other equipment containing dielectric or hydraulic fluid shall be accompanied by the manufacturer's certification that all materials/equipment (e.g., ballasts, capacitors, transformers, Paper Insulated Lead Covered (PILC) Cable, etc.) contain non-PCB dielectric fluid. All Non-PCB certificates shall be submitted to the 718 CES/CEIE and CEOFE with the list of the new material/equipment information prior to transformer or equipment installation. Newly procured transformers and equipment shall have permanent plate affixed stating they are PCB-free in Japanese and in English. The plate should be a 200mm long and 50mm wide acrylic plate with heat treated paint. Base color should be green and letter color should be white. The plate must state "PCB Free by Japanese standard; date of installation, manufacturer, and serial number" in both Japanese and English. The plate should be affixed to the outside of transformer or cubicle enclosure in an easily identifiable location.

Follow the JEGS regarding the strict handling, storage, marking, transportation, disposal, and recordkeeping for all equipment containing less than 0.5 ppm PCB.

All transformers or any other equipment containing dielectric/hydraulic fluid shall be considered and treated as PCB containing unless information to the contrary exists. All suspected PCB materials removed for disposal must be sampled for analysis. The Contractor shall identify if the material contains PCB by using Japanese approved method required by the Ministry of the Environment. After the material is confirmed as PCB, the Contractor shall provide the sample result to 718 CES/CEIE. In addition, all sample results, regardless of PCB contamination, must be provided to 718 CES/CEIE. Material and equipment which cannot be sampled such as those hermetically sealed shall be treated as PCB containing. If the material is confirmed as Non-PCB, the Contractor is able to dispose of it as a regular industrial waste or recycle it by following applicable local Japanese regulations. Government will provide PCB sampling results to the Contractor, if available. If the material is confirmed as PCB, the Contractor shall turn in the material to 718 CES/CEIE for proper disposal. All PCB containing materials must be containerized in appropriate UN certified drum or placed in a metal secondary containment which holds 125 percent of total oil volume. PCB item(s) and secondary containment must be placed on a pallet and secured with metal strap bands. Prior to turn in, the Contractor shall provide a list of PCB items to 718 CES/CEIE. The list must contain country of origin, name of manufacturer, year of manufacture, serial number, KVA, PCB concentration, oil volume, weight of item, weight of drum/secondary containment, weight

of pallet and total weight. The list needs to be reviewed by 718 CES/CEIE prior to turn in. The Contractor must not remove any PCB identification label affixed on the material, and ensure the PCB items do not spill during handling, storage and transportation.

3.13.1 Removal Or Installation

Work on PCB equipment is forbidden using this supplement alone. If the project involves the disturbance, relocation, installation, or removal of products that have historically contained PCBs, follow the instructions in the supplement to the specification that pertains to that product's installation or removal (Transformer Replacement, Lighting Replacement, etc.). Should the project scope of work require the installation of a new transformer or the removal of an existing transformer, the Contractor shall fill and submit attachments D and E, respectively.

] [3.14 CONTROL AND MANAGEMENT OF LIGHTING BALLAST AND LAMPS CONTAINING PCBS

Manage and dispose of contaminated waste in accordance with the JEGS and installation requirements. Refer to Section 02 84 16 HANDLING OF LIGHTING BALLASTS AND LAMPS CONTAINING PCBS AND MERCURY

Lighting ballasts removed under Section 02 84 16 HANDLING OF LIGHTING BALLASTS AND LAMPS CONTAINING PCBS AND MERCURY(if applicable) shall be turned in at Kadena Air Base Building 3625 on any Friday (except for federal holidays) between the hours of 0800 and 1200. The Contractor must complete a Ballast Turn-in Inventory form that can be obtained by calling the Kadena Air Base Environmental Section at DSN 634-2600.

The Contractor shall identify if lighting fixture/ballast(s) contain PCBs. Japanese lighting fixtures/ballasts may be identified with the manufacturer's website. All U.S. lighting fixtures/ballasts shall be treated as PCBs unless there is a label stating as Non-PCB. All U.S. made ballasts with or without a label stating PCB Free shall be treated as PCB containing materials in Japan unless information to the contrary exists. U.S. made ballasts without label may contain greater than 500ppm PCBs, and those with "PCB Free" label may contain less than 50ppm PCBs. If the lighting fixture/ballast is confirmed as Non-PCB item, the Contractor is able to dispose of it as a regular industrial waste or recycle it following local Japanese regulations. If the lighting fixture/ballast is confirmed as to contain PCBs, the Contractor shall segregate ballasts by country and manufacturer, and shrink wrap each individually for turn in to the 718 CES/CEIE. Prior to turn in, the Contractor shall submit a list of PCB waste including country of origin, name of manufacturer, serial number, year of manufacture, weight and quantity of ballasts to 718 CES/CEIE for proper disposal .

Small quantities can be brought in "as-is". However, quantities over 100 must be stacked on pallets and sorted by manufacturer type. Assistance unloading pallets may be available, but Contractor is required to call to make an appointment to ensure the service is available.

Do not attempt to open sealed ballasts or perform an oil analysis. If ballasts are leaking when removed, notify the Contracting Officer and call the Kadena Air Base Environmental Section immediately for instructions on how to proceed.

]3.15 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING

POL products include flammable or combustible liquids, such as gasoline, diesel, lubricating oil, used engine oil, hydraulic oil, mineral oil, and cooking oil. Store POL products and fuel equipment and motor vehicles in a manner that affords the maximum protection against spills into the environment. Manage and store POL products in accordance with the JEGS and installation requirements. Use secondary containments, dikes, curbs, and other barriers, to prevent POL products from spilling and entering the ground, storm or sewer drains, stormwater ditches or canals, or navigable waters of Japan. Describe in the EPP (see paragraph ENVIRONMENTAL PROTECTION PLAN) how POL tanks and containers must be stored, managed, and inspected and what protections must be provided. Storage of fuel on the project site must be in accordance with the JEGS and installation requirements, and local laws and regulations and paragraph OIL STORAGE INCLUDING FUEL TANKS.

3.15.1 Used Oil Management

"Used oil," means any oil or other waste petroleum, oil, or lubricant (POL) product that has been refined from crude oil, or is synthetic oil, has been used and as a result of such use, is contaminated by physical or chemical impurities, or is off specification and cannot be used as intended. Although used oil may exhibit the characteristics of reactivity, toxicity, ignitability, or corrosivity, it is still considered used oil, unless it has been mixed with hazardous waste. Manage used oil generated on site in accordance with the JEGS and installation requirements. Determine if any used oil generated while onsite exhibits a characteristic of hazardous waste. Used oil containing 1,000 parts per million of solvents is considered a hazardous waste and disposed of at the Contractor's expense. Used oil mixed with a hazardous waste is also considered a hazardous waste. Dispose in accordance with paragraph HAZARDOUS WASTE DISPOSAL.

3.15.2 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection for oil storage tanks. A berm used to provide secondary containment must be of sufficient size and strength to contain the contents of the tanks plus 12 centimeters freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks. If tanks and containers with an aggregate aboveground capacity greater than 5000 liter will be used onsite (only containers with a capacity of 208 liter or greater are counted), provide and implement a SPCC plan meeting the requirements of 40 CFR 112. Do not bring underground storage tanks to the installation for Contractor use during a project. Submit the SPCC plan to the Contracting Officer for approval.

Monitor and remove any rainwater that accumulates in open containment dikes or berms. Inspect the accumulated rainwater prior to draining from a containment dike to the environment, to determine there is no oil sheen present.

3.15.3 Temporary Fuel Storage Tanks

The use of temporary fuel storage tanks on base as part of a project must be approved by [718 CES/CEIE][374 CES/CEIE][35 CES/CEIE]. The Contractor shall submit a Temporary Fuel Storage Tank Approval Request to 718 CES/CEIE. The approval request must contain justification for the use of a temporary tank and design details, including a layout diagram indicating the proposed location of the tank within the project site.

If approved, temporary fuel storage tanks shall be installed in accordance with all applicable regulation (JEGS, AFIs, UFCs, etc.) and the 18WG AST standards, which are maintained by 718 CES/CEIE and are available upon request. In addition, a site-specific Spill Contingency Plan (SSCP) must be developed. After construction/installation of a temporary fuel storage tank, the Contractor shall contact 718 CES/CEIE at DSN 634-2600 and the Fuels Management Flight (18 LRS/LGRF) at DSN 634-7039 or 634-7037 to coordinate an initial inspection of the tank before it is put in service.

3.15.4 Aboveground Storage Tank (AST) Installation

If the Contractor is installing a permanent aboveground fuel storage tanks as part of the project, it must be approved by 718 CES/CEIE, Storage Tank Program Manager. Tank approval request must contain justification for the installation tank and design details, including a layout diagram indicating the proposal location of the tank within the project site. the tanks shall be installed in accordance with all applicable regulations (JEGS, AFIs, UFCs, etc.) and the 18WG AST standards, which are maintained by 718 CES/CEIE and are available upon request. The Contractor is responsible for conducting all operational checks, including providing fuel for the tests, prior to turnover to the Installation. The Contractor shall ensure that design drawings contain adequate detail to demonstrate that the AST will be installed in accordance with all applicable regulations.

718 CES/CEIE will assign a Tank ID Number to each AST that will be used on tank signage and in project documentation. After construction/installation of an AST, the Contractor shall contact 718 CES/CEIE at DSN 634-2600 and the Fuels Management Flight (18 LRS/LGRF) at DSN 634-7039 or 634-7037 to coordinate an Initial Inspection of the tank before it's put in-service in accordance with AFI 23-204 KADENAABSUP.

3.15.5 Underground Storage Tank (UST) Installation

As per PACAF UST Policy, USTs shall not be installed. Fuel storage tanks shall be installed above ground, unless overriding requirements exist against above ground installation such as safety, fire protection measures, or force protection measures. Installation of a UST requires the approval of 718 CES/CEIE and approval will only be granted if adequate justification of overriding requirements against installation of an AST is provided.

To request approval for installation of a UST, the Contractor shall submit an Underground Fuel Storage Tank Approval Request to 718 CES/CEIE. The approval request must contain a detailed justification of the overriding requirements against installation of an AST. If a UST is approved for installation as part of a project, the Contractor shall ensure that design drawings contain adequate detail to demonstrate that the UST will be installed in accordance with all applicable regulations. After construction/installation of a UST, the Contractor shall contact 718

CES/CEIE at DSN 634-2600 and the Fuels Management Flight (18 LRS/LGRF) at DSN 634-7039 or 634-7037 to coordinate an initial inspection of the tank in accordance with AFI 23-204 KADENAABSUP before it's put in-service.

3.15.6 AST Removal

When a facility with a fuel storage tank(s) is demolished, demolition of the tank(s) and associated infrastructure, including above ground piping, underground piping to the building serviced by the tank, indoor piping to the equipment serviced by the tank, secondary containment curbing, tank saddles, tank pads, Oil Water Separators (OWS), grounding stations, signage, fencing around the AST, and sidewalks leading to the AST shall be included in the scope of the project. The Contractor shall comply with applicable regulations, including, but not limited to AFMAN 32-1067, UFC 3-460-01, and 33 01 50.55 CLEANING OF PETROLEUM STORAGE TANKS.

3.15.7 UST Removal

The following standards shall be applied when removing a UST, as required by the UST Removal Procedures appendix of the PACAF UST Policy. Additional UST standards shall also be applied, including but not limited to the JEGS, AFMAN 32-1067, UFC 3-460-01, Section 02 65 00 UNDERGROUND STORAGE TANK REMOVAL,, and 33 01 50.55 CLEANING OF PETROLEUM STORAGE TANKS.:

- a. Tank Removal. Notify the Environmental Representative at least 24 hours before removing a tank. Deactivate any electrical power connected to the tank or associated infrastructure. Clean the tank (triple rinse) and render tank inert (vapor must not exceed 10 percent Lower Explosive Limit for the substance(s) that were in the tank).
- b. Excavation. Place barricades around all excavations, excavated soils, removed tanks, and equipment until completion. The Contractor must have an approved AF IMT 103 prior to any excavation.
- c. Tank Cutting. Cut a 0.6 m by 0.6 m opening on each side of the tank to prevent reuse, paint the Tank ID Number (obtained from 718 CES/CEIE) and date of removal on the tank, and take photos for documentation purposes.
- d. Soil Removal. If exposed free product and/or obviously contaminated soil is encountered during UST removal, contact 718 CES/CEIE immediately. When a leaking UST is removed, exposed free product and/or obviously contaminated soil in the immediate vicinity of the tank will be removed and appropriately disposed of in accordance with the JEGS. If soil contamination is caused by the Contractor's mismanagement, the Contractor is responsible for cleanup and disposal in accordance with spill response procedures outlined in this specification and local applicable regulations.

If additional contaminated soil is present beyond the immediate vicinity of the tank, 718 CES/CEIE will provide additional guidance on necessary actions. Segregate clean soil from the contaminated soil. Only clean soil shall be used as backfill. If the soil is contaminated with Petroleum, Oil, and Lubricants (POL), it shall be disposed of in accordance with local regulations. If it is contaminated with hazardous substances, the Contractor shall dispose of it in accordance with Paragraph CHAPTER 5: HAZARDOUS WASTE and the JEGS. If additional information is needed, consult the Environmental Representative.

3.16 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS WASTES

If petroleum-contaminated soil, or suspected hazardous waste is found during construction that was not identified in the Contract documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.17 PEST MANAGEMENT

In order to minimize impacts to existing fauna and flora, coordinate with the [Installation Pest Management Consultant (IPMC) and the Natural Resources Manager], through the Contracting Officer, at the earliest possible time prior to pesticide application. Discuss integrated pest management strategies with the IPC and receive concurrence from the IPM through the Contracting Officer prior to the application of any pesticide associated with these specifications. Provide Installation Project Office Pest Management personnel the opportunity to be present at meetings concerning treatment measures for pest or disease control and during application of the pesticide. The use and management of pesticides are regulated under the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations and installation requirements.

3.17.1 Application

Apply pesticides using a DoD-certified or equivalent Japanese-certified pesticide applicator in accordance with guidance given on the pesticide label. Labels will bear the appropriate use instructions and precautionary message based on the toxicity category of the pesticide. The certified applicator must wear clothing and personal protective equipment as specified on the pesticide label. If local nationals will be using the pesticides, the precautionary messages and use instructions shall be in English and Japanese. The Contracting Officer shall designate locations for water used in formulating. Do not allow the equipment to overflow. Inspect equipment for leaks, clogging, wear, or damage and repair prior to application of pesticide.

3.17.2 Pesticide Treatment Plan

Include and update a pesticide treatment plan, as information becomes available. Include in the plan the sequence of treatment, dates, times, locations, pesticide trade name, EPA registration numbers if applicable, authorized uses, chemical composition, formulation, original and applied concentration, application rates of active ingredient (that is, pounds of active ingredient applied), equipment used for application, and calibration of equipment. Comply with the JEGS or any applicable Federal GOJ national or prefectural and installation requirements.

3.18 PESTICIDES IN SOIL

Evaluate excess soils and concrete foundation debris generated during the demolition of housing units or other wooden structures for the presence of pesticides prior to reuse or final disposal.

3.19 SOUND INTRUSION

Make the maximum use of low-noise emission products, as certified by the EPA or Japanese equivalent. Blasting or use of explosives are not permitted without written permission from the Contracting Officer, and

then only during the designated times. Confine pile-driving operations to the period between 0800 and 1600, Monday through Friday, exclusive of holidays, unless otherwise specified.

Keep construction activities under surveillance and control to minimize environment damage by noise. Comply with the provisions of the 29 CFR 1910, and applicable G0J, U.S. Federal, and local rules and regulations.

3.20 POST CONSTRUCTION CLEANUP

Clean up areas used for construction in accordance with Contract Clause CLEANING UP. Unless otherwise instructed in writing by the Contracting Officer, remove traces of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. Grade and seed disturbed areas to conform to surrounding contours.

-- End of Section --

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ENVIRONMENTAL PROTECTION

1. GENERAL

1.1 SECTION INCLUDES: This specification covers the requirements for environmental protection during construction activities.

1.2 REFERENCES:

The Contractor shall comply with applicable laws and regulations of the Governments of the United States and Japan, however denominated, including those applicable political subdivisions, departments, and other entities, to include but not be limited to, the most current versions of the following:

- A. DOD Japan Environmental Governing Standards (JEGS), Headquarters, U.S. Forces Japan
- B. Air Force Instructions (AFIs)
- C. 374AWI 32-7001 Solid Waste Management
- D. 374AWI 32-7002 Water Pollution Control Facilities
- E. Other applicable regulations and management plans

1.3 ENVIRONMENTAL PROTECTION REQUIREMENTS:

A. The Contractor shall minimize environmental pollution and damage as the result of contractor's performance. The control of environmental pollution and damage requires consideration of land, water, and air, and includes management of visual aesthetics, solid waste, as well as other pollutants. The environmental resources within the project boundaries and those affected outside the limits of permanent work shall be protected during the entire duration of this contract.

B. Subcontractors: The Contractor shall ensure compliance with this section by their subcontractors.

C. Environmental Protection Plan: The Contractor shall submit an Environmental Protection Plan (EPP) within 15 days after receipt of the notice to proceed. Approval of the Contractor's plan will not relieve the Contractor of responsibility for adequate and continuing control of pollutants and other environmental protection measures. The EPP shall include, but not be limited to, the statement under 1.3.1 through 1.3.13:

D. The Project Manager (or designee) shall be responsible for the execution of the EPP. His or her responsibilities are, but not limited to, as follows:

1. Ensuring adherence to the Environmental Protection Plan;
2. Ensuring all appropriate documents are submitted to Environmental Element (374 CES/CEIE);
3. Appointing an emergency response manager for environmental emergencies, and;
4. Training personnel in accordance with the requirements of this plan.

E. The use of the following materials is prohibited:

1. Do not use materials containing asbestos unless deemed absolutely necessary. If necessary, asbestos concentration shall be limited to less than 0.1 percent by weight.

2. Do not use materials containing lead-based paint greater than 0.06 percent lead by weight.
3. Do not use materials containing polychlorinated biphenyl's (PCB) greater than 0.5mg/kg.
4. Do not use class 1 ozone depleting substances (ODS) listed in JEGS, Chapter 2.

1.3.1. ENVIRONMENTAL MANAGEMENT SYSTEM

- A. All contractor personnel shall be aware of and comply with the Yokota Air Base Environmental Commitment Statement. The contractor shall also comply with the EMS program, including all base environmental plans, instructions and the Japan Environmental Governing Standards (JEGS). This also includes instructions/comments provided on planning documents such as 332's (Base Civil Engineer Work Request), and 813s (Request for Environmental Impact Analysis).
- B. Communicate with Environmental Program Managers (EPMs) to be aware of any operational controls (permits, plans, requirements to purchase recycled content, bio-based, or energy efficient products).
- C. Provide copies of environmental records as requested by EPMs to meet recordkeeping requirements.
- D. Ensure that personnel are properly trained in accordance with applicable statutes and regulations.
- E. Immediately report all hazardous waste or hazardous material releases to the installation emergency response activity, and fully cooperate with any emergency response.

1.3.2. AIR QUALITY

- A. Equipment operation and activities or processes performed by the Contractor in accomplishing the specified construction shall be in accordance with all applicable Federal and local laws and standards.
- B. Class I Ozone Depleting Substances (ODS): Class I ODS listed in JEGS, Chapter 2 are prohibited from being used in construction, remodeling, or maintenance. Contractor must provide certifications that refrigerant utilized during the project do not contain Class I ODS. The purchase of Class I ODS for air conditioning and refrigeration equipment for ground applications is also prohibited by AFI 32-7086 on all new and/or refurbishing projects. NOTE: Limited exceptions exist for using Class I ODS for airborne equipment with prior approval. Contact 374 CES/CEIE for details.
- C. Class II ODS Approval Request: Class II ODS listed in JEGS, Chapter 2 must first be approved by 374 CES/CEIE prior to use.
- D. All Refrigerant Types: Ensure a weatherproof data plate is permanently attached to refrigeration (or air conditioning) equipment, in a location visible to maintenance workers, showing refrigerant type and full charge quantity.
- E. Generators, Incinerator, and Boilers: If generators, incinerator or boilers are to be installed or replaced as part of the project, the contractor shall provide 374 CES/CEIE stack height with:
- 1) Fuel Combustion Rate (L/hr) for Generators;
 - 2) Grate area (m2), Rated Capacity (t/day), Incineration rate(kg/h) for Incinerator;
 - 3) Heating area (m2), Burner combustion rate (L/hr), Heat Input rating (MMbtu/hr) for Boilers.

F. Particulates: Dust particles; aerosols and gaseous by-products from construction activities; and processing and preparation of materials, such as from asphaltic batch plants; shall be controlled at all times, including weekends, holidays and hours when work is not in progress. The Contractor shall maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates which would cause the air pollution standards to be exceeded or which would cause a hazard or a nuisance. Sprinkling, chemical treatment of an approved type, light bituminous treatment, baghouse, scrubbers, electrostatic precipitators or other methods will be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp at all times. The contractor shall coordinate with 374 CES/CEIE before the commencement of any activities. Even sprinkling of water should be reconsidered from resource conservation standpoint. The Contractor must have sufficient, competent equipment available to accomplish these tasks. Particulate control shall be performed as the work proceeds and whenever a particulate nuisance or hazard occurs.

G. Hydrocarbons and Carbon Monoxide: Hydrocarbons and carbon monoxide emissions from equipment shall be controlled to allowable limits at all times.

H. Odors: Odors shall be controlled at all times for all construction activities, processing and preparation of materials.

I. Sound Intrusions: The Contractor shall control noise to minimize environment damage.

1.3.3.WATER RESOURCES

A. The Contractor shall manage their activities to avoid pollution of surface and ground waters. Toxic or hazardous chemicals shall not cause contamination of the fresh water reserve. Monitoring of water areas affected by construction shall be the Contractor's responsibility. All water areas affected by construction activities shall be monitored by the Contractor.

B. Washing and Curing Water: Wastewaters originating from construction activities shall not be allowed to enter water areas.

C. Fish and Wildlife: The Contractor shall minimize interference with, disturbance to, and damage of fish and wildlife. Species that require specific attention along with measures for their protection shall be listed by the Contractor prior to beginning of construction operations.

D. Wastewater Management: The Contractor is not authorized sewage holding tanks on base and is not authorized to dispose of waste from chemical toilets/porta-potties via the installation sewer system. When using chemical toilets, the contractor shall plan for procuring a porta-potty/chemical toilet service contract. The porta-potty/chemical toilet service contract must include the correct maintenance, waste collection, transportation and disposal of all porta-potty/chemical toilets and content to include provisions for pest control and elimination of odors. If the project generates wastewater from rinsing tanks, dewatering sites, etc, the contractor shall ensure proper disposal after CEIE's approval. The Contractor may be required to provide further analysis and treatment based on the condition of wastewater.

1.3.4. HAZARDOUS MATERIALS

A. Approval is required before bringing any hazardous materials (HAZMAT) on to the installation. Contractors seeking approval to bring HAZMAT onto the installation must submit AF Form 3000 along

with Safety Data Sheet (SDS) in English and Japanese. The contractor is also required to provide the estimated amount of HAZMAT to be used during the project. If the HAZMAT is suspected to contain lead or asbestos, such as paint, primer, caulking, a certificate from the manufacturer has to be provided to prove that the HAZMAT is free of asbestos/lead. The AF Form 3000 package has to be reviewed by Bioenvironmental (374 AMDS/SGPB), Safety (374 AW/SE) and Environmental (374 CES/CEIE). The Contractor should use non or less -hazardous chemicals when available. All excess materials and empty containers are the responsibility of the contractor and shall be removed at the end of the contract. When the occupants of the facility request the contractor leave the excess HAZMAT, the contractor has to coordinate with and be approved by 374 CES/CEIE.

B. Storage: All containers of HAZMAT in the work place should be labeled, tagged or marked in both English and Japanese with the identity of the products. The contractor shall ensure adequate handling and storage in accordance with JEGS where the HAZMAT is stored. Storage of oils, greases, chemicals, fuels or other liquids will require spill prevention and security. Contractor is not allowed to store HAZMAT on base for over 24 hours without authorization. If on-base HAZMAT storage areas are needed, Contractor shall submit the list of all items that are to be stored on-base and estimated quantities to 374 CES/CEIE, 374 CES/CEF and 374 AW/SE for approval. Hazardous materials storage areas may be inspected by 374 CES/CEIE, 374 CES/CEF and 374 AW/SE. SDSs must be available at storage and on site at all times.

1.3.5. WASTE DISPOSAL

A. Every effort shall be made to use recycling and/or reuse (diversion) measures to reduce the amount of waste generated under the Contract. Waste minimization and disposal plan shall be included in the EPP. In addition, a copy of contract(s) for waste collection/transportation and disposal permits issued from metropolitan/prefectural governments shall be submitted with AF Form 3000 to 374 CES/CEIE separate from the EPP. The plan also needs to list waste that will be generated from the contract and how they are treated (recycled/incinerated/landfilled). After the disposal, waste manifests shall be submitted to 374 CES/CEIE along with recycling report that shows the amount of waste recycled, incinerated/landfilled (Atch 1).

B. Solid Wastes: Solid wastes (excluding clearing debris) shall be placed in containers which are emptied on a regular schedule. Handling and disposal shall be conducted to prevent contamination. Segregation measures shall be employed so that no hazardous or toxic waste will become co-mingled with solid waste. The Contractor shall transport solid waste off government property and dispose of it in compliance with applicable federal and local requirements for solid waste disposal. The Contractor shall comply with site procedures Federal and local laws and regulations pertaining to the use of landfill areas. At a minimum, the following items will be recycled: scrap metal, cardboard, concrete, asphalt, scrap wood, wooden pallets, glass and plastic. Green wastes should be mulched or composted.

C. Hazardous Waste (HW): All HW generated by the contractor shall be disposed/recycled off base in accordance with Japanese regulation, Article 3 of “Law of Disposal and Clearing of Waste *“Haikibutsu no shori oyobi seisou ni kansuru horitsu”* and local regulations. US/foreign made ionization smoke detectors shall be turned in to 374 CES/CEOFA (Alarm Shop). The Contractor shall not leave any hazardous waste at work site areas at the end of duty hours. If temporary hazardous waste storage is needed to accumulate hazardous waste on base, the contractor shall submit spill response plan with site map, the list of all hazardous waste to be stored on-base and estimated quantities to 374 CES/CEIE, 374 CES/CEF, 374 AMDS/SGPB and 374 AW/SE for approval.

E. Burning: Open burning is prohibited for the purpose of reducing waste.

1.3.6. FUEL STORAGE TANKS

A. Temporary Fuel Storage Tanks: The use of temporary fuel storage tanks with capacities greater than 55 gallons as part of a project must be approved by 374 CES/CEIE. The approval request must contain justification for the use of a temporary tank and design details, including a layout diagram indicating the proposed location of the tank within the project site. If a temporary fuel storage tank is approved for use, a Site-specific Spill Contingency Plan (SSCP) must be developed. After construction/installation of a temporary fuel storage tank, the Contractor shall contact project manager or 374 CES/CEIE at 225-5440 to inform Fuels Management Flight (374 LRS/LGRF) to coordinate an initial inspection (AFI 23-204).

B. Aboveground Storage Tank (AST) Installation: If the contractor is installing a permanent aboveground fuel storage tanks as part of the project, it shall be designed and installed in accordance with all applicable regulations (JEGS, AFIs, UFCs, etc). The contractor is responsible to conduct all operation check, including providing fuel for the test, before hand over to the base. AST will be assigned a Tank ID Number by 374CES/CEIAP and this Tank ID Number shall be used on tank signage and in project documentation. After construction/installation of an AST, the Contractor shall contact project manager or 374CES/CEIE at 225-5440 and the Fuels Management Flight (374 LRS/LGRF) at 225-4617 to coordinate an initial inspection (AFI 23-204).

C. Underground Storage Tank (UST) Installation: In general, USTs should not be installed. Fuel storage tanks shall be installed aboveground, unless overriding requirements exist against aboveground installation such as safety, fire protection measures, or force protection measures. The contractor shall report justification of overriding requirements against installation of an AST to 374 CES/CEIE. If a UST is approved for installation as part of a project, the Contractor shall ensure that design drawings contain adequate detail to demonstrate that the UST will be installed in accordance with all applicable regulations. The contractor is responsible to conduct all operation checks, including providing fuel for the test, before hand over to the base. After construction/installation of an AST, the Contractor shall contact project manager or 374CES/CEIE at 225-5440 and the Fuels Management Flight (374 LRS/LGRF) at 225-4617 to coordinate an initial inspection (AFI 23-204).

D. AST Removal: When a facility with a fuel storage tank(s) is demolished, demolition of the tank(s) and associated infrastructure, including above-ground piping, underground piping to the building serviced by the tank, indoor piping to the equipment serviced by the tank, secondary containment curbing, tank saddles, tank pads, Oil Water Separators (OWS), grounding stations, signage, fencing around the AST, associated equipment and sidewalks leading to the AST shall be included in the scope of the project. Contractor shall ensure to comply with applicable regulations such as AFI32-7044, UFC 3-460-01 and UFGS 33 01 50.01, 33 65 00.

E. UST Removal: The following standards of UST removal shall be applied as required including but not limited to applicable JEGS Chapter 19, AFI32-7044, UFC 3-460-01, USGF 02 65 00, 33 01 50.01, 33 65 00 the UST Removal Procedures appendix of the PACAF UST Policy: Contractor shall ensure to comply with above regulations.

F. Contaminated Soil Removal: If exposed free product and/or obviously contaminated soil is encountered during UST removal, contact 374CES/CEIE immediately. If soil contamination is caused by the contractor's mismanagement, the contractor is responsible for cleanup and disposal in accordance with spill response procedures outlined in this specification and local applicable regulations.

G. Fuel Storage Tank Installation/Removal Report: If an AST/UST is installed or an AST/UST is removed as part of the project, the Contractor shall prepare a Fuel Storage Tank Installation/Removal Report and provide it to 374CES/CEIE within 30 days of the completion of site work. At a minimum, the report shall include the following:

- 1) Summary of work conducted;
- 2) Removal date for each removed storage tank;
- 3) Date placed in service, capacity, manufacturer name, model number, and serial number for each new AST/UST;
- 4) As-built drawings showing the removed AST/USTs and new ASTs and associated infrastructure;
- 5) Leak check report for buried piping associated with storage tank at time of installation, modification, construction, relocation or replacement.
- 6) Manufacturer Operations & Maintenance Manuals and product information;
- 7) Manufacturer Warranty Cards(Copy);
- 8) Photo documentation of construction activities, including AST/UST removals and/or new AST installations.

1.3.7. POLYCHLORINATED BIPHENYLS (PCBs)

A. Procurement of No PCB Containing Materials: Materials that contain PCBs are prohibited from being used under this Contract. All procurement of transformers or any other equipment containing dielectric or hydraulic fluid shall be accompanied by the manufacturer's certification that all materials/equipment (e.g., ballasts, capacitors, transformers, Paper Insulated Lead Covered (PILC) Cable, etc.) contain non-PCB dielectric fluid. All Non-PCB certificates shall be submitted to the 374 CES/CEIE and CEOFE with the list of the new material/equipment information prior to transformer or equipment installation (Atch 2). Newly procured transformers and equipment shall have permanent plate affixed stating they are PCB-free in Japanese and in English. The plate should be a 200mm long and 50mm wide acrylic plate with heat treated paint. Base color should be green and letter color should be white. The plate must state "PCB Free by Japanese standard; date of installation, manufacturer, and serial number" in both Japanese and English. The sample of PCB Free Plate is attached (Atch 3). The plate should be affixed to the outside of transformer or cubicle enclosure in an easily identifiable location.

B. Lighting Fixture/Ballast Disposal: The contractor shall identify if lighting fixture/ballast(s) contain-PCBs. Japanese lighting fixtures/ballasts may be identified with the manufacturer's website. All US lighting fixtures/ballasts shall be treated as PCBs unless there is a label stating as Non-PCB. All US made ballasts with or without a label stating PCB Free shall be treated as PCB containing materials in Japan unless information to the contrary exists. US made ballasts without label may contain >500ppm PCBs, and those with "PCB Free" label may contain <50ppm PCBs. If the lighting fixture/ballast is confirmed as Non-PCB item, the contractor is able to dispose of it as a regular industrial waste or recycle it following local Japanese regulations. If the lighting fixture/ballast is confirmed as to contain PCBs, the Contractor shall segregate ballasts by country and manufacturer, and shrink wrap each individually for turn in to the 374 CES/CEIE. Prior to turn in, the contractor shall submit a list of PCB waste including country of origin, name of manufacturer, serial number, year of manufacture, weight and quantity of ballasts to 374 CES/CEIE for proper disposal.

C. PCB Transformer/Equipment Disposal: All transformers or any other equipment containing dielectric/hydraulic fluid shall be considered and treated as PCB containing unless information to the contrary exists. All suspected PCB materials removed for disposal must be sampled for analysis. The Contractor shall identify if the material contains PCB by using Japanese approved method required by the Ministry of the Environment. After the material is confirmed as PCB, the Contractor shall provide the sample result to 374 CES/CEIE. In addition, all sample results, regardless of PCB contamination, must be provided to 374 CES/CEIE. Material and equipment which cannot be sampled such as those hermetically sealed shall be treated as PCB containing. If the material is confirmed as Non-PCB, the Contractor is able to dispose of it as a regular industrial waste or recycle it by following applicable local Japanese regulations. Government will provide PCB sampling results to the contractor, if available. If the material is confirmed as PCB, the Contractor shall turn in the material to 374 CES/CEIE for proper disposal. All PCB containing materials must be containerized in appropriate UN certified drum or placed in a metal secondary containment which holds 125 percent of total oil volume. PCB item(s) and secondary containment must be placed on a pallet and secured with metal strap bands. Prior to turn in, the Contractor shall provide a list of PCB items to 374 CES/CEIE (Atch 2). The list must contain country of origin, name of manufacturer, year of manufacture, serial number, KVA, PCB concentration, oil volume, weight of item, weight of drum/secondary containment, weight of pallet and total weight. The list needs to be reviewed by 374 CES/CEIE prior to turn in. The Contractor must remove any PCB identification label affixed on the material, and ensure the PCB items do not spill during handling, storage and transportation.

1.3.8. ASBESTOS

A. No Asbestos Containing Materials (ACM): Material that contains asbestos is prohibited from being used in construction, remodeling, or maintenance unless deemed absolutely necessary. If necessary, asbestos concentration shall be limited to less than 0.1% by weight. The Contractor must provide manufacturer certifications that state materials utilized do not contain asbestos or justification for ACM and relevant certification.

B. Asbestos Sampling and Analysis Plan (SAP): If historical asbestos sampling data is not available for the building materials to be disturbed, asbestos sampling may be required prior to the start of the project. If the contractor is required to collect samples, an asbestos Sampling and Analysis Plan must be submitted and approved by 374 CES/CEIE prior to sample collection. The SAP must adhere to Asbestos Hazard Emergency Response Act (AHERA) sampling requirements.

C. Asbestos Abatement & Disposal Plan: An Asbestos Abatement and Disposal Plan must be submitted and approved by 374 CES/CEIE prior to starting asbestos abatement.

D. Asbestos Accident Prevention Plan: If asbestos abatement work is required, the Contractor shall prepare a written comprehensive site-specific Accident Prevention Plan (APP).

E. Notification of Unexpected Discovery of Asbestos During Work: If the Contractor discovers previously untested building components suspected to contain asbestos and located in areas impacted by the work, the Contractor shall not disturb the suspect material. In such instances, the Contractor shall notify the CO and 374 CES/CEIE immediately for direction.

F. Asbestos Disposal Manifests: A copy of the asbestos disposal manifest indicating final disposal

must be provided to 374 CES/CEIE and the CO within 60 days after issuance. It will be kept for a minimum of five years. If a copy of the manifest is not received by the CO within 60 days after issuance, the government will investigate the Contractor's work for transportation or disposal and report the findings to USFJ/J42E.

1.3.9. LEAD-BASED PAINT

A. Lead-Based Paints: Paint containing lead is prohibited from being used in construction, remodeling, or maintenance. Contractor must provide manufacturer certifications that materials utilized are lead free per JEGS, Chapter 17. The contractor shall furnish a certificate for paint and paint related materials proposed for use attesting that the paint is lead free unless deemed necessary. If so, the concentration of lead shall be no more than 0.06 percent lead.

B. Lead-Based Paint Sampling Plan: If historical lead-based paint sampling data is not available for the building materials to be disturbed, lead-based paint sampling may be required prior to the start of the project. If the contractor is required to collect samples, Lead-Based Paint Sampling and Analysis Plan must be submitted and approved by 374 CES/CEIE prior to sample collection.

C. Lead-Based Paint Abatement & Disposal Plan: A Lead-Based Paint Abatement and Disposal Plan must be submitted and approved by 374 CES/CEIE and 374 AMDS/SGPB prior to starting lead-based paint abatement activities.

D. Lead-Based Paint Clearance Sampling Results: Clearance sampling must be coordinated with 374 AMDS/SGPB for approval prior to tearing down the containment enclosure. The contractor will provide sample analysis results to the CO, 374 CES/CEIE, and 374 AMDS/SGPB. 374 AMDS /SGPB will provide approval for containment tear-down. The on-site QA will ensure the site is returned to the base in proper condition and will notify the CO of the site status.

E. Lead Waste Disposal: The contractor must collect, characterize for possible hazardous toxicity, and dispose of in accordance with applicable local regulations.

1.3.10. HISTORICAL, ARCHAEOLOGICAL, AND CULTURAL RESOURCES

A. Existing historical, archaeological, and cultural resources within the contractor's work area will be so designated by the Contracting Officer if any has been identified. The contractor shall take precautions to preserve all such resources as they existed at the time they were first pointed out. The contractor shall provide and install protection for these resources and be responsible for their preservation during the life of the contract. If any previously unidentified or unanticipated resources are discovered during the project, stop any type of work, which is anticipated, to damage the resource and report the fact to the Contracting Officer. Resources covered by this paragraph include but are not limited to: any human skeletal remains or burials; artifacts; shell, midden, bone, charcoal, or other deposits; rocks or coral alignments, pavings, wall, or other constructed features; and any indication of agricultural or other human activities. Upon such discovery or find, the Contractor shall immediately notify the Contracting Officer.

1.3.11. LAND RESOURCES

A. The contractor's field offices, staging areas, stockpile storage, and temporary buildings shall be placed in areas designated on the drawings or as directed by the Contracting Officer. Temporary

movement or relocation of contractor facilities shall be made only when approved. The contractor shall manage to minimize erosion and to prevent sediment from entering nearby waters. The contractor shall confine all activities to areas defined by the drawings and specifications. Prior to the beginning of any construction, the contractor shall identify the land resources to be preserved within the work area. Except in areas indicated on the drawings or specified to be cleared, the contractor shall not remove, cut, deface, injure, or destroy land resources including trees, shrubs, vines, grasses, topsoil, and land forms without permission. No ropes, cables, or guys shall be fastened to or attached to any trees for anchorage unless specifically authorized. Where such emergency use is permitted, the contractor shall provide effective protection for land and vegetation resources at all times as defined in the following subparagraphs. Stone, earth or other material displaced into uncleared areas shall be removed.

B. Work Area Limits: Prior to any construction, the contractor shall mark the areas that need not be disturbed under this contract. Isolated areas within the general work area which are to be saved and protected shall also be marked or fenced. Monuments and markers shall be protected before construction operations commence.

C. Landscape: Vegetation resources, land forms and other landscape features indicated and defined on the drawings to be preserved shall be clearly identified by marking, fencing, or wrapping with boards, or any other approved techniques. Any anticipated vegetation disturbance needs to be coordinated with the Government before it occurs.

D. Unprotected Erodible Soils: All earthwork shall be planned and conducted to minimize the duration of exposure of unprotected soils.

E. Disturbed Areas: The Contractor shall effectively prevent erosion and control sedimentation. Control methods should be coordinated with the Government.

F. Preconstruction Survey: Prior to starting any onsite construction activities, the Contractor and the Contracting Officer shall make a joint condition survey after which the Contractor shall prepare a brief report indicating on a layout plan the condition of trees, shrubs and grassed areas immediately adjacent to work sites and adjacent to the assigned storage area and access routes as applicable. This report will be signed by both the Contracting Officer and the Contractor upon mutual agreement as to its accuracy and completeness.

1.3.12. ENVIRONMENTAL EMERGENCY PROCEDURES

A. Drawing: Drawings showing locations of any proposed temporary facilities, hazardous materials storage, hazardous and solid waste collection point, excavations, embankments, existing utility lines to include storm drains shall be attached to EPP and submitted to 374 CES/CEIE for review.

B. Notification: The Contracting Officer will notify the Contractor in writing of any observed noncompliance with the previously mentioned Federal or local laws or regulations, permits, and other elements of the Contractor's environmental protection plan. The Contractor shall, after receipt of such notice, inform the Contracting Officer of proposed corrective action and take such action when approved. If the Contractor fails to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions shall be granted or costs or damages allowed to the Contractor for any such suspensions.

C. Spill Prevention and Response:

1) The Contractor shall conduct all operations in a manner that prevents spills of POL and Hazardous Substances. The Contractor is required to familiarize personnel with spill prevention and response procedures, fire suppression systems, and SDSs for all materials used and/or stored on the project site. The Contractor shall provide and maintain spill equipment, sufficient in both type and quantity, at all sites involving the storage, use, or handling of POL and/or hazardous substances.

2) Site-specific Spill Contingency Plan (SSCP): An SSCP must be developed for any project at which POL or Hazardous Substances are used and/or stored at the project site. The SSCP shall include spill response procedures, an inventory of POL and/or hazardous substances, probable spill routes, a project site layout diagram, and a spill response equipment inventory. A template SSCP can be obtained from 374CES/CEIE. The Contractor shall train employees on the contents of the SSCP and the use of spill response equipment and shall document such training.

3) Spill Reporting: The Contractor shall immediately report all POL or Hazardous Substances spills to the Fire Emergency Services (374 CES/CEF) at 911 or Commercial 042-507-6560 (direct number to Yokota Fire Dispatch).

4) Spill Cleanup:

a. The Contractor is responsible for clean-up of POL and/or hazardous substance spills and disposal of clean-up material. The Contractor shall determine, as quickly as possible, the nature of the spilled substance and implement necessary safety precautions to protect both human health and the environment at no additional cost to the US Government, and shall only be performed by personnel adequately trained in spill response and cleanup techniques for the severity of the spill incident.

b. Costs incurred from any Contractor spills are the responsibility of the Contractor. If the US Government has to perform emergency spill response and cleanup, due to the non-availability of designated personnel or if the spill is beyond the capability of designated personnel, the Contractor shall be held liable for all costs associated with performing said work.

1.3.13. OTHERS

A. Previously Used Equipment: The Contractor shall thoroughly clean all construction equipment previously used at other sites before it is brought into the work areas, ensuring that soil residuals are removed and that egg deposits from plant pests are not present.

B. Post Construction Cleanup: The contractor shall clean up all areas used for construction.

C. Restoration of Landscape Damage: The Contractor shall restore landscape features damaged or destroyed during construction operations.

D. Training of Contractor Personnel: The Contractor's personnel shall be trained in all phases of environmental protection. The training shall include methods of detecting and avoiding pollution, familiarization with pollution standards, both statutory and contractual, and installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental pollution control.

Attachments:

1. Construction Debris Disposal and Recycling Data
2. New & Existing Transformer List
3. PCB Free Plate
4. PCB Lighting Fixture/Ballast List

END OF SECTION

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Disposal and Recycling Data Sheet

Project Number					Name of Contractor					
Title of Project					Name of Project Manager					
Date Project Started					Phone Number of Contractor					
Date of Final Inspection					Name of Inspector/Office Symbol					
Date of Submission					Phone Number of Inspector					

Items	Concrete Debris	Recycle Amount	Metals	Recycle Amount	Asphalt Debris	Recycle Amount	Wood	Recycle Amount	Others (soil not included)	Recycle Amount
Generated Weight (ton)										
Volume (m3)										
Recycle Rate (%)										
Remarks										

1. Fill out and submit this form to CE Environmental through government representative before final inspection is completed.
2. Specify method of recycling in "Remarks".
3. If any other wastes are recycled, list them below.
4. Specify how the waste get disposed of (incinerated, landfilled etc.)

Existing Transformer List (to-be-removed)

既存の変圧器リスト(撤去予定)

Project # (プロジェクト番号) :

Project Title (プロジェクト名) :

No.	Bldg # 建物番号	Facilty Name 施設名	Trans ID トランス I D	Type 型式	KVA 容量	Phase 相数	Serial# 製造番号	MFR Name 製造元	MFG Date 製造年・月	Oil (L) 油量	Total Weight (Kg) 総重量	Non PCB Certificate/Analyt ical Results (Y/N) P C B 不含証明書/分 析結果の有・無
1			T-									
2			T-									
3			T-									
4			T-									
5			T-									
6			T-									
7			T-									
8			T-									
9			T-									
10			T-									

Please submit this list to CE Environmental Office (DSN: 225-5440) with Non-PCB Certificate prior to install transformer(s) listed above.
本リストは上記の変圧器の撤去前に C E 環境課(内線 225-5440)へ P C B 不含証明書もしくは分析結果とともにご提出願います。

New Transformer List (to-be-installed)

新規変圧器リスト(設置予定)

Project # (プロジェクト番号) :

Project Title (プロジェクト名) :

No.	Bldg # 建物番号	Facilty Name 施設名	Trans ID トランス I D	Type 型式	KVA 容量	Phase 相数	Serial# 製造番号	MFR Name 製造元	MFG Date 製造年・月	Oil (L) 油量	Total Weight (Kg) 総重量	Non PCB Certificate/Analyt ical Results (Y/N) P C B 不含証明書/分 析結果の有・無
1			T-									
2			T-									
3			T-									
4			T-									
5			T-									
6			T-									
7			T-									
8			T-									
9			T-									
10			T-									

Please submit this list to CE Environmental Office (DSN: 225-5440) with Non-PCB Certificate prior to install transformer(s) listed above.
本リストは上記の変圧器の設置前に C E 環境課(内線 225-5440)へ P C B 不含証明書もしくは分析結果とともにご提出願います。

Attachment 3: Example of PCB Free Plate

PCB Free by Japanese Standard	日本基準準拠PCB 不含変圧器
Date of Installation: 31 Dec 2012	設置年月日: 2012年12月31日
Manufacturer: XXXXXXXXXXXXX	製造所: XXXXXXXXXXXXX
Serial Number: XXXXXXXXXXXXX	製造番号: XXXXXXXXXXXXX

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Project # (プロジェクト番号) :

Project Title (プロジェクト名):

Bldg (建物番号) :

	MFR Name 製造元	Serial/Catalog# 製造・カタログ番号	Power Factor 力率	Country of Origin 生産国	MFG Date 製造年	Qty 数量	Method of PCB Identification (PCB判別方法)
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							



HAZARDOUS MATERIALS INSPECTION REPORT

**PN77833 VEHICLE MAINTENANCE SHOP,
KADENA AIR BASE, OKINAWA, JAPAN**

**USACE JED CONTRACT NO. W912HV-19-D-0010
TASK ORDER W912HV23F0041**

Prepared for:

**U.S Army Engineer District, Japan
CEPOJ-PP-AF
Unit 45010
APO, AP 96343-5010**

By:

OYO CORPORATION, PACIFIC

Okinawa Times Bldg. 2F, 1-12-24 Goya
Okinawa-shi, Okinawa, Japan 904-0021

OYO Project No. 232061

November 2025

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E	Sampling Photos
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HAZARDOUS MATERIALS INSPECTION REPORT

PN77833 VEHICLE MAINTENANCE SHOP

KADENA AIR BASE, OKINAWA, JAPAN

USACE JED CONTRACT NO. W912HV-19-D-0010, TO W912HV23F0041

1.0 INTRODUCTION

OYO Corporation, Pacific (OYO) was contracted by SSFM International, Inc. to conduct sampling, laboratory analysis, and compliant reporting of all suspected hazardous materials associated with the PN77833 Vehicle Maintenance Shop (VMS) project at Kadena Air Base, Okinawa, Japan. OYO performed a hazardous materials (Hazmat) survey of existing buildings 4040, 4041, 4042, 4043, 4044, 4045, 4046, 4047, and 4048 to supplement information provided in prior surveys, to specifically address the existence of hazardous materials requiring specialized abatement and/or disposal considerations during future facility improvement construction activities. The results from this survey will identify the locations of hazardous materials in support of future design of hazardous material abatement specifications and drawings.

The survey identified the types and locations of all hazardous materials in support of the project “PN77833 Vehicle Maintenance Shop, Kadena Air Base, Okinawa, Japan”, under Contract No: W912HV19D0010 Task Order W912HV23F0041.

The primary inspection was conducted from 30 November 2023 to 12 January 2024. The activities under this project were to:

- Review available inspection reports and building as-builts provided by the Government.
- Collect samples of building materials for asbestos confirmational analysis.
- This inspection report uses previous test results for lead (Pb) for all identified painted or coated surfaces. Those same surfaces were re-sampled under this investigation and tested for cadmium (Cd) and chromium (Cr) in accordance to project scope of work requirements.
- Inventory fluorescent light fixtures for ballasts and other electrical equipment containing polychlorinated biphenyls (PCBs) and mercury light tubes.
- Inventory smoke detectors or lamps for radioactive sources.
- Identify the existence of Chlorofluorocarbon (CFC) refrigerants.
- Identify any other hazardous materials that would require special abatement, treatment, transport, or disposal considerations during the construction phase of the project.

A follow-up inspection was conducted on 9 May 2025 at the Swing Space area on Kadena Air Base. The activities under this additional inspection were to:

- Research and review any available inspection reports provided by the Government (none were available at the time of survey).

- Collect additional samples of construction materials for asbestos, lead, cadmium, and chromium analysis.

1.1 Work Scope and Locations

The hazardous materials investigation/survey was conducted in Buildings 4040, 4041, 4042, 4043, 4044, 4045, 4046, 4047 and 4048 of Kadena Air Base (See Figure 1). All of these buildings are planned for full demolition in support of the construction of new VMS facilities.

Additional hazardous materials investigation/survey was conducted at the Swing Space on Kadena Air Base during the May 2025 follow up inspection (See Figure 2).

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FIGURE 1: LOCATION OF SITE INVESTIGATION

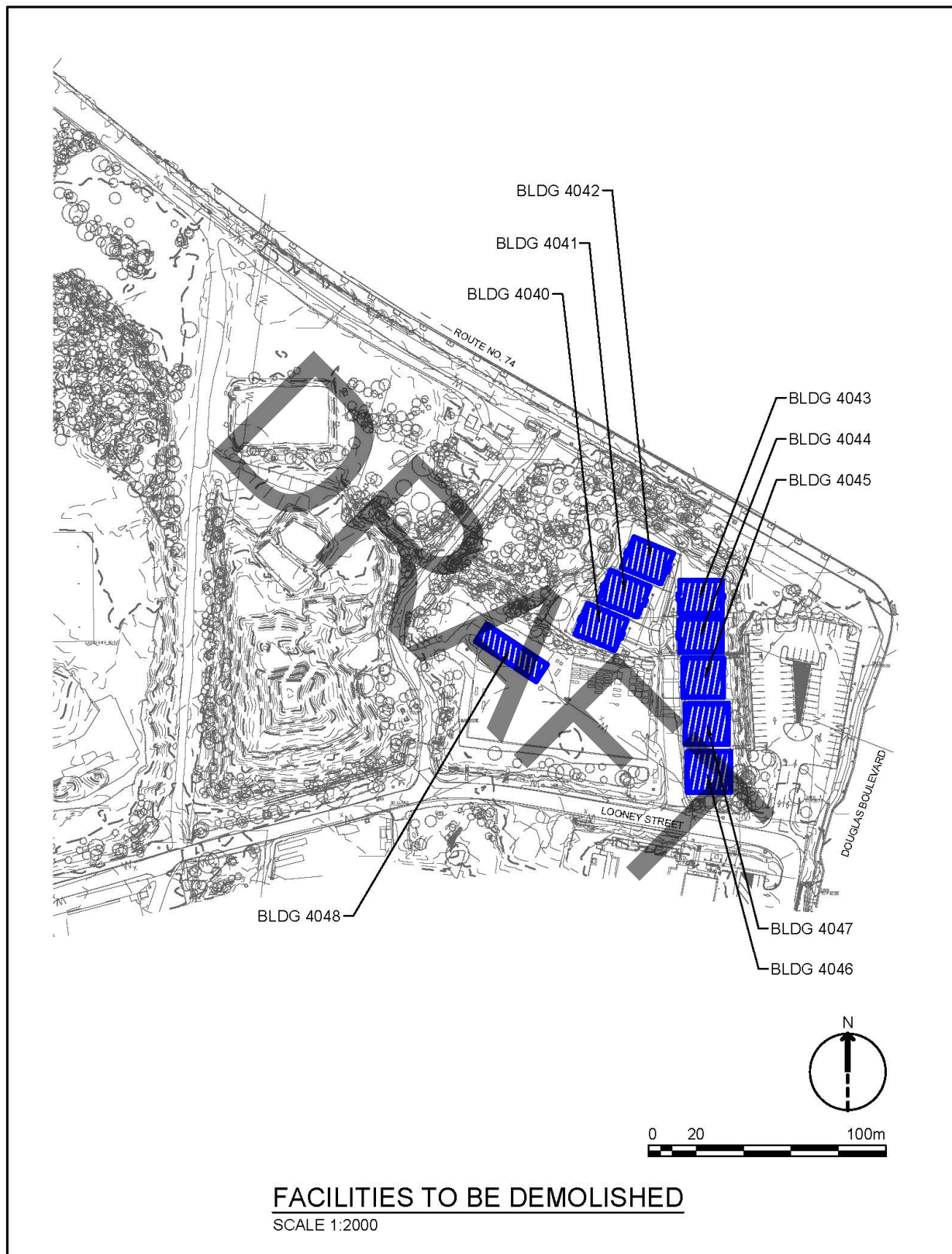
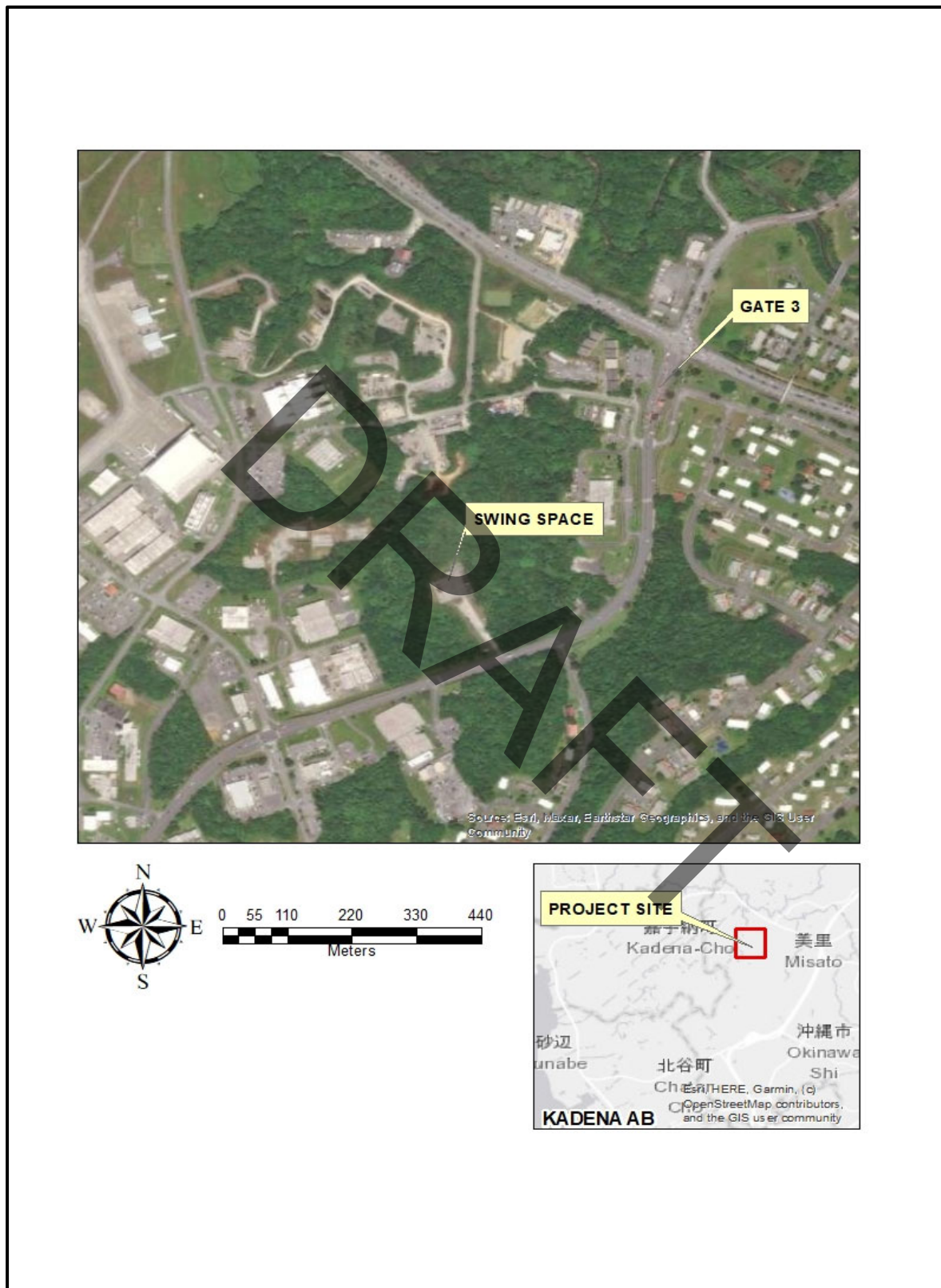


FIGURE 2: LOCATION OF ADDITIONAL SITE INVESTIGATION



2.0 ASBESTOS

Bulk samples were collected from suspect ACM homogeneous layers to include layers previously identified as non-ACM in past inspection report(s). From the total number of samples, the following number of ACM samples were taken at each of the 9 buildings planned for demolition along with the total estimated number of homogeneous areas (HAs) sampled from:

- Building 4040 – 24 samples from 12 HAs
- Building 4041 – 22 samples from 12 HAs
- Building 4042 – 27 samples from 14 HAs
- Building 4043 – 22 samples from 11 HAs
- Building 4044 – 28 samples from 16 HAs
- Building 4045 – 41 samples from 21 HAs
- Building 4046 – 24 samples from 13 HAs
- Building 4047 – 21 samples from 12 HAs
- Building 4048 – 6 samples from 2 HAs
- Swing Space – 7 samples from 3 HAs

Homogeneous areas already identified as ACM were resampled in cases where previous samples were not taken in accordance to AHERA prescribed recommendations (i.e. minimum of 3 samples for misc. materials, 7 samples for surfacing materials greater than 5000 ft² in area, etc.). ACM designations previously determined for ACM HAs were not resampled and the ACM determination will still apply. In addition, any additional homogeneous areas noted by the building inspector during the survey which was not previously identified were sampled and analyzed for ACM.

2.1 Sampling of Suspect Materials

A detailed previous survey reports in 2023 (Environmental Science Corporation., 2023) identified a total of one hundred thirty-six (136) HAs in Buildings 4040, 4041, 4042, 4043, 4044, 4045, 4046, and 4047 (See Appendix F for all past reports). Out of these HAs, one (1) building material was determined to be ACM. No historical ACM sampling and analysis data was available for Building 4048.

The OYO certified AHERA Building Inspector conducted an additional two hundred and nine (209) confirmational samples to augment and update the previous sampling and testing data and six (6) additional samples at Building 4048. The following original homogeneous areas were reanalyzed for asbestos and twenty-three (23) new homogeneous areas were added as shown in Tables 1 through 9 of this section. No new ACM materials were identified.

2.2 Methodology

A visual inspection for suspect ACM and homogeneous areas (areas that have uniform color, texture, and appearance) was conducted. Suspect materials were divided into three Environmental Protection Agency (EPA) categories:

- Surfacing Materials (sprayed or troweled-on materials)
- Thermal Systems Insulations (materials generally applied to various mechanical systems)
- Miscellaneous Materials (any materials which do not fit in the above categories)

Sampling methodology generally followed the procedures presented in US EPA 40 CFR 763 *Asbestos-Containing Materials in School* of the Code of Federal Regulations (CFR). Sampling locations are included as Appendix D. Certifications of the surveying personnel and the laboratory used for analysis are included in Appendix G.

Chapter 5 of the 2024 JEGS defines asbestos-containing materials as any material containing more than 0.1% asbestos by weight. JEGS also categorizes ACM as either being a friable material, a Category I non-friable material or a Category II non-friable material. Friable asbestos are materials containing more than 0.1% asbestos fibers that, when dry, can be crumbled, pulverized, or reduced to powder by hand pressure. Category I Nonfriable ACM are asbestos containing packings, gaskets, resilient floor covering, and asphalt roofing products containing more than 0.1% asbestos. Category II Nonfriable ACM are materials, excluding Category I nonfriable ACM, containing more than 0.1% asbestos that, when dry, cannot be crumbled, pulverized, or reduced to powder by hand pressure.

TABLE 1: HOMOGENEOUS AREAS AND PRE-IDENTIFIED ACM (BLDG. 4040)

HA ID#	Material Description	Condition	Friable (Y/N)	ACM (Y/N)	Est. Quantity
H1 ^{*1}	30x30cm gray vinyl floor tile w/ light gray mastic	Good	N	N	40 SM
H2	10cm gray cove base w/ light gray mastic	Good	N	N	45 LM
H3	Gypsum board wall, smooth white	Good	N	N	265 SM
H4 ^{*1}	60x60cm ceiling tile, off white, all patterns	Good	N	N	55 SM
H5 ^{*1}	Duct TSI, yellow fiberglass w/ silver backing	Good	Y	N	N/A
H6 ^{*1}	30x30cm grey vinyl floor tile w/ tan mastic	Good	N	N	20 SM
H7 ^{*1}	20x20cm beige ceramic floor tile w/ grey cement	Good	N	N	20 SM
H8 ^{*1}	10x10cm off white ceramic wall tile w/ white grout and grey mastic	Good	N	N	40 SM
H9 ^{*1}	Gypsum ceiling board, white	Good	N	N	20 SM
H10 ^{*1}	Flange gasket, black	Good	N	N	N/A
H11	Wall joint sealant, light grey mastic	Good	N	N	N/A
H12	Concrete wall coating, off white, textured	Good	N	N	220 SM
H13 ^{*1}	Window joint sealant, brown mastic	Good	N	N	N/A
H14 ^{*1}	Roofing material (concrete)	Good	N	N	280 SM
H15 ^{*1}	Roof vent sheeting material	Good	N	N	2 SM
H16 ^{*1}	Building insulation behind walls	Good	N	N	N/A

Notes:

*1 – Previously identified homogeneous areas (HA's) observed as having less than three (3) historic analysis results were sampled for additional lab analysis by OYO Pacific Asbestos building inspectors during the field investigation period, Nov 2023 to Jan 2024. Newly identified HA's are indicated in BLUE. ACM detected samples are shown in RED.

Abbreviations:

HA – Homogeneous Area

LM – linear meters

SM – square meters

TABLE 2: HOMOGENEOUS AREAS AND PRE-IDENTIFIED ACM (BLDG. 4041)

HA ID#	Material Description	Condition	Friable (Y/N)	ACM (Y/N)	Est. Quantity
H1	Flange gasket, black	Good	N	N	N/A
H2	Concrete wall coating, off white, textured	Good	N	N	250 SM
H3 ^{*1}	Wall joint sealant, light gray mastic	Good	N	N	150 LM
H4 ^{*1}	Window joint sealant, brown mastic	Good	N	N	30 LM
H5	30x30cm grey vinyl floor tile w/ tan & light gray mastic	Good	N	N	220 SM
H6	10 cm gray vinyl cove base w/ white mastic	Good	N	N	150 LM
H7 ^{*1}	Gypsum wall board, gray	Good	N	N	350 SM
H8 ^{*1}	60x60cm ceiling tile, off-white, all patterns	Good	N	N	220 SM
H9 ^{*1}	Duct TSI, yellow fiberglass w/ silver backing	Good	Y	N	N/A
H10 ^{*1}	Gypsum board wall, white	Good	N	N	350 SM
H11 ^{*1}	Ceiling board, white	Good	N	N	30 SM
H12 ^{*1}	20x20cm beige ceramic floor tile w/ grey grout	Good	N	N	30 SM
H13 ^{*1}	10x10cm off white ceramic wall tile w/ white grout	Good	N	N	60 SM
H14	Sink joint sealant, white	Good	N	N	2LM
H15 ^{*1}	Pipe TSI (straight), yellow fiberglass w/ silver	Good	Y	N	N/A
H16	Pipe TSI (elbow), yellow fiberglass w/ silver	Good	Y	N	N/A
H17 ^{*1}	Roofing material (concrete)	Good	N	N	280 SM
H18 ^{*1}	Roof vent sheeting material	Good	N	N	2 SM

Notes:

*1 – Previously identified homogeneous areas (HA's) observed as having less than three (3) historic analysis results were sampled for additional lab analysis by OYO Pacific Asbestos building inspectors during the field investigation period, Nov 2023 to Jan 2024. Newly identified HA's are indicated in BLUE.
ACM detected samples are shown in RED.

Abbreviations:

HA – Homogeneous Area LM – linear meters SM – square meters

TABLE 3: HOMOGENEOUS AREAS AND PRE-IDENTIFIED ACM (BLDG. 4042)

HA ID#	Material Description	Condition	Friable (Y/N)	ACM (Y/N)	Est. Quantity
H1 ^{*1}	Flange gasket, black	Good	N	N	N/A
H2	Concrete wall coating, off white, textured	Good	N	N	250 SM
H3 ^{*1}	Wall joint sealant, light gray mastic	Good	N	N	150 LM
H4 ^{*1}	Window joint sealant, dark brown mastic	Good	N	N	30 LM
H5	30x30cm grey vinyl floor tile w/ off-white mastic	Good	N	N	220 SM
H6	Gypsum board wall, gray, smooth	Good	N	N	350 SM
H7	10 cm gray vinyl cove base w/ tan & white mastic	Good	N	N	150 LM
H8 ^{*1}	60x60cm ceiling tile, off-white, all patterns	Good	N	N	220 SM
H9 ^{*1}	Duct TSI, yellow fiberglass w/ silver backing	Good	Y	N	N/A
H10 ^{*1}	Pipe TSI (elbow), yellow fiberglass w/ silver	Good	N	N	N/A
H11 ^{*1}	Pipe TSI (straight), yellow fiberglass w/ silver	Good	N	N	N/A
H12 ^{*1}	Concrete floor coating, gray	Good	N	N	2 SM
H13 ^{*1}	Concrete wall coating, off-white, smooth	Good	N	N	5 SM
H14 ^{*1}	10x10cm off white ceramic wall tile w/ white grout	Good	N	N	60 SM
H15 ^{*1}	20x20cm beige ceramic floor tile w/ grey grout	Good	N	N	30 SM
H16	Sink joint sealant, white mastic	Good	N	N	2 LM
H17 ^{*1}	Gypsum ceiling board, off-white, smooth	Good	N	N	30 SM
H18 ^{*1}	Wall joint sealant, white mastic	Good	N	N	N/A
H19 ^{*1}	Roofing material (concrete)	Good	N	N	280 SM
H20 ^{*1}	Roof vent sheet	Good	N	N	2 SM

Notes:

*1 – Previously identified homogeneous areas (HA's) observed as having less than three (3) historic analysis results were sampled for additional lab analysis by OYO Pacific Asbestos building inspectors during the field investigation period, Nov 2023 to Jan 2024. Newly identified HA's are indicated in BLUE. ACM detected samples are shown in RED.

Abbreviations:

HA – Homogeneous Area LM – linear meters SM – square meters

TABLE 4: HOMOGENEOUS AREAS AND PRE-IDENTIFIED ACM (BLDG. 4043)

HA ID#	Material Description	Condition	Friable (Y/N)	ACM (Y/N)	Est. Quantity
H1	30x30cm grey vinyl floor tile w/ tan & light gray mastic	Good	N	N	220 SM
H2	10cm gray cove base w/ white mastic	Good	N	N	150 LM
H3	Gypsum board wall, gray	Good	N	N	350 SM
H4 ^{*1}	60x60cm ceiling tile, off-white, all patterns	Good	N	N	220 SM
H5 ^{*1}	Duct TSI, yellow fiberglass w/ silver backing	Good	Y	N	N/A
H6	Sink joint sealant, off white	Good	N	N	2 LM
H7 ^{*1}	Ceiling board, off white	Good	N	N	30 SM
H8 ^{*1}	10x10cm off white ceramic wall tile w/ white grout	Good	N	N	60 SM
H9 ^{*1}	20x20cm beige ceramic floor tile w/ grey grout	Good	N	N	30 SM
H10 ^{*1}	Pipe TSI (elbow), yellow fiberglass w/ silver	Good	N	N	N/A
H11 ^{*1}	Pipe TSI (straight), yellow fiberglass w/ silver	Good	N	N	N/A
H12 ^{*1}	Flange gasket, black	Good	N	N	N/A
H13	Concrete wall coating, off-white, textured	Good	N	N	250 SM
H14	Wall joint sealant, off white	Good	N	N	150 LM
H15 ^{*1}	Window joint sealant, brown	Good	N	N	30 LM
H16 ^{*1}	Roofing material	Good	N	N	280 SM
H17 ^{*1}	Roof vent sheeting material	Good	N	N	2 SM

Notes:

*1 – Previously identified homogeneous areas (HA's) observed as having less than three (3) historic analysis results were sampled for additional lab analysis by OYO Pacific Asbestos building inspectors during the field investigation period, Nov 2023 to Jan 2024. Newly identified HA's are indicated in BLUE. ACM detected samples are shown in RED.

Abbreviations:

HA – Homogeneous Area

LM – linear meters

SM – square meters

TABLE 5: HOMOGENEOUS AREAS AND PRE-IDENTIFIED ACM (BLDG. 4044)

HA ID#	Material Description	Condition	Friable (Y/N)	ACM (Y/N)	Est. Quantity
H1	30x30cm gray vinyl floor tile w/ light gray mastic	Good	N	N	60 SM
H2	10cm gray cove base w/ brown mastic	Good	N	N	45 LM
H3 ^{*1}	Gypsum board wall, gray, smooth	Good	N	N	265 SM
H4 ^{*1}	Pipe TSI (straight), yellow fiberglass w/ silver	Good	Y	N	N/A
H5 ^{*1}	Pipe TSI (elbow), yellow fiberglass w/ silver	Good	Y	N	N/A
H6 ^{*1}	60x60cm ceiling tile, off-white, pin hole pattern w/ fissures	Good	N	N	55 SM
H7 ^{*1}	Concrete wall coating, gray, smooth	Good	N	N	20 SM
H8 ^{*1}	Gypsum board wall, off-white, smooth	Good	N	N	30 SM
H9 ^{*1}	Door joint sealant, off white	Good	N	N	N/A
H10 ^{*1}	Concrete floor coating, gray	Good	N	N	10 SM
H11 ^{*1}	10x10cm off white ceramic wall tile w/ white grout	Good	N	N	40 SM
H12 ^{*1}	20x20cm beige ceramic floor tile w/ grey grout	Good	N	N	20 SM
H13	Sink joint sealant, white mastic	Good	N	N	N/A
H14 ^{*1}	Ceiling board, off white, smooth	Good	N	N	20 SM
H15 ^{*1}	Flange gasket, black	Good	N	N	N/A
H16	Concrete wall coating, off white, textured	Good	N	N	220 SM
H17	Wall joint sealant, gray	Good	N	N	N/A
H18 ^{*1}	Window joint sealant, brown	Good	N	N	N/A
H19 ^{*1}	Floor joint sealant, light gray	Good	N	N	N/A
H20 ^{*1}	Door joint sealant, off white	Good	N	N	N/A
H21 ^{*1}	Roofing material	Good	N	N	280 SM
H22 ^{*1}	Roof vent sheeting material	Good	N	N	2 SM

Notes:

*1 – Previously identified homogeneous areas (HA's) observed as having less than three (3) historic analysis results were sampled for additional lab analysis by OYO Pacific Asbestos building inspectors during the field investigation period, Nov 2023 to Jan 2024. Newly identified HA's are indicated in BLUE. ACM detected samples are shown in RED.

Abbreviations:

HA – Homogeneous Area LM – linear meters SM – square meters

TABLE 6: HOMOGENEOUS AREAS AND PRE-IDENTIFIED ACM (BLDG. 4045)

HA ID#	Material Description	Condition	Friable (Y/N)	ACM (Y/N)	Est. Quantity
H1	30x30cm gray vinyl floor tile w/ gray & tan mastic	Good	N	N	250 SM
H2	10cm light gray cove base w/ gray & tan mastic	Good	N	N	100 LM
H3 ^{*1}	Concrete wall coating, off white, smooth	Good	N	N	20 SM
H4 ^{*1}	Gypsum board wall, off white, smooth	Good	N	N	200 SM
H5 ^{*1}	Duct TSI, yellow fiberglass w/ silver backing	Good	Y	N	N/A
H6 ^{*1}	Ceiling insulation, yellow fiberglass w/ silver backing	Good	N	N	N/A
H7 ^{*1}	60x60cm ceiling tile, off-white, fissured	Good	N	N	280 SM
H8	Concrete wall coating, above ceiling, black	Good	N	N	N/A
H9 ^{*1}	Concrete block wall coating, gray painted	Good	N	N	80 SM
H10 ^{*1}	Pipe TSI (straight), yellow fiberglass w/ silver	Good	Y	N	N/A
H11 ^{*1}	Pipe TSI (elbow), yellow fiberglass w/ silver	Good	Y	N	N/A
H12 ^{*1}	20x20cm gray ceramic floor tile w/ grey grout	Good	N	N	30 SM
H13 ^{*1}	10x10cm off white ceramic wall tile w/ white grout & gray cement	Good	N	N	50 SM
H14 ^{*1}	Gypsum ceiling board, off white	Good	N	N	30 SM
H15	Joint sealant (door) (sink)	Good	N	N	3 LM
H16 ^{*1}	Door joint sealant, white mastic	Good	N	N	15 LM
H17	Exterior wall joint sealant, white mastic	Good	N	N	150 LM
H18 ^{*1}	Exterior concrete block wall coating, off white, textured	Good	N	N	250 SM
H19 ^{*1}	Exterior concrete wall base coating, red	Good	N	N	30 SM
H20 ^{*1}	Exterior door joint sealant, gray mastic	Good	N	N	6 LM
H21 ^{*1}	Exterior flex sheet, black	Damaged	N	N	N/A
H22	Exterior Duct TSI, black foam w/ tan mastic	Damaged	N	N	N/A
H23	Exterior duct mastic, gray	Good	N	N	N/A
H24	Exterior concrete base coating, AHU, grey	Good	N	N	5 SM
H25 ^{*1}	Exterior floor joint sealant, solid black	Good	N	N	50 LM
H26 ^{*1}	Exterior window joint sealant, black mastic	Good	N	N	20 LM
H27 ^{*1}	Roofing material, asphalt	Good	N	N	280 SM
H28 ^{*1}	Roof vent sealant	Good	N	N	2 SM
H29 ^{*1}	Roofing caulking	Good	N	N	5 SM

Notes:

*1 – Previously identified homogeneous areas (HA's) observed as having less than three (3) historic analysis results were sampled for additional lab analysis by OYO Pacific Asbestos building inspectors during the field investigation period, Nov 2023 to Jan 2024. Newly identified HA's are indicated in BLUE. ACM detected samples are shown in RED.

Abbreviations: HA – Homogeneous Area LM – linear meters SM – square meters

TABLE 7: HOMOGENEOUS AREAS AND PRE-IDENTIFIED ACM (BLDG. 4046)

HA ID#	Material Description	Condition	Friable (Y/N)	ACM (Y/N)	Est. Quantity
H1	10cm gray cove base w/ off white mastic	Good	N	N	90 LM
H2* ¹	1 Gypsum board wall, off white, smooth	Good	N	N	175 SM
H3	Concrete floor coating, gray	Good	N	N	220 SM
H4	Floor joint sealant, gray mastic	Good	N	N	N/A
H5* ¹	Concrete block wall coating, off white, smooth	Good	N	N	175 SM
H6* ¹	Door joint mastic, white mastic	Good	N	N	15 LM
H7* ¹	Pipe TSI (elbow), yellow fiberglass w/ silver	Good	Y	N	N/A
H8* ¹	Pipe TSI (straight), yellow fiberglass w/ silver	Good	Y	N	N/A
H9	ACU insulation, black foam w/ tan mastic	Good	N	N	N/A
H10* ¹	10x10cm pale purple ceramic wall tile w/ gray grout	Good	N	N	15 SM
H11* ¹	20x20cm gray ceramic floor tile w/ grey grout	Good	N	N	6 SM
H12	Sink joint sealant, white mastic	Good	N	N	6 LM
H13* ¹	Concrete wall coating, off white, smooth	Good	N	N	20 SM
H14* ¹	Exterior floor joint sealant, gray mastic	Good	N	N	N/A
H15	Exterior conc wall coating, off white, textured	Good	N	N	245 SM
H16	Exterior wall joint sealant, white mastic	Good	N	N	60 LM
H17* ¹	TSI behind wall, building insulation, yellow fiberglass	Good	Y	N	N/A
H18* ¹	Moisture barrier, black mastic	Good	N	N	N/A
H19* ¹	Roofing material	Good	N	N	280 SM
H20* ¹	Roof vent sheeting material	Good	N	N	2 SM

Notes:

*1 – Previously identified homogeneous areas (HA's) observed as having less than three (3) historic analysis results were sampled for additional lab analysis by OYO Pacific Asbestos building inspectors during the field investigation period, Nov 2023 to Jan 2024. Newly identified HA's are indicated in BLUE. ACM detected samples are shown in RED.

TABLE 8: HOMOGENEOUS AREAS AND PRE-IDENTIFIED ACM (BLDG. 4047)

HA ID#	Material Description	Condition	Friable (Y/N)	ACM (Y/N)	Est. Quantity
H1	Concrete floor coating, gray	Good	N	N	280 SM
H2	10cm gray cove base w/ white mastic	Good	N	Y	100 LM
H3	Gypsum board wall, off white, smooth	Good	N	N	250 SM
H4 ^{*1}	Gypsum wall coating, off white, smooth	Good	N	N	50 SM
H5	Floor joint sealant, gray mastic	Good	N	N	100 LM
H6 ^{*1}	Door joint sealant, white mastic	Good	N	N	15 LM
H6 ^{*2}	Concrete block wall coating, off white, smooth	Good	N	N	50 SM
H7 ^{*1}	10x10cm pale purple ceramic wall tile w/ gray grout	Good	N	N	15 SM
H8 ^{*1}	10x10cm beige ceramic floor tile w/ gray grout	Good	N	N	20 SM
H9	Sink joint sealant, white mastic	Good	N	N	1 LM
H10 ^{*1}	Exterior floor joint sealant, gray mastic	Good	N	N	70 LM
H11	Exterior concrete wall coating, off white, textured	Good	N	N	280 SM
H12	Exterior wall joint sealant, white mastic	Good	N	N	150 LM
H13 ^{*1}	Exterior concrete wall base coating, gray painted orange	Good	N	N	50 SM
H14 ^{*1}	Fire piping gasket	Good	N	N	N/A
H15 ^{*1}	HVAC pipe conduit TSI	Good	Y	N	N/A
H16 ^{*1}	Water pipe, TSI	Good	Y	N	N/A
H17 ^{*1}	Roofing material	Good	N	N	280 SM
H18 ^{*1}	Roof vent sheeting material	Good	N	N	2 SM

Notes:

*1 – Previously identified homogeneous areas (HA's) observed as having less than three (3) historic analysis results were sampled for additional lab analysis by OYO Pacific Asbestos building inspectors during the field investigation period, Nov 2023 to Jan 2024. Newly identified HA's are indicated in BLUE. ACM detected samples are shown in RED.

*2 – Previous hazmat sampling report shows duplicate HA id numbers for different materials.

Abbreviations:

LM – linear meters SM – square meters

TABLE 9: HOMOGENEOUS AREAS ACM (BLDG. 4048)

HA ID#	Material Description	Condition	Friable (Y/N)	ACM (Y/N)	Est. Quantity
H1	Concrete floor	Good	N	N	360 SM
H2	Concrete wall	Good	N	N	297 SM

Notes:

No previous sampling and analysis data was available for this building. ACM detected samples are shown in RED.

Abbreviations:

LM – linear meters SM – square meters

TABLE 10: HOMOGENEOUS AREAS ACM (SWING SPACE)

HA ID#	Material Description	Condition	Friable (Y/N)	ACM (Y/N)	Est. Quantity
H1	Concrete pavement	Good	N	N	2200 SM
H2	Concrete expansion joint material	Good	N	N	1030 LM
H3	Pedestal Concrete	Good	N	N	1 SM

Notes:

No previous sampling and analysis data was available for this building. ACM detected samples are shown in RED.

Abbreviations:

LM – linear meters SM – square meters

2.3 Laboratory Analysis

For determination of asbestos content by testing methods, installations must analyze bulk samples by a National Institute of Standards and Technology / National Voluntary Laboratory Accreditation Program (NIST/NVLAP) accredited laboratory using polarized light microscopy (PLM) following U.S. Environmental Protection Agency (USEPA) methods for the determination of asbestos. The laboratory must have a working definition of “trace” amounts of asbestos, and the laboratory must report any detectable amount of asbestos in a bulk sample that is less than the PLM Limit of Quantitation (LOQ) of 1 percent as a “trace” concentration. If PLM does not detect the presence of asbestos (e.g., “non-detect”), consider the material to be less than 0.1 percent asbestos. If PLM analysis detects asbestos in any discernible amount (to include “trace” or “less than 1.0 percent”), consider the material to be greater than 0.1 percent asbestos unless proven to be non-ACM by the use of testing methods capable of achieving an LOQ of less than 0.1 percent, such as Transmission Electron Microscopy (TEM) or 1000-point counting. Samples were submitted to Hawaii Analytical Laboratory and Eurofins EPK Built Environment Testing, LLC at the following addresses:

Hawaii Analytical Laboratory

3615 Harding Ave., Suite 308
Honolulu, HI 96816-3759 USA

Eurofins EPK Built Environment Testing, LLC

2841 Dow Avenue, Suite 300
Tustin, CA 92780 USA

Hawaii Analytical Laboratory is a National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory under NVLAP Lab Code 200655-0. Eurofins is accredited under NVLAP Lab Code 500031-0. The samples were analyzed by polarized-light microscopy, by EPA Method 600/R-93-116, *Visual Area Estimation*.

None of the specifically targeted material layers containing trace amounts of asbestos were 1000-point counted for this report as there were no samples shown containing asbestos at trace concentrations above 0.1%. Refer to the accompanying appendices for details on material sample summary tables, sample locations, results, and photographs.

3.0 LEAD, CHROMIUM, AND CADMIUM

It is necessary to inspect coatings on surfaces that may be disturbed during the proposed project. The previous hazardous materials report did not provide analysis for cadmium and chromium, therefore all previously identified homogeneous materials sampled for lead were re-sampled and examined in the laboratory for the presence of cadmium and chromium over regulatory limits which present an occupational health risk during any planned future renovation and demolition work.

The survey was performed by certified Lead Building Inspectors, and specific sampling locations and sample collection materials were determined while on site. Sampling locations are included as Appendix D. Certifications of the surveying personnel and the laboratory used for analysis are included in Appendix G.

3.1 Sampling of Suspect Materials

Prior to sample collection, painted surfaces are categorized into areas of homogeneity by paint type, substrate, building material and apparent age. Paint-chip samples are taken in which approximately one-square-inch portion of sample material is collected using a utility knife and then placed into a plastic bag. The sampling bag is immediately sealed and labeled, and sampling equipment cleaned in between each sampling event to avoid cross-contamination between samples. All samples are logged and recorded following strict COC procedures.

3.2 Methodology

Sampling methodology generally followed the procedures presented in the USEPA 40 CFR 745 Lead-Based Paint Poisoning Prevention in Certain Residential Structures but adapted to a non-residential and commercial or industrial building and by taking paint chip samples rather than utilizing an X-Ray Fluorescence (XRF) instrument.

3.3 Lead Based Paint Vs. Lead Containing Paint

It is important to differentiate lead-based paint (LBP) from lead-containing paint (LCP), which is paint containing lead that is not regulated by EPA but does fall under OSHA exposure regulations. Lead-Based Paint (LBP): Paint, varnish, shellac or other coatings on surfaces that contain 1.0 milligram per square centimeter (1.0 mg/cm²) or more lead or **0.5% or more lead by weight**. The concentration 0.5% is equivalent to:

- a) 5,000 parts per million (5,000 ppm) or
- b) 5,000 milligrams per kilogram (5,000 mg/kg)

Lead-Containing Paint (LCP): Paint, varnish, shellac or other coatings on surfaces that contain measurable concentrations less than 1.0 milligrams per square centimeter (1.0 mg/cm²) lead or **less than 0.5% or more lead by weight** including equivalents less than 5,000 ppm and 5,000

mg/kg. This does not include paint, varnish, shellac or other coatings on surfaces where the concentration is below the detection limit of the respective laboratory method / device.

3.4 Resampling for Chromium and Cadmium

During a previous investigations at Building 4040, 4041, 4042, 4043, 4044, 4045, 4046, and 4047 in 2023 the following eighty-six (86) substrate and coating combinations were sampled and identified as lead-based paint (LBP), as shown in the following Tables 10 through 18 of this section. These substrate and coating combinations were re-sampled and analyzed additionally for cadmium and chromium in accordance with the project scope of work. Any additional coating combinations identified during the field survey were sampled and analyzed for lead, cadmium and chromium. The quantity shown in the report are for the purposes of identifying Homogeneous Materials. It represents the quantity observed at the time of the survey and should not be utilized for purposes of developing a bid for future work on the facility.

TABLE 11: PRE-IDENTIFIED LBP RE-SAMPLED FOR CR & Cd (BLDG. 4040)

ID#	Material Description	Condition	LBP (Y/N)	LCP (Y/N)	Cr (Y/N)	Cd (Y/N)
S1	Interior gypsum board wall, white	Used	N	N	N	N
S2	Interior metal pipe (above ceiling), red	Used	N	N	N	N
S3	Interior gypsum board wall, gray	Used	N	Y	N	N
S4	Interior gypsum ceiling board, white	Used	N	N	N	N
S5	Exterior metal pipe, red over gray and off white	Used	N	N	N	N
S6	Exterior metal flange, light blue over blue	Used	N	N	N	N
S7	Exterior concrete wall, off white	Used	N	N	N	N
S8	Exterior concrete base, orange	Used	N	N	N	N

Notes:

- 1) Results for LBP (lead >0.5% by weight) shown in **RED** (regulated by JEGS).
- 2) Results for lead, chromium, and cadmium to any detectable level shown in **BLUE** (regulated by OSHA).

TABLE 12: PRE-IDENTIFIED LBP RE-SAMPLED FOR CR & Cd (BLDG. 4041)

ID#	Material Description	Condition	LBP (Y/N)	LCP (Y/N)	Cr (Y/N)	Cd (Y/N)
S1	Exterior metal flange, light blue	Used	N	N	N	N
S2	Exterior metal pipe, red over white	Used	N	N	N	Y
S3	Exterior concrete wall, off white	Used	N	N	N	N
S4	Exterior concrete wall base, orange	Used	N	Y	N	N
S5	Interior gypsum board wall, gray	Used	N	N	N	N
S6	Interior metal pipe (above ceiling), red	Used	N	N	N	N
S7	Interior gypsum board wall, white	Used	N	N	N	N
S8	Interior wooden door, black	Used	N	Y	N	N
S9	Interior ceiling board, white	Used	N	N	N	N
S10	Interior gypsum board wall, red	Used	N	N	N	N
S11	Interior wooden door, white	Used	N	N	N	N

Notes:

- 1) Results for LBP (lead >0.5% by weight) shown in **RED** (regulated by JEGS).
- 2) Results for lead, chromium, and cadmium to any detectable level shown in **BLUE** (regulated by OSHA).

TABLE 13: PRE-IDENTIFIED LBP RE-SAMPLED FOR CR & Cd (BLDG. 4042)

ID#	Material Description	Condition	LBP (Y/N)	LCP (Y/N)	Cr (Y/N)	Cd (Y/N)
S1	Exterior metal pipe, red	Used	N	N	N	N
S2	Exterior metal flange, light blue	Used	N	N	N	N
S3	Exterior concrete wall, off white	Used	N	N	N	N
S4	Interior gypsum board wall, gray	Used	N	N	N	N
S5	Interior wooden door frame, red over white	Used	N	Y	N	N
S6	Interior wooden door, off white over black	Used	N	N	N	N
S7	Interior concrete wall, off white	Used	N	N	N	N
S8	Interior gypsum ceiling board, white	Used	N	N	N	N
S9	Interior gypsum board, off white	Used	N	N	N	N

Notes:

- 1) Results for LBP (lead >0.5% by weight) shown in **RED** (regulated by JEGS).
- 2) Results for lead, chromium, and cadmium to any detectable level shown in **BLUE** (regulated by OSHA).

TABLE 14: PRE-IDENTIFIED LBP RE-SAMPLED FOR CR & Cd (BLDG. 4043)

ID#	Material Description	Condition	LBP (Y/N)	LCP (Y/N)	Cr (Y/N)	Cd (Y/N)
S1	Interior gypsum board wall, gray	Used	N	N	N	N
S2	Interior metal pipe (above ceiling), red	Used	N	N	N	N
S3	Interior ceiling board, off white	Used	N	N	N	N
S4	Interior metal door frame, off white over red and black	Used	Y	Y	N	N
S5	Interior wooden door, black over white	Used	N	N	N	N
S6	Exterior metal pipe, red over white and black	Used	N	N	N	N
S7	Exterior metal flange, light blue	Used	N	N	N	N
S8	Exterior concrete wall, off white	Used	N	N	N	N
S9	Exterior concrete wall base, light brown	Used	N	N	N	N

Notes:

- 1) Results for LBP (lead >0.5% by weight) shown in **RED** (regulated by JEGS).
- 2) Results for lead, chromium, and cadmium to any detectable level shown in **BLUE** (regulated by OSHA).

TABLE 15: PRE-IDENTIFIED LBP RE-SAMPLED FOR CR & Cd (BLDG. 4044)

ID#	Material Description	Condition	LBP (Y/N)	LCP (Y/N)	Cr (Y/N)	Cd (Y/N)
S1	Interior gypsum board wall, gray	Used	N	N	N	N
S2	Interior metal pipe, red over white	Used	N	N	N	N
S3	Interior wooden door, green over off white	Used	N	Y	N	N
S4	Interior wooden door frame, black over red and off white	Used	N	Y	N	N
S5	Interior gypsum board wall, off white	Used	N	N	N	N
S6	Interior concrete wall, off white	Used	N	N	N	N
S7	Interior ceiling board, off white	Used	N	N	N	N
S8	Exterior metal pipe, red over off white	Used	N	N	N	N
S9	Exterior metal flange, light blue	Used	N	N	N	N
S10	Exterior concrete wall, off white	Used	N	N	N	N
S11	Exterior concrete wall base, light brown	Used	N	Y	N	N
S12	Exterior metal pipe, red	Used	N	N	N	N

Notes:

- 1) Results for LBP (lead >0.5% by weight) shown in **RED** (regulated by JEGS).
- 2) Results for lead, chromium, and cadmium to any detectable level shown in **BLUE** (regulated by OSHA).

TABLE 16: PRE-IDENTIFIED LBP RE-SAMPLED FOR CR & Cd (BLDG. 4045)

ID#	Material Description	Condition	LBP (Y/N)	LCP (Y/N)	Cr (Y/N)	Cd (Y/N)
S1	Interior concrete wall, off white	Used	N	Y	N	N
S2	Interior gypsum board wall, off white	Used	N	N	N	N
S3	Interior concrete wall (above ceiling), black	Used	N	N	N	N
S4	Interior concrete block wall, grey	Used	N	Y	N	N
S5	Interior concrete block wall, gray and off white	Used	N	N	N	N
S6	Interior gyp ceiling board, off white	Used	N	N	N	N
S7	Interior metal pipe (elec), off white	Used	N	N	N	Y
S8	Interior gypsum board wall, off white	Used	N	N	N	N
S9	Exterior concrete wall base, red	Used	N	N	N	N
S4*1	Exterior conc. block wall, off white	Used	N	Y	N	N
S10	Exterior metal pipe (elec), off white	Used	N	Y	N	Y
S11	Exterior plastic drain pipe, off white	Used	N	N	N	N

Notes:

- 1) Results for LBP (lead >0.5% by weight) shown in **RED** (regulated by JEGS).
- 2) Results for lead, chromium, and cadmium to any detectable level shown in **BLUE** (regulated by OSHA).

*1 – S4 appears to have been mistakenly repeated for a different substrate-paint combination. Information taken from previous report (ESC, 2023)

TABLE 17: PRE-IDENTIFIED LBP RE-SAMPLED FOR CR & Cd (BLDG. 4046)

ID#	Material Description	Condition	LBP (Y/N)	LCP (Y/N)	Cr (Y/N)	Cd (Y/N)
S1	Interior gypsum board wall, off white	Used	N	Y	N	N
S2	Interior metal door frame, light brown	Used	N	Y	N	N
S3	Interior metal pipe, red	Used	N	N	N	N
S4	Interior metal pipe, off white	Used	N	N	N	N
S5	Interior concrete floor coating, red	Used	N	Y	N	N
S6	Interior metal door frame, off white	Used	N	N	N	N
S7	Exterior telephone box, yellow	Used	N	N	N	N
S8	Exterior concrete wall, off white	Used	N	N	N	N
S9	Exterior metal pipe, red over white	Used	N	N	N	N
S10	Exterior metal flange, light blue	Used	N	N	N	N

Notes:

- 1) Results for LBP (lead >0.5% by weight) shown in **RED** (regulated by JEGS).
- 2) Results for lead, chromium, and cadmium to any detectable level shown in **BLUE** (regulated by OSHA).

TABLE 18: PRE-IDENTIFIED LBP RE-SAMPLED FOR CR & Cd (BLDG. 4047)

ID#	Material Description	Condition	LBP (Y/N)	LCP (Y/N)	Cr (Y/N)	Cd (Y/N)
S1	Interior gypsum board wall, off white	Used	N	Y	N	N
S2	Interior metal door frame, light brown	Used	N	Y	N	N
S3	Interior metal pipe, red	Used	N	N	N	N
S4	Interior metal pipe, off white	Used	N	N	N	Y
S5	Interior concrete floor coating, red	Used	N	N	N	N
S6	Exterior telephone box, yellow	Used	N	N	N	N
S7	Exterior concrete wall, off-white	Used	N	Y	N	N
S8	Exterior metal pipe, red	Used	N	N	N	N
S9	Exterior metal flange, light blue	Used	N	N	N	N
S10	Exterior metal pipe, off white	Used	N	Y	N	Y

Notes:

- 1) Results for LBP (lead >0.5% by weight) shown in **RED** (regulated by JEGS).
- 2) Results for lead, chromium, and cadmium to any detectable level shown in **BLUE** (regulated by OSHA).

TABLE 19: PRE-IDENTIFIED LBP RE-SAMPLED FOR CR & Cd (BLDG. 4048)

ID#	Material Description	Condition	LBP (Y/N)	LCP (Y/N)	Cr (Y/N)	Cd (Y/N)
S1	Interior concrete wall, white	Used	-	-	N	N
S2	Interior metal conduit pipe, white	Used	-	-	N	N
S3	Exterior concrete wall, white	Used	-	-	N	N
S4	Exterior metal bollard, beige	Used	-	-	Y	N
S5	Exterior metal wall, white	Used	-	-	N	Y
S6	Exterior metal roof, beige	Used	-	-	N	Y

Notes:

- 1) Results for LBP (lead >0.5% by weight) shown in **RED** (regulated by JEGS).
- 2) Results for lead, chromium, and cadmium to any detectable level shown in **BLUE** (regulated by OSHA).

TABLE 20: MATERIALS SAMPLED FOR Pb, Cr & Cd (SWING SPACE)

ID#	Material Description	Condition	LBP (Y/N)	LCP (Y/N)	Cr (Y/N)	Cd (Y/N)
S1	Pedestal #1 electrical panel, grey	Used	N	N	N	N
S2	Paint on bollards, black & yellow	Used	N	N	N	N
S3	NE side guard railing, white	Used	N	N	N	N

Notes:

- 3) Results for LBP (lead >0.5% by weight) shown in **RED** (regulated by JEGS).
 4) Results for lead, chromium, and cadmium to any detectable level shown in **BLUE** (regulated by OSHA).

3.5 Laboratory Analysis

Samples for cadmium and chromium analysis were submitted to Hawaii Analytical Laboratory located in Honolulu, Hawaii at the following address:

Hawaii Analytical Laboratory
 3615 Harding Ave., Suite 308
 Honolulu, HI 96816-3759 USA

Hawaii Analytical Laboratory is an American Industrial Hygiene Association (AIHA) accredited laboratory under Laboratory ID LAP-101812. Paint chip and coating samples were analyzed for cadmium and chromium samples were analyzed by EPA Method 7000B *Flame Atomic Absorption Spectrophotometry*.

Based on the previous laboratory analytical reports, one (1) of the one hundred and five (105) paint samples have lead concentrations exceeding the EPA guidelines for lead in paint. EPA defines lead-based paint (LBP) as paint or other coatings containing lead in equal to or in excess of 0.5% by weight. Lead at concentrations below the EPA guidelines was also detected. These paints have a lead concentration of less than 0.5% by weight and are identified as lead-containing paint (LCP).

Chromium was detected in one (1) paint sample, while cadmium was detected in seven (7) paint samples. While there are no specific EPA regulations addressing the remediation of cadmium or chromium, the Occupational Safety and Health Administration (OSHA) does have regulations to ensure a safe and healthful working environment for workers.

Refer to the accompanying appendices for details on material summary tables, sample locations, results, and photographs.

4.0 POLYCHLORINATED BIPHEYNYLS

4.1 Lighting Ballasts

4.1.1 Methodology

A visual inspection for electrical transformers and fluorescent light fixtures. Light fixtures were selected as randomly as feasible to be a representative cross section of the work areas and the manufacturer's label reviewed on each observed ballast. The light fixtures were dismantled to expose the ballast for inspection. If the ballast was marked with a US manufacturer's label of "NO PCB", the ballast is not presumed to be PCB free, since in Japan the PCB free standard is 0.5ppm while the US manufacturer PCB free standard is 50 ppm. Ballasts with or without the PCB free label shall be presumed to contain PCBs unless information from the manufacturer's product data can be reviewed and the contents fully confirmed. It was not part of this work scope to examine every light fixture in the building.

4.1.2 Results

Approximately 5-10% of all lighting ballasts were visually checked. All ballasts examined were a newer electronic type of ballast, manufactured after 1972. PCBs are contained within the fluorescent light ballast (FLB) capacitors and interior potting material of old, magnetic T12 lighting fixtures.

PCB containing ballasts were not observed. However, if during any renovation or demolition work, a ballast manufactured prior to 1972 is observed, the ballasts must be collected and dispossessed of in accordance with Chapter 7 of the JEGS and EPA 40 CFR 761 Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions and the JEGS.

At this time Kadena 718 CES/CEIE requires that all lighting ballasts (whether they contain PCBs or not) slated to be removed shall be transported and delivered to the Kadena Environmental PCB Storage Facility at Building 3625 by the contractor responsible for construction. 718 CES/CEIE requires the contractor to segregate the ballasts by US manufactured and foreign manufactured types.

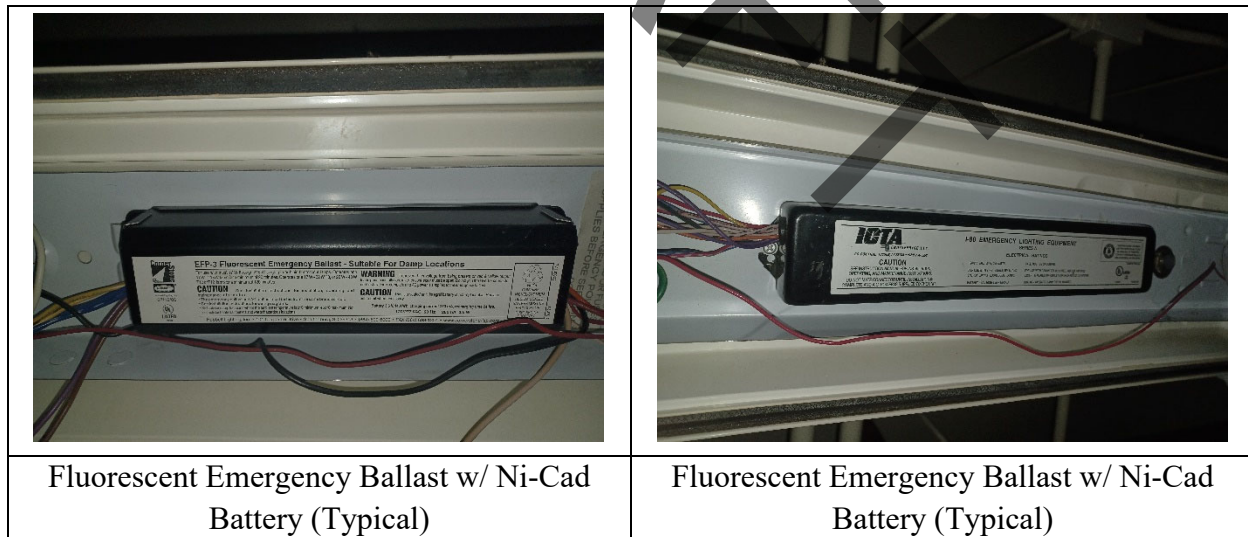
The following criteria provide help to identify Fluorescent Light Ballast (FLBs), this information comes from the EPA:

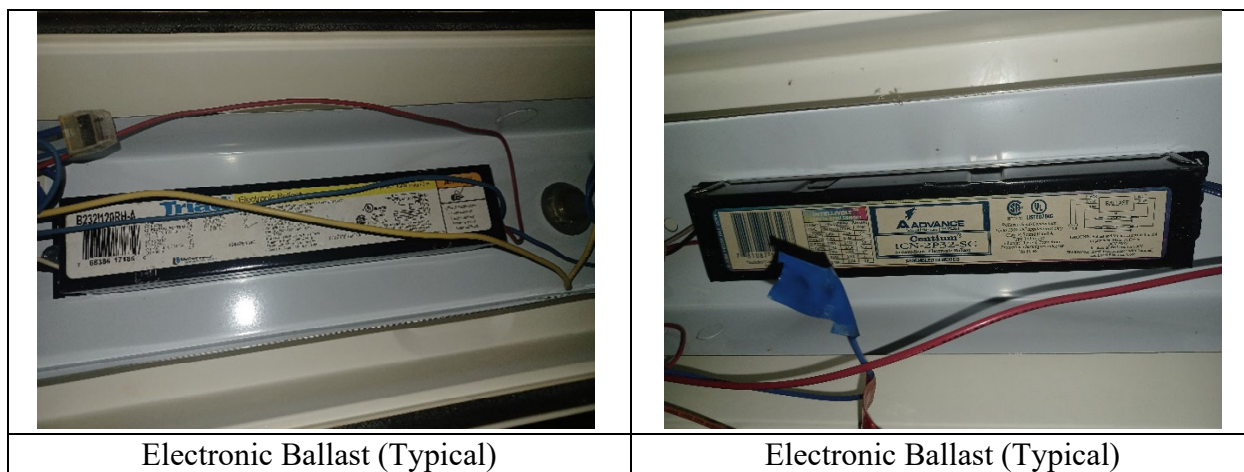
- FLBs manufactured before July 1, 1979, may contain PCBs
- FLBs manufactured between July 1, 1979 and July 1, 1998, that do not contain PCBs must be labeled "No PCBs", otherwise they may contain PCBs
- FLBs manufactured after 1998 are not required to be labeled

TABLE 21: LIGHTING BALLASTS, FIXTURES AND FLUORESCENT TUBES

Building No.	Approx. Quantity (Ballasts)	Approx. Quantity (Lighting Fixtures)	Approx. Quantity (Fluorescent Tubes)
4040	26	26	63
4041	28	28	71
4042	30	30	62
4043	30	30	62
4044	30	30	62
4045	24	24	72
4046	21	11	21
4047	21	11	21
4048	5	5	10
TOTAL	215	195	444

FIGURE 2: TYPICAL LIGHTING BALLASTS AT KADENA BUILDINGS 4040-4048





Please note that fluorescent emergency ballasts contain a lithium battery which should be recycled or disposed of through designated battery recycling programs or hazardous waste disposal facilities. Follow JEGS and local regulations and guidelines for the proper disposal of lithium batteries to ensure they are handled safely and responsibly.

4.2 Transformers and Other Electric Equipment

Three (3) electrical transformers (SET-4044, Circuit T-1), (SET-4046, Circuit T-1) and (SET-871A, Circuit T-1) were observed in the project area which will be disturbed (removal is required) due to the planned construction. Another electrical transformer (SET-4049, Circuit T-1) is nearby the project but outside of the development boundary. 718 CES/CEIE Environmental has not provided historical testing data on the oils associated with these electrical transformers. The transformers were not placarded at the time of survey.

The Contractor responsible for handling the transformers for future demolition or replacement projects shall assume the transformers contain PCBs, unless proven otherwise through valid sampling and analytical protocols.

Kadena 718 CES/CEIE requires that any electrical transformers to be demolished be transported and delivered to the Kadena Electrical shop warehouse at Building 1465 by the contractor responsible for construction. The Contractor is not allowed to recycle or dispose of electrical transformers off base, and instead must return equipment containing oils to the 718 Electrical Shop at Building 1465. The Contractor shall contact the 718 CES/CEIE PCB Manager or HW Manager at DSN 634-3253, to coordinate transformer turn-in to Building 1465.

FIGURE 3: ELECTRICAL OIL-FILLED TRANSFORMERS ON SITE

SET-4044 Circuit T-1 Electrical Transformer (1 EA)	SET-4046 Circuit T-1 Electrical Transformer (1 EA)
SET-871A Circuit T-1 Electrical Transformer (1 EA)	SET-4049 Circuit T-1 Electrical Transformer (1 EA) Note *1

Note *1): Transformer at SET-4049 Circuit T01 is nearby the site but likely will not be impacted by the construction activity. Photo provided for informational purposes only

5.0 HAZARDOUS WASTES

5.1 Mercury

Fluorescent light bulbs and light tubes typically contain a small amount of mercury. Mercury is used in fluorescent bulbs to facilitate the process of producing ultraviolet (UV) light, which in turn stimulates the phosphor coating inside the bulb to emit visible light. While the amount of mercury in each bulb is relatively small, handling and disposal of fluorescent bulbs properly to prevent environmental contamination and worker safety.

Specially controlled municipal waste (SCMW) and specially controlled industrial waste (SCIW) are specific categories of hazardous waste under the Japanese waste regulations and Chapter 16 of the JEGS. It includes certain types of common hazardous wastes, such as fluorescent lamps (including compact fluorescent lamps or CFLs), and some other mercury-containing devices.

If the light tubes are to be removed, the Contractor shall collect, transport, and recycle or dispose of them in accordance with the JEGS Chapter 16 and the Japan Waste Management and Public Cleaning Law.

The estimated volume of fluorescent light bulbs/tubes inside Buildings 4040, 4041, 4042, 4043, 4044, 4045, 4046, 4047 and 4048 are provided in Table 20, included in Section 4.1 of this report.

5.2 Radioactive Materials

5.2.1 Smoke Detectors

According to the US EPA and Nuclear Regulatory Commission, common ionization chamber smoke detectors, the most common type, use small amounts of radioactive material to detect smoke. These ionization smoke detectors use a small radioactive isotope, typically Americium-241, as part of their sensing mechanism. The Americium-241 emits alpha particles, which ionize the air inside the smoke detector's sensing chamber. When smoke particles enter the chamber, they disrupt the flow of the ionized air, triggering the detector's alarm.

It's important to note that the amount of radioactive material in ionization smoke detectors is very small and tightly sealed within the detector. When properly functioning and not tampered with, the radiation exposure from smoke detectors is minimal and well within safety limits established by regulatory agencies such as the U.S. Nuclear Regulatory Commission (NRC).

Modern smoke detectors often use alternative technologies, such as photoelectric or dual-sensor detectors, which do not rely on radioactive materials. Photoelectric smoke detectors use a light source and a photosensitive sensor to detect smoke particles, while dual-sensor detectors combine both ionization and photoelectric technologies for enhanced sensitivity to different types of fires.

5.2.2 Emergency Exit Signs

Some emergency exit signs, particularly older models, may contain radioactive materials, specifically tritium (a radioactive isotope of hydrogen). These signs are known as "self-luminous" or "glow-in-the-dark" exit signs. Tritium signs were used as a reliable and maintenance-free solution for illuminating exit signs in low-light conditions.

Tritium exit signs work by encapsulating a small amount of tritium gas in a phosphorescent coating. The radioactive decay of tritium emits low-energy beta particles, which interact with the phosphorescent material, causing it to glow. This self-illumination allows the exit sign to remain visible even during power outages or in areas with minimal lighting.

The amount of tritium used in these signs is relatively small, and the radiation levels are typically well within safety limits established by regulatory authorities, such as the U.S. NRC. In recent years, many facilities have shifted to using LED-based exit signs, which are more energy-efficient and do not contain radioactive materials. LED exit signs are considered safer and more environmentally friendly alternatives to tritium signs.

TABLE 22: POTENTIAL RADIOACTIVE MATERIALS

Building No.	Approx. Quantity (Smoke Detectors)	Approx. Quantity (Emergency Signs)	Notes
4040	1	2	Smoke detectors shall be checked for presence of Americium-241 and disposal appropriately. Emergency exit signs shall be checked for presence of Tritium and disposal appropriately.
4041	1	2	
4042	1	2	
4043	1	2	
4044	1	2	
4045	1	2	
4046	1	2	
4047	1	2	
4048	0	0	
TOTAL	8	16	

Note that some emergency signs, particularly those that are illuminated or use LED technology, may contain lithium batteries. These batteries are often used to power the lights in emergency

signs because they offer high energy density, long lifespan, and reliability. Follow JEGS and local regulations and guidelines for the proper disposal of lithium batteries to ensure they are handled safely and responsibly.

5.3 Oily Waste / Sludge

There were no visual indications of oily waste/sludge stored in or near Buildings 4040, 4041, 4042, 4043, 4044, 4045, 4046, 4047 and 4048 during the time of our hazardous materials survey.

5.4 Hazardous Waste Accumulation Points / Storage Areas

No hazardous waste accumulation points (HWAPs) or hazardous waste storage areas were observed in the project site during the time of our hazardous materials survey.

5.5 Arsenic Containing Materials

There have been instances in past renovation and demolition projects at Kadena Air Base and other US military projects where arsenic is found in gypsum board wall materials. Gypsum board, commonly known as drywall or sheetrock, is made from gypsum, a naturally occurring mineral that contains small amounts of various elements, including arsenic. However, the levels of arsenic found in gypsum board are typically very low and considered safe for use in construction.

When gypsum board is manufactured, steps are taken to minimize the release of arsenic and other potentially harmful substances. Additionally, gypsum board is coated with paper on both sides, which helps to contain any potential emissions. However, during the demolition of gypsum board wall materials, consideration to the extent of arsenic shall be considered.

Sampling and analysis associated with arsenic was not in the scope of work for the investigation.

6.0 POL STORAGE TANKS

There were no aboveground or underground petroleum, oil, or lubricant (POL) tanks noted in the project site during the time of our hazardous materials survey.

One POL storage tank was observed in the general vicinity of the site; however, it is not expected to be impacted by future site development for this project. The tank information and photographs are provided below for informational purposes only.

FIGURE 4: POL TANK NEAR PROJECT SITE

	
Building # 4040 Diesel FJ-1 fuel / Capacity: 500 gallons Convault, Inc. CE-AST (UL 2085)	Building # 4040 Diesel FJ-1 fuel / Capacity: 500 gallons Convault, Inc. CE-AST (UL 2085)

7.0 CHLOROFLUOROCARBON (CFC)-CONTAINING EQUIPMENT

7.1 Methodology

Older HVAC units have a higher potential of utilizing ozone depleting refrigerants. A visual inspection was conducted for CFC containing equipment and labels that identify the type of refrigerants being used for each piece of HVAC equipment.

7.2 Results

Based on direct observation of equipment labeling the refrigerant types and volumes being used currently at Buildings 4040, 4041, 4042, 4043, 4044, 4045, 4046, 4047 and 4048 are summarized in the following Table 22.

TABLE 23: EST. QUANTITY & TYPES OF REFRIGERANTS IN BLDGS. 4040-4048

Building No.	(A/C Unit #) / Manufacturer	Refrigerant Type	Volume Stored
4040	Daikin	R32	1.35 kg
	Daikin	R410A	5.7 kg
4041	Daikin	R32	0.6 kg
	Daikin	R410A	5.7 kg
4042	Daikin	R410A	5.7 kg
	Daikin	R32	0.6 kg
4043	(ACCU-1) / Daikin	R410A	5.7 kg
	(ACCU-2) / Daikin	R32	0.48 kg
4044	Daikin	R32	0.47 kg
	Daikin	R410A	5.7 kg
4045	Carrier	R410A	15.1 kg
4046	Daikin	R410A	5.2 kg
4047	(AC-1) / Hitachi	R410A	1.9 kg
	(AC-2) / Hitachi	R410A	1.9 kg
	(AC-3) / Hitachi	R410A	1.9 kg
	(AC-4) / Hitachi	R410A	1.9 kg
4048	N/A		
TOTAL (R32)			3.5 kg
TOTAL (R410A)			57.0 kg

R32 and R410A are not considered ozone-depleting substances. Both are hydrofluorocarbon (HFC) refrigerants commonly used in air conditioning and heat pump systems. Therefore, these are not considered to be CFC containing equipment.

For refrigerant containing pieces of equipment with CFCs that will be disturbed by future projects are required be collected and recycled or disposed of in accordance with the JEGS and following Kadena base procures. In most cases, turn-in of CFCs to the Kadena HVAC shop will be required (reference Section 9.7 of this report).

DRAFT

8.0 AQUEOUS FIRE FIGHTING FOAM (AFFF)

8.1 Background

Aqueous Film Forming Foam (AFFF), is a fire-fighting agent used commercially and by the DoD, including the Air Force. Most commonly used to combat petroleum fires in aircraft accidents, hangars and during live-fire training exercises, AFFF was found to contain perfluorooctanesulfonic acid (PFOS) and perfluorooctanoic acid (PFOA) - two perfluorinated compounds that persist in the environment and are not known to degrade by any natural process. The US EPA has classified these compounds as emerging contaminants due to inconclusive human health risks and evolving regulatory standards.

Signed at the end of 2019, the 2020 National Defense Authorization Act (NDAA) responded to increased public concern surrounding PFAS contamination by issuing several directives including a provision requiring the DoD to phase out AFFF at all military installations by Oct. 1, 2024. In response, the Air Force established an AFFF Sundown Policy outlining the service's plan to lockout and tagout all AFFF hangar systems. The policy details how the foam systems will be "locked out" at the tanks by closing and locking the valves that connect the foam to the hangar fire suppression systems. The tanks will then be tagged for removal with AFFF signage.

The Air Force began replacing legacy AFFF stockpiles with a new formula in August 2016. The legacy AFFF formula contains long-chain fluorosurfactants while the new formula contains shorter chain (six carbon chain forms) molecules.

8.2 Methodology

Fire suppression systems were visually inspected for AFFF related systems during the field survey.

8.3 Results

There were no visual indications of AFFF stored in or near Buildings 4040, 4041, 4042, 4043, 4044, 4045, 4046, 4047 and 4048 during the time of our hazardous materials survey.

9.0 SUMMARY OF RESULTS

9.1 Identified Asbestos Containing Materials

The listed materials were identified as ACM.

TABLE 24: MATERIALS IDENTIFIED AS ACM

Bldg. No.	HA ID	Material Description / Layer Description	Condition	Asbestos Result
4047	H2 (Note *1)	10cm gray cove base w/ white mastic	Good	2% Chrysotile (gray mastic layer)

Notes:

*1 - Identified as ACM in previous FY21 Comprehensive ACM and LBP Surveys Report by ESC, 2023
(See Appendix F)

The ACM were found to be in good condition and no immediate abatement action is necessary. The ACM material must be removed prior to the demolition activity. All removal must be completed by a certified asbestos abatement contractor under controlled conditions in accordance with EPA and JEGS regulations. Work should also be monitored by an independent industrial hygiene professional.

9.2 Identified Materials Containing Lead, Chromium, and Cadmium

The listed paints were identified as Pb, Cr and Cd.

TABLE 25: MATERIALS IDENTIFIED AS PB, CR, AND CD CONTAINING

Bldg. No.	ID#	Material Description	Condition	Pb	Cr (Y/N)	Cd (Y/N)
4040	S3	Interior gypsum board wall, gray	Used	LCP	N	N
4041	S2	Exterior metal pipe, red over white	Used	N	N	Y
	S4	Exterior concrete wall base, orange	Used	LCP	N	N
	S8	Interior wood door, black	Used	LCP	N	N
4042	S5	Interior wooden door frame, red over white	Used	LCP	N	N
4043	S4	Interior metal door frame, off white over red and black	Used	LBP	N	N
4044	S3	Interior wooden door, green over off white	Used	LCP	N	N
	S4	Interior wooden door frame, black over red and off white	Used	LCP	N	N
	S11	Exterior concrete wall base, light brown	Used	LCP	N	N
4045	S1	Interior concrete wall, off white	Used	LCP	N	N
	S4	Interior concrete block wall, gray	Used	LCP	N	N
	S4*	Exterior concrete block wall, off white	Used	LCP	N	N
	S7	Interior metal pipe (electrical), off white	Used	N	N	Y
	S10	Exterior metal pipe (electrical), off white	Used	LCP	N	Y

Bldg. No.	ID#	Material Description	Condition	Pb	Cr (Y/N)	Cd (Y/N)
4046	S1	Interior gypsum board wall, off white	Used	LCP	N	N
	S2	Interior metal door frame, light brown	Used	LCP	N	N
	S5	Interior concrete floor coating, red	Used	LCP	N	N
4047	S1	Interior gypsum board wall, off white	Used	LCP	N	N
	S2	Interior metal door frame, light brown	Used	LCP	N	N
	S4	Interior metal pipe, off white	Used	N	N	Y
	S7	Exterior concrete wall, off white	Used	LCP	N	N
	S10	Exterior metal pipe, off white	Used	LCP	N	Y
4048	S4	Exterior metal bollard, beige	Used	-	Y	N
	S5	Exterior metal wall, white	Used	-	N	Y
	S6	Exterior metal roof, beige	Used	-	N	Y

Notes:

LBP – Lead based paint (Pb > 0.5% by weight) LCP – Lead containing paint (Pb > 0.06% by weight)

EPA defines LBP as paint or other coatings containing lead in equal to or in excess of 1.0 mg/cm² by XRF or 0.5% by weight. Lead at concentrations of less than 1.0 mg/cm² or 0.5% by weight but having detectable levels of lead above 0.06% by weight are identified as lead-containing paint (LCP).

Disturbance to any lead, chromium, and cadmium containing paints must be conducted in accordance with OSHA 29 CFR 1926.55 Gases, Vapors, Fumes, Dusts and Mists and specifically for lead, chromium, and cadmium; 29 CFR 1926.62 Lead; 29 CFR 1926.1126 Chromium and 29 CFR 1926.1127 Cadmium. If any untested paints are disturbed, they should be assumed to contain lead, cadmium, and chromium. The Contractor shall employ a Certified Industrial Hygienist or Japanese equivalent safety professional to assess and provide a plan for worker protection when there is a potential disturbance of materials containing lead, chromium or cadmium in the course of demolition, reform, or construction work.

Prior to the disturbance of any paints, the contractor's employees disturbing the painted material must be informed that it contains lead, chromium, and cadmium and must have received training under OSHA 29 CFR 1926.62 *Lead*; 29 CFR 1926.1126 *Chromium (VI)*; and 29 CFR 1926.1127 *Cadmium*.

9.2.1 Lead, Chromium, & Cadmium as Specially Controlled Industrial Waste

It should be noted that JEGS and off-base GOJ regulations have the following regulatory limits for lead, chromium and cadmium for treatment as a specially controlled industrial waste (SCIW) are listed in Table 19, below.

TABLE 26: SCIW DISPOSAL CRITERIA FOR Pb, Cr, Cd UNDER JEGS TABLE 16.7

JEGS HW No.	Contaminant	CAS No.	Concentration Level (mg/L)
J006	Cadmium	7440-43-9	0.09
J007	Lead	7439-92-1	0.3
J009	Chromium (Hexavalent)	18540-29-9	1.5

The regulatory limits stipulated in Table 24 are for waste sludge, acid, and alkali, as determined by toxicity characteristic leaching procedure (TCLP) laboratory analysis. Although TCLP testing was not part of the scope of this survey, it is recommended that TCLP analysis be performed on lead, chromium, and cadmium positive materials (particularly for those materials of high surface area) to determine whether the paint in question should be classified as a SCIW for disposal purposes.

9.3 Polychlorinated Biphenyls

Based on our inspection, approximately 444 fluorescent lighting ballasts (FLBs) across approximately 215 lighting fixtures at Buildings 4040, 4041, 4042, 4043, 4044, 4045, 4046, 4047, and will need be removed as a part of the demolition of existing buildings prior to site development.

If during any renovation work, additional suspect ballasts are observed or otherwise uncovered, the ballasts must be collected and dispossessed of in accordance with Chapter 7 of the JEGS and EPA 40 CFR 761 Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions and the JEGS. At this time Kadena Air Base 718 CES/CEIE requires that all FLBs, slated to be removed shall be transported and delivered to the Kadena Environmental PCB Storage Facility at Building 3625 by the contractor responsible for construction. This applies to all FLBs, whether they contain PCBs or not.

718 CES/CEIE Environmental has indicated to OYO investigators that there is currently no available historical testing data on the oils associated with the three electrical transformers at the project site. Therefore, the Contractor responsible for handling the transformers for future demolition or replacement projects shall assume the transformers contain PCBs, unless proven otherwise through valid sampling and analytical protocols in accordance with JEGS.

Kadena 718 CES/CEIE requires that any electrical transformers or other electrical equipment (i.e. capacitors, switch gear, etc.) planned for demolition shall be transported and delivered by the

Contractor to the Kadena Electrical shop warehouse at Building 1465. The Contractor is **NOT ALLOWED** to recycle or dispose of electrical equipment off base. The Contractor in charge of construction shall transport the electrical equipment containing oils to the 718 Electrical Shop at Building 1465 at which time the Contractor will be relinquished of responsibility for the item.

The Contractor shall contact the 718 CES/CEIE PCB Manager or HW Manager at DSN 634-3253, to coordinate transformer turn-in to Building 1465.

718 CES/CEIE will take custody of the item, test the oils for PCBs and process the item for disposal with Defense Logistics Agency Disposition Services (DLA-DS) or recycling through the base Quality Recycling Program (QRP).

9.4 Hazardous Wastes

9.4.1 Mercury

The estimated volume of fluorescent light bulbs/tubes inside Buildings 4040, 4041, 4042, 4043, 4044, 4045, 4046, 4047 and 4048 are provided in Table 20, included in Section 4.1 of this report.

Prior to demolition work, fluorescent lights (potentially containing mercury) indicated by the results of this survey or otherwise uncovered, must be collected, and recycled or disposed of in accordance with JEGS Chapter 16 and applicable local environmental regulations (i.e., the Japan Waste Management and Public Cleansing Law and related statutes). The contractor in charge of construction shall consult with the 718 CES/CEIE is necessary to decide whether to classify these lamps as hazardous waste, and specific disposal guidelines might apply.

9.4.2 Batteries

Any mercury, nickel-cadmium, lithium, and lead-acid batteries which are associated with building facilities that are removed during demolition or renovation activities shall be disposed of by the Contractor in charge of Construction in accordance with JEGS Chapter 16 and local Environmental regulations (i.e., the Japan Waste Management and Public Cleansing Law and related statutes).

9.4.3 Radioactive Materials

Photoelectric and ionization smoke detectors were observed. All detectors should be assumed to contain americium-214 (unless proven otherwise) and be collected and disposed of in accordance with JEGS and local Environmental regulations (i.e., the Japan Waste Management and Public Cleansing Law and related statutes).

Additionally, emergency exit signs shall be checked for presence of Tritium and disposed of appropriately, in accordance with the JEGS and local Environmental regulations.

9.4.4 Arsenic

There have been instances in past renovation and demolition projects at Kadena Air Base and other US military projects where arsenic was found in gypsum board wall materials. Prior to demolition of gypsum board wall materials, consideration to the extent of arsenic shall be considered.

Sampling and analysis associated with arsenic was not in the scope of work for the investigation. Therefore, we recommend further sampling and testing during the construction stage prior to the demolition or impacts to walls and wall surfaces.

9.5 Above Ground Storage Tanks / Petroleum, Oils, and Lubricants (POLs)

If future renovation or demolition work impacts POL tanks, the fuel and fuel containments must be collected, handled and disposed (recycled) in accordance with JEGS Chapter 16 and applicable local environmental regulations (i.e., the Japan Waste Management and Public Cleansing Law and related statutes).

9.6 CFC-Containing Equipment

For this site, CFC containing equipment was not observed in any of the facilities planned for demolition under this project.

If in the course of the demolition previously unidentified suspect CFC-containing equipment is discovered, it shall be recovered from the HVAC system prior to demolition and turned into the Kadena HVAC shop for appropriate disposition. CFC refrigerant gas shall not be disposed of by the Contractor. Instead, the refrigerant gas shall be recovered into DOT certified containers and transportation to the Kadena HVAC shop for turn-in. Please refer to the Environmental Project specification for project specific details on CFC refrigerant turn in procedures.

9.7 AFFF/PFOS/PFOA

In the case of this specific site investigation, AFFF containing equipment or PFOS/PFOA impacted materials was not observed in any of the facilities planned for demolition under this project.

If in the course of the demolition previously unidentified suspect AFFF equipment is discovered, contact the Contracting Officer and report to 718 CES/CEIE PCB Manager or HW Manager at DSN 634-3253 for further direction.

10.0 REFERENCES

Code of Federal Regulation (CFR) Title 29 Subtitle B Chapter 17, Part 1926 Subpart D. CFR 1926.55 Gases, Vapors, Fumes, Dusts and Mists

CFR Title 29 Subtitle B Chapter 17, Part 1926. CFR 1926.62 Lead.

CFR Title 29 Subtitle B Chapter 17, Part 1926 Subpart Z. CFR 1926.1126 Chromium (VI).

CFR Title 29 Subtitle B Chapter 17, Part 1926. CFR 1926.1127 Cadmium in Construction Standard.

Department of the Army. (2014). Safety and Health Requirements Manual, EM 385-1-1, November 30, 2014.

Environmental Science Corporation. (2023). Comprehensive ACM & LBP Bulk Surveys, Building 4040, 6 March 2023.

Environmental Science Corporation. (2023). Comprehensive ACM & LBP Bulk Surveys, Building 4041, 25 February 2023.

Environmental Science Corporation. (2023). Comprehensive ACM & LBP Bulk Surveys, Building 4042, 25 February 2023.

Environmental Science Corporation. (2023). Comprehensive ACM & LBP Bulk Surveys, Building 4043, 25 February 2023.

Environmental Science Corporation. (2023). Comprehensive ACM & LBP Bulk Surveys, Building 4044, 20 March 2023.

Environmental Science Corporation. (2023). Comprehensive ACM & LBP Bulk Surveys, Building 4045, 6 March 2023.

Environmental Science Corporation. (2023). Comprehensive ACM & LBP Bulk Surveys, Building 4046, 9 December 2022.

Environmental Science Corporation. (2023). Comprehensive ACM & LBP Bulk Surveys, Building 4040, 6 March 2023.

Government of Japan, Ministry of the Environment (Amended 2001). Waste Management and Public Cleansing Law, Law No. 137 of 1970. Amended by Law No. 71 of 1974, No. 47 and No. 68 of 1976, No. 43 of 1983, No. 87 of 1987, No. 95 of 1991, No. 105 of 1992, No. 89 and No. 92 of 1993, No. 84 of 1994, No. 91 of 1995, No. 85 of 1997, No.54 of 1998, No.64, No.87, No.151 and No.160 of 1999, No.91 and No.105 of 2000, No.66 of 2001

Headquarters US Forces Japan. (2020). Supplementary Guidance for the Disposal of Waste Containing Perfluorooctane Sulfonate (PFOS) and Perfluorooctanoic Acid (PFOA). Dated 29 May 2020.

Headquarters US Forces Japan. (2024). Japan Environmental Governing Standards.

US Army Corps of Engineers, Japan Engineering District. (2023). Supplemental Statement of Work (SSOW), PN77833 Vehicle Maintenance Shop, Kadena AB, Okinawa, Japan. Dated 29 June 2023

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APPENDIX A

Sampling and Analysis Results Summary Tables

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ASBESTOS SAMPLE SUMMARY TABLE (BLDG. 4040)							
Sample ID	HA# / Material Description	Friable / Non-Friable	Type	Condition	Sample Locations	Photo No.	Result
4040-001	H1 / 30x30cm gray vinyl floor tile w/ gray mastic	Non-friable	Misc.	Good	Entrance corridor	A001-A002	ND
4040-002				Good	Conference room		ND
4040-003	H4 / 16x16cm off-white ceiling tile, all patterns	Non-friable	Misc.	Good	Entrance corridor	A003-A004	ND
4040-004				Good	Entrance corridor		ND
4040-005	H5 / Duct TSI, yellow fiberglass w/ silver backing	Friable	TSI	Good	Entrance corridor	A005-A006	ND
4040-006				Good	Entrance corridor		ND
4040-007	H6 / 30x30cm gray vinyl floor tile w/ tan mastic	Non-friable	Misc.	Good	Supply room	A007	ND
4040-008	H7 / 20x20cm beige ceramic floor tile w/ gray grout	Non-friable	Misc.	Good	Men's restroom	A008-A009	ND
4040-009				Good	Women's restroom		ND
4040-010	H8 / 10x10cm off-white ceramic wall tile w/ white grout	Non-friable	Misc.	Good	Men's restroom	A010-A011	ND
4040-011				Good	Women's restroom		ND
4040-012	H9 / gypsum ceiling board white	Non-friable	Misc.	Good	Men's restroom	A012-A013	ND
4040-013				Good	Women's restroom		ND
4040-14	H10 / black flange gasket	Non-friable	Misc.	Good	Exterior, near front entrance	A014-A015	ND
4040-15				Good	Exterior, near front entrance		ND
4040-16	H13 / Under joint sealant, /brown mastic	Non-friable	Misc.	Good	Exterior, north side	A016-A017	ND
4010-17				Good	Exterior, south side		ND
4040 018	Roofing material (concrete)	Non-friable	Misc.	Good	Roof	A018-A020	ND
4040 019				Good	Roof		ND
4040 020				Good	Roof		ND
4040-021	Roof vent sheeting material	Non friable	Misc.	Good	Roof	A021	ND
4040-022	Building insulation behind walls	Friable	TSI	Good	Entry corridor, near rear door	A022-A024	ND
4040-023				Good	Entry corridor, near rear door		ND
4040-024				Good	Entry corridor, near entrance		ND

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4040 (ESC, 6 March 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4040 (ESC, 6 March 2023) (see Appendix F)

Type - Surf. = surfacing; TSI = thermal system insulation; Misc. = Miscellaneous

Condition – Good, Damaged, Significantly Damaged

“ND” = None Detected, It means assuming contains less than 0.1% of Asbestos unless further analysis is conducted

ASBESTOS SAMPLE SUMMARY TABLE (BLDG. 4041)							
Sample ID	HA# / Material Description	Friable / Non-Friable	Type	Condition	Sample Locations	Photo No.	Result
4041-003	H3 / Wall joint sealant, / light grey mastic	Non-friable	Misc.	Good	Southeast side exterior wall	A025	ND
				Good			
4041-004	H4 / Window joint sealant, brown mastic	Non-friable	Misc.	Good	Southeast corner window	A026-A027	ND
4041-005				Good	South side, center window		ND
4041-006	H7 / Gypsum board wall, / gray	Non-friable	Misc.	Good	Conference room, north wall	A028-A029	ND
4041-007				Good	Conference room, south wall		ND
4041-008	H8 / 60x60 off white acoustical ceiling tile, all patterns	Non-friable	Misc.	Good	Conference room, west end	A030-A031	ND
4041-009				Good	Conference room, east end		ND
4041-010	H9 / Duct TSI, yellow fiberglass w/ silver backing	Friable	TSI	Good	Reception room, northwest	A032-A033	ND
4041-011				Good	Reception room, northwest		ND
4041-012	H10 / Gypsum board wall	Non-friable	Misc.	Good	Men's restroom	A034	ND
4041-013	H11 / Ceiling board, white	Non-friable	Misc.	Good	Women's restroom	A035-A036	ND
4041-014				Good	Men's restroom		ND
4041-015	H12 / 20x20cm Beige ceramic floor tile w/ gray grout	Non-friable	Misc.	Good	Men's restroom	A037-A038	ND
4041-016				Good	Women's restroom		ND
4041-017	H13 / 10x10 Off white ceramic wall tile w/ white grout	Non-friable	Misc.	Good	Men's restroom	A039-A040	ND
4041-018				Good	Women's restroom		ND
4041-020	H15 Pipe TSI (straight), yellow fiberglass w/ silver backing	Friable	TSI	Good	Janitors room	A041-A042	ND
4041-021				Good	Janitors room		ND
4041-024	Roofing material (concrete) (NI)	Non-friable	Misc.	Good	Roof	A043-A045	ND
4041-025				Good	Roof		ND
4041-026				Good	Roof		ND
4041-027	Roof vent sheeting material	Non-friable	Misc.	Good	Roof	A046	ND

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4041 (ESC, 25 February 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4041 (ESC, 25 February 2023) (see Appendix F)

Type - Surf. = surfacing; TSI = thermal system insulation; Misc. = Miscellaneous

Condition – Good, Damaged, Significantly Damaged

“ND” = None Detected, It means assuming contains less than 0.1% of Asbestos unless further analysis is conducted

ASBESTOS SAMPLE SUMMARY TABLE (BLDG. 4042)							
Sample ID	HA# / Material Description	Friable / Non-Friable	Type	Condition	Sample Locations	Photo No.	Result
4042-001	H1 Flange gasket, black	Non-friable	Misc.	Good	Exterior, south of main entrance	A047-A048	ND
4042-002				Good	Exterior, south of main entrance		ND
4042-003	H3 Wall joint sealant, light grey mastic	Non-friable	Misc.	Good	Exterior, north of main entrance	A049	ND
4042-004	H4 Window joint sealant, dark brown mastic	Non-friable	Misc.	Good	Exterior, southeast window	A050-A051	ND
4042-005				Good	Exterior, south side window		ND
4042-006	H8 Ceiling tile, 60x60 off white, hole pattern	Non-friable	Misc.	Good	Entrance corridor, east side	A052-A053	ND
4042-007				Good	2 nd room from entrance, N side		ND
4042-008	H9 Duct D1, yellow fiberglass w/ silver backing	Friable	TSI	Good	Entrance corridor, east side	A054-A055	ND
4042-009				Good	Entrance corridor, east side		ND
4042-010	H10 Pipe TSI (elbow), yellow fiberglass w/ silver backing	Friable	TSI	Good	Janitor's room	A056-A057	ND
4042-011				Good	Janitor's room		ND
4042-012	H11 Pipe TSI (run), yellow fiberglass w/ silver backing	Friable	TSI	Good	Janitor's room	A058-A059	ND
4042-013				Good	Janitor's room		ND
4042-014	H12 Concrete floor coating, gray	Non-friable	Misc.	Good	Janitor's room	A060-A061	ND
4042-015				Good	Janitor's room		ND
4042-016	H13 Concrete wall coating, off white, smooth	Non-friable	Misc.	Good	Janitor's room	A062-A063	ND
4042-017				Good	Janitor's room		ND
4042-018	H14 Ceramic wall tile, 10x10 off white tile w/ white grout	Non-friable	Misc.	Good	Men's room	A064-A065	ND
4042-019				Good	Men's room		ND
4042-020	H15 Ceramic floor tile, 20x20 beige tile w/ gray grout	Non-friable	Misc.	Good	Men's room	A066-A067	ND
4042-021				Good	Men's room		ND
4042-022	H17 Gypsum ceiling board, off white smooth	Non-friable	Misc.	Good	Men's room	A068-A069	ND
4042-023				Good	Men's room		ND
4042-026	Roofing material (concrete)	Non-friable	Misc.	Good	Roof, south side	A070-A072	ND
4042-027				Good	Roof, east side		ND
4042-028				Good	Roof, north side		ND
4042-029	Roof vent sheet	Non-friable	Misc.	Good	Roof, west side	A073	ND

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4042 (ESC, 25 February 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4042 (ESC, 25 February 2023) (see Appendix F)

Type - Surf. = surfacing; TSI = thermal system insulation; Misc. = Miscellaneous

Condition – Good, Damaged, Significantly Damaged

“ND” = None Detected, It means assuming contains less than 0.1% of Asbestos unless further analysis is conducted

ASBESTOS SAMPLE SUMMARY TABLE (BLDG. 4043)							
Sample ID	HA# / Material Description	Friable / Non-Friable	Type	Condition	Sample Locations	Photo No.	Result
4043-001	H14 60x60cm off white, all patterns ceiling tile	Non-friable	Misc.	Good	1 st room south of entrance	A074-A075	ND
4043-002				Good	East side of main corridor front		ND
4043-003	H15 Duct TSI, yellow fiberglass w/ silver backing	Friable	TSI	Good	East side of main corridor front	A076-A077	ND
4043-004				Good	1 st room south of entrance		ND
4043-005	H7 Ceiling board, off white	Non-friable	Misc.	Good	Men's room	A078-A079	ND
4043-006				Good	Women's room		ND
4043-007	H8 10x10cm ceramic wall tile, w/ white grout	Non-friable	Misc.	Good	Men's room	A080-A081	ND
4043-008				Good	Women's room		ND
4043-009	H9 20x20cm ceramic floor tile, beige w/ gray grout	Non-friable	Misc.	Good	Men's room	A082-A083	ND
4043-010				Good	Women's room		ND
4043-011	H10 Pipe TSI (elbow) yellow fiberglass w/ silver backing	Friable	TSI	Good	1 st room south of entrance	A084-A085	ND
4043-012				Good	Main corridor, rear		ND
4043-013	H11 Pipe TSI (run) yellow fiberglass w/ silver backing	Friable	TSI	Good	1 st room south of entrance	A086-A087	ND
4043-014				Good	Main corridor, rear		ND
4043-015	H12 Flange gasket, black (exterior)	Non-friable	Misc.	Good	Exterior, north of main entrance	A088-A089	ND
4043-016				Good	Exterior, north of main entrance		ND
4043-017	H15 Window joint sealant, brown (Ext)	Non-friable	Misc.	Good	Exterior, northwest window	A090-A091	ND
4043-018				Good	Exterior, southwest window		ND
4043-019	Roofing materials	Non-friable	Misc.	Good	Roof, north side	A092-A094	ND
4043-020				Good	Roof, south side		ND
4043-021				Good	Roof, west side		ND
4043-022	Roof vent sheeting material	Non-friable	Misc.	Good	Roof, east side	A095	ND

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4043 (ESC, 25 February 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4043 (ESC, 25 February 2023) (see Appendix F)

Type - Surf. = surfacing; TSI = thermal system insulation; Misc. = Miscellaneous

Condition – Good, Damaged, Significantly Damaged

“ND” = None Detected, It means assuming contains less than 0.1% of Asbestos unless further analysis is conducted

ASBESTOS SAMPLE SUMMARY TABLE (BLDG. 4044)							
Sample ID	HA# / Material Description	Friable / Non-Friable	Type	Condition	Sample Locations	Photo No.	Result
4044-001	H3 Gypsum board wall, gray, smooth	Non-friable	Misc.	Good	Entrance corridor, south wall	A096	ND
4044-002	H4 Pipe TSI (run), yellow fiberglass w/ silver backing	Friable	TSI	Good	Entrance corridor, near 1 st room doorway	A097	ND
4044-003	H5 Pipe TSI (elbow), yellow fiberglass w/ silver backing	Friable	TSI	Good	Entrance corridor, west side	A098	ND
4044-004	H6 Ceiling tile, 60x60 w/ hole pattern	Non-friable	Misc.	Good	2 nd room from entrance, south side	A099-A100	ND
4044-005				Good	2nd room from entrance (storage room), north side		ND
4044-006	H7 Concrete wall coating, off white, smooth	Non-friable	Misc.	Good	3rd room from entrance, north side	A101	ND
4044-007	H8 gypsum board wall, off white, smooth	Non-friable	Misc.	Good	3rd room from entrance, north side	A102-A103	ND
4044-008				Good	2nd room from entrance (storage room), north side		ND
4044-009	H9 Door joint sealant, off white	Non-friable	Misc.	Good	Men's room door	A104-A105	ND
4044-010				Good	Janitor's room door		ND
4044-011	H10 Concrete floor coating, gray	Non-friable	Misc.	Good	Janitor's room, west side of door	A106-A107	ND
4044-012				Good	Janitor's room, east side of door		ND
4044-013	H11 10x10 ceramic wall tile, off white w/ white grout	Non-friable	Misc.	Good	Men's room, near door	A108-A109	ND
4044-014				Good	Women's room, near door		ND
4044-015	H12 20x20 ceramic floor tile, beige w/ grey grout	Non-friable	Misc.	Good	Men's room, near door	A110-A111	ND
4044-016				Good	Women's room, near door		ND
4044-017	H14 Ceiling board, off white, smooth	Non-friable	Misc.	Good	Men's room, west side	A112-A113	ND
4044-018				Good	Women's room, west side		ND
4044-019	H15 Flange gasket, black	Non-friable	Misc.	Good	Exterior, north of entrance	A114-A115	ND
4044-020				Good	Exterior, north of entrance		ND
4044-021	H18 Window joint sealant, brown	Non-friable	Misc.	Good	Exterior, north side window	A116-A117	ND
4044-022				Good	Exterior, south side window		ND
4044-023	H19 Floor joint sealant, light gray	Non-friable	Misc.	Good	Exterior, near rear entrance	A118	ND
4044-025	H20 Door joint sealant, off white	Non-friable	Misc.	Good	Exterior, near rear entrance	A119	ND
4044-027	Roofing material (concrete)	Non-friable	Misc.	Good	Roof, north side	A120-A122	ND
4044-028				Good	Roof, south side		ND
4044-029				Good	Roof west side		ND

ASBESTOS SAMPLE SUMMARY TABLE (BLDG. 4044)							
Sample ID	HA# / Material Description	Friable / Non-Friable	Type	Condition	Sample Locations	Photo No.	Result
4044-030	Roof vent sheeting material	Non-friable	Misc.	Good	Roof, east side	A123	ND

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4044 (ESC, 20 March 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4044 (ESC, 20 March 2023) (see Appendix F)

Type - Surf. = surfacing; TSI = thermal system insulation; Misc. = Miscellaneous

Condition – Good, Damaged, Significantly Damaged

“ND” = None Detected, It means assuming contains less than 0.1% of Asbestos unless further analysis is conducted

ASBESTOS SAMPLE SUMMARY TABLE (BLDG. 4045)							
Sample ID	HA# / Material Description	Friable / Non-Friable	Type	Condition	Sample Locations	Photo No.	Result
4045-001	H3 Concrete wall coating, off white, smooth	Non-friable	Misc.	Good	North side office, N of corridor	A124-A125	ND
4045-002				Good	North side office, N of corridor		ND
4045-003	H4 Gypsum board wall, off white, smooth	Non-friable	Misc.	Good	North side office, N of corridor	A126	ND
4045-004	H5 Duct TSI, yellow fiberglass w/ silver backing	Friable	TSI	Good	Classroom, above ceiling	A127-A128	ND
4045-005				Good	Classroom, above ceiling		ND
4045-006	H6 Ceiling insulation, yellow fiberglass w/ silver backing	Friable	TSI	Good	Classroom, above ceiling	A129-130	ND
4045-007				Good	Classroom, above ceiling		ND
4045-008	H7 60x60cm off white fissured acoustical ceiling tile	Non-friable	Misc.	Good	Classroom, above ceiling	A131-A132	ND
4045-009				Good	Classroom, above ceiling		ND
4045-012	H9 Concrete block wall coating, gray painted	Non-friable	Misc.	Good	Men's room, near door	A133-A134	ND
4045-013				Good	Main corridor, southeast end		ND
4045-014	H10 Pipe TSI (run), yellow fiberglass w/ silver backing	Friable	TSI	Good	Classroom storage, above ceiling	A135-A136	ND
4045-015				Good	Classroom storage, above ceiling		ND
4045-016	H11 Pipe TSI (elbow), yellow fiberglass w/ silver backing	Friable	TSI	Good	Classroom storage, above ceiling	A137-A138	ND
4045-017				Good	Classroom storage, above ceiling		ND
4045-018	H12 20x20 ceramic floor tile, gray w/ gray grout	Non-friable	Misc.	Good	Men's room	A139-A140	ND
4045-019				Good	Men's room		ND
4045-020	H13 10x10 ceramic wall tile, white w/ white grout	Non-friable	Misc.	Good	Classroom south adj. room	A141-A142	ND
4045-021				Good	Classroom south adj. room		ND
4045-022	H14 gypsum ceiling board, off white	Non-friable	Misc.	Good	Men's room	A143-A144	ND
4045-023				Good	Classroom south adj. room		ND
4045-024	H16 Joint sealant (door) white mastic	Non-friable	Misc.	Good	Men's room	A145-A146	ND
4045-025				Good	Women's room		ND
4045-026	H18 Exterior concrete block wall, off white, textured	Non-friable	Misc.	Good	Exterior, east side wall	A147	ND
4045-027	H19 Exterior concrete wall base coating (red)	Non-friable	Misc.	Good	Exterior, east side wall	A148-A149	ND
4045-028				Good	Exterior, south side wall		ND
4045-029	H20 Exterior door joint sealant, gray mastic	Non-friable	Misc.	Good	Exterior, entrance door	A150-A151	ND
4045-030				Good	Exterior, rear door		ND
4045-031	H21 Exterior black flex sheet	Non-friable	Misc.	Good	Exterior, north side	A152-A153	ND
4045-032				Good	Exterior, north side		ND
4045-033	H25 Floor joint sealant (exterior), solid black	Non-friable	Misc.	Good	Exterior, near rear door	A154-A155	ND
4045-034				Good	Exterior, near rear door		ND
4045-035	H26 Window joint sealant, black mastic	Non-friable	Misc.	Good	Exterior, north side	A156-A157	ND
4045-036				Good	Exterior, south side		ND

ASBESTOS SAMPLE SUMMARY TABLE (BLDG. 4045)							
Sample ID	HA# / Material Description	Friable / Non-Friable	Type	Condition	Sample Locations	Photo No.	Result
4045-037	Roofing material (asphalt)		Misc.	Good	Roof, south side	A158-A160	ND
4045-038				Good	Roof, north side		ND
4045-039				Good	Roof, west side		ND
4045-040	Roof vent sealant		Misc.	Good	Roof, south side	A161	ND
4045-041	Roof Caulking		Misc.	Good	Roof, south side	A162-A164	ND
4045-042				Good	Roof, west side		ND
4045-043				Good	Roof, north side		ND

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4045 (ESC, 6 March 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4045 (ESC, 6 March 2023) (see Appendix F)

Type - Surf. = surfacing; TSI = thermal system insulation; Misc. = Miscellaneous

Condition – Good, Damaged, Significantly Damaged

“ND” = None Detected, It means assuming contains less than 0.1% of Asbestos unless further analysis is conducted

ASBESTOS SAMPLE SUMMARY TABLE (BLDG. 4046)							
Sample ID	HA# / Material Description	Friable / Non-Friable	Type	Condition	Sample Locations	Photo No.	Result
4046-001	H2 Gypsum board wall, off white smooth	Non-friable	Misc.	Good	Rm 101, south side west door	A165	ND
4046-002	H5 Concrete block wall coating, off white smooth	Non-friable	Misc.	Good	Rm 101, south side, near restroom door	A166	ND
4046-003	H6 Door sealant, white mastic	Non-friable	Misc.	Good	Rm. 101, south restroom door	A167-A168	ND
4046-004				Good	Rm. 101, south, door to Rm. 104		ND
4046-005	H7 Pipe TSI (elbow), yellow fiberglass w/ silver backing	Friable	TSI	Good	Rm. 104, south side	A169-A170	ND
4046-006				Good	Rm. 104, north side		ND
4046-007	H8 Pipe TSI (run), yellow fiberglass w/ silver backing	Friable	TSI	Good	Rm. 104, south side	A171-A172	ND
4046-008				Good	Rm. 104, north side		ND
4046-011	H10 Ceramic wall tile, 10x10 pale purple w/ gray grout	Non-friable	Misc.	Good	Restroom, west wall	A173-A174	ND
4046-012				Good	Restroom, north wall		ND
4046-013	H11 10x10 Gray ceramic floor tile w/gray grout	Non-friable	Misc.	Good	Restroom, west side	A175-A176	ND
4046-014				Good	Restroom, north side		ND
4046-015	H13 Concrete wall coating, off white smooth	Non-friable	Misc.	Good	Rm. 101, south side, east wall	A177-A178	ND
4046-016				Good	Rm. 101, north side, east wall		ND
4046-017	H14 Floor joint sealant, gray mastic	Non-friable	Misc.	Good	Rm. 101, south side slab joint	A179-A180	ND
4046-018				Good	Rm. 101, north side slab joint		ND
4046-019	TSI behind wall, building insulation	Friable	TSI	Good	Rm. 101, north side	A181-A182	ND
4046-020				Good	Rm. 101, south side		ND
4046-021	Moisture barrier, black mastic	Non-friable	Misc.	Good	Rm. 101, north side	A183-A184	ND
4046-022				Good	Rm. 101, south side		ND
4046-023	Roofing material	Non-friable	Misc.	Good	Roof, north side	A185-A187	ND
4046-024				Good	Roof, south side		ND
4046-025				Good	Roof, west side		ND
4046-026	Roof vent sheeting material	Non-friable	Misc.	Good	Roof, north side	A188	ND

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4046 (ESC, 9 December 2022)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4046 (ESC, 9 December 2022) (see Appendix F)

Type - Surf. = surfacing; TSI = thermal system insulation; Misc. = Miscellaneous

Condition – Good, Damaged, Significantly Damaged

“ND” = None Detected, It means assuming contains less than 0.1% of Asbestos unless further analysis is conducted

ASBESTOS SAMPLE SUMMARY TABLE (BLDG. 4047)							
Sample ID	HA# / Material Description	Friable / Non-Friable	Type	Condition	Sample Locations	Photo No.	Result
4047-001	H4 Concrete wall coating, off white, smooth	Non-friable	Misc.	Good	Main entrance hallway, east wall on north side	A189	ND
4047-002	H6 Door joint sealant, white mastic	Non-friable	Misc.	Good	Main entrance hall, middle door	A190-A191	ND
4047-003				Good	Main entrance hall, north door		ND
4047-004	H6 Concrete block wall coating, off white, smooth	Non-friable	Misc.	Good	Main entrance hall, east side wall	A192-A193	ND
4047-005				Good	North storage room, west wall		ND
4047-006	H7 10x10cm Pale purple ceramic wall tiles w/ gray grout	Non-friable	Misc.	Good	Restroom, northwest corner	A194-A195	ND
4047-007				Good	Restroom, north side wall		ND
4047-008	H8 10x10cm beige ceramic floor tile, w/ gray grout	Non-friable	Misc.	Good	Restroom, northwest corner	A196-A197	ND
4047-009				Good	Restroom, north side wall		ND
4047-010	H10 Floor joint sealant, gray mastic	Non-friable	Misc.	Good	Main entrance hall, east side	A198-A199	ND
4047-011				Good	South storage room, west side		ND
4047-012	H13 Concrete wall base coating, gray painted	Non-friable	Misc.	Good	Exterior, north side wall	A200-A201	ND
4047-013				Good	Exterior, east side wall		ND
4047-014	Fire piping gasket	Non-friable	Misc.	Good	North side storage room, interior	A202	ND
4047-015	HVAC pipe TSI	Friable	TSI	Good	Main entrance hall, north (run)	A203-A204	ND
4047-016				Good	Main entrance hall, north (elbow)		ND
4047-017	Water pipe, TSI	Friable	TSI	Good	Exterior, south side water pipe	A205	ND
4047-018	Roofing material	Non-friable	Misc.	Good	Roof, south side	A206-A208	ND
4047-020				Good	Roof, north side		ND
4047-021				Good	Roof, east side		ND
4047-019	Roof vent sheeting material	Non-friable	Misc.	Good	Roof, south side	A209	ND

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4047 (ESC, 6 March 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4047 (ESC, 6 March 2023) (see Appendix F)

Type - Surf. = surfacing; TSI = thermal system insulation; Misc. = Miscellaneous

Condition – Good, Damaged, Significantly Damaged

“ND” = None Detected, It means assuming contains less than 0.1% of Asbestos unless further analysis is conducted

ASBESTOS SAMPLE SUMMARY TABLE (BLDG. 4048)							
Sample ID	HA# / Material Description	Friable / Non-Friable	Type	Condition	Sample Locations	Photo No.	Result
4048-001	Concrete floor	Non-friable	Misc.	Good	Inside main bay, middle of north side	A210-A212	ND
4048-002				Good	Inside main bay, near east entrance door		ND
4048-003				Good	Inside main bay, on south side		ND
4048-004	Concrete wall	Non-friable	Misc.	Good	Exterior, southeast corner	A213-A215	ND
4048-005				Good	Exterior, southwest side		ND
4048-006				Good	Interior, north side wall		ND

Notes:

Type - Surf. = surfacing; TSI = thermal system insulation; Misc. = Miscellaneous

Condition – Good, Damaged, Significantly Damaged

“ND” = None Detected, It means assuming contains less than 0.1% of Asbestos unless further analysis is conducted

ASBESTOS SAMPLE SUMMARY TABLE (SWING SPACE)							
Sample ID	HA# / Material Description	Friable / Non-Friable	Type	Condition	Sample Locations	Photo No.	Result
SS-001	Concrete pavement	Non-friable	Misc.	Good	NE center portion of conc. paved area	SS-001	ND
SS-002	Concrete pavement				North edge of conc. paved area	SS-002	ND
SS-003	Concrete pavement				South side of conc. paved area	SS-003	ND
SS-004	Concrete expansion joint material				NE center portion of conc. paved area	SS-004	ND
SS-005	Concrete expansion joint material				Northern edge of of conc. paved area	SS-005	ND
SS-006	Concrete expansion joint material				South side of conc. paved area	SS-006	ND
SS-007	Concrete from pedestal #1				Concrete from pedestal #1	SS-007	ND

Notes:

Type - Surf. = surfacing; TSI = thermal system insulation; Misc. = Miscellaneous

Condition – Good, Damaged, Significantly Damaged

“ND” = None Detected, It means assuming contains less than 0.1% of Asbestos unless further analysis is conducted

LEAD, CHROMIUM, CADMIUM SAMPLE SUMMARY TABLE (BLDG. 4040)									
Sample ID	Paint Color	Interior / Exterior	Substrate	Component	Photo No.	LBP ^{*1} (Y/N) [%]	LCP ^{*1} (Y/N) [%]	Cr (Y/N) [mg/kg]	Cd (Y/N) [mg/kg]
4040-S1	White	Interior	Gypsum board	Wall	M001	N	N	N	N
4040-S2	Red	Interior	Metal	Pipe	M002	N	N	N	N
4040-S3	Gray	Interior	Gypsum board	Wall	M003	N	N	N	N
4040-S5	White	Interior	Gypsum board	Ceiling	M004	N	N	N	N
4040-S6	Red over gray	Exterior	Metal	Pipe	M005	N	N	N	N
4040-S7	Light blue over blue	Exterior	Metal	Pipe flange	M006	N	N	N	N
4040-S8	Off white	Exterior	Concrete	Wall	M007	N	N	N	N
4040-S9	Light brown	Exterior	Concrete	Wall	M008	N	Y [0.010%]	N	N

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4040 (ESC, 6 March 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4040 (ESC, 6 March 2023) (see Appendix F)

LBP: Lead-Based Paint => *Lead Concentration equal or higher than 0.5% by weight*

LCP: Lead Containing Paint => *Laboratory Detection Limit*

Chromium Containing Paint (Cr) => *Laboratory Detection Limit*

Cadmium Containing Paint (Cd) => *Laboratory Detection Limit*

LEAD, CHROMIUM, CADMIUM SAMPLE SUMMARY TABLE (BLDG. 4041)									
Sample ID	Paint Color	Interior / Exterior	Substrate	Component	Photo No.	LBP ^{*1} (Y/N) [%]	LCP ^{*1} (Y/N) [%]	Cr (Y/N) [mg/kg]	Cd (Y/N) [mg/kg]
4041-S1	Light blue	Exterior	Metal	Flange	M009	N	N	N	N
4041-S2	Red over white	Exterior	Metal	Pipe	M010	N	N	Y [83 mg/kg]	N
4041-S3	Off white	Exterior	Concrete	Wall	M011	N	N	N	N
4041-S5	Grey	Interior	Gypsum board	Wall	M012	N	N	N	N
4041-S6	Red	Interior	Metal	Pipe	M013	N	N	N	N
4041-S7	White	Interior	Gypsum board	Wall	M014	N	N	N	N
4041-S8	White over black	Interior	Wood	Door	M015	N	Y [0.02%]	N	N
4041-S9	White	Interior	Ceiling board	Ceiling	M016	N	N	N	N
4041-S10	Light brown	Exterior	Concrete	Wall	M017	N	N	N	N

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4041 (ESC, 25 February 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4041 (ESC, 25 February 2023) (see Appendix F)

LBP: Lead-Based Paint => *Lead Concentration equal or higher than 0.5% by weight*

LCP: Lead Containing Paint => *Laboratory Detection Limit*

Chromium Containing Paint (Cr) => *Laboratory Detection Limit*

Cadmium Containing Paint (Cd) => *Laboratory Detection Limit*

LEAD, CHROMIUM, CADMIUM SAMPLE SUMMARY TABLE (BLDG. 4042)									
Sample ID	Paint Color	Interior / Exterior	Substrate	Component	Photo No.	LBP ^{*1} (Y/N) [%]	LCP ^{*1} (Y/N) [%]	Cr (Y/N) [mg/kg]	Cd (Y/N) [mg/kg]
4042-S1	Red	Exterior	Metal	Pipe	M018	N	N	N	N
4042-S2	Light blue	Exterior	Metal	Flange	M019	N	N	N	N
4042-S3	Off white	Exterior	Concrete	Wall	M020	N	N	N	N
4042-S4	Blue	Interior	Gypsum board	Wall	M021	N	N	N	N
4042-S5	Red over white	Interior	Wood	Door frame	M021	N	Y [0.21%]	N	N
4042-S6	Off white	Interior	Wood	Door	M022	N	N	N	N
4042-S7	Off white Smooth	Interior	Concrete	Wall	M023	N	N	N	N
4042-S8	Off white	Interior	Gypsum ceiling board	Ceiling	M023	N	N	N	N
4042-S9	Off white	Interior	Gypsum board	Wall	M024	N	N	N	N
4042-S10	Light brown	Exterior	Concrete	Wall	M025	N	N	N	N

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4041 (ESC, 25 February 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4041 (ESC, 25 February 2023) (see Appendix F)

LBP: Lead-Based Paint => *Lead Concentration equal or higher than 0.5% by weight*

LCP: Lead Containing Paint => *Laboratory Detection Limit*

Chromium Containing Paint (Cr) => *Laboratory Detection Limit*

Cadmium Containing Paint (Cd) => *Laboratory Detection Limit*

LEAD, CHROMIUM, CADMIUM SAMPLE SUMMARY TABLE (BLDG. 4043)									
Sample ID	Paint Color	Interior / Exterior	Substrate	Component	Photo No.	LBP *1 (Y/N) [%]	LCP *1 (Y/N) [%]	Cr (Y/N) [mg/kg]	Cd (Y/N) [mg/kg]
4043-S1	Gray	Interior	Gypsum board	Wall	M026	N	N	N	N
4043-S2	Red	Interior	Metal	Pipe	M027	N	N	N	N
4043-S3	Off white	Interior	Gypsum board	Ceiling	M028	N	N	N	N
4043-S4	Off white over red & black	Interior	Metal	Door frame	M029	Y [1.3%]	Y [1.3%]	N	N
4043-S5	Black over white	Interior	Wood	Door	M030	N	N	N	N
4043-S6	Red over white & black	Exterior	Metal	Pipe	M031	N	N	N	N
4043-S7	Light blue	Exterior	Metal	Flange	M032	N	N	N	N
4043-S8	Off white Textured	Exterior	Concrete	Wall	M033	N	N	N	N
4043-S9	Light brown	Exterior	Concrete	Wall base	M034	N	Y [0.008%]	N	N
4043-S10	Light brown	Exterior	Concrete	Wall	M035	N	N	N	N

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4043 (ESC, 25 February 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4043 (ESC, 25 February 2023) (see Appendix F)

LBP: Lead-Based Paint => *Lead Concentration equal or higher than 0.5% by weight*

LCP: Lead Containing Paint => *Laboratory Detection Limit*

Chromium Containing Paint (Cr) => *Laboratory Detection Limit*

Cadmium Containing Paint (Cd) => *Laboratory Detection Limit*

LEAD, CHROMIUM, CADMIUM SAMPLE SUMMARY TABLE (BLDG. 4044)

Sample ID	Paint Color	Interior / Exterior	Substrate	Component	Photo No.	LBP *1 (Y/N) [%]	LCP *1 (Y/N) [%]	Cr (Y/N) [mg/kg]	Cd (Y/N) [mg/kg]
4044-S1	Gray	Interior	Gypsum board	Wall	M036	N	N	N	N
4044-S2	Red over white	Interior	Metal	Pipe	M037	N	N	N	N
4044-S3	Green over off white	Interior	Wood	Door	M038	N	Y [0.008%]	N	N
4044-S4	Black over red & off white	Interior	Wood	Door frame	M039	N	Y [0.008%]	N	N
4044-S5	Off white	Interior	Gypsum board	Wall	M040	N	N	N	N
4044-S6	Gray	Interior	Concrete	Wall	M041	N	N	N	N
4044-S7	Off white	Interior	Gypsum board	Ceiling	M042	N	N	N	N
4044-S8	Red over off white	Exterior	Metal	Pipe	M043	N	N	N	N
4044-S9	Light blue	Exterior	Metal	Flange	M044	N	N	N	N
4044-S10	Off white	Exterior	Concrete	Wall	M045	N	N	N	N
4044-S11	Light brown	Exterior	Concrete	Wall base	M046	N	Y [0.010%]	N	N
4044-S12	Red	Exterior	Concrete	Pipe	M047	N	N	N	N
4044-S13	Light brown	Exterior	Concrete	Wall	M048	N	N	N	N

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4044 (ESC, 20 March 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4044 (ESC, 20 March 2023) (see Appendix F)

LBP: Lead-Based Paint => *Lead Concentration equal or higher than 0.5% by weight*

LCP: Lead Containing Paint => *Laboratory Detection Limit*

Chromium Containing Paint (Cr) => *Laboratory Detection Limit*

Cadmium Containing Paint (Cd) => *Laboratory Detection Limit*

LEAD, CHROMIUM, CADMIUM SAMPLE SUMMARY TABLE (BLDG. 4045)

Sample ID	Paint Color	Interior / Exterior	Substrate	Component	Photo No.	LBP *1 (Y/N) [%]	LCP *1 (Y/N) [%]	Cr (Y/N) [mg/kg]	Cd (Y/N) [mg/kg]
4045-S1	Off white	Interior	Concrete	Wall	M049	N	Y [0.011%]	N	N
4045-S2	Off white	Interior	Gypsum board	Wall	M050	N		N	N
4045-S4	Gray / Off white	Interior	Concrete CMU	CMU wall	M051	N	Y [0.022%]	N	N
4045-S6	Off white	Interior	Gypsum	Ceiling	M052	N		N	N
4045-S7	Off white	Interior	Metal	Pipe (electrical)	M053	N		Y [5400 mg/kg]	N
4045-S8	Off white	Interior	Gypsum board	Wall	M054	N		N	N
4045-S9	Red brown	Exterior	Concrete	Wall base	M055	N		N	N
4045-S10	Off white	Exterior	Metal	Pipe (electrical)	M056	N	Y [0.053%]	Y [1400 mg/kg]	N
4045-S11	Off white	Exterior	Plastic	Drain pipe	M057	N		N	N

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4045 (ESC, 6 March 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4045 (ESC, 6 March 2023) (see Appendix F)

LBP: Lead-Based Paint => *Lead Concentration equal or higher than 0.5% by weight*

LCP: Lead Containing Paint => *Laboratory Detection Limit*

Chromium Containing Paint (Cr) => *Laboratory Detection Limit*

Cadmium Containing Paint (Cd) => *Laboratory Detection Limit*

LEAD, CHROMIUM, CADMIUM SAMPLE SUMMARY TABLE (BLDG. 4046)

Sample ID	Paint Color	Interior / Exterior	Substrate	Component	Photo No.	LBP ^{*1} (Y/N) [%]	LCP ^{*1} (Y/N) [%]	Cr (Y/N) [mg/kg]	Cd (Y/N) [mg/kg]
4046-S1	Off white	Interior	Gypsum board	Wall	M058	N	Y [0.014%]	N	N
4046-S2	Light brown	Interior	Metal	Door frame	M059	N	Y [0.017%]	N	N
4046-S3	Red	Interior	Metal	Pipe	M060	N	N	N	N
4046-S4	Off white	Interior	Metal	Pipe	M061	N	N	N	N
4046-S5	Red	Interior	Concrete	Floor	M062	N	Y [0.009%]	N	N
4046-S6	Off white	Interior	Metal	Door frame	M063	N	N	N	N
4046-S7	Yellow	Exterior	Metal	Telephone box	M064	N	N	N	N
4046-S8	Off white	Exterior	Concrete	Wall	M065	N	N	N	N
4046-S9	Red	Exterior	Metal	Pipe	M066	N	N	N	N
4046-S10	Light blue	Exterior	Metal	Flange	M067	N	N	N	N
4046-S11	Light brown	Exterior	Concrete	Wall	M068	N	N	N	N

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4046 (ESC, 9 December 2022)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4046 (ESC, 9 December 2022) (see Appendix F)

LBP: Lead-Based Paint => *Lead Concentration equal or higher than 0.5% by weight*

LCP: Lead Containing Paint => *Laboratory Detection Limit*

Chromium Containing Paint (Cr) => *Laboratory Detection Limit*

Cadmium Containing Paint (Cd) => *Laboratory Detection Limit*

LEAD, CHROMIUM, CADMIUM SAMPLE SUMMARY TABLE (BLDG. 4047)

Sample ID	Paint Color	Interior / Exterior	Substrate	Component	Photo No.	LBP ^{*1} (Y/N) [%]	LCP ^{*1} (Y/N) [%]	Cr (Y/N) [mg/kg]	Cd (Y/N) [mg/kg]
4047-S1	Off white	Int	Gypsum board	Wall	M069	N	Y [0.007%]	N	N
4047-S2	Light brown	Int	Metal	Door frame	M070	N	Y [0.21%]	N	N
4047-S3	Red	Int	Metal	Pipe	M071	N	N	N	N
4047-S4	Off white	Int	Metal	Pipe	M072	N	N	Y [110 mg/kg]	N
4047-S6	Yellow	Ext	Metal	Telephone box	M073	N	N	N	N
4047-S7	Off white	Ext	Concrete	Wall	M074	N	Y [0.007%]	N	N
4047-S8	Red	Ext	Metal	Pipe	M075	N	N	N	N
4047-S9	Light blue	Ext	Metal	Flange	M076	N	N	N	N
4047-S10	Off white	Ext	Metal	Pipe	M077	N	Y [0.042%]	Y [2100 mg/kg]	N
4047-S11	Light brown	Ext	Concrete	Wall	M078	N	N	N	N

Notes:

*1 – The results shown are intended to augment the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4047 (ESC, 6 March 2023)

*2 – Photos can be referenced in the previous FY21 Comprehensive ACM & LBP Surveys Report, building 4047 (ESC, 6 March 2023) (see Appendix F)

LBP: Lead-Based Paint => *Lead Concentration equal or higher than 0.5% by weight*

LCP: Lead Containing Paint => *Laboratory Detection Limit*

Chromium Containing Paint (Cr) => *Laboratory Detection Limit*

Cadmium Containing Paint (Cd) => *Laboratory Detection Limit*

LEAD, CHROMIUM, CADMIUM SAMPLE SUMMARY TABLE (BLDG. 4048)

Sample ID	Paint Color	Interior / Exterior	Substrate	Component	Photo No.	LBP *1 (Y/N) [%]	LCP *1 (Y/N) [%]	Cr (Y/N) [mg/kg]	Cd (Y/N) [mg/kg]
4048-S1	White	Int	Concrete	Wall	M079	-	-	N	N
4048-S2	White	Int	Metal	Conduit pipe	M080	-	-	N	N
4048-S3	White	Ext	Concrete	Wall	M081	-	-	N	N
4048-S4	Beige	Ext	Metal	Bollard	M082	-	-	N	Y [11 mg/kg]
4048-S5	White	Ext	Metal	Wall	M083	-	-	Y [410 mg/kg]	N
4048-S6	Beige	Ext	Metal	Roof	M084	-	-	Y [690 mg/kg]	N

Notes:-

LBP: Lead-Based Paint => *Lead Concentration equal or higher than 0.5% by weight*

LCP: Lead Containing Paint => *Laboratory Detection Limit*

Chromium Containing Paint (Cr) => *Laboratory Detection Limit*

Cadmium Containing Paint (Cd) => *Laboratory Detection Limit*

LEAD, CHROMIUM, CADMIUM SAMPLE SUMMARY TABLE (SWING SPACE)									
Sample ID	Paint Color	Interior / Exterior	Substrate	Component	Photo No.	LBP ^{*1} (Y/N) [%]	LCP ^{*1} (Y/N) [%]	Cr (Y/N) [mg/kg]	Cd (Y/N) [mg/kg]
SS-S1	Grey	Ext	Metal	Elec. Panel	SS-S1	N	N	N	N
SS-S2	Black & yellow	Ext	Concrete	Bollard	SS-S2	N	N	N	N
SS-S3	White	Ext	Metal	Guard rail	SS-S3	N	N	N	N

Notes:-

LBP: Lead-Based Paint => *Lead Concentration equal or higher than 0.5% by weight*

LCP: Lead Containing Paint => *Laboratory Detection Limit*

Chromium Containing Paint (Cr) => *Laboratory Detection Limit*

Cadmium Containing Paint (Cd) => *Laboratory Detection Limit*

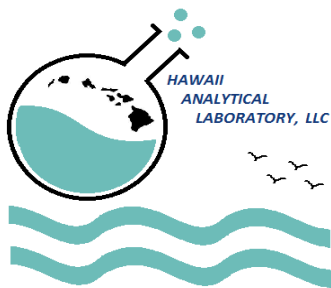
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APPENDIX B

Asbestos Laboratory Analytical Reports & Chains-of-Custody

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Hawaii Analytical Laboratory ANALYTICAL REPORT

Monday, January 8, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314103

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey, BLDG. 4040, 12/20/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202398012	4040-001		NONE DETECTED		None detected		Calcite + binder	1/5/2024
	<u>Layer</u> <u>Gray mastic</u>							
	<u>Comments</u>							
202398012	4040-001		NONE DETECTED		None detected		Calcite + vinyl	1/5/2024
	<u>Layer</u> <u>Gray vinyl floor tile</u>							
	<u>Comments</u>							
202398013	4040-002		NONE DETECTED		None detected		Calcite + vinyl	1/5/2024
	<u>Layer</u> <u>Gray vinyl floor tile</u>							
	<u>Comments</u>							
202398013	4040-002		NONE DETECTED		None detected		Calcite + binder	1/5/2024
	<u>Layer</u> <u>Tan mastic</u>							
	<u>Comments</u>							
202398014	4040-003		NONE DETECTED		Fibrous glass (amorphous) + cellulose (undulose)	60	Calcite + perlite + other	1/5/2024
	<u>Layer</u> <u>Off-white ceiling tile</u>							
	<u>Comments</u>							
202398015	4040-004		NONE DETECTED		Fibrous glass (amorphous) + cellulose (undulose)	60	Calcite + perlite + other	1/5/2024
	<u>Layer</u> <u>Off-white ceiling tile</u>							
	<u>Comments</u>							

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314103

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey, BLDG. 4040, 12/20/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202398016	4040-005		NONE DETECTED		Fibrous glass (amorphous)	40 Foil	1/5/2024
Layer	Silver wrap						
Comments							
202398016	4040-005		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	1/5/2024
Layer	Yellow insulation						
Comments							
202398017	4040-006		NONE DETECTED		Fibrous glass (amorphous)	40 Foil	1/5/2024
Layer	Silver wrap						
Comments							
202398017	4040-006		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	1/5/2024
Layer	Yellow insulation						
Comments							
202398018	4040-007		NONE DETECTED		None detected	Calcite + vinyl	1/5/2024
Layer	Gray vinyl floor tile						
Comments							
202398018	4040-007		NONE DETECTED		None detected	Calcite + binder	1/5/2024
Layer	Tan mastic						
Comments							
202398019	4040-008		NONE DETECTED		None detected	Ceramic + quartz	1/5/2024
Layer	Beige ceramic tile						
Comments							
202398019	4040-008		NONE DETECTED		None detected	Calcite + quartz	1/5/2024
Layer	Gray grout						
Comments							

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aledezmaes@oyopacific.com;

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Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202398020	4040-009		NONE DETECTED		None detected	Ceramic + quartz	1/5/2024
Layer	Beige ceramic tile						
Comments							
202398020	4040-009		NONE DETECTED		None detected	Calcite + quartz	1/5/2024
Layer	Gray cementitious material						
Comments							
202398020	4040-009		NONE DETECTED		None detected	Calcite + quartz	1/5/2024
Layer	Gray grout						
Comments							
202398021	4040-010		NONE DETECTED		None detected	Calcite + quartz	1/5/2024
Layer	Gray cementitious material						
Comments							
202398021	4040-010		NONE DETECTED		None detected	Ceramic + quartz	1/5/2024
Layer	Off-white ceramic tile						
Comments							
202398021	4040-010		NONE DETECTED		None detected	Calcite	1/5/2024
Layer	White grout						
Comments							
202398022	4040-011		NONE DETECTED		None detected	Calcite + quartz	1/5/2024
Layer	Gray cementitious material						
Comments							
202398022	4040-011		NONE DETECTED		None detected	Ceramic + quartz	1/5/2024
Layer	Off-white ceramic tile						
Comments							

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aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314103

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey, BLDG. 4040, 12/20/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202398022	4040-011		NONE DETECTED		None detected	Calcite	1/5/2024
<u>Layer</u>	<u>White grout</u>						
<u>Comments</u>							
202398023	4040-012		NONE DETECTED		Cellulose (undulose)	15 Gypsum	1/5/2024
<u>Layer</u>	<u>White gypsum board</u>						
<u>Comments</u>							
202398024	4040-013		NONE DETECTED		Cellulose (undulose)	15 Gypsum	1/5/2024
<u>Layer</u>	<u>White gypsum board</u>						
<u>Comments</u>							
202398025	4040-014		NONE DETECTED		None detected	Paint + other	1/5/2024
<u>Layer</u>	<u>Brown rust / red paint</u>						
<u>Comments</u>							
202398026	4040-015		NONE DETECTED		None detected	Paint + other	1/5/2024
<u>Layer</u>	<u>Brown rust / red paint</u>						
<u>Comments</u>							
202398027	4040-016		NONE DETECTED		None detected	Calcite + binder	1/5/2024
<u>Layer</u>	<u>White joint sealant</u>						
<u>Comments</u>							
202398028	4040-017		NONE DETECTED		None detected	Calcite + binder	1/5/2024
<u>Layer</u>	<u>Brown joint sealant</u>						
<u>Comments</u>							
202398029	4040-018		NONE DETECTED		None detected	Calcite + quartz	1/5/2024
<u>Layer</u>	<u>Gray concrete</u>						
<u>Comments</u>							

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Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314103

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey, BLDG. 4040, 12/20/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202398030	4040-019		NONE DETECTED		None detected		Calcite + quartz	1/5/2024
Layer	Gray concrete							
Comments								
202398031	4040-020		NONE DETECTED		None detected		Calcite + quartz	1/5/2024
Layer	Gray concrete							
Comments								
202398032	4040-021		NONE DETECTED		Fibrous glass (amorphous)	15	Calcite + quartz + other	1/5/2024
Layer	Black/gray roof vent sheet material							
Comments								
202398033	4040-022		NONE DETECTED		Fibrous glass (amorphous)	> 99	None detected	1/5/2024
Layer	Yellow insulation							
Comments								
202398034	4040-023		NONE DETECTED		Fibrous glass (amorphous)	85	Foil	1/5/2024
Layer	Yellow insulation / silver foil							
Comments								
202398035	4040-024		NONE DETECTED		Fibrous glass (amorphous)	> 99	None detected	1/5/2024
Layer	Yellow insulation							
Comments								

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Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314103

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey, BLDG. 4040, 12/20/23

General Comments

The bulk sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures outlined in the United States Environmental Protection Agency's "Interim Method for the Determination of Asbestos in Bulk Insulation Samples" (EPA-600/M4-82-020, Dec. 1982) and / or "Method for Determination of Asbestos in bulk Building Materials" (EPA-600/R-93-116, July 1993). The analysis of each bulk sample relates only to the material examined, and may or may not represent the overall composition of its original source. Floor tile and other resinously bound materials, when analyzed by the EPA methods referenced above may yield false negative results because of limitations in separating closely bound fibers and in detecting fibers of small length and diameter. Alternative methods of identification, including Transmission Electron Microscopy (TEM) may or may not be applicable. We utilize calibrated visual area estimation on a routine basis and do not conduct point counting unless specifically requested to do so. Estimated error for the visual determinations presented are 75% relative (1 to 2%), 50% relative (3 to 5%); 25% relative (6 to 25%) and 20% (>26% v/v). We will not separate layers which in our opinion are not readily discernable. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report must not be used by the client to claim product certification, approval or endorsement by NVLAP, NIST or any agency of Federal Government. Unless otherwise indicated, the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

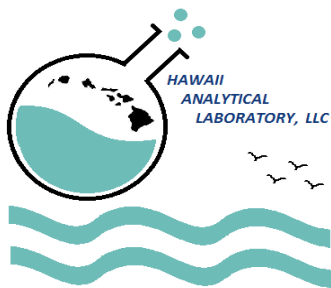
< This testing result is less than the numerical value listed.

None Detected = asbestos was not observed in the sample. If trace amount of asbestos was detected below our quantifiable limits of 1.0%, <1% (trace) would be indicated and the asbestos type listed. Point counting, where applicable, are recommended to improve accuracy.



Eva Skogsberg
Laboratory Supervisor

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Hawaii Analytical Laboratory ANALYTICAL REPORT

Monday, January 8, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314094

Date Submitted: 12/29/2023

Your Project: 232061, PN7833 Kadena VMS Hazmat Survey Bldg. 4041, 12/8/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202397901	4041-003	NONE DETECTED			None detected		Calcite + binder	1/4/2024
Layer	Gray sealant							
Comments								
202397901	4041-003	NONE DETECTED			Cellulose (undulose) + fibrous glass (amorphous)	15	Paint + calcite	1/4/2024
Layer	White textured paint							
Comments								
202397902	4041-004	NONE DETECTED			None detected		Calcite + binder	1/4/2024
Layer	White sealant							
Comments								
202397902	4041-004	NONE DETECTED			Cellulose (undulose) + fibrous glass (amorphous)	15	Paint + calcite	1/4/2024
Layer	White textured paint							
Comments								
202397903	4041-005	NONE DETECTED			None detected		Calcite + binder	1/4/2024
Layer	White sealant							
Comments								
202397903	4041-005	NONE DETECTED			Cellulose (undulose) + fibrous glass (amorphous)	15	Paint + calcite	1/4/2024
Layer	White textured paint							
Comments								

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Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314094

Date Submitted: 12/29/2023

Your Project: 232061, PN7833 Kadena VMS Hazmat Survey Bldg. 4041, 12/8/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202397904	4041-006		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum + paint	1/4/2024
<u>Layer</u>	<u>White gypsum board / gray paint</u>							
Comments								
202397905	4041-007		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum + paint	1/4/2024
<u>Layer</u>	<u>White gypsum board / gray paint</u>							
Comments								
202397906	4041-008		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	65	Calcite + perlite + glass beads	1/4/2024
<u>Layer</u>	<u>Off-white acoustic ceiling tile</u>							
Comments								
202397907	4041-009		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	65	Calcite + perlite + glass beads	1/4/2024
<u>Layer</u>	<u>Off-white acoustic ceiling tile</u>							
Comments								
202397908	4041-010		NONE DETECTED		Fibrous glass (amorphous)	40	Foil	1/4/2024
<u>Layer</u>	<u>Silver wrap</u>							
Comments								
202397908	4041-010		NONE DETECTED		Fibrous glass (amorphous)	> 99	None detected	1/4/2024
<u>Layer</u>	<u>Yellow insulation</u>							
Comments								
202397909	4041-011		NONE DETECTED		Fibrous glass (amorphous)	40	Foil + other	1/4/2024
<u>Layer</u>	<u>Silver wrap</u>							
Comments								
202397909	4041-011		NONE DETECTED		Fibrous glass (amorphous)	> 99	None detected	1/4/2024
<u>Layer</u>	<u>Yellow insulation</u>							
Comments								

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Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314094

Date Submitted: 12/29/2023

Your Project: 232061, PN7833 Kadena VMS Hazmat Survey Bldg. 4041, 12/8/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202397910	4041-012		NONE DETECTED		Cellulose (undulose)	2	Calcite + paint	1/4/2024
Layer	Off-white joint compound / beige paint							
Comments								
202397910	4041-012		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum	1/4/2024
Layer	White gypsum board							
Comments								
202397911	4041-013		NONE DETECTED		Cellulose (undulose)	2	Calcite + paint	1/4/2024
Layer	Off-white joint compound / white paint							
Comments								
202397911	4041-013		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum	1/4/2024
Layer	White gypsum board							
Comments								
202397912	4041-014		NONE DETECTED		Cellulose (undulose)	2	Calcite + paint	1/4/2024
Layer	Off-white joint compound / white paint							
Comments								
202397912	4041-014		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum	1/4/2024
Layer	White gypsum board							
Comments								
202397913	4041-015		NONE DETECTED		None detected		Ceramic	1/4/2024
Layer	Beige ceramic tile							
Comments								

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aledezmaes@oyopacific.com;

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Date Submitted: 12/29/2023

Your Project: 232061, PN7833 Kadena VMS Hazmat Survey Bldg. 4041, 12/8/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202397913	4041-015		NONE DETECTED		None detected		Calcite + quartz	1/4/2024
Layer	Gray grout							
Comments								
202397914	4041-016		NONE DETECTED		None detected		Ceramic	1/4/2024
Layer	Beige ceramic tile							
Comments								
202397914	4041-016		NONE DETECTED		None detected		Calcite + quartz	1/4/2024
Layer	Gray grout							
Comments								
202397915	4041-017		NONE DETECTED		None detected		Ceramic	1/4/2024
Layer	Off-white ceramic tile							
Comments								
202397916	4041-018		NONE DETECTED		None detected		Ceramic	1/4/2024
Layer	Off-white ceramic tile							
Comments								
202397916	4041-018		NONE DETECTED		None detected		Calcite + quartz	1/4/2024
Layer	White grout (limited)							
Comments								
202397917	4041-020		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	40	Foil + other	1/4/2024
Layer	Silver wrap							
Comments								
202397917	4041-020		NONE DETECTED		Fibrous glass (amorphous)	> 99	None detected	1/4/2024
Layer	Yellow insulation							
Comments								

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Date Submitted: 12/29/2023

Your Project: 232061, PN7833 Kadena VMS Hazmat Survey Bldg. 4041, 12/8/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202397918	4041-021		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	40	Foil	1/5/2024
Layer	Silver wrap							
Comments								
202397918	4041-021		NONE DETECTED		Fibrous glass (amorphous)	> 99	None detected	1/5/2024
Layer	Yellow insulation							
Comments								
202397919	4041-024		NONE DETECTED		None detected		Calcite + quartz	1/5/2024
Layer	Gray roofing material							
Comments								
202397920	4041-025		NONE DETECTED		None detected		Calcite + quartz	1/5/2024
Layer	Gray roofing material							
Comments								
202397921	4041-026		NONE DETECTED		None detected		Calcite + quartz	1/5/2024
Layer	Gray roofing material							
Comments								
202397922	4041-027		NONE DETECTED		None detected		Calcite + quartz	1/5/2024
Layer	Gray concrete							
Comments								
202397922	4041-027		NONE DETECTED		Fibrous glass (amorphous)	20	Calcite + quartz + binder + other	1/5/2024
Layer	Gray roof vent sheeting material							
Comments								

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Okinawa Japan .

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Lab Job No: 202314094

Date Submitted: 12/29/2023

Your Project: 232061, PN7833 Kadena VMS Hazmat Survey Bldg. 4041, 12/8/23

General Comments

The bulk sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures outlined in the United States Environmental Protection Agency's "Interim Method for the Determination of Asbestos in Bulk Insulation Samples" (EPA-600/M4-82-020, Dec. 1982) and / or "Method for Determination of Asbestos in bulk Building Materials" (EPA-600/R-93-116, July 1993). The analysis of each bulk sample relates only to the material examined, and may or may not represent the overall composition of its original source. Floor tile and other resinously bound materials, when analyzed by the EPA methods referenced above may yield false negative results because of limitations in separating closely bound fibers and in detecting fibers of small length and diameter. Alternative methods of identification, including Transmission Electron Microscopy (TEM) may or may not be applicable. We utilize calibrated visual area estimation on a routine basis and do not conduct point counting unless specifically requested to do so. Estimated error for the visual determinations presented are 75% relative (1 to 2%), 50% relative (3 to 5%); 25% relative (6 to 25%) and 20% (>26% v/v). We will not separate layers which in our opinion are not readily discernable. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report must not be used by the client to claim product certification, approval or endorsement by NVLAP, NIST or any agency of Federal Government. Unless otherwise indicated, the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

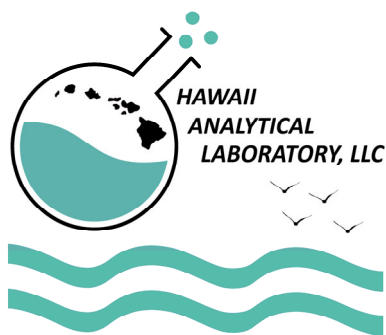
< This testing result is less than the numerical value listed.

None Detected = asbestos was not observed in the sample. If trace amount of asbestos was detected below our quantifiable limits of 1.0%, <1% (trace) would be indicated and the asbestos type listed. Point counting, where applicable, are recommended to improve accuracy.



Eva Skogsberg
Laboratory Supervisor

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Hawaii Analytical Laboratory

3615 Harding Avenue, Suite 308, Honolulu, Hawaii, 96816

Tel: (808) 735-0422 – Fax: (808) 735-0047

February 5, 2024

OYO Corporation, Pacific
Okinawa Times Building. 2F
1-12-24 Goya, Okinawa-shi,
Okinawa-ken, 904-0021 Japan

Project Name: 232061 PN77833 Kadena VMS Survey BLDG. 4042
Date collected: N/A
Date received: 1/26/2024
HAL #: 202401038

To Whom it May Concern,

Enclosed are the analytical results for the samples received by our laboratory on January 26, 2024. The samples on the chain of custody were received in good condition unless otherwise noted.

The twenty-seven (27) bulk samples submitted for PLM analysis were subcontracted to Eurofins EMLab P&K in Tustin, CA (a NVLAP accredited Laboratory, NVLAP LAB CODE: 200757-0.). Its report is enclosed in its entirety.

Results in this report are based on the sampling data provided by the client and refer only to the samples as they were received at the laboratory. Unless otherwise requested, all samples will be discarded 14 days from the date of this report.

Quality assurance data is collected to accompany all analyses and to ensure that results generated meet Hawaii Analytical Laboratory's quality standards. This data is available upon request.

Please contact us at (808) 735-0422 if you have questions.

Thank you for using Hawaii Analytical Laboratory and have a great day!

Anne Antin
Quality Manager
Hawaii Analytical Laboratory
3615 Harding Ave. Ste. 308
Honolulu, HI 96816
Phone: (808) 735-0422
E-mail: aantin@analyzehawaii.com

Report for:

Anne Antin
Hawaii Analytical Lab
3615 Harding Ave
Suite 308
Honolulu, HI 96816

Regarding: Eurofins EPK Built Environment Testing, LLC
Project: 202401038; 232061 PN77833 Kadena VMS Hazmat Survey BLDG, 4042
EML ID: 3524110

Approved by:



Approved Signatory
Danny Li

Dates of Analysis:
Asbestos PLM: 02-05-2024

Service SOPs: Asbestos PLM (EPA 40CFR App E to Sub E of Part 763 & EPA METHOD 600/R-93-116, SOP EM-AS-S-1267)
NVLAP Lab Code 200757-0

All samples were received in acceptable condition unless noted in the Report Comments portion in the body of the report. The results relate only to the samples as received and tested. The results include an inherent uncertainty of measurement associated with estimating percentages by polarized light microscopy. Measurement uncertainty data for sample results with >1% asbestos concentration can be provided when requested.

Eurofins EPK Built Environment Testing, LLC ("the Company"), a member of the Eurofins Built Environment Testing group of companies, shall have no liability to the client or the client's customer with respect to decisions or recommendations made, actions taken or courses of conduct implemented by either the client or the client's customer as a result of or based upon the Test Results. In no event shall the Company be liable to the client with respect to the Test Results except for the Company's own willful misconduct or gross negligence nor shall the Company be liable for incidental or consequential damages or lost profits or revenues to the fullest extent such liability may be disclaimed by law, even if the Company has been advised of the possibility of such damages, lost profits or lost revenues. In no event shall the Company's liability with respect to the Test Results exceed the amount paid to the Company by the client therefor.

Eurofins EPK Built Environment Testing, LLC

2841 Dow Avenue, Suite 300, Tustin, CA 92780

(800) 651-4802 www.eurofinsus.com/Built

Client: Hawaii Analytical Lab

C/O: Anne Antin

Re: 202401038; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG, 4042

Date of Submittal: 01-30-2024

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Total Samples Submitted:** 27**Total Samples Analyzed:** 27**Total Samples with Layer Asbestos Content > 1%:** 0**Location: 4042-001, H1-Flange gasket, black**

Lab ID-Version‡: 17216413-1

Sample Layers	Asbestos Content
Black Gasket	ND
Sample Composite Homogeneity: Good	

Location: 4042-002, H1-Flange gasket, black

Lab ID-Version‡: 17216414-1

Sample Layers	Asbestos Content
Black Gasket	ND
Sample Composite Homogeneity: Good	

Location: 4042-003, H3-Wall joint sealant, light grey mastic

Lab ID-Version‡: 17216415-1

Sample Layers	Asbestos Content
Gray Sealant	ND
Light Gray Mastic	ND
Sample Composite Homogeneity: Moderate	

Location: 4042-004, H4-Window point sealant, dark brown mastic

Lab ID-Version‡: 17216416-1

Sample Layers	Asbestos Content
Brown Sealant	ND
Sample Composite Homogeneity: Good	

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C/O: Anne Antin

Re: 202401038; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG, 4042

Date of Submittal: 01-30-2024

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Location: 4042-005, H4-Window point sealant, dark brown mastic**

Lab ID-Version‡: 17216417-1

Sample Layers	Asbestos Content
Brown Sealant	ND
Sample Composite Homogeneity:	Good

Location: 4042-006, H8-Ceiling tile, 60x60 offwhite, hole pattern

Lab ID-Version‡: 17216418-1

Sample Layers	Asbestos Content
Gray Ceiling Tile	ND
Composite Non-Asbestos Content:	40% Cellulose 40% Glass Fibers
Sample Composite Homogeneity:	Good

Location: 4042-007, H8-Ceiling tile, 60x60 offwhite, hole pattern

Lab ID-Version‡: 17216419-1

Sample Layers	Asbestos Content
Gray Ceiling Tile	ND
Composite Non-Asbestos Content:	40% Cellulose 40% Glass Fibers
Sample Composite Homogeneity:	Good

Location: 4042-008, H9-Duct D1, yellow fiberglass with silver backing

Lab ID-Version‡: 17216420-1

Sample Layers	Asbestos Content
Yellow Insulation	ND
Silver Foil	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Moderate

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Client: Hawaii Analytical Lab

C/O: Anne Antin

Re: 202401038; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG, 4042

Date of Submittal: 01-30-2024

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Location: 4042-009, H9-Duct D1, yellow fiberglass with silver backing**

Lab ID-Version‡: 17216421-1

Sample Layers	Asbestos Content
Yellow Insulation	ND
Silver Foil	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Moderate

Location: 4042-010, H10-Pipe TSI (elbow), yellow fiberglass with silver backing

Lab ID-Version‡: 17216422-1

Sample Layers	Asbestos Content
Yellow Insulation	ND
Silver Foil	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Moderate

Location: 4042-011, H10-Pipe TSI (elbow), yellow fiberglass with silver backing

Lab ID-Version‡: 17216423-1

Sample Layers	Asbestos Content
Yellow Insulation	ND
Silver Foil	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Moderate

Location: 4042-012, H11-Pipe TSI (run), yellow fiberglass with silver backing

Lab ID-Version‡: 17216424-1

Sample Layers	Asbestos Content
Yellow Insulation	ND
Silver Foil	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Moderate

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Client: Hawaii Analytical Lab

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Re: 202401038; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG, 4042

Date of Submittal: 01-30-2024

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Location: 4042-013, H11-Pipe TSI (run), yellow fiberglass with silver backing**

Lab ID-Version‡: 17216425-1

Sample Layers	Asbestos Content
Yellow Insulation	ND
Silver Foil	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Moderate

Location: 4042-014, H12-Concrete floor coating, gray

Lab ID-Version‡: 17216426-1

Sample Layers	Asbestos Content
Gray Coating	ND
Sample Composite Homogeneity:	Good

Location: 4042-015, H12-Concrete floor coating, gray

Lab ID-Version‡: 17216427-1

Sample Layers	Asbestos Content
Gray Coating	ND
Sample Composite Homogeneity:	Good

Location: 4042-016, H13-Concrete wall coating, off-white, smooth

Lab ID-Version‡: 17216428-1

Sample Layers	Asbestos Content
Off-White Coating	ND
Sample Composite Homogeneity:	Good

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Client: Hawaii Analytical Lab
C/O: Anne Antin
Re: 202401038; 232061 PN77833 Kadena VMS
Hazmat Survey BLDG, 4042

Date of Submittal: 01-30-2024
Date of Receipt: 01-31-2024
Date of Report: 02-05-2024

ASBESTOS PLM REPORT

Location: 4042-017, H13-Concrete wall coating, off-white, smooth

Lab ID-Version‡: 17216429-1

Sample Layers	Asbestos Content
Off-White Coating	ND
Sample Composite Homogeneity: Good	

Location: 4042-018, H14-Ceramic wall tile, 10x10 off-white tile with white grout

Lab ID-Version‡: 17216430-1

Sample Layers	Asbestos Content
Off-White Ceramic Tile	ND
White Grout	ND
Sample Composite Homogeneity: Moderate	

Location: 4042-019, H14-Ceramic wall tile, 10x10 off-white tile with white grout

Lab ID-Version‡: 17216431-1

Sample Layers	Asbestos Content
Off-White Ceramic Tile	ND
White Grout	ND
Sample Composite Homogeneity: Moderate	

Location: 4042-020, H15-Ceramic floor tile, 20x20 beige tile with gray grout

Lab ID-Version‡: 17216432-1

Sample Layers	Asbestos Content
Beige Ceramic Tile	ND
Gray Grout	ND
Sample Composite Homogeneity: Moderate	

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Client: Hawaii Analytical Lab
C/O: Anne Antin
Re: 202401038; 232061 PN77833 Kadena VMS
Hazmat Survey BLDG, 4042

Date of Submittal: 01-30-2024
Date of Receipt: 01-31-2024
Date of Report: 02-05-2024

ASBESTOS PLM REPORT

Location: 4042-021, H15-Ceramic floor tile, 20x20 beige tile with gray grout

Lab ID-Version‡: 17216433-1

Sample Layers	Asbestos Content
Beige Ceramic Tile	ND
Gray Grout	ND
Sample Composite Homogeneity:	Moderate

Location: 4042-022, H17-Gypson ceiling board, off-white smooth

Lab ID-Version‡: 17216434-1

Sample Layers	Asbestos Content
White Drywall with Brown Paper	ND
Composite Non-Asbestos Content:	2% Cellulose
Sample Composite Homogeneity:	Good

Location: 4042-023, H17-Gypson ceiling board, off-white smooth

Lab ID-Version‡: 17216435-1

Sample Layers	Asbestos Content
Off-White Joint Compound with Paint	ND
White Drywall with Brown Paper	ND
Composite Non-Asbestos Content:	2% Cellulose
Sample Composite Homogeneity:	Good

Location: 4042-026, Roofing material (concrete) (N)

Lab ID-Version‡: 17216438-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity:	Good

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Client: Hawaii Analytical Lab

C/O: Anne Antin

Re: 202401038; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG, 4042

Date of Submittal: 01-30-2024

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Location: 4042-027, Roofing material (concrete) (N)**

Lab ID-Version‡: 17216439-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity: Good	

Location: 4042-028, Roofing material (concrete) (N)

Lab ID-Version‡: 17216440-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity: Good	

Location: 4042-029, Roof vent sheet (V2)

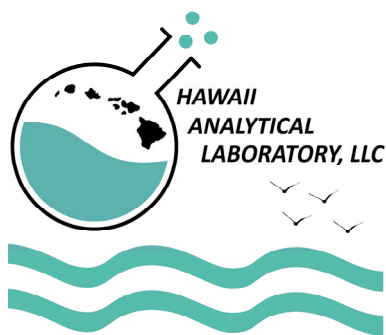
Lab ID-Version‡: 17216441-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity: Good	

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Hawaii Analytical Laboratory

3615 Harding Avenue, Suite 308, Honolulu, Hawaii, 96816

Tel: (808) 735-0422 – Fax: (808) 735-0047

February 6, 2024

OYO Corporation, Pacific
Okinawa Times Building. 2F
1-12-24 Goya, Okinawa-shi,
Okinawa-ken, 904-0021 Japan

Project Name: 232061 PN77833 Kadena VMS Survey BLDG. 4043
Date collected: 12/21/2023
Date received: 1/26/2024
HAL #: 202401039

To Whom it May Concern,

Enclosed are the analytical results for the samples received by our laboratory on January 26, 2024. The samples on the chain of custody were received in good condition unless otherwise noted.

The twenty-two (22) bulk samples submitted for PLM analysis were subcontracted to Eurofins EMLab P&K in Tustin, CA (a NVLAP accredited Laboratory, NVLAP LAB CODE: 200757-0.). Its report is enclosed in its entirety.

Results in this report are based on the sampling data provided by the client and refer only to the samples as they were received at the laboratory. Unless otherwise requested, all samples will be discarded 14 days from the date of this report.

Quality assurance data is collected to accompany all analyses and to ensure that results generated meet Hawaii Analytical Laboratory's quality standards. This data is available upon request.

Please contact us at (808) 735-0422 if you have questions.

Thank you for using Hawaii Analytical Laboratory and have a great day!

Anne Antin
Quality Manager
Hawaii Analytical Laboratory
3615 Harding Ave. Ste. 308
Honolulu, HI 96816
Phone: (808) 735-0422
E-mail: aantin@analyzehawaii.com

Report for:

Anne Antin
Hawaii Analytical Lab
3615 Harding Ave
Suite 308
Honolulu, HI 96816

Regarding: Eurofins EPK Built Environment Testing, LLC
Project: 202401039; 232061 PN77833 Kadena VMS Hazmat Survey BLDG. 4043
EML ID: 3524142

Approved by:



Approved Signatory
Danny Li

Dates of Analysis:
Asbestos PLM: 02-05-2024

Service SOPs: Asbestos PLM (EPA 40CFR App E to Sub E of Part 763 & EPA METHOD 600/R-93-116, SOP EM-AS-S-1267)
NVLAP Lab Code 200757-0

All samples were received in acceptable condition unless noted in the Report Comments portion in the body of the report. The results relate only to the samples as received and tested. The results include an inherent uncertainty of measurement associated with estimating percentages by polarized light microscopy. Measurement uncertainty data for sample results with >1% asbestos concentration can be provided when requested.

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(800) 651-4802 www.eurofinsus.com/Built

Client: Hawaii Analytical Lab

C/O: Anne Antin

Re: 202401039; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG. 4043

Date of Sampling: 12-21-2023

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Total Samples Submitted:** 22**Total Samples Analyzed:** 22**Total Samples with Layer Asbestos Content > 1%:** 0**Location: 4043-001, Center of the first room from the entrance**

Lab ID-Version‡: 17216951-1

Sample Layers	Asbestos Content
Gray Ceiling Tile with White Surface	ND
Composite Non-Asbestos Content:	40% Glass Fibers 20% Cellulose
Sample Composite Homogeneity:	Good

Location: 4043-002, East of main colidor, in front of the room from the entrance

Lab ID-Version‡: 17216952-1

Sample Layers	Asbestos Content
Gray Ceiling Tile with White Surface	ND
Composite Non-Asbestos Content:	40% Glass Fibers 20% Cellulose
Sample Composite Homogeneity:	Good

Location: 4043-003, East of main colidor, in from of the first room door from the entrance

Lab ID-Version‡: 17216953-1

Sample Layers	Asbestos Content
Yellow Insulation	ND
Silver Foil	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Good

Location: 4043-004, the entrance of the first room on the east from the building entrance

Lab ID-Version‡: 17216954-1

Sample Layers	Asbestos Content
Yellow Insulation	ND
Silver Foil	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Good

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Client: Hawaii Analytical Lab
C/O: Anne Antin
Re: 202401039; 232061 PN77833 Kadena VMS
Hazmat Survey BLDG. 4043

Date of Sampling: 12-21-2023
Date of Receipt: 01-31-2024
Date of Report: 02-05-2024

ASBESTOS PLM REPORT

Location: 4043-005, Men's room north side

Lab ID-Version‡: 17216955-1

Sample Layers	Asbestos Content
Off-White Drywall with Brown/Green Paper	ND
White Texture with Paint	ND
Composite Non-Asbestos Content:	5% Cellulose
Sample Composite Homogeneity:	Good

Location: 4043-006, Women's room north east corner

Lab ID-Version‡: 17216956-1

Sample Layers	Asbestos Content
Off-White Drywall with Brown/Green Paper	ND
White Texture with Paint	ND
Composite Non-Asbestos Content:	5% Cellulose
Sample Composite Homogeneity:	Good

Location: 4043-007, South east corner of men's room

Lab ID-Version‡: 17216957-1

Sample Layers	Asbestos Content
Gray Mortar	ND
White Grout	ND
White Ceramic Tile	ND
Sample Composite Homogeneity:	Poor

Location: 4043-008, North east corner (behind door) of women's room

Lab ID-Version‡: 17216958-1

Sample Layers	Asbestos Content
Gray Mortar	ND
White Grout	ND
White Ceramic Tile	ND
Sample Composite Homogeneity:	Poor

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Re: 202401039; 232061 PN77833 Kadena VMS
Hazmat Survey BLDG. 4043

Date of Sampling: 12-21-2023
Date of Receipt: 01-31-2024
Date of Report: 02-05-2024

ASBESTOS PLM REPORT

Location: 4043-009, South east corner of men's room

Lab ID-Version‡: 17216959-1

Sample Layers	Asbestos Content
Gray Grout	ND
Beige Ceramic Tile	ND
Sample Composite Homogeneity:	Moderate

Location: 4043-010, North east corner (behind door) of women's room

Lab ID-Version‡: 17216960-1

Sample Layers	Asbestos Content
Gray Grout	ND
Beige Ceramic Tile	ND
Sample Composite Homogeneity:	Moderate

Location: 4043-011, Center of the first room on east from the entrance

Lab ID-Version‡: 17216961-1

Sample Layers	Asbestos Content
Yellow Insulation	ND
Silver Foil	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Good

Location: 4043-012, East of back colidor

Lab ID-Version‡: 17216962-1

Sample Layers	Asbestos Content
Yellow Insulation	ND
Silver Foil	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Good

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Client: Hawaii Analytical Lab

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Re: 202401039; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG. 4043

Date of Sampling: 12-21-2023

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Location: 4043-013, Center of the first room on east from the entrance**

Lab ID-Version‡: 17216963-1

Sample Layers	Asbestos Content
Yellow Insulation	ND
Silver Foil	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Good

Location: 4043-014, East of back colidor

Lab ID-Version‡: 17216964-1

Sample Layers	Asbestos Content
Yellow Insulation	ND
Silver Foil	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Good

Location: 4043-015, Fire oppression valve gasket in from of the building

Lab ID-Version‡: 17216965-1

Sample Layers	Asbestos Content
Black Gasket with Paint	ND
Sample Composite Homogeneity:	Good

Location: 4043-016, Fire oppression valve gasket in from of the building

Lab ID-Version‡: 17216966-1

Sample Layers	Asbestos Content
Black Gasket with Paint	ND
Sample Composite Homogeneity:	Good

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Client: Hawaii Analytical Lab

C/O: Anne Antin

Re: 202401039; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG. 4043

Date of Sampling: 12-21-2023

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Location: 4043-017, South west window of the building**

Lab ID-Version‡: 17216967-1

Sample Layers	Asbestos Content
Brown Sealant	ND
Sample Composite Homogeneity: Good	

Location: 4043-018, South east window of the building

Lab ID-Version‡: 17216968-1

Sample Layers	Asbestos Content
Brown Sealant	ND
Sample Composite Homogeneity: Good	

Location: 4043-019, West side roof

Lab ID-Version‡: 17216969-1

Sample Layers	Asbestos Content
Gray Cementitious Material	ND
Sample Composite Homogeneity: Good	

Location: 4043-020, East side roof

Lab ID-Version‡: 17216970-1

Sample Layers	Asbestos Content
Gray Cementitious Material	ND
Sample Composite Homogeneity: Good	

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Inhomogeneous samples are separated into homogeneous subsamples and analyzed individually. ND means no fibers were detected. When detected, the minimum detection and reporting limit is less than 1% unless point counting is performed. Floor tile samples may contain large amounts of interference material and it is recommended that the sample be analyzed by gravimetric point count analysis to lower the detection limit and to aid in asbestos identification.

‡ A "Version" indicated by "-x" after the Lab ID# with a value greater than 1 indicates a sample with amended data. The revision number is reflected by the value of "x".

Eurofins EPK Built Environment Testing, LLC

2841 Dow Avenue, Suite 300, Tustin, CA 92780

(800) 651-4802 www.eurofinsus.com/Built

Client: Hawaii Analytical Lab

C/O: Anne Antin

Re: 202401039; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG. 4043

Date of Sampling: 12-21-2023

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Location: 4043-021, South side roof**

Lab ID-Version‡: 17216971-1

Sample Layers	Asbestos Content
Gray Cementitious Material	ND
Sample Composite Homogeneity: Good	

Location: 4043-022, North side roof

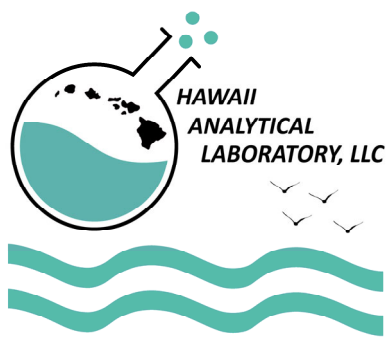
Lab ID-Version‡: 17216972-1

Sample Layers	Asbestos Content
Gray Semi-Fibrous Material	ND
Gray Debris	ND
Composite Non-Asbestos Content:	30% Glass Fibers
Sample Composite Homogeneity: Good	

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Inhomogeneous samples are separated into homogeneous subsamples and analyzed individually. ND means no fibers were detected. When detected, the minimum detection and reporting limit is less than 1% unless point counting is performed. Floor tile samples may contain large amounts of interference material and it is recommended that the sample be analyzed by gravimetric point count analysis to lower the detection limit and to aid in asbestos identification.

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Hawaii Analytical Laboratory

3615 Harding Avenue, Suite 308, Honolulu, Hawaii, 96816

Tel: (808) 735-0422 – Fax: (808) 735-0047

February 5, 2024

OYO Corporation, Pacific
Okinawa Times Building. 2F
1-12-24 Goya, Okinawa-shi,
Okinawa-ken, 904-0021 Japan

Project Name: 232061 PN77833 Kadena VMS Survey BLDG. 4044
Date collected: 12/21/2023
Date received: 1/26/2024
HAL #: 202401040

To Whom it May Concern,

Enclosed are the analytical results for the samples received by our laboratory on January 26, 2024. The samples on the chain of custody were received in good condition unless otherwise noted.

The twenty-eight (28) bulk samples submitted for PLM analysis were subcontracted to Eurofins EMLab P&K in Tustin, CA (a NVLAP accredited Laboratory, NVLAP LAB CODE: 200757-0.). Its report is enclosed in its entirety.

Results in this report are based on the sampling data provided by the client and refer only to the samples as they were received at the laboratory. Unless otherwise requested, all samples will be discarded 14 days from the date of this report.

Quality assurance data is collected to accompany all analyses and to ensure that results generated meet Hawaii Analytical Laboratory's quality standards. This data is available upon request.

Please contact us at (808) 735-0422 if you have questions.

Thank you for using Hawaii Analytical Laboratory and have a great day!

Anne Antin
Quality Manager
Hawaii Analytical Laboratory
3615 Harding Ave. Ste. 308
Honolulu, HI 96816
Phone: (808) 735-0422
E-mail: aantin@analyzehawaii.com

Report for:

Anne Antin
Hawaii Analytical Lab
3615 Harding Ave
Suite 308
Honolulu, HI 96816

Regarding: Eurofins EPK Built Environment Testing, LLC
Project: 202401040; 232061 PN77833 Kadena VMS Hazmat Survey BLDG. 4044
EML ID: 3524151

Approved by:



Approved Signatory
Danny Li

Dates of Analysis:
Asbestos PLM: 02-05-2024

Service SOPs: Asbestos PLM (EPA 40CFR App E to Sub E of Part 763 & EPA METHOD 600/R-93-116, SOP EM-AS-S-1267)
NVLAP Lab Code 200757-0

All samples were received in acceptable condition unless noted in the Report Comments portion in the body of the report. The results relate only to the samples as received and tested. The results include an inherent uncertainty of measurement associated with estimating percentages by polarized light microscopy. Measurement uncertainty data for sample results with >1% asbestos concentration can be provided when requested.

Eurofins EPK Built Environment Testing, LLC ("the Company"), a member of the Eurofins Built Environment Testing group of companies, shall have no liability to the client or the client's customer with respect to decisions or recommendations made, actions taken or courses of conduct implemented by either the client or the client's customer as a result of or based upon the Test Results. In no event shall the Company be liable to the client with respect to the Test Results except for the Company's own willful misconduct or gross negligence nor shall the Company be liable for incidental or consequential damages or lost profits or revenues to the fullest extent such liability may be disclaimed by law, even if the Company has been advised of the possibility of such damages, lost profits or lost revenues. In no event shall the Company's liability with respect to the Test Results exceed the amount paid to the Company by the client therefor.

Eurofins EPK Built Environment Testing, LLC

2841 Dow Avenue, Suite 300, Tustin, CA 92780

(800) 651-4802 www.eurofinsus.com/Built

Client: Hawaii Analytical Lab

C/O: Anne Antin

Re: 202401040; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG. 4044

Date of Sampling: 12-21-2023

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Total Samples Submitted:** 28**Total Samples Analyzed:** 28**Total Samples with Layer Asbestos Content > 1%:** 0**Location: 4044-001, East wall of entrance colidore by electrical pannel**

Lab ID-Version‡: 17217090-1

Sample Layers	Asbestos Content
White Joint Compound	ND
White Drywall with Brown Paper	ND
Composite Non-Asbestos Content:	5% Cellulose
Sample Composite Homogeneity:	Moderate

Location: 4044-002, East side of entrance colidor in front of the room door of the first room from the entrance (above ceiling)

Lab ID-Version‡: 17217091-1

Sample Layers	Asbestos Content
Silver Foil	ND
Yellow Insulation	ND
Composite Non-Asbestos Content:	75% Glass Fibers
Sample Composite Homogeneity:	Moderate

Location: 4044-003, Center of entrance near the first room from the entrance (above ceiling)

Lab ID-Version‡: 17217092-1

Sample Layers	Asbestos Content
Silver Foil	ND
Yellow Insulation	ND
Composite Non-Asbestos Content:	75% Glass Fibers
Sample Composite Homogeneity:	Moderate

Location: 4044-004, By the door of 2nd room from the entrance on the east

Lab ID-Version‡: 17217093-1

Sample Layers	Asbestos Content
White Ceiling Tile	ND
Composite Non-Asbestos Content:	65% Cellulose 25% Glass Fibers
Sample Composite Homogeneity:	Good

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Inhomogeneous samples are separated into homogeneous subsamples and analyzed individually. ND means no fibers were detected. When detected, the minimum detection and reporting limit is less than 1% unless point counting is performed. Floor tile samples may contain large amounts of interference material and it is recommended that the sample be analyzed by gravimetric point count analysis to lower the detection limit and to aid in asbestos identification.

‡ A "Version" indicated by "-x" after the Lab ID# with a value greater than 1 indicates a sample with amended data. The revision number is reflected by the value of "x".

Client: Hawaii Analytical Lab
C/O: Anne Antin
Re: 202401040; 232061 PN77833 Kadena VMS
Hazmat Survey BLDG. 4044

Date of Sampling: 12-21-2023
Date of Receipt: 01-31-2024
Date of Report: 02-05-2024

ASBESTOS PLM REPORT

Location: 4044-005, By the door of 2nd room from the entrance on the west

Lab ID-Version‡: 17217094-1

Sample Layers	Asbestos Content
White Ceiling Tile	ND
Composite Non-Asbestos Content:	65% Cellulose 25% Glass Fibers
Sample Composite Homogeneity:	Good

Location: 4044-006, H7-Concrete wall coating, off-white, smooth

Lab ID-Version‡: 17217095-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity:	Good

Location: 4044-007, By the north side of the door frame of the storage

Lab ID-Version‡: 17217096-1

Sample Layers	Asbestos Content
White Paint	ND
White Drywall with Brown Paper	ND
Composite Non-Asbestos Content:	5% Cellulose
Sample Composite Homogeneity:	Good

Location: 4044-008, The south wall of storage by the door

Lab ID-Version‡: 17217097-1

Sample Layers	Asbestos Content
White Paint	ND
White Drywall with Brown Paper	ND
Composite Non-Asbestos Content:	5% Cellulose
Sample Composite Homogeneity:	Good

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Eurofins EPK Built Environment Testing, LLC

2841 Dow Avenue, Suite 300, Tustin, CA 92780

(800) 651-4802 www.eurofinsus.com/Built

Client: Hawaii Analytical Lab

C/O: Anne Antin

Re: 202401040; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG. 4044

Date of Sampling: 12-21-2023

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Location: 4044-009, Men's room door joint on the colidor side**

Lab ID-Version‡: 17217098-1

Sample Layers	Asbestos Content
White Non-Fibrous Material Joint	ND
Sample Composite Homogeneity: Good	

Location: 4044-010, Storage room door joint in the janitor room side

Lab ID-Version‡: 17217099-1

Sample Layers	Asbestos Content
White Non-Fibrous Material Joint\	ND
Sample Composite Homogeneity: Good	

Location: 4044-011, South side of boiler/janitor room by the door

Lab ID-Version‡: 17217100-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity: Good	

Location: 4044-012, Boiler foundation

Lab ID-Version‡: 17217101-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity: Good	

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Client: Hawaii Analytical Lab
C/O: Anne Antin
Re: 202401040; 232061 PN77833 Kadena VMS
Hazmat Survey BLDG. 4044

Date of Sampling: 12-21-2023
Date of Receipt: 01-31-2024
Date of Report: 02-05-2024

ASBESTOS PLM REPORT

Location: 4044-013, Men's room by the door hinge

Lab ID-Version‡: 17217102-1

Sample Layers	Asbestos Content
White Ceramic Tile	ND
Gray Grout	ND
White Mortar	ND
Sample Composite Homogeneity: Good	

Location: 4044-014, Women's room by the door hinge

Lab ID-Version‡: 17217103-1

Sample Layers	Asbestos Content
White Ceramic Tile	ND
Gray Grout	ND
White Mortar	ND
Sample Composite Homogeneity: Good	

Location: 4044-015, Men's room by the door hinge

Lab ID-Version‡: 17217104-1

Sample Layers	Asbestos Content
White Ceramic Tile	ND
Gray Grout	ND
White Mortar	ND
Sample Composite Homogeneity: Good	

Location: 4044-016, Women's room by the door hinge

Lab ID-Version‡: 17217105-1

Sample Layers	Asbestos Content
White Ceramic Tile	ND
Gray Grout	ND
White Mortar	ND
Sample Composite Homogeneity: Good	

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Client: Hawaii Analytical Lab

C/O: Anne Antin

Re: 202401040; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG. 4044

Date of Sampling: 12-21-2023

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Location: 4044-017, Men's room north side**

Lab ID-Version‡: 17217106-1

Sample Layers	Asbestos Content
White Paint	ND
White Drywall with Brown Paper	ND
Composite Non-Asbestos Content:	5% Cellulose
Sample Composite Homogeneity:	Good

Location: 4044-018, Women's room north side

Lab ID-Version‡: 17217107-1

Sample Layers	Asbestos Content
White Paint	ND
White Drywall with Brown Paper	ND
Composite Non-Asbestos Content:	5% Cellulose
Sample Composite Homogeneity:	Good

Location: 4044-019, Fire piping gasket, exterior

Lab ID-Version‡: 17217108-1

Sample Layers	Asbestos Content
Black Gasket	ND
Sample Composite Homogeneity:	Good

Location: 4044-020, Fire piping gasket, exterior

Lab ID-Version‡: 17217109-1

Sample Layers	Asbestos Content
Black Gasket	ND
Sample Composite Homogeneity:	Good

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Client: Hawaii Analytical Lab

C/O: Anne Antin

Re: 202401040; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG. 4044

Date of Sampling: 12-21-2023

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Location: 4044-021, H18-Window joint sealant, brown**

Lab ID-Version‡: 17217110-1

Sample Layers	Asbestos Content
Black Sealant	ND
Sample Composite Homogeneity: Good	

Location: 4044-022, H18-Window joint sealant, brown

Lab ID-Version‡: 17217111-1

Sample Layers	Asbestos Content
Black Sealant	ND
Sample Composite Homogeneity: Good	

Location: 4044-023, On stoop; joint near rear exit door

Lab ID-Version‡: 17217112-1

Sample Layers	Asbestos Content
Gray Sealant	ND
Sample Composite Homogeneity: Good	

Location: 4044-025, rear exit, bottom of door frame

Lab ID-Version‡: 17217114-1

Sample Layers	Asbestos Content
Gray Sealant	ND
Sample Composite Homogeneity: Good	

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Client: Hawaii Analytical Lab

C/O: Anne Antin

Re: 202401040; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG. 4044

Date of Sampling: 12-21-2023

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Location: 4044-027, West side of room**

Lab ID-Version‡: 17217116-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity: Good	

Location: 4044-028, East side of room

Lab ID-Version‡: 17217117-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity: Good	

Location: 4044-029, South side of room

Lab ID-Version‡: 17217118-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity: Good	

Location: 4044-030, North side of room

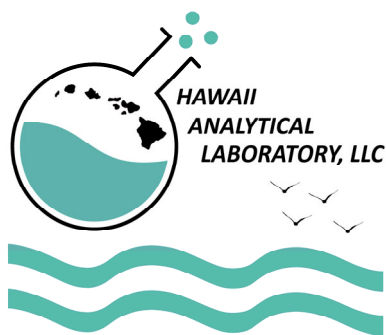
Lab ID-Version‡: 17217119-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Composite Non-Asbestos Content: 5% Glass Fibers	
Sample Composite Homogeneity: Good	

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Hawaii Analytical Laboratory

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Tel: (808) 735-0422 – Fax: (808) 735-0047

February 5, 2024

OYO Corporation, Pacific
Okinawa Times Building. 2F
1-12-24 Goya, Okinawa-shi,
Okinawa-ken, 904-0021 Japan

Project Name: 232061 PN77833 Kadena VMS Survey BLDG. 4045
Date collected: N/A
Date received: 1/26/2024
HAL #: 202401041

To Whom it May Concern,

Enclosed are the analytical results for the samples received by our laboratory on January 26, 2024. The samples on the chain of custody were received in good condition unless otherwise noted.

The forty-one (41) bulk samples submitted for PLM analysis were subcontracted to Eurofins EMLab P&K in Tustin, CA (a NVLAP accredited Laboratory, NVLAP LAB CODE: 200757-0.). Its report is enclosed in its entirety.

Results in this report are based on the sampling data provided by the client and refer only to the samples as they were received at the laboratory. Unless otherwise requested, all samples will be discarded 14 days from the date of this report.

Quality assurance data is collected to accompany all analyses and to ensure that results generated meet Hawaii Analytical Laboratory's quality standards. This data is available upon request.

Please contact us at (808) 735-0422 if you have questions.

Thank you for using Hawaii Analytical Laboratory and have a great day!

Anne Antin
Quality Manager
Hawaii Analytical Laboratory
3615 Harding Ave. Ste. 308
Honolulu, HI 96816
Phone: (808) 735-0422
E-mail: aantin@analyzehawaii.com

Report for:

Anne Antin
Hawaii Analytical Lab
3615 Harding Ave
Suite 308
Honolulu, HI 96816

Regarding: Eurofins EPK Built Environment Testing, LLC
Project: 202401041; 232061 PN77833 Kadena VMS Hazmat Survey BLDG. 4045
EML ID: 3524160

Approved by:



Approved Signatory
Amin Suliman

Dates of Analysis:
Asbestos PLM: 02-05-2024

Service SOPs: Asbestos PLM (EPA 40CFR App E to Sub E of Part 763 & EPA METHOD 600/R-93-116, SOP EM-AS-S-1267)
NVLAP Lab Code 200728-0

All samples were received in acceptable condition unless noted in the Report Comments portion in the body of the report. The results relate only to the samples as received and tested. The results include an inherent uncertainty of measurement associated with estimating percentages by polarized light microscopy. Measurement uncertainty data for sample results with >1% asbestos concentration can be provided when requested.

Eurofins EPK Built Environment Testing, LLC ("the Company"), a member of the Eurofins Built Environment Testing group of companies, shall have no liability to the client or the client's customer with respect to decisions or recommendations made, actions taken or courses of conduct implemented by either the client or the client's customer as a result of or based upon the Test Results. In no event shall the Company be liable to the client with respect to the Test Results except for the Company's own willful misconduct or gross negligence nor shall the Company be liable for incidental or consequential damages or lost profits or revenues to the fullest extent such liability may be disclaimed by law, even if the Company has been advised of the possibility of such damages, lost profits or lost revenues. In no event shall the Company's liability with respect to the Test Results exceed the amount paid to the Company by the client therefor.

Client: Hawaii Analytical Lab
C/O: Anne Antin
Re: 202401041; 232061 PN77833 Kadena VMS
Hazmat Survey BLDG. 4045

Date of Submittal: 01-31-2024
Date of Receipt: 01-31-2024
Date of Report: 02-05-2024

ASBESTOS PLM REPORT

Total Samples Submitted: 41

Total Samples Analyzed: 41

Total Samples with Layer Asbestos Content > 1%: 0

Location: 4045-001, H3- Concrete Wall Coating, Off-White, Smooth

Lab ID-Version‡: 17217125-1

Sample Layers	Asbestos Content
Gray Cementitious Material with Paint	ND
Sample Composite Homogeneity:	Good

Location: 4045-002, H3- Concrete Wall Coating, Off-White, Smooth

Lab ID-Version‡: 17217126-1

Sample Layers	Asbestos Content
Gray Cementitious Material with Paint	ND
Sample Composite Homogeneity:	Good

Location: 4045-003, H4- Gypsum Board Wall, Off-White, Smooth

Lab ID-Version‡: 17217127-1

Sample Layers	Asbestos Content
Off-White Joint Compound	ND
Off-White Drywall with Brown Paper	ND
Composite Non-Asbestos Content:	10% Cellulose
Sample Composite Homogeneity:	Moderate

Location: 4045-004, H5- Duct TSI, Yellow Fiberglass w/ Silver Backing

Lab ID-Version‡: 17217128-1

Sample Layers	Asbestos Content
Silver Foil	ND
Transparent Mastic	ND
Cream Tape	ND
Tan Insulation	ND
Composite Non-Asbestos Content:	90% Glass Fibers 5% Cellulose
Sample Composite Homogeneity:	Moderate

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Client: Hawaii Analytical Lab
C/O: Anne Antin
Re: 202401041; 232061 PN77833 Kadena VMS
Hazmat Survey BLDG. 4045

Date of Submittal: 01-31-2024
Date of Receipt: 01-31-2024
Date of Report: 02-05-2024

ASBESTOS PLM REPORT

Location: 4045-005, H5- Duct TSI, Yellow Fiberglass w/ Silver Backing

Lab ID-Version‡: 17217129-1

Sample Layers	Asbestos Content
Silver Foil	ND
Transparent Mastic	ND
Cream Tape	ND
Tan Insulation	ND
Composite Non-Asbestos Content:	90% Glass Fibers 5% Cellulose
Sample Composite Homogeneity:	Moderate

DRAFT

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Client: Hawaii Analytical Lab
C/O: Anne Antin
Re: 202401041; 232061 PN77833 Kadena VMS
Hazmat Survey BLDG. 4045

Date of Submittal: 01-31-2024
Date of Receipt: 01-31-2024
Date of Report: 02-05-2024

ASBESTOS PLM REPORT

Location: 4045-006, H6- Ceiling Insulation, Yellow Fiberglass w/ Silver Backing

Lab ID-Version‡: 17217130-1

Sample Layers	Asbestos Content
Silver Foil Tape	ND
Tan Insulation	ND
Composite Non-Asbestos Content:	90% Glass Fibers 3% Cellulose
Sample Composite Homogeneity:	Moderate

Location: 4045-007, H6- Ceiling Insulation, Yellow Fiberglass w/ Silver Backing

Lab ID-Version‡: 17217131-1

Sample Layers	Asbestos Content
Silver Foil Tape	ND
Tan Insulation	ND
Composite Non-Asbestos Content:	90% Glass Fibers 3% Cellulose
Sample Composite Homogeneity:	Moderate

Location: 4045-008, H7- 60x60 cm Off-White Fissured Acoustical Ceiling Tile

Lab ID-Version‡: 17217132-1

Sample Layers	Asbestos Content
Off-White Ceiling Tile with White Surface	ND
Composite Non-Asbestos Content:	45% Cellulose 45% Glass Fibers
Sample Composite Homogeneity:	Good

Location: 4045-009, H7- 60x60 cm Off-White Fissured Acoustical Ceiling Tile

Lab ID-Version‡: 17217133-1

Sample Layers	Asbestos Content
Off-White Ceiling Tile with White Surface	ND
Composite Non-Asbestos Content:	45% Cellulose 45% Glass Fibers
Sample Composite Homogeneity:	Good

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Client: Hawaii Analytical Lab
C/O: Anne Antin
Re: 202401041; 232061 PN77833 Kadena VMS
Hazmat Survey BLDG. 4045

Date of Submittal: 01-31-2024
Date of Receipt: 01-31-2024
Date of Report: 02-05-2024

ASBESTOS PLM REPORT

Location: 4045-012, H9- Concrete Block Wall Coating, Grey Painted

Lab ID-Version‡: 17217136-1

Sample Layers	Asbestos Content
Gray Paint	ND
Off-White Non-Fibrous Material	ND
Sample Composite Homogeneity: Good	

Location: 4045-013, H9- Concrete Block Wall Coating, Grey Painted

Lab ID-Version‡: 17217137-1

Sample Layers	Asbestos Content
Gray Paint	ND
Off-White Non-Fibrous Material	ND
Sample Composite Homogeneity: Good	

Location: 4045-014, H10- Pipe TSI (Run), Yellow Fiberglass w/ Silver Backing

Lab ID-Version‡: 17217138-1

Sample Layers	Asbestos Content
Silver Foil	ND
Transparent Mastic	ND
Off-White Paper	ND
Tan Insulation	ND
Composite Non-Asbestos Content: 90% Glass Fibers 5% Cellulose	
Sample Composite Homogeneity: Poor	

Location: 4045-015, H10- Pipe TSI (Run), Yellow Fiberglass w/ Silver Backing

Lab ID-Version‡: 17217139-1

Sample Layers	Asbestos Content
Silver Foil	ND
Transparent Mastic	ND
Off-White Paper	ND
Tan Insulation	ND
Composite Non-Asbestos Content: 90% Glass Fibers 5% Cellulose	
Sample Composite Homogeneity: Poor	

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Client: Hawaii Analytical Lab
 C/O: Anne Antin
 Re: 202401041; 232061 PN77833 Kadena VMS
 Hazmat Survey BLDG. 4045

Date of Submittal: 01-31-2024
 Date of Receipt: 01-31-2024
 Date of Report: 02-05-2024

ASBESTOS PLM REPORT

Location: 4045-016, H11- Pipe TSI (Elbow), Yellow Fiberglass w/ Silver Backing

Lab ID-Version‡: 17217140-1

Sample Layers	Asbestos Content
Silver Foil	ND
Transparent Mastic	ND
Tan Insulation	ND
Composite Non-Asbestos Content:	90% Glass Fibers
Sample Composite Homogeneity:	Poor

Location: 4045-017, H11- Pipe TSI (Elbow), Yellow Fiberglass w/ Silver Backing

Lab ID-Version‡: 17217141-1

Sample Layers	Asbestos Content
Silver Foil	ND
Transparent Mastic	ND
Off-White Paper	ND
Tan Insulation	ND
Composite Non-Asbestos Content:	90% Glass Fibers 5% Cellulose
Sample Composite Homogeneity:	Poor

Location: 4045-018, H12- 20x20 Ceramic Floor Tile, Grey w/ Grey Grout

Lab ID-Version‡: 17217142-1

Sample Layers	Asbestos Content
Gray Ceramic Tile	ND
Sample Composite Homogeneity:	Good

Location: 4045-019, H12- 20x20 Ceramic Floor Tile, Grey w/ Grey Grout

Lab ID-Version‡: 17217143-1

Sample Layers	Asbestos Content
Gray Ceramic Tile	ND
Sample Composite Homogeneity:	Good

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ASBESTOS PLM REPORT

Location: 4045-020, H13- 10x10 Ceramic Wall Tile, White w/ White Grout

Lab ID-Version‡: 17217144-1

Sample Layers	Asbestos Content
White Ceramic Tile	ND
Sample Composite Homogeneity: Good	

Location: 4045-0021, H13- 10x10 Ceramic Wall Tile, White w/ White Grout

Lab ID-Version‡: 17217145-1

Sample Layers	Asbestos Content
White Ceramic Tile	ND
Sample Composite Homogeneity: Good	

Location: 4045-022, H14- Gypsum Ceiling Board, Off-White

Lab ID-Version‡: 17217146-1

Sample Layers	Asbestos Content
Off-White Joint Compound	ND
Off-White Drywall with Brown Paper	ND
Composite Non-Asbestos Content:	10% Cellulose
Sample Composite Homogeneity: Moderate	

Location: 4045-023, H14- Gypsum Ceiling Board, Off-White

Lab ID-Version‡: 17217147-1

Sample Layers	Asbestos Content
Off-White Joint Compound	ND
Off-White Drywall with Brown Paper	ND
Composite Non-Asbestos Content:	10% Cellulose
Sample Composite Homogeneity: Moderate	

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ASBESTOS PLM REPORT

Location: 4045-024, H16- Joint Sealant (Door) White Mastic

Lab ID-Version‡: 17217148-1

Sample Layers	Asbestos Content
White Mastic	ND
Sample Composite Homogeneity: Good	

Location: 4045-025, H16- Joint Sealant (Door) White Mastic

Lab ID-Version‡: 17217149-1

Sample Layers	Asbestos Content
White Mastic	ND
Sample Composite Homogeneity: Good	

Location: 4045-026, H18- Exterior Concrete Block Wall, Off-White, Textured

Lab ID-Version‡: 17217150-1

Sample Layers	Asbestos Content
Off-White Paint	ND
Off-White Non-Fibrous Material	ND
Sample Composite Homogeneity: Good	

Location: 4045-027, H19- Exterior Concrete Wall Base Coating (Red)

Lab ID-Version‡: 17217151-1

Sample Layers	Asbestos Content
Gray Cementitious Material	ND
Sample Composite Homogeneity: Good	

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ASBESTOS PLM REPORT

Location: 4045-028, H19- Exterior Concrete Wall Base Coating (Red)

Lab ID-Version‡: 17217152-1

Sample Layers	Asbestos Content
Gray Cementitious Material	ND
Sample Composite Homogeneity: Good	

Location: 4045-029, H20- Exterior Door Joint Sealant, Grey Mastic

Lab ID-Version‡: 17217153-1

Sample Layers	Asbestos Content
Gray Mastic	ND
Sample Composite Homogeneity: Good	

Location: 4045-030, H20- Exterior Door Joint Sealant, Grey Mastic

Lab ID-Version‡: 17217154-1

Sample Layers	Asbestos Content
Gray Mastic	ND
Sample Composite Homogeneity: Good	

Location: 4045-031, H21- Exterior Black Flex Sheet

Lab ID-Version‡: 17217155-1

Sample Layers	Asbestos Content
Black Foam	ND
Sample Composite Homogeneity: Good	

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ASBESTOS PLM REPORT

Location: 4045-032, H21- Exterior Black Flex Sheet

Lab ID-Version‡: 17217156-1

Sample Layers	Asbestos Content
Black Foam	ND
Tan Mastic	ND
Sample Composite Homogeneity: Good	

Location: 4045-033, H25- Floor Joint Sealant (Exterior), Solid Black

Lab ID-Version‡: 17217157-1

Sample Layers	Asbestos Content
Black Sealant	ND
Sample Composite Homogeneity: Good	

Location: 4045-034, H25- Floor Joint Sealant (Exterior), Solid Black

Lab ID-Version‡: 17217158-1

Sample Layers	Asbestos Content
Black Sealant	ND
Sample Composite Homogeneity: Good	

Location: 4045-035, H26- Window Joint Sealant, Black Mastic

Lab ID-Version‡: 17217159-1

Sample Layers	Asbestos Content
Black Mastic	ND
Sample Composite Homogeneity: Good	

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ASBESTOS PLM REPORT

Location: 4045-036, H26- Window Joint Sealant, Black Mastic

Lab ID-Version‡: 17217160-1

Sample Layers	Asbestos Content
Black Mastic	ND
Sample Composite Homogeneity: Good	

Location: 4045-037, Roofing Material (Asphalt); North Side Roof

Lab ID-Version‡: 17217161-1

Sample Layers	Asbestos Content
Black Roofing Material	ND
Composite Non-Asbestos Content:	60% Glass Fibers
Sample Composite Homogeneity: Moderate	

Location: 4045-038, Roofing Material (Asphalt); South Side Roof

Lab ID-Version‡: 17217162-1

Sample Layers	Asbestos Content
Black Roofing Material	ND
Composite Non-Asbestos Content:	60% Glass Fibers
Sample Composite Homogeneity: Moderate	

Location: 4045-039, Roofing Material (Asphalt); East Side Roof

Lab ID-Version‡: 17217163-1

Sample Layers	Asbestos Content
Black Roofing Material	ND
Composite Non-Asbestos Content:	60% Glass Fibers
Sample Composite Homogeneity: Moderate	

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Hazmat Survey BLDG. 4045

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ASBESTOS PLM REPORT

Location: 4045-040, Roof Vent Sealant (N2); North Side Roof

Lab ID-Version‡: 17217164-1

Sample Layers	Asbestos Content
Black Roofing Material	ND
Off-White Woven Material	ND
Off-White Non-Fibrous Material	ND
Silver Foil	ND
Composite Non-Asbestos Content:	65% Glass Fibers
Sample Composite Homogeneity:	Moderate

Location: 4045-041, Roof Caulking (N3); North Side Roof

Lab ID-Version‡: 17217165-1

Sample Layers	Asbestos Content
Silver Caulk	ND
Sample Composite Homogeneity:	Good

Location: 4045-042, Roof Caulking (N3); East Side Roof

Lab ID-Version‡: 17217166-1

Sample Layers	Asbestos Content
Silver Caulk	ND
Sample Composite Homogeneity:	Good

Location: 4045-043, Roof Caulking (N3); South Side Roof

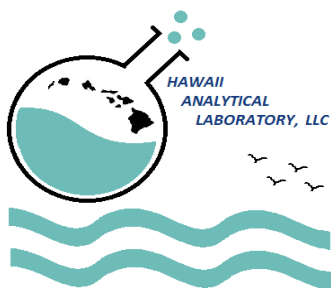
Lab ID-Version‡: 17217167-1

Sample Layers	Asbestos Content
Silver Caulk	ND
Sample Composite Homogeneity:	Good

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Hawaii Analytical Laboratory ANALYTICAL REPORT

Monday, January 8, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314100

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4046, 11/30/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202397979	4046-001	NONE DETECTED			Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum + mica + paint	1/3/2024
	<u>Layer</u> White gypsum board / beige paint							
	Comments							
202397980	4046-002	NONE DETECTED			None detected		Calcite + quartz + paint	1/3/2024
	<u>Layer</u> Gray concrete coating / off-white paint							
	Comments							
202397981	4046-003	NONE DETECTED			None detected		Calcite + binder + paint	1/3/2024
	<u>Layer</u> White sealant / off-white paint							
	Comments							
202397982	4046-004	NONE DETECTED			None detected		Calcite + binder + paint	1/3/2024
	<u>Layer</u> White sealant / off-white paint							
	Comments							
202397983	4046-005	NONE DETECTED			Fibrous glass (amorphous)	25	Paint + other	1/3/2024
	<u>Layer</u> Gray wrap							
	Comments							
202397983	4046-005	NONE DETECTED			Fibrous glass (amorphous)	40	Foil	1/3/2024
	<u>Layer</u> Silver wrap							
	Comments							

Hawaii Analytical Laboratory is a NIST NVLAP accredited laboratory (NVLAP Lab Code 200655-0) and is accredited in accordance with the recognized ISO/ IEC 17025:2017. Controlled doc.: Asbestos Report, rev. 3 – 20200630

OYO Corporation, Pacific

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Lab Job No: 202314100

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4046, 11/30/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202397983	4046-005		NONE DETECTED		Fibrous glass (amorphous)	> 99	Glass beads	1/3/2024
<u>Layer</u>	<u>Yellow insulation</u>							
<u>Comments</u>								
202397984	4046-006		NONE DETECTED		Fibrous glass (amorphous)	25	Paint + other	1/3/2024
<u>Layer</u>	<u>Gray wrap</u>							
<u>Comments</u>								
202397984	4046-006		NONE DETECTED		Fibrous glass (amorphous)	40	Foil	1/3/2024
<u>Layer</u>	<u>Silver wrap</u>							
<u>Comments</u>								
202397984	4046-006		NONE DETECTED		Fibrous glass (amorphous)	> 99	Glass beads	1/3/2024
<u>Layer</u>	<u>Yellow insulation</u>							
<u>Comments</u>								
202397985	4046-007		NONE DETECTED		Fibrous glass (amorphous)	25	Paint + other	1/3/2024
<u>Layer</u>	<u>Dark gray wrap</u>							
<u>Comments</u>								
202397985	4046-007		NONE DETECTED		Fibrous glass (amorphous)	40	Foil	1/3/2024
<u>Layer</u>	<u>Silver wrap</u>							
<u>Comments</u>								
202397985	4046-007		NONE DETECTED		Fibrous glass (amorphous)	> 99	Glass beads	1/3/2024
<u>Layer</u>	<u>Yellow insulation</u>							
<u>Comments</u>								
202397986	4046-008		NONE DETECTED		Fibrous glass (amorphous)	25	Paint + other + foil	1/3/2024
<u>Layer</u>	<u>Dark gray wrap</u>							
<u>Comments</u>								

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Lab Job No: 202314100

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4046, 11/30/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202397986	4046-008		NONE DETECTED		Fibrous glass (amorphous)	40	Foil	1/3/2024
<u>Layer</u>	<u>Silver wrap</u>							
<u>Comments</u>								
202397986	4046-008		NONE DETECTED		Fibrous glass (amorphous)	> 99	Glass beads	1/3/2024
<u>Layer</u>	<u>Yellow insulation</u>							
<u>Comments</u>								
202397987	4046-011		NONE DETECTED		None detected		Calcite + quartz	1/3/2024
<u>Layer</u>	<u>Gray grout</u>							
<u>Comments</u>								
202397987	4046-011		NONE DETECTED		None detected		Ceramic	1/3/2024
<u>Layer</u>	<u>Pale purple ceramic tile</u>							
<u>Comments</u>								
202397987	4046-011		NONE DETECTED		None detected		Calcite + binder	1/3/2024
<u>Layer</u>	<u>White caulk</u>							
<u>Comments</u>								
202397988	4046-012		NONE DETECTED		None detected		Calcite + quartz	1/3/2024
<u>Layer</u>	<u>Gray grout</u>							
<u>Comments</u>								
202397988	4046-012		NONE DETECTED		None detected		Ceramic	1/3/2024
<u>Layer</u>	<u>Pale purple ceramic tile</u>							
<u>Comments</u>								
202397988	4046-012		NONE DETECTED		None detected		Calcite + binder	1/3/2024
<u>Layer</u>	<u>White caulk</u>							
<u>Comments</u>								

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Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202397989	4046-013		NONE DETECTED		None detected	Ceramic	1/3/2024
Layer	Gray ceramic tile						
Comments							
202397989	4046-013		NONE DETECTED		None detected	Calcite + quartz	1/3/2024
Layer	Gray grout						
Comments							
202397990	4046-014		NONE DETECTED		None detected	Ceramic	1/3/2024
Layer	Gray ceramic tile						
Comments							
202397990	4046-014		NONE DETECTED		None detected	Calcite + quartz	1/3/2024
Layer	Gray grout						
Comments							
202397991	4046-015		NONE DETECTED		None detected	Calcite + quartz + paint	1/3/2024
Layer	Gray concrete coating / off-white paint						
Comments							
202397992	4046-016		NONE DETECTED		None detected	Calcite + quartz + paint	1/3/2024
Layer	Gray concrete coating / off-white paint						
Comments							
202397993	4046-017		NONE DETECTED		None detected	Calcite + binder	1/3/2024
Layer	Gray sealant						
Comments							
202397994	4046-018		NONE DETECTED		None detected	Calcite + binder	1/3/2024
Layer	Gray sealant						
Comments							

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314100

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4046, 11/30/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202397995	4046-019		NONE DETECTED		Fibrous glass (amorphous)	> 99	None detected	1/3/2024
Layer	Yellow insulation							
Comments								
202397996	4046-020		NONE DETECTED		Fibrous glass (amorphous)	> 99	None detected	1/3/2024
Layer	Yellow insulation							
Comments								
202397997	4046-021		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	3	Tar + calcite + quartz	1/3/2024
Layer	Black mastic							
Comments								
202397997	4046-021		NONE DETECTED		None detected		Calcite + quartz	1/3/2024
Layer	Gray cementitious material							
Comments								
202397998	4046-022		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	3	Tar + calcite + quartz	1/3/2024
Layer	Black mastic							
Comments								
202397998	4046-022		NONE DETECTED		None detected		Calcite + quartz	1/3/2024
Layer	Gray cementitious material							
Comments								
202397999	4046-023		NONE DETECTED		None detected		Calcite + quartz	1/3/2024
Layer	Gray roofing material							
Comments								
202398000	4046-024		NONE DETECTED		None detected		Calcite + quartz	1/3/2024
Layer	Gray roofing material							
Comments								

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314100

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4046, 11/30/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202398001	4046-025		NONE DETECTED		None detected		Calcite + quartz	1/3/2024
<u>Layer</u>	<u>Gray roofing material</u>							
<u>Comments</u>								
202398002	4046-026		NONE DETECTED		None detected		Calcite + binder	1/3/2024
<u>Layer</u>	<u>White caulk (1)</u>							
<u>Comments</u>								
202398002	4046-026		NONE DETECTED		None detected		Calcite + binder	1/3/2024
<u>Layer</u>	<u>White caulk (2)</u>							
<u>Comments</u>								
202398002	4046-026		NONE DETECTED		Fibrous glass (amorphous)	> 99	None detected	1/3/2024
<u>Layer</u>	<u>White mesh</u>							
<u>Comments</u>								

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314100

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4046, 11/30/23

General Comments

The bulk sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures outlined in the United States Environmental Protection Agency's "Interim Method for the Determination of Asbestos in Bulk Insulation Samples" (EPA-600/M4-82-020, Dec. 1982) and / or "Method for Determination of Asbestos in bulk Building Materials" (EPA-600/R-93-116, July 1993). The analysis of each bulk sample relates only to the material examined, and may or may not represent the overall composition of its original source. Floor tile and other resinously bound materials, when analyzed by the EPA methods referenced above may yield false negative results because of limitations in separating closely bound fibers and in detecting fibers of small length and diameter. Alternative methods of identification, including Transmission Electron Microscopy (TEM) may or may not be applicable. We utilize calibrated visual area estimation on a routine basis and do not conduct point counting unless specifically requested to do so. Estimated error for the visual determinations presented are 75% relative (1 to 2%), 50% relative (3 to 5%); 25% relative (6 to 25%) and 20% (>26% v/v). We will not separate layers which in our opinion are not readily discernable. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report must not be used by the client to claim product certification, approval or endorsement by NVLAP, NIST or any agency of Federal Government. Unless otherwise indicated, the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

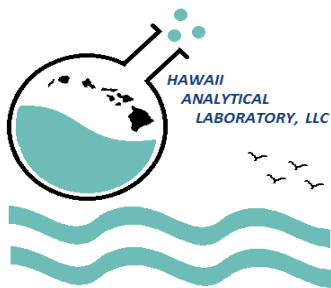
< This testing result is less than the numerical value listed.

None Detected = asbestos was not observed in the sample. If trace amount of asbestos was detected below our quantifiable limits of 1.0%, <1% (trace) would be indicated and the asbestos type listed. Point counting, where applicable, are recommended to improve accuracy.



Eva Skogsberg
Laboratory Supervisor

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Hawaii Analytical Laboratory ANALYTICAL REPORT

Monday, January 8, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314098

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4047, 11/30/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202397948	4047-001	NONE DETECTED			None detected	Calcite + quartz + paint	1/3/2024
Layer	Gray concrete coating / off-white paint						
Comments							
202397949	4047-002	NONE DETECTED			None detected	Calcite + binder + paint	1/3/2024
Layer	White sealant / off-white paint						
Comments							
202397950	4047-003	NONE DETECTED			None detected	Calcite + binder + paint	1/3/2024
Layer	White sealant / off-white paint						
Comments							
202397951	4047-004	NONE DETECTED			None detected	Calcite + quartz + paint	1/3/2024
Layer	Gray concrete coating / off-white paint						
Comments							
202397952	4047-005	NONE DETECTED			None detected	Calcite + quartz + paint	1/3/2024
Layer	Gray concrete coating / off-white paint						
Comments							
202397953	4047-006	NONE DETECTED			None detected	Calcite + quartz	1/3/2024
Layer	Gray grout						
Comments							

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314098

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4047, 11/30/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202397953	4047-006		NONE DETECTED		None detected	Ceramic	1/3/2024
<u>Layer</u>	<u>Pale purple ceramic tile</u>						
<u>Comments</u>							
202397954	4047-007		NONE DETECTED		None detected	Calcite + quartz	1/3/2024
<u>Layer</u>	<u>Gray grout</u>						
<u>Comments</u>							
202397954	4047-007		NONE DETECTED		None detected	Ceramic	1/3/2024
<u>Layer</u>	<u>Pale purple ceramic tile</u>						
<u>Comments</u>							
202397955	4047-008		NONE DETECTED		None detected	Calcite + quartz	1/3/2024
<u>Layer</u>	<u>Gray grout</u>						
<u>Comments</u>							
202397955	4047-008		NONE DETECTED		None detected	Ceramic	1/3/2024
<u>Layer</u>	<u>Pale purple ceramic tile</u>						
<u>Comments</u>							
202397955	4047-008		NONE DETECTED		None detected	Ceramic	1/3/2024
<u>Layer</u>	<u>White ceramic tile</u>						
<u>Comments</u>							
202397956	4047-009		NONE DETECTED		None detected	Calcite + quartz	1/3/2024
<u>Layer</u>	<u>Gray grout</u>						
<u>Comments</u>							
202397956	4047-009		NONE DETECTED		None detected	Ceramic	1/3/2024
<u>Layer</u>	<u>Pale purple ceramic tile</u>						
<u>Comments</u>							

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314098

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4047, 11/30/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202397956	4047-009		NONE DETECTED		None detected		Ceramic	1/3/2024
Layer	White ceramic tile							
Comments								
202397957	4047-010		NONE DETECTED		None detected		Calcite + binder	1/3/2024
Layer	Gray sealant							
Comments								
202397958	4047-011		NONE DETECTED		None detected		Calcite + binder	1/4/2024
Layer	Gray sealant							
Comments								
202397958	4047-011		NONE DETECTED		None detected		Calcite + binder	1/4/2024
Layer	White sealant							
Comments								
202397959	4047-012		NONE DETECTED		None detected		Calcite + quartz + paint	1/4/2024
Layer	Orange paint / gray concrete coating							
Comments								
202397960	4047-013		NONE DETECTED		None detected		Calcite + quartz + paint	1/4/2024
Layer	Orange paint / gray concrete coating							
Comments								
202397961	4047-014		NONE DETECTED		Synthetic fibers (undulose) + fibrous glass (amorphous)	40	Calcite + paint	1/4/2024
Layer	Blue gasket / red paint							
Comments								
202397962	4047-015		NONE DETECTED		Fibrous glass (amorphous)	40	Foil + other	1/4/2024
Layer	Silver wrap							
Comments								

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Okinawa Japan .

Phone Number:

Facsimile:

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aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314098

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4047, 11/30/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202397962	4047-015		NONE DETECTED		Fibrous glass (amorphous)	25	Paint + binder + other	1/4/2024
Layer	White wrap / yellow mastic							
Comments								
202397962	4047-015		NONE DETECTED		Fibrous glass (amorphous)	> 99	Glass beads	1/4/2024
Layer	Yellow insulation							
Comments								
202397963	4047-016		NONE DETECTED		Fibrous glass (amorphous)	25	Paint + foil + other	1/4/2024
Layer	Dark gray wrap							
Comments								
202397963	4047-016		NONE DETECTED		Fibrous glass (amorphous)	40	Foil	1/4/2024
Layer	Silver wrap							
Comments								
202397963	4047-016		NONE DETECTED		Fibrous glass (amorphous)	> 99	Glass beads	1/4/2024
Layer	Yellow insulation							
Comments								
202397964	4047-017		NONE DETECTED		Fibrous glass (amorphous)	40	Foil	1/4/2024
Layer	Silver wrap							
Comments								
202397964	4047-017		NONE DETECTED		Fibrous glass (amorphous)	> 99	None detected	1/4/2024
Layer	Yellow insulation							
Comments								
202397965	4047-018		NONE DETECTED		None detected		Calcite + quartz	1/4/2024
Layer	Gray roofing material							
Comments								

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314098

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4047, 11/30/23

Bulk Asbestos Determination

Sample No.	Your Sample Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202397966	4047-020		NONE DETECTED		None detected		Calcite + quartz	1/4/2024
<u>Layer</u>	<u>Gray roofing material</u>							
<u>Comments</u>								
202397967	4047-021		NONE DETECTED		None detected		Calcite + quartz	1/4/2024
<u>Layer</u>	<u>Gray roofing material</u>							
<u>Comments</u>								
202397968	4047-019		NONE DETECTED		None detected		Calcite + binder	1/4/2024
<u>Layer</u>	<u>White caulk (1)</u>							
<u>Comments</u>								
202397968	4047-019		NONE DETECTED		None detected		Calcite + binder	1/4/2024
<u>Layer</u>	<u>White caulk (2)</u>							
<u>Comments</u>								
202397968	4047-019		NONE DETECTED		Fibrous glass (amorphous)	> 99	None detected	1/4/2024
<u>Layer</u>	<u>White mesh</u>							
<u>Comments</u>								

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Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314098

Date Submitted: 12/29/2023

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4047, 11/30/23

General Comments

The bulk sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures outlined in the United States Environmental Protection Agency's "Interim Method for the Determination of Asbestos in Bulk Insulation Samples" (EPA-600/M4-82-020, Dec. 1982) and / or "Method for Determination of Asbestos in bulk Building Materials" (EPA-600/R-93-116, July 1993). The analysis of each bulk sample relates only to the material examined, and may or may not represent the overall composition of its original source. Floor tile and other resinously bound materials, when analyzed by the EPA methods referenced above may yield false negative results because of limitations in separating closely bound fibers and in detecting fibers of small length and diameter. Alternative methods of identification, including Transmission Electron Microscopy (TEM) may or may not be applicable. We utilize calibrated visual area estimation on a routine basis and do not conduct point counting unless specifically requested to do so. Estimated error for the visual determinations presented are 75% relative (1 to 2%), 50% relative (3 to 5%); 25% relative (6 to 25%) and 20% (>26% v/v). We will not separate layers which in our opinion are not readily discernable. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report must not be used by the client to claim product certification, approval or endorsement by NVLAP, NIST or any agency of Federal Government. Unless otherwise indicated, the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

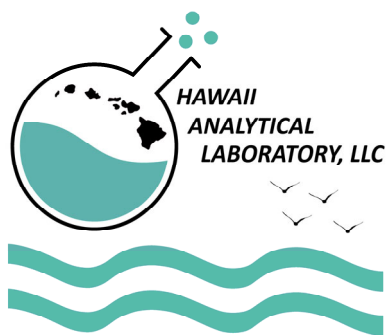
< This testing result is less than the numerical value listed.

None Detected = asbestos was not observed in the sample. If trace amount of asbestos was detected below our quantifiable limits of 1.0%, <1% (trace) would be indicated and the asbestos type listed. Point counting, where applicable, are recommended to improve accuracy.



Eva Skogsberg
Laboratory Supervisor

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Hawaii Analytical Laboratory

3615 Harding Avenue, Suite 308, Honolulu, Hawaii, 96816

Tel: (808) 735-0422 – Fax: (808) 735-0047

February 6, 2024

OYO Corporation, Pacific
Okinawa Times Building. 2F
1-12-24 Goya, Okinawa-shi,
Okinawa-ken, 904-0021 Japan

Project Name: 232061 PN77833 Kadena VMS Survey BLDG. 4048
Date collected: 1/12/2024
Date received: 1/26/2024
HAL #: 202401042

To Whom it May Concern,

Enclosed are the analytical results for the samples received by our laboratory on January 26, 2024. The samples on the chain of custody were received in good condition unless otherwise noted.

The six (6) bulk samples submitted for PLM analysis were subcontracted to Eurofins EMLab P&K in Tustin, CA (a NVLAP accredited Laboratory, NVLAP LAB CODE: 200757-0.). Its report is enclosed in its entirety.

Results in this report are based on the sampling data provided by the client and refer only to the samples as they were received at the laboratory. Unless otherwise requested, all samples will be discarded 14 days from the date of this report.

Quality assurance data is collected to accompany all analyses and to ensure that results generated meet Hawaii Analytical Laboratory's quality standards. This data is available upon request.

Please contact us at (808) 735-0422 if you have questions.

Thank you for using Hawaii Analytical Laboratory and have a great day!

Anne Antin
Quality Manager
Hawaii Analytical Laboratory
3615 Harding Ave. Ste. 308
Honolulu, HI 96816
Phone: (808) 735-0422
E-mail: aantin@analyzehawaii.com

Report for:

Anne Antin
Hawaii Analytical Lab
3615 Harding Ave
Suite 308
Honolulu, HI 96816

Regarding: Eurofins EPK Built Environment Testing, LLC
Project: 202401042; 232061 PN77833 Kadena VMS Hazmat Survey BLDG. 4048
EML ID: 3524167

Approved by:



Approved Signatory
Danny Li

Dates of Analysis:
Asbestos PLM: 02-05-2024

Service SOPs: Asbestos PLM (EPA 40CFR App E to Sub E of Part 763 & EPA METHOD 600/R-93-116, SOP EM-AS-S-1267)
NVLAP Lab Code 200757-0

All samples were received in acceptable condition unless noted in the Report Comments portion in the body of the report. The results relate only to the samples as received and tested. The results include an inherent uncertainty of measurement associated with estimating percentages by polarized light microscopy. Measurement uncertainty data for sample results with >1% asbestos concentration can be provided when requested.

Eurofins EPK Built Environment Testing, LLC ("the Company"), a member of the Eurofins Built Environment Testing group of companies, shall have no liability to the client or the client's customer with respect to decisions or recommendations made, actions taken or courses of conduct implemented by either the client or the client's customer as a result of or based upon the Test Results. In no event shall the Company be liable to the client with respect to the Test Results except for the Company's own willful misconduct or gross negligence nor shall the Company be liable for incidental or consequential damages or lost profits or revenues to the fullest extent such liability may be disclaimed by law, even if the Company has been advised of the possibility of such damages, lost profits or lost revenues. In no event shall the Company's liability with respect to the Test Results exceed the amount paid to the Company by the client therefor.

Eurofins EPK Built Environment Testing, LLC

2841 Dow Avenue, Suite 300, Tustin, CA 92780

(800) 651-4802 www.eurofinsus.com/Built

Client: Hawaii Analytical Lab

C/O: Anne Antin

Re: 202401042; 232061 PN77833 Kadena VMS

Hazmat Survey BLDG. 4048

Date of Sampling: 01-12-2024

Date of Receipt: 01-31-2024

Date of Report: 02-05-2024

ASBESTOS PLM REPORT**Total Samples Submitted:** 6**Total Samples Analyzed:** 6**Total Samples with Layer Asbestos Content > 1%:** 0**Location: 4048-001, Inside main bay, middle of north side**

Lab ID-Version‡: 17217174-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity: Good	

Location: 4048-002, Inside main bay, near east entrance door

Lab ID-Version‡: 17217175-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity: Good	

Location: 4048-003, Inside main bay, on south side

Lab ID-Version‡: 17217176-1

Sample Layers	Asbestos Content
Gray Concrete	ND
Sample Composite Homogeneity: Good	

The test report shall not be reproduced except in full, without written approval of the laboratory. The report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government. The Company reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified.

Inhomogeneous samples are separated into homogeneous subsamples and analyzed individually. ND means no fibers were detected. When detected, the minimum detection and reporting limit is less than 1% unless point counting is performed. Floor tile samples may contain large amounts of interference material and it is recommended that the sample be analyzed by gravimetric point count analysis to lower the detection limit and to aid in asbestos identification.

‡ A "Version" indicated by "-x" after the Lab ID# with a value greater than 1 indicates a sample with amended data. The revision number is reflected by the value of "x".

Client: Hawaii Analytical Lab
C/O: Anne Antin
Re: 202401042; 232061 PN77833 Kadena VMS
Hazmat Survey BLDG. 4048

Date of Sampling: 01-12-2024
Date of Receipt: 01-31-2024
Date of Report: 02-05-2024

ASBESTOS PLM REPORT

Location: 4048-004, Concrete wall (exterior) with textured paint SE corner

Lab ID-Version‡: 17217177-1

Sample Layers	Asbestos Content
White Paint	ND
Gray Concrete	ND
Sample Composite Homogeneity: Good	

Location: 4048-005, Concrete wall (exterior) with texture paint SW side

Lab ID-Version‡: 17217178-1

Sample Layers	Asbestos Content
White Paint	ND
Gray Concrete	ND
Sample Composite Homogeneity: Good	

Location: 4048-006, Concrete wall (interior), north side interior side wall

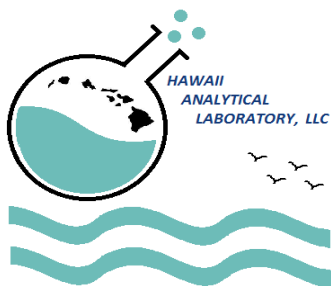
Lab ID-Version‡: 17217179-1

Sample Layers	Asbestos Content
White Paint	ND
Gray Concrete	ND
Sample Composite Homogeneity: Good	

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Inhomogeneous samples are separated into homogeneous subsamples and analyzed individually. ND means no fibers were detected. When detected, the minimum detection and reporting limit is less than 1% unless point counting is performed. Floor tile samples may contain large amounts of interference material and it is recommended that the sample be analyzed by gravimetric point count analysis to lower the detection limit and to aid in asbestos identification.

‡ A "Version" indicated by "-x" after the Lab ID# with a value greater than 1 indicates a sample with amended data. The revision number is reflected by the value of "x".



Hawaii Analytical Laboratory ANALYTICAL REPORT

Tuesday, May 27, 2025

OYO Corporation, Pacific
Okinawa Times Building. 2F
1-12-24 Goya, Okinawa-shi,
Okinawa-ken, 904-0021 Japan
Okinawa Japan.

Phone Number:

Facsimile:

Email: aledezmaes@oyopacific.com

Lab Job No: 202504596

Date Submitted: 5/20/2025

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey, Swing Space, 5/9/25

Bulk Asbestos Determination

Lab Sple No.	Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202532687	SS-001	NONE DETECTED			None detected	Binder	5/22/2025
	<u>Layer</u> <u>Black rubbery material</u>						
	Comments						
202532687	SS-001	NONE DETECTED			None detected	Cementitious + other	5/22/2025
	<u>Layer</u> <u>Gray concrete</u>						
	Comments						
202532688	SS-002	NONE DETECTED			None detected	Binder	5/22/2025
	<u>Layer</u> <u>Black rubbery material</u>						
	Comments						
202532688	SS-002	NONE DETECTED			None detected	Cementitious + other	5/22/2025
	<u>Layer</u> <u>Gray concrete</u>						
	Comments						
202532689	SS-003	NONE DETECTED			None detected	Binder	5/22/2025
	<u>Layer</u> <u>Black rubbery material</u>						
	Comments						
202532689	SS-003	NONE DETECTED			None detected	Cementitious + other	5/22/2025
	<u>Layer</u> <u>Gray concrete</u>						
	Comments						

Hawaii Analytical Laboratory is a NIST NVLAP accredited laboratory (NVLAP Lab Code 200655-0) and is accredited in accordance with the recognized ISO/IEC 17025:2017. Controlled doc.: Asbestos Report, rev. 5 – 20241127

OYO Corporation, Pacific
Okinawa Times Building. 2F
1-12-24 Goya, Okinawa-shi,
Okinawa-ken , 904-0021 Japan
Okinawa Japan .

Phone Number:
Facsimile:
Email: aledezmaes@oyopacific.com

Lab Job No: 202504596
Date Submitted: 5/20/2025
Your Project: 232061, PN77833 Kadena VMS Hazmat Survey, Swing Space, 5/9/25

Bulk Asbestos Determination

Lab Sple No.	Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202532690	SS-004		NONE DETECTED		None detected	Binder	5/22/2025
	<u>Layer</u> <u>Black expansion joint</u>						
	Comments						
202532690	SS-004		NONE DETECTED		None detected	Cementitious + other	5/22/2025
	<u>Layer</u> <u>Gray concrete</u>						
	Comments						
202532690	SS-004		NONE DETECTED		None detected	Foam	5/22/2025
	<u>Layer</u> <u>White foam</u>						
	Comments						
202532691	SS-005		NONE DETECTED		None detected	Binder	5/22/2025
	<u>Layer</u> <u>Black expansion joint</u>						
	Comments						
202532691	SS-005		NONE DETECTED		None detected	Foam	5/22/2025
	<u>Layer</u> <u>White foam</u>						
	Comments						
202532692	SS-006		NONE DETECTED		None detected	Binder	5/22/2025
	<u>Layer</u> <u>Black expansion joint</u>						
	Comments						
202532692	SS-006		NONE DETECTED		Wood fiber / cellulose (undulose)	70 Tar + other	5/22/2025
	<u>Layer</u> <u>Black fibrous material</u>						
	Comments						
202532693	SS-007		NONE DETECTED		None detected	Cementitious + other	5/22/2025
	<u>Layer</u> <u>Gray concrete</u>						
	Comments						

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OYO Corporation, Pacific
Okinawa Times Building. 2F
1-12-24 Goya, Okinawa-shi,
Okinawa-ken , 904-0021 Japan
Okinawa Japan .

Phone Number:
Facsimile:
Email: aledezmaes@oyopacific.com

Lab Job No: 202504596

Date Submitted: 5/20/2025

Your Project: 232061, PN77833 Kadena VMS Hazmat Survey, Swing Space, 5/9/25

General Comments

The bulk sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures outlined in the United States Environmental Protection Agency's "Interim Method for the Determination of Asbestos in Bulk Insulation Samples" (EPA-600/M4-82-020, Dec. 1982) and / or "Method for Determination of Asbestos in bulk Building Materials" (EPA-600/R-93-116, July 1993). The analysis of each bulk sample relates only to the material examined, and may or may not represent the overall composition of its original source. Floor tile and other resinously bound materials, when analyzed by the EPA methods referenced above may yield false negative results because of limitations in separating closely bound fibers and in detecting fibers of small length and diameter. Gravimetric treatment, which HAL does not offer, may also be appropriate for certain NOB (non-friable organically bound) materials. Unless specifically requested by clients, NOB samples can be subcontracted to a NVLAP accredited lab, or else, they will be analyzed by HAL using regular PLM technique. In addition, alternative methods of identification, including Transmission Electron Microscopy (TEM) may or may not be applicable. We utilize calibrated visual area estimation on a routine basis and do not conduct point counting unless specifically requested to do so. Estimated error for the visual determinations presented are 75% relative (<1 to 10%), 65% relative (11 to 19%), 50% relative (20 to 34%); 40% relative (35 to 50%), 35% relative (51 to 60%), and 25% relative (>60% v/v). We will not separate layers which in our opinion are not readily discernable. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report must not be used by the client to claim product certification, approval or endorsement by NVLAP, NIST or any agency of Federal Government. Unless otherwise indicated, the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

< This testing result is less than the numerical value listed.

None Detected = asbestos was not observed in the sample. If trace amount of asbestos was detected below our quantifiable limits of 1.0%, <1% (trace) would be indicated and the asbestos type listed. Point counting, where applicable, are recommended to improve accuracy.



Jennifer Hsu Liao
Laboratory Manager

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DRAFT

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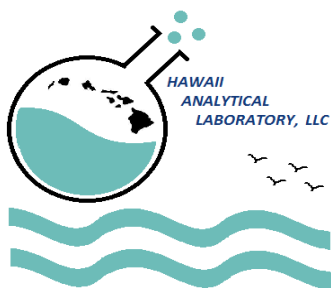
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APPENDIX C

Cadmium, & Chromium Laboratory Analytical Reports & Chains-of-Custody

DRAFT

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Hawaii Analytical Laboratory ANALYTICAL REPORT

Monday, January 8, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314102

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, 12/20/23

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202398004	4040-S1	< 11	mg/kg	1/8/2024
Comments:	MRL = 11 mg/kg			
202398005	4040-S2	< 17	mg/kg	1/8/2024
Comments:	MRL = 17 mg/kg			
202398006	4040-S3	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202398007	4040-S5	< 12	mg/kg	1/8/2024
Comments:	MRL = 12 mg/kg			
202398008	4040-S6	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202398009	4040-S7	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202398010	4040-S8	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202398011	4040-S9	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			

All Quality Control data are acceptable unless otherwise noted.

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314102

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, 12/20/23

Total Recoverable Chromium

EPA Method: 3051m / 7000Bm

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202398004	4040-S1	< 86	mg/kg	1/8/2024
Comments:	MRL = 86 mg/kg			
202398005	4040-S2	< 140	mg/kg	1/8/2024
Comments:	MRL = 140 mg/kg			
202398006	4040-S3	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202398007	4040-S5	< 93	mg/kg	1/8/2024
Comments:	MRL = 93 mg/kg			
202398008	4040-S6	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202398009	4040-S7	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202398010	4040-S8	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202398011	4040-S9	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			

All Quality Control data are acceptable unless otherwise noted.

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OYO Corporation, Pacific

Phone Number:

Okinawa Japan .

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314102

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, 12/20/23

General Comments

The sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures associated with the "analytical method" referenced above. Modifications to this methodology may have been made based upon the analyst's professional judgment and / or sample matrix effects encountered. The analysis of sample relates only to the sample analyzed, and may or may not be representative of the original source of the material submitted for our analysis. All analysts participate in interlaboratory quality control testing to continuously document proficiency. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report should not be construed as an endorsement for a product or a service by the AIHA LAP, LLC or any affiliated organizations. Sample and associated sampling / collection data is reported as provided by client. TWA values have been calculated based on information supplied by the client that the laboratory has not independently verified. Results have not been corrected for blank determinations unless noted in remarks. Unless otherwise indicated the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

< This testing result is less than the numerical value listed.

= Analytical methods marked with an "#" are not within our AIHA LAP, LLC Scope of Accreditation.

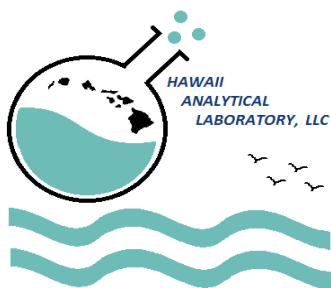
MRL = Method Reporting Limit.



Anne Antin

Quality Control Manager

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Hawaii Analytical Laboratory ANALYTICAL REPORT

Monday, January 8, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314093

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, 12/20/23

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202397891	4041-S1	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397892	4041-S2	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397893	4041-S3	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397895	4041-S5	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397896	4041-S6	< 12	mg/kg	1/8/2024
Comments:	MRL = 12 mg/kg			
202397897	4041-S7	< 11	mg/kg	1/8/2024
Comments:	MRL = 11 mg/kg			
202397898	4041-S8	< 16	mg/kg	1/8/2024
Comments:	MRL = 16 mg/kg			
202397899	4041-S9	< 11	mg/kg	1/8/2024
Comments:	MRL = 11 mg/kg			
202397900	4041-S10	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			

All Quality Control data are acceptable unless otherwise noted.

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OYO Corporation, Pacific

Phone Number:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Okinawa Japan .

Lab Job No: 202314093

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, 12/20/23

Total Recoverable Chromium

EPA Method: 3051m / 7000Bm

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202397891	4041-S1	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397892	4041-S2	83	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397893	4041-S3	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397895	4041-S5	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397896	4041-S6	< 93	mg/kg	1/8/2024
Comments:	MRL = 93 mg/kg			
202397897	4041-S7	< 84	mg/kg	1/8/2024
Comments:	MRL = 84 mg/kg			
202397898	4041-S8	< 130	mg/kg	1/8/2024
Comments:	MRL = 130 mg/kg			
202397899	4041-S9	< 85	mg/kg	1/8/2024
Comments:	MRL = 85 mg/kg			
202397900	4041-S10	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			

All Quality Control data are acceptable unless otherwise noted.

Hawaii Analytical Laboratory (101812) is accredited by the AIHA LAP, LLC in the EMLAP, IHLAP, and ELLAP programs for the scope of work listed on www.aihaaccreditedlabs.org, in accordance with the recognized ISO/ IEC 17025:2017 and participates in the CAPT proficiency testing program. Controlled doc.: Analytical Report, rev. 2 - 20160829

OYO Corporation, Pacific

Phone Number:

Okinawa Japan .

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314093

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, 12/20/23

General Comments

The sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures associated with the "analytical method" referenced above. Modifications to this methodology may have been made based upon the analyst's professional judgment and / or sample matrix effects encountered. The analysis of sample relates only to the sample analyzed, and may or may not be representative of the original source of the material submitted for our analysis. All analysts participate in interlaboratory quality control testing to continuously document proficiency. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report should not be construed as an endorsement for a product or a service by the AIHA LAP, LLC or any affiliated organizations. Sample and associated sampling / collection data is reported as provided by client. TWA values have been calculated based on information supplied by the client that the laboratory has not independently verified. Results have not been corrected for blank determinations unless noted in remarks. Unless otherwise indicated the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

< This testing result is less than the numerical value listed.

= Analytical methods marked with an "#" are not within our AIHA LAP, LLC Scope of Accreditation.

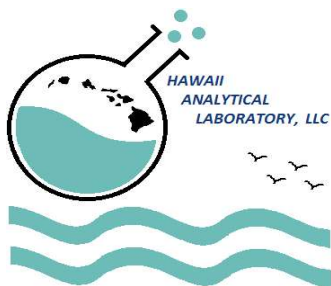
MRL = Method Reporting Limit.



Anne Antin

Quality Control Manager

Hawaii Analytical Laboratory (101812) is accredited by the AIHA LAP, LLC in the EMLAP, IHLAP, and ELLAP programs for the scope of work listed on www.aihaaccreditedlabs.org, in accordance with the recognized ISO/ IEC 17025:2017 and participates in the CAPT proficiency testing program. Controlled doc.: Analytical Report, rev. 2 - 20160829



Hawaii Analytical Laboratory ANALYTICAL REPORT

Friday, February 2, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401043

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg. 4042

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407895	4042-51	< 28	mg/kg	1/30/2024
Comments	MRL = 28 mg/kg			
202407896	4042-52	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			
202407897	4042-53	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			
202407898	4042-54	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			
202407899	4042-55	< 27	mg/kg	1/30/2024
Comments	MRL = 27 mg/kg			
202407900	4042-56	< 42	mg/kg	1/30/2024
Comments	MRL = 42 mg/kg			
202407901	4042-57	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			
202407902	4042-58	< 17	mg/kg	1/30/2024
Comments	MRL = 17 mg/kg			
202407903	4042-59	< 12	mg/kg	1/30/2024
Comments	MRL = 12 mg/kg			

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3615 Harding Avenue, Ste. 308, Honolulu, HI 96816 - Telephone: (808) 735-0422 - Fax: (808) 735-0047

Page 1 of 4

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401043

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg. 4042

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407904	4042-510	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			

DRAFT

Hawaii Analytical Laboratory (101812) is accredited by the AIHA LAP, LLC in the EMLAP, IHLAP, and ELLAP programs for the scope of work listed on www.aihaaccreditedlabs.org, in accordance with the recognized ISO/ IEC 17025:2017. AIHA LAP, LLC is a NLLAP recognized accrediting body. Controlled doc.: Lead Report, rev. 3 - 20181015

3615 Harding Avenue, Ste. 308, Honolulu, HI 96816 - Telephone: (808) 735-0422 - Fax: (808) 735-0047

Page 2 of 4

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401043

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg. 4042

Total Recoverable Chromium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407895	4042-51	< 220	mg/kg	1/30/2024
Comments	MRL = 220 mg/kg			
202407896	4042-52	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			
202407897	4042-53	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			
202407898	4042-54	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			
202407899	4042-55	< 210	mg/kg	1/30/2024
Comments	MRL = 220 mg/kg			
202407900	4042-56	< 340	mg/kg	1/30/2024
Comments	MRL = 340 mg/kg			
202407901	4042-57	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			
202407902	4042-58	< 140	mg/kg	1/30/2024
Comments	MRL = 140 mg/kg			
202407903	4042-59	< 97	mg/kg	1/30/2024
Comments	MRL = 97 mg/kg			
202407904	4042-510	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			

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3615 Harding Avenue, Ste. 308, Honolulu, HI 96816 - Telephone: (808) 735-0422 - Fax: (808) 735-0047

Page 3 of 4

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401043

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg. 4042

All Quality Control data are acceptable unless otherwise noted.

MRL for lead air is 5ug.

MRL for lead wipe is 10ug.

MRL for lead paint or soil is 40 mg/kg for a 0.25g sample.

General Comments

The sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures associated with the "analytical method" referenced above. Modifications to this methodology may have been made based upon the analyst's professional judgment and / or sample matrix effects encountered. The analysis of sample relates only to the sample analyzed, and may or may not be representative of the original source of the material submitted for our analysis. All analysts participate in interlaboratory quality control testing to continuously document proficiency. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report should not be construed as an endorsement for a product or a service by the AIHA LAP, LLC or any affiliated organizations. Sample and associated sampling / collection data is reported as provided by client. TWA values have been calculated based on information supplied by the client that the laboratory has not independently verified. Results have not been corrected for blank determinations unless noted in remarks. Unless otherwise indicated the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

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MRL = Method Reporting Limit.



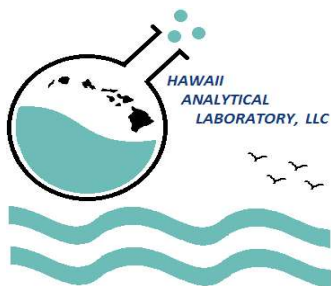
Anne Antin

Quality Control Manager

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3615 Harding Avenue, Ste. 308, Honolulu, HI 96816 - Telephone: (808) 735-0422 - Fax: (808) 735-0047

Page 4 of 4



Hawaii Analytical Laboratory ANALYTICAL REPORT

Friday, February 2, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401044

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg. 4043

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407905	4043-51	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			
202407906	4043-52	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			
202407907	4043-53	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			
202407908	4043-54	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			
202407909	4043-55	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			
202407910	4043-56	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			
202407911	4043-57	< 17	mg/kg	1/30/2024
Comments	MRL = 17 mg/kg			
202407912	4043-58	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			
202407913	4043-59	< 20	mg/kg	1/30/2024
Comments	MRL = 20 mg/kg			

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Page 1 of 4

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401044

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg. 4043

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407914	4043-510	< 10	mg/kg	1/30/2024
Comments	MRL = 10 mg/kg			

DRAFT

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Okinawa Japan .

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Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401044

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg. 4043

Total Recoverable Chromium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407905	4043-51	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			
202407906	4043-52	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			
202407907	4043-53	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			
202407908	4043-54	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			
202407909	4043-55	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			
202407910	4043-56	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			
202407911	4043-57	< 140	mg/kg	1/30/2024
Comments	MRL = 140 mg/kg			
202407912	4043-58	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			
202407913	4043-59	< 160	mg/kg	1/30/2024
Comments	MRL = 160 mg/kg			
202407914	4043-510	< 80	mg/kg	1/30/2024
Comments	MRL = 80 mg/kg			

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Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401044

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg. 4043

All Quality Control data are acceptable unless otherwise noted.

MRL for lead air is 5ug.

MRL for lead wipe is 10ug.

MRL for lead paint or soil is 40 mg/kg for a 0.25g sample.

General Comments

The sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures associated with the "analytical method" referenced above. Modifications to this methodology may have been made based upon the analyst's professional judgment and / or sample matrix effects encountered. The analysis of sample relates only to the sample analyzed, and may or may not be representative of the original source of the material submitted for our analysis. All analysts participate in interlaboratory quality control testing to continuously document proficiency. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report should not be construed as an endorsement for a product or a service by the AIHA LAP, LLC or any affiliated organizations. Sample and associated sampling / collection data is reported as provided by client. TWA values have been calculated based on information supplied by the client that the laboratory has not independently verified. Results have not been corrected for blank determinations unless noted in remarks. Unless otherwise indicated the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

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MRL = Method Reporting Limit.



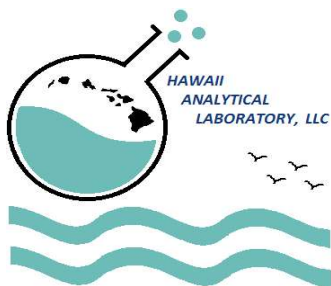
Anne Antin

Quality Control Manager

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Hawaii Analytical Laboratory ANALYTICAL REPORT

Friday, February 2, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401045

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg. 4044

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407915	4044-51	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407916	4044-52	< 16	mg/kg	2/1/2024
Comments	MRL = 16 mg/kg			
202407917	4044-53	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407918	4044-54	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407920	4044-56	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407921	4044-57	< 21	mg/kg	2/1/2024
Comments	MRL = 21 mg/kg			
202407922	4044-58	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407923	4044-59	< 17	mg/kg	2/1/2024
Comments	MRL = 17 mg/kg			
202407924	4044-510	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401045

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg. 4044

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407925	4044-511	< 14	mg/kg	2/1/2024
Comments	MRL = 14 mg/kg			
202407926	4044-512	< 23	mg/kg	2/1/2024
Comments	MRL = 23 mg/kg			
202407927	4044-513	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			

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Phone Number:

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Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401045

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg. 4044

Total Recoverable Chromium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407915	4044-51	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407916	4044-52	< 130	mg/kg	2/1/2024
Comments	MRL = 130 mg/kg			
202407917	4044-53	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407918	4044-54	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407920	4044-56	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407921	4044-57	< 160	mg/kg	2/1/2024
Comments	MRL = 160 mg/kg			
202407922	4044-58	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407923	4044-59	< 140	mg/kg	2/1/2024
Comments	MRL = 140 mg/kg			
202407924	4044-510	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407925	4044-511	< 110	mg/kg	2/1/2024
Comments	MRL = 110 mg/kg			
202407926	4044-512	< 180	mg/kg	2/1/2024
Comments	MRL = 180 mg/kg			
202407927	4044-513	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			

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aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401045

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg. 4044

All Quality Control data are acceptable unless otherwise noted.

MRL for lead air is 5ug.

MRL for lead wipe is 10ug.

MRL for lead paint or soil is 40 mg/kg for a 0.25g sample.

General Comments

The sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures associated with the "analytical method" referenced above. Modifications to this methodology may have been made based upon the analyst's professional judgment and / or sample matrix effects encountered. The analysis of sample relates only to the sample analyzed, and may or may not be representative of the original source of the material submitted for our analysis. All analysts participate in interlaboratory quality control testing to continuously document proficiency. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report should not be construed as an endorsement for a product or a service by the AIHA LAP, LLC or any affiliated organizations. Sample and associated sampling / collection data is reported as provided by client. TWA values have been calculated based on information supplied by the client that the laboratory has not independently verified. Results have not been corrected for blank determinations unless noted in remarks. Unless otherwise indicated the sample condition at the time of receipt was acceptable.

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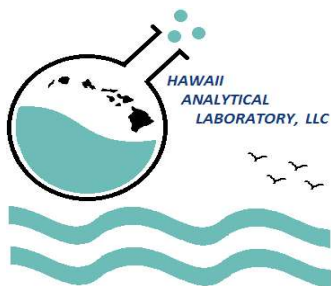
Anne Antin

Quality Control Manager

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Hawaii Analytical Laboratory ANALYTICAL REPORT

Friday, February 2, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401046

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survery, Kadena Air Base, Bldg 4045

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407928	4045-51	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407929	4045-52	< 17	mg/kg	2/1/2024
Comments	MRL = 17 mg/kg			
202407930	4045-54	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407931	4045-56	< 16	mg/kg	2/1/2024
Comments	MRL = 16 mg/kg			
202407932	4045-57	< 15	mg/kg	2/1/2024
Comments	MRL = 15 mg/kg			
202407933	4045-58	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407934	4045-59	< 20	mg/kg	2/1/2024
Comments	MRL = 20 mg/kg			
202407935	4045-510	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407936	4045-511	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			

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Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401046

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survery, Kadena Air Base, Bldg 4045

Total Recoverable Chromium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407928	4045-51	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407929	4045-52	< 140	mg/kg	2/1/2024
Comments	MRL = 140 mg/kg			
202407930	4045-54	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407931	4045-56	< 130	mg/kg	2/1/2024
Comments	MRL = 130 mg/kg			
202407932	4045-57	5400	mg/kg	2/1/2024
Comments	MRL = 120 mg/kg			
202407933	4045-58	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407934	4045-59	< 160	mg/kg	2/1/2024
Comments	MRL = 160 mg/kg			
202407935	4045-510	1400	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407936	4045-511	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			

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Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401046

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survery, Kadena Air Base, Bldg 4045

All Quality Control data are acceptable unless otherwise noted.

MRL for lead air is 5ug.

MRL for lead wipe is 10ug.

MRL for lead paint or soil is 40 mg/kg for a 0.25g sample.

General Comments

The sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures associated with the "analytical method" referenced above. Modifications to this methodology may have been made based upon the analyst's professional judgment and / or sample matrix effects encountered. The analysis of sample relates only to the sample analyzed, and may or may not be representative of the original source of the material submitted for our analysis. All analysts participate in interlaboratory quality control testing to continuously document proficiency. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report should not be construed as an endorsement for a product or a service by the AIHA LAP, LLC or any affiliated organizations. Sample and associated sampling / collection data is reported as provided by client. TWA values have been calculated based on information supplied by the client that the laboratory has not independently verified. Results have not been corrected for blank determinations unless noted in remarks. Unless otherwise indicated the sample condition at the time of receipt was acceptable.

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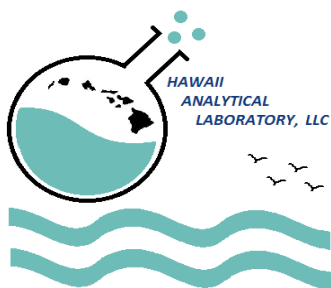
Anne Antin

Quality Control Manager

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Hawaii Analytical Laboratory ANALYTICAL REPORT

Monday, January 8, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314099

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4046, 11/30/23

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202397969	4046-S1	< 13	mg/kg	1/8/2024
Comments:	MRL = 13 mg/kg			
202397970	4046-S2	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397971	4046-S3	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397972	4046-S4	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397973	4046-S5	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397973.1	4046-S6	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397974	4046-S7	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397975	4046-S8	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397976	4046-S9	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314099

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4046, 11/30/23

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202397977	4046-S10	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397978	4046-S11	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			

All Quality Control data are acceptable unless otherwise noted.

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OYO Corporation, Pacific

Phone Number:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Okinawa Japan .

Lab Job No: 202314099

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4046, 11/30/23

Total Recoverable Chromium

EPA Method: 3051m / 7000Bm

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202397969	4046-S1	< 110	mg/kg	1/8/2024
Comments:	MRL = 110 mg/kg			
202397970	4046-S2	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397971	4046-S3	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397972	4046-S4	< 83	mg/kg	1/8/2024
Comments:	MRL = 83 mg/kg			
202397973	4046-S5	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397973.1	4046-S6	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397974	4046-S7	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397975	4046-S8	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397976	4046-S9	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397977	4046-S10	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397978	4046-S11	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			

All Quality Control data are acceptable unless otherwise noted.

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314099

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4046, 11/30/23

General Comments

The sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures associated with the "analytical method" referenced above. Modifications to this methodology may have been made based upon the analyst's professional judgment and / or sample matrix effects encountered. The analysis of sample relates only to the sample analyzed, and may or may not be representative of the original source of the material submitted for our analysis. All analysts participate in interlaboratory quality control testing to continuously document proficiency. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report should not be construed as an endorsement for a product or a service by the AIHA LAP, LLC or any affiliated organizations. Sample and associated sampling / collection data is reported as provided by client. TWA values have been calculated based on information supplied by the client that the laboratory has not independently verified. Results have not been corrected for blank determinations unless noted in remarks. Unless otherwise indicated the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

< This testing result is less than the numerical value listed.

= Analytical methods marked with an "#" are not within our AIHA LAP, LLC Scope of Accreditation.

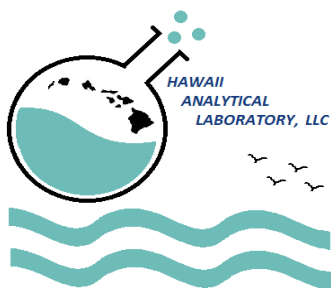
MRL = Method Reporting Limit.



Anne Antin

Quality Control Manager

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Hawaii Analytical Laboratory ANALYTICAL REPORT

Monday, January 8, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314097

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4047, 11/30/23

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202397937	4047-S1	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397938	4047-S2	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397939	4047-S3	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397940	4047-S4	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397942	4047-S6	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397943	4047-S7	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397944	4047-S8	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397945	4047-S9	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			
202397946	4047-S10	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314097

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4047, 11/30/23

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202397947	4047-S11	< 10	mg/kg	1/8/2024
Comments:	MRL = 10 mg/kg			

All Quality Control data are acceptable unless otherwise noted.

DRAFT

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OYO Corporation, Pacific

Phone Number:

Email:

aledesmaes@oyopacific.com;

kbull@oyopacific.com

Okinawa Japan .

Lab Job No: 202314097

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4047, 11/30/23

Total Recoverable Chromium

EPA Method: 3051m / 7000Bm

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202397937	4047-S1	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397938	4047-S2	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397939	4047-S3	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397940	4047-S4	110	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397942	4047-S6	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397943	4047-S7	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397944	4047-S8	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397945	4047-S9	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397946	4047-S10	2100	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			
202397947	4047-S11	< 80	mg/kg	1/8/2024
Comments:	MRL = 80 mg/kg			

All Quality Control data are acceptable unless otherwise noted.

Hawaii Analytical Laboratory (101812) is accredited by the AIHA LAP, LLC in the EMLAP, IHLAP, and ELLAP programs for the scope of work listed on www.aihaaccreditedlabs.org, in accordance with the recognized ISO/ IEC 17025:2017 and participates in the CAPT proficiency testing program. Controlled doc.: Analytical Report, rev. 2 - 20160829

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Email:

aledezmaes@oyopacific.com;

kbull@oyopacific.com

Lab Job No: 202314097

Date Submitted: 12/29/2023

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey Bldg. 4047, 11/30/23

General Comments

The sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures associated with the "analytical method" referenced above. Modifications to this methodology may have been made based upon the analyst's professional judgment and / or sample matrix effects encountered. The analysis of sample relates only to the sample analyzed, and may or may not be representative of the original source of the material submitted for our analysis. All analysts participate in interlaboratory quality control testing to continuously document proficiency. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report should not be construed as an endorsement for a product or a service by the AIHA LAP, LLC or any affiliated organizations. Sample and associated sampling / collection data is reported as provided by client. TWA values have been calculated based on information supplied by the client that the laboratory has not independently verified. Results have not been corrected for blank determinations unless noted in remarks. Unless otherwise indicated the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

< This testing result is less than the numerical value listed.

= Analytical methods marked with an "#" are not within our AIHA LAP, LLC Scope of Accreditation.

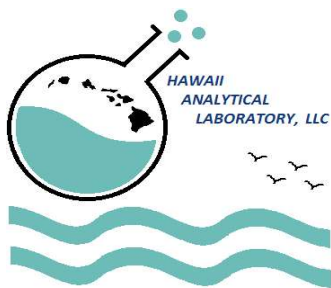
MRL = Method Reporting Limit.



Anne Antin

Quality Control Manager

Hawaii Analytical Laboratory (101812) is accredited by the AIHA LAP, LLC in the EMLAP, IHLAP, and ELLAP programs for the scope of work listed on www.aihaaccreditedlabs.org, in accordance with the recognized ISO/ IEC 17025:2017 and participates in the CAPT proficiency testing program. Controlled doc.: Analytical Report, rev. 2 - 20160829



Hawaii Analytical Laboratory ANALYTICAL REPORT

Friday, February 2, 2024

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401047

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg 4048

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407937	4048-5001	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407938	4048-5002	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407939	4048-5003	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407940	4048-5004	11	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407941	4048-5005	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			
202407942	4048-5006	< 10	mg/kg	2/1/2024
Comments	MRL = 10 mg/kg			

Hawaii Analytical Laboratory (101812) is accredited by the AIHA LAP, LLC in the EMLAP, IHLAP, and ELLAP programs for the scope of work listed on www.aihaaccreditedlabs.org, in accordance with the recognized ISO/ IEC 17025:2017. AIHA LAP, LLC is a NLLAP recognized accrediting body. Controlled doc.: Lead Report, rev. 3 - 20181015

3615 Harding Avenue, Ste. 308, Honolulu, HI 96816 - Telephone: (808) 735-0422 - Fax: (808) 735-0047

Page 1 of 3

OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401047

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg 4048

Total Recoverable Chromium

EPA Method: 3051m / 7000Bm

Sample No.	Your Sample ID / Description	Results	Units	Date Analyzed
202407937	4048-5001	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407938	4048-5002	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407939	4048-5003	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407940	4048-5004	< 80	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407941	4048-5005	410	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			
202407942	4048-5006	690	mg/kg	2/1/2024
Comments	MRL = 80 mg/kg			

Hawaii Analytical Laboratory (101812) is accredited by the AIHA LAP, LLC in the EMLAP, IHLAP, and ELLAP programs for the scope of work listed on www.aihaaccreditedlabs.org, in accordance with the recognized ISO/ IEC 17025:2017. AIHA LAP, LLC is a NLLAP recognized accrediting body. Controlled doc.: Lead Report, rev. 3 - 20181015

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OYO Corporation, Pacific

Okinawa Japan .

Phone Number:

Facsimile:

Email:

aledezmaes@oyopacific.com;
kbull@oyopacific.com,
khirota@oyopacific.com (1/16/24 TS)

Lab Job No: 202401047

Date Submitted: 1/26/2024

Project Name: 232061, PN77833 Kadena VMS Hazmat Survey, Kadena Air Base, Bldg 4048

All Quality Control data are acceptable unless otherwise noted.

MRL for lead air is 5ug.

MRL for lead wipe is 10ug.

MRL for lead paint or soil is 40 mg/kg for a 0.25g sample.

General Comments

The sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures associated with the "analytical method" referenced above. Modifications to this methodology may have been made based upon the analyst's professional judgment and / or sample matrix effects encountered. The analysis of sample relates only to the sample analyzed, and may or may not be representative of the original source of the material submitted for our analysis. All analysts participate in interlaboratory quality control testing to continuously document proficiency. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report should not be construed as an endorsement for a product or a service by the AIHA LAP, LLC or any affiliated organizations. Sample and associated sampling / collection data is reported as provided by client. TWA values have been calculated based on information supplied by the client that the laboratory has not independently verified. Results have not been corrected for blank determinations unless noted in remarks. Unless otherwise indicated the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

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< This testing result is less than the numerical value listed.

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MRL = Method Reporting Limit.



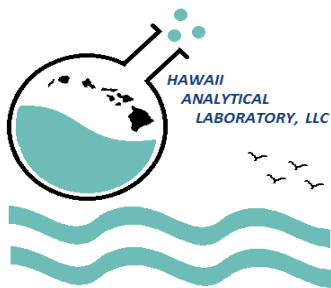
Anne Antin

Quality Control Manager

Hawaii Analytical Laboratory (101812) is accredited by the AIHA LAP, LLC in the EMLAP, IHLAP, and ELLAP programs for the scope of work listed on www.aihaaccreditedlabs.org, in accordance with the recognized ISO/ IEC 17025:2017. AIHA LAP, LLC is a NLLAP recognized accrediting body. Controlled doc.: Lead Report, rev. 3 - 20181015

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Page 3 of 3



Hawaii Analytical Laboratory ANALYTICAL REPORT

Wednesday, May 28, 2025

OYO Corporation, Pacific
Okinawa Times Building. 2F
1-12-24 Goya, Okinawa-shi,
Okinawa-ken, 904-0021 Japan
Okinawa Japan.

Phone Number:
Email: aledeszmaes@oyopacific.com

Lab Job No: 202504595
Total Analyzed: 3
Date Collected: 5/9/2025
Date Submitted: 5/20/2025
Project Name: 232061, PN77833 Kadena VMS Hazmat Survey (Swing Space)

Total Lead (paint chips)

NIOSH Method: 7082m LEAD by FAAS

Lab Sple No.	Sample ID / Description	Results	Units	Date Analyzed
202532684	SS-S1 MRL = 40 mg/kg	< 40	mg/kg	5/22/2025
202532685	SS-S2 MRL = 40 mg/kg	320	mg/kg	5/22/2025
202532686	SS-S3 MRL = 40 mg/kg	< 40	mg/kg	5/22/2025

All Quality Control data are acceptable unless otherwise noted.

Total Recoverable Cadmium

EPA Method: 3051m / 7000Bm

Lab Sple No.	Sample ID / Description	Results	Units	Date Analyzed
202532684	SS-S1 MRL = 10 mg/kg	< 10	mg/kg	5/22/2025
202532685	SS-S2 MRL = 10 mg/kg	< 10	mg/kg	5/22/2025
202532686	SS-S3 MRL = 10 mg/kg	< 10	mg/kg	5/22/2025

All Quality Control data are acceptable unless otherwise noted.

Hawaii Analytical Laboratory (101812) is accredited by the AIHA LAP, LLC in the EMLAP, IHLAP, and ELLAP programs for the scope of work listed on www.aihaaccreditedlabs.org, in accordance with the recognized ISO/IEC 17025:2017. AIHA LAP, LLC is a NLLAP recognized accrediting body. Controlled doc.: Analytical Report, rev. 6 - 20250123

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Page 1 of 2

OYO Corporation, Pacific
Okinawa Times Building. 2F
1-12-24 Goya, Okinawa-shi,
Okinawa-ken , 904-0021 Japan
Okinawa Japan .

Phone Number:
Email: aledezmaes@oyopacific.com

Lab Job No: 202504595
Total Analyzed: 3
Date Collected: 5/9/2025
Date Submitted: 5/20/2025
Project Name: 232061, PN77833 Kadena VMS Hazmat Survey (Swing Space)

Total Recoverable Chromium

EPA Method: 3051m / 7000Bm

Lab Sple No.	Sample ID / Description	Results	Units	Date Analyzed
202532684	SS-S1 MRL = 80 mg/kg	< 80	mg/kg	5/22/2025
202532685	SS-S2 MRL = 80 mg/kg	< 80	mg/kg	5/22/2025
202532686	SS-S3 MRL = 80 mg/kg	< 80	mg/kg	5/22/2025

All Quality Control data are acceptable unless otherwise noted.

General Comments

The sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures associated with the "analytical method" referenced above. The analysis of sample relates only to the sample analyzed, and may or may not be representative of the original source of the material submitted for our analysis. All analysts participate in interlaboratory quality control testing to continuously document proficiency. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report should not be construed as an endorsement for a product or a service by the AIHA LAP, LLC or any affiliated organizations. Sample and associated sampling / collection data (e.g. air volume or surface area) is reported as provided by client. TWA values have been calculated based on information supplied by the client that the laboratory has not independently verified. Results have not been corrected for blank determinations unless noted in remarks. Unless otherwise indicated the sample condition at the time of receipt was acceptable. Measurement of uncertainty for lead in paint, dust, airborne particulates, and soil taken from and around buildings and related structures is available upon request. MRL for lead air is 5ug; MRL for lead wipe is 5ug; MRL for lead paint or soil is 40 mg/kg for a 0.25g

Results and Symbols Definitions

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< This testing result is less than the numerical value listed.

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MRL = Method Reporting Limit



Anne Antin
Quality Control Manager

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Page 2 of 2

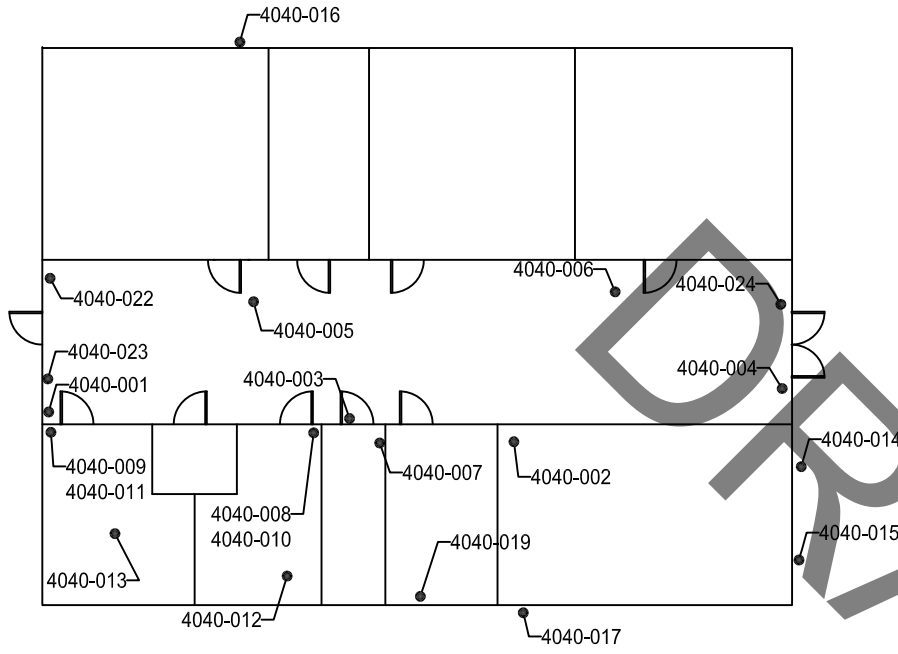
DRAFT

APPENDIX D

Sample Location Drawings

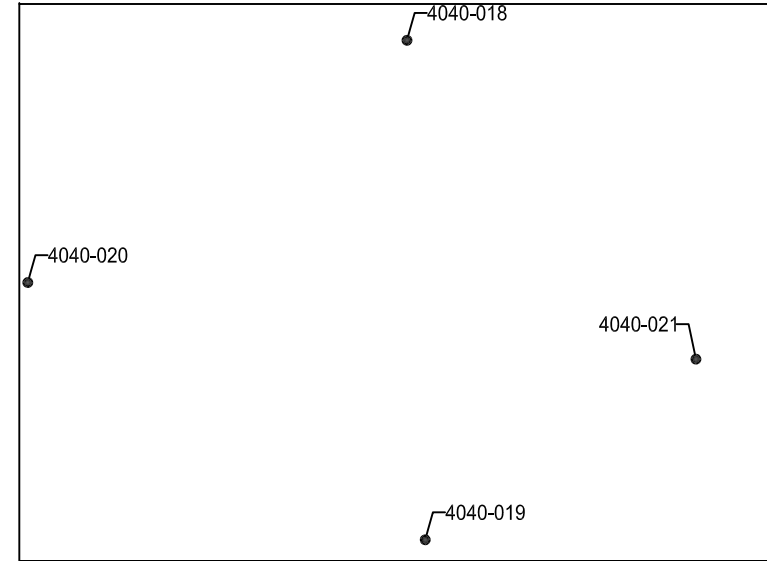
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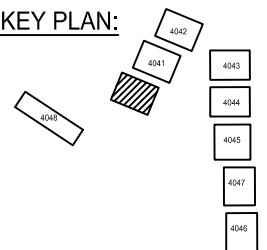
N.T.S.



ASBESTOS SAMPLING LOCATIONS
(ROOF)

N.T.S.

KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4040

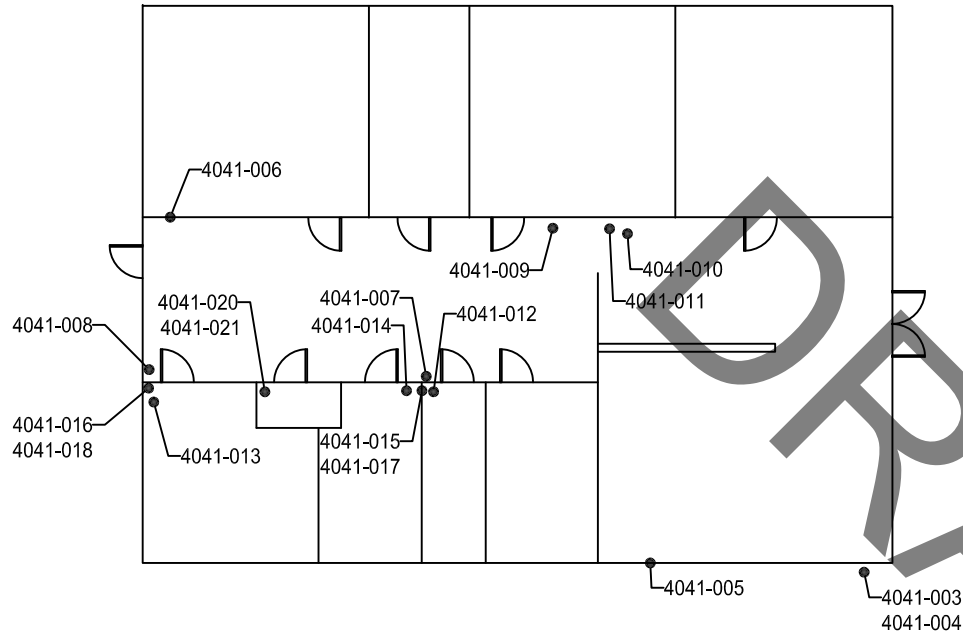
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1

OF 18

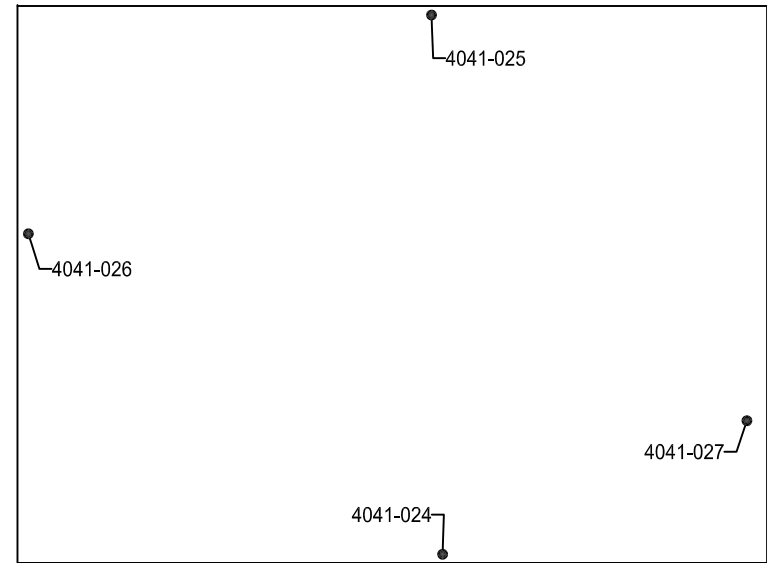


OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



ASBESTOS SAMPLING LOCATIONS
(1st FLOOR)

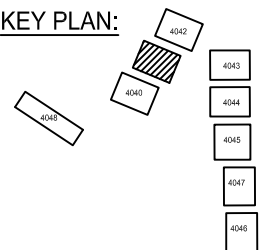
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ASBESTOS SAMPLING LOCATIONS
(ROOF)

N.T.S.

KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4041

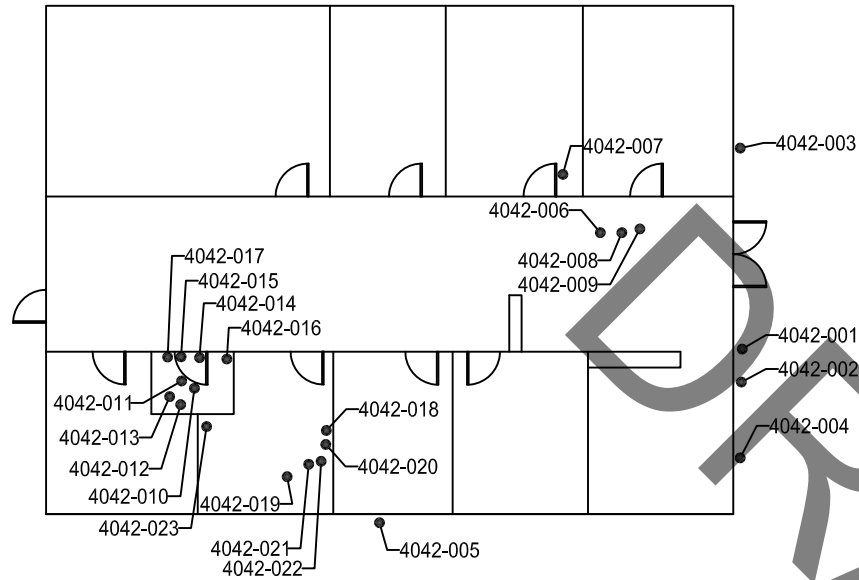
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OF 18

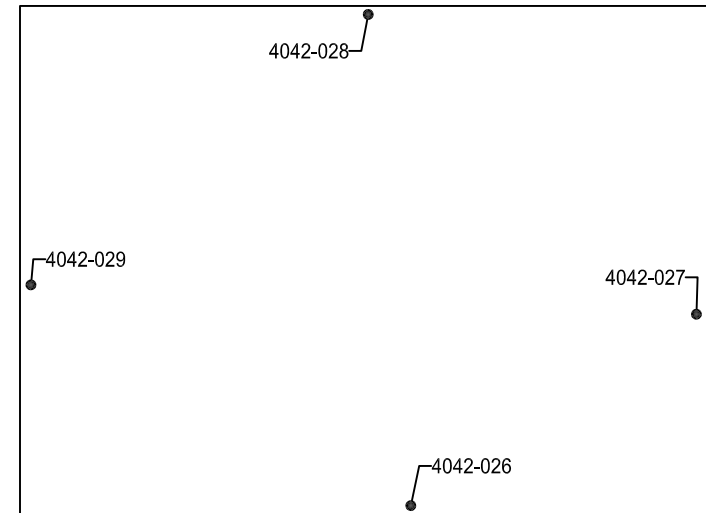


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1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



ASBESTOS SAMPLING LOCATIONS
(1st FLOOR)

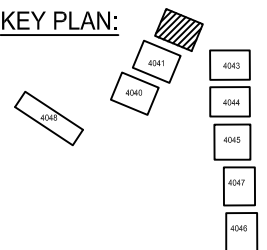
N.T.S.



ASBESTOS SAMPLING LOCATIONS
(ROOF)

N.T.S.

KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4042

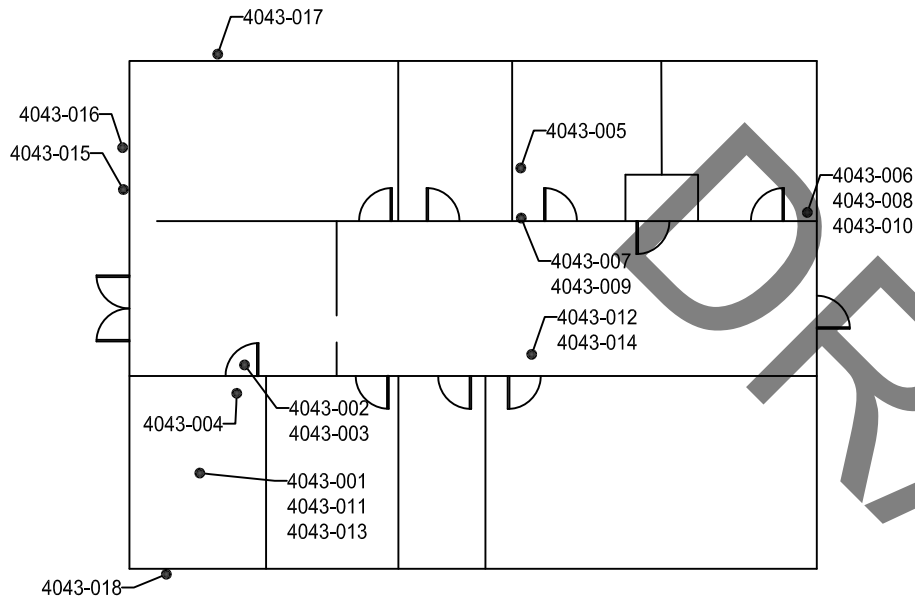
SHEET

3

OF 18

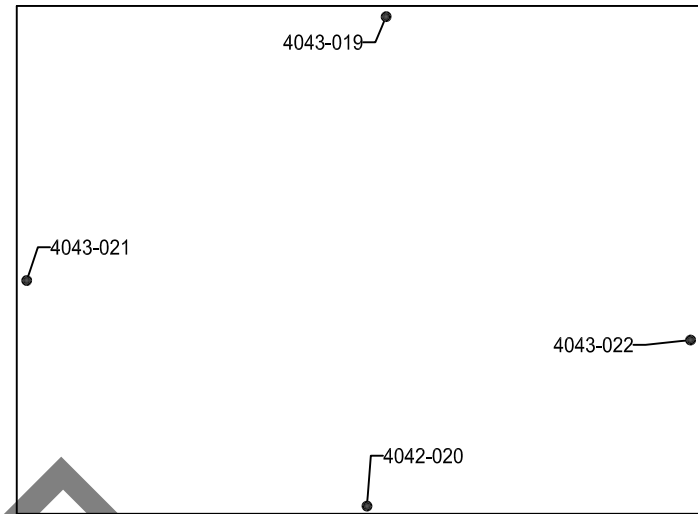


OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



ASBESTOS SAMPLING LOCATIONS
(1st FLOOR)

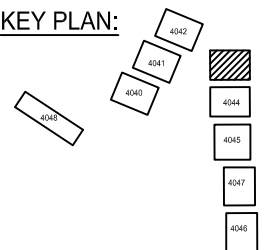
N.T.S.



ASBESTOS SAMPLING LOCATIONS
(ROOF)

N.T.S.

KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4043

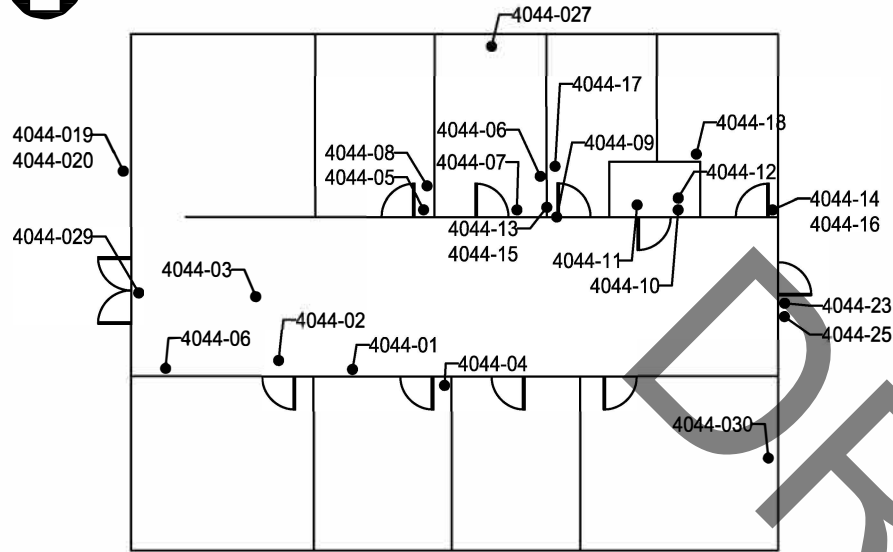
SHEET

4

OF 18

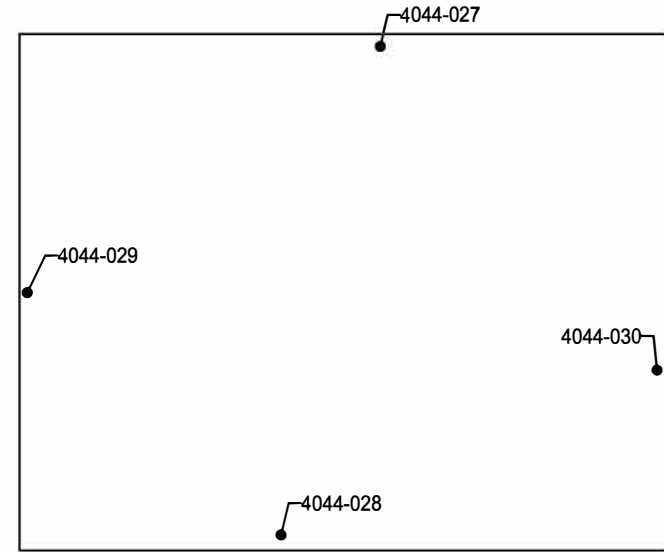


OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



ASBESTOS SAMPLING LOCATIONS
(1st FLOOR)

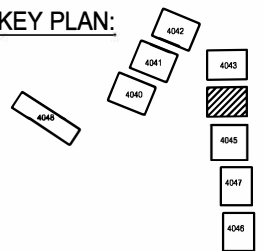
N.T.S.



ASBESTOS SAMPLING LOCATIONS
(ROOF)

N.T.S.

KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4044

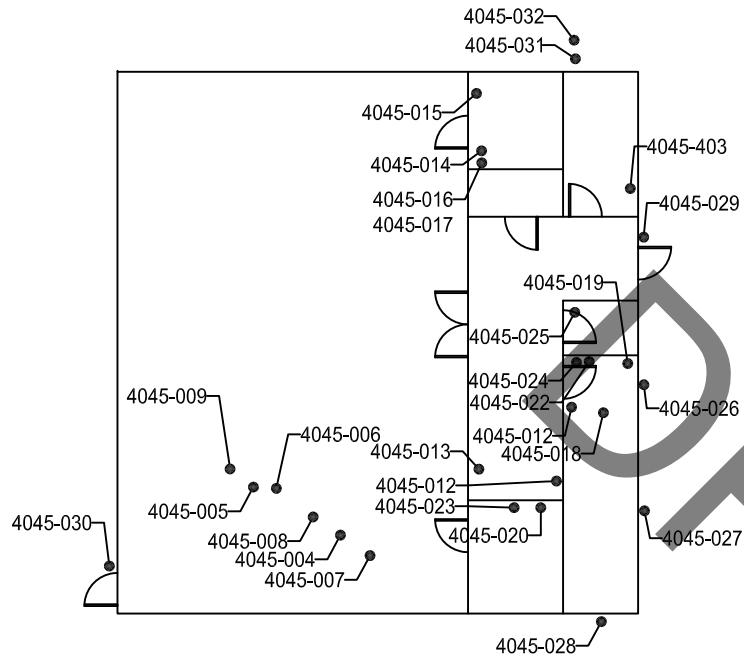
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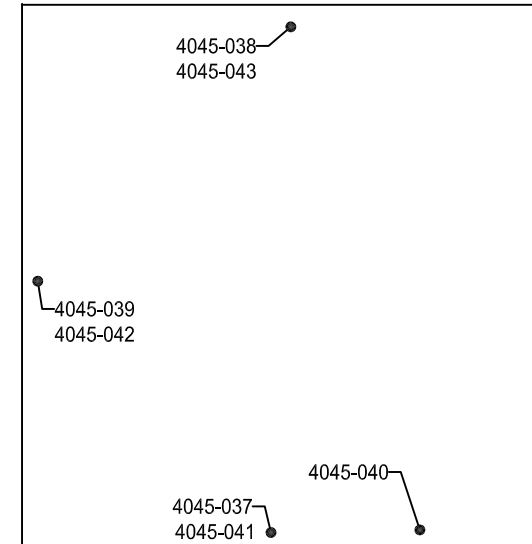


OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



ASBESTOS SAMPLING LOCATIONS
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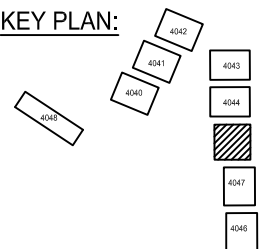
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ASBESTOS SAMPLING LOCATIONS
(ROOF)

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KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4045

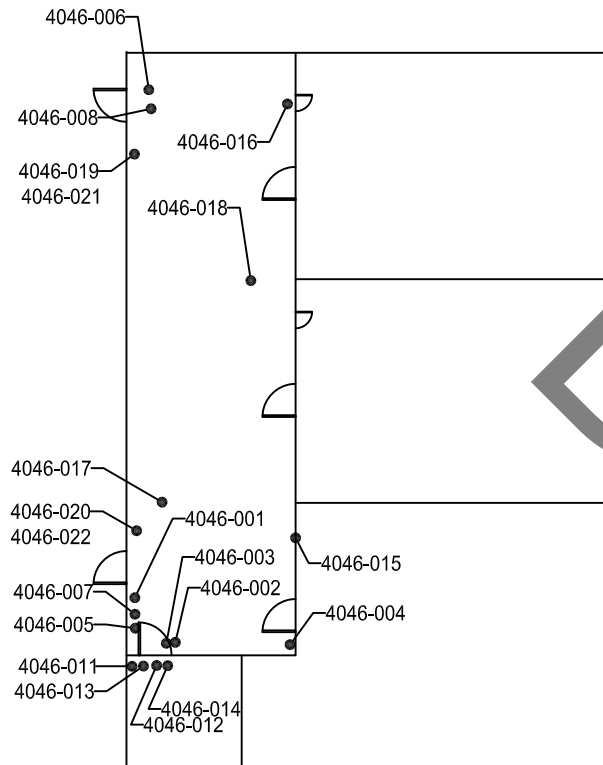
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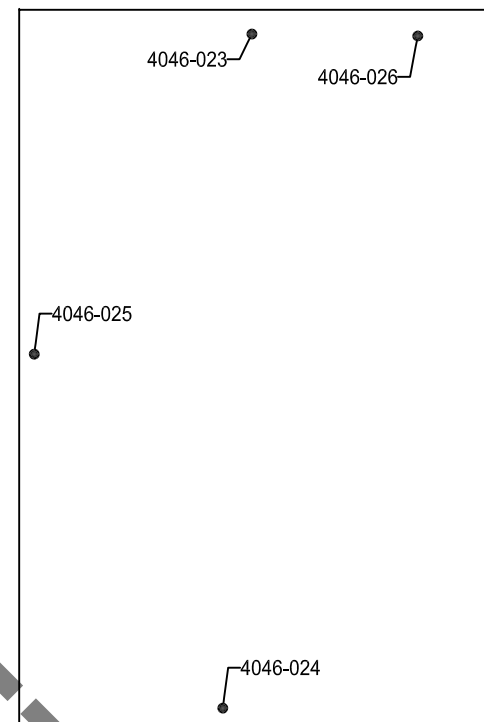


OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



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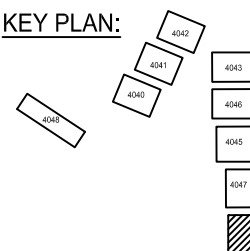
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KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4046

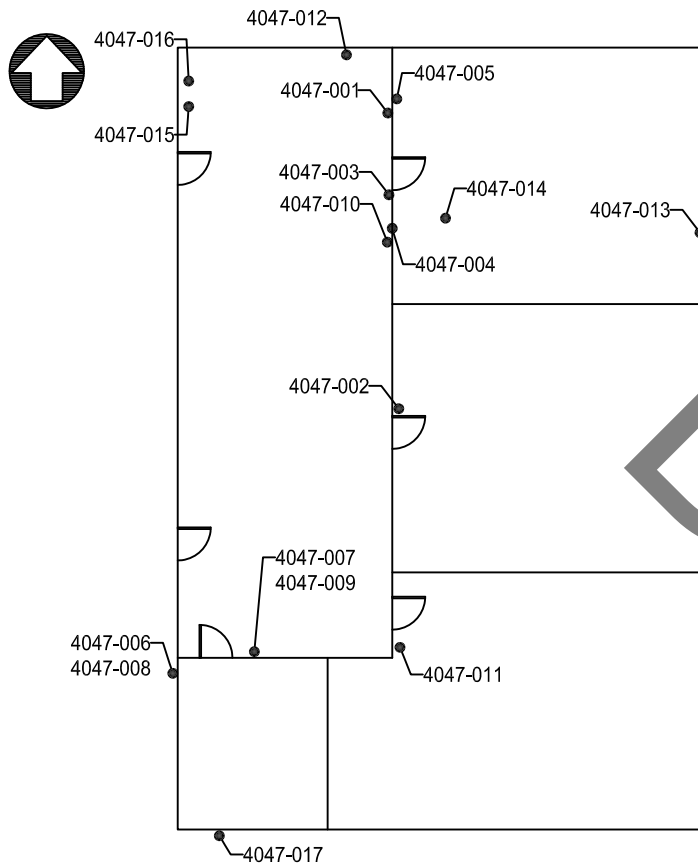
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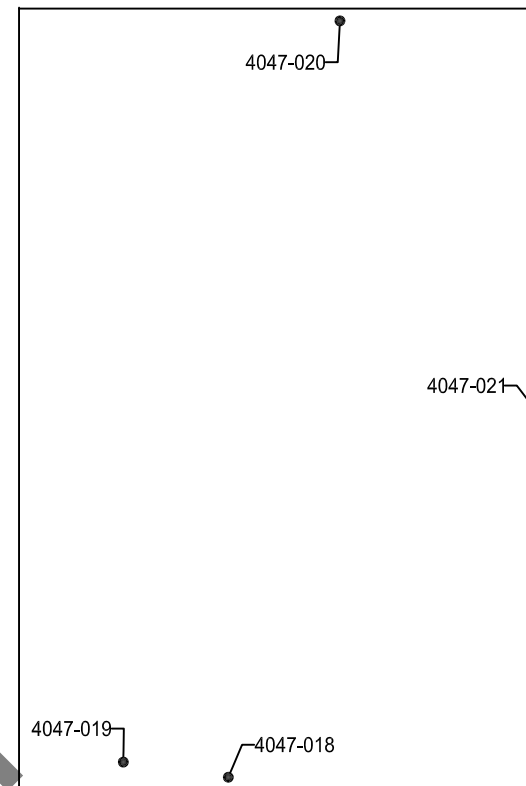


OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



ASBESTOS SAMPLING LOCATIONS
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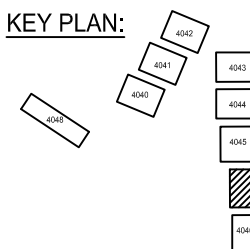
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ASBESTOS SAMPLING LOCATIONS
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KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4047

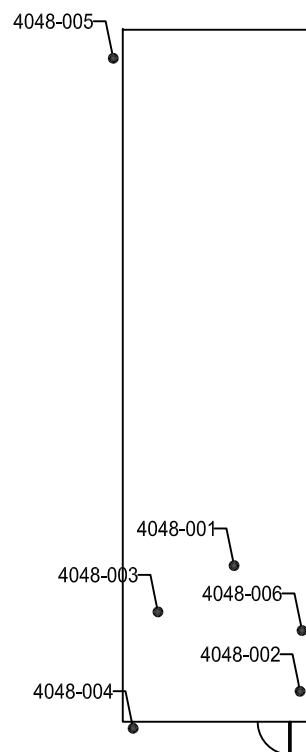
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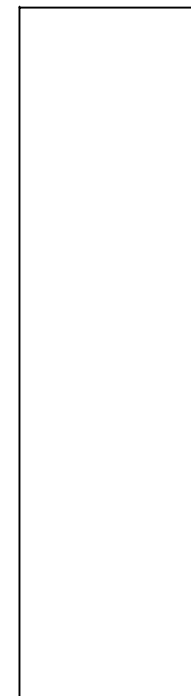


OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



ASBESTOS SAMPLING LOCATIONS
(1st FLOOR)

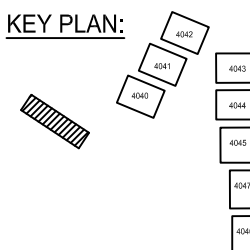
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(ROOF)

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KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4048

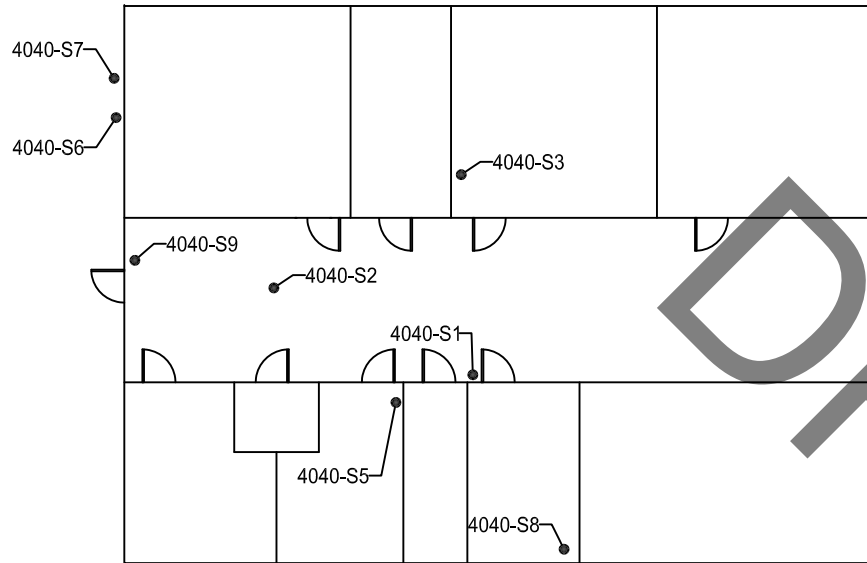
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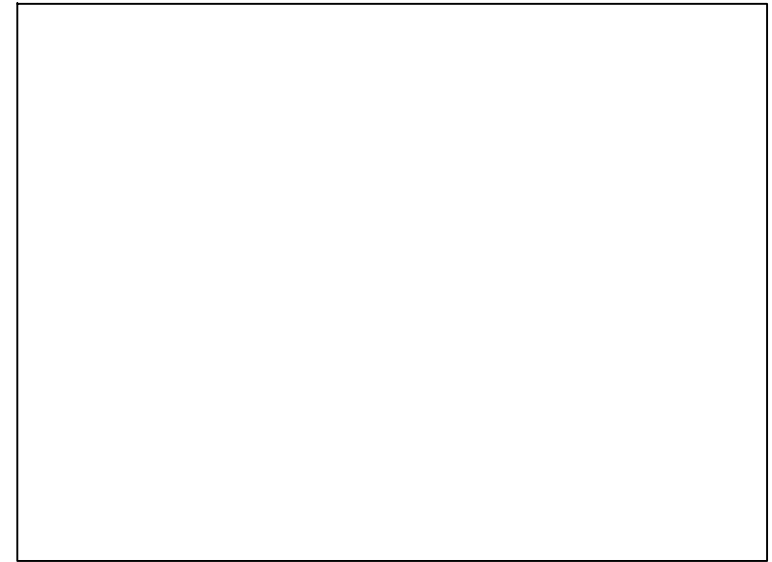


OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



CADMIUM / CHROMIUM SAMPLING LOCATIONS
(1st FLOOR)

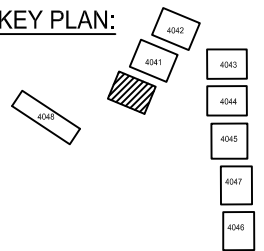
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KEY PLAN:



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LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4040

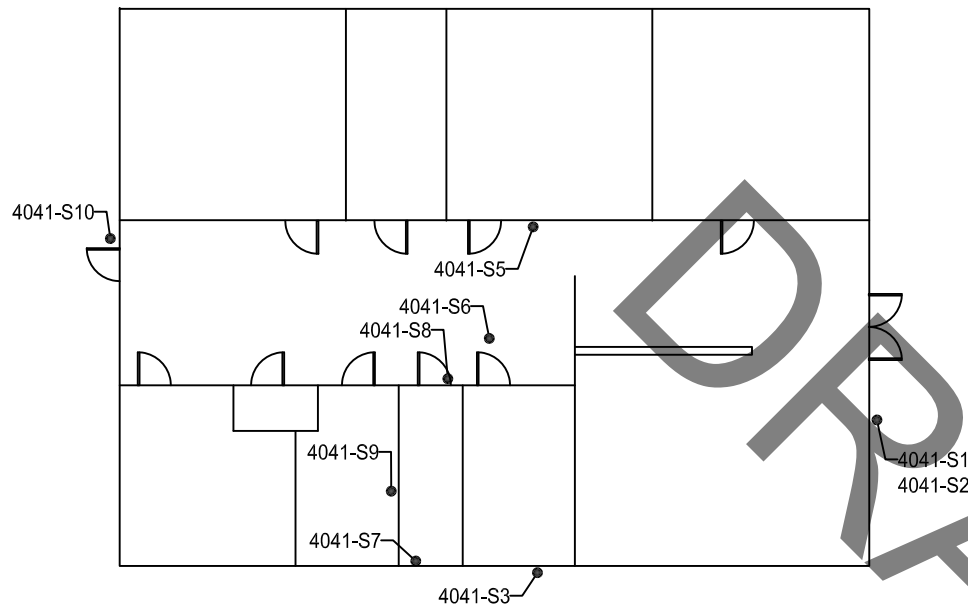
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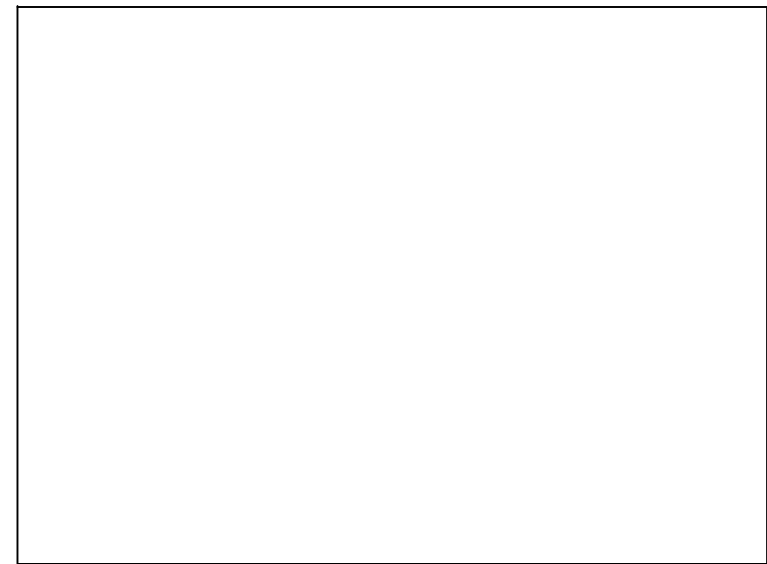


OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



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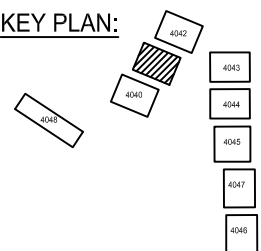
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KEY PLAN:



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LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4041

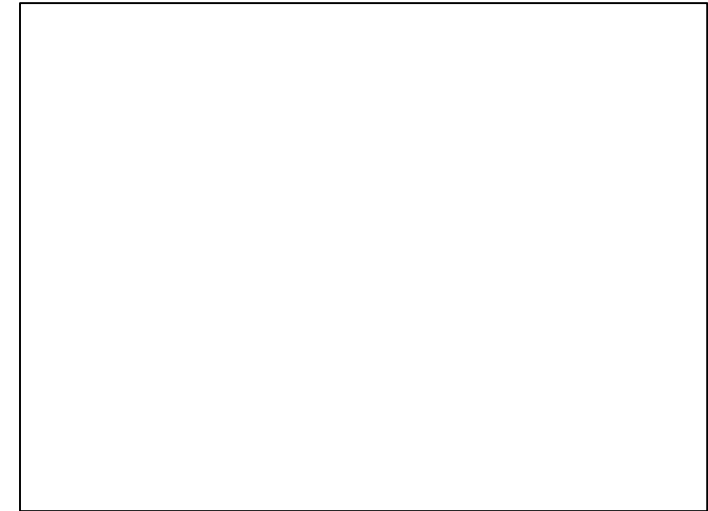
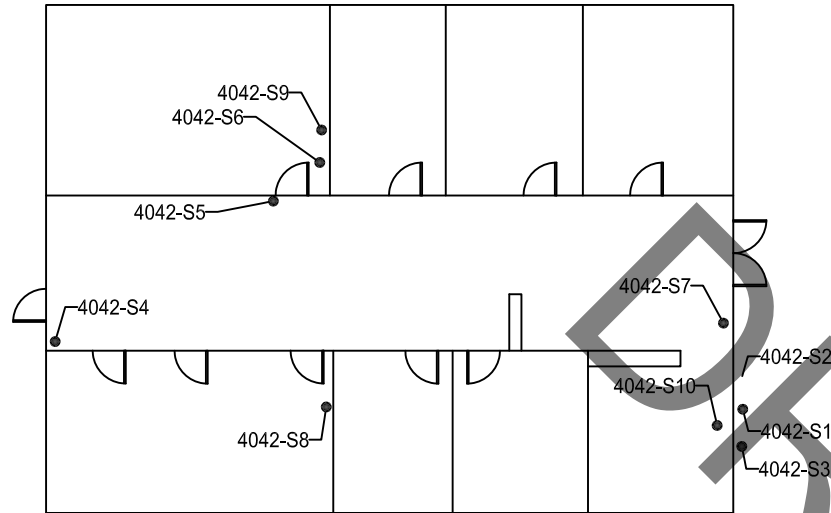
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OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



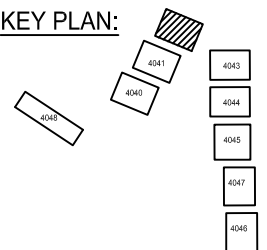
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CADMIUM / CHROMIUM SAMPLING LOCATIONS
(ROOF)

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KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4042

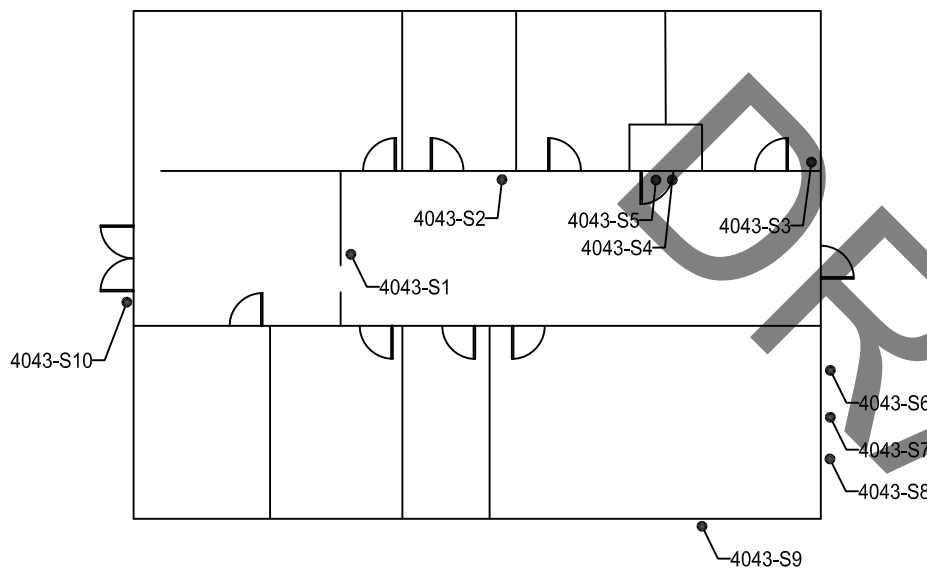
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OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



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(1st FLOOR)

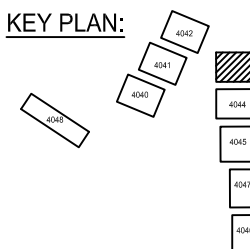
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(ROOF)

N.T.S.

KEY PLAN:



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LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4043

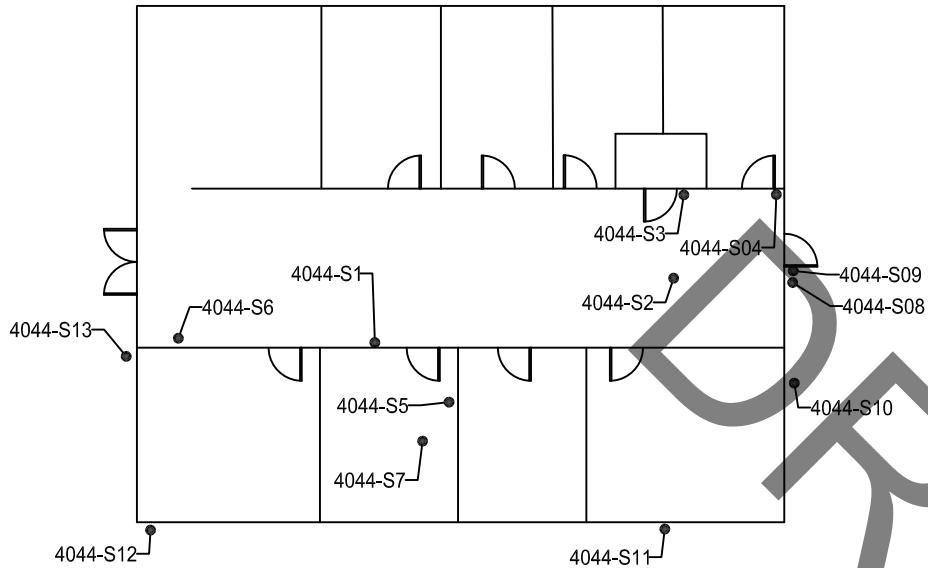
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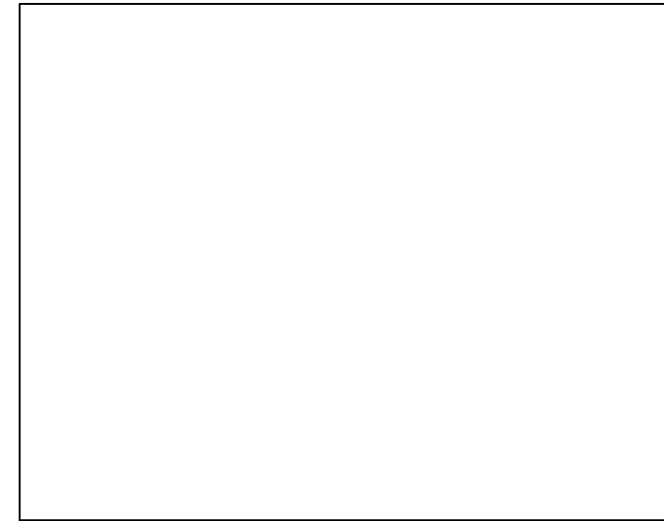


OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



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(1st FLOOR)

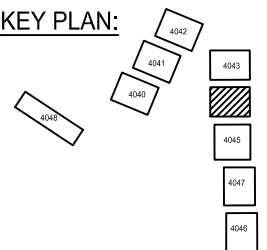
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(ROOF)

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KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4044

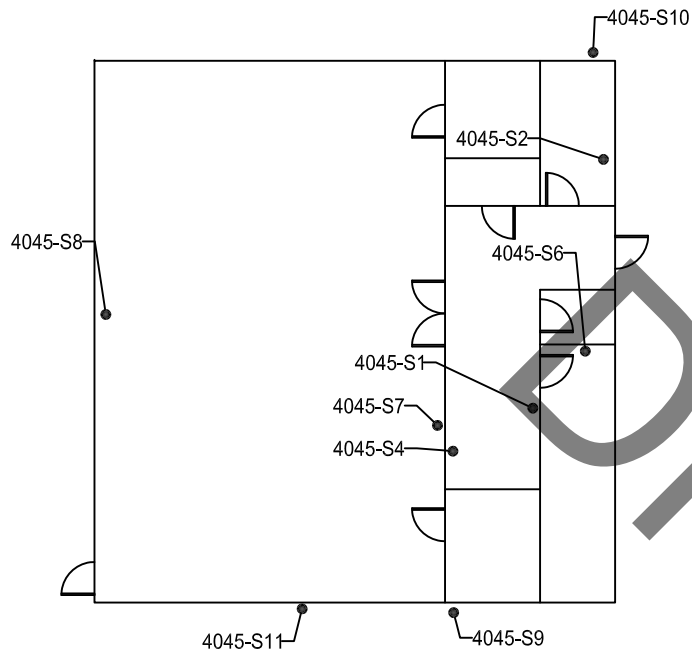
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OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



CADMIUM / CHROMIUM SAMPLING LOCATIONS
(1st FLOOR)

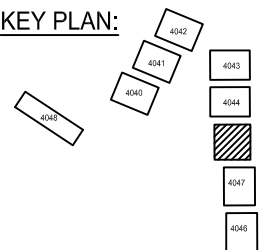
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CADMIUM / CHROMIUM SAMPLING LOCATIONS
(ROOF)

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KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4045

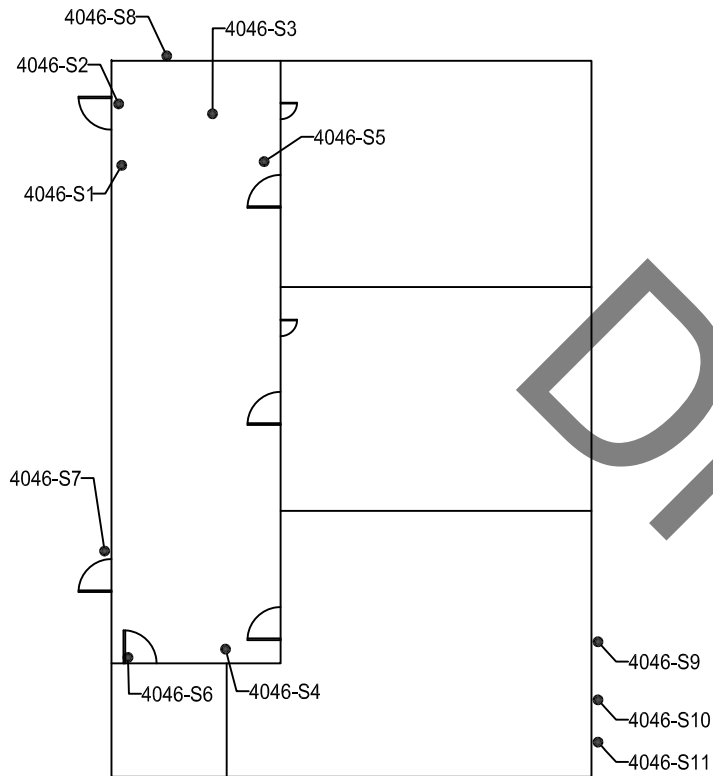
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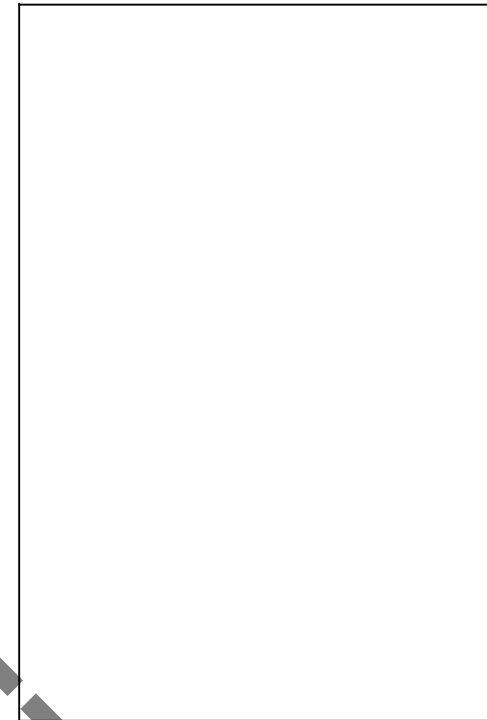


OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



CADMIUM / CHROMIUM SAMPLING LOCATIONS
(1st FLOOR)

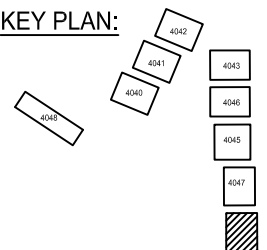
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CADMIUM / CHROMIUM SAMPLING LOCATIONS
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N.T.S.

KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4046

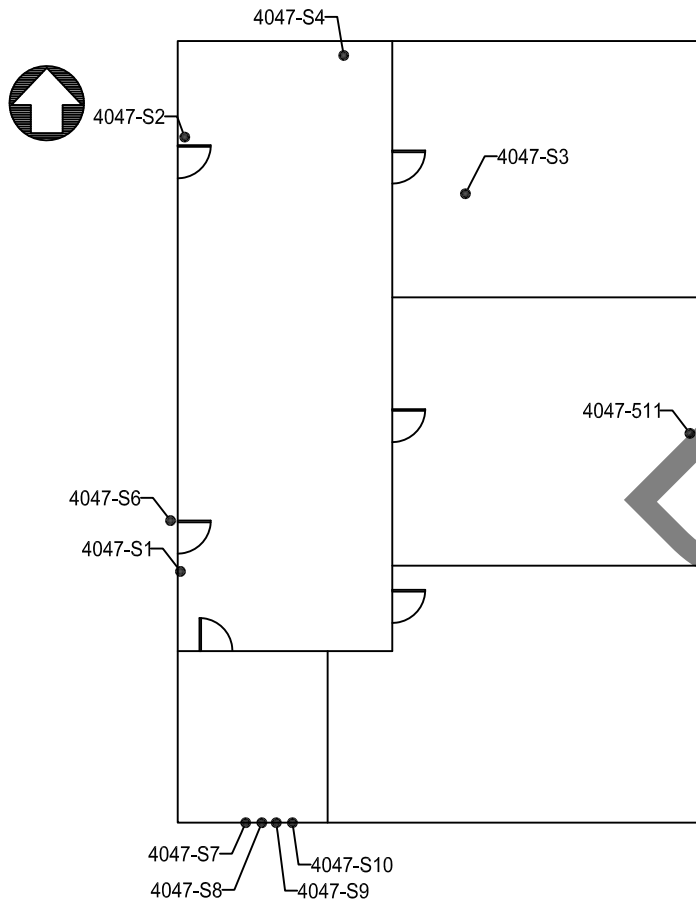
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OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



CADMIUM / CHROMIUM SAMPLING LOCATIONS
(1st FLOOR)

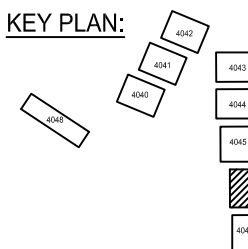
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CADMIUM / CHROMIUM SAMPLING LOCATIONS
(ROOF)

N.T.S.

KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4047

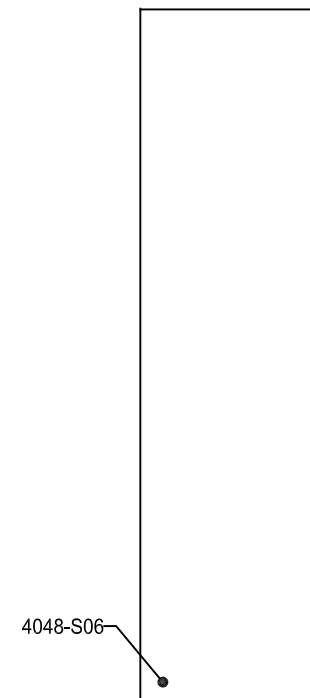
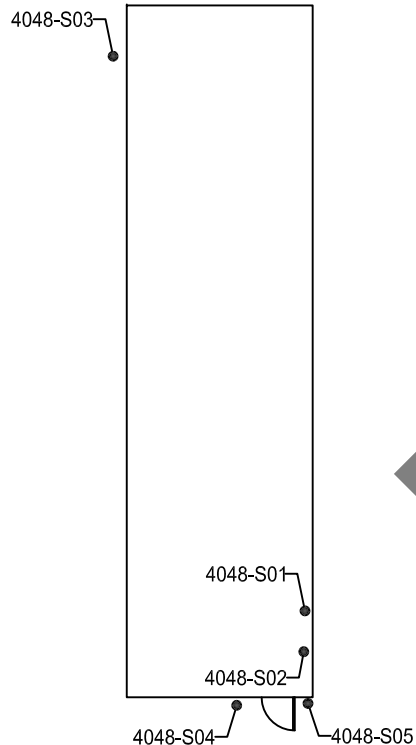
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OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021



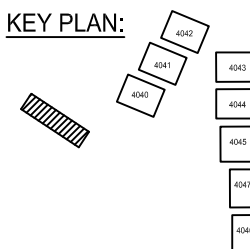
CADMIUM / CHROMIUM SAMPLING LOCATIONS
(1st FLOOR)

N.T.S.

CADMIUM / CHROMIUM SAMPLING LOCATIONS
(ROOF)

N.T.S.

KEY PLAN:



PROJECT: PN77833 KADENA VMS HAZMAT SURVEY

LOCATION: KADENA AIR BASE, OKINAWA, JAPAN

BUILDING: 4048

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OF 18



OKINAWA TIMES BLDG 2F
1-12-24 GOYA, OKINAWA-SHI,
OKINAWA JAPAN 904-0021

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APPENDIX E
Sampling Photos

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**ASBESTOS
SAMPLING PHOTOS**

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METALS IN PAINTS/COATINGS (Pb, Cr, Cd)
SAMPLING PHOTOS

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APPENDIX F

Previous ACM & LBP Bulk Sampling Reports

Prepared by Environmental Science Corporation
Dates: From 9 December 2022 to 20 March 2023

DRAFT

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NOTE: The following report has been excerpted from the original version to provide the information pertinent to the execution of this project.

FY21 Comprehensive ACM & LBP Surveys Kadena Air Base, Okinawa, Japan

Building 4040

Date: 6 March 2023
Prepared for: Kadena Air Base 718th CES/CENM
Prepared by: Environmental Science Corporation (ESC)

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APPENDICES

A	ASBESTOS / LEAD PAINT SAMPLING LOCATIONS
B	ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
C	LBP SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
D	PHOTOGRAPHIC RECORDS OF BULK SAMPLING
E	BUILDING INSPECTOR & LABORATORY CERTIFICATIONS

ACRONYMS AND ABBREVIATIONS

ACBM	Asbestos Containing Building Material
ACM	Asbestos Containing Material
AHERA	Asbestos Hazard Emergency Response Act
BLDG	Building
CFR	Code of Federal Regulations
cm	Centimeter
COC	Chain of Custody
EPA	Environmental Protection Agency
ESC	Environmental Science Corporation
GOJ	Government of Japan
HAZMAT	Hazardous Materials
JEGS	Japan Environmental Governing Standards
KAB	Kadena Air Base
LBP	Lead Based Paint
LCP	Lead Containing Paint
MAP	Model Accreditation Program
mg/L	Milligrams per liter
N/A	Not Applicable
N/D	None Detected
NESHAP	National Emission Standard for Asbestos
NIST	National Institute of Standards and Technologies
NVLAP	National Voluntary Laboratory Accreditation Program
PLM	Polarized Light Microscopy
PPM	Parts per million
Rm	Room
TEM	Transmission Electron Microscopy
TSI	Thermal System Insulation
US	United States of America
USFJ	United States Forces Japan

1.0 SUMMARY

Environmental Science Corporation (ESC) conducted a hazardous material inspection at Building 4040, KAB, Okinawa, Japan. For materials suspected as possible Asbestos Containing Materials (ACM) and Lead-Based Paint (LBP), bulk sampling was conducted, and samples were sent for subsequent analysis at an accredited analytical laboratory.

The following tasks were conducted as part of the Hazardous Materials Survey:

- Evaluate Government-provided records for validity, conduct on-site sampling and analysis of materials/surfaces where Government-provided records were invalid, conduct new sampling and analysis for areas/surfaces where no records existed, report the evaluation results and new sampling laboratory analysis results in separate summary tables for Asbestos and Lead-Based Paint.

Summary data of all suspect asbestos and lead based paint samples collected from Government provided past records and/or within this survey are located in Table 1 below.

2.0 ASBESTOS SAMPLING SUMMARY

Materials assumed, reported, or confirmed to be ACM was not found at the site during this survey investigation. Out of twenty-two (22) total asbestos samples collected for this survey, no samples were determined to be ACM.

The following additional caveats apply to this supplementary hazardous materials survey for ACM materials:

- Per the Japan Environmental Governing Standards (JEGS), any polarized light microscopy (PLM) sample result of “trace” is assumed to be $> 0.1\%$ asbestos and therefore ACM. For any “trace” sample re-analyzed using 1000-point counting and found to contain less than 0.1% asbestos, the associated homogenous area is considered non-ACM, as long as all other samples of that homogenous area are also less than 0.1% asbestos.
- The highest analytical result from all samples representative of each homogeneous area is used to recommend management of that material as a whole.
- Although not observed at the site during this survey, transite piping, used for older underground sewer and storm drain pipelines, shall be presumed ACM if encountered, unless verified non-ACM by a certified asbestos building inspector.
- Past Government provided ND records analyzed under a 1% laboratory Limit of Detection (LOD) shall also be treated as an invalid record per JEGS 2022 definition of ACM.

2.1 ASBESTOS SAMPLING METHODS/ANALYSIS

JEGS defines ACM as any material containing more than 0.1 percent asbestos by weight. All samples were provided to the laboratory accompanied with a copy of paragraph 5.8 from JEGS 2022 as analysis and reporting instructions. The laboratory was instructed to conduct 1000-point counting for all asbestos trace detections. Laboratory result reporting complies with the JEGS criteria.

3.0 LEAD BASED PAINT SAMPLING SUMMARY

The US Environmental Protection Agency (EPA) and JEGS define LBP as paint that contains equal to or greater than 0.5 percent lead by weight. Out of ten (10) total paint chip samples collected for this survey, no paint samples were determined to be lead-based paint.

Paint chip samples were collected in accordance with ASTM E1729-05, *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination* (ASTM, 2005). The samples were delivered to SGS Forensic Laboratories, Carson, CA under chain of custody for analysis by EPA Method 3050B/7000B, Atomic Absorption Spectroscopy (EPA, 1996). Laboratory reports are included in Appendix C. Sample locations are shown in Appendix A.

3.1 LBP PAINT AS A GOJ SPECIALLY-CONTROLLED INDUSTRIAL WASTE

It should be noted that GOJ regulatory limit for lead as a specially-controlled industrial waste is 0.3 milligrams per liter (mg/L) for waste sludge, acid, and alkali, as determined by toxicity characteristic leaching procedure (TCLP) laboratory analysis. Although TCLP testing was not part of the scope of this survey, it is recommended that TCLP analysis be performed on LCP-positive materials (particularly for those materials of high surface area) to determine whether the paint in question should be classified as a specially controlled industrial waste for disposal purposes.

3.2 PAINT WITH LEAD

Paint with lead (PWL) can be categorized as lead based paint (LBP) or lead containing paint (LCP). LBP is defined as paint or other surface coatings that contain lead equal to or greater than one milligram per square centimeter or more than 0.5 percent by weight. Lead-containing paint is paint containing detectable concentrations of lead, but not at sufficient concentration to be classified as LBP. The JEGS define lead containing as paint or other similar surface coating materials containing lead or lead compounds in which the lead content (calculated as lead material) is more than 0.009 percent by weight of the total nonvolatile content of the paint or the weight of the dried paint film (USFJ, 2022).

Table 1 Summary Table of Asbestos and Lead Paint Sampling

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1 BLDG NUMBER	2 ROOM OR FUNCTIONAL AREA	3 HOMOGENEOUS /SURFACE NUMBER	4 MATERIAL NAME /APPLICATION	5 DESCRIPTION (Size, color, other)	6 FRIABLE	7 ACCESS	8 CONDITION	9 HA QTY	10 GOV T	11 C T R	12 LAB/ SAMPLE ID NO.	13 CTR COC SAMPLE ID NO.	14 LAB ANALYSIS DATE	15 RESULT	16 VALIDITY DETERMINATION	17 NOTES / INVALID OR UNKNOWN REASONING
ASBESTSOS SAMPLES																
4040	Reception	H1	Floor tile	30x30 gray tile w/light gray mastic	N	Y	Good	40 SM		Y	51553798	KAB-4040-A1	2022-09-19	ND	Valid	
4040	Reception	H2	Cove base	10cm gray w/light gray mastic	N	Y	Good	45 LM		Y	51553799	KAB-4040-A2	2022-09-19	ND	Valid	
4040	Reception	H3	Gypsum board wall	Smooth white	N	Y	Good	265 SM		Y	51553800	KAB-4040-A3	2022-09-19	ND	Valid	
4040	Reception	H4	Ceiling tile	16x16 off white all patterns	N	N	Good	55 SM		Y	51553801	KAB-4040-A4	2022-09-19	ND	Valid	
4040	Reception	H5	Duct TSI	Yellow fiberglass w/silver backing	Y	N	Good	N/A		Y	51553802	KAB-4040-A5	2022-09-19	ND	Valid	
4040	Executive office	H6	Floor tile	30x30 gray tile w/tan mastic	N	Y	Good	20 SM		Y	51553803	KAB-4040-A6	2022-09-19	ND	Valid	
4040	Executive office	H2	Cove base	10cm gray w/light gray mastic	N	Y	Good	45 LM		Y	51553804	KAB-4040-A7	2022-09-19	ND	Valid	
4040	Executive office	H3	Gypsum board wall	Smooth white	N	Y	Good	265 SM		Y	51553805	KAB-4040-A8	2022-09-19	ND	Valid	
4040	Storage	H6	Floor tile	30x30 gray tile w/tan mastic	N	Y	Good	20 SM		Y	51553806	KAB-4040-A9	2022-09-19	ND	Valid	
4040	Storage	H2	Cove base	10cm gray w/light gray mastic	N	Y	Good	45 LM		Y	51553807	KAB-4040- A10	2022-09-19	ND	Valid	
4040	Storage	H3	Gypsum board wall	Smooth white	N	Y	Good	265 SM		Y	51553808	KAB-4040- A11	2022-09-19	ND	Valid	
4040	Men's bathroom	H7	Ceramic floor tile	20x20 beige tile gray guard gray cement	N	Y	Good	20 SM		Y	51553809	KAB-4040- A12	2022-09-19	ND	Valid	
4040	Men's bathroom	H8	Ceramic wall tile	10x10 off white tile w/ white grout and gray mastic	N	Y	Good	40 SM		Y	51553810	KAB-4040- A13	2022-09-19	ND	Valid	
4040	Men's bathroom	H9	Gypsum ceiling board	white	N	N	Good	20 SM		Y	51553811	KAB-4040- A14	2022-09-19	ND	Valid	
4040	Exterior	H10	Flange gasket	Black	N	Y	Good	N/A		Y	51553812	KAB-4040- A15	2022-09-19	ND	Valid	
4040	Exterior	H11	Wall joint sealant	Light gray mastic	N	Y	Good	N/A		Y	51553813	KAB-4040- A16	2022-09-19	ND	Valid	
4040	Exterior	H12	Concrete wall coating	Off white,textured	N	Y	Good	220 SM		Y	51553814	KAB-4040- A17	2022-09-19	ND	Valid	
4040	Exterior	H12	Concrete wall coating	Off white,textured	N	Y	Good	220 SM		Y	51553815	KAB-4040- A18	2022-09-19	ND	Valid	
4040	Exterior	H11	Wall joint sealant	Light gray mastic	N	Y	Good	N/A		Y	51553816	KAB-4040- A19	2022-09-19	ND	Valid	
4040	Exterior	H13	Window joint sealant	Brown mastic	N	Y	Good	N/A		Y	51553817	KAB-4040- A20	2022-09-19	ND	Valid	
4040	Exterior	H12	Concrete wall coating	Off white,textured	N	Y	Good	220 SM		Y	51553818	KAB-4040- A21	2022-09-19	ND	Valid	
4040	Exterior	H11	Wall joint sealant	Light gray mastic	N	Y	Good	N/A		Y	51553819	KAB-4040- A22	2022-09-19	ND	Valid	

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1 BLDG NUMBER	2 ROOM OR FUNCTIONAL AREA	3 HOMOGENEOUS /SURFACE NUMBER	4 MATERIAL NAME /APPLICATION	5 DESCRIPTION (Size, color, other)	6 FRIABLE	7 ACCESS	8 CONDITION	9 HA QTY	10 GOV T	11 C T R	12 LAB/ SAMPLE ID NO.	13 CTR COC SAMPLE ID NO.	14 LAB ANALYSIS DATE	15 RESULT	16 VALIDITY DETERMINATION	17 NOTES / INVALID OR UNKNOWN REASONING
LEAD PAINT SAMPLES																
4040	Reception	S1	Gypsum board wall	White		Y	Used			Y	LM220520	KAB-4040-L1	2022-06-24	<0.007	Valid	
4040	Reception	S2	Metal pipe (above ceiling)	Red		N	Used			Y	LM220521	KAB-4040-L2	2022-06-24	<0.009	Valid	
4040	Executive office	S3	Gypsum board wall	Gray		Y	Used			Y	LM220522	KAB-4040-L3	2022-06-24	<0.007	Valid	
4040	Storage	S1	Gypsum board wall	White		Y	Used			Y	LM220523	KAB-4040-L4	2022-06-24	<0.006	Valid	
4040	Storage	S4	Gypsum ceiling board	White		N	Used			Y	LM220524	KAB-4040-L5	2022-06-24	<0.007	Valid	
4040	Exterior	S5	Metal pipe	Red over gray and off white		Y	Used			Y	LM220525	KAB-4040-L6	2022-06-24	<0.01	Valid	
4040	Exterior	S6	Metal flange	Light blue over blue		Y	Used			Y	LM220526	KAB-4040-L7	2022-06-24	<0.02	Valid	
4040	Exterior	S7	Concrete wall	Off white		Y	Used			Y	LM220527	KAB-4040-L8	2022-06-24	<0.006	Valid	
4040	Exterior	S8	Concrete base	Orange		Y	Used			Y	LM220528	KAB-4040-L9	2022-06-24	0.010	Valid	
4040	Exterior	S7	Concrete wall	Off white		Y	Used			Y	LM220529	KAB-4040-L10	2022-06-24	<0.007	Valid	

Notes;
The quantities provided in this table represents rough order quantities for the purpose to aid in identifying the homogeneous material identified by the CBI at the time of the survey. This estimate should not be utilized as a basis to develop an estimate for the purpose of formulating a bid for any future work involving all or part of such material.

RED HIGHLIGHT – Asbestos Positive
BLUE HIGHLIGHT – Lead Based Paint Positive
GREEN HIGHLIGHT – Unknown, Trace result with Insufficient Material for Additional Analysis

4.0 REFERENCES

40 CFR Part 745, Lead-Based Paint Poisoning Prevention in Certain Residential Structures.

ASTM, 2005. *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination.*

Government of Japan, 2001. *Japan Waste Management and Public Cleansing Law of 1970, updated 2001.*

United States Code 40 CFR Part 745. *Lead-Based Paint Poisoning Prevention in Certain Residential Structures.*

US EPA, 1996. *Method 3050B/Method No. 7420, Atomic Absorption Spectroscopy.*

US EPA, 1993. *Method for the Determination of Asbestos in Bulk Building Materials*

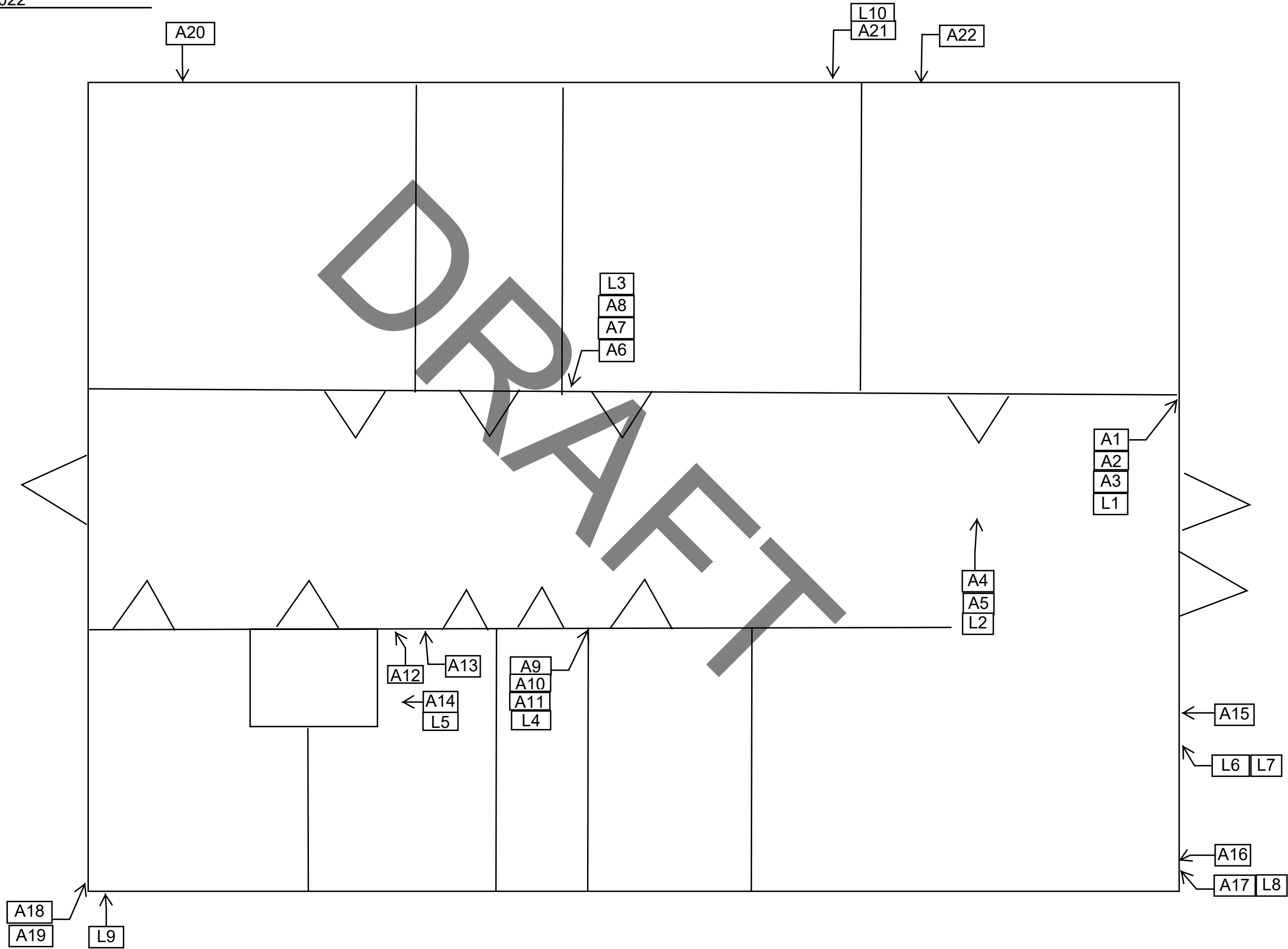
USFJ, 2022. *Japan Environmental Governing Standards (JEGS).* April 2022.

DRAFT

APPENDIX A
ASBESTOS / LEAD PAINT SAMPLING LOCATIONS



ASBESTOS & LEAD PAINT SAMPLES



APPENDIX B
ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 25 April 2022
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day	
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer ☒ PLM: ☒ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435		
Site Name: Kadena Air Base, Okinawa, Japan		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)		
Site Location: Bldg. 4040		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project		
Comments: See attached sample logs.		☐ Metals Analysis Matrix: Method: Analytes:		
		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only		

Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4040-A1-A22		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.			A P C				

Sampled By: Sean R Dudney	Date/Time: 4/5/22	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS
Relinquished By: Sean R Dudney	Relinquished By:	Relinquished By:
Date / Time: 25 April 2022	Date / Time:	Date / Time:
Received By: Japan Post Office	Received By: <i>Alvin M...</i>	Received By:
Date / Time: 25 April 2022	Date / Time: 5/6/22 14:00 USPT	Date / Time:
Condition Acceptable? ☒ Yes ☐ No	Condition Acceptable? ☒ Yes ☐ No	Condition Acceptable? ☐ Yes ☐ No

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
 Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
 Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA							
ROOM OR SECTION	HOMOGENEOUS / SURFACE	MATERIAL NAME / APPLICATION	DESCRIPTION (Size, color, other)	FRI	ACC	CONDITION	G O	C T	LAB/ SAMPLE ID	CTR COC	LAB ANALYSIS	RESULT	VALIDITY DETERMINATION	NOTES / INVALID OR UNKNOWN REASONING	
Reception	H1	Floor tile	30x30 gray tile w/light gray mastic	N	Y	Use				KAB-4040-A1					
	H2	Cove base	10cm gray w/light gray mastic	N	Y	Use				KAB-4040-A2					
	H3	Gypsum board wall	Smooth white	N	Y	Use				KAB-4040-A3					
	H4	Ceiling tile	16x16 off white all patterns	N	N	Use				KAB-4040-A4					
	H5	Duct TSI	Yellow fiberglass w/silver backing	Y	N	Use				KAB-4040-A5					
Executive office	H6	Floor tile	30x30 gray tile w/tan mastic	N	Y	Use				KAB-4040-A6					
	H2	Cove base	10cm gray w/light gray mastic	N	Y	Use				KAB-4040-A7					
	H3	Gypsum board wall	Smooth white	N	Y	Use				KAB-4040-A8					
Storage	H6	Floor tile	30x30 gray tile w/tan mastic	N	Y	Use				KAB-4040-A9					
	H2	Cove base	10cm gray w/light gray mastic	N	Y	Use				KAB-4040-A10					
	H3	Gypsum board wall	Smooth white	N	Y	Use				KAB-4040-A11					
Men's bathroom	H7	Ceramic floor tile	20x20 beige tile gray guard gray cement	N	Y	Use				KAB-4040-A12					
	H8	Ceramic wall tile	10x10 off white tile w/ white grout and gray mastic	N	Y	Use				KAB-4040-A13					
	H9	Gypsum ceiling board	white	N	N	Use				KAB-4040-A14					
Exterior	H10	Flange gasket	Black	N	Y	Use				KAB-4040-A15					
	H11	Wall joint sealant	Light gray mastic	N	Y	Used				KAB-4040-A16					
	H12	Concrete wall	Off white, textured	N	Y	Used				KAB-4040-A17					
	H12	Concrete wall	Off white, textured	N	Y	Used				KAB-4040-A18					
	H11	Wall joint sealant	Light gray mastic	N	Y	Used				KAB-4040-A19					
	H13	Window joint sealant	Brown mastic	N	Y	Used				KAB-4040-A20					
	H12	Concrete wall	Off white, textured	N	Y	Used				KAB-4040-A21					
Exterior	H11	Wall joint sealant	Light gray mastic	N	Y	Used				KAB-4040-A22					

Bulk Asbestos Analysis

(EPA Method 40CFR, Part 763, Appendix E to Subpart E and EPA 600/R-93-116, Visual Area Estimation)
NVLAP Lab Code: 101459-1

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: B333885
Date Received: 05/26/22
Date Analyzed: 09/19/22
Date Printed: 09/19/22
First Reported: 09/19/22

Job ID/Site: ESC.7315; KAB Comprehensive Hazmat, Kadena Air Base, Okinawa, Japan,
Bldg. 4040

Date(s) Collected: 04/05/2022

SGSFL Job ID: L1232-1
Total Samples Submitted: 22
Total Samples Analyzed: 22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4040-A1	51553798						
Layer: Grey Tile			ND				
Layer: Light Grey Mastic			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4040-A2	51553799						
Layer: Grey Non-Fibrous Material			ND				
Layer: White Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4040-A3	51553800						
Layer: Off-White Drywall			ND				
Layer: Paint			ND				
Layer: Grey Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (2 %) Fibrous Glass (Trace)							
KAB-4040-A4	51553801						
Layer: Beige Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (35 %) Fibrous Glass (45 %)							
KAB-4040-A5	51553802						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace) Fibrous Glass (80 %)							
KAB-4040-A6	51553803						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333885

Date Printed: 09/19/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4040-A7	51553804						
Layer: Grey Non-Fibrous Material			ND				
Layer: Tan Mastic			ND				
Layer: Tan Fibrous Backing			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4040-A8	51553805						
Layer: Off-White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (20 %)							
KAB-4040-A9	51553806						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4040-A10	51553807						
Layer: Grey Non-Fibrous Material			ND				
Layer: Tan Mastic			ND				
Layer: Paint			ND				
Layer: White Drywall			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4040-A11	51553808						
Layer: White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (20 %) Fibrous Glass (Trace)							
KAB-4040-A12	51553809						
Layer: Beige Ceramic Tile			ND				
Layer: Grey Grout			ND				
Layer: Beige Mortar			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4040-A13	51553810						
Layer: White Ceramic Tile			ND				
Layer: Light Grey Mastic			ND				
Layer: Grey Grout			ND				
Layer: Off-White Cementitious Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333885

Date Printed: 09/19/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4040-A14	51553811						
Layer: White Drywall			ND				
Layer: Off-White Joint Compound			ND				
Layer: White Woven Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %)	Fibrous Glass (Trace)						
KAB-4040-A15	51553812						
Layer: Black Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4040-A16	51553813						
Layer: Light Grey Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4040-A17	51553814						
Layer: Grey Cementitious Material			ND				
Layer: White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4040-A18	51553815						
Layer: Grey Cementitious Material			ND				
Layer: White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4040-A19	51553816						
Layer: Light Grey Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4040-A20	51553817						
Layer: Brown Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333885

Date Printed: 09/19/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4040-A21	51553818						
Layer: Grey Cementitious Material			ND				
Layer: White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4040-A22	51553819						
Layer: Light Grey Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

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Tiffani Ludd, Laboratory Supervisor, Carson Laboratory

Note: Limit of Quantification ('LOQ') = 1%. 'Trace' denotes the presence of asbestos below the LOQ. 'ND' = 'None Detected'.

Analytical results and reports are generated by SGS Forensic Laboratories (SGSFL) at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGSFL to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGSFL. The client is solely responsible for the use and interpretation of test results and reports requested from SGSFL. SGSFL is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. All samples were received in acceptable condition unless otherwise noted.

APPENDIX C
LBP SAMPLING CHAIN OF CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 25 April 2022			
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day				
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer					
Site Name: Kadena Air Base, Okinawa, Japan		☐ PLM: ☐ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435					
Site Location: Bldg. 4040		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)					
Comments: See attached sample logs.		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project					
		☒ Metals Analysis Matrix: Paint Chip Method: Analytes: Lead					
		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only					
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4040-L1-L10		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.			A P C				
Sampled By: Sean R Dudney		Date/Time: 4/5/22	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS				
Relinquished By: Sean R Dudney		Relinquished By:	Relinquished By:				
Date / Time: 25 April 2022		Date / Time:	Date / Time:				
Received By: Japan Post Office		Received By: <i>Chir Minoy</i>	Received By:				
Date / Time: 25 April 2022		Date / Time: 5/6/22 14:00 USPS	Date / Time:				
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No	Condition Acceptable? ☐ Yes ☐ No				

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Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA							
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BLDG NUMBE	ROOM OR FUNCTIONA	HOMOGENEOU S /SURFACE	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRI A	ACC	CONDITIO N	G O	C T	LAB/ SAMPLE ID	CTR COC	LAB ANALYSIS	RESULT	VALIDITY DETERMINATI	NOTES / INVALID OR UNKNOWN REASONING
4040	Reception	S1	Gypsum board wall	White		Y	Use				KAB- 4040-L1				
		S2	Metal pipe (above ceiling)	Red		N	Use				KAB- 4040-L2				
	Executive office	S3	Gypsum board wall	Gray		Y	Use				KAB- 4040-L3				
	Storage	S1	Gypsum board wall	White		Y	Use				KAB- 4040-L4				
		S4	Gypsum ceiling board	White		N	Use				KAB- 4040-L5				
	Exterior	S5	Metal pipe	Red over gray and off white		Y	Use				KAB- 4040-L6				
		S6	Metal flange	Light blue over blue		Y	Use				KAB- 4040-L7				
		S7	Concrete wall	Off white		Y	Use				KAB- 4040-L8				
		S8	Concrete base	Orange		Y	Used				KAB- 4040-L9				
		S7	Concrete wall	Off white		Y	Use				KAB- 4040- L10				

Metals Analysis of Paints

(AIHA-LAP, LLC Accreditation, Lab ID #101762)

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: M242183
Date Received: 05/26/22
Date Analyzed: 06/24/22
Date Printed: 06/24/22
First Reported: 06/24/22

Job ID / Site: ESC.7315; KAB Comprehensive Hazmat, Kadens Air Base, Okinawa, Japan,
Bldg. 4040

Date(s) Collected: 04/05/22

SGSFL Job ID: L1232-1

Total Samples Submitted: 10

Total Samples Analyzed: 10

Sample Number	Lab Number	Analyte	Result	Result Units	Reporting Limit*	Method Reference
KAB-4040-L1	LM220520	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4040-L2	LM220521	Pb	< 0.009	wt%	0.009	EPA 3050B/7000B
KAB-4040-L3	LM220522	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4040-L4	LM220523	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4040-L5	LM220524	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4040-L6	LM220525	Pb	< 0.01	wt%	0.01	EPA 3050B/7000B
KAB-4040-L7	LM220526	Pb	< 0.02	wt%	0.02	EPA 3050B/7000B
KAB-4040-L8	LM220527	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4040-L9	LM220528	Pb	0.010	wt%	0.009	EPA 3050B/7000B
KAB-4040-L10	LM220529	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B

* The Reporting Limit represents the lowest amount of analyte that the laboratory can confidently detect in the sample, and is not a regulatory level. The Units for the Reporting Limit are the same as the Units for the Final Results.



Kevin Poon, Laboratory Analyst, Hayward Laboratory

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Note* Sampling data used in this report was provided by the client as noted on the associated chain of custody form.

ASBESTOS SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

HazMat Survey

Bldg. 4040



Photo Record Sheet

Asbestos Sampling
Kadena Air Base, Bldg.4040



KAB-4040-A1



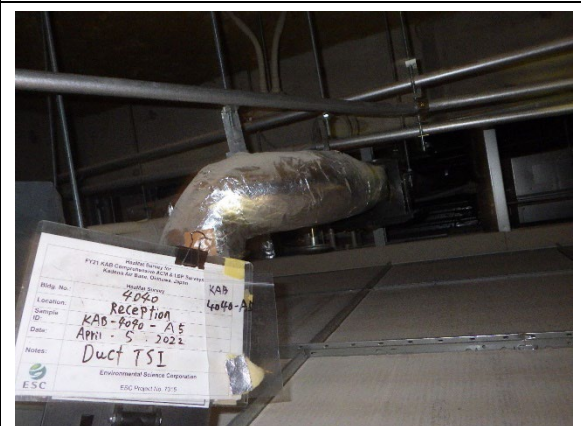
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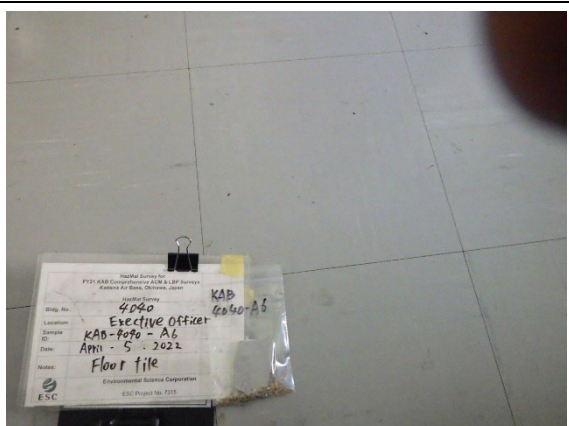
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KAB-4040-A4



KAB-4040-A5



KAB-4040-A6

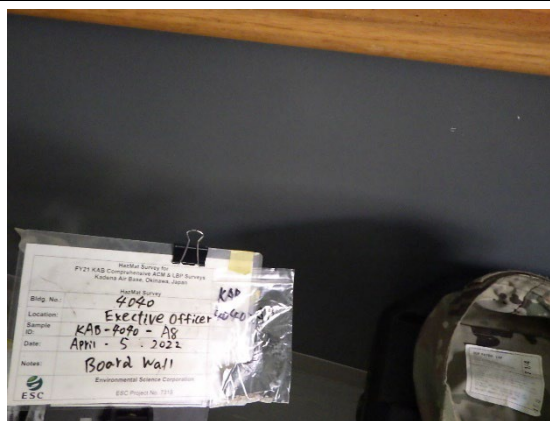


Photo Record Sheet

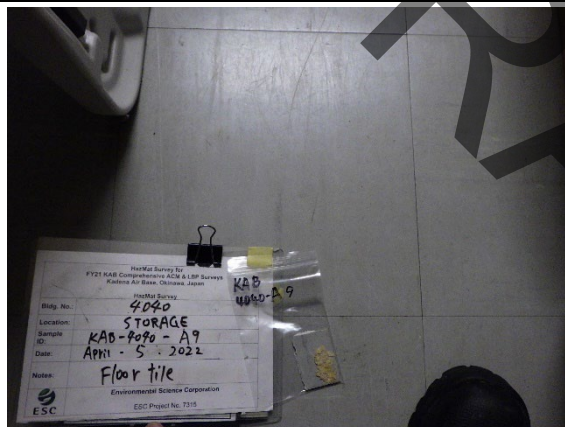
Asbestos Sampling
Kadena Air Base, Bldg.4040



KAB-4040-A7



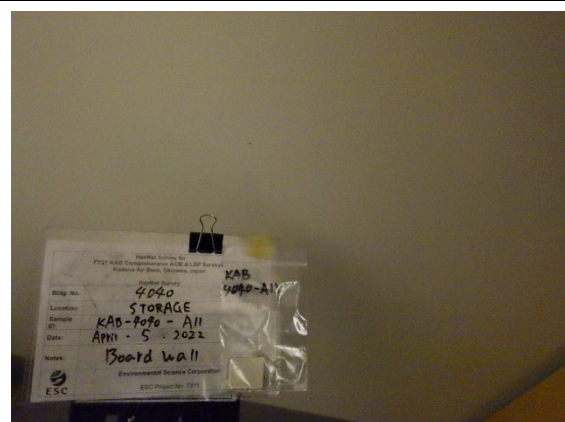
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KAB-4040-A9



KAB-4040-A10



KAB-4040-A11

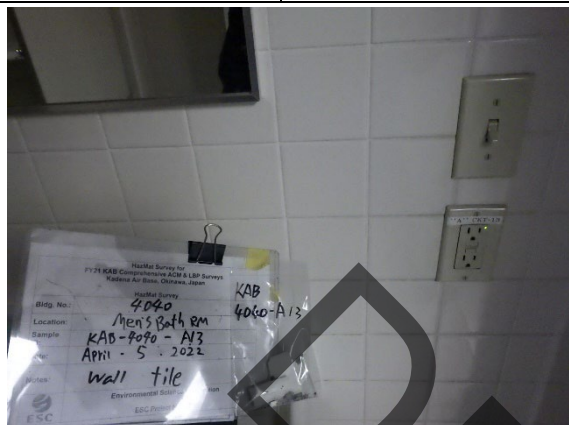


KAB-4040-A12



Photo Record Sheet

Asbestos Sampling
Kadena Air Base, Bldg.4040



KAB-4040-A13



KAB-4040-A14



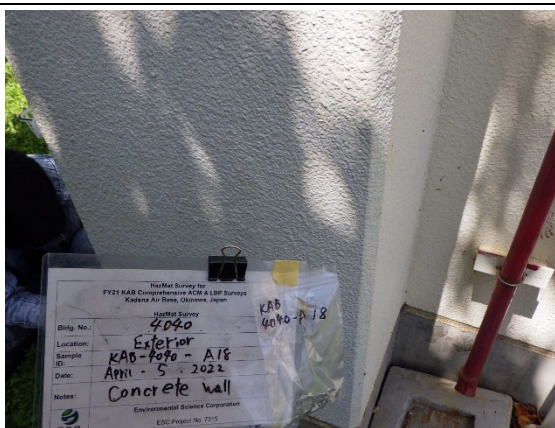
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KAB-4040-A16



KAB-4040-A17



KAB-4040-A18

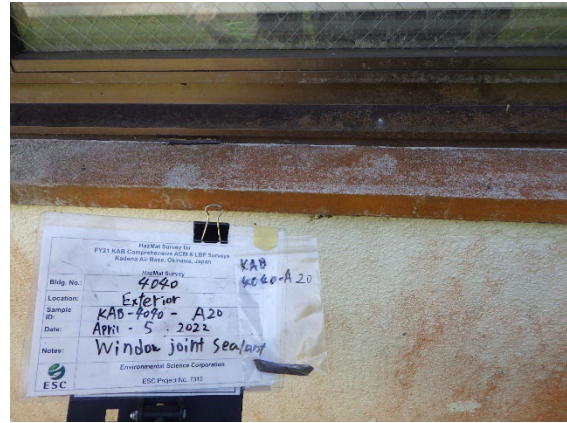


Photo Record Sheet

Asbestos Sampling
Kadena Air Base, Bldg.4040



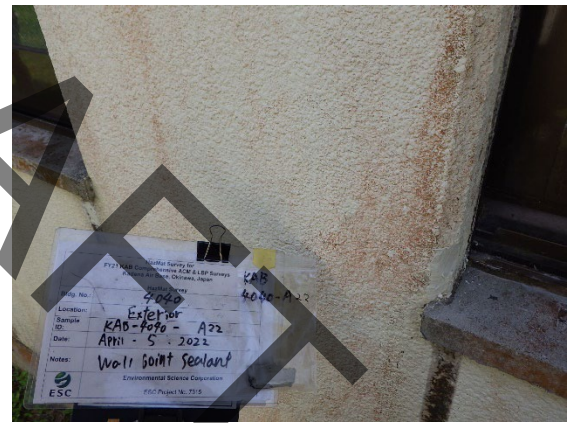
KAB-4040-A19



KAB-4040-A20



KAB-4040-A21



KAB-4040-A22

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LEAD BASED PAINT SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

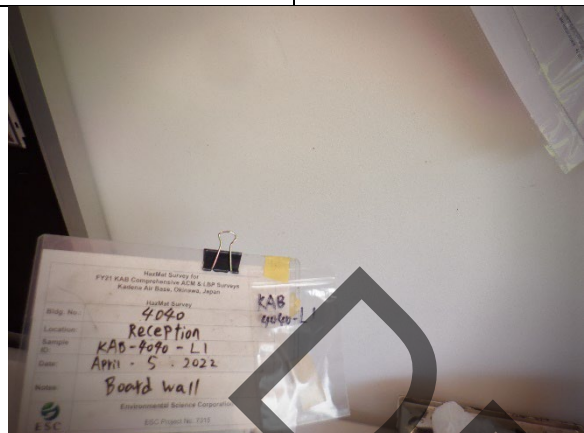
HazMat Survey

Bldg. 4040



Photo Record Sheet

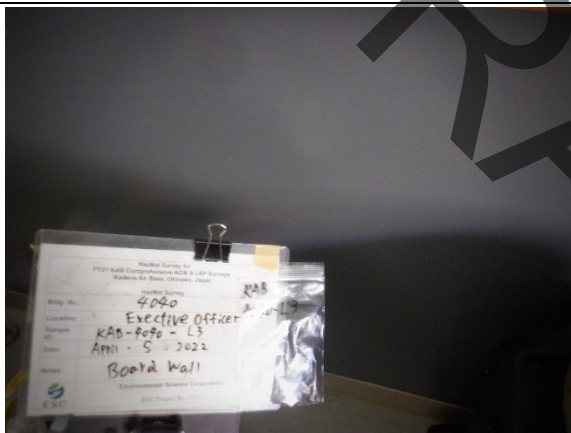
Lead Based Paint Sampling Kadena Air Base, Bldg.4040



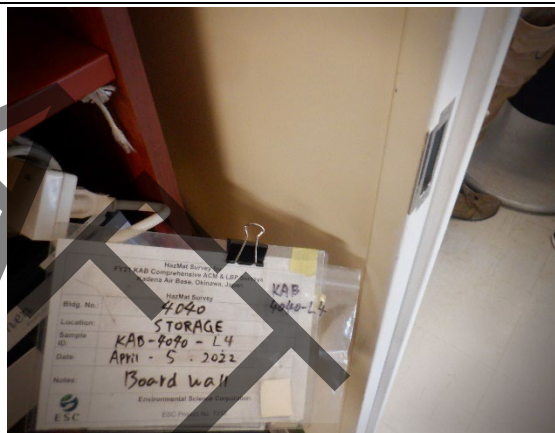
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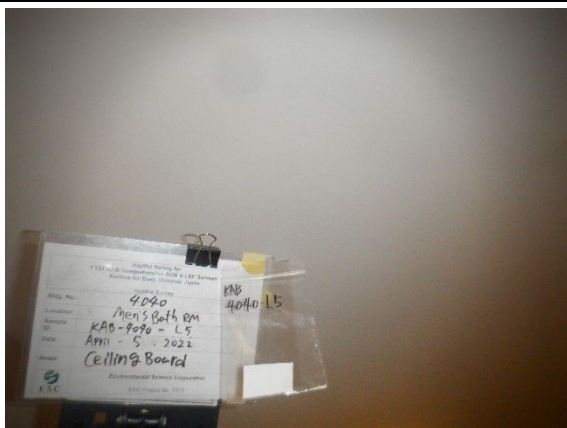
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KAB-4040-L3



KAB-4040-L4



KAB-4040-L5



KAB-4040-L6



Photo Record Sheet

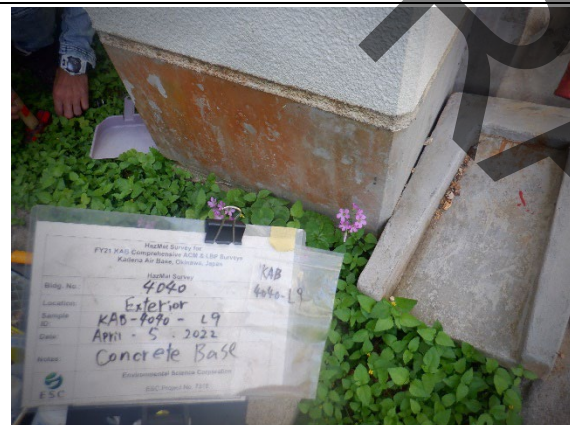
Lead Based Paint Sampling
Kadena Air Base, Bldg.4040



KAB-4040-L7



KAB-4040-L8



KAB-4040-L9



KAB-4040-L10

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NOTE: The following report has been excerpted from the original version to provide the information pertinent to the execution of this project.

FY21 Comprehensive ACM & LBP Surveys
Kadena Air Base, Okinawa, Japan

Building 4041

Date: 25 February 2023
Prepared for: Kadena Air Base 718th CES/CENM
Prepared by: Environmental Science Corporation (ESC)

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APPENDICES

A	ASBESTOS / LEAD PAINT SAMPLING LOCATIONS
B	ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
C	LBP SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
D	PHOTOGRAPHIC RECORDS OF BULK SAMPLING
E	BUILDING INSPECTOR & LABORATORY CERTIFICATIONS

1.0 SUMMARY

Environmental Science Corporation (ESC) conducted a hazardous material inspection at Building 4041, KAB, Okinawa, Japan. For materials suspected as possible Asbestos Containing Materials (ACM) and Lead-Based Paint (LBP), bulk sampling was conducted, and samples were sent for subsequent analysis at an accredited analytical laboratory.

The following tasks were conducted as part of the Hazardous Materials Survey:

- Evaluate Government-provided records for validity, conduct on-site sampling and analysis of materials/surfaces where Government-provided records were invalid, conduct new sampling and analysis for areas/surfaces where no records existed, report the evaluation results and new sampling laboratory analysis results in separate summary tables for Asbestos and Lead-Based Paint.

Summary data of all suspect asbestos and lead based paint samples collected from Government provided past records and/or within this survey are located in Table 1 below.

2.0 ASBESTOS SAMPLING SUMMARY

Materials assumed, reported, or confirmed to be ACM was not found at the site during this survey investigation. Out of twenty-five (25) total asbestos samples collected for this survey, no samples were determined to be ACM.

The following additional caveats apply to this supplementary hazardous materials survey for ACM materials:

- Per the Japan Environmental Governing Standards (JEGS), any polarized light microscopy (PLM) sample result of “trace” is assumed to be $> 0.1\%$ asbestos and therefore ACM. For any “trace” sample re-analyzed using 1000-point counting and found to contain less than 0.1% asbestos, the associated homogenous area is considered non-ACM, as long as all other samples of that homogenous area are also less than 0.1% asbestos.
- The highest analytical result from all samples representative of each homogeneous area is used to recommend management of that material as a whole.
- Although not observed at the site during this survey, transite piping, used for older underground sewer and storm drain pipelines, shall be presumed ACM if encountered, unless verified non-ACM by a certified asbestos building inspector.
- Past Government provided ND records analyzed under a 1% laboratory Limit of Detection (LOD) shall also be treated as an invalid record per JEGS 2022 definition of ACM.

2.1 ASBESTOS SAMPLING METHODS/ANALYSIS

JEGS defines ACM as any material containing more than 0.1 percent asbestos by weight. All samples were provided to the laboratory accompanied with a copy of paragraph 5.8 from JEGS 2022 as analysis and reporting instructions. The laboratory was instructed to conduct 1000-point counting for all asbestos trace detections. Laboratory result reporting complies with the JEGS criteria.

3.0 LEAD BASED PAINT SAMPLING SUMMARY

The US Environmental Protection Agency (EPA) and JEGS define LBP as paint that contains equal to or greater than 0.5 percent lead by weight. Out of thirteen (13) total paint chip samples collected for this survey, no paint samples were determined to be lead-based paint.

Paint chip samples were collected in accordance with ASTM E1729-05, *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination* (ASTM, 2005). The samples were delivered to SGS Forensic Laboratories, Carson, CA under chain of custody for analysis by EPA Method 3050B/7000B, Atomic Absorption Spectroscopy (EPA, 1996). Laboratory reports are included in Appendix C. Sample locations are shown in Appendix A.

3.1 LBP PAINT AS A GOJ SPECIALLY-CONTROLLED INDUSTRIAL WASTE

It should be noted that GOJ regulatory limit for lead as a specially-controlled industrial waste is 0.3 milligrams per liter (mg/L) for waste sludge, acid, and alkali, as determined by toxicity characteristic leaching procedure (TCLP) laboratory analysis. Although TCLP testing was not part of the scope of this survey, it is recommended that TCLP analysis be performed on LCP-positive materials (particularly for those materials of high surface area) to determine whether the paint in question should be classified as a specially controlled industrial waste for disposal purposes.

3.2 PAINT WITH LEAD

Paint with lead (PWL) can be categorized as lead based paint (LBP) or lead containing paint (LCP). LBP is defined as paint or other surface coatings that contain lead equal to or greater than one milligram per square centimeter or more than 0.5 percent by weight. Lead-containing paint is paint containing detectable concentrations of lead, but not at sufficient concentration to be classified as LBP. The JEGS define lead containing as paint or other similar surface coating materials containing lead or lead compounds in which the lead content (calculated as lead material) is more than 0.009 percent by weight of the total nonvolatile content of the paint or the weight of the dried paint film (USFJ, 2022).

Table 1 Summary Table of Asbestos and Lead Paint Sampling

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1 BLDG NUMBER	2 ROOM OR FUNCTIONAL AREA	3 HOMOGENEOUS /SURFACE NUMBER	4 MATERIAL NAME /APPLICATION	5 DESCRIPTION (Size, color, other)	6 FRIABLE	7 ACCESS	8 CONDITION	9 HA QTY	10 GOV T	11 C T R	12 LAB/ SAMPLE ID NO.	13 CTR COC SAMPLE ID NO.	14 LAB ANALYSIS DATE	15 RESULT	16 VALIDITY DETERMINATION	17 NOTES / INVALID OR UNKNOWN REASONING
ASBESTOS SAMPLES																
4041	Exterior	H1	Flange gasket	Black	N	Y	Good	N/A		Y	51553745	KAB-4041-A1	2022-08-18	ND	Valid	
4041	Exterior	H2	Concrete wall coating	Off white, textured	N	Y	Good	250 SM		Y	51553746	KAB-4041-A2	2022-08-18	ND	Valid	
4041	Exterior	H3	Wall joint sealant	Light gray mastic	N	Y	Good	150 LM		Y	51553747	KAB-4041-A3	2022-08-18	ND	Valid	
4041	Exterior	H2	Concrete wall coating	Off white, textured	N	Y	Good	250 SM		Y	51553748	KAB-4041-A4	2022-08-18	ND	Valid	
4041	Exterior	H3	Wall joint sealant	Light gray mastic	N	Y	Good	150 LM		Y	51553749	KAB-4041-A5	2022-08-18	ND	Valid	
4041	Exterior	H4	Window joint sealant	Brown mastic	N	Y	Good	30 LM		Y	51553750	KAB-4041-A6	2022-08-18	ND	Valid	
4041	Exterior	H2	Concrete wall Coating	Off white, textured	N	Y	Good	250 SM		Y	51553751	KAB-4041-A7	2022-08-18	ND	Valid	
4041	Exterior	H4	Window joint sealant	Brown mastic	N	Y	Good	30 LM		Y	51553752	KAB-4041-A8	2022-08-18	ND	Valid	
4041	Reception	H5	Vinyl floor tile	30x30 gray tile w/tan, light gray mastic	N	Y	Good	220 SM		Y	51553753	KAB-4041-A9	2022-08-18	ND	Valid	
4041	Reception	H6	Vinyl cove base	10cm gray w/white mastic	N	Y	Good	150 LM		Y	51553754	KAB-4041-A10	2022-08-18	ND	Valid	
4041	Reception	H7	Gypsum board wall	Gray	N	Y	Good	350 SM		Y	51553755	KAB-4041-A11	2022-08-18	ND	Valid	
4041	Reception	H8	Ceiling tile	60x60 off white all patterns	N	N	Good	220 SM		Y	51553756	KAB-4041-A12	2022-08-18	ND	Valid	
4041	Reception	H9	Duct TSI	Yellow fiberglass w/silver backing	Y	N	Good	N/A		Y	51553757	KAB-4041-A13	2022-08-18	ND	Valid	
4041	XO RM	H5	Vinyl floor tile	30x30 gray tile w/tan mastic	N	Y	Good	220 SM		Y	51553758	KAB-4041-A14	2022-08-18	ND	Valid	
4041	XO RM	H6	Vinyl cove base	10cm gray w/white mastic	N	Y	Good	150 LM		Y	51553759	KAB-4041-A15	2022-08-18	ND	Valid	
4041	XO RM	H10	Gypsum board wall	White	N	Y	Good	350 SM		Y	51553760	KAB-4041-A16	2022-08-18	ND	Valid	
4041	Men's bathroom	H11	Ceiling board	White	N	N	Good	30 SM		Y	51553761	KAB-4041-A17	2022-08-18	ND	Valid	
4041	Men's bathroom	H12	Ceramic floor tile	20x20 beige w/gray grout	N	Y	Good	30 SM		Y	51553762	KAB-4041-A18	2022-08-18	ND	Valid	
4041	Men's bathroom	H13	Wall tile	10x10 off white w/white grout	N	Y	Good	60 SM		Y	51553763	KAB-4041-A19	2022-08-18	ND	Valid	
4041	Men's bathroom	H14	Sink joint sealant	White	N	Y	Good	2 LM		Y	51553764	KAB-4041-A20	2022-08-18	ND	Valid	
4041	Conference Rm.	H5	Vinyl floor tile	30x30 gray tile w/tan, light gray mastic	N	Y	Good	220 SM		Y	51553765	KAB-4041-A21	2022-08-18	ND	Valid	
4041	Conference Rm.	H10	Gypsum board wall	Red	N	Y	Good	350 SM		Y	51553766	KAB-4041-A22	2022-08-18	ND	Valid	
4041	Conference Rm.	H6	Vinyl cove base	10cm gray w/white mastic	N	Y	Good	150 LM		Y	51553767	KAB-4041-A23	2022-08-18	ND	Valid	

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1 BLDG NUMBER	2 ROOM OR FUNCTIONAL AREA	3 HOMOGENEOUS /SURFACE NUMBER	4 MATERIAL NAME /APPLICATION	5 DESCRIPTION (Size, color, other)	6 FRIABLE	7 ACCESS	8 CONDITION	9 HA QTY	10 GOV T	11 C T R	12 LAB/ SAMPLE ID NO.	13 CTR COC SAMPLE ID NO.	14 LAB ANALYSIS DATE	15 RESULT	16 VALIDITY DETERMINATION	17 NOTES / INVALID OR UNKNOWN REASONING
4041	Janitor	H15	Pipe TSI(straight)	Yellow fiberglass w/silver backing	Y	N	Good	N/A		Y	51553768	KAB-4041-A24	2022-08-18	ND	Valid	
4041	Janitor	H16	Pipe TSI(elbow)	Yellow fiberglass w/silver backing	Y	N	Good	N/A		Y	51553769	KAB-4041-A25	2022-08-18	ND	Valid	
LEAD PAINT SAMPLES																
4041	Exterior	S1	Metal flange	Light blue		Y	Used			Y	LM220646	KAB-4041-L1	2022-06-22	<0.007	Valid	
4041	Exterior	S2	Metal pipe	Red over white		Y	Used			Y	LM220647	KAB-4041-L2	2022-06-22	<0.007	Valid	
4041	Exterior	S3	Concrete wall	Off white		Y	Used			Y	LM220648	KAB-4041-L3	2022-06-22	<0.006	Valid	
4041	Exterior	S4	Concrete wall base	Orange		Y	Used			Y	LM220649	KAB-4041-L4	2022-06-22	0.028	Valid	
4041	Exterior	S3	Concrete wall	Off white		Y	Used			Y	LM220650	KAB-4041-L5	2022-06-22	<0.007	Valid	
4041	Exterior	S3	Concrete wall	Off white		Y	Used			Y	LM220651	KAB-4041-L6	2022-06-22	<0.007	Valid	
4041	Reception	S5	Gypsum board wall	Gray		Y	Used			Y	LM220652	KAB-4041-L7	2022-06-22	<0.007	Valid	
4041	Reception	S6	Metal pipe (above ceiling)	Red		Y	Used			Y	LM220653	KAB-4041-L8	2022-06-22	<0.01	Valid	
4041	XO rm	S7	Gypsum board wall	White		Y	Used			Y	LM220654	KAB-4041-L9	2022-06-22	<0.006	Valid	
4041	XO rm	S8	Wooden door	Black		Y	Used			Y	LM220655	KAB-4041-L10	2022-06-22	0.02	Valid	
4041	Men's bathroom	S9	Ceiling board	White		N	Used			Y	LM220656	KAB-4041-L11	2022-06-22	<0.006	Valid	
4041	Men's bathroom	S10	Gypsum board wall	Red		Y	Used			Y	LM220657	KAB-4041-L12	2022-06-22	<0.007	Valid	
4041	Storage	S11	Wooden door	White		Y	Used			Y	LM220658	KAB-4041-L13	2022-06-22	<0.006	Valid	

Notes;
The quantities provided in this table represents rough order quantities for the purpose to aid in identifying the homogeneous material identified by the CBI at the time of the survey. This estimate should not be utilized as a basis to develop an estimate for the purpose of formulating a bid for any future work involving all or part of such material.

RED HIGHLIGHT – Asbestos Positive
BLUE HIGHLIGHT – Lead Based Paint Positive
GREEN HIGHLIGHT – Unknown, Trace result with Insufficient Material for Additional Analysis

4.0 REFERENCES

40 CFR Part 745, Lead-Based Paint Poisoning Prevention in Certain Residential Structures.

ASTM, 2005. *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination.*

Government of Japan, 2001. *Japan Waste Management and Public Cleansing Law of 1970, updated 2001.*

United States Code 40 CFR Part 745. *Lead-Based Paint Poisoning Prevention in Certain Residential Structures.*

US EPA, 1996. *Method 3050B/Method No. 7420, Atomic Absorption Spectroscopy.*

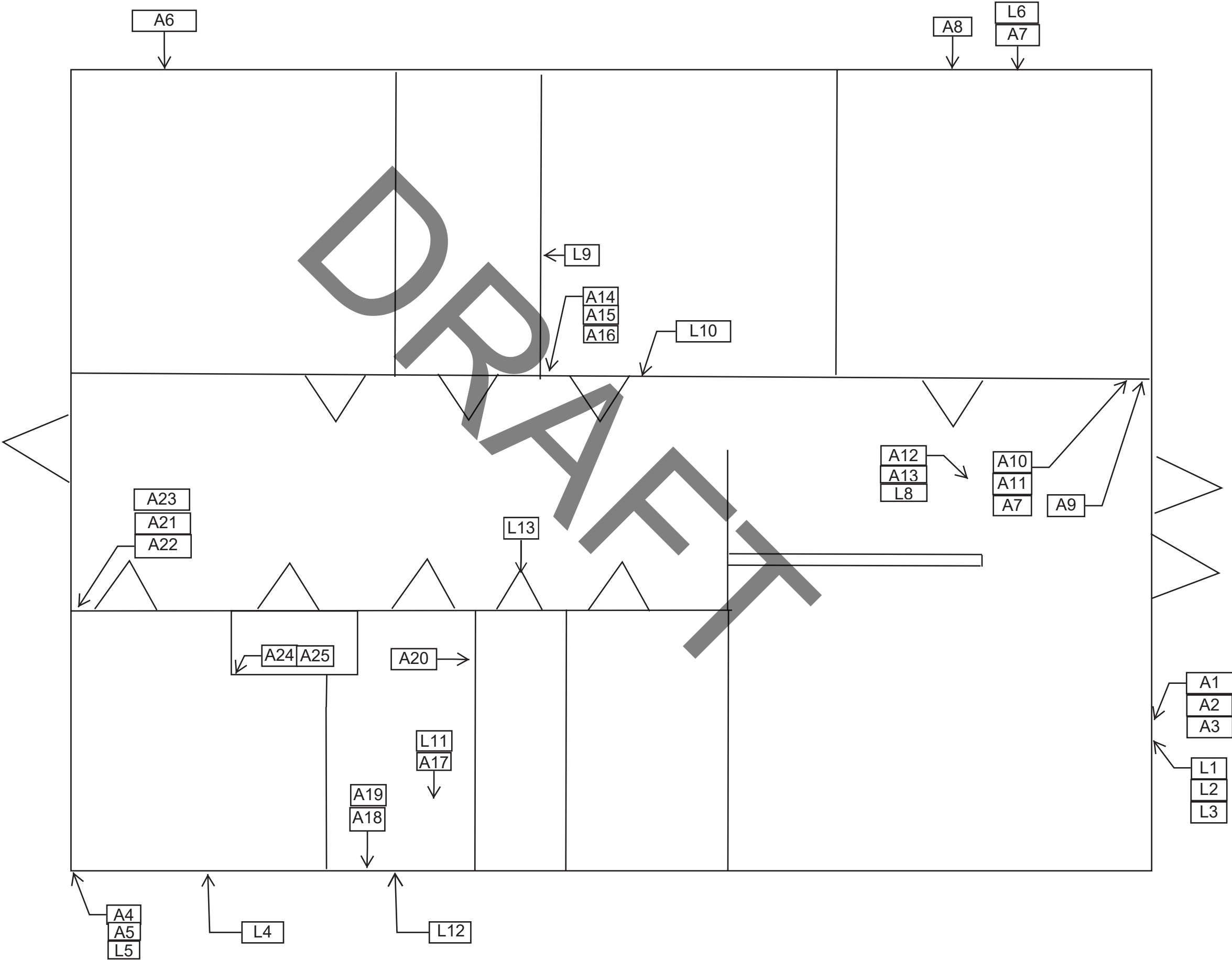
US EPA, 1993. *Method for the Determination of Asbestos in Bulk Building Materials*

USFJ, 2022. *Japan Environmental Governing Standards (JEGS).* April 2022.

APPENDIX A
ASBESTOS / LEAD PAINT SAMPLING LOCATIONS



ASBESTOS & LEAD PAINT SAMPLES



APPENDIX B
ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 25 April 2022			
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: ☐ Same Day / ☐ 1Day / ☐ 2Day / ☐ 3Day / ☐ 4Day / ☒ 5Day				
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer ☒ PLM: ☒ Standard / ☐ Point Count ☐ 400 / ☐ 1000 / ☐ CARB 435					
Site Name: Kadena Air Base, Okinawa, Japan		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)					
Site Location: Bldg. 4041		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project					
Comments: See attached sample logs.		☐ Metals Analysis Matrix: Method: Analytes: ☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only					
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4041-A1-A25		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.			A P C				
Sampled By: Sean R Dudney		Date/Time: 4/5/22	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS				
Relinquished By: Sean R Dudney		Relinquished By:		Relinquished By:			
Date / Time: 25 April 2022		Date / Time:		Date / Time:			
Received By: Japan Post Office		Received By: <i>Alvin M...</i>		Received By:			
Date / Time: 25 April 2022		Date / Time: 5-26-22 14:00 USPT		Date / Time:			
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No			

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
 Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
 Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

SUMMARY REPORTING SHEET (V2.0)

=A2:AC29HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA							
ROOM OR LOCATION	HOMOGENEOUS SURFACE	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRI	ACC	CONDITION	GCOT	LAB/SAMPLE ID	CTR COC	LAB ANALYSIS	RESULT	VALIDITY DETERMINATION	NOTES / INVALID OR UNKNOWN REASONING		
Corridor	H1	Flange gasket	Black	N	Y	Use			KAB-4041-A1						
	H2	Concrete wall	Off white, textured	N	Y	Use			KAB-4041-A2						
	H3	Wall joint sealant	Light gray mastic	N	Y	Used			KAB-4041-A3						
	H2	Concrete wall	Off white, textured	N	Y	Use			KAB-4041-A4						
	H3	Wall joint sealant	Light gray mastic	N	Y	Used			KAB-4041-A5						
	H4	Window joint sealant	Brown mastic	N	Y	Used			KAB-4041-A6						
	H2	Concrete wall	Off white, textured	N	Y	Use			KAB-4041-A7						
	H4	Window joint sealant	Brown mastic	N	Y	Used			KAB-4041-A8						
Reception	H5	Floor tile	30x30 gray tile w/tan, light gray mastic	N	Y	Use			KAB-4041-A9						
	H6	Vinyl cove base	10cm gray w/white mastic	N	Y	Use			KAB-4041-A10						
	H7	Gypsum board wall	Gray	N	Y	Use			KAB-4041-A11						
	H8	Ceiling tile	60x60 off white all patterns	N	N	Use			KAB-4041-A12						
Office RM	H9	Duct TSI	Yellow fiberglass w/silver backing	N	N	Use			KAB-4041-A13						
	H5	Floor tile	30x30 gray tile w/tan mastic	N	Y	Use			KAB-4041-A14						
	H6	Vinyl cove base	10cm gray w/white mastic	N	Y	Use			KAB-4041-A15						
	H10	Gypsum board wall	White	N	Y	Use			KAB-4041-A16						
Men's Restroom	H11	Ceiling board	White	N	N	Use			KAB-4041-A17						
	H12	Ceramic floor tile	20x20 beige w/gray grout	N	Y	Use			KAB-4041-A18						
	H13	Wall tile	10x10 off white w/white grout	N	Y	Use			KAB-4041-A19						
	H14	Sink joint sealant	White	N	Y	Used			KAB-4041-A20						
	H5	Floor tile	30x30 gray tile w/tan, light gray mastic	N	Y	Use			KAB-4041-A21						
Conference Rm.	H10	Gypsum board wall	Red	N	Y	Use			KAB-4041-A22						
	H6	Vinyl cove base	10cm gray w/white mastic	N	Y	Use			KAB-4041-A23						
Monitor	H15	Pipe TSI(straight)	Yellow fiberglass w/silver backing	N	N	Use			KAB-4041-A24						
	H16	Pipe TSI(elbow)	Yellow fiberglass w/silver backing	N	N	Use			KAB-4041-A25						

Bulk Asbestos Analysis

(EPA Method 40CFR, Part 763, Appendix E to Subpart E and EPA 600/R-93-116, Visual Area Estimation)

NVLAP Lab Code: 101459-1

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: B333883
Date Received: 05/26/22
Date Analyzed: 08/18/22
Date Printed: 09/06/22
First Reported: 09/06/22

Job ID/Site: ESC.7315; KAB Comprehensive Hazmat, Kadena Air Base, Okinawa, Japan,
Bldg. 4041

Date(s) Collected: 04/05/2022

SGSFL Job ID: L1232-1
Total Samples Submitted: 25
Total Samples Analyzed: 25

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4041-A1	51553745						
Layer: Black Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4041-A2	51553746						
Layer: Grey Cementitious Material			ND				
Layer: Paint/Coating			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4041-A3	51553747						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint/Coating			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4041-A4	51553748						
Layer: Grey Cementitious Material			ND				
Layer: Paint/Coating			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4041-A5	51553749						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint/Coating			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4041-A6	51553750						
Layer: Black Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333883

Date Printed: 09/06/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4041-A7	51553751						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint/Coating			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4041-A8	51553752						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint/Coating			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4041-A9	51553753						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4041-A10	51553754						
Layer: Paint			ND				
Layer: Grey Non-Fibrous Material			ND				
Layer: Tan Mastic			ND				
Layer: Paint			ND				
Layer: Fibrous Backing			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (3 %)							
KAB-4041-A11	51553755						
Layer: Off-White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (20 %) Fibrous Glass (2 %)							
KAB-4041-A12	51553756						
Layer: Beige Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (35 %) Fibrous Glass (45 %)							
KAB-4041-A13	51553757						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (2 %) Fibrous Glass (90 %)							
KAB-4041-A14	51553758						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333883

Date Printed: 09/06/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4041-A15	51553759						
Layer: Paint			ND				
Layer: Grey Non-Fibrous Material			ND				
Layer: Tan Mastic			ND				
Layer: Paint			ND				
Layer: Fibrous Backing			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (3 %)							
KAB-4041-A16	51553760						
Layer: Off-White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %) Fibrous Glass (2 %)							
KAB-4041-A17	51553761						
Layer: Off-White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %) Fibrous Glass (2 %)							
KAB-4041-A18	51553762						
Layer: Beige Ceramic Tile			ND				
Layer: Grey Grout			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4041-A19	51553763						
Layer: Off-White Ceramic Tile			ND				
Layer: Grey Grout			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4041-A20	51553764						
Layer: White Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4041-A21	51553765						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4041-A22	51553766						
Layer: Off-White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %) Fibrous Glass (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333883

Date Printed: 09/06/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4041-A23	51553767						
Layer: Grey Non-Fibrous Material			ND				
Layer: Tan Mastic			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4041-A24	51553768						
Layer: Yellow Fibrous Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace) Fibrous Glass (90 %)							
KAB-4041-A25	51553769						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace) Fibrous Glass (90 %)							



Tiffani Ludd, Laboratory Supervisor, Carson Laboratory

Note: Limit of Quantification ('LOQ') = 1%. 'Trace' denotes the presence of asbestos below the LOQ. 'ND' = 'None Detected'.

Analytical results and reports are generated by SGS Forensic Laboratories (SGSFL) at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGSFL to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGSFL. The client is solely responsible for the use and interpretation of test results and reports requested from SGSFL. SGSFL is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. All samples were received in acceptable condition unless otherwise noted.

APPENDIX C
LBP SAMPLING CHAIN OF CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 25 April 2022			
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day ☒				
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer					
Site Name: Kadena Air Base, Okinawa, Japan		☐ PLM: ☐ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435					
Site Location: Bldg. 4041		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)					
Comments: See attached sample logs.		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project					
		☒ Metals Analysis Matrix: Paint Chip Method: Analytes: Lead					
		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only					
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4041-L1-L13		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.			A P C				
Sampled By: Sean R Dudney		Date/Time: 4/5/22	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS				
Relinquished By: Sean R Dudney		Relinquished By:		Relinquished By:			
Date / Time: 25 April 2022		Date / Time:		Date / Time:			
Received By: Japan Post Office		Received By: <i>all m...</i>		Received By:			
Date / Time: 25 April 2022		Date / Time: 5-26-22 14:00 USPT		Date / Time:			
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No			

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

SUMMARY REPORTING SHEET (V2.0)

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA							
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BLDG NUMBE	ROOM OR FUNCTIONA	HOMOGENEOU S /SURFACE	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRIA	ACC	CONDITIO N	G O	C T	LAB/ SAMPLE ID	CTR COC	LAB ANALYSIS	RESULT	VALIDITY DETERMINATI	NOTES / INVALID OR UNKNOWN REASONING
4041	Exterior	S1	Metal flange	Light blue		Y	Used				KAB-4041-L1				
		S2	Metal pipe	Red over white		Y	Use				KAB-4041-L2				
		S3	Concrete wall	Off white		Y	Use				KAB-4041-L3				
		S4	Concrete wall base	Orange		Y	Used				KAB-4041-L4				
		S3	Concrete wall	Off white		Y	Use				KAB-4041-L5				
		S3	Concrete wall	Off white		Y	Use				KAB-4041-L6				
	Reception	S5	Gypsum board wall	Gray		Y	Use				KAB-4041-L7				
		S6	Metal pipe (above ceiling)	Red		Y	Use				KAB-4041-L8				
	XO rm	S7	Gypsum board wall	White		Y	Use				KAB-4041-L9				
		S8	Wooden door	Black		Y	Use				KAB-4041-L10				
	Men's bathroom	S9	Ceiling board	White		N	Use				KAB-4041-L11				
		S10	Gypsum board wall	Red		Y	Use				KAB-4041-L12				
	Storage	S11	Wooden door	White		Y	Used				KAB-4041-L13				

Metals Analysis of Paints

(AIHA-LAP, LLC Accreditation, Lab ID #101762)

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: M242210
Date Received: 05/26/22
Date Analyzed: 06/22/22
Date Printed: 06/22/22
First Reported: 06/22/22

Job ID / Site: ESC.7315; KAB Comprehensive Hazmat, Kadena Air Base, Okinawa, Japan,
Bldg. 4041

SGSFL Job ID: L1232-1

Date(s) Collected: 04/05/22

Total Samples Submitted: 13

Total Samples Analyzed: 13

Sample Number	Lab Number	Analyte	Result	Result Units	Reporting Limit*	Method Reference
KAB-4041-L1	LM220646	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4041-L2	LM220647	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4041-L3	LM220648	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4041-L4	LM220649	Pb	0.028	wt%	0.007	EPA 3050B/7000B
KAB-4041-L5	LM220650	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4041-L6	LM220651	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4041-L7	LM220652	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4041-L8	LM220653	Pb	< 0.01	wt%	0.01	EPA 3050B/7000B
KAB-4041-L9	LM220654	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4041-L10	LM220655	Pb	< 0.02	wt%	0.02	EPA 3050B/7000B
KAB-4041-L11	LM220656	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4041-L12	LM220657	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4041-L13	LM220658	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B

* The Reporting Limit represents the lowest amount of analyte that the laboratory can confidently detect in the sample, and is not a regulatory level. The Units for the Reporting Limit are the same as the Units for the Final Results.



Kevin Poon, Laboratory Analyst, Hayward Laboratory

Analytical results and reports are generated by SGS Forensic Laboratories at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGS Forensic Laboratories to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGS Forensic Laboratories. The client is solely responsible for the use and interpretation of test results and reports requested from SGS Forensic Laboratories. SGS Forensic Laboratories is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. Any modifications that have been made to referenced test methods are documented in SGS Forensic Laboratories' Standard Operating Procedures Manual. Sample results have not been blank corrected. Quality control and sample receipt condition were acceptable unless otherwise noted.

Note* Sampling data used in this report was provided by the client as noted on the associated chain of custody form.

ASBESTOS SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

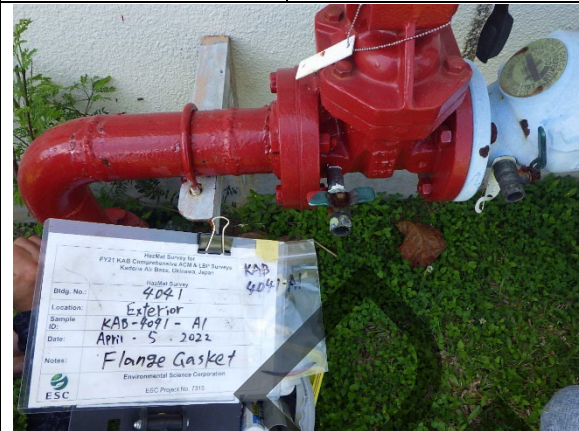
HazMat Survey

Bldg. 4041



Photo Record Sheet

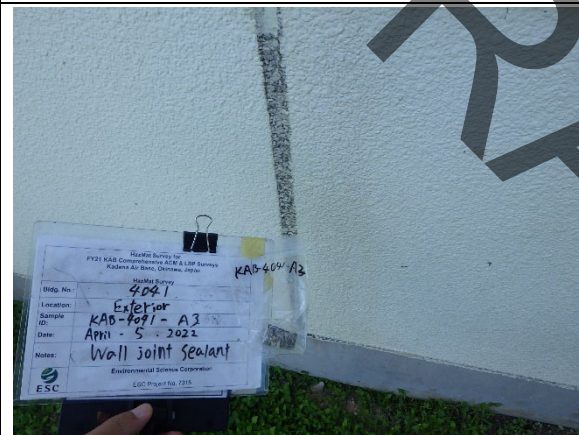
Asbestos Sampling
Kadena Air Base, Bldg.4041



KAB-4041-A1



KAB-4041-A2



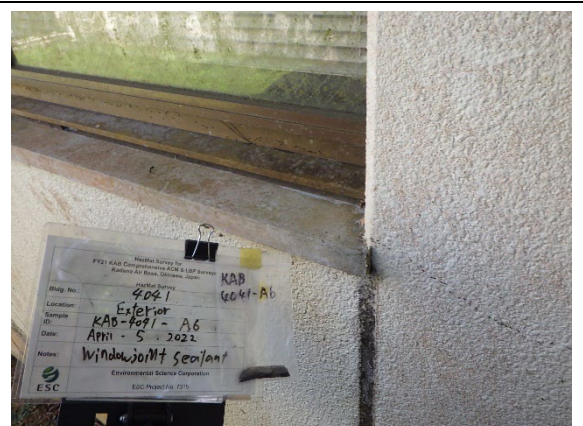
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KAB-4041-A4



KAB-4041-A5



KAB-4041-A6

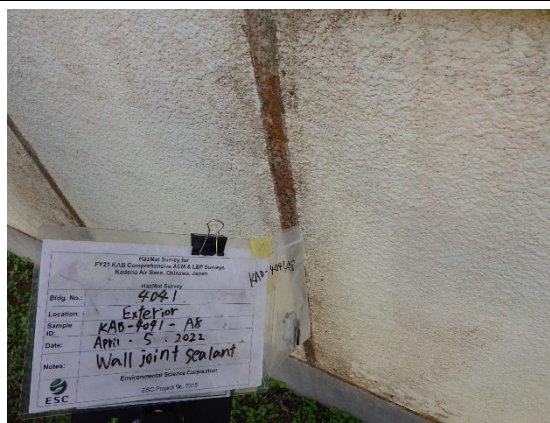


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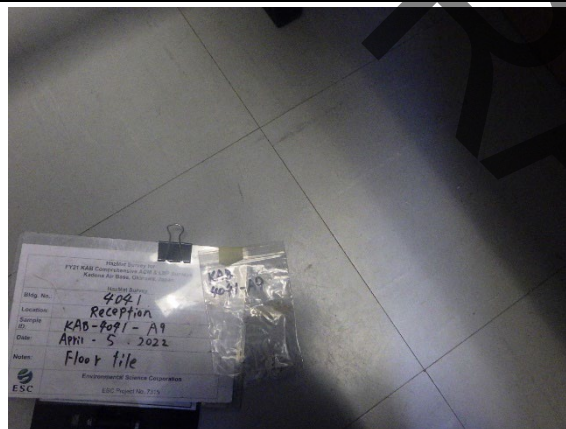
Asbestos Sampling
Kadena Air Base, Bldg.4041



KAB-4041-A7



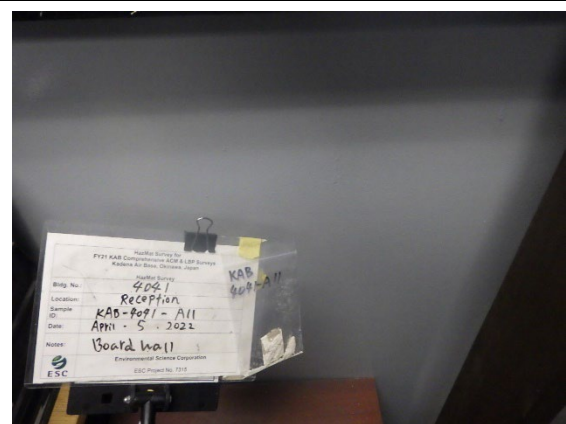
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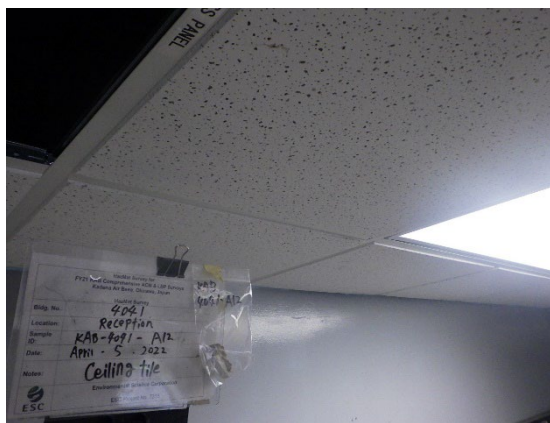
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KAB-4041-A10



KAB-4041-A11

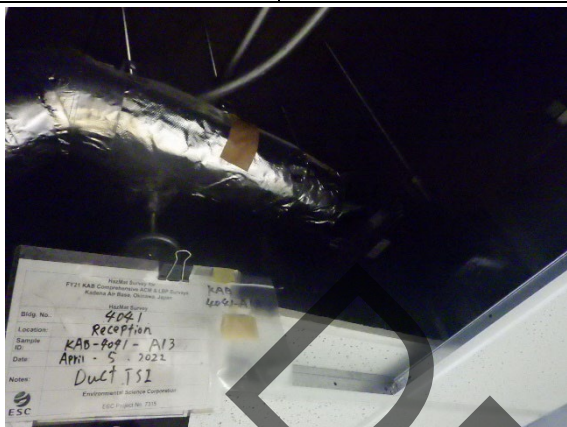


KAB-4041-A12

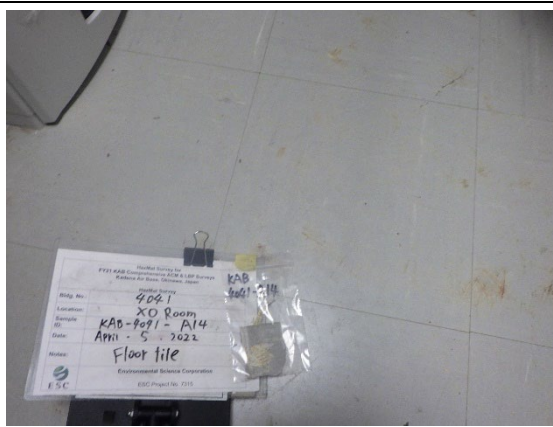


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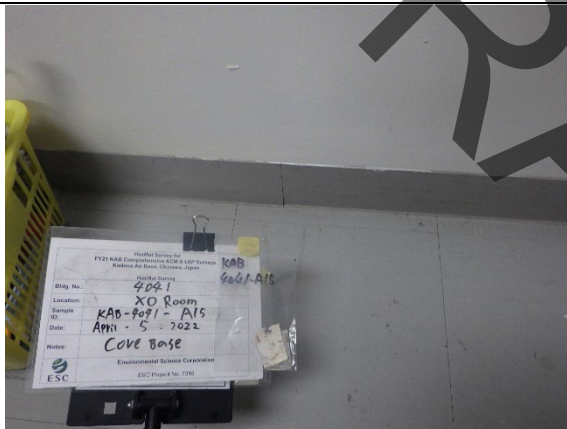
Asbestos Sampling
Kadena Air Base, Bldg.4041



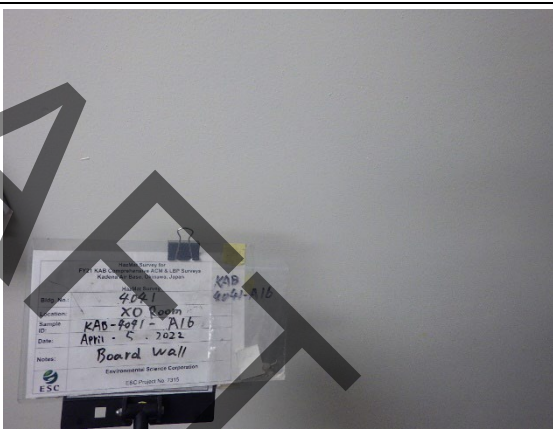
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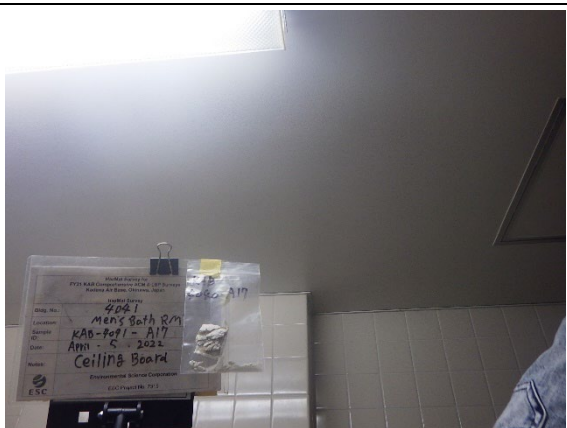
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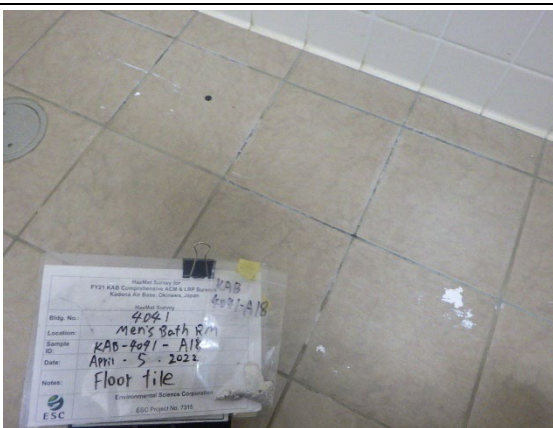
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KAB-4041-A16



KAB-4041-A17



KAB-4041-A18